Mathematics

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Contents

Notation List 1

I Linear and Operator Structures 6

1 Structural Linear Algebra 6
1.1 Vector Spaces and Linear Structure . . . 6
1.2 Direct Sums, Quotients, and Decomposition . . . 11
1.3 Linear Maps and Representations . . . 14
1.4 Duality and Multilinear Structure . . . 26

2 Geometric Linear Algebra 31
2.1 Inner Product Geometry . . . 31
2.2 Orthogonal Complements and Projections . . . 39
2.3 Least Squares and Metric Optimization . . . 42
2.4 Adjoints and Orthogonal Operators . . . 44

3 Spectral and Operator Structure 50
3.1 Eigenstructure and Invariant Subspaces . . . 50
3.2 Polynomial Structure of Operators . . . 57
3.3 Jordan and Canonical Forms . . . 68
3.4 Spectral Theory of Structured Operators . . . 78
3.5 Operator Functions . . . 86

4 Matrix Analysis and Numerical Structure 88
4.1 Matrix Classes and Invariants . . . 89
4.2 Matrix Decompositions . . . 107
4.3 Matrix Norms and Conditioning . . . 113
4.4 Perturbation and Stability . . . 118
4.5 Numerical Linear Algebra . . . 124

II Algebraic Structures 129

5 Algebraic Language and Categories 129
5.1 Sets, Maps, Relations, and Universal Properties . . . 129
5.2 Categories and Functors . . . 129
5.3 Algebraic Structures as Objects . . . 129
5.4 Exactness and Diagrams . . . 129

6 Group Theory 129
6.1 Groups, Subgroups, and Homomorphisms . . . 129
6.2 Group Actions . . . 129
6.3 Normality and Structure Theorems . . . 129
6.4 Sylow Theory and Finite Groups . . . 129
6.5 Representation-Theoretic Preview . . . 129

7 Rings, Ideals, and Modules 129
7.1 Rings and Ideals . . . 129
7.2 Polynomial Rings and Factorization . . . 129
7.3 Modules . . . 129
7.4 Structure Theorems . . . 129
7.5 Linear Operators as Modules . . . 129

8 Fields and Galois Theory 129
8.1 Field Extensions . . . 129
8.2 Splitting Fields and Normality . . . 129
8.3 Galois Groups . . . 129
8.4 Classical Applications . . . 129
8.5 Scalar Extension in Linear Algebra . . . 129

9 Tensor, Exterior, and Homological Algebra **129**
9.1 Tensor Products and Universal Properties . . . 129
9.2 Symmetric and Exterior Algebras . . . 129
9.3 Multilinear Algebra Revisited . . . 129
9.4 Chain Complexes and Homology . . . 129
9.5 Derived Functors Preview . . . 129

III Algebraic Geometry and Commutative Algebra **130**

10 Commutative Algebra **130**
10.1 Prime Ideals and Spectra . . . 130
10.2 Localization and Local Rings . . . 130
10.3 Noetherian Rings and Modules . . . 130
10.4 Integral Dependence and Dimension . . . 130
10.5 Regularity and Singularities Preview . . . 130

11 Algebraic Geometry **130**
11.1 Affine Algebraic Sets . . . 130
11.2 Morphisms and Coordinate Rings . . . 130
11.3 Projective Geometry . . . 130
11.4 Schemes . . . 130
11.5 Sheaves and Cohomology Preview . . . 130
11.6 Smoothness, Tangent Spaces, and Dimension . . . 130
11.7 Divisors, Line Bundles, and Curves . . . 130

IV Analysis and Continuous Structure **131**

12 Foundations of Analysis **131**
12.1 Mathematical Language, Sets, and Functions . . . 131
12.2 Number Systems and Completeness . . . 135
12.3 Cardinality and Infinite Sets . . . 142
12.4 Real Sequences and Completeness Methods . . . 146

13 Metric and Topological Structure **150**
13.1 Metric Spaces and Euclidean Topology . . . 150
13.2 Compactness and Completeness . . . 156
13.3 Connectedness and Continuity . . . 166
13.4 Selected General Topology . . . 174

14 Convergence and Function Spaces **175**
14.1 Infinite Series and Summability . . . 175
14.2 Uniform Convergence . . . 181
14.3 Function-Space Compactness . . . 184
14.4 Power Series and Approximation . . . 188
14.5 Fourier Structure . . . 193

15 Differential and Geometric Analysis **195**
15.1 One-Variable Differentiation and Integration . . . 195
15.2 Several-Variable Differentiation . . . 208
15.3 Inverse, Implicit, and Rank Theorems . . . 213
15.4 Differential Forms and Manifolds . . . 215
15.5 Stokes Theory . . . 220

16 Measure and Integration Structure **222**
16.1 Sigma-Algebras and Measures . . . 222
16.2 Measurable Functions . . . 225
16.3 Lebesgue Integration . . . 227
16.4 Convergence Theorems . . . 229
16.5 Product Measures and Fubini Theory . . . 229
16.6 Radon-Nikodym and Differentiation . . . 231

16.7 Lp Spaces . . . 233

17 Functional and Operator Analysis **234**
17.1 Normed and Banach Spaces . . . 234
17.2 Hilbert Spaces . . . 235
17.3 Core Functional Analytic Theorems . . . 236
17.4 Weak Topologies and Duality . . . 236
17.5 Compact Operators . . . 236
17.6 Spectral Theory of Operators . . . 237
17.7 Distributions and Sobolev Preview . . . 237

V Probability and Stochastic Structure **238**

18 Probability Spaces and Convergence **238**
18.1 Probability Spaces and Random Variables . . . 238
18.2 Expectation and Integration in Probability . . . 238
18.3 Conditional Probability and Conditional Expectation . . . 238
18.4 Modes of Probabilistic Convergence . . . 238
18.5 Laws of Large Numbers . . . 238
18.6 Central Limit Theory . . . 238
18.7 Probability on Metric Spaces . . . 238

19 Stochastic Processes **238**
19.1 Process-Level Probability . . . 238
19.2 Filtrations and Adapted Processes . . . 238
19.3 Martingales in Discrete and Continuous Time . . . 238
19.4 Markov Chains and Markov Processes . . . 238
19.5 Stationary Processes and Ergodicity . . . 238
19.6 Gaussian Processes . . . 238
19.7 Point Processes and Renewal Structure . . . 238

20 Stochastic Calculus and Dynamics **238**
20.1 Brownian Motion . . . 238
20.2 Stochastic Integration . . . 238
20.3 Itô Calculus . . . 238
20.4 Stochastic Differential Equations . . . 238
20.5 Change of Measure and Girsanov Theory . . . 238
20.6 Diffusions and PDE Connections . . . 238
20.7 Stochastic Flows and Advanced Dynamics . . . 238

VI Dynamical and PDE Systems **239**

21 Dynamical Systems and ODEs **239**
21.1 ODE Foundations and Flows . . . 239
21.2 Linear Systems and Stability . . . 239
21.3 Nonlinear Stability Theory . . . 239
21.4 Bifurcation Theory . . . 239
21.5 Dynamical Systems Structure . . . 239
21.6 Semigroups and Evolution Viewpoint . . . 239

22 PDE and Evolution Equations **239**
22.1 First-Order PDE and Characteristics . . . 239
22.2 Elliptic Equations . . . 239
22.3 Parabolic Equations . . . 239
22.4 Hyperbolic Equations . . . 239
22.5 Sobolev Spaces for PDE . . . 239
22.6 Weak Solutions and Variational Methods . . . 239
22.7 Evolution Equations . . . 239

VII Numerical and Computational Mathematics **240**

23 Numerical Analysis **240**
23.1 Floating Point and Error Analysis . . . 240
23.2 Nonlinear Equations and Optimization Basics . . . 240
23.3 Interpolation and Approximation . . . 240
23.4 Numerical Differentiation and Quadrature . . . 240
23.5 Numerical Stability and Convergence . . . 240

24 Numerical Differential Systems **240**
24.1 Numerical ODEs . . . 240
24.2 Geometric Numerical Integration . . . 240
24.3 Finite Difference PDE Methods . . . 240
24.4 Finite Element Methods . . . 240
24.5 Numerical SDEs . . . 240
24.6 Computational Stability Across ODE/PDE/SDE . . . 240

25 Monte Carlo and Stochastic Computation **240**
25.1 Monte Carlo Foundations . . . 240
25.2 Variance Reduction . . . 240
25.3 Markov Chain Monte Carlo . . . 240
25.4 Sequential Monte Carlo . . . 240
25.5 Rare-Event Simulation . . . 240
25.6 Stochastic Optimization . . . 240

VIII Statistical and Learning Systems **241**

26 Statistical Learning **241**
26.1 Statistical Inference Foundations . . . 241
26.2 Risk Minimization . . . 241
26.3 Linear and Generalized Linear Models . . . 241
26.4 Regularization and Model Selection . . . 241
26.5 Kernels and Nonlinear Methods . . . 241
26.6 Trees, Ensembles, and Boosting . . . 241
26.7 Generalization Theory . . . 241

27 Deep Learning and Representation Systems **241**
27.1 Neural Network Foundations . . . 241
27.1.1 Deep Feedforward Networks . . . 241
27.2 Optimization for Deep Learning . . . 247
27.3 Regularization and Generalization . . . 255
27.4 Representation Learning . . . 260
27.5 Sequence Models . . . 260
27.6 Generative Models . . . 260
27.7 Modern Architecture and Systems View . . . 260
27.7.1 Convolutional Neural Networks . . . 260

28 Time Series and Sequential Systems **262**
28.1 Time Series Foundations . . . 262
28.1.1 Stationary Time Series . . . 262
28.1.2 Univariate Time Series Models . . . 262
28.2 ARMA and ARIMA Structure . . . 264
28.3 Spectral Methods . . . 264
28.4 State-Space Models . . . 264
28.5 Nonlinear and Volatility Models . . . 264
28.6 Long Memory and Multiscale Structure . . . 264
28.7 Sequential Inference and Filtering . . . 264

Notation List

General Notation

Convention. Notation is scoped by subject, argument type, font, and any local definition in force. When a glyph is reused, the distinguishing subscript, superscript, font, or argument is recorded below.

Set, Function, and Logical Notation

$x \in E,\ x \notin E$	membership / non-membership
$E \subseteq F,\ \emptyset$	subset relation / empty set
$X \times Y,\ f : X \to Y$	Cartesian product / function from X to Y
$\Longrightarrow,\ \Longleftrightarrow$	implies / if and only if
$:=$	defined to be equal to

Number, Size, and Scalar Notation

$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	standard number systems
$\lvert x\rvert,\ \lVert x\rVert$	absolute value or complex modulus for scalars / norm of a vector, function, or operator
a, b, c	generic scalars or interval endpoints, with meaning determined by context

Linear Algebra Notation

Vector Spaces and Lists

$\mathbb{F}$	field of scalars, usually $\mathbb{R}$ or $\mathbb{C}$
$\mathbb{F}^{(\mathbb{N})}$	space of finitely nonzero sequences over $\mathbb{F}$
$\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{0}$	vectors and the zero vector
$\mathcal{U}, \mathcal{V}, \mathcal{W}$	vector spaces or subspaces, depending on context
$I_{\mathcal{V}},\ I_n,\ 0$	identity operator on $\mathcal{V}$, identity matrix, and zero map or matrix; not the cell I^k or interval I_R
$\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$	finite ordered list, often a basis
$(),\ \beta_{\widehat{j}}$	empty list / list β with the jth vector omitted
$(\beta, \mathbf{v}),\ \beta_{j \leftarrow \mathbf{v}}$	list with $\mathbf{v}$ appended / list with jth entry replaced by $\mathbf{v}$
$\operatorname{span}(E),\ \dim(\mathcal{V})$	span of E / dimension of $\mathcal{V}$
$\mathcal{U} + \mathcal{W},\ \mathcal{U} \oplus \mathcal{W},\ \mathcal{V}/\mathcal{U}$	sum, direct sum, quotient space
$\mathcal{V}_1 \times \cdots \times \mathcal{V}_m$	product vector space with componentwise operations
$\mathcal{V} \cong \mathcal{W},\ \mathbf{e}_{j,k}$	isomorphic vector spaces / basis vector placed in one product component
$\mathbf{v} + \mathcal{U},\ \pi(\mathbf{v})$	coset modulo $\mathcal{U}$ / quotient map $\mathcal{V} \to \mathcal{V}/\mathcal{U}$ when introduced

Linear Maps and Duality

$T : \mathcal{V} \to \mathcal{W},\ \mathcal{L}(\mathcal{V}, \mathcal{W})$	linear map / space of linear maps
$ST,\ T\lvert_{\mathcal{U}},\ T/\mathcal{U}$	composition $S \circ T$, restriction, and quotient operator
$T^{-1},\ T^m,\ p(T)$	inverse of a bijective map or operator, operator power, and polynomial applied to an operator
$\ker(T),\ \operatorname{range}(T)$	kernel and range of T
$\mathcal{V}_{\mathbb{C}},\ T_{\mathbb{C}}$	complexification of a real vector space / complexification of a real operator
$\operatorname{graph}(T),\ [S, T]$	graph of a map / commutator $ST - TS$
$\operatorname{Comm}(\mathcal{F}),\ \operatorname{Comm}(A)$	commutant of a matrix family / commutant of one matrix
$\mathcal{V}',\ \beta',\ T'$	dual space, dual basis, and dual map; prime here is not derivative notation
$\varphi, \psi,\ \mathcal{U}^0$	linear functionals / annihilator
$\mathcal{V}'',\ \Lambda_{\mathcal{V}}$	double dual / natural evaluation map
$[\mathbf{v}]_\beta,\ [T]_{\gamma \leftarrow \beta}$	coordinate vector with respect to basis β / matrix representation
$C_{\gamma \leftarrow \beta}$	change-of-basis matrix from β-coordinates to γ-coordinates

Matrices and Entrywise Structure

$A, B, C,\ \mathrm{Mat}(m, n, \mathbb{F})$	matrix variables in linear-algebra contexts / $m \times n$ matrices over $\mathbb{F}$
$A_{ij},\ A_{j,\cdot},\ A_{\cdot,k}$	matrix entry, jth row, kth column
$[a_{ij}],\ E_{j,k},\ A\mathbf{x}$	matrix written by entries / matrix unit / matrix action
$A^{-1},\ A^t,\ A^T,\ A^*$	inverse, transpose, alternate transpose notation, and adjoint or conjugate transpose
$\mathrm{diag}(\lambda_1, \ldots, \lambda_n)$	diagonal matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$
$\overline{z},\ \mathrm{Re}\, z,\ \mathrm{Im}\, z$	complex conjugate, real part, and imaginary part; overline on sets means closure
$\mathrm{tr}(A), \det(A)$	trace and determinant
$\mathfrak{S}_n,\ \mathrm{sign}(\sigma),\ \mathrm{adj}(A)$	permutation group, permutation sign, adjugate matrix
$M_{ij},\ C_{ij},\ A_j(\mathbf{b})$	minor matrix, cofactor, and the matrix obtained from A by replacing column j with $\mathbf{b}$
$\mathrm{col}(A),\ \mathrm{row}(A)$	column space and row space of A
$\mathrm{rank}(A),\ \mathrm{nullity}(A)$	rank and nullity of A
$A \oplus B,\ A \otimes B$	matrix direct sum / Kronecker product; vector-space direct sums use subspace notation
$A \otimes I_q + I_n \otimes B$	Kronecker sum of square matrices A and B in compatible dimensions
$A \circ B,\ \mathrm{vec}(A)$	Hadamard product / column-stacking vectorisation
$M/A,\ M/D$	Schur complement of the block A or D in M; not quotient-space notation
$B,\ L = B^tB$	directed incidence matrix and incidence Laplacian when a graph context is active

Inner Product Geometry

$\|\mathbf{v}\|,\ \|T\|_{\mathrm{op}}$	vector norm / operator norm
$\|\cdot\|_1,\ \|\cdot\|_2,\ \|\cdot\|_\infty$	coordinate one-norm, Euclidean norm on vectors or spectral norm on matrices/operators, and sup norm
$\|\cdot\|_F$	Frobenius norm
$\langle \mathbf{u}, \mathbf{v} \rangle,\ \langle A, B \rangle_F$	inner product / Frobenius inner product
$\mathbf{u} \perp \mathbf{v},\ \mathcal{U}^\perp,\ \delta_{ij}$	orthogonal vectors, orthogonal complement, Kronecker delta
$\mathrm{proj}_{\mathbf{v}}(\mathbf{u}),\ P_{\mathcal{U}}$	projection of $\mathbf{u}$ onto the line spanned by $\mathbf{v}$ / orthogonal projection onto $\mathcal{U}$
$d(\mathbf{v}, \mathcal{U})$	distance from $\mathbf{v}$ to the set or subspace $\mathcal{U}$
$\mathrm{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$	Gram matrix of $\mathbf{u}_1, \ldots, \mathbf{u}_n$
$\mathrm{GramDet}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$	Gram determinant
$P_{\mathbf{u}},\ U_{\mathbf{w}}$	rank-one projection matrix / Householder matrix, both determined by the vector subscript
$\gamma = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$	often an orthonormal basis
$\mathbf{e} = (1, \ldots, 1)^t$	all-ones vector
$\mathrm{Ray}_A(\mathbf{x}),\ \mathbf{x}^* A \mathbf{x}$	Rayleigh quotient notation or unit-vector form / Hermitian quadratic form

Spectral and Polynomial Structure

$E_\lambda(T),\ \mathrm{spec}(T)$	eigenspace for eigenvalue λ / spectrum
$\lambda_i(A),\ m_A(\lambda)$	eigenvalues in the locally declared order / algebraic multiplicity of λ as a root of p_A, not measure $m(E)$
$\mathcal{G}_\lambda(T),\ J_k(\lambda)$	generalized eigenspace / Jordan block of size k
$p_A,\ q_T$	matrix characteristic polynomial / operator characteristic polynomial
$\mu_A,\ \mu_T,\ w_p(\lambda, A)$	minimal polynomials subscripted by matrix or operator / Weyr rank-drop count; not measure $\mu(E)$
$P_j,\ T = \sum_j \mu_j P_j$	spectral projections in a spectral resolution / corresponding decomposition with distinct eigenvalues μ_j
$\sqrt{T},\ A^{1/2},\ \rho(A)$	positive square root / spectral radius of A, not the interval parameter in K_ρ
$\mathbb{F}[z]$	polynomials in z over $\mathbb{F}$
e^{tA}	matrix exponential, defined by the convergent power series in tA

Matrix Analysis and Numerical Structure

$A \succeq 0,\ A \succ 0$	positive semidefinite and positive definite matrices
$\operatorname{in}(A),\ n_+(A), n_-(A), n_0(A)$	inertia and positive, negative, zero eigenvalue counts
$A = QR,\ A = LL^*,\ A = LDL^t$	QR factorization, Cholesky factorization, and symmetric LDL^t factorization
$A = V\Sigma W^*,\ A = X\Sigma_r Y^*$	singular value decomposition / compact SVD
$\sigma_j(A),\ \sigma_{\max}(A),\ \sigma_{\min}(A)$	singular values and extreme singular values
$\lvert T\rvert,\ \lvert A\rvert,\ \operatorname{vol}(\Omega)$	operator or matrix modulus / volume of a set; scalar absolute value remains $\lvert x\rvert$
$A^\dagger,\ A_k$	Moore–Penrose pseudoinverse / rank-k truncated SVD approximation
$\kappa_2(A),\ \Delta A,\ \Delta\mathbf{x}$	spectral condition number / matrix and solution perturbations
$\Delta(A)$	defect from normality, not a perturbation of A
$R_k(A),\ R_k^{\text{off}}(A),\ R_{\max}(A)$	absolute row sum, deleted row sum, and maximum absolute row sum
$\operatorname{Ger}_k(A),\ \operatorname{Ger}(A)$	Gersgorin disk and Gersgorin union
$\operatorname{tridiag}(a, b, c)$	tridiagonal matrix with subdiagonal a, diagonal b, and superdiagonal c
$w,\ F_n,\ Q_n,\ \omega_n$	half-bandwidth / Fourier matrix / normalized Fourier matrix / primitive nth root used for DFT
$\mathcal{K}_m(A, \mathbf{b})$	Krylov subspace generated by $(A, \mathbf{b})$
$Q_m,\ H_m$	Krylov basis matrix / projected Hessenberg or tridiagonal matrix
$A = S - T,\ B = S^{-1}T,\ \mathbf{e}_k$	matrix splitting, iteration matrix, and stationary-method error vector
$A = D + L + U,\ B_J,\ B_{\text{GS}}$	diagonal/lower/upper splitting and Jacobi/Gauss–Seidel iteration matrices
$\omega,\ S_\omega,\ T_\omega,\ B_\omega$	SOR parameter, SOR splitting matrices, and SOR iteration matrix
$M,\ M^{-1}A,\ AM^{-1}$	preconditioner and left/right preconditioned matrices

Analysis Notation

Sets and Functions in Analysis

$\mathcal{P}(X)$	power set of X; projection symbols use $P_{\mathcal{U}}$ or P_j
$f(A),\ f^{-1}(B)$	image of A / preimage of B under f
$\operatorname{range}(f),\ f\lvert_A$	range of f / restriction of f to A
$\{A_i\}_{i\in I},\ \bigcup_{i\in I} A_i,\ \bigcap_{i\in I} A_i$	indexed family of sets / indexed union / indexed intersection
$g \circ f,\ \operatorname{id}_X$	composition of functions / identity function on X
$\#E,\ A \sim_{\#} B,\ [x]$	cardinality of a finite set / same cardinality / equivalence class of x for a specified relation

Order and Completeness

$\sup E,\ \inf E$	supremum / infimum of an ordered subset
$\lvert x\rvert,\ \lvert a - b\rvert$	real or complex absolute value / distance on the real line
$\overline{\mathbb{R}}$	extended real line $\mathbb{R} \cup \{-\infty, +\infty\}$; the bar is conventional, not topological closure
$x < y,\ x \le y$	strict order / non-strict order

Sequences and Series

$(a_n),\ a_n \to a$	sequence / convergence of a sequence
$(a_{n_k}),\ n_1 < n_2 < \cdots$	subsequence indexed by an increasing sequence of positive integers
$s_n = \sum_{k=1}^{n} a_k$	partial sums of a series
$\limsup a_n = s^*,\ \liminf a_n = s_*$	upper / lower limits of a real sequence
$\sum_{n=1}^{\infty} a_n,\ \prod_{n=1}^{\infty} a_n$	infinite series / infinite product

Metric and Topological Notation

$d(x,y),\ B_r(x)$	metric distance / italic open ball of radius r centered at x
$B_\varepsilon(a) = V_\varepsilon(a)$	real-line open ball $(a-\varepsilon, a+\varepsilon)$; $V_\varepsilon(a)$ is Abbott's notation
$\overline{B}_r(x),\ \{G_\alpha\}_{\alpha\in A}$	closed ball / indexed open cover
$d_E,\ d_{\max},\ d_{\text{sum}}$	Euclidean, maximum, and sum product metrics
$E^\circ,\ \overline{E},\ E^{\text{acc}}$	interior, topological closure, and limit-point set of E
$\text{bd}(E),\ E^c$	topological boundary and complement of E
$\text{diam}\, E$	diameter of a metric set
$X \simeq Y$	homeomorphic metric spaces
$A \cap \overline{B} = \emptyset,\ \overline{A} \cap B = \emptyset$	separation condition for sets A and B
$F_\sigma,\ G_\delta,$ meager/thin/first category	countable union of closed sets / countable intersection of open sets / countable union of nowhere-dense sets
residual/thick, generic	set containing a countable intersection of dense open sets / property holding on a residual set

Functions, Differentiation, and Integration

$\lim_{x\to p} f(x),\ f(x+),\ f(x-)$	function limit / right and left limits
$C(X),\ C([a,b]),\ C_b(X),\ \|f\|_\infty$	continuous function spaces named by domain / bounded continuous functions / supremum norm
$f_n \to f,\ f_n \rightrightarrows f$	pointwise convergence / uniform convergence
$f'(x),\ Df(a),\ D_j f_i,\ \partial_i f,\ J_f(x)$	one-variable derivative, total derivative, jth partial of the ith component, scalar partial derivative, and Jacobian determinant
$\nabla f,\ D_{\mathbf{u}} f,\ C^r,\ C^\infty,\ C^\omega$	gradient, directional derivative, r times continuously differentiable, smooth, and analytic classes
$\Pi = \{x_0, \dots, x_n\},\ c_i,\ \Delta x_i,\ \Delta\alpha_i$	partition points, tags, and interval/integrator increments
$R(f, \Pi, \{c_i\}),\ \delta,\ \delta(x)$	tagged Riemann sum / constant mesh bound / gauge; this uppercase $R(\cdots)$ is local to Riemann sums
$U(\Pi, f, \alpha),\ L(\Pi, f, \alpha)$	upper and lower Riemann–Stieltjes sums
$f \in \mathcal{R},\ f \in \mathcal{R}(\alpha)$	Riemann integrability class / Riemann–Stieltjes integrability class with respect to α
$\int_a^b f,\ \int_a^b f\, d\alpha,\ \int_E f\, d\mu$	Riemann integral, Riemann–Stieltjes integral, and measure integral
$\int_a^\infty f,\ \int_{-\infty}^\infty f$	one-sided and two-sided improper integrals, defined by truncation limits
$\ell(\Pi, \gamma),\ \ell(\gamma)$	polygonal length for a partition / length of a curve
$I^k,\ Q^k,\ \text{supp}(f)$	k-cell, standard k-simplex, and support of f; I^k is not an identity operator
$dx_I,\ dx_I \wedge dx_J$	standard basic form for a multi-index I / wedge product of basic forms
$\omega,\ d\omega,\ T^*\omega,\ \partial_{\text{ch}}\Psi$	differential form, exterior derivative, pullback by T, and boundary of a chain
$\int_\Psi \omega,\ \partial_{\text{ch}}[\mathbf{p}_0, \dots, \mathbf{p}_k]$	integration of a form over a chain / oriented simplex boundary
$dV,\ dA,\ ds$	volume, area, and arc-length elements

Power Series, Fourier, and Special Functions

$\sum c_n(x-a)^n,\ R$	power series about a / radius of convergence in this context, unrelated to $R(\cdots)$ or the R in QR
$I_R,\ K_\rho$	power-series interval $I_R = (-R, R)$ / closed inner interval $K_\rho = [-\rho, \rho] \subset I_R$
$\widehat{f}(m),\ s_N(f;x),\ s_N(f)$	trigonometric Fourier coefficient / Nth Fourier partial sum value or function
$f \sim_{\text{coef}} \sum \widehat{f}(n)e^{inx},\ f \sim_{\text{coef}} \sum c_n\phi_n$	formal Fourier or orthonormal expansion associated to coefficients
$D_N(t)$	Dirichlet kernel $\sum_{m=-N}^N e^{imt}$; uppercase D_N is not derivative notation
$\mathsf{E},\ \mathsf{L},\ \mathsf{C},\ \mathsf{S}$	power-series exponential, logarithm inverse, cosine, and sine
$\Gamma(x),\ \mathrm{B}(x,y)$	gamma function / roman beta function, distinct from balls $B_r(x)$ and matrix variables B

Measure and Function Spaces

$E \in \mathcal{M},\ E \in \mathcal{M}(\mu)$	measurable set in a measure space / Rudin measurable set generated by μ
$\mathcal{B}(\mathbb{R}^n),\ \mathcal{L}^n$	Borel σ-algebra on $\mathbb{R}^n$ / Lebesgue measurable subsets of $\mathbb{R}^n$
$\mathcal{E},\ \mathcal{M}_F(\mu),\ A \triangle B,\ d_\mu(A,B)$	elementary sets, finitely μ-measurable sets, symmetric difference, and Rudin set distance on sets
$\mu^*(E),\ m^*(E),\ \mathbf{1}_E$	outer measure of E / Lebesgue outer measure / indicator function of E
$\mathrm{vol}_n(B),\ m_n^*(\Omega)$	n-dimensional box volume / box-cover outer measure $\inf\{\sum_{j\in J} \mathrm{vol}_n(B_j) : (B_j)_{j\in J} \text{ covers } \Omega,\ J \text{ at most countable}\}$
$m(E),\ \mu(E)$	elementary volume or Lebesgue measure for set arguments / measure of E; not algebraic multiplicity $m_A(\lambda)$ or minimal polynomial μ_A
$\bowtie,\ \{f \bowtie a\}$	one of $>, \geq, <, \leq$ / corresponding level set
$f^+,\ f^-,\ \mathcal{S}^+(f),\ \mathcal{I}_E(s)$	positive/negative parts, simple minorants, and simple-function integral
$f \sim_{\text{a.e.}} g$ on E, a.e.	equal except on a measure-zero subset of E / almost everywhere
$\mathscr{L}^1(\mu)$	Lebesgue-integrable functions with respect to μ
$L^p(\mu),\ L^p(E),\ \|f\|_p$	Lebesgue space with respect to a measure or domain, using the ambient measure when only E is shown / L^p norm

Part I
Linear and Operator Structures

Linear algebra moves from structure to geometry, spectra, and computation: vector spaces, maps, and duality; inner products, projections, least squares, and adjoints; invariant subspaces, polynomial structure, and spectral theorems; decompositions, norms, perturbation, and numerics.
References: Garcia and Horn (2017), Axler (2024).

1 Structural Linear Algebra

Structural linear algebra develops the coordinate-free language of vector spaces, decompositions, linear maps, and duality.

1.1 Vector Spaces and Linear Structure

Vector spaces are objects that can be added and scaled. Subspaces, spans, bases, dimension, and coordinates describe how they are generated and measured.

Definition 1.1.1. *Vector Space*
Let $\mathbb{F}$ be a field. A vector space over $\mathbb{F}$ is a set $\mathcal{V}$ together with operations of vector addition and scalar multiplication such that, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$ and all $a, b \in \mathbb{F}$, the following axioms hold:

i. vector addition is commutative: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$;
ii. vector addition is associative: $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$;
iii. there exists an additive identity $\mathbf{0} \in \mathcal{V}$ such that $\mathbf{0} + \mathbf{v} = \mathbf{v}$;
iv. for each $\mathbf{v} \in \mathcal{V}$ there exists an additive inverse $-\mathbf{v} \in \mathcal{V}$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$;
v. scalar multiplication has identity: $1\mathbf{v} = \mathbf{v}$;
vi. scalar multiplication is compatible with multiplication in $\mathbb{F}$: $(ab)\mathbf{v} = a(b\mathbf{v})$;
vii. scalar multiplication distributes over vector addition: $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$;
viii. scalar multiplication distributes over addition in $\mathbb{F}$: $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$.

If $\mathcal{V} = \{\mathbf{0}\}$, then $\mathcal{V}$ is a *zero vector space*; if $\mathcal{V} \neq \{\mathbf{0}\}$, then $\mathcal{V}$ is a *nonzero vector space*.

Definition 1.1.2. *Finitely Nonzero Sequence Space*
Let $\mathbb{F}^{(\mathbb{N})}$ denote the set of all sequences $(x_1, x_2, \ldots)$ of elements of $\mathbb{F}$ such that $x_j = 0$ for all but finitely many $j \in \mathbb{N}$. Addition and scalar multiplication are defined componentwise.

Proposition 1.1.3. *Finitely Nonzero Sequences Form a Vector Space*
The set $\mathbb{F}^{(\mathbb{N})}$ is a vector space over $\mathbb{F}$ under componentwise addition and scalar multiplication.

Proof. The zero sequence belongs to $\mathbb{F}^{(\mathbb{N})}$. If $\mathbf{x} = (x_1, x_2, \ldots)$ and $\mathbf{y} = (y_1, y_2, \ldots)$ are in $\mathbb{F}^{(\mathbb{N})}$ and $a \in \mathbb{F}$, then the nonzero entries of $a\mathbf{x} + \mathbf{y}$ can occur only among the finitely many indices where $\mathbf{x}$ or $\mathbf{y}$ is nonzero. Thus $a\mathbf{x} + \mathbf{y} \in \mathbb{F}^{(\mathbb{N})}$, and the vector space axioms are inherited componentwise from $\mathbb{F}$.
Structural Commentary: Finite support is preserved by finite linear combinations.
Key Insight: Componentwise operations cannot create infinitely many new nonzero entries. □

Theorem 1.1.4. *Basic Consequences of Vector Space Axioms*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. Then the additive identity and additive inverses are unique. Moreover, for all $a \in \mathbb{F}$ and $\mathbf{v} \in \mathcal{V}$, $0\mathbf{v} = \mathbf{0}$, $a\mathbf{0} = \mathbf{0}$, and $a\mathbf{v} = \mathbf{0}$ if and only if $a = 0$ or $\mathbf{v} = \mathbf{0}$.

Proof. If $\mathbf{0}$ and $\mathbf{0}'$ are additive identities, then $\mathbf{0} = \mathbf{0} + \mathbf{0}' = \mathbf{0}'$, so the additive identity is unique.
If $\mathbf{w}$ and $\mathbf{w}'$ are additive inverses of $\mathbf{v}$, then $\mathbf{w} = \mathbf{w} + \mathbf{0} = \mathbf{w} + (\mathbf{v} + \mathbf{w}') = (\mathbf{w} + \mathbf{v}) + \mathbf{w}' = \mathbf{0} + \mathbf{w}' = \mathbf{w}'$, so additive inverses are unique. Since $0\mathbf{v} = (0 + 0)\mathbf{v} = 0\mathbf{v} + 0\mathbf{v}$, cancellation gives $0\mathbf{v} = \mathbf{0}$.
Similarly, $a\mathbf{0} = a(\mathbf{0} + \mathbf{0}) = a\mathbf{0} + a\mathbf{0}$, so $a\mathbf{0} = \mathbf{0}$. If $a = 0$ or $\mathbf{v} = \mathbf{0}$, then the preceding identities give $a\mathbf{v} = \mathbf{0}$. Conversely, suppose $a\mathbf{v} = \mathbf{0}$ and $a \neq 0$. Then a^{-1} exists in $\mathbb{F}$, and $\mathbf{v} = 1\mathbf{v} = (a^{-1}a)\mathbf{v} = a^{-1}(a\mathbf{v}) = a^{-1}\mathbf{0} = \mathbf{0}$.
Structural Commentary: These identities are the basic algebraic hygiene of vector spaces. They let later arguments use cancellation, zero vectors, and inverses without reopening the axioms each time.
Key Insight: The vector space axioms force the zero and inverse behaviour to be rigid, not optional structure. □

Corollary 1.1.5. *Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. Then $(-1)\mathbf{v} = -\mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$.*

Proof. Let $\mathbf{v} \in \mathcal{V}$. We show that $(-1)\mathbf{v} + \mathbf{v} = \mathbf{0}$. By the vector space axioms and Theorem 1.1.4,

$$(-1)\mathbf{v} + \mathbf{v} = (-1)\mathbf{v} + 1\mathbf{v} = (-1+1)\mathbf{v} = 0\mathbf{v} = \mathbf{0}$$

Thus $(-1)\mathbf{v}$ is the additive inverse of $\mathbf{v}$, so $(-1)\mathbf{v} = -\mathbf{v}$.
Structural Commentary: Scalar multiplication by -1 is exactly the same operation as taking an additive inverse.
Key Insight: Negation is not an extra operation; it is already encoded by the field action. □

Definition 1.1.6. A *subspace* of a vector space $\mathcal{V}$ over $\mathbb{F}$ is a subset $\mathcal{U} \subseteq \mathcal{V}$ that is itself a vector space over $\mathbb{F}$ with the same vector addition and scalar multiplication as in $\mathcal{V}$.

Theorem 1.1.7. *Subspace Test*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and let $\mathcal{U}$ be a nonempty subset of $\mathcal{V}$. Then $\mathcal{U}$ is a subspace of $\mathcal{V}$ if and only if $c\mathbf{u} + \mathbf{v} \in \mathcal{U}$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ and all $c \in \mathbb{F}$.

Proof.

($\Rightarrow$) If $\mathcal{U}$ is a subspace, then $c\mathbf{u} + \mathbf{v} \in \mathcal{U}$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ and all $c \in \mathbb{F}$ because $\mathcal{U}$ is closed under scalar multiplication and vector addition.

($\Leftarrow$) Suppose that $c\mathbf{u} + \mathbf{v} \in \mathcal{U}$ whenever $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ and $c \in \mathbb{F}$. We verify that $\mathcal{U}$ is a vector space with the operations inherited from $\mathcal{V}$. Since $\mathcal{U}$ is nonempty, let $\mathbf{u} \in \mathcal{U}$. By Corollary 1.1.5, $\mathbf{0} = (-1)\mathbf{u} + \mathbf{u} \in \mathcal{U}$. If $\mathbf{u} \in \mathcal{U}$, then $-\mathbf{u} = (-1)\mathbf{u} + \mathbf{0} \in \mathcal{U}$. If $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ and $c \in \mathbb{F}$, then $c\mathbf{u} = c\mathbf{u} + \mathbf{0} \in \mathcal{U}$ and $\mathbf{u} + \mathbf{v} = 1\mathbf{u} + \mathbf{v} \in \mathcal{U}$. Thus $\mathcal{U}$ contains $\mathbf{0}$ and is closed under additive inverses, scalar multiplication, and vector addition; the remaining vector space axioms are inherited from $\mathcal{V}$. Hence $\mathcal{U}$ is a subspace.

Structural Commentary: The theorem compresses the subspace axioms into one closure condition. This is the practical test used whenever a subset is proposed as a linear space.
Key Insight: To prove a subset is a subspace, prove it survives one affine-looking linear combination. □

Method 1.1.8. *Subspace Diagnostic*

(i) Let $S \subseteq V$. Check that $S \neq \varnothing$.
(ii) For arbitrary $u, v \in S$ and $c \in \mathbb{F}$, prove $cu + v \in S$. Then S is a subspace.
(iii) To disprove subspacehood, find $u, v \in S$ and $c \in \mathbb{F}$ s.t. $cu + v \notin S$. Having $0 \in S$ alone is not sufficient.
(iv) If S fails the test, the smallest subspace containing it is $\operatorname{span}(S) = \left\{ \sum_{i=1}^{n} a_i s_i : n \in \mathbb{N},\ a_i \in \mathbb{F},\ s_i \in S \right\}$.

Proposition 1.1.9. *Intersection of Subspaces*
Let $\mathcal{U}$ and $\mathcal{W}$ be subspaces of $\mathcal{V}$. Then $\mathcal{U} \cap \mathcal{W} = \{\mathbf{v} : \mathbf{v} \in \mathcal{U}$ and $\mathbf{v} \in \mathcal{W}\}$ is a subspace of $\mathcal{V}$.

Proof. Since $\mathbf{0} \in \mathcal{U}$ and $\mathbf{0} \in \mathcal{W}$, we have $\mathbf{0} \in \mathcal{U} \cap \mathcal{W}$, so the intersection is nonempty. If $\mathbf{u}, \mathbf{v} \in \mathcal{U} \cap \mathcal{W}$ and $c \in \mathbb{F}$, then $c\mathbf{u} + \mathbf{v}$ lies in both $\mathcal{U}$ and $\mathcal{W}$ because both are subspaces.
Hence $c\mathbf{u} + \mathbf{v} \in \mathcal{U} \cap \mathcal{W}$, and Theorem 1.1.7 shows that $\mathcal{U} \cap \mathcal{W}$ is a subspace of $\mathcal{V}$.
Structural Commentary: Intersections inherit shared closure.
Key Insight: Common vectors remain closed under linear combinations. □

Definition 1.1.10. *Lists, Linear Combinations, and Span*
A *list* of vectors in $\mathcal{V}$ is a finite ordered sequence $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ in $\mathcal{V}$; the empty list is denoted $()$.
If $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is nonempty and $a_1, \ldots, a_m \in \mathbb{F}$, then $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m$ is a *linear combination* of β.
For $E \subseteq \mathcal{V}$, the *span* of E is $\operatorname{span}(E) := \{a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m : m \in \mathbb{N},\ a_j \in \mathbb{F},\ \mathbf{v}_j \in E\}$. $\operatorname{span}(\emptyset) = \{\mathbf{0}\}$.
If $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is a nonempty list, then $\operatorname{span}(\beta) = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_m) := \operatorname{span}(\{\mathbf{v}_1, \ldots, \mathbf{v}_m\})$.
A subset $E \subseteq \mathcal{V}$ *spans* $\mathcal{V}$, or is a *spanning set* for $\mathcal{V}$, if $\operatorname{span}(E) = \mathcal{V}$.
A list β in $\mathcal{V}$ *spans* $\mathcal{V}$, or is a *spanning list* for $\mathcal{V}$, if $\operatorname{span}(\beta) = \mathcal{V}$.

Theorem 1.1.11. *Span is the Smallest Containing Subspace*
Let $E \subseteq \mathcal{V}$. Then $\operatorname{span}(E)$ *is the smallest subspace of $\mathcal{V}$ containing E.*

Proof. If $E = \emptyset$, then $\operatorname{span}(E) = \{\mathbf{0}\}$, which is a subspace. Suppose $E \neq \emptyset$. If $\mathbf{x}, \mathbf{y} \in \operatorname{span}(E)$ and $c \in \mathbb{F}$, then $\mathbf{x}$ and $\mathbf{y}$ are finite linear combinations of vectors in E. Hence $c\mathbf{x} + \mathbf{y}$ is also a finite linear combination of vectors in E, so $c\mathbf{x} + \mathbf{y} \in \operatorname{span}(E)$. By Theorem 1.1.7, $\operatorname{span}(E)$ is a subspace of $\mathcal{V}$.
Each $\mathbf{v} \in E$ satisfies $\mathbf{v} = 1\mathbf{v} \in \operatorname{span}(E)$, so $E \subseteq \operatorname{span}(E)$. If $\mathcal{U}$ is a subspace of $\mathcal{V}$ with $E \subseteq \mathcal{U}$, then closure under vector addition and scalar multiplication forces every finite linear combination of vectors in E to lie in $\mathcal{U}$. Thus $\operatorname{span}(E) \subseteq \mathcal{U}$. Therefore $\operatorname{span}(E)$ is the smallest subspace of $\mathcal{V}$ containing E.
Structural Commentary: Span takes a set of vectors and closes it under all finite linear combinations. The result is precisely the smallest subspace forced by those vectors.
Key Insight: Nothing beyond linear closure is added. Every containing subspace must already contain span. □

Proposition 1.1.12. *Monotonicity of Span*
Let $\mathcal{U}$ and $\mathcal{W}$ be subsets of $\mathcal{V}$. If $\mathcal{U} \subseteq \mathcal{W}$, then $\operatorname{span}(\mathcal{U}) \subseteq \operatorname{span}(\mathcal{W})$.

Proof. If $\mathcal{U} = \emptyset$, then $\operatorname{span}(\mathcal{U}) = \{\mathbf{0}\} \subseteq \operatorname{span}(\mathcal{W})$.
Suppose $\mathcal{U} \neq \emptyset$. Every vector in $\operatorname{span}(\mathcal{U})$ is a finite linear combination of vectors in $\mathcal{U}$; since $\mathcal{U} \subseteq \mathcal{W}$, the same expression is a finite linear combination of vectors in $\mathcal{W}$. Hence $\operatorname{span}(\mathcal{U}) \subseteq \operatorname{span}(\mathcal{W})$.
Structural Commentary: Enlarging the generating set can only enlarge the collection of linear combinations.
Key Insight: Span is monotone with respect to inclusion of generating sets. □

Definition 1.1.13. *Linear Independence*
A list $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ in $\mathcal{V}$ is *linearly independent* if $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$ implies $a_1 = \cdots = a_m = 0$. The empty list is linearly independent. Any list that is not linearly independent is linearly dependent.

Proposition 1.1.14. *Unique Coefficients for Independent Lists*
Let $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be a linearly independent list in $\mathcal{V}$.
Then $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = b_1\mathbf{v}_1 + \cdots + b_m\mathbf{v}_m$ if and only if $a_j = b_j$ for each $j = 1, \ldots, m$.

Proof.
($\Rightarrow$) Subtracting the two sides gives $(a_1 - b_1)\mathbf{v}_1 + \cdots + (a_m - b_m)\mathbf{v}_m = \mathbf{0}$.
Linear independence forces $a_j - b_j = 0$ for every j, so $a_j = b_j$.
($\Leftarrow$) If $a_j = b_j$ for every j, then the two linear combinations have identical coefficients and are equal.
Structural Commentary: Linear independence makes coefficients in a linear combination unique.
Key Insight: Independent vectors allow no hidden coefficient cancellation. □

Proposition 1.1.15. *Appending to a Dependent List*
Let $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be a linearly dependent list in $\mathcal{V}$.
Then for every $\mathbf{v} \in \mathcal{V}$, the list $(\mathbf{v}_1, \ldots, \mathbf{v}_m, \mathbf{v})$ is linearly dependent.

Proof. Since $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly dependent, $\exists$ scalars $a_1, \ldots, a_m$, not all zero, s.t. $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$.
For any $\mathbf{v} \in \mathcal{V}$, $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m + 0\mathbf{v} = \mathbf{0}$ and the scalars $a_1, \ldots, a_m, 0$ are not all zero.
Structural Commentary: Longer lists inherit dependence; set new coefficients to 0.
Key Insight: Once a list has a redundancy, adding more vectors cannot remove it. □

Definition 1.1.16. *Omitting a Vector from a List*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be a list in $\mathcal{V}$ with $m \geq 2$.
For $j \in \{1, \ldots, m\}$, the list obtained from β by omitting $\mathbf{v}_j$ is denoted $(\mathbf{v}_1, \ldots, \widehat{\mathbf{v}_j}, \ldots, \mathbf{v}_m)$.

Proposition 1.1.17. *Omitting from an Independent List*
If $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent and $m \geq 2$, then $(\mathbf{v}_1, \ldots, \widehat{\mathbf{v}_j}, \ldots, \mathbf{v}_m)$ is linearly independent for every $j \in \{1, \ldots, m\}$.

Proof. Fix j, and suppose $\sum_{\substack{i=1 \\ i \neq j}}^{m} a_i\mathbf{v}_i = \mathbf{0}$. Set $a_j = 0$.

Then $\sum_{i=1}^{m} a_i\mathbf{v}_i = \mathbf{0}$, so independence of β gives $a_i = 0$ for every $i \neq j$.
Structural Commentary: Shorter lists inherit independence; deletion cannot create cancellation.
Key Insight: Every sublist of an independent list remains independent. □

Proposition 1.1.18. *Adjoining and Removing Vectors from Lists*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be a list in $\mathcal{V}$. Write $(\beta, \mathbf{v}) := (\mathbf{v}_1, \ldots, \mathbf{v}_m, \mathbf{v})$ and $\beta_{\widehat{j}} := (\mathbf{v}_1, \ldots, \widehat{\mathbf{v}_j}, \ldots, \mathbf{v}_m)$.

(i) If β is linearly independent and $\mathbf{v} \notin \operatorname{span}(\beta)$, then $(\beta, \mathbf{v})$ is linearly independent.

(ii) If $\sum_{i=1}^{m} a_i\mathbf{v}_i = \mathbf{0}$ is nontrivial, $a_j \neq 0$, then $\operatorname{span}(\beta_{\widehat{j}}) = \operatorname{span}(\beta)$. Hence $\operatorname{span}(\beta) = \mathcal{V}$ implies $\operatorname{span}(\beta_{\widehat{j}}) = \mathcal{V}$.

Proof.
(i) If $\sum_{i=1}^{m} a_i\mathbf{v}_i + a\mathbf{v} = \mathbf{0}$ and $a \neq 0$, then $\mathbf{v} = -a^{-1}\sum_{i=1}^{m} a_i\mathbf{v}_i \in \operatorname{span}(\beta)$, a contradiction.
(ii) Since $a_j \neq 0$, $\mathbf{v}_j = -\sum_{\substack{i=1 \\ i \neq j}}^{m} \frac{a_i}{a_j}\mathbf{v}_i \in \operatorname{span}(\beta_{\widehat{j}})$. Hence $\operatorname{span}(\beta) \subseteq \operatorname{span}(\beta_{\widehat{j}})$; the reverse inclusion is immediate.

Structural Commentary: New directions preserve independence; redundant vectors preserve span.
Key Insight: Add new directions; remove redundant ones. □

Lemma 1.1.19. *Replacement Lemma*
Let $\mathcal{V} \neq \{\mathbf{0}\}$, let $\beta = (\mathbf{u}_1, \ldots, \mathbf{u}_r)$ span $\mathcal{V}$, and let $\mathbf{0} \neq \mathbf{v} = \sum_{i=1}^{r} c_i \mathbf{u}_i$.
For $j \in \{1, \ldots, r\}$, set $\beta_{j \leftarrow \mathbf{v}} := (\mathbf{u}_1, \ldots, \mathbf{u}_{j-1}, \mathbf{v}, \mathbf{u}_{j+1}, \ldots, \mathbf{u}_r)$.

(i) $c_j \neq 0$ for some j.
(ii) If $c_j \neq 0$, then $\operatorname{span}(\beta_{j \leftarrow \mathbf{v}}) = \mathcal{V}$; if β is a basis, then $\beta_{j \leftarrow \mathbf{v}}$ is a basis.
(iii) If $r \geq 2$, β is a basis, and $\mathbf{v} \notin \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$ for some $1 \leq k < r$, then $\beta_{j \leftarrow \mathbf{v}}$ is a basis for some $j > k$.

Proof.

(i) If all $c_j = 0$, then $\mathbf{v} = \mathbf{0}$.

(ii) Since $\mathbf{u}_j = c_j^{-1}\left(\mathbf{v} - \sum_{i \neq j} c_i \mathbf{u}_i\right)$, every vector of β lies in $\operatorname{span}(\beta_{j \leftarrow \mathbf{v}})$, while every vector of $\beta_{j \leftarrow \mathbf{v}}$ lies in $\operatorname{span}(\beta)$. Hence $\operatorname{span}(\beta_{j \leftarrow \mathbf{v}}) = \mathcal{V}$. If β is a basis and $\sum_{i<j} b_i \mathbf{u}_i + a\mathbf{v} + \sum_{i>j} b_i \mathbf{u}_i = \mathbf{0}$, then substitution gives $\sum_{i \neq j} (b_i + ac_i)\mathbf{u}_i + ac_j \mathbf{u}_j = \mathbf{0}$. Linear independence of β gives $ac_j = 0$ and $b_i + ac_i = 0$ for $i \neq j$.
Since $c_j \neq 0$, $a = 0$, and then every $b_i = 0$. Thus $\beta_{j \leftarrow \mathbf{v}}$ is also linearly independent.

(iii) If $c_i = 0 \ \forall i > k$, then $\mathbf{v} \in \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$, contradiction. Thus some $c_j \neq 0$ with $j > k$, and (ii) applies.

Structural Commentary: Replacement preserves span.
Key Insight: Replace where the coefficient is nonzero. □

Lemma 1.1.20. *Linear Dependence Lemma*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be linearly dependent with $\mathbf{v}_1 \neq \mathbf{0}$, and set $\beta_{\widehat{j}} := (\mathbf{v}_1, \ldots, \widehat{\mathbf{v}_j}, \ldots, \mathbf{v}_m)$.
Then $\exists j \in \{2, \ldots, m\}$ such that $\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1})$ and $\operatorname{span}(\beta_{\widehat{j}}) = \operatorname{span}(\beta)$.

Proof. Choose a nontrivial relation $\sum_{i=1}^{m} a_i \mathbf{v}_i = \mathbf{0}$ and let $j = \max\{i : a_i \neq 0\}$.
If $j = 1$, then $a_1 \mathbf{v}_1 = \mathbf{0}$ with $a_1 \neq 0$, so $\mathbf{v}_1 = \mathbf{0}$, contrary to hypothesis. Hence $j \geq 2$.
Since $a_i = 0$ for $i > j$, we have $\mathbf{v}_j = -\sum_{i=1}^{j-1} \frac{a_i}{a_j} \mathbf{v}_i$, so $\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1})$.
Therefore every linear combination of β can be rewritten without $\mathbf{v}_j$, which gives $\operatorname{span}(\beta) \subseteq \operatorname{span}(\beta_{\widehat{j}})$.
The reverse inclusion is immediate because $\beta_{\widehat{j}}$ is a sublist of β.
Structural Commentary: The last nonzero coefficient gives the removable vector.
Key Insight: Dependence gives a later vector removable without losing span. □

Theorem 1.1.21. *Independent Lists Are No Longer Than Spanning Lists*
If $\mathcal{V}$ is finite-dimensional, $\alpha = (\mathbf{u}_1, \ldots, \mathbf{u}_m)$ is linearly independent, and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ spans $\mathcal{V}$, then $m \leq n$.

Proof. Suppose $m > n$. If $n = 0$, then $\operatorname{span}(\gamma) = \{\mathbf{0}\}$, so $\mathcal{V} = \{\mathbf{0}\}$ and no nonempty independent list exists; hence $n \geq 1$. Starting from γ, we construct for each $0 \leq k \leq n$ a spanning list of length n containing $\mathbf{u}_1, \ldots, \mathbf{u}_k$. The case $k = 0$ is γ. Suppose the list has been constructed for some $k < n$. Write $\mathbf{u}_{k+1}$ as a linear combination of its entries. If all coefficients of entries other than $\mathbf{u}_1, \ldots, \mathbf{u}_k$ were zero, then $\mathbf{u}_{k+1} \in \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$, contradicting the linear independence of α. Thus some remaining entry has nonzero coefficient; replacing that entry by $\mathbf{u}_{k+1}$ preserves span by Lemma 1.1.19(ii). This proves the induction.
For $k = n$, the resulting spanning list has length n and contains $\mathbf{u}_1, \ldots, \mathbf{u}_n$; hence it is exactly these vectors, up to order. Thus $\operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_n) = \mathcal{V}$.
Since $m > n$, this puts $\mathbf{u}_{n+1}$ in $\operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$, contradicting the linear independence of α.
Structural Commentary: Independent length is bounded by spanning length.
Key Insight: Independent lists cannot be longer than spanning lists in finite dimensions. □

Definition 1.1.22. *Basis*
A list β in $\mathcal{V}$ is a *basis* of $\mathcal{V}$ if it is linearly independent and $\operatorname{span}(\beta) = \mathcal{V}$.

Proposition 1.1.23. *Unique Coordinate Representation*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$. Then β is a basis of $\mathcal{V}$ iff each $\mathbf{v} \in \mathcal{V}$ has a unique $(a_1, \ldots, a_n) \in \mathbb{F}^n$ with $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{v}_j$.

Proof.

($\Rightarrow$) Spanning gives existence; Proposition 1.1.14 gives uniqueness.

($\Leftarrow$) Existence gives $\operatorname{span}(\beta) = \mathcal{V}$. If $\sum_{j=1}^{n} a_j \mathbf{v}_j = \mathbf{0}$, uniqueness of zero representation gives $a_1 = \cdots = a_n = 0$.

Structural Commentary: A basis is exactly existence and uniqueness of coordinates.
Key Insight: Coordinates are basis coefficients. □

Theorem 1.1.24. *Reducing Spanning Lists*
Every spanning list in a finite-dimensional vector space contains a sublist that is a basis.

Proof. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a spanning list of $\mathcal{V}$. If β is linearly independent, it is a basis. Otherwise, remove one vector without changing the span: if $\mathbf{v}_1 = \mathbf{0}$, omit $\mathbf{v}_1$; if $\mathbf{v}_1 \neq \mathbf{0}$, Lemma 1.1.20 gives a removable $\mathbf{v}_j$. The new list is a sublist of β, still spans $\mathcal{V}$, and has length one less.
Repeating, the length strictly decreases, so the process stops with a spanning independent sublist, hence a basis.
Structural Commentary: Delete redundant vectors while preserving span.
Key Insight: Spanning gives enough vectors; reduction deletes exactly the unnecessary ones. □

Theorem 1.1.25. *Extending Independent Lists*
Every linearly independent list in a finite-dimensional vector space can be extended to a basis.

Proof. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be linearly independent in $\mathcal{V}$, and choose a spanning list $(\mathbf{w}_1, \ldots, \mathbf{w}_n)$ of $\mathcal{V}$.
Starting with $\mathcal{B} = \beta$, inspect $\mathbf{w}_1, \ldots, \mathbf{w}_n$ in order and append $\mathbf{w}_j$ exactly when $\mathbf{w}_j \notin \operatorname{span}(\mathcal{B})$. Each append preserves linear independence by Proposition 1.1.18.
After the inspection, each $\mathbf{w}_j$ lies in $\operatorname{span}(\mathcal{B})$: it was either already there when inspected, or it was appended to $\mathcal{B}$. Since the $\mathbf{w}_j$ span $\mathcal{V}$, every vector of $\mathcal{V}$ is a linear combination of vectors in $\operatorname{span}(\mathcal{B})$, and hence lies in $\operatorname{span}(\mathcal{B})$. Thus $\mathcal{B}$ is a spanning independent list, so it is a basis.
Structural Commentary: An independent list has no redundancy, but may miss directions. Appending only vectors outside its span fills those directions while preserving independence.
Key Insight: A finite-dimensional independent list becomes a basis once enough new directions are added. □

Corollary 1.1.26. *Every finite-dimensional vector space has a basis, and every subspace of a finite-dimensional vector space is finite-dimensional.*

Proof. Since $\mathcal{V}$ is finite-dimensional, it has a finite spanning list; Theorem 1.1.24 reduces it to a basis.
Let $\mathcal{U} \subseteq \mathcal{V}$ be a subspace. If $\mathcal{U} = \{\mathbf{0}\}$, then the empty list is a basis of $\mathcal{U}$.
Otherwise, begin with any nonzero vector of $\mathcal{U}$, and repeatedly append a vector of $\mathcal{U}$ outside the span of the current list if such a vector exists. Proposition 1.1.18 keeps the list linearly independent. Theorem 1.1.21 bounds its length by the length of any spanning list of $\mathcal{V}$, so this finite process must terminate. When it terminates, no vector of $\mathcal{U}$ lies outside the span of the list; hence the list spans $\mathcal{U}$ and is a basis of $\mathcal{U}$.
Structural Commentary: Finite-dimensional spaces have finite coordinate systems, and subspaces cannot contain longer independent lists than the ambient space permits.
Key Insight: Finite-dimensionality passes downward to subspaces. □

Theorem 1.1.27. *Basis Length Is Well-Defined*
Any two bases of a finite-dimensional vector space have the same length.

Proof. Let $\beta_1 = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ and $\beta_2 = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ be bases of $\mathcal{V}$. Since β_1 is linearly independent and β_2 spans $\mathcal{V}$, Theorem 1.1.21 gives $m \leq n$. Reversing the roles of the two bases gives $n \leq m$. Hence $m = n$.
Structural Commentary: Bases are simultaneously independent and spanning, so the comparison theorem traps their lengths in both directions. This makes dimension well-defined.
Key Insight: Dimension is an intrinsic number, not an artefact of a particular basis. □

Definition 1.1.28. *Dimension*
The zero vector space has *dimension* 0. If $\mathcal{V}$ has a basis of positive integer length n, then $\mathcal{V}$ is *n-dimensional* and $\dim(\mathcal{V}) = n$. A vector space is *finite-dimensional* if it has finite dimension; otherwise it is *infinite-dimensional*.

Proposition 1.1.29. *Dimension Bounds and Basis Criteria*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U} \subseteq \mathcal{V}$ is a subspace, then $\dim(\mathcal{U}) \leq \dim(\mathcal{V})$, with equality if and only if $\mathcal{U} = \mathcal{V}$. Moreover, a spanning list of length $\dim(\mathcal{V})$ is automatically a basis, and a linearly independent list of length $\dim(\mathcal{V})$ is automatically a basis.

Proof. Let $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ be a basis of $\mathcal{U}$ and $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ a basis of $\mathcal{V}$.
Theorem 1.1.21 gives $m \leq n$, so $\dim(\mathcal{U}) \leq \dim(\mathcal{V})$. If $\mathcal{U} = \mathcal{V}$, equality is clear.
Conversely, if $m = n$, then the basis of $\mathcal{U}$ is an independent list in $\mathcal{V}$ of length $\dim(\mathcal{V})$; extending it to a basis of $\mathcal{V}$ adds no vectors, so it already spans $\mathcal{V}$. Hence $\mathcal{U} = \mathcal{V}$. Let γ be a spanning list of length $n = \dim(\mathcal{V})$.
By Theorem 1.1.24, γ contains a sublist that is a basis of $\mathcal{V}$. Any proper sublist has length $< n$, impossible by Theorem 1.1.27; hence no vector was removed and γ is itself a basis.
Let α be a linearly independent list of length n. By Theorem 1.1.25, α extends to a basis of $\mathcal{V}$.

Adding any vector would produce a basis of length $> n$, again contradicting Theorem 1.1.27; hence no vector was added and α is itself a basis.
Structural Commentary: Dimension bounds subspaces; equality forces fullness.
Key Insight: In dimension n, n independent or n spanning vectors already form a basis. □

1.2 Direct Sums, Quotients, and Decomposition

Direct sums split spaces into complementary pieces; quotients collapse selected directions. Dimension formulas keep both constructions controlled.

Definition 1.2.1. *Sums of Subspaces*
If $\mathcal{U}_1, \ldots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$, define $\mathcal{U}_1 + \cdots + \mathcal{U}_m := \{\mathbf{u}_1 + \cdots + \mathbf{u}_m : \mathbf{u}_j \in \mathcal{U}_j\}$.
In particular, for subspaces $\mathcal{U}, \mathcal{W} \subseteq \mathcal{V}$, $\mathcal{U} + \mathcal{W} = \operatorname{span}(\mathcal{U} \cup \mathcal{W}) = \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in \mathcal{U},\ \mathbf{w} \in \mathcal{W}\}$.
If every vector in this sum has a unique such decomposition, then the sum is direct and we write $\mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$.
For two subspaces, $\mathcal{U} + \mathcal{W}$ is direct if and only if $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$; in this case it is denoted $\mathcal{U} \oplus \mathcal{W}$.

Perspective 1.2.2. *Sum Is Not Union*
The union $\mathcal{U} \cup \mathcal{W}$ keeps vectors from either subspace; the sum $\mathcal{U} + \mathcal{W}$ adds vectors from both. Unless one subspace contains the other, $\mathcal{U} \cup \mathcal{W}$ is not closed under addition.

Proposition 1.2.3. *Uniqueness in a Direct Sum*
Let $\mathcal{U}$ and $\mathcal{W}$ be subspaces of $\mathcal{V}$ with $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$. Then each vector in $\mathcal{U} \oplus \mathcal{W}$ has a unique representation $\mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$.

Proof. Existence follows from the definition of $\mathcal{U} + \mathcal{W}$. For uniqueness, suppose $\mathbf{u}_1 + \mathbf{w}_1 = \mathbf{u}_2 + \mathbf{w}_2$, $\mathbf{u}_1, \mathbf{u}_2 \in \mathcal{U}$, $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W}$. Then $\mathbf{u}_1 - \mathbf{u}_2 = \mathbf{w}_2 - \mathbf{w}_1 \in \mathcal{U}, \mathcal{W}$, hence equals $\mathbf{0}$. Thus $\mathbf{u}_1 = \mathbf{u}_2$ and $\mathbf{w}_1 = \mathbf{w}_2$.
Structural Commentary: Trivial intersection prevents cancellation between summands.
Key Insight: Directness means unique decomposition. □

Proposition 1.2.4. *Sum of Subspaces Is the Smallest Containing Subspace*
Suppose $\mathcal{U}_1, \ldots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is the smallest subspace of $\mathcal{V}$ containing $\mathcal{U}_1, \ldots, \mathcal{U}_m$.

Proof. Since $\mathbf{0} \in \mathcal{U}_j$ for each j, we have $\mathbf{0} \in \mathcal{U}_1 + \cdots + \mathcal{U}_m$. If $\mathbf{x} = \mathbf{u}_1 + \cdots + \mathbf{u}_m$ and $\mathbf{y} = \mathbf{v}_1 + \cdots + \mathbf{v}_m$ lie in $\mathcal{U}_1 + \cdots + \mathcal{U}_m$, then for any $c \in \mathbb{F}$, $c\mathbf{x} + \mathbf{y} = (c\mathbf{u}_1 + \mathbf{v}_1) + \cdots + (c\mathbf{u}_m + \mathbf{v}_m) \in \mathcal{U}_1 + \cdots + \mathcal{U}_m$, because $c\mathbf{u}_j + \mathbf{v}_j \in \mathcal{U}_j$ for each j. By Theorem 1.1.7, $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is a subspace of $\mathcal{V}$.
Each $\mathcal{U}_j$ is contained in $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ by taking all other summands to be $\mathbf{0}$. If $\mathcal{W}$ is any subspace of $\mathcal{V}$ containing $\mathcal{U}_1, \ldots, \mathcal{U}_m$, then closure under addition gives $\mathbf{u}_1 + \cdots + \mathbf{u}_m \in \mathcal{W}$ whenever $\mathbf{u}_j \in \mathcal{U}_j$. Hence $\mathcal{U}_1 + \cdots + \mathcal{U}_m \subseteq \mathcal{W}$, so the sum is the smallest such subspace.
Structural Commentary: Sums of subspaces are obtained by closing several subspaces under finite addition. Minimality follows because every containing subspace is already closed under those sums.
Key Insight: The sum adds no extra structure beyond the linear combinations forced by the given subspaces. □

Proposition 1.2.5. *Equivalent Conditions for a Direct Sum*
For subspaces $\mathcal{U}_1, \ldots, \mathcal{U}_m \subseteq \mathcal{V}$, the following are equivalent:

i. $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$;
ii. *every $\mathbf{v} \in \mathcal{V}$ has a unique representation $\mathbf{v} = \mathbf{u}_1 + \cdots + \mathbf{u}_m$ with $\mathbf{u}_j \in \mathcal{U}_j$;*
iii. $\mathcal{V} = \mathcal{U}_1 + \cdots + \mathcal{U}_m$, *and if $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ with $\mathbf{u}_j \in \mathcal{U}_j$, then $\mathbf{u}_1 = \cdots = \mathbf{u}_m = \mathbf{0}$.*

Proof.

(*i*) $\Rightarrow$ (*ii*) This follows from the definition of direct sum.
(*ii*) $\Rightarrow$ (*iii*) Every vector lies in $\mathcal{U}_1 + \cdots + \mathcal{U}_m$, so $\mathcal{V} = \mathcal{U}_1 + \cdots + \mathcal{U}_m$. If $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$, then this representation of $\mathbf{0}$ must equal $\mathbf{0} + \cdots + \mathbf{0}$ by uniqueness, so each $\mathbf{u}_j = \mathbf{0}$.
(*iii*) $\Rightarrow$ (*i*) Since $\mathcal{V} = \mathcal{U}_1 + \cdots + \mathcal{U}_m$, every $\mathbf{v} \in \mathcal{V}$ has at least one representation. If $\mathbf{v} = \mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{w}_1 + \cdots + \mathbf{w}_m$, then $(\mathbf{u}_1 - \mathbf{w}_1) + \cdots + (\mathbf{u}_m - \mathbf{w}_m) = \mathbf{0}$. By iii., $\mathbf{u}_j - \mathbf{w}_j = \mathbf{0}$ for every j, so the representation is unique. Hence $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$.

Structural Commentary: Direct sums require both spanning by the summands and no cancellation among them. The zero-representation condition isolates the no-cancellation part.
Key Insight: Uniqueness is equivalent to saying the only way the summands add to zero is trivially. □

Definition 1.2.6. *Product of Vector Spaces*
Let $\mathcal{V}_1, \ldots, \mathcal{V}_m$ be vector spaces over $\mathbb{F}$. Their *product* is $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m := \{(\mathbf{v}_1, \ldots, \mathbf{v}_m) : \mathbf{v}_j \in \mathcal{V}_j\}$.
Addition, scalar multiplication defined componentwise: $(\mathbf{u}_1, \ldots, \mathbf{u}_m) + (\mathbf{v}_1, \ldots, \mathbf{v}_m) := (\mathbf{u}_1 + \mathbf{v}_1, \ldots, \mathbf{u}_m + \mathbf{v}_m)$ and $a(\mathbf{v}_1, \ldots, \mathbf{v}_m) := (a\mathbf{v}_1, \ldots, a\mathbf{v}_m)$.

Proposition 1.2.7. *Products Are Vector Spaces*
If $\mathcal{V}_1, \dots, \mathcal{V}_m$ are vector spaces over $\mathbb{F}$, then $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m$ is a vector space under componentwise operations.

Proof. The additive identity is $(\mathbf{0}_1, \dots, \mathbf{0}_m)$, where $\mathbf{0}_j$ is the zero vector of $\mathcal{V}_j$, and additive inverse of $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ is $(-\mathbf{v}_1, \dots, -\mathbf{v}_m)$. All remaining vector space axioms hold componentwise because they hold in each $\mathcal{V}_j$.
Structural Commentary: Product spaces collect several vector spaces into one coordinatewise structure.
Key Insight: Every vector space axiom reduces to the same axiom in each component. □

Theorem 1.2.8. *Dimension of a Product*
If $\mathcal{V}_1, \dots, \mathcal{V}_m$ are finite-dimensional vector spaces over $\mathbb{F}$, then $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m$ is finite-dimensional and

$$\dim(\mathcal{V}_1 \times \cdots \times \mathcal{V}_m) = \dim(\mathcal{V}_1) + \cdots + \dim(\mathcal{V}_m)$$

Proof. For each j, let $(\mathbf{v}_{j,1}, \dots, \mathbf{v}_{j,n_j})$ be a basis of $\mathcal{V}_j$. For every j and k, let $\mathbf{e}_{j,k}$ denote the element of $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m$ whose jth component is $\mathbf{v}_{j,k}$ and whose other components are zero.
The list of all $\mathbf{e}_{j,k}$ spans the product because each component can be expanded in its own basis. If a linear combination of the $\mathbf{e}_{j,k}$ is zero, then looking at the jth component gives a linear relation among $(\mathbf{v}_{j,1}, \dots, \mathbf{v}_{j,n_j})$, so all coefficients in that component are zero. Hence the list is linearly independent and therefore is a basis of the product. Its length is $n_1 + \cdots + n_m$.
Structural Commentary: A basis for a product is formed by placing each component basis into its own slot.
Key Insight: Product dimension counts the independent directions contributed by each factor. □

Proposition 1.2.9. *Direct Sum Map*
Suppose $\mathcal{U}_1, \dots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$. Define $\Gamma : \mathcal{U}_1 \times \cdots \times \mathcal{U}_m \to \mathcal{U}_1 + \cdots + \mathcal{U}_m$ by $\Gamma(\mathbf{u}_1, \dots, \mathbf{u}_m) = \mathbf{u}_1 + \cdots + \mathbf{u}_m$. Then $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is a direct sum if and only if Γ is injective.

Proof. The map Γ is linear, and its kernel consists of all $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ such that $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$.

($\Rightarrow$) If $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is a direct sum and $\Gamma(\mathbf{u}_1, \dots, \mathbf{u}_m) = \mathbf{0}$, uniqueness of the zero decomposition forces each $\mathbf{u}_j = \mathbf{0}$. Hence $\ker(\Gamma) = \{\mathbf{0}\}$, so Γ is injective.

($\Leftarrow$) If Γ is injective and $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$, then $(\mathbf{u}_1, \dots, \mathbf{u}_m) \in \ker(\Gamma)$, so every $\mathbf{u}_j = \mathbf{0}$. Thus the only zero decomposition is trivial, which is the direct-sum condition.

Structural Commentary: The sum map packages all decompositions into one linear map.
Key Insight: Directness is injectivity of the map that adds the components. □

Theorem 1.2.10. *Dimension Formula for a Sum*
If $\mathcal{U}, \mathcal{W}$ are finite-dimensional subspaces of $\mathcal{V}$, then $\dim(\mathcal{U} + \mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{U} \cap \mathcal{W})$.

Proof. Choose a basis $(\mathbf{e}_1, \dots, \mathbf{e}_r)$ of $\mathcal{U} \cap \mathcal{W}$, extend it to a basis $(\mathbf{e}_1, \dots, \mathbf{e}_r, \mathbf{u}_1, \dots, \mathbf{u}_p)$ of $\mathcal{U}$, and to a basis $(\mathbf{e}_1, \dots, \mathbf{e}_r, \mathbf{w}_1, \dots, \mathbf{w}_q)$ of $\mathcal{W}$. Then $(\mathbf{e}_1, \dots, \mathbf{e}_r, \mathbf{u}_1, \dots, \mathbf{u}_p, \mathbf{w}_1, \dots, \mathbf{w}_q)$ spans $\mathcal{U} + \mathcal{W}$ because every vector in the sum is the sum of one vector from $\mathcal{U}$ and one vector from $\mathcal{W}$.
To prove linear independence, suppose $a_1\mathbf{e}_1 + \cdots + a_r\mathbf{e}_r + b_1\mathbf{u}_1 + \cdots + b_p\mathbf{u}_p + c_1\mathbf{w}_1 + \cdots + c_q\mathbf{w}_q = \mathbf{0}$. Then $a_1\mathbf{e}_1 + \cdots + a_r\mathbf{e}_r + b_1\mathbf{u}_1 + \cdots + b_p\mathbf{u}_p$ lies in $\mathcal{U} \cap \mathcal{W}$. Since $(\mathbf{e}_1, \dots, \mathbf{e}_r)$ is a basis of $\mathcal{U} \cap \mathcal{W}$ and $(\mathbf{e}_1, \dots, \mathbf{e}_r, \mathbf{u}_1, \dots, \mathbf{u}_p)$ is a basis of $\mathcal{U}$, we get $b_1 = \cdots = b_p = 0$. The remaining relation lies in the basis $(\mathbf{e}_1, \dots, \mathbf{e}_r, \mathbf{w}_1, \dots, \mathbf{w}_q)$ of $\mathcal{W}$, so $a_1 = \cdots = a_r = c_1 = \cdots = c_q = 0$. Thus the displayed list is a basis of $\mathcal{U} + \mathcal{W}$.
Therefore $\dim(\mathcal{U} + \mathcal{W}) = r + p + q = (r + p) + (r + q) - r$.
Structural Commentary: The formula corrects double-counting: vectors in $\mathcal{U} \cap \mathcal{W}$ are counted once in each subspace. Extending an intersection basis separates shared directions from new directions.
Key Insight: The dimension of a sum is the sum of dimensions minus the overlap. □

Corollary 1.2.11. *Dimension of a Direct Sum*
If $\mathcal{U}$ and $\mathcal{W}$ are finite-dimensional subspaces of $\mathcal{V}$ with $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$, then $\dim(\mathcal{U} \oplus \mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W})$.

Proof. Since $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$, the sum is direct and $\dim(\mathcal{U} \cap \mathcal{W}) = 0$. The result follows from Theorem 1.2.10.
Structural Commentary: When the overlap is trivial, there is no double-counting in the dimension formula. Direct sums are exactly the case where the overlap term vanishes.
Key Insight: Dimensions add exactly for direct sums. □

Theorem 1.2.12. *Direct Sums and Dimension*
Suppose $\mathcal{V}$ is finite-dimensional and $\mathcal{U}_1, \dots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is a direct sum if and only if $\dim(\mathcal{U}_1 + \cdots + \mathcal{U}_m) = \dim(\mathcal{U}_1) + \cdots + \dim(\mathcal{U}_m)$.

Proof. For each j, choose a basis β_j of $\mathcal{U}_j$, and concatenate these bases into a list β.
This list spans $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ and has length $\dim(\mathcal{U}_1) + \cdots + \dim(\mathcal{U}_m)$.

($\Rightarrow$) If the sum is direct, any linear relation among the vectors in β gives a zero decomposition $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ with $\mathbf{u}_j \in \mathcal{U}_j$. Directness gives $\mathbf{u}_j = \mathbf{0}$ for every j, and independence of each β_j gives all coefficients zero. Thus β is a basis of the sum, and the dimensions add.

($\Leftarrow$) If the dimensions add, then β is a spanning list whose length equals the dimension of the sum, so Proposition 1.1.29 makes β a basis. A zero decomposition $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ expands to a zero linear relation in this basis, so every $\mathbf{u}_j = \mathbf{0}$. Hence the sum is direct.

Structural Commentary: Direct sums add dimensions without hidden overlap.
Key Insight: Equality of dimensions means the concatenated bases have no linear redundancy. □

Corollary 1.2.13. *Large Subspaces Must Intersect*
Let $\mathcal{U}$ and $\mathcal{W}$ be subspaces of a finite-dimensional vector space $\mathcal{V}$, and let k be a positive integer.

i. If $\dim(\mathcal{U}) + \dim(\mathcal{W}) > \dim(\mathcal{V})$, then $\mathcal{U} \cap \mathcal{W}$ contains a nonzero vector.
ii. If $\dim(\mathcal{U}) + \dim(\mathcal{W}) \geq \dim(\mathcal{V}) + k$, then $\mathcal{U} \cap \mathcal{W}$ contains k linearly independent vectors.

Proof. Since $\mathcal{U} + \mathcal{W} \subseteq \mathcal{V}$, Theorem 1.2.10 gives

$$\dim(\mathcal{U} \cap \mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{U} + \mathcal{W}) \geq \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{V})$$

If $\dim(\mathcal{U}) + \dim(\mathcal{W}) > \dim(\mathcal{V})$, the right-hand side is at least 1, so the intersection contains a nonzero vector. If $\dim(\mathcal{U}) + \dim(\mathcal{W}) \geq \dim(\mathcal{V}) + k$, then $\dim(\mathcal{U} \cap \mathcal{W}) \geq k$, so intersection contains k linearly independent vectors.
Structural Commentary: Excess dimension appears as overlap.
Key Insight: Two large subspaces cannot fit inside $\mathcal{V}$ without sharing directions. □

Proposition 1.2.14. *Complements Exist in Finite Dimensions*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U} \subseteq \mathcal{V}$ is a subspace, then there exists a subspace $\mathcal{W}$ such that $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.

Proof. Choose a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ of $\mathcal{U}$ and extend it to a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_n)$ of $\mathcal{V}$.
Let $\mathcal{W} = \operatorname{span}(\mathbf{w}_1, \ldots, \mathbf{w}_n)$. Every vector in $\mathcal{V}$ is a linear combination of this extended basis, so $\mathcal{V} = \mathcal{U} + \mathcal{W}$. If $\mathbf{v} \in \mathcal{U} \cap \mathcal{W}$, then $\mathbf{v}$ has one representation using only $\mathbf{u}_1, \ldots, \mathbf{u}_m$ and another using only $\mathbf{w}_1, \ldots, \mathbf{w}_n$. Uniqueness of coefficients in the basis of $\mathcal{V}$ forces $\mathbf{v} = \mathbf{0}$. Hence $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.
Structural Commentary: Complements split a space into a subspace and the remaining basis directions.
Key Insight: Extend a basis of $\mathcal{U}$ to split $\mathcal{V}$. □

Definition 1.2.15. *Affine Subset and Parallel Subspaces*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$. For $\mathbf{v} \in \mathcal{V}$, the subset $\mathbf{v} + \mathcal{U} := \{\mathbf{v} + \mathbf{u} : \mathbf{u} \in \mathcal{U}\}$ is an *affine subset* of $\mathcal{V}$ parallel to $\mathcal{U}$.

Proposition 1.2.16. *Parallel Affine Subsets Are Equal or Disjoint*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$ and let $\mathbf{v}, \mathbf{w} \in \mathcal{V}$. The following are equivalent:

(i) $\mathbf{v} - \mathbf{w} \in \mathcal{U}$;
(ii) $\mathbf{v} + \mathcal{U} = \mathbf{w} + \mathcal{U}$;
(iii) $(\mathbf{v} + \mathcal{U}) \cap (\mathbf{w} + \mathcal{U}) \neq \emptyset$.

Proof.

(i) $\Rightarrow$ (ii) If $\mathbf{v} - \mathbf{w} \in \mathcal{U}$ and $\mathbf{u} \in \mathcal{U}$, then $\mathbf{v} + \mathbf{u} = \mathbf{w} + ((\mathbf{v} - \mathbf{w}) + \mathbf{u}) \in \mathbf{w} + \mathcal{U}$. Thus $\mathbf{v} + \mathcal{U} \subseteq \mathbf{w} + \mathcal{U}$, and the reverse inclusion is similar.

(ii) $\Rightarrow$ (iii) Equal affine subsets have the same elements.

(iii) $\Rightarrow$ (i) If $\mathbf{x} \in (\mathbf{v} + \mathcal{U}) \cap (\mathbf{w} + \mathcal{U})$, then $\mathbf{x} = \mathbf{v} + \mathbf{u}_1 = \mathbf{w} + \mathbf{u}_2$ for some $\mathbf{u}_1, \mathbf{u}_2 \in \mathcal{U}$. Hence $\mathbf{v} - \mathbf{w} = \mathbf{u}_2 - \mathbf{u}_1 \in \mathcal{U}$.

Structural Commentary: Parallel affine subsets either coincide or do not meet.
Key Insight: Intersection is controlled by the difference of base points. □

Definition 1.2.17. *Quotient Space*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$.
The *quotient space* $\mathcal{V}/\mathcal{U}$ is the set of all affine subsets of $\mathcal{V}$ parallel to $\mathcal{U}$: $\mathcal{V}/\mathcal{U} := \{\mathbf{v} + \mathcal{U} : \mathbf{v} \in \mathcal{V}\}$.

Proposition 1.2.18. *Structure of the Quotient*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$. Addition and scalar multiplication on $\mathcal{V}/\mathcal{U}$ defined by $(\mathbf{v} + \mathcal{U}) + (\mathbf{w} + \mathcal{U}) := (\mathbf{v} + \mathbf{w}) + \mathcal{U}$ and $a(\mathbf{v} + \mathcal{U}) := a\mathbf{v} + \mathcal{U}$ are well-defined, and with these operations $\mathcal{V}/\mathcal{U}$ is a vector space.

Proof. If $\mathbf{v}+\mathcal{U}=\mathbf{v}'+\mathcal{U}$ and $\mathbf{w}+\mathcal{U}=\mathbf{w}'+\mathcal{U}$, then $\mathbf{v}-\mathbf{v}'\in\mathcal{U}$ and $\mathbf{w}-\mathbf{w}'\in\mathcal{U}$.
Since $\mathcal{U}$ is a subspace, $(\mathbf{v}+\mathbf{w})-(\mathbf{v}'+\mathbf{w}')\in\mathcal{U}$, and $a\mathbf{v}-a\mathbf{v}'\in\mathcal{U}$ so the operations do not depend on the chosen representatives. The vector space axioms are inherited from $\mathcal{V}$ under these operations.
Structural Commentary: Quotients preserve operations modulo $\mathcal{U}$.
Key Insight: Cosets identify vectors differing by elements of $\mathcal{U}$. □

Definition 1.2.19. *Quotient Map*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$. The *quotient map* is the map $\pi:\mathcal{V}\to\mathcal{V}/\mathcal{U}$ defined by $\pi(\mathbf{v})=\mathbf{v}+\mathcal{U}$.

Proposition 1.2.20. *Properties of the Quotient Map*
The quotient map π is linear, surjective, and $\ker(\pi)=\mathcal{U}$.

Proof. For $\mathbf{v},\mathbf{w}\in\mathcal{V}$ and $a\in\mathbb{F}$, $\pi(a\mathbf{v}+\mathbf{w})=(a\mathbf{v}+\mathbf{w})+\mathcal{U}=a(\mathbf{v}+\mathcal{U})+(\mathbf{w}+\mathcal{U})=a\pi(\mathbf{v})+\pi(\mathbf{w})$ so π is linear. It is surjective because every coset in $\mathcal{V}/\mathcal{U}$ has the form $\mathbf{v}+\mathcal{U}=\pi(\mathbf{v})$. Finally, $\ker(\pi)=\{\mathbf{v}\in\mathcal{V}:\mathbf{v}+\mathcal{U}=\mathcal{U}\}=\mathcal{U}$.
Structural Commentary: The quotient map collapses $\mathcal{U}$ to zero and records cosets.
Key Insight: Quotient maps realise subspaces as kernels. □

Definition 1.2.21. *Induced Map on a Quotient*
For $T\in\mathcal{L}(\mathcal{V},\mathcal{W})$, define $\widetilde{T}:\mathcal{V}/\ker(T)\to\mathcal{W}$ by $\widetilde{T}(\mathbf{v}+\ker(T))=T\mathbf{v}$.

Theorem 1.2.22. *Quotient by the Kernel*
Let $T\in\mathcal{L}(\mathcal{V},\mathcal{W})$. Then:

(i) $\widetilde{T}$ *is a well-defined linear map from* $\mathcal{V}/\ker(T)$ *to* $\mathcal{W}$;
(ii) $\widetilde{T}$ *is injective;*
(iii) $\operatorname{range}(\widetilde{T})=\operatorname{range}(T)$;
(iv) $\mathcal{V}/\ker(T)\cong\operatorname{range}(T)$.

Proof.

(i) If $\mathbf{v}+\ker(T)=\mathbf{v}'+\ker(T)$, then $\mathbf{v}-\mathbf{v}'\in\ker(T)$, so $T\mathbf{v}=T\mathbf{v}'$. Thus $\widetilde{T}$ is well-defined.
For cosets and $a\in\mathbb{F}$, $\widetilde{T}\big(a(\mathbf{v}+\ker(T))+(\mathbf{v}'+\ker(T))\big)=T(a\mathbf{v}+\mathbf{v}')=aT\mathbf{v}+T\mathbf{v}'$, so $\widetilde{T}$ is linear.

(ii) If $\widetilde{T}(\mathbf{v}+\ker(T))=\mathbf{0}$, then $T\mathbf{v}=\mathbf{0}$, so $\mathbf{v}\in\ker(T)$ and $\mathbf{v}+\ker(T)=\ker(T)$, the zero coset.
Hence $\ker(\widetilde{T})=\{\ker(T)\}$ and $\widetilde{T}$ is injective.

(iii) For every coset $\mathbf{v}+\ker(T)$, the value $\widetilde{T}(\mathbf{v}+\ker(T))=T\mathbf{v}$ lies in $\operatorname{range}(T)$.
Conversely, every vector in $\operatorname{range}(T)$ has the form $T\mathbf{v}=\widetilde{T}(\mathbf{v}+\ker(T))$. Hence the ranges are equal.

(iv) Let $\widehat{T}:\mathcal{V}/\ker(T)\to\operatorname{range}(T)$ be the same formula as $\widetilde{T}$ with codomain restricted to $\operatorname{range}(T)$.
(i)–(iii) show that $\widehat{T}$ is linear, injective, and surjective, so Proposition 1.3.26 gives $\mathcal{V}/\ker(T)\cong\operatorname{range}(T)$.

Structural Commentary: Quotienting by the kernel removes exactly the ambiguity of T.
Key Insight: Every linear map factors as a quotient followed by an isomorphism onto its range. □

Theorem 1.2.23. *Dimension of a Quotient*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U}$ is a subspace, then $\dim(\mathcal{V}/\mathcal{U})=\dim(\mathcal{V})-\dim(\mathcal{U})$.

Proof. Choose a basis $(\mathbf{u}_1,\dots,\mathbf{u}_m)$ of $\mathcal{U}$ and extend it to a basis $(\mathbf{u}_1,\dots,\mathbf{u}_m,\mathbf{w}_1,\dots,\mathbf{w}_k)$ of $\mathcal{V}$.
The cosets $\mathbf{w}_1+\mathcal{U},\dots,\mathbf{w}_k+\mathcal{U}$ span $\mathcal{V}/\mathcal{U}$ because the $\mathbf{u}_j$-terms vanish in the quotient.
If $a_1(\mathbf{w}_1+\mathcal{U})+\cdots+a_k(\mathbf{w}_k+\mathcal{U})=\mathcal{U}$, then $a_1\mathbf{w}_1+\cdots+a_k\mathbf{w}_k\in\mathcal{U}$, so a linear relation in the extended basis forces $a_1=\cdots=a_k=0$. Hence these cosets form a basis of $\mathcal{V}/\mathcal{U}$, and its length is $k=\dim(\mathcal{V})-\dim(\mathcal{U})$.
Structural Commentary: Quotiening removes the directions in $\mathcal{U}$ and keeps the added basis directions.
Key Insight: Passing to $\mathcal{V}/\mathcal{U}$ subtracts the collapsed dimension. □

This block develops two complementary ways to organise a vector space: build from smaller pieces by spans and direct sums, or collapse a chosen subspace by passing to a quotient. Bases and dimension measure exactly what is added, split off, or made invisible.

1.3 Linear Maps and Representations

Linear maps preserve vector-space structure. Kernels, ranges, rank-nullity, matrices, change of basis, and similarity describe them without making coordinates primary.

Definition 1.3.1. *Linear Map*
Let $\mathcal{V}$ and $\mathcal{W}$ be vector spaces over the same field $\mathbb{F}$.
A function $T:\mathcal{V}\to\mathcal{W}$ is *linear* if for all $\mathbf{u},\mathbf{v}\in\mathcal{V}$ and all $a,b\in\mathbb{F}$, $T(a\mathbf{u}+b\mathbf{v})=aT(\mathbf{u})+bT(\mathbf{v})$.
Set of all linear maps from $\mathcal{V}$ to $\mathcal{W}$ is denoted $\mathcal{L}(\mathcal{V},\mathcal{W})$. If $\mathcal{V}=\mathcal{W}$, then elements of $\mathcal{L}(\mathcal{V})$ are *operators* on $\mathcal{V}$.

Definition 1.3.2. *Zero and Identity Maps*
The *zero map* from $\mathcal{V}$ to $\mathcal{W}$ is the function $0_{\mathcal{V},\mathcal{W}} : \mathcal{V} \to \mathcal{W}$ defined by $0_{\mathcal{V},\mathcal{W}}\mathbf{v} = \mathbf{0}_{\mathcal{W}}$.
The *identity map* on $\mathcal{V}$ is $I_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}$ defined by $I_{\mathcal{V}}\mathbf{v} = \mathbf{v}$.

Proposition 1.3.3. *Linear Maps Send Zero to Zero*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $T\mathbf{0}_{\mathcal{V}} = \mathbf{0}_{\mathcal{W}}$.

Proof. By linearity, $T\mathbf{0}_{\mathcal{V}} = T(\mathbf{0}_{\mathcal{V}} + \mathbf{0}_{\mathcal{V}}) = T\mathbf{0}_{\mathcal{V}} + T\mathbf{0}_{\mathcal{V}}$. Adding the additive inverse of $T\mathbf{0}_{\mathcal{V}}$ gives $T\mathbf{0}_{\mathcal{V}} = \mathbf{0}_{\mathcal{W}}$.
Structural Commentary: Linearity preserves zero.
Key Insight: A linear map cannot move zero away from zero. □

Definition 1.3.4. *Operations on Linear Maps*
For $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$, define $(S+T)\mathbf{v} := S\mathbf{v} + T\mathbf{v}$, $(aT)\mathbf{v} := a(T\mathbf{v})$ for every $\mathbf{v} \in \mathcal{V}$.

Proposition 1.3.5. *Linear Maps Form a Vector Space*
With the operations above, $\mathcal{L}(\mathcal{V}, \mathcal{W})$ is a vector space over $\mathbb{F}$.

Proof. If $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$, then for $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $c \in \mathbb{F}$,

$$(aS+T)(c\mathbf{u}+\mathbf{v}) = a(cS\mathbf{u}+S\mathbf{v}) + (cT\mathbf{u}+T\mathbf{v}) = c(aS+T)\mathbf{u} + (aS+T)\mathbf{v}.$$

Thus $\mathcal{L}(\mathcal{V}, \mathcal{W})$ is closed under linear combinations. The zero map is the additive identity, $-T$ is the additive inverse of T, and the remaining vector space axioms hold pointwise because they hold in $\mathcal{W}$.
Structural Commentary: Linear map operations are pointwise.
Key Insight: Pointwise sums and scalar multiples preserve linearity. □

Definition 1.3.6. *Product of Linear Maps*
If $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the *product* $ST : \mathcal{U} \to \mathcal{W}$ is composition: $(ST)\mathbf{u} := S(T\mathbf{u})$ for every $\mathbf{u} \in \mathcal{U}$.

Proposition 1.3.7. *Algebra of Linear Map Products*
Products of linear maps are linear and satisfy associativity, identity laws, and distributive laws whenever the domains and codomains make the expressions meaningful.

Proof. If $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$, $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, $\mathbf{u}, \mathbf{x} \in \mathcal{U}$, and $a \in \mathbb{F}$, then $(ST)(a\mathbf{u}+\mathbf{x}) = S(aT\mathbf{u}+T\mathbf{x}) = a(ST)\mathbf{u}+(ST)\mathbf{x}$, so ST is linear. The identities $(T_1T_2)T_3 = T_1(T_2T_3)$, $TI_{\mathcal{U}} = I_{\mathcal{V}}T = T$, $(S_1+S_2)T = S_1T + S_2T$, and $S(T_1+T_2) = ST_1 + ST_2$ follow by evaluating both sides on an arbitrary vector and using associativity and distributivity in the relevant codomain.
Structural Commentary: Composition chains linear actions, while addition distributes through evaluation.
Key Insight: Linear map algebra is function composition plus pointwise vector-space algebra. □

Definition 1.3.8. *Graph of a Function*
For a function $T : \mathcal{V} \to \mathcal{W}$, the *graph* of T is the subset of $\mathcal{V} \times \mathcal{W}$ defined by $\text{graph}(T) := \{(\mathbf{v}, T\mathbf{v}) : \mathbf{v} \in \mathcal{V}\}$.

Proposition 1.3.9. *Linearity and Graphs*
A function $T : \mathcal{V} \to \mathcal{W}$ is linear if and only if graph(T) *is a subspace of $\mathcal{V} \times \mathcal{W}$.*

Proof.
($\Rightarrow$) If T is linear, then $(\mathbf{0}, \mathbf{0}) \in \text{graph}(T)$.
If $(\mathbf{u}, T\mathbf{u}), (\mathbf{v}, T\mathbf{v}) \in \text{graph}(T)$ and $a \in \mathbb{F}$, then $a(\mathbf{u}, T\mathbf{u}) + (\mathbf{v}, T\mathbf{v}) = (a\mathbf{u}+\mathbf{v}, T(a\mathbf{u}+\mathbf{v})) \in \text{graph}(T)$.
Thus graph(T) is a subspace.
($\Leftarrow$) If graph(T) is a subspace, then for $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $a \in \mathbb{F}$, $a(\mathbf{u}, T\mathbf{u}) + (\mathbf{v}, T\mathbf{v}) = (a\mathbf{u}+\mathbf{v}, aT\mathbf{u}+T\mathbf{v})$ lies in graph(T). Hence $T(a\mathbf{u}+\mathbf{v}) = aT\mathbf{u}+T\mathbf{v}$, so T is linear.

Structural Commentary: A graph records a function as a subset of a product space.
Key Insight: Closure of the graph is exactly linearity of the function. □

Principle 1.3.10. *Coordinate-Free Principle*
A linear map is completely determined by what it does to a basis. Coordinates do not create the map; they only encode it once bases have been chosen.

Proposition 1.3.11. *Linear Maps Are Determined by a Basis*
If $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$ and $\mathbf{w}_1, \ldots, \mathbf{w}_n \in \mathcal{W}$ are arbitrary, then there exists a unique linear map $T : \mathcal{V} \to \mathcal{W}$ such that $T(\mathbf{v}_j) = \mathbf{w}_j$ for $j = 1, \ldots, n$.

Proof. Every $\mathbf{v} \in \mathcal{V}$ has a unique expression $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$. Define $T(\mathbf{v}) := a_1\mathbf{w}_1 + \cdots + a_n\mathbf{w}_n$.
This is well-defined as basis coordinates are unique, and linearity follows by applying the definition to coordinate expansions. Any linear map with the prescribed values on the basis must agree with this formula, so it is unique.
Structural Commentary: A basis reduces a linear map to finitely many choices.
Key Insight: Once basis images are fixed, linearity determines every other value. □

Definition 1.3.12. *Kernel and Range*
For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, define $\ker(T) := \{\mathbf{v} \in \mathcal{V} : T(\mathbf{v}) = \mathbf{0}\}$, $\text{range}(T) := \{T(\mathbf{v}) : \mathbf{v} \in \mathcal{V}\}$.
The kernel is also called the *nullspace* of T.

Definition 1.3.13. *Injective and Surjective Linear Maps*
A function $T : \mathcal{V} \to \mathcal{W}$ is *injective* if $T\mathbf{u} = T\mathbf{v}$ implies $\mathbf{u} = \mathbf{v}$. It is *surjective* if $\text{range}(T) = \mathcal{W}$.

Proposition 1.3.14. *Kernel and Range Are Subspaces*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $\ker(T)$ is a subspace of $\mathcal{V}$ and $\text{range}(T)$ is a subspace of $\mathcal{W}$.

Proof. Since $T\mathbf{0} = \mathbf{0}$, the kernel is nonempty.
If $\mathbf{u}, \mathbf{v} \in \ker(T)$ and $a \in \mathbb{F}$, then $T(a\mathbf{u} + \mathbf{v}) = aT\mathbf{u} + T\mathbf{v} = \mathbf{0}$ so $a\mathbf{u} + \mathbf{v} \in \ker(T)$.
For the range, if $T\mathbf{u}, T\mathbf{v} \in \text{range}(T)$, then $aT\mathbf{u} + T\mathbf{v} = T(a\mathbf{u} + \mathbf{v}) \in \text{range}(T)$. Thus both sets are subspaces.
Structural Commentary: The kernel and range are the two canonical subspaces attached to a linear map.
Key Insight: Linearity makes both preimages of zero and images of vectors closed under linear combinations. □

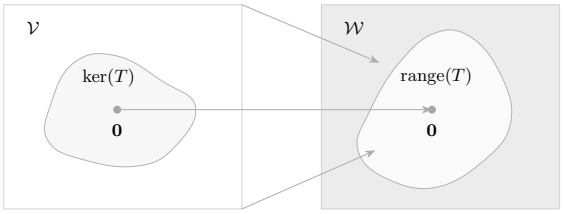


Figure 1: For a linear map $T : \mathcal{V} \to \mathcal{W}$, the range of T may be a proper subspace of $\mathcal{W}$, and the kernel of T may contain nonzero vectors. $\text{range}(T) = \mathcal{W}$ and $\ker(T) = \{\mathbf{0}\}$ characterise surjectivity and injectivity, respectively.

Proposition 1.3.15. *Shift Operators on Finitely Nonzero Sequences*
Define $R, S : \mathbb{F}^{(\mathbb{N})} \to \mathbb{F}^{(\mathbb{N})}$ by $R(x_1, x_2, \ldots) = (0, x_1, x_2, \ldots)$, $S(x_1, x_2, \ldots) = (x_2, x_3, \ldots)$. Then $R, S \in \mathcal{L}(\mathbb{F}^{(\mathbb{N})})$. Also, $\ker(R) = \{\mathbf{0}\}$, $\text{range}(R) = \{(x_1, x_2, \ldots) \in \mathbb{F}^{(\mathbb{N})} : x_1 = 0\}$, $\ker(S) = \text{span}((1, 0, 0, \ldots))$, $\text{range}(S) = \mathbb{F}^{(\mathbb{N})}$.

Proof. Both R and S preserve finite support, so they map $\mathbb{F}^{(\mathbb{N})}$ into itself.
Since both maps act componentwise by shifting entries, $R(a\mathbf{x} + \mathbf{y}) = aR\mathbf{x} + R\mathbf{y}$ and $S(a\mathbf{x} + \mathbf{y}) = aS\mathbf{x} + S\mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{F}^{(\mathbb{N})}$ and $a \in \mathbb{F}$. Thus R and S are linear.
The map R has zero kernel as $R\mathbf{x} = \mathbf{0}$ forces every entry of $\mathbf{x}$ to be zero, and its range is exactly the set of finitely nonzero sequences with first entry zero. For S, $S\mathbf{x} = \mathbf{0}$ forces $x_2 = x_3 = \cdots = 0$, so $\ker(S) = \text{span}((1, 0, 0, \ldots))$.
Finally, for any $(y_1, y_2, \ldots) \in \mathbb{F}^{(\mathbb{N})}$, we have $S(0, y_1, y_2, \ldots) = (y_1, y_2, \ldots)$, so S is surjective.
Structural Commentary: Shift operators expose behavior that finite-dimensional maps cannot have.
Key Insight: The right shift is injective but not surjective, while the left shift is surjective but not injective. □

Proposition 1.3.16. *Injectivity and the Kernel*
A linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is injective if and only if $\ker(T) = \{\mathbf{0}\}$.

Proof.
($\Rightarrow$) If T is injective and $\mathbf{v} \in \ker(T)$, then $T\mathbf{v} = \mathbf{0} = T\mathbf{0}$, so $\mathbf{v} = \mathbf{0}$.
($\Leftarrow$) If $\ker(T) = \{\mathbf{0}\}$ and $T\mathbf{v} = T\mathbf{w}$, then $T(\mathbf{v} - \mathbf{w}) = \mathbf{0}$. Hence $\mathbf{v} - \mathbf{w} \in \ker(T)$, so $\mathbf{v} = \mathbf{w}$.
Structural Commentary: The kernel measures exactly where a linear map collapses vectors to zero.
Key Insight: Injectivity is equivalent to collapsing only the zero vector. □

Theorem 1.3.17. *Range from Basis Images*
Let $\mathcal{V}$ be finite-dimensional with basis $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then $\text{range}(T) = \text{span}(T\mathbf{v}_1, \ldots, T\mathbf{v}_n)$. In particular, $\text{range}(T)$ is finite-dimensional and $\dim(\text{range}(T)) \leq n$.

Proof. If $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, then $T\mathbf{v} = c_1T\mathbf{v}_1 + \cdots + c_nT\mathbf{v}_n$.
Thus every vector in $\text{range}(T)$ lies in $\text{span}(T\mathbf{v}_1, \ldots, T\mathbf{v}_n)$, and the reverse inclusion is immediate from the definition of range. The dimension bound follows because this spanning list has length n.
Structural Commentary: The range is generated by the images of any basis of the domain.
Key Insight: Linear maps carry a basis-spanning formula directly into a range-spanning formula. □

Definition 1.3.18. *Rank and Nullity*
For a finite-dimensional domain, the rank of T is $\dim(\text{range}(T))$ and the nullity of T is $\dim(\ker(T))$.

Theorem 1.3.19. *Rank-Nullity Theorem*
If $\mathcal{V}$ is finite-dim and $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $\text{range}(T)$ is finite-dim and $\dim(\mathcal{V}) = \dim(\ker(T)) + \dim(\text{range}(T))$.

Proof. By Proposition 1.3.14 and Corollary 1.1.26, $\ker(T)$ has a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m)$.
Extend it to a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_r)$ of $\mathcal{V}$ by Theorem 1.1.25.
If $\mathbf{v} = \sum_{i=1}^{m} a_i \mathbf{u}_i + \sum_{j=1}^{r} b_j \mathbf{w}_j$, then $T\mathbf{v} = \sum_{j=1}^{r} b_j T\mathbf{w}_j$, because $T\mathbf{u}_i = \mathbf{0}$. Hence $(T\mathbf{w}_1, \dots, T\mathbf{w}_r)$ spans $\text{range}(T)$.
If $\sum_{j=1}^{r} c_j T\mathbf{w}_j = \mathbf{0}$, then $\sum_{j=1}^{r} c_j \mathbf{w}_j \in \ker(T)$, so for some scalars d_i, $\sum_{j=1}^{r} c_j \mathbf{w}_j = \sum_{i=1}^{m} d_i \mathbf{u}_i$.
Thus $\sum_{j=1}^{r} c_j \mathbf{w}_j - \sum_{i=1}^{m} d_i \mathbf{u}_i = \mathbf{0}$, and linear independence of extended basis forces every c_j and d_i to be zero.
Hence $(T\mathbf{w}_1, \dots, T\mathbf{w}_r)$ is a basis of $\text{range}(T)$. Thus $\dim(\ker(T)) = m$, $\dim(\text{range}(T)) = r$, and $\dim(\mathcal{V}) = m + r$.
Structural Commentary: An adapted basis splits $\mathcal{V}$ into kernel directions and directions detected by T.
Key Insight: Rank-nullity says every domain dimension is either killed by T or appears in its range. □

Corollary 1.3.20. *Maps to Smaller Spaces Are Not Injective*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces with $\dim(\mathcal{V}) > \dim(\mathcal{W})$.
Then no linear map from $\mathcal{V}$ to $\mathcal{W}$ is injective.

Proof. Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. By Theorem 1.3.19, $\dim(\ker(T)) = \dim(\mathcal{V}) - \dim(\text{range}(T)) \geq \dim(\mathcal{V}) - \dim(\mathcal{W}) > 0$.
Thus $\ker(T)$ contains a nonzero vector, so T is not injective by Proposition 1.3.16.
Structural Commentary: Excess domain dimension forces collapse.
Key Insight: The dimension deficit must appear inside the kernel. □

Corollary 1.3.21. *Maps to Larger Spaces Are Not Surjective*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces with $\dim(\mathcal{V}) < \dim(\mathcal{W})$.
Then no linear map from $\mathcal{V}$ to $\mathcal{W}$ is surjective.

Proof. Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. By Theorem 1.3.19, $\dim(\text{range}(T)) = \dim(\mathcal{V}) - \dim(\ker(T)) \leq \dim(\mathcal{V}) < \dim(\mathcal{W})$.
Hence $\text{range}(T)$ is a proper subspace of $\mathcal{W}$, so T is not surjective.
Structural Commentary: A smaller domain cannot cover a larger codomain.
Key Insight: The range can never have dimension larger than the domain. □

Corollary 1.3.22. *Equal-Dimension Kernel-Range Criterion*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces with $\dim(\mathcal{V}) = \dim(\mathcal{W})$, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
Then $\ker(T) = \{\mathbf{0}\}$ if and only if $\text{range}(T) = \mathcal{W}$. Equivalently, T is injective if and only if T is surjective.

Proof.
($\Rightarrow$) If $\ker(T) = \{\mathbf{0}\}$, then rank-nullity gives $\dim(\text{range}(T)) = \dim(\mathcal{V}) = \dim(\mathcal{W})$.
Since $\text{range}(T)$ is a subspace of $\mathcal{W}$ with the same dimension, $\text{range}(T) = \mathcal{W}$.
($\Leftarrow$) If $\text{range}(T) = \mathcal{W}$, then rank-nullity gives $\dim(\ker(T)) = \dim(\mathcal{V}) - \dim(\mathcal{W}) = 0$, hence $\ker(T) = \{\mathbf{0}\}$.
Structural Commentary: Equal finite dimensions make injectivity and surjectivity equivalent.
Key Insight: In equal dimensions, having no kernel loss is exactly having full range. □

Definition 1.3.23. *Invertible Linear Map and Isomorphism*
A linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is *invertible* if there exists $S \in \mathcal{L}(\mathcal{W}, \mathcal{V})$ such that $ST = I_{\mathcal{V}}$ and $TS = I_{\mathcal{W}}$.
An invertible linear map is called an *isomorphism*, and if one exists we write $\mathcal{V} \cong \mathcal{W}$.

Proposition 1.3.24. *Inverse Is Unique*
An invertible linear map has a unique inverse. Thus if T is invertible, its inverse is denoted T^{-1}.

Proof. Suppose $S_1, S_2 \in \mathcal{L}(\mathcal{W}, \mathcal{V})$ are inverses of $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
Then $S_1 = S_1 I_{\mathcal{W}} = S_1(TS_2) = (S_1 T)S_2 = I_{\mathcal{V}} S_2 = S_2$. Hence the inverse is unique.
Structural Commentary: A two-sided inverse is forced by the composition identities.
Key Insight: Associativity lets the two candidate inverses meet around T and collapse to the same map. □

Proposition 1.3.25. *Inverse of a Product Reverses Order*
Let $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ be invertible. Then ST is invertible and $(ST)^{-1} = T^{-1}S^{-1}$.

Proof. Since $T^{-1} \in \mathcal{L}(\mathcal{V}, \mathcal{U})$ and $S^{-1} \in \mathcal{L}(\mathcal{W}, \mathcal{V})$, the product $T^{-1}S^{-1} \in \mathcal{L}(\mathcal{W}, \mathcal{U})$ is defined.
Moreover, $(T^{-1}S^{-1})(ST) = T^{-1}(S^{-1}S)T = T^{-1}T = I_{\mathcal{U}}$ and $(ST)(T^{-1}S^{-1}) = S(TT^{-1})S^{-1} = SS^{-1} = I_{\mathcal{W}}$.
Thus $T^{-1}S^{-1}$ is a two-sided inverse of ST, so $(ST)^{-1} = T^{-1}S^{-1}$.
Structural Commentary: To undo a composition, undo the last map first.
Key Insight: Composition order reverses under inversion. □

Proposition 1.3.26. *Invertibility and Bijectivity*
A linear map is invertible if and only if it is both injective and surjective.

Proof.

($\Rightarrow$) Let S be an inverse of T. If $T\mathbf{v} = T\mathbf{w}$, then $\mathbf{v} = ST\mathbf{v} = ST\mathbf{w} = \mathbf{w}$, so T is injective. Also every $\mathbf{w}$ in the codomain satisfies $\mathbf{w} = T(S\mathbf{w})$, so T is surjective.

($\Leftarrow$) Suppose T is injective and surjective. Define $S\mathbf{w}$ to be the unique vector $\mathbf{v}$ with $T\mathbf{v} = \mathbf{w}$. Since both sides below map under T to $a\mathbf{w}_1 + \mathbf{w}_2$, $S(a\mathbf{w}_1 + \mathbf{w}_2) = aS\mathbf{w}_1 + S\mathbf{w}_2$. Thus S is linear and satisfies $ST = I$ and $TS = I$, so T is invertible.

Structural Commentary: Bijectivity gives a unique reverse map, and linearity makes it a linear inverse.
Key Insight: A linear map is invertible exactly when it is one-to-one and onto. □

Theorem 1.3.27. *Finite-Dimensional Invertibility Criteria*
If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$, then the following are equivalent:

i. T is invertible;
ii. T is injective;
iii. T is surjective.

Proof.

(i) $\Rightarrow$ (ii) follows from Proposition 1.3.26.

(ii) $\Rightarrow$ (iii): If T is injective, then $\ker(T) = \{\mathbf{0}\}$. Rank-nullity gives $\dim(\text{range}(T)) = \dim(\mathcal{V})$, hence $\text{range}(T) = \mathcal{V}$, so T is surjective.

(iii) $\Rightarrow$ (i): If T is surjective, rank-nullity gives $\dim(\ker(T)) = 0$, so T is injective. Thus T is bijective, hence invertible by Proposition 1.3.26.

Structural Commentary: For operators on finite-dimensional spaces, kernel and range dimensions trade off exactly. Injectivity and surjectivity therefore force each other.
Key Insight: In equal finite dimensions, losing no information is the same as reaching the whole space. □

Perspective 1.3.28. *Kernel Obstruction to Invertibility*
By Theorem 1.3.27 and Proposition 1.3.16, a finite-dimensional operator is invertible exactly when its kernel is zero. Any nonzero $\mathbf{v} \in \ker T$ therefore blocks invertibility.

Theorem 1.3.29. *Isomorphism and Dimension*
Two finite-dimensional vector spaces are isomorphic if and only if they have the same dimension.

Proof.

($\Rightarrow$) If $T : \mathcal{V} \to \mathcal{W}$ is an isomorphism and $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$, then $(T\mathbf{v}_1, \ldots, T\mathbf{v}_n)$ is a basis of $\mathcal{W}$.

($\Leftarrow$) If both spaces have dimension n, choose bases $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ and $(\mathbf{w}_1, \ldots, \mathbf{w}_n)$. By Proposition 1.3.11, there is a linear map T with $T\mathbf{v}_j = \mathbf{w}_j$ for every j. Then $T(\sum a_j\mathbf{v}_j) = \sum a_j\mathbf{w}_j$, so $\ker(T) = \{\mathbf{0}\}$ and $\text{range}(T) = \mathcal{W}$. Hence T is an isomorphism.

Structural Commentary: Dimension classifies finite-dimensional spaces up to isomorphism.
Key Insight: Matching bases give an isomorphism; any isomorphism carries bases to bases. □

Definition 1.3.30. *Matrices, Rows, and Columns*
For positive integers m, n, $\text{Mat}(m, n, \mathbb{F})$ denotes the set of all $m \times n$ matrices with entries in $\mathbb{F}$.
If $A = [A_{j,k}] \in \text{Mat}(m, n, \mathbb{F})$, then $A_{j,k}$ is the entry in row j and column k. The jth row is denoted $A_{j,\cdot}$, and the kth column is denoted $A_{\cdot,k}$. The transpose of A is $A^t \in \text{Mat}(n, m, \mathbb{F})$, defined by $(A^t)_{k,j} = A_{j,k}$.

Definition 1.3.31. *Matrix Operations*
Matrices of same size, addition and scalar multiplication defined by $(A+C)_{j,k} = A_{j,k}+C_{j,k}$ and $(aA)_{j,k} = aA_{j,k}$.
If $A \in \text{Mat}(m, n, \mathbb{F})$ and $B \in \text{Mat}(n, p, \mathbb{F})$, then $AB \in \text{Mat}(m, p, \mathbb{F})$ is defined by $(AB)_{j,k} := \sum_{r=1}^{n} A_{j,r}B_{r,k}$.

Proposition 1.3.32. *Matrix Spaces*
For positive integers m, n, $\text{Mat}(m, n, \mathbb{F})$ is a vector space over $\mathbb{F}$ with dimension mn.

Proof. Entrywise addition and scalar multiplication satisfy the vector space axioms because $\mathbb{F}$ does. For $1 \le j \le m$ and $1 \le k \le n$, let $E_{j,k}$ be the matrix with 1 in entry (j, k) and 0 elsewhere.
Every $A = [A_{j,k}]$ has the unique expansion $A = \sum_{j=1}^{m} \sum_{k=1}^{n} A_{j,k}E_{j,k}$.
Thus the mn matrices $E_{j,k}$ form a basis, so $\dim(\text{Mat}(m, n, \mathbb{F})) = mn$.
Structural Commentary: A matrix is a finite array of independent scalar slots.
Key Insight: The standard entry matrices isolate those slots one at a time. □

Proposition 1.3.33. *Matrix Product Entries Are Row Times Column*
If $A \in \text{Mat}(m, n, \mathbb{F})$ and $B \in \text{Mat}(n, p, \mathbb{F})$, then $(AB)_{j,k} = A_{j,\cdot}B_{\cdot,k}$ for $1 \le j \le m$ and $1 \le k \le p$.

Proof. The row-column product $A_{j,\cdot}B_{\cdot,k}$ is exactly $\sum_{r=1}^{n} A_{j,r}B_{r,k}$, which is the definition of $(AB)_{j,k}$.
Structural Commentary: Matrix multiplication is assembled entry-by-entry from row-column pairings.
Key Insight: Each entry of AB sees one row of A and one column of B. □

Proposition 1.3.34. *Matrix-Vector Products as Column Combinations*
If $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m, n, \mathbb{F})$ *and* $\mathbf{x} = (x_1, \ldots, x_n)^t \in \mathbb{F}^n$, *then* $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$.

Proof. The jth entry of $A\mathbf{x}$ is $\sum_{k=1}^{n} A_{j,k}x_k$, which is also the jth entry of $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$.
Structural Commentary: Multiplying by a column selects a linear combination of the columns of A.
Key Insight: The coordinates of $\mathbf{x}$ are precisely the column coefficients. □

Proposition 1.3.35. *Solvability and Complete Solution Sets*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m, n, \mathbb{F})$ *and let* $\mathbf{b} \in \mathbb{F}^m$. *Then* $A\mathbf{x} = \mathbf{b}$ *is solvable if and only if* $\mathbf{b} \in \mathrm{col}(A)$.
If $\mathbf{x}_p$ *is one solution, then every solution is of the form* $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_0$, *where* $\mathbf{x}_0 \in \ker(A)$.

Proof.

($\Rightarrow$) If $A\mathbf{x} = \mathbf{b}$ for some $\mathbf{x} = (x_1, \ldots, x_n)^T$, then $\mathbf{b} = A\mathbf{x} = \sum_{j=1}^{n} x_j\mathbf{a}_j \in \mathrm{col}(A)$.

($\Leftarrow$) If $\mathbf{b} \in \mathrm{col}(A)$, then $\mathbf{b} = \sum_{j=1}^{n} x_j\mathbf{a}_j$ for some scalars x_j. Hence, for $\mathbf{x} = (x_1, \ldots, x_n)^T$, we have $A\mathbf{x} = \mathbf{b}$.

Now suppose $A\mathbf{x}_p = \mathbf{b}$. For any $\mathbf{x}$, $A\mathbf{x} = \mathbf{b} \iff A(\mathbf{x} - \mathbf{x}_p) = \mathbf{0} \iff \mathbf{x} - \mathbf{x}_p \in \ker(A)$.
Thus every solution is exactly of the form $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_0$, where $\mathbf{x}_0 \in \ker(A)$.
Structural Commentary: A consistent system is an affine shift of $\ker(A)$ by any particular solution.
Key Insight: Column space decides existence; kernel decides nonuniqueness. □

Proposition 1.3.36. *Matrix Products by Columns*
If $A \in \mathrm{Mat}(m, r, \mathbb{F})$ *and* $B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_n] \in \mathrm{Mat}(r, n, \mathbb{F})$, *then* $AB = [A\mathbf{b}_1 \ \cdots \ A\mathbf{b}_n]$.

Proof. For each k, the kth column of AB has entries $\sum_{\ell=1}^{r} A_{j,\ell}B_{\ell,k}$, which are exactly the entries of $A\mathbf{b}_k$.
Structural Commentary: Matrix products apply A to columns of B.
Key Insight: The columns of AB are obtained by applying A to the columns of B. □

Proposition 1.3.37. *Matrix Products as Sums of Outer Products*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_r] \in \mathrm{Mat}(m, r, \mathbb{F})$, *and write* $B \in \mathrm{Mat}(r, n, \mathbb{F})$ *by rows as*

$$B = \begin{bmatrix} \mathbf{b}_1^t \\ \vdots \\ \mathbf{b}_r^t \end{bmatrix}.$$

Then $AB = \mathbf{a}_1\mathbf{b}_1^t + \cdots + \mathbf{a}_r\mathbf{b}_r^t$.

Proof. The (j, k) entry of $\mathbf{a}_\ell\mathbf{b}_\ell^t$ is $A_{j,\ell}B_{\ell,k}$. Summing over $\ell = 1, \ldots, r$ gives $\sum_{\ell=1}^{r} A_{j,\ell}B_{\ell,k} = (AB)_{j,k}$.
Structural Commentary: Column-row products decompose a matrix product into single column-row pieces.
Key Insight: Each shared index ℓ contributes one outer product. □

Proposition 1.3.38. *Transpose Reverses Products*
If $A \in \mathrm{Mat}(m, n, \mathbb{F})$ *and* $B \in \mathrm{Mat}(n, p, \mathbb{F})$, *then* $(AB)^t = B^tA^t$.

Proof. For $1 \le j \le p$ and $1 \le k \le m$, $((AB)^t)_{j,k} = (AB)_{k,j} = \sum_{\ell=1}^{n} A_{k,\ell}B_{\ell,j} = \sum_{\ell=1}^{n} (B^t)_{j,\ell}(A^t)_{\ell,k} = (B^tA^t)_{j,k}$.
Thus the matrices are equal entrywise.
Structural Commentary: Transposition swaps the order of composition encoded by matrix multiplication.
Key Insight: Rows become columns, so the product order reverses. □

Corollary 1.3.39. *Matrix Inverses and Transposes Reverse Products*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ *be invertible. Then* AB, A^t *are invertible, and* $(AB)^{-1} = B^{-1}A^{-1}$; $(A^t)^{-1} = (A^{-1})^t$.

Proof. First, $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I_n$; $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I_n$.
Hence $(AB)^{-1} = B^{-1}A^{-1}$. By Proposition 1.3.38, $(A^{-1})^tA^t = (AA^{-1})^t = I_n$; $A^t(A^{-1})^t = (A^{-1}A)^t = I_n$.
Thus $(A^t)^{-1} = (A^{-1})^t$.
Structural Commentary: Inversion and transposition both reverse multiplication order.
Key Insight: Apply the defining inverse equations and reverse products. □

Definition 1.3.40. *Coordinate Vector*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of a finite-dimensional vector space $\mathcal{V}$.
If $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, then the *$\beta$-coordinate vector* of $\mathbf{v}$ is

$$[\mathbf{v}]_\beta := \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{F}^n.$$

The function $[\cdot]_\beta : \mathcal{V} \to \mathbb{F}^n$ is the *β-basis representation function*.

Proposition 1.3.41. *Coordinate Maps Are Linear Bijections*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of a finite-dimensional vector space $\mathcal{V}$.
Then the coordinate map $[\cdot]_\beta : \mathcal{V} \to \mathbb{F}^n$ is linear and bijective.

Proof. If $\mathbf{u} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$ and $\mathbf{v} = b_1\mathbf{v}_1 + \cdots + b_n\mathbf{v}_n$, then for $c \in \mathbb{F}$,

$$[c\mathbf{u} + \mathbf{v}]_\beta = \begin{bmatrix} ca_1 + b_1 \\ \vdots \\ ca_n + b_n \end{bmatrix} = c[\mathbf{u}]_\beta + [\mathbf{v}]_\beta$$

Thus $[\cdot]_\beta$ is linear. It is injective because $[\mathbf{v}]_\beta = \mathbf{0}$ forces all basis coefficients of $\mathbf{v}$ to be zero, and it is surjective because every column $(c_1, \dots, c_n)^t \in \mathbb{F}^n$ equals $[c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n]_\beta$.
Structural Commentary: A basis converts abstract vectors into coordinate columns without losing structure.
Key Insight: Coordinates give a structure-preserving copy of the vector space, not extra information. □

Definition 1.3.42. *Matrix Representation*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$ and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_m)$ a basis of $\mathcal{W}$. For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the matrix of T relative to (β, γ) is $[T]_{\gamma \leftarrow \beta} := \begin{bmatrix} [T(\mathbf{v}_1)]_\gamma & \cdots & [T(\mathbf{v}_n)]_\gamma \end{bmatrix} \in \mathrm{Mat}(m, n, \mathbb{F})$.

Definition 1.3.43. *Column Space, Matrix Rank, and Nullity*
Let $A = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix} \in \mathrm{Mat}(m, n, \mathbb{F})$.

(i) $\mathrm{col}(A) := \mathrm{span}(\mathbf{a}_1, \dots, \mathbf{a}_n) \subseteq \mathbb{F}^m$, and $\mathrm{row}(A) := \mathrm{col}(A^t) \subseteq \mathbb{F}^n$.
(ii) $\mathrm{rank}(A) := \dim(\mathrm{col}(A))$; column rank is $\dim(\mathrm{col}(A))$, and row rank is $\dim(\mathrm{row}(A))$.
(iii) A has *full rank* if $\mathrm{rank}(A) = \min(m, n)$, *full column rank* if $\mathrm{rank}(A) = n$, and *full row rank* if $\mathrm{rank}(A) = m$.
(iv) $\ker(A) := \{\mathbf{x} \in \mathbb{F}^n : A\mathbf{x} = \mathbf{0}\}$ is the *nullspace*, and $\mathrm{nullity}(A) := \dim(\ker(A))$.

Proposition 1.3.44. *Stacks Intersect Kernels, Concatenations Sum Columns*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$, $B \in \mathrm{Mat}(p, n, \mathbb{F})$, $C \in \mathrm{Mat}(m, r, \mathbb{F})$, and $D \in \mathrm{Mat}(m, s, \mathbb{F})$.
Then $\ker \begin{bmatrix} A \\ B \end{bmatrix} = \ker(A) \cap \ker(B)$*; and* $\mathrm{col} \begin{bmatrix} C & D \end{bmatrix} = \mathrm{col}(C) + \mathrm{col}(D)$.

Proof. For $\mathbf{x} \in \mathbb{F}^n$, $\mathbf{x} \in \ker \begin{bmatrix} A \\ B \end{bmatrix} \iff \begin{bmatrix} A \\ B \end{bmatrix} \mathbf{x} = \mathbf{0} \iff A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0} \iff \mathbf{x} \in \ker(A) \cap \ker(B)$.

For $\mathbf{y} \in \mathbb{F}^m$, $\mathbf{y} \in \mathrm{col} \begin{bmatrix} C & D \end{bmatrix} \iff \exists\, \mathbf{u} \in \mathbb{F}^r,\ \mathbf{v} \in \mathbb{F}^s$ such that $\mathbf{y} = \begin{bmatrix} C & D \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ \mathbf{v} \end{bmatrix}$
$\iff \exists\, \mathbf{u}, \mathbf{v}$ such that $\mathbf{y} = C\mathbf{u} + D\mathbf{v} \iff \mathbf{y} \in \mathrm{col}(C) + \mathrm{col}(D)$.
Structural Commentary: Stacking intersects constraints; concatenation adds column directions.
Key Insight: Stacks encode intersections of nullspaces, while concatenations encode sums of column spaces. □

Proposition 1.3.45. *Matrices Respect Linear Combinations of Maps*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$ and γ be a basis of $\mathcal{W}$.
If $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$, then $[aS + T]_{\gamma \leftarrow \beta} = a[S]_{\gamma \leftarrow \beta} + [T]_{\gamma \leftarrow \beta}$.

Proof. The jth column of $[aS + T]_{\gamma \leftarrow \beta}$ is $[(aS + T)\mathbf{v}_j]_\gamma = [aS\mathbf{v}_j + T\mathbf{v}_j]_\gamma = a[S\mathbf{v}_j]_\gamma + [T\mathbf{v}_j]_\gamma$, which is the jth column of $a[S]_{\gamma \leftarrow \beta} + [T]_{\gamma \leftarrow \beta}$.
Structural Commentary: Matrix representation records basis images column-by-column.
Key Insight: Linear combinations of maps become entrywise linear combinations of their matrices. □

Definition 1.3.46. *Linear Map Induced by a Matrix*
For $A \in \mathrm{Mat}(m, n, \mathbb{F})$, the *linear map induced by A* is the map $T_A : \mathbb{F}^n \to \mathbb{F}^m$ defined by $T_A(\mathbf{x}) = A\mathbf{x}$.

Proposition 1.3.47. *Matrix-Induced Maps Are Linear*
If $A \in \mathrm{Mat}(m, n, \mathbb{F})$, then $T_A : \mathbb{F}^n \to \mathbb{F}^m$ is linear.

Proof. For $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$ and $a \in \mathbb{F}$, $T_A(a\mathbf{x}+\mathbf{y}) = A(a\mathbf{x}+\mathbf{y}) = aA\mathbf{x}+A\mathbf{y} = aT_A(\mathbf{x})+T_A(\mathbf{y})$.
Structural Commentary: Matrix multiplication preserves addition and scalar multiplication.
Key Insight: Every matrix defines a canonical linear map between coordinate spaces. □

Corollary 1.3.48. *Matrix Rank-Nullity*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$. *Then* $\dim(\ker(A)) + \dim(\mathrm{col}(A)) = n$ *and* $\mathrm{nullity}(A) + \mathrm{rank}(A) = n$.
If $m = n$, *then* $\ker(A) = \{\mathbf{0}\}$ *if and only if* $\mathrm{col}(A) = \mathbb{F}^n$.

Proof. Apply Theorem 1.3.19 to the induced linear map $T_A : \mathbb{F}^n \to \mathbb{F}^m$. By definition, $\ker(T_A) = \ker(A)$ and $\mathrm{range}(T_A) = \mathrm{col}(A)$, giving the displayed identities. When $m = n$, the remaining equivalence follows as:
($\Rightarrow$) If $\ker(A) = \{\mathbf{0}\}$, then $\dim(\mathrm{col}(A)) = n$. Since $\mathrm{col}(A) \subseteq \mathbb{F}^n$, Proposition 1.1.29 gives $\mathrm{col}(A) = \mathbb{F}^n$.
($\Leftarrow$) If $\mathrm{col}(A) = \mathbb{F}^n$, then $\dim(\mathrm{col}(A)) = n$, so the identity gives $\dim(\ker(A)) = 0$. Hence $\ker(A) = \{\mathbf{0}\}$.
Structural Commentary: Matrix rank-nullity is ordinary rank-nullity written in coordinates.
Key Insight: Kernel loss plus column-space dimension equals the number of columns. □

Theorem 1.3.49. *Row Rank Equals Column Rank*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $\dim(\mathrm{row}(A)) = \dim(\mathrm{col}(A))$. *Equivalently,* $\mathrm{rank}(A^t) = \mathrm{rank}(A)$.

Proof. Let $r = \mathrm{rank}(A)$. If $r = 0$, then $\mathrm{col}(A) = \{\mathbf{0}\}$, so every column of A is zero and $A = 0$; the result follows. Suppose $r > 0$. Choose columns $\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_r}$ forming a basis of $\mathrm{col}(A)$, and set $B = [\mathbf{a}_{j_1} \ \cdots \ \mathbf{a}_{j_r}] \in \mathrm{Mat}(m,r,\mathbb{F})$. For each column $\mathbf{a}_k$ of A, there is a unique $\mathbf{x}_k \in \mathbb{F}^r$ such that $\mathbf{a}_k = B\mathbf{x}_k$. With $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_n] \in \mathrm{Mat}(r,n,\mathbb{F})$, we have $A = BX$. Therefore $A^t = X^tB^t$, so every column of A^t is a linear combination of the r columns of X^t. Hence $\mathrm{rank}(A^t) \le r = \mathrm{rank}(A)$. Applying the same argument to A^t gives $\mathrm{rank}(A) \le \mathrm{rank}(A^t)$.
Structural Commentary: Column rank controls row rank by factoring A through a basis of its column space.
Key Insight: Rows and columns measure the same intrinsic rank, viewed from transposed coordinates. □

Theorem 1.3.50. *Four Fundamental Matrix Subspaces*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ *and let* $r = \mathrm{rank}(A)$. *Then* $\mathrm{row}(A), \ker(A) \subseteq \mathbb{F}^n$; $\mathrm{col}(A), \ker(A^t) \subseteq \mathbb{F}^m$,
and $\dim(\mathrm{col}(A)) = \dim(\mathrm{row}(A)) = r$; $\dim(\ker(A)) = n - r$; $\dim(\ker(A^t)) = m - r$.

Proof. The inclusions follow from the ambient dimensions of the rows and columns, and from the domains of A and A^t. By definition, $\dim(\mathrm{col}(A)) = r$; by Theorem 1.3.49, $\dim(\mathrm{row}(A)) = r$.
Rank-nullity gives $\dim(\ker(A)) = n - \mathrm{rank}(A) = n - r$. Since $\mathrm{rank}(A^t) = \dim(\mathrm{col}(A^t)) = \dim(\mathrm{row}(A)) = r$, rank-nullity for $A^t : \mathbb{F}^m \to \mathbb{F}^n$ gives $\dim(\ker(A^t)) = m - \mathrm{rank}(A^t) = m - r$.
Structural Commentary: One rank determines all four fundamental dimensions.
Key Insight: Rank counts what survives; nullity counts what is killed. □

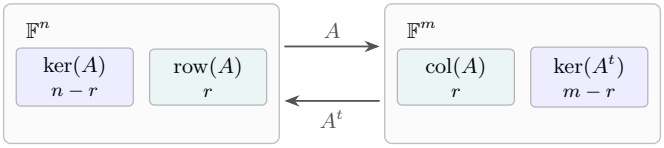


Figure 2: Four fundamental subspaces by host coordinate space.

Corollary 1.3.51. *Homogeneous Systems with More Variables Than Equations*
If $A \in \mathrm{Mat}(m,n,\mathbb{F})$ *with* $n > m$, *then the homogeneous system* $A\mathbf{x} = \mathbf{0}$ *has a nonzero solution.*

Proof. The induced map $T_A : \mathbb{F}^n \to \mathbb{F}^m$ has domain dimension larger than codomain dimension, so Corollary 1.3.20 shows that T_A is not injective.
Hence $\ker(A) = \ker(T_A)$ contains a nonzero vector $\mathbf{x}$, and this vector satisfies $A\mathbf{x} = \mathbf{0}$.
Structural Commentary: Extra variables force a nontrivial kernel.
Key Insight: A nontrivial kernel is exactly a nonzero homogeneous solution. □

Corollary 1.3.52. *Inhomogeneous Systems with More Equations Than Variables*
If $A \in \mathrm{Mat}(m,n,\mathbb{F})$ *with* $m > n$, *then there exists* $\mathbf{b} \in \mathbb{F}^m$ *such that* $A\mathbf{x} = \mathbf{b}$ *has no solution.*

Proof. The induced map $T_A : \mathbb{F}^n \to \mathbb{F}^m$ has domain dimension smaller than codomain dimension, so Corollary 1.3.21 shows that T_A is not surjective. Choose $\mathbf{b} \in \mathbb{F}^m \setminus \mathrm{range}(T_A)$. Then no $\mathbf{x} \in \mathbb{F}^n$ satisfies $A\mathbf{x} = \mathbf{b}$.
Structural Commentary: Too few variables cannot reach every possible right-hand side.
Key Insight: A missing range vector becomes an inconsistent inhomogeneous system. □

Proposition 1.3.53. *Matrix Representation Acts on Coordinates*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$, let γ be a basis of $\mathcal{W}$, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
Then for every $\mathbf{v} \in \mathcal{V}$, $[T(\mathbf{v})]_\gamma = [T]_{\gamma \leftarrow \beta}[\mathbf{v}]_\beta$.

Proof. Write $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$. Then linearity gives $T(\mathbf{v}) = a_1T(\mathbf{v}_1) + \cdots + a_nT(\mathbf{v}_n)$ and the right-hand side is exactly the linear combination of the columns of $[T]_{\gamma \leftarrow \beta}$ with coefficients $a_1, \dots, a_n$.
Structural Commentary: Matrix representation turns applying T into multiplying coordinate columns.
Key Insight: The columns of $[T]_{\gamma \leftarrow \beta}$ are the coordinate images of the basis vectors. □

Theorem 1.3.54. *Columns of an Invertible Matrix Form a Basis*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(n, n, \mathbb{F})$ be invertible. Then $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ is a basis of $\mathbb{F}^n$.

Proof. The columns span $\mathbb{F}^n$ because for any $\mathbf{y} \in \mathbb{F}^n$, writing $\mathbf{x} = A^{-1}\mathbf{y}$ gives $\mathbf{y} = A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$.
They are linearly independent because if $c_1\mathbf{a}_1 + \cdots + c_n\mathbf{a}_n = \mathbf{0}$, then $A\mathbf{c} = \mathbf{0}$, so $\mathbf{c} = A^{-1}\mathbf{0} = \mathbf{0}$.
Structural Commentary: An invertible matrix is a coordinate isomorphism of $\mathbb{F}^n$. Its columns are the images of the standard coordinate directions.
Key Insight: Invertibility forces the columns to span and prevents nontrivial column relations. □

Lemma 1.3.55. *Uniqueness of Two-Sided Inverses*
Let $A, B, C \in \mathrm{Mat}(n, n, \mathbb{F})$. If $AB = I$ and $BC = I$, then $A = C$.

Proof. Using $BC = I$ and $AB = I$, $A = AI = A(BC) = (AB)C = IC = C$.
Structural Commentary: A common middle inverse forces the left and right inverses to coincide.
Key Insight: Associativity turns $AB = I = BC$ into $A = C$. □

Theorem 1.3.56. *One-Sided Inverses of Square Matrices*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$. If $AB = I$, then $BA = I$.

Proof. If $B\mathbf{x} = \mathbf{0}$, then $\mathbf{x} = AB\mathbf{x} = \mathbf{0}$, so the induced map $T_B : \mathbb{F}^n \to \mathbb{F}^n$ is injective. By Theorem 1.3.27, T_B is surjective. For any $\mathbf{y} \in \mathbb{F}^n$, choose $\mathbf{x}$ with $\mathbf{y} = B\mathbf{x}$. Then $BA\mathbf{y} = BAB\mathbf{x} = B\mathbf{x} = \mathbf{y}$.
Thus BA acts as the identity on every vector of $\mathbb{F}^n$, so $BA = I$.
Structural Commentary: In finite dimensions, injectivity and surjectivity coincide for square linear maps. A left inverse makes B injective, hence onto, which forces the missing inverse relation.
Key Insight: A one-sided inverse of a square matrix is automatically two-sided. □

Proposition 1.3.57. *Composition and Matrix Multiplication*
If $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $[ST]_{\delta \leftarrow \alpha} = [S]_{\delta \leftarrow \beta}[T]_{\beta \leftarrow \alpha}$ for any bases α of $\mathcal{U}$, β of $\mathcal{V}$, and δ of $\mathcal{W}$.

Proof. For every $\mathbf{u} \in \mathcal{U}$, $[(ST)\mathbf{u}]_\delta = [S]_{\delta \leftarrow \beta}[T\mathbf{u}]_\beta = [S]_{\delta \leftarrow \beta}[T]_{\beta \leftarrow \alpha}[\mathbf{u}]_\alpha$.
Since coordinate columns $[\mathbf{u}]_\alpha$ range over all of $\mathbb{F}^{\dim(\mathcal{U})}$, the representing matrices are equal.
Structural Commentary: Matrix multiplication is the coordinate shadow of composition of linear maps.
Key Insight: Applying T then S becomes multiplying first by the matrix of T, then by the matrix of S. □

Theorem 1.3.58. *Linear Maps and Matrices Are the Same Data*
If $\dim(\mathcal{V}) = n$ and $\dim(\mathcal{W}) = m$, then choosing bases of $\mathcal{V}$ and $\mathcal{W}$ gives isomorphism $\mathcal{L}(\mathcal{V}, \mathcal{W}) \cong \mathrm{Mat}(m, n, \mathbb{F})$.
Consequently, $\dim(\mathcal{L}(\mathcal{V}, \mathcal{W})) = mn$.

Proof. Fix bases β of $\mathcal{V}$ and γ of $\mathcal{W}$. The map $T \mapsto [T]_{\gamma \leftarrow \beta}$ is linear. It is injective because a linear map whose matrix is zero sends every basis vector of $\mathcal{V}$ to $\mathbf{0}$, hence is the zero map. It is surjective because, for any $A = [a_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$, there is a unique linear map satisfying $T(\mathbf{v}_j) = a_{1j}\mathbf{w}_1 + \cdots + a_{mj}\mathbf{w}_m$, $j = 1, \dots, n$.
Thus $\mathcal{L}(\mathcal{V}, \mathcal{W}) \cong \mathrm{Mat}(m, n, \mathbb{F})$, whose standard entry matrices form a basis of length mn.
Structural Commentary: Choosing bases converts every linear map into one and only one array of scalars.
Key Insight: A matrix is exactly the list of coordinate images of the domain basis vectors. □

Motivation 1.3.59. *Coordinate Encoding*
This is the key bridge between abstraction and computation. The linear map is basis-free; the matrix is what the map looks like after coordinates have been chosen.

Definition 1.3.60. *Change of Basis Matrix*
Let β and γ be two bases of the same vector space $\mathcal{V}$. The change-of-basis matrix from β-coordinates to γ-coordinates is $C_{\gamma \leftarrow \beta} := [I_\mathcal{V}]_{\gamma \leftarrow \beta}$. Thus $[\mathbf{v}]_\gamma = C_{\gamma \leftarrow \beta}[\mathbf{v}]_\beta$ for every $\mathbf{v} \in \mathcal{V}$.

Theorem 1.3.61. *Change-of-Basis Matrices*
Let $\mathcal{V}$ be n-dimensional and let β be a basis of $\mathcal{V}$.

(i) If γ is a basis of $\mathcal{V}$, then $C_{\gamma \leftarrow \beta}$ is invertible and $(C_{\gamma \leftarrow \beta})^{-1} = C_{\beta \leftarrow \gamma}$.

(ii) If $S \in \mathrm{Mat}(n,n,\mathbb{F})$ is invertible, then there is a basis γ of $\mathcal{V}$ such that $S = C_{\gamma\leftarrow\beta}$.

Proof.

(i) For any $\mathbf{v} \in \mathcal{V}$, $C_{\beta\leftarrow\gamma}C_{\gamma\leftarrow\beta}[\mathbf{v}]_\beta = [\mathbf{v}]_\beta$, $C_{\gamma\leftarrow\beta}C_{\beta\leftarrow\gamma}[\mathbf{v}]_\gamma = [\mathbf{v}]_\gamma$.
Since $[\cdot]_\beta$ and $[\cdot]_\gamma$ are onto $\mathbb{F}^n$, both products equal I_n. Hence $C_{\gamma\leftarrow\beta}$ is invertible with inverse $C_{\beta\leftarrow\gamma}$.

(ii) Write $S^{-1} = [\mathbf{s}_1 \ \cdots \ \mathbf{s}_n]$. For each j, choose the unique $\mathbf{w}_j \in \mathcal{V}$ with $[\mathbf{w}_j]_\beta = \mathbf{s}_j$, and set $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$.
Since columns of S^{-1} form a basis of $\mathbb{F}^n$ and $[\cdot]_\beta$ is an isomorphism, γ is a basis of $\mathcal{V}$.
Moreover $C_{\beta\leftarrow\gamma} = S^{-1}$, so $C_{\gamma\leftarrow\beta} = S$.

Structural Commentary: Change of basis is an invertible coordinate rewrite.
Key Insight: Every basis change is reversible, and every invertible matrix is one. □

Corollary 1.3.62. *Basis Columns Give an Invertible Matrix*
If $(\mathbf{a}_1, \ldots, \mathbf{a}_n)$ is a basis of $\mathbb{F}^n$, then $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(n,n,\mathbb{F})$ is invertible.

Proof. Let $\epsilon = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be the standard basis and $\alpha = (\mathbf{a}_1, \ldots, \mathbf{a}_n)$. Then $C_{\epsilon\leftarrow\alpha} = \begin{bmatrix}[\mathbf{a}_1]_\epsilon & \cdots & [\mathbf{a}_n]_\epsilon\end{bmatrix} = A$.
Since α is a basis, Theorem 1.3.61 gives that A is invertible.
Structural Commentary: Basis columns form a change-of-basis matrix.
Key Insight: Changing coordinates is reversible. □

Theorem 1.3.63. *Completing a Full-Column-Rank Matrix*
If $X \in \mathrm{Mat}(m,n,\mathbb{F})$ has rank $n < m$, then $\exists\ X' \in \mathrm{Mat}(m, m-n, \mathbb{F})$ s.t. $\begin{bmatrix}X & X'\end{bmatrix} \in \mathrm{Mat}(m,m,\mathbb{F})$ is invertible.

Proof. Write $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_n]$. Since $\mathrm{rank}(X) = n$, the columns $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are linearly independent in $\mathbb{F}^m$. By Theorem 1.1.25, extend this list to a basis $(\mathbf{x}_1, \ldots, \mathbf{x}_n, \mathbf{x}_{n+1}, \ldots, \mathbf{x}_m)$ of $\mathbb{F}^m$.
Set $X' = [\mathbf{x}_{n+1} \ \cdots \ \mathbf{x}_m]$. Corollary 1.3.62 shows that $\begin{bmatrix}X & X'\end{bmatrix}$ is invertible.
Structural Commentary: Full column rank means the existing columns are independent directions. Extending them to a basis turns the rectangular matrix into an invertible square matrix.
Key Insight: Independent columns can always be completed to a basis in finite dimensions. □

Corollary 1.3.64. *Invertibility and Full Rank*
For $A \in \mathrm{Mat}(n,n,\mathbb{F})$, A is invertible $\iff \mathrm{rank}(A) = n$.

Proof.

($\Rightarrow$) If A is invertible, its columns form a basis of $\mathbb{F}^n$ by Theorem 1.3.54; hence $\mathrm{rank}(A) = n$.

($\Leftarrow$) If $\mathrm{rank}(A) = n$, then $\mathrm{col}(A) = \mathbb{F}^n$. Thus the columns of A form a basis by Proposition 1.1.29, so Corollary 1.3.62 gives that A is invertible.

Structural Commentary: Full rank is square-matrix invertibility.
Key Insight: Invertibility means the columns fill $\mathbb{F}^n$. □

Corollary 1.3.65. *Transpose Preserves Invertibility*
For $A \in \mathrm{Mat}(n,n,\mathbb{F})$, the matrix A is invertible if and only if A^t is invertible.

Proof. By Theorem 1.3.49, $\mathrm{rank}(A) = \mathrm{rank}(A^t)$. Thus $\mathrm{rank}(A) = n$ if and only if $\mathrm{rank}(A^t) = n$.
Corollary 1.3.64 gives the result.
Structural Commentary: Transposition exchanges rows and columns without changing rank.
Key Insight: A square matrix is invertible exactly when either its rows or columns have full rank. □

Theorem 1.3.66. *Change of Basis Formula*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let β be a basis of $\mathcal{V}$.

(i) If γ is a basis of $\mathcal{V}$ and $S = C_{\gamma\leftarrow\beta}$, then $[T]_{\gamma\leftarrow\gamma} = S[T]_{\beta\leftarrow\beta}S^{-1}$.
(ii) If $S \in \mathrm{Mat}(n,n,\mathbb{F})$ is invertible, then there is a basis γ of $\mathcal{V}$ such that $[T]_{\gamma\leftarrow\gamma} = S[T]_{\beta\leftarrow\beta}S^{-1}$.

Proof.

(i) Suppose γ is a basis and $S = C_{\gamma\leftarrow\beta}$. By Theorem 1.3.61, $S^{-1} = C_{\beta\leftarrow\gamma}$.
For every $\mathbf{v} \in \mathcal{V}$, $[T\mathbf{v}]_\gamma = S[T\mathbf{v}]_\beta = S[T]_{\beta\leftarrow\beta}[\mathbf{v}]_\beta = S[T]_{\beta\leftarrow\beta}S^{-1}[\mathbf{v}]_\gamma$.
Since $[\cdot]_\gamma$ is onto $\mathbb{F}^n$, the representing matrices are equal.

(ii) If S is invertible, Theorem 1.3.61 supplies a basis γ with $S = C_{\gamma\leftarrow\beta}$, and part (i) applies.

Structural Commentary: Changing basis conjugates operator matrices.
Key Insight: Similarity is a coordinate rewrite. □

Definition 1.3.67. *Similarity*
Matrices $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$ are similar if there exists an invertible matrix P such that $B = PAP^{-1}$.

Perspective 1.3.68. *Similarity as Same Operator, New Coordinates*
Similarity does not mean two operators are merely related; it means they are the same operator written in two different bases. This is the structural content behind change of basis.

Remark 1.3.69. *Similarity Convention*
The forms PAP^{-1} and $M^{-1}AM$ are the same convention with $P = M^{-1}$. In Theorem 1.3.66, this means choosing opposite coordinate-change directions.

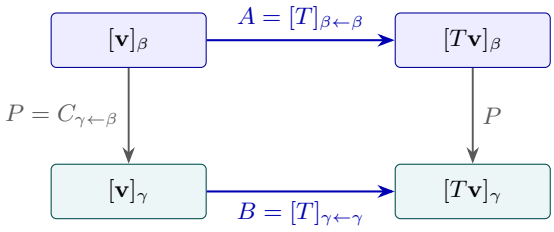


Figure 3: One operator in two bases: $PA = BP$, so $B = PAP^{-1}$.

Corollary 1.3.70. *Similarity as One Operator in Two Bases*
For $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$, the following are equivalent:

(i) A and B are similar.
(ii) There are an n-dim vector space $\mathcal{V}$, bases β, γ of $\mathcal{V}$, and $T \in \mathcal{L}(\mathcal{V})$ such that $A = [T]_{\beta\leftarrow\beta}$ and $B = [T]_{\gamma\leftarrow\gamma}$.

Proof.

(i) $\Rightarrow$ (ii): Suppose $B = PAP^{-1}$ with P invertible.
Let $\mathcal{V} = \mathbb{F}^n$, let β be the standard basis, and let $T_A(\mathbf{x}) = A\mathbf{x}$. By Theorem 1.3.61, choose γ with $P = C_{\gamma\leftarrow\beta}$. Then $[T_A]_{\beta\leftarrow\beta} = A$, and Theorem 1.3.66 gives $[T_A]_{\gamma\leftarrow\gamma} = B$.
(ii) $\Rightarrow$ (i): If $A = [T]_{\beta\leftarrow\beta}$ and $B = [T]_{\gamma\leftarrow\gamma}$, then $B = C_{\gamma\leftarrow\beta}AC_{\beta\leftarrow\gamma}$ with $C_{\gamma\leftarrow\beta}$ invertible.

Structural Commentary: Similar matrices are one operator in different bases.
Key Insight: Conjugation records a change of basis. □

Theorem 1.3.71. *Similarity and Scalar Shifts*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) If A and B are similar, then $A - \lambda I_n$ and $B - \lambda I_n$ are similar for every $\lambda \in \mathbb{F}$.
(ii) If $A - \lambda I_n$ and $B - \lambda I_n$ are similar for some $\lambda \in \mathbb{F}$, then A and B are similar.

Proof.

(i) If $B = PAP^{-1}$ with P invertible, then $B - \lambda I_n = PAP^{-1} - \lambda PP^{-1} = P(A - \lambda I_n)P^{-1}$.
(ii) If $B - \lambda I_n = P(A - \lambda I_n)P^{-1}$ with P invertible, then adding $\lambda I_n = \lambda PP^{-1}$ gives $B = PAP^{-1}$.

Structural Commentary: Scalar shifts commute with change of basis.
Key Insight: The identity matrix is basis-independent. □

Definition 1.3.72. *Equivalence Relation*
A relation is an equivalence relation if it is reflexive, symmetric, and transitive.

Proposition 1.3.73. *Similarity Is an Equivalence Relation*
Similarity is an equivalence relation on $\mathrm{Mat}(n, n, \mathbb{F})$.

Proof. For any A, $A = I_nAI_n^{-1}$, so similarity is reflexive. If $B = PAP^{-1}$, then $A = P^{-1}BP$, so it is symmetric. If $B = PAP^{-1}$ and $C = QBQ^{-1}$, then $C = Q(PAP^{-1})Q^{-1} = (QP)A(QP)^{-1}$ so it is transitive.
Structural Commentary: Similarity groups matrices by change of basis.
Key Insight: Identity, inverse, and composition give the three equivalence properties. □

Definition 1.3.74. *Block Matrices*
Suppose $\mathcal{V} = \mathcal{U}_1 \oplus \mathcal{U}_2$ and $\mathcal{W} = \mathcal{Z}_1 \oplus \mathcal{Z}_2$. If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, relative to bases adapted to these decompositions, the matrix of T has block form $[T] = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, where each block records the part of T mapping one summand of the domain into one summand of the codomain.

Definition 1.3.75. *Conformal Block Partitions*
For a block matrix $A = [A_{ij}]$, the submatrices A_{ij} are called *blocks*. Two block partitions are *conformal for addition* when corresponding blocks have the same sizes. They are *conformal for multiplication* when every block product appearing in the usual row-column block multiplication is defined.

Proposition 1.3.76. *Block Matrix Arithmetic*
For conformal 2×2 block matrices,

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} + \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11}+B_{11} & A_{12}+B_{12} \\ A_{21}+B_{21} & A_{22}+B_{22} \end{bmatrix}$$

and

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11}B_{11}+A_{12}B_{21} & A_{11}B_{12}+A_{12}B_{22} \\ A_{21}B_{11}+A_{22}B_{21} & A_{21}B_{12}+A_{22}B_{22} \end{bmatrix}.$$

Proof. Each displayed block consists of the same entries obtained by ordinary entrywise addition or ordinary row-column multiplication, with the common indices grouped according to the partition.
Structural Commentary: This is ordinary matrix arithmetic with indices grouped into rectangular packages.
Key Insight: Conformality is exactly the condition that the grouped row-column products make sense. □

Proposition 1.3.77. *Block Centrosymmetry and Commutation*
Let $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \in \mathrm{Mat}(2n, 2n, \mathbb{F})$, $L_{2n} = \begin{bmatrix} 0 & I_n \\ I_n & 0 \end{bmatrix}$.
Then M has block centrosymmetric form $A = D$ and $B = C$ if and only if $L_{2n}M = ML_{2n}$.

Proof. Block multiplication gives $L_{2n}M = \begin{bmatrix} C & D \\ A & B \end{bmatrix}$ and $ML_{2n} = \begin{bmatrix} B & A \\ D & C \end{bmatrix}$.
The two matrices are equal exactly when $A = D$ and $B = C$.
Structural Commentary: The matrix L_{2n} swaps the two block coordinates. Commuting with that swap means M treats the two block positions symmetrically.
Key Insight: Compute both block products; equality forces matching opposite blocks. □

Definition 1.3.78. *Direct Sum of Matrices*
For $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $B \in \mathrm{Mat}(p, q, \mathbb{F})$, the *direct sum* of A and B is $A \oplus B := \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \in \mathrm{Mat}(m+p, n+q, \mathbb{F})$.

Proposition 1.3.79. *Algebra of Matrix Direct Sums*
Whenever the expressions are defined,

(i) $a(A \oplus B) + b(C \oplus D) = (aA + bC) \oplus (aB + bD)$;
(ii) $(A \oplus B)(C \oplus D) = AC \oplus BD$;
(iii) if A and B are square, then $(A \oplus B)^k = A^k \oplus B^k$ for every positive integer k;
(iv) if A and B are square and p is a polynomial, then $p(A \oplus B) = p(A) \oplus p(B)$.

Proof. Block addition gives $a(A \oplus B) + b(C \oplus D) = \begin{bmatrix} aA+bC & 0 \\ 0 & aB+bD \end{bmatrix} = (aA + bC) \oplus (aB + bD)$.
Block multiplication gives $(A \oplus B)(C \oplus D) = \begin{bmatrix} AC & 0 \\ 0 & BD \end{bmatrix} = AC \oplus BD$.
For powers, the case $k = 1$ is the definition. If $(A \oplus B)^k = A^k \oplus B^k$, then part (ii) gives $(A \oplus B)^{k+1} = A^{k+1} \oplus B^{k+1}$. Hence (iii) follows by induction. If $p(z) = \sum_{j=0}^{d} c_j z^j$, with the $j = 0$ term interpreted as $c_j I$,
then $p(A \oplus B) = \sum_{j=0}^{d} c_j (A^j \oplus B^j) = \left(\sum_{j=0}^{d} c_j A^j \right) \oplus \left(\sum_{j=0}^{d} c_j B^j \right) = p(A) \oplus p(B)$.
Structural Commentary: A direct sum matrix keeps the two blocks independent under linear combinations, products, powers, and polynomials.
Key Insight: Block diagonal matrices behave componentwise. □

Proposition 1.3.80. *Diagonal Multiplication Scales Rows and Columns*
Let $A = [a_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$, let $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, and let $M = \mathrm{diag}(\mu_1, \ldots, \mu_m)$. Then $A\Lambda = [\lambda_j a_{ij}]$ and $MA = [\mu_i a_{ij}]$. Thus right multiplication by Λ scales columns of A, left multiplication by M scales rows of A.

Proof. The (i, j) entry of $A\Lambda$ is $\sum_{r=1}^{n} a_{ir}\Lambda_{rj} = a_{ij}\lambda_j$, and the (i, j) entry of MA is $\sum_{r=1}^{m} M_{ir}a_{rj} = \mu_i a_{ij}$.
Structural Commentary: Diagonal matrices act by independent scalar weights on coordinate directions.
Key Insight: Multiplication on the right scales columns, while multiplication on the left scales rows. □

Proposition 1.3.81. *Inverse of an Upper Block-Triangular Matrix*
If Y and Z are square invertible matrices, X has compatible size, then $\begin{bmatrix} Y & X \\ 0 & Z \end{bmatrix}^{-1} = \begin{bmatrix} Y^{-1} & -Y^{-1}XZ^{-1} \\ 0 & Z^{-1} \end{bmatrix}$.

Proof. Let $M = \begin{bmatrix} Y & X \\ 0 & Z \end{bmatrix}$ and $N = \begin{bmatrix} Y^{-1} & -Y^{-1}XZ^{-1} \\ 0 & Z^{-1} \end{bmatrix}$. Proposition 1.3.76 gives

$$MN = \begin{bmatrix} I & -XZ^{-1} + XZ^{-1} \\ 0 & I \end{bmatrix} = I, \qquad NM = \begin{bmatrix} I & Y^{-1}X - Y^{-1}X \\ 0 & I \end{bmatrix} = I$$

Thus $N = M^{-1}$.
Structural Commentary: Upper block-triangular inversion preserves the zero lower-left block. The off-diagonal term is forced by cancelling the original upper-right block.
Key Insight: Inverting a block triangular matrix inverts diagonal blocks and corrects the coupling block. □

Theorem 1.3.82. *Inverse of an Upper Triangular Matrix*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$ be upper triangular with nonzero diagonal entries. Then A is invertible, A^{-1} is upper triangular, and the diagonal entries of A^{-1} are $a_{11}^{-1}, \ldots, a_{nn}^{-1}$.

Proof. We argue by induction on n. The case $n = 1$ is immediate. For $n > 1$, write $A = \begin{bmatrix} B & \mathbf{x} \\ 0 & a_{nn} \end{bmatrix}$, where B is $(n-1) \times (n-1)$ upper triangular with nonzero diagonal entries.
By induction, B^{-1} is upper triangular with diagonal entries $a_{11}^{-1}, \ldots, a_{n-1,n-1}^{-1}$.
Proposition 1.3.81 gives $A^{-1} = \begin{bmatrix} B^{-1} & -B^{-1}\mathbf{x}a_{nn}^{-1} \\ 0 & a_{nn}^{-1} \end{bmatrix}$, which is upper triangular and has the asserted diagonal.
Structural Commentary: An upper triangular matrix splits into a smaller upper triangular block and one final scalar block. The block inverse formula preserves this triangular structure.
Key Insight: Nonzero diagonals make blockwise triangular inversion possible. □

Proposition 1.3.83. *Upper Block-Triangular Similarity Shift*
Let $B \in \mathrm{Mat}(m, m, \mathbb{F})$, let $D \in \mathrm{Mat}(n, n, \mathbb{F})$, and let $C, X \in \mathrm{Mat}(m, n, \mathbb{F})$.
Then $\begin{bmatrix} B & C \\ 0 & D \end{bmatrix}$ *is similar to* $\begin{bmatrix} B & C + XD - BX \\ 0 & D \end{bmatrix}$.

Proof. The matrices $\begin{bmatrix} I & X \\ 0 & I \end{bmatrix}$ and $\begin{bmatrix} I & -X \\ 0 & I \end{bmatrix}$ are inverses, and block multiplication gives

$$\begin{bmatrix} I & X \\ 0 & I \end{bmatrix} \begin{bmatrix} B & C \\ 0 & D \end{bmatrix} \begin{bmatrix} I & -X \\ 0 & I \end{bmatrix} = \begin{bmatrix} B & C + XD - BX \\ 0 & D \end{bmatrix}.$$

Structural Commentary: Unit upper block-triangular conjugation changes only the coupling block.
Key Insight: Similarity can adjust the connecting block while preserving the diagonal blocks. □

1.4 Duality and Multilinear Structure

Duality studies vectors through scalar-valued tests. Functionals, dual maps, annihilators, transposes, bilinear forms, and tensor products extend linear structure into multilinear language.

Definition 1.4.1. *Linear Functional and Dual Space*
A *linear functional* on $\mathcal{V}$ is a linear map from $\mathcal{V}$ to $\mathbb{F}$. The *dual space* of $\mathcal{V}$ is $\mathcal{V}' := \mathcal{L}(\mathcal{V}, \mathbb{F})$.

Definition 1.4.2. *Dual Basis*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$. The *dual basis* of β is the list $\beta' = (\varphi_1, \ldots, \varphi_n)$ in $\mathcal{V}'$ defined by

$$\varphi_j(\mathbf{v}_k) = \begin{cases} 1, & k = j, \\ 0, & k \neq j. \end{cases}$$

Existence and uniqueness of each φ_j follow from Proposition 1.3.11.

Theorem 1.4.3. *Dual Basis and Dimension of the Dual Space*
Let $\mathcal{V}$ be finite-dimensional with basis $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$, and let $\beta' = (\varphi_1, \ldots, \varphi_n)$ be the dual basis. Then β' is a basis of $\mathcal{V}'$. In particular, $\dim(\mathcal{V}') = \dim(\mathcal{V})$.

Proof. First suppose $a_1\varphi_1 + \cdots + a_n\varphi_n = 0$. Evaluating at $\mathbf{v}_j$ gives $a_j = 0$ for each j, so β' is linearly independent. Now let $\varphi \in \mathcal{V}'$. For $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, $\left(\sum_{j=1}^{n} \varphi(\mathbf{v}_j)\varphi_j\right)(\mathbf{v}) = \sum_{j=1}^{n} \varphi(\mathbf{v}_j)c_j = \varphi(\mathbf{v})$.

Thus $\varphi = \sum_{j=1}^{n} \varphi(\mathbf{v}_j)\varphi_j$, so β' spans $\mathcal{V}'$.
Structural Commentary: A basis of $\mathcal{V}$ creates coordinate-reading functionals on $\mathcal{V}$. These functionals form the coordinate system for the dual space.
Key Insight: A linear functional is determined exactly by its values on a basis. □

Definition 1.4.4. *Dual Map*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the *dual map* of T is the map $T' : \mathcal{W}' \to \mathcal{V}'$ defined by $T'\varphi := \varphi T$ for every $\varphi \in \mathcal{W}'$.
Equivalently, $(T'\varphi)(\mathbf{v}) = \varphi(T\mathbf{v})$.

Proposition 1.4.5. *Dual Maps Are Linear*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $T' \in \mathcal{L}(\mathcal{W}', \mathcal{V}')$.

Proof. For $\varphi, \psi \in \mathcal{W}'$, $a \in \mathbb{F}$, and $\mathbf{v} \in \mathcal{V}$, $(T'(a\varphi + \psi))(\mathbf{v}) = (a\varphi+\psi)(T\mathbf{v}) = a\varphi(T\mathbf{v})+\psi(T\mathbf{v}) = (aT'\varphi + T'\psi)(\mathbf{v})$.
Structural Commentary: The dual map sends a test on $\mathcal{W}$ to the pulled-back test on $\mathcal{V}$. Linearity is inherited from pointwise linearity of functionals.
Key Insight: Dual maps reverse direction because a functional can test $T\mathbf{v}$ only after T is applied. □

Theorem 1.4.6. *Algebra of Dual Maps*
The dual operation has the following properties:

(i) *If $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$, then $(aS + T)' = aS' + T'$.*
(ii) *If $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $(ST)' = T'S'$.*
(iii) *$(I_{\mathcal{V}})' = I_{\mathcal{V}'}$.*

Proof.

(i) For $\varphi \in \mathcal{W}'$ and $\mathbf{v} \in \mathcal{V}$, $((aS + T)'\varphi)(\mathbf{v}) = \varphi(aS\mathbf{v}+T\mathbf{v}) = a(S'\varphi)(\mathbf{v})+(T'\varphi)(\mathbf{v})$, so $(aS+T)' = aS'+T'$.
(ii) For $\varphi \in \mathcal{W}'$ and $\mathbf{u} \in \mathcal{U}$, $((ST)'\varphi)(\mathbf{u}) = \varphi(S(T\mathbf{u})) = (T'(S'\varphi))(\mathbf{u})$, so $(ST)' = T'S'$.
(iii) For $\varphi \in \mathcal{V}'$ and $\mathbf{v} \in \mathcal{V}$, $(I'_{\mathcal{V}}\varphi)(\mathbf{v}) = \varphi(I_{\mathcal{V}}\mathbf{v}) = \varphi(\mathbf{v})$, so $(I_{\mathcal{V}})' = I_{\mathcal{V}'}$.

Structural Commentary: Duality preserves linear combinations but reverses products. This reversal is the same order reversal seen in transposes.
Key Insight: Pullbacks compose in the opposite order. □

Definition 1.4.7. *Annihilator*
If $\mathcal{U} \subseteq \mathcal{V}$, the *annihilator* of $\mathcal{U}$ is $\mathcal{U}^0 := \{\varphi \in \mathcal{V}' : \varphi(\mathbf{u}) = 0 \text{ for every } \mathbf{u} \in \mathcal{U}\}$.

Proposition 1.4.8. *Annihilators Are Subspaces*
If $\mathcal{U} \subseteq \mathcal{V}$, then $\mathcal{U}^0$ is a subspace of $\mathcal{V}'$.

Proof. The zero functional lies in $\mathcal{U}^0$. If $\varphi, \psi \in \mathcal{U}^0$, $a \in \mathbb{F}$, and $\mathbf{u} \in \mathcal{U}$, then $(a\varphi + \psi)(\mathbf{u}) = a\varphi(\mathbf{u}) + \psi(\mathbf{u}) = 0$.
Thus $a\varphi + \psi \in \mathcal{U}^0$, and Theorem 1.1.7 shows that $\mathcal{U}^0$ is a subspace of $\mathcal{V}'$.
Structural Commentary: The annihilator collects all linear tests that vanish on a chosen set. Closure follows because vanishing is preserved by linear combinations of tests.
Key Insight: Annihilators are kernels of evaluation constraints. □

Theorem 1.4.9. *Dimension of the Annihilator*
Let $\mathcal{V}$ be finite-dimensional and let $\mathcal{U}$ be a subspace of $\mathcal{V}$. Then $\dim(\mathcal{U}) + \dim(\mathcal{U}^0) = \dim(\mathcal{V})$.

Proof. Choose a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ of $\mathcal{U}$ and extend it to a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_k)$ of $\mathcal{V}$.
Let $(\varphi_1, \ldots, \varphi_m, \psi_1, \ldots, \psi_k)$ be the corresponding dual basis.
By Theorem 1.4.3, every $\varphi \in \mathcal{V}'$ has the expansion $\varphi = \sum_{j=1}^{m} \varphi(\mathbf{u}_j)\varphi_j + \sum_{\ell=1}^{k} \varphi(\mathbf{w}_\ell)\psi_\ell$.
If $\varphi \in \mathcal{U}^0$, then $\varphi(\mathbf{u}_j) = 0$ for every j, so $\varphi \in \operatorname{span}(\psi_1, \ldots, \psi_k)$.
Conversely, each ψ_ℓ vanishes on every $\mathbf{u}_j$, hence on $\mathcal{U} = \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_m)$. Therefore $\mathcal{U}^0 = \operatorname{span}(\psi_1, \ldots, \psi_k)$.
The list $(\psi_1, \ldots, \psi_k)$ is linearly independent as a sublist of the dual basis, so $\dim(\mathcal{U}^0) = k = \dim(\mathcal{V}) - \dim(\mathcal{U})$.
Structural Commentary: Extending a basis of $\mathcal{U}$ separates directions inside $\mathcal{U}$ from complementary directions. The annihilator is dual to the complementary part.
Key Insight: To vanish on $\mathcal{U}$, a functional can only use dual coordinates attached to directions outside $\mathcal{U}$. □

Theorem 1.4.10. *Kernel and Range of the Dual Map*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then

(i) $\ker(T') = (\operatorname{range}(T))^0$;
(ii) $\dim(\ker(T')) = \dim(\ker(T)) + \dim(\mathcal{W}) - \dim(\mathcal{V})$;
(iii) $\operatorname{range}(T') = (\ker(T))^0$;
(iv) $\dim(\operatorname{range}(T')) = \dim(\operatorname{range}(T))$.

Proof.

(i) For $\varphi \in \mathcal{W}'$, $T'\varphi = 0$ if and only if $\varphi(T\mathbf{v}) = 0$ for every $\mathbf{v} \in \mathcal{V}$, which is equivalent to $\varphi \in (\text{range}(T))^0$.
(ii) By Theorems 1.4.9 and 1.3.19, $\dim(\ker(T')) = \dim(\mathcal{W}) - \dim(\text{range}(T)) = \dim(\mathcal{W}) - \dim(\mathcal{V}) + \dim(\ker(T))$.
(iii) If $\varphi = T'\psi$ and $\mathbf{v} \in \ker(T)$, then $\varphi(\mathbf{v}) = \psi(T\mathbf{v}) = 0$, so $\text{range}(T') \subseteq (\ker(T))^0$,
Conversely, suppose $\varphi \in (\ker(T))^0$. Define $\widetilde{\varphi} : \text{range}(T) \to \mathbb{F}$ by $\widetilde{\varphi}(T\mathbf{v}) = \varphi(\mathbf{v})$.
This is well-defined because $T\mathbf{v} = T\mathbf{u}$ implies $\mathbf{v} - \mathbf{u} \in \ker(T)$, hence $\varphi(\mathbf{v} - \mathbf{u}) = 0$ and $\varphi(\mathbf{v}) = \varphi(\mathbf{u})$.
It is linear since $\widetilde{\varphi}\big(aT\mathbf{v} + T\mathbf{u}\big) = \widetilde{\varphi}\big(T(a\mathbf{v} + \mathbf{u})\big) = \varphi(a\mathbf{v} + \mathbf{u}) = a\varphi(\mathbf{v}) + \varphi(\mathbf{u})$.
Choose a basis of $\text{range}(T)$ and extend it to a basis of $\mathcal{W}$. Assign to the basis vectors in $\text{range}(T)$ the values prescribed by $\widetilde{\varphi}$, assign 0 to the added basis vectors, and use Proposition 1.3.11 to obtain $\psi \in \mathcal{W}'$ extending $\widetilde{\varphi}$. For every $\mathbf{v} \in \mathcal{V}$, $(T'\psi)(\mathbf{v}) = \psi(T\mathbf{v}) = \widetilde{\varphi}(T\mathbf{v}) = \varphi(\mathbf{v})$, so $\varphi \in \text{range}(T')$. Hence $\text{range}(T') = (\ker(T))^0$.
(iv) By rank-nullity applied to T' and by (i),
$\dim(\text{range}(T')) = \dim(\mathcal{W}') - \dim(\ker(T')) = \dim(\mathcal{W}) - \dim((\text{range}(T))^0) = \dim(\text{range}(T))$,
where the last equality uses Theorems 1.4.3 and 1.4.9.

Structural Commentary: Dual kernels are annihilators of ranges, and dual ranges are annihilators of kernels. This is the rank-nullity geometry of T seen through scalar tests.
Key Insight: A functional dies after pulling back by T exactly when it vanishes on everything T can reach. □

Theorem 1.4.11. *Range Membership Dual Alternative*
Let $\mathcal{V}, \mathcal{W}$ be finite-dimensional, $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, and $\mathbf{w} \in \mathcal{W}$. Exactly one holds:

(i) $\mathbf{w} \in \text{range}(T)$;
(ii) there exists $\psi \in \mathcal{W}'$ such that $\psi T = 0$ and $\psi(\mathbf{w}) \neq 0$.

Proof.

(i) The alternatives are incompatible: if $\mathbf{w} = T\mathbf{v}$ and $\psi T = 0$, then $\psi(\mathbf{w}) = \psi(T\mathbf{v}) = 0$.
(ii) Suppose $\mathbf{w} \notin \text{range}(T)$. Let $\mathcal{U} = \text{range}(T)$, choose a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_r)$ of $\mathcal{U}$, and extend $(\mathbf{u}_1, \ldots, \mathbf{u}_r, \mathbf{w})$ to a basis of $\mathcal{W}$. Define $\psi \in \mathcal{W}'$ by $\psi(\mathbf{u}_j) = 0$, $\psi(\mathbf{w}) = 1$, and $\psi = 0$ on the added basis vectors.
Then ψ vanishes on $\text{range}(T)$, so $\psi T = 0$, while $\psi(\mathbf{w}) = 1$.

Structural Commentary: A missing range vector is detected by a dual test that vanishes on the whole range.
Key Insight: Either a vector is reached by the map, or a functional separates it from the range. □

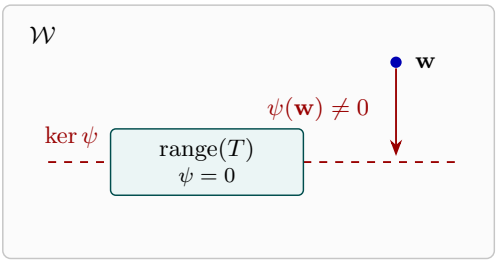


Figure 4: A dual certificate detects a vector outside the range.

Corollary 1.4.12. *Matrix Equation Alternative*
Let $A \in \text{Mat}(m, n, \mathbb{F})$ and $\mathbf{b} \in \mathbb{F}^m$. Exactly one holds:

(i) $A\mathbf{x} = \mathbf{b}$ for some $\mathbf{x} \in \mathbb{F}^n$;
(ii) $\mathbf{y}^t A = \mathbf{0}^t$ and $\mathbf{y}^t\mathbf{b} \neq 0$ for some $\mathbf{y} \in \mathbb{F}^m$.

Proof.

(i) By Theorem 1.4.11, for $T_A : \mathbb{F}^n \to \mathbb{F}^m$ and $\mathbf{w} = \mathbf{b}$, $\mathbf{b} \in \text{range}(T_A)$ is exactly $A\mathbf{x} = \mathbf{b}$ for some $\mathbf{x} \in \mathbb{F}^n$.
(ii) Under the standard dual identification, $\psi \in (\mathbb{F}^m)'$ has the form $\psi(\mathbf{z}) = \mathbf{y}^t\mathbf{z}$.
Thus $\psi T_A = 0$ iff $\mathbf{y}^t A = \mathbf{0}^t$, and $\psi(\mathbf{b}) \neq 0$ iff $\mathbf{y}^t\mathbf{b} \neq 0$.

Structural Commentary: Inconsistency is certified by a left-null vector.
Key Insight: Left null vectors are the coordinate form of dual separating functionals. □

Corollary 1.4.13. *Surjectivity and Dual Injectivity*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then T is surjective if and only if T' is injective.

Proof.

($\Rightarrow$) Suppose T is surjective. Then $\text{range}(T) = \mathcal{W}$, so Theorem 1.4.10 gives $\ker(T') = \mathcal{W}^0 = \{0\}$.

($\Leftarrow$) Suppose T' is injective. Then $\ker(T') = \{0\}$, so Theorem 1.4.10 gives $(\text{range}(T))^0 = \{0\}$. By Theorem 1.4.9, $\dim(\text{range}(T)) = \dim(\mathcal{W})$, hence $\text{range}(T) = \mathcal{W}$.

Structural Commentary: A map is onto exactly when no nonzero functional can vanish on its range. Dual injectivity detects this absence of invisible codomain directions.
Key Insight: Surjectivity of T becomes injectivity of the pullback map T'. □

Corollary 1.4.14. *Injectivity and Dual Surjectivity*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then T is injective if and only if T' is surjective.

Proof.
($\Rightarrow$) Suppose T is injective. Then $\ker(T) = \{\mathbf{0}\}$, so Theorem 1.4.10 gives $\text{range}(T') = (\ker(T))^0 = \mathcal{V}'$.
($\Leftarrow$) Suppose T' is surjective. Then $\text{range}(T') = \mathcal{V}'$, so Theorem 1.4.10 gives $(\ker(T))^0 = \mathcal{V}'$. By Theorem 1.4.9, this forces $\dim(\ker(T)) = 0$, hence $\ker(T) = \{\mathbf{0}\}$.

Structural Commentary: Injectivity says there are no nonzero domain directions killed by T. The dual statement says every test on the domain comes by pulling back a test from the codomain.
Key Insight: No kernel directions means no obstruction to extending functionals backward through T. □

Theorem 1.4.15. *Matrix of the Dual Map Is the Transpose*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, and let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ be bases of $\mathcal{V}$ and $\mathcal{W}$. If β' and γ' are the corresponding dual bases, then $[T']_{\beta' \leftarrow \gamma'} = ([T]_{\gamma \leftarrow \beta})^t$.

Proof. Let $A = [T]_{\gamma \leftarrow \beta}$ and $C = [T']_{\beta' \leftarrow \gamma'}$. Write $\gamma' = (\psi_1, \ldots, \psi_m)$ and $\beta' = (\varphi_1, \ldots, \varphi_n)$.
For $1 \le j \le m$, $T'\psi_j = \sum_{r=1}^{n} C_{r,j}\varphi_r$. Evaluating at $\mathbf{v}_k$ gives $C_{k,j} = (T'\psi_j)(\mathbf{v}_k) = \psi_j(T\mathbf{v}_k)$.
Since $T\mathbf{v}_k = \sum_{r=1}^{m} A_{r,k}\mathbf{w}_r$, we get $\psi_j(T\mathbf{v}_k) = A_{j,k}$. Hence $C_{k,j} = A_{j,k}$ for all j, k, so $C = A^t$.
Structural Commentary: Duality turns the matrix of a map into the matrix of its pullback on coordinate functionals. The row-column indexing is exactly transposition.
Key Insight: Coefficients of $T\mathbf{v}_k$ become values of the dual basis functionals on $T\mathbf{v}_k$. □

Definition 1.4.16. *Bilinear Forms and Matrices*
A map $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ is *bilinear* if it is linear in each variable separately.
For bases $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ of $\mathcal{V}$ and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ of $\mathcal{W}$, the *matrix of B in the bases β, γ* is $[B]_{\beta,\gamma} = [B(\mathbf{v}_i, \mathbf{w}_j)]$. If $\mathbf{x} = [\mathbf{v}]_\beta$ and $\mathbf{y} = [\mathbf{w}]_\gamma$, then $B(\mathbf{v}, \mathbf{w}) = \mathbf{x}^t[B]_{\beta,\gamma}\mathbf{y}$.

Proposition 1.4.17. *Bilinear Forms Linearize through Duals*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ be bilinear, and define $\Phi_B : \mathcal{V} \to \mathcal{W}'$ where $(\Phi_B\mathbf{v})(\mathbf{w}) = B(\mathbf{v}, \mathbf{w})$, and $\Psi_B : \mathcal{W} \to \mathcal{V}'$ where $(\Psi_B\mathbf{w})(\mathbf{v}) = B(\mathbf{v}, \mathbf{w})$. Then Φ_B, Ψ_B are linear. For bases β, γ and $M = [B]_{\beta,\gamma}$, $[\Phi_B]_{\gamma' \leftarrow \beta} = M^t$; and $[\Psi_B]_{\beta' \leftarrow \gamma} = M$.

Proof. For fixed $\mathbf{v}$ and $\mathbf{w}$, bilinearity gives $\Phi_B\mathbf{v} \in \mathcal{W}'$ and $\Psi_B\mathbf{w} \in \mathcal{V}'$. For $\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{V}$, $a \in \mathbb{F}$, and every $\mathbf{w} \in \mathcal{W}$, $(\Phi_B(a\mathbf{v}_1 + \mathbf{v}_2))(\mathbf{w}) = B(a\mathbf{v}_1 + \mathbf{v}_2, \mathbf{w}) = a(\Phi_B\mathbf{v}_1)(\mathbf{w}) + (\Phi_B\mathbf{v}_2)(\mathbf{w})$, so Φ_B is linear.
For $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W}$, $a \in \mathbb{F}$, and every $\mathbf{v} \in \mathcal{V}$, $(\Psi_B(a\mathbf{w}_1 + \mathbf{w}_2))(\mathbf{v}) = B(\mathbf{v}, a\mathbf{w}_1 + \mathbf{w}_2) = a(\Psi_B\mathbf{w}_1)(\mathbf{v}) + (\Psi_B\mathbf{w}_2)(\mathbf{v})$, so Ψ_B is linear. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$.
The jth γ'-coordinate of $\Phi_B(\mathbf{v}_i)$ is $(\Phi_B\mathbf{v}_i)(\mathbf{w}_j) = B(\mathbf{v}_i, \mathbf{w}_j) = M_{i,j}$, so column i of $[\Phi_B]_{\gamma' \leftarrow \beta}$ is row i of M transposed, hence $[\Phi_B]_{\gamma' \leftarrow \beta} = M^t$. Likewise, the ith β'-coordinate of $\Psi_B(\mathbf{w}_j)$ is $M_{i,j}$, so $[\Psi_B]_{\beta' \leftarrow \gamma} = M$.
Structural Commentary: A bilinear form becomes linear once one input is frozen. Tracking both freezes gives two dual-valued linear maps whose matrices are transposes of one another.
Key Insight: Bilinearity is linearity into a dual space. □

Perspective 1.4.18. *Bilinear Forms before Tensor Products*
The maps Φ_B and Ψ_B place bilinear forms inside dual-space and transpose machinery. The tensor model below is only a finite-coordinate tool; the full tensor product belongs later.

Perspective 1.4.19. *Sums, Products, and Tensor Scaffolds*
Direct sums and products separate coordinates; the tensor scaffold below separates paired basis directions as rank-one matrix slots.

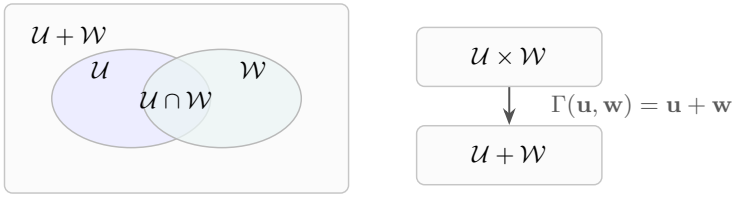


Figure 5: Intersection, sum, and product of subspaces.

Definition 1.4.20. *Finite-Coordinate Tensor Product Model*
Let $\mathcal{V},\mathcal{W}$ have ordered bases $\beta=(\mathbf{v}_1,\dots,\mathbf{v}_m)$ and $\gamma=(\mathbf{w}_1,\dots,\mathbf{w}_n)$.
The *coordinate tensor model* for (β,γ) is $\mathrm{Mat}(m,n,\mathbb{F})$, with $\mathbf{v}_i\otimes_{\beta,\gamma}\mathbf{w}_j$ represented by $E_{i,j}$.
For $\mathbf{v}=\sum_i x_i\mathbf{v}_i$ and $\mathbf{w}=\sum_j y_j\mathbf{w}_j$, set $\mathbf{v}\otimes_{\beta,\gamma}\mathbf{w}:=\sum_{i=1}^{m}\sum_{j=1}^{n}x_iy_jE_{i,j}\in\mathrm{Mat}(m,n,\mathbb{F})$.

Proposition 1.4.21. *Tensor Model Basis and Bilinear Evaluation*
In the coordinate tensor model associated to bases $\beta=(\mathbf{v}_1,\dots,\mathbf{v}_m)$ and $\gamma=(\mathbf{w}_1,\dots,\mathbf{w}_n)$:

(i) the matrices $E_{i,j}=\mathbf{v}_i\otimes_{\beta,\gamma}\mathbf{w}_j$ form a basis;
(ii) the map $(\mathbf{v},\mathbf{w})\mapsto\mathbf{v}\otimes_{\beta,\gamma}\mathbf{w}$ is bilinear;
(iii) every bilinear form $B:\mathcal{V}\times\mathcal{W}\to\mathbb{F}$ determines a unique linear functional Θ_B on $\mathrm{Mat}(m,n,\mathbb{F})$ such that $\Theta_B(\mathbf{v}\otimes_{\beta,\gamma}\mathbf{w})=B(\mathbf{v},\mathbf{w})$ for all $\mathbf{v}\in\mathcal{V}$ and $\mathbf{w}\in\mathcal{W}$.

Proof.

(i) This is the standard entry-matrix basis of $\mathrm{Mat}(m,n,\mathbb{F})$ from Proposition 1.3.32.
(ii) The coordinates of $a\mathbf{v}_1+\mathbf{v}_2$ are the same linear combination of the coordinates of $\mathbf{v}_1$ and $\mathbf{v}_2$, and the displayed formula is linear in those coordinates. The same argument applies in the second variable.
(iii) Define $\Theta_B(E_{i,j})=B(\mathbf{v}_i,\mathbf{w}_j)$ on the basis from (i), and extend linearly by Proposition 1.3.11. If $\mathbf{v}=\sum_i x_i\mathbf{v}_i$ and $\mathbf{w}=\sum_j y_j\mathbf{w}_j$, then $\Theta_B(\mathbf{v}\otimes_{\beta,\gamma}\mathbf{w})=\sum_{i=1}^{m}\sum_{j=1}^{n}x_iy_jB(\mathbf{v}_i,\mathbf{w}_j)=B(\mathbf{v},\mathbf{w})$, by bilinearity of B. Uniqueness follows since the values on basis matrices $E_{i,j}$ are forced by $\mathbf{v}_i,\mathbf{w}_j$.

Structural Commentary: The coordinate tensor model stores each pair of basis directions in its own matrix slot, so a bilinear form can be evaluated by a single linear functional on those slots.
Key Insight: Tensor coordinates turn products of coordinates into linear coordinates. □

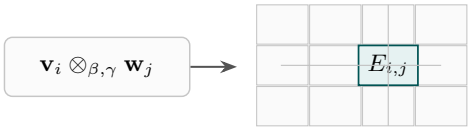


Figure 6: A basis pair occupies the (i,j) matrix slot.

Definition 1.4.22. *Double Dual Space*
The *double dual space* of $\mathcal{V}$ is $\mathcal{V}''=(\mathcal{V}')'$. Define $\Lambda_{\mathcal{V}}:\mathcal{V}\to\mathcal{V}''$ by $(\Lambda_{\mathcal{V}}\mathbf{v})(\varphi)=\varphi(\mathbf{v}),\mathbf{v}\in\mathcal{V},\ \varphi\in\mathcal{V}'$.

Theorem 1.4.23. *Natural Map to the Double Dual*
The map $\Lambda_{\mathcal{V}}:\mathcal{V}\to\mathcal{V}''$ satisfies:

(i) $\Lambda_{\mathcal{V}}$ is linear;
(ii) if $T\in\mathcal{L}(\mathcal{V},\mathcal{W})$ and $T''=(T')'$, then $T''\Lambda_{\mathcal{V}}=\Lambda_{\mathcal{W}}T$;
(iii) if $\mathcal{V}$ is finite-dimensional, then $\Lambda_{\mathcal{V}}$ is an isomorphism.

Proof.

(i) For fixed $\mathbf{v}\in\mathcal{V}$, the map $\varphi\mapsto\varphi(\mathbf{v})$ is linear on $\mathcal{V}'$, so $\Lambda_{\mathcal{V}}\mathbf{v}\in\mathcal{V}''$.
For $\mathbf{u},\mathbf{v}\in\mathcal{V}$, $a\in\mathbb{F}$, and $\varphi\in\mathcal{V}'$, $(\Lambda_{\mathcal{V}}(a\mathbf{u}+\mathbf{v}))(\varphi)=\varphi(a\mathbf{u}+\mathbf{v})=a(\Lambda_{\mathcal{V}}\mathbf{u})(\varphi)+(\Lambda_{\mathcal{V}}\mathbf{v})(\varphi)$, so $\Lambda_{\mathcal{V}}$ is linear.
(ii) If $\psi\in\mathcal{W}'$, then $(T''(\Lambda_{\mathcal{V}}\mathbf{v}))(\psi)=(\Lambda_{\mathcal{V}}\mathbf{v})(T'\psi)=(T'\psi)(\mathbf{v})=\psi(T\mathbf{v})=(\Lambda_{\mathcal{W}}(T\mathbf{v}))(\psi)$; $T''\Lambda_{\mathcal{V}}=\Lambda_{\mathcal{W}}T$.
(iii) Suppose $\mathcal{V}$ is finite-dimensional. If $\mathbf{v}\neq\mathbf{0}$, extend $(\mathbf{v})$ to a basis of $\mathcal{V}$ and let φ be the first functional of the corresponding dual basis. Then $\varphi(\mathbf{v})=1$, so $(\Lambda_{\mathcal{V}}\mathbf{v})(\varphi)=1$ and $\Lambda_{\mathcal{V}}\mathbf{v}\neq0$. Hence $\ker(\Lambda_{\mathcal{V}})=\{\mathbf{0}\}$ and $\Lambda_{\mathcal{V}}$ is injective. Since $\dim(\mathcal{V}'')=\dim(\mathcal{V}')=\dim(\mathcal{V})$, Corollary 1.3.22 shows that $\Lambda_{\mathcal{V}}$ is an isomorphism.

Structural Commentary: The double-dual map sends a vector to its evaluation action on functionals. It is natural because it commutes with every linear map.
Key Insight: Finite-dimensional vectors are completely recovered by how all linear functionals evaluate them. □

Perspective 1.4.24. *Why the Double Dual Is Natural*
A finite-dimensional space $\mathcal{V}$ is isomorphic to $\mathcal{V}'$, but such an isomorphism usually requires choosing a basis. The map $\Lambda_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}''$ requires no choice of basis.

2 Geometric Linear Algebra

Geometric linear algebra adds inner products, orthogonality, projections, least squares, adjoints, and isometries to the linear story.

2.1 Inner Product Geometry

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$. Inner products turn linear spaces into geometric spaces with norms, orthogonality, orthonormal bases, and Gram matrices.

Definition 2.1.1. *Preliminary Terminology*

(i) For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the *dot product* is $\mathbf{x} \cdot \mathbf{y} := x_1 y_1 + \cdots + x_n y_n$.
(ii) For $z = a + bi \in \mathbb{C}$, the *complex conjugate* is $\overline{z} := a - bi$, and the *modulus* is $|z| := \sqrt{a^2 + b^2}$.
(iii) A map $f : \mathcal{V} \to \mathbb{F}$ is *conjugate linear* if $f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v})$ and $f(\lambda \mathbf{v}) = \overline{\lambda} f(\mathbf{v})$.
(iv) A function $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ is *sesquilinear* if it is linear in one slot and conjugate linear in the other.

Motivation 2.1.2. *From Dot Products to Inner Products*
For $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$, the Euclidean length is $\|\mathbf{x}\| = \sqrt{x_1^2 + \cdots + x_n^2}$.
The expression under the square root is the dot product of $\mathbf{x}$ with itself. Over $\mathbb{C}$, conjugates are needed to keep squared length real and nonnegative: $\|\mathbf{z}\|^2 = |z_1|^2 + \cdots + |z_n|^2 = z_1\overline{z_1} + \cdots + z_n\overline{z_n}$.
Thus the standard inner product on $\mathbb{F}^n$ is $\langle \mathbf{x}, \mathbf{y} \rangle := x_1\overline{y_1} + \cdots + x_n\overline{y_n}$.

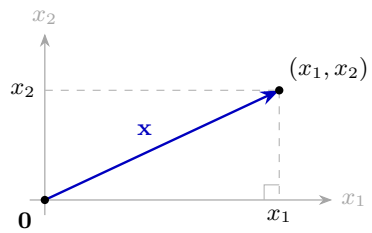


Figure 7: The norm is vector length, with $\|\mathbf{x}\|^2 = \langle \mathbf{x}, \mathbf{x} \rangle$.

Definition 2.1.3. *Inner Product*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. An *inner product* on $\mathcal{V}$ is a function that assigns to each pair $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ a scalar $\langle \mathbf{u}, \mathbf{v} \rangle \in \mathbb{F}$ satisfying, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$ and all $\lambda \in \mathbb{F}$,

i. *positivity*: $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$;
ii. *definiteness*: $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$;
iii. *additivity in the first slot*: $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$;
iv. *homogeneity in the first slot*: $\langle \lambda \mathbf{u}, \mathbf{v} \rangle = \lambda \langle \mathbf{u}, \mathbf{v} \rangle$;
v. *conjugate symmetry*: $\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle}$.

In the positivity axiom, $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$ means that $\langle \mathbf{v}, \mathbf{v} \rangle$ is a real nonnegative scalar.

Definition 2.1.4. *Inner-Product Space*
An *inner-product space* is a vector space $\mathcal{V}$ together with an inner product on $\mathcal{V}$.

Example 2.1.5. *Standard Inner Product on $\mathbb{F}^n$*
For $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, the standard inner product is $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{j=1}^{n} x_j \overline{y_j}$. When $\mathbb{F} = \mathbb{R}$, this is the usual dot product.

Example 2.1.6. *Function and Matrix Inner Products*
If $\mathcal{V} = C_{\mathbb{F}}[a, b]$, then $\langle f, g \rangle = \int_a^b f(t)\overline{g(t)}\, dt$ is an inner product.
If $\mathcal{V} = \mathrm{Mat}(m, n, \mathbb{F})$, then the *Frobenius inner product* is $\langle A, B \rangle_F := \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}\overline{b_{ij}}$.

Proposition 2.1.7. *Basic Inner-Product Identities*
Let $\mathcal{V}$ be an inner-product space. Then for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$ and $\lambda \in \mathbb{F}$,

i. for fixed $\mathbf{u}$, the map $\mathbf{v} \mapsto \langle \mathbf{v}, \mathbf{u} \rangle$ is linear;

ii. $\langle \mathbf{0}, \mathbf{u} \rangle = 0$ *and* $\langle \mathbf{u}, \mathbf{0} \rangle = 0$*;*
iii. $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$*;*
iv. $\langle \mathbf{u}, \lambda \mathbf{v} \rangle = \overline{\lambda} \langle \mathbf{u}, \mathbf{v} \rangle$*.*

Proof.

(i) This is exactly additivity and homogeneity in the first slot.
(ii) A linear map sends $\mathbf{0}$ to 0, so $\langle \mathbf{0}, \mathbf{u} \rangle = 0$. Conjugate symmetry gives $\langle \mathbf{u}, \mathbf{0} \rangle = \overline{\langle \mathbf{0}, \mathbf{u} \rangle} = 0$.
(iii) By conjugate symmetry and additivity, $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \overline{\langle \mathbf{v} + \mathbf{w}, \mathbf{u} \rangle} = \overline{\langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{u} \rangle} = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$.
(iv) By conjugate symmetry and homogeneity, $\langle \mathbf{u}, \lambda \mathbf{v} \rangle = \overline{\langle \lambda \mathbf{v}, \mathbf{u} \rangle} = \overline{\lambda \langle \mathbf{v}, \mathbf{u} \rangle} = \overline{\lambda} \langle \mathbf{u}, \mathbf{v} \rangle$.

Structural Commentary: First-slot linearity and conjugate symmetry force conjugate linearity in second slot.
Key Insight: Once one slot is linear, the other slot is forced to be conjugate linear. □

Definition 2.1.8. *Norm Induced by an Inner Product*
For $\mathbf{v} \in \mathcal{V}$, the *norm* of $\mathbf{v}$ is $\|\mathbf{v}\| := \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.

Example 2.1.9. *Norms from Standard Inner Products*

In $\mathbb{F}^n$, the norm from the standard inner product is the *Euclidean norm* $\|(x_1, \ldots, x_n)\|_2 = \sqrt{|x_1|^2 + \cdots + |x_n|^2}$.

In $C_{\mathbb{F}}[a, b]$, with the integral inner product, the derived norm is the *L^2 norm* $\|f\|_2 = \sqrt{\int_a^b |f(t)|^2 \, dt}$.

In $\mathrm{Mat}(m, n, \mathbb{F})$, with the Frobenius inner product, the *Frobenius norm* is $\|A\|_F = \sqrt{\langle A, A \rangle_F} = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2}$, $A = [a_{ij}]$. The Frobenius norm is also called the *Schur norm* or *Hilbert-Schmidt norm*.

Proposition 2.1.10. *Basic Properties of the Derived Norm*
Let $\mathcal{V}$ be an inner-product space with derived norm $\|\cdot\|$. For all $\mathbf{v} \in \mathcal{V}$ and $\lambda \in \mathbb{F}$,

i. $\|\mathbf{v}\|$ *is real and nonnegative;*
ii. $\|\mathbf{v}\| = 0$ *if and only if* $\mathbf{v} = \mathbf{0}$*;*
iii. $\|\lambda \mathbf{v}\| = |\lambda| \, \|\mathbf{v}\|$*.*

Proof.

(i) By positivity, $\langle \mathbf{v}, \mathbf{v} \rangle$ is real and nonnegative, so its square root is real and nonnegative.
(ii) By definition, $\|\mathbf{v}\| = 0$ if and only if $\langle \mathbf{v}, \mathbf{v} \rangle = 0$, which holds if and only if $\mathbf{v} = \mathbf{0}$ by definiteness.
(iii) Compute $\|\lambda \mathbf{v}\|^2 = \langle \lambda \mathbf{v}, \lambda \mathbf{v} \rangle = \lambda \overline{\lambda} \langle \mathbf{v}, \mathbf{v} \rangle = |\lambda|^2 \|\mathbf{v}\|^2$. Taking nonnegative square roots, $\|\lambda \mathbf{v}\| = |\lambda| \, \|\mathbf{v}\|$.

Structural Commentary: Positivity gives length; sesquilinearity gives scaling.
Key Insight: Squared length is easier to manipulate than length itself. □

Theorem 2.1.11. *Cauchy-Schwarz Inequality*
For all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|$. Equality holds if and only if one of $\mathbf{u}, \mathbf{v}$ is a scalar multiple of the other.

Proof. If $\mathbf{v} = \mathbf{0}$, both sides are 0. Assume $\mathbf{v} \neq \mathbf{0}$.
For any $\lambda \in \mathbb{F}$, positivity gives $0 \leq \|\mathbf{u} - \lambda \mathbf{v}\|^2 = \|\mathbf{u}\|^2 - \lambda \langle \mathbf{v}, \mathbf{u} \rangle - \overline{\lambda} \langle \mathbf{u}, \mathbf{v} \rangle + |\lambda|^2 \|\mathbf{v}\|^2$.
Choose $\lambda = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}$. Since $\langle \mathbf{v}, \mathbf{u} \rangle = \overline{\langle \mathbf{u}, \mathbf{v} \rangle}$, the inequality becomes $0 \leq \|\mathbf{u}\|^2 - \frac{|\langle \mathbf{u}, \mathbf{v} \rangle|^2}{\|\mathbf{v}\|^2}$.
Multiplying by $\|\mathbf{v}\|^2$ and taking square roots proves the inequality. Equality in this argument is equivalent to $\|\mathbf{u} - \lambda \mathbf{v}\| = 0$, hence to $\mathbf{u} = \lambda \mathbf{v}$ for the chosen λ.
Conversely, if $\mathbf{u} = \lambda \mathbf{v}$, then $|\langle \mathbf{u}, \mathbf{v} \rangle| = |\lambda| \, \langle \mathbf{v}, \mathbf{v} \rangle = \|\lambda \mathbf{v}\| \, \|\mathbf{v}\| = \|\mathbf{u}\| \, \|\mathbf{v}\|$, so equality holds.
The case $\mathbf{v} = \mathbf{0}$ is also covered because $\mathbf{v} = 0\mathbf{u}$.
Structural Commentary: Positivity of squared length controls the mixed inner-product term.
Key Insight: The best scalar comparison between two vectors bounds their inner product. □

Theorem 2.1.12. *Triangle Inequality*
For all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.
Equality holds if and only if one of $\mathbf{u}, \mathbf{v}$ is a nonnegative scalar multiple of the other.

Proof. We compute

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2 \operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle \\ &\leq \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2 |\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2 \|\mathbf{u}\| \, \|\mathbf{v}\| = \big(\|\mathbf{u}\| + \|\mathbf{v}\|\big)^2. \end{aligned}$$

Taking square roots gives the inequality. If either vector is $\mathbf{0}$, the equality condition is immediate.
Otherwise, equality in the two displayed inequalities holds exactly when $\operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle = |\langle \mathbf{u}, \mathbf{v} \rangle|$ and equality holds in Cauchy-Schwarz. The first condition says $\langle \mathbf{u}, \mathbf{v} \rangle$ is real nonnegative, and the second says $\mathbf{u} = a\mathbf{v}$ for some $a \in \mathbb{F}$. Then $a \|\mathbf{v}\|^2 = \langle \mathbf{u}, \mathbf{v} \rangle$ is real nonnegative, so $a \geq 0$.

Conversely, if $\mathbf{u} = a\mathbf{v}$ with $a \geq 0$, then $\|\mathbf{u}+\mathbf{v}\| = (a+1)\|\mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$.
Structural Commentary: Square the norm and bound the cross term by Cauchy-Schwarz.
Key Insight: Equality means the two vectors point in the same direction. □

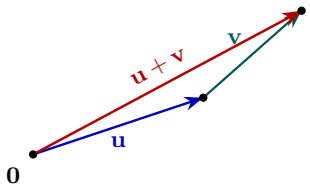


Figure 8: Triangle inequality: $\|\mathbf{u}+\mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

Definition 2.1.13. *Norm and Normed Vector Space*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. A *norm* on $\mathcal{V}$ is a function $\|\cdot\| : \mathcal{V} \to [0,\infty)$ such that, $\forall \mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $\lambda \in \mathbb{F}$,

i. $\|\mathbf{u}\| \geq 0$;
ii. $\|\mathbf{u}\| = 0$ if and only if $\mathbf{u} = \mathbf{0}$;
iii. $\|\lambda\mathbf{u}\| = |\lambda|\,\|\mathbf{u}\|$;
iv. $\|\mathbf{u}+\mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

A *normed vector space* is a vector space equipped with a norm. Its *unit ball* is $\{\mathbf{v} \in \mathcal{V} : \|\mathbf{v}\| \leq 1\}$.

Theorem 2.1.14. *Parallelogram Identity*
For all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2\big(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2\big)$.

Proof. Expanding both squared norms gives

$$\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{v}, \mathbf{u}\rangle + \|\mathbf{v}\|^2, \quad \|\mathbf{u}-\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - \langle \mathbf{u}, \mathbf{v}\rangle - \langle \mathbf{v}, \mathbf{u}\rangle + \|\mathbf{v}\|^2.$$

Adding the two identities cancels the cross terms and yields the result.
Structural Commentary: Opposite cross terms cancel when the two diagonal norms are added.
Key Insight: Inner-product norms remember parallelogram geometry through cancellation. □

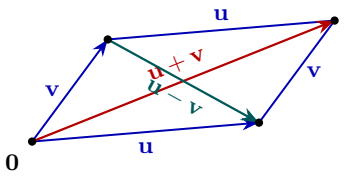


Figure 9: Parallelogram identity: $\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2)$.

Proposition 2.1.15. *Coordinate Norms on* $\mathbb{F}^n$
For $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{F}^n$, $\|\mathbf{x}\|_1 = \sum_{j=1}^{n} |x_j|$, $\|\mathbf{x}\|_2 = \left(\sum_{j=1}^{n} |x_j|^2\right)^{1/2}$, $\|\mathbf{x}\|_\infty = \max_{1\leq j\leq n} |x_j|$ *are norms on* $\mathbb{F}^n$.

Proof.

(i) For $\|\cdot\|_1$, nonnegativity, definiteness, and homogeneity follow from the corresponding facts for modulus. The triangle inequality follows from $\|\mathbf{x}+\mathbf{y}\|_1 = \sum_{j=1}^{n} |x_j + y_j| \leq \sum_{j=1}^{n} \big(|x_j| + |y_j|\big) = \|\mathbf{x}\|_1 + \|\mathbf{y}\|_1$.
(ii) For $\|\cdot\|_2$, the axioms follow from its standard inner product via Proposition 2.1.10 and Theorem 2.1.12.
(iii) For $\|\cdot\|_\infty$, nonnegativity, definiteness, and homogeneity are immediate from the maximum of finitely many moduli. If k is chosen so that $\|\mathbf{x}+\mathbf{y}\|_\infty = |x_k + y_k|$, then $\|\mathbf{x}+\mathbf{y}\|_\infty \leq |x_k| + |y_k| \leq \|\mathbf{x}\|_\infty + \|\mathbf{y}\|_\infty$.

Structural Commentary: The ℓ_1 norm sums coordinate magnitudes, ℓ_∞ keeps only the largest coordinate magnitude, and ℓ_2 comes from the standard inner product.
Key Insight: All three obey the same abstract axioms, even though their unit balls have different shapes. □

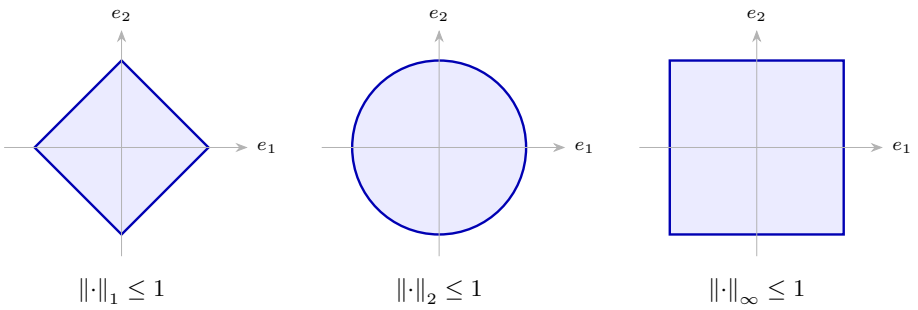


Figure 10: The unit balls of $\|\cdot\|_1$, $\|\cdot\|_2$, and $\|\cdot\|_\infty$ in $\mathbb{R}^2$.

Corollary 2.1.16. *ℓ_1 and ℓ_∞ Are Not Inner-Product Norms*
If $n \ge 2$, then the norms $\|\cdot\|_1$ and $\|\cdot\|_\infty$ on $\mathbb{F}^n$ are not induced by any inner product.

Proof. Let $\mathbf{e}_1, \mathbf{e}_2$ be the first two standard coordinate vectors.
For $\|\cdot\|_1$, $\|\mathbf{e}_1 + \mathbf{e}_2\|_1^2 + \|\mathbf{e}_1 - \mathbf{e}_2\|_1^2 = 4 + 4 = 8$, but $2\|\mathbf{e}_1\|_1^2 + 2\|\mathbf{e}_2\|_1^2 = 4$.
For $\|\cdot\|_\infty$, $\|\mathbf{e}_1 + \mathbf{e}_2\|_\infty^2 + \|\mathbf{e}_1 - \mathbf{e}_2\|_\infty^2 = 1 + 1 = 2$ but $2\|\mathbf{e}_1\|_\infty^2 + 2\|\mathbf{e}_2\|_\infty^2 = 4$.
Both norms violate Theorem 2.1.14, which every inner-product norm must satisfy.
Structural Commentary: The parallelogram identity detects inner-product norms.
Key Insight: Different unit-ball geometry can rule out hidden inner-product structure. □

Theorem 2.1.17. *Polarization Identities*
Let $\mathcal{V}$ be an inner-product space and let $\mathbf{u}, \mathbf{v} \in \mathcal{V}$.

i. If $\mathbb{F} = \mathbb{R}$, then $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}\left(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2\right)$.

ii. If $\mathbb{F} = \mathbb{C}$, then $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}\left(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 + i\|\mathbf{u} + i\mathbf{v}\|^2 - i\|\mathbf{u} - i\mathbf{v}\|^2\right)$.

Proof. Let $z := \langle \mathbf{u}, \mathbf{v} \rangle$.

(i) If $\mathbb{F} = \mathbb{R}$, then expanding squared norms gives $\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 = 4 \operatorname{Re} z = 4z$, proving (i).

(ii) If $\mathbb{F} = \mathbb{C}$, the first expansion gives $\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 = 4 \operatorname{Re} z$. The same expansion with $i\mathbf{v}$ gives $\|\mathbf{u} + i\mathbf{v}\|^2 - \|\mathbf{u} - i\mathbf{v}\|^2 = 4 \operatorname{Im} z$. Combining the real and imaginary parts gives (ii).

Structural Commentary: Two norm differences extract the real and imaginary parts.
Key Insight: The norm determines the inner product because it determines both parts of every cross term. □

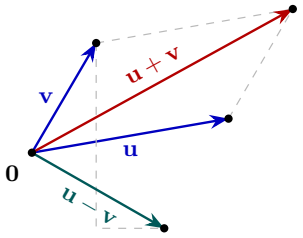


Figure 11: Polarization identity: $\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 = 4 \operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle$.

Principle 2.1.18. *Geometry from Algebra*
Most geometric statements in an inner-product space come from expanding $\|\mathbf{u} \pm \mathbf{v}\|^2$ and then interpreting the cross terms $\langle \mathbf{u}, \mathbf{v} \rangle$ and $\langle \mathbf{v}, \mathbf{u} \rangle$.

Definition 2.1.19. *Unit Vector and Normalization*
A vector $\mathbf{u} \in \mathcal{V}$ is a *unit vector* if $\|\mathbf{u}\| = 1$. If $\mathbf{v} \neq \mathbf{0}$, then $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ is called the *normalization* of $\mathbf{v}$.

Proposition 2.1.20. *Normalization Gives a Unit Vector*
If $\mathbf{v} \in \mathcal{V}$ and $\mathbf{v} \neq \mathbf{0}$, then $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ is a unit vector in the same direction as $\mathbf{v}$.

Proof. By Proposition 2.1.10, $\|\mathbf{v}\| > 0$ and $\left\| \frac{\mathbf{v}}{\|\mathbf{v}\|} \right\| = \frac{1}{\|\mathbf{v}\|} \|\mathbf{v}\| = 1$.
The vector $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ is a positive scalar multiple of $\mathbf{v}$, so it lies in the same one-dimensional direction. *Structural Commentary*: Normalization changes scale but preserves direction.
Key Insight: Divide by the length to make the length equal to one. □

Definition 2.1.21. *Orthogonal Vectors*
Vectors $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ are *orthogonal*, written $\mathbf{u} \perp \mathbf{v}$, if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Proposition 2.1.22. *Orthogonality and the Zero Vector*
Let $\mathcal{V}$ be an inner-product space.

i. $\mathbf{u} \perp \mathbf{v}$ *if and only if* $\mathbf{v} \perp \mathbf{u}$.
ii. $\mathbf{0}$ *is orthogonal to every vector in* $\mathcal{V}$.
iii. If $\mathbf{v}$ *is orthogonal to itself, then* $\mathbf{v} = \mathbf{0}$.

Proof.
(i) If $\langle \mathbf{u}, \mathbf{v} \rangle = 0$, then conjugate symmetry gives $\langle \mathbf{v}, \mathbf{u} \rangle = \overline{0} = 0$, and the converse is identical.
(ii) This follows from Proposition 2.1.7, which gives $\langle \mathbf{0}, \mathbf{u} \rangle = 0$ and $\langle \mathbf{u}, \mathbf{0} \rangle = 0$.
(iii) If $\mathbf{v} \perp \mathbf{v}$, then $\langle \mathbf{v}, \mathbf{v} \rangle = 0$, hence $\mathbf{v} = \mathbf{0}$ by definiteness.

Structural Commentary: Conjugate symmetry makes orthogonality symmetric, and definiteness isolates $\mathbf{0}$.
Key Insight: Conjugate symmetry preserves the equation "equals zero." □

Proposition 2.1.23. *Mutually Orthogonal Nonzero Vectors Are Independent*
Mutually orthogonal nonzero vectors $\mathbf{v}_1, \ldots, \mathbf{v}_m$ in an inner-product space are linearly independent.

Proof. If $\sum_{j=1}^{m} c_j \mathbf{v}_j = \mathbf{0}$, then for each k, $0 = \left\langle \sum_{j=1}^{m} c_j \mathbf{v}_j, \mathbf{v}_k \right\rangle = c_k \langle \mathbf{v}_k, \mathbf{v}_k \rangle$.
Since $\langle \mathbf{v}_k, \mathbf{v}_k \rangle > 0$, $c_k = 0$. Thus all coefficients vanish.
Structural Commentary: Orthogonality isolates one coefficient at a time.
Key Insight: Pair a dependence relation with one vector from the list; every cross term disappears. □

Definition 2.1.24. *Orthonormal List*
A list $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ of vectors in an inner-product space $\mathcal{V}$ is *orthonormal* if

$$\langle \mathbf{e}_j, \mathbf{e}_k \rangle = \begin{cases} 1, & j = k, \\ 0, & j \neq k. \end{cases}$$

Equivalently, every $\mathbf{e}_j$ has norm 1 and distinct vectors in the list are orthogonal.

Proposition 2.1.25. *Norm of an Orthonormal Linear Combination*
If $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ is an orthonormal list in $\mathcal{V}$, then $\|a_1 \mathbf{e}_1 + \cdots + a_m \mathbf{e}_m\|^2 = |a_1|^2 + \cdots + |a_m|^2$ for all $a_1, \ldots, a_m \in \mathbb{F}$.

Proof. Using orthonormality, $\left\| \sum_{j=1}^{m} a_j \mathbf{e}_j \right\|^2 = \left\langle \sum_{j=1}^{m} a_j \mathbf{e}_j, \sum_{k=1}^{m} a_k \mathbf{e}_k \right\rangle = \sum_{j=1}^{m} \sum_{k=1}^{m} a_j \overline{a_k} \langle \mathbf{e}_j, \mathbf{e}_k \rangle = \sum_{j=1}^{m} |a_j|^2$.
Structural Commentary: Orthonormality deletes every cross term in the squared-norm expansion.
Key Insight: In orthonormal coordinates, length is computed coefficient-by-coefficient. □

Corollary 2.1.26. *Orthonormal Lists Are Linearly Independent*
Every orthonormal list of vectors in an inner-product space is linearly independent.

Proof. If $a_1 \mathbf{e}_1 + \cdots + a_m \mathbf{e}_m = \mathbf{0}$, then Proposition 2.1.25 gives $0 = \|\mathbf{0}\|^2 = \|a_1 \mathbf{e}_1 + \cdots + a_m \mathbf{e}_m\|^2 = \sum_{j=1}^{m} |a_j|^2$.
Hence each $a_j = 0$, so the list is linearly independent.
Structural Commentary: The norm identity converts a linear relation into a sum of nonnegative scalars.
Key Insight: A zero orthonormal linear combination can only have zero coefficients. □

Definition 2.1.27. *Orthonormal Basis*
An *orthonormal basis* of $\mathcal{V}$ is a basis of $\mathcal{V}$ that is an orthonormal list.

Proposition 2.1.28. *Right-Length Orthonormal Lists Are Bases*
If $\mathcal{V}$ is finite-dimensional and $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ is an orthonormal list in $\mathcal{V}$ with $n = \dim(\mathcal{V})$, then $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ is an orthonormal basis of $\mathcal{V}$.

Proof. By Corollary 2.1.26, the list is linearly independent.
Since its length is $\dim(\mathcal{V})$, Proposition 1.1.29 implies that it is a basis.
Structural Commentary: Orthonormality gives independence; the correct length forces spanning.
Key Insight: In finite dimensions, n independent vectors in an n-dimensional space already form a basis. □

Theorem 2.1.29. *Gram–Schmidt Process*
Let $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ *be a linearly independent list in an inner-product space* $\mathcal{V}$.
Define $\mathbf{u}_1 = \mathbf{v}_1$, $\mathbf{e}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|}$ *and, for* $2 \le j \le m$, $\mathbf{u}_j = \mathbf{v}_j - \sum_{k=1}^{j-1} \langle \mathbf{v}_j, \mathbf{e}_k \rangle \mathbf{e}_k$, $\mathbf{e}_j = \frac{\mathbf{u}_j}{\|\mathbf{u}_j\|}$.
Then $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ *is orthonormal and* $\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_j) = \operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_j)$.

Proof. Proceed by induction on j. The case $j = 1$ is normalization of the nonzero vector $\mathbf{v}_1$.
Assume the result holds through $j-1$. Since $\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}) = \operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_{j-1})$ and the $\mathbf{v}$-list is linearly independent, $\mathbf{v}_j \notin \operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_{j-1})$. Hence $\mathbf{u}_j \neq \mathbf{0}$, because $\mathbf{u}_j = \mathbf{0}$ would put $\mathbf{v}_j$ in that span.
For $k < j$, $\langle \mathbf{u}_j, \mathbf{e}_k \rangle = \langle \mathbf{v}_j, \mathbf{e}_k \rangle - \sum_{i=1}^{j-1} \langle \mathbf{v}_j, \mathbf{e}_i \rangle \langle \mathbf{e}_i, \mathbf{e}_k \rangle = \langle \mathbf{v}_j, \mathbf{e}_k \rangle - \langle \mathbf{v}_j, \mathbf{e}_k \rangle = 0$.
Thus $\mathbf{e}_j$ is a unit vector orthogonal to $\mathbf{e}_1, \ldots, \mathbf{e}_{j-1}$.
Also $\mathbf{e}_j$ lies in $\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_j)$ because it is a nonzero scalar multiple of $\mathbf{u}_j$, and $\mathbf{u}_j$ is built from $\mathbf{v}_j$ and the previous $\mathbf{e}_k$, whose span agrees with $\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1})$.
Conversely, $\mathbf{v}_j = \mathbf{u}_j + \sum_{k=1}^{j-1} \langle \mathbf{v}_j, \mathbf{e}_k \rangle \mathbf{e}_k$ lies in $\operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_j)$.
Combining this with the induction hypothesis gives equality of the two spans at step j.
Structural Commentary: Subtract the previous-span component, then normalize.
Key Insight: Gram–Schmidt changes the basis vectors but preserves every initial span. □

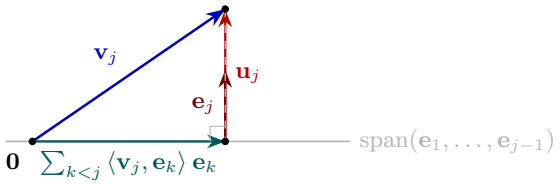


Figure 12: One Gram–Schmidt step: subtract the previous-span component, then normalize.

Lemma 2.1.30. *Gram–Schmidt on Orthogonal Lists*
If $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ *is a nonzero mutually orthogonal list, then Gram–Schmidt produces* $\mathbf{e}_j = \frac{\mathbf{v}_j}{\|\mathbf{v}_j\|}$, $j = 1, \ldots, m$.
In particular, if the original list is orthonormal, Gram–Schmidt leaves it unchanged.

Proof. Assume $\mathbf{e}_i = \mathbf{v}_i / \|\mathbf{v}_i\|$ for $i < j$. Since $\mathbf{v}_j \perp \mathbf{v}_i$, we have $\langle \mathbf{v}_j, \mathbf{e}_i \rangle = 0$ for every $i < j$.
Hence the Gram–Schmidt remainder at step j is $\mathbf{u}_j = \mathbf{v}_j$, so $\mathbf{e}_j = \mathbf{v}_j / \|\mathbf{v}_j\|$.
If the original list is orthonormal, then $\|\mathbf{v}_j\| = 1$ for every j.
Structural Commentary: Orthogonal input has no previous-span component to subtract.
Key Insight: Gram–Schmidt only removes nonorthogonal overlap. □

Corollary 2.1.31. *Existence of Orthonormal Bases*
Every finite-dimensional inner-product space has an orthonormal basis.

Proof. Choose any basis of $\mathcal{V}$ and apply Theorem 2.1.29.
The resulting orthonormal list has the same span as the original basis, hence spans $\mathcal{V}$.
Structural Commentary: Gram–Schmidt orthonormalizes a basis without changing its span.
Key Insight: Finite-dimensional inner-product spaces always admit geometry-friendly coordinates. □

Corollary 2.1.32. *Extending an Orthonormal List*
Every orthonormal list in a finite-dimensional inner-product space can be extended to an orthonormal basis.

Proof. Let $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ be orthonormal. It is linearly independent by Corollary 2.1.26, so extend it to a basis $(\mathbf{e}_1, \ldots, \mathbf{e}_m, \mathbf{v}_{m+1}, \ldots, \mathbf{v}_n)$ by Theorem 1.1.25.
During Gram–Schmidt, the first m remainders are the original $\mathbf{e}_j$ because the list is already orthonormal; after normalization they remain unchanged. The remaining vectors are orthonormalized by Theorem 2.1.29, and the final list spans $\mathcal{V}$ because Gram–Schmidt preserves the total span.
Structural Commentary: Extend to a basis, then orthonormalize the new part.
Key Insight: Existing orthonormal directions are already cleaned up, so Gram–Schmidt does not move them. □

Theorem 2.1.33. *Coordinates in an Orthonormal Basis*
Let $\mathcal{V}$ be finite-dimensional and let $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be an orthonormal basis of $\mathcal{V}$.
Then for every $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$, $\|\mathbf{v}\|^2 = \sum_{j=1}^{n} \left|\langle \mathbf{v}, \mathbf{e}_j \rangle\right|^2$ and

$$[\mathbf{v}]_\beta = \begin{bmatrix} \langle \mathbf{v}, \mathbf{e}_1 \rangle \\ \vdots \\ \langle \mathbf{v}, \mathbf{e}_n \rangle \end{bmatrix}.$$

Proof. Since β is a basis, write $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{e}_j$. Taking inner product with $\mathbf{e}_k$ gives $\langle \mathbf{v}, \mathbf{e}_k \rangle = \sum_{j=1}^{n} a_j \langle \mathbf{e}_j, \mathbf{e}_k \rangle = a_k$. Thus the displayed expansion and coordinate column follow.
Applying Proposition 2.1.25 to the same expansion gives $\|\mathbf{v}\|^2 = \sum_{j=1}^{n} |a_j|^2 = \sum_{j=1}^{n} \left|\langle \mathbf{v}, \mathbf{e}_j \rangle\right|^2$. *Structural Commentary*: An orthonormal basis makes coordinate extraction an inner-product computation.
Key Insight: Pairing with $\mathbf{e}_k$ isolates the kth coordinate. □

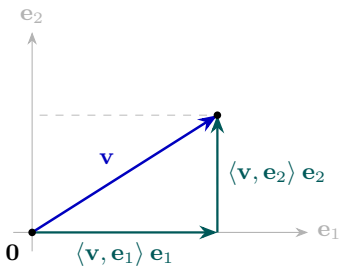


Figure 13: An orthonormal basis resolves a vector into perpendicular coordinate components.

Remark 2.1.34. *Cross Terms*
In an expansion such as $\|\mathbf{u} + \mathbf{v}\|^2$, the terms involving both vectors, $\langle \mathbf{u}, \mathbf{v} \rangle$ and $\langle \mathbf{v}, \mathbf{u} \rangle$, are called *cross terms*. The terms $\|\mathbf{u}\|^2$ and $\|\mathbf{v}\|^2$ are the pure squared-length terms.

Theorem 2.1.35. *Pythagorean Theorem*
If $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $\mathbf{u} \perp \mathbf{v}$, then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Proof. Since $\mathbf{u} \perp \mathbf{v}$, Proposition 2.1.22 gives $\langle \mathbf{v}, \mathbf{u} \rangle = 0$. Thus

$$\|\mathbf{u} + \mathbf{v}\|^2 = \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$

Structural Commentary: Orthogonality cancels the cross terms in the squared-norm expansion.
Key Insight: Perpendicular components add in squared length, not in length. □

Theorem 2.1.36. *Bessel's Inequality*
Let $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ be an orthonormal list in an inner-product space $\mathcal{V}$.
Then, for every $\mathbf{v} \in \mathcal{V}$, $\sum_{j=1}^{m} \left|\langle \mathbf{v}, \mathbf{e}_j \rangle\right|^2 \leq \|\mathbf{v}\|^2$.

Proof. Set $\mathbf{u} = \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$. For each k, orthonormality gives $\langle \mathbf{v} - \mathbf{u}, \mathbf{e}_k \rangle = \langle \mathbf{v}, \mathbf{e}_k \rangle - \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \langle \mathbf{e}_j, \mathbf{e}_k \rangle = 0$.
Thus $\mathbf{v} - \mathbf{u}$ is orthogonal to $\mathbf{u}$. By Pythagorean Theorem 2.1.35 and Proposition 2.1.25,

$$\|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v} - \mathbf{u}\|^2 \geq \|\mathbf{u}\|^2 = \sum_{j=1}^{m} \left|\langle \mathbf{v}, \mathbf{e}_j \rangle\right|^2.$$

Structural Commentary: The chosen directions give an orthogonal part of $\mathbf{v}$; the residual is perpendicular.
Key Insight: Partial orthonormal coordinates cannot contain more energy than the whole vector. □

Corollary 2.1.37. *Coefficients Against an Orthonormal Sequence Vanish*
Let $(\mathbf{e}_1, \mathbf{e}_2, \ldots)$ be an orthonormal sequence in an inner-product space $\mathcal{V}$, meaning every finite sublist is orthonormal. Then, for every $\mathbf{v} \in \mathcal{V}$, $\lim_{n \to \infty} \langle \mathbf{v}, \mathbf{e}_n \rangle = 0$.

Proof. Fix $\epsilon > 0$. By Theorem 2.1.36, every finite set of indices satisfying $\left|\langle \mathbf{v}, \mathbf{e}_j \rangle\right| \geq \epsilon$ has cardinality at most $\|\mathbf{v}\|^2 / \epsilon^2$. Hence only finitely many j have $\left|\langle \mathbf{v}, \mathbf{e}_j \rangle\right| \geq \epsilon$. This holds for every $\epsilon > 0$, so $\langle \mathbf{v}, \mathbf{e}_n \rangle \to 0$.
Structural Commentary: Bessel bounds the number of large orthonormal coefficients.
Key Insight: Infinitely many large coefficients would force infinite energy. □

Theorem 2.1.38. *Parseval's Identity*
Let $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be an orthonormal basis of an inner-product space $\mathcal{V}$.
Then, for all $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \overline{\langle \mathbf{w}, \mathbf{e}_j \rangle} = \langle [\mathbf{v}]_\beta, [\mathbf{w}]_\beta \rangle_{\mathbb{F}^n}$.

Proof. By Theorem 2.1.33, $\mathbf{v} = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$ and $\mathbf{w} = \sum_{k=1}^{n} \langle \mathbf{w}, \mathbf{e}_k \rangle \mathbf{e}_k$. By linearity in first slot and conjugate linearity in second, $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{j=1}^{n} \sum_{k=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \overline{\langle \mathbf{w}, \mathbf{e}_k \rangle} \langle \mathbf{e}_j, \mathbf{e}_k \rangle = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \overline{\langle \mathbf{w}, \mathbf{e}_j \rangle}$.
The last expression is exactly the standard inner product of the coordinate columns in $\mathbb{F}^n$.
Structural Commentary: Orthonormal coordinates make inner products coordinatewise.
Key Insight: Orthonormal coordinates preserve angles and lengths, not just linear combinations. □

Theorem 2.1.39. *Matrix Entries from an Orthonormal Codomain Basis*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces over $\mathbb{F}$, with $\mathcal{W}$ an inner-product space. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$ and let $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ be an orthonormal basis of $\mathcal{W}$. If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then

$$[T]_{\gamma \leftarrow \beta} = \begin{bmatrix} \langle T\mathbf{v}_1, \mathbf{w}_1 \rangle & \cdots & \langle T\mathbf{v}_n, \mathbf{w}_1 \rangle \\ \vdots & \ddots & \vdots \\ \langle T\mathbf{v}_1, \mathbf{w}_m \rangle & \cdots & \langle T\mathbf{v}_n, \mathbf{w}_m \rangle \end{bmatrix}.$$

Conversely, for each $A \in \mathrm{Mat}(m, n, \mathbb{F})$, there is a unique $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ such that $[T]_{\gamma \leftarrow \beta} = A$.

Proof. The jth column of $[T]_{\gamma \leftarrow \beta}$ is $[T\mathbf{v}_j]_\gamma$. Since γ is orthonormal, Theorem 2.1.33 gives

$$[T\mathbf{v}_j]_\gamma = \begin{bmatrix} \langle T\mathbf{v}_j, \mathbf{w}_1 \rangle \\ \vdots \\ \langle T\mathbf{v}_j, \mathbf{w}_m \rangle \end{bmatrix}.$$

This proves the displayed matrix formula. The converse is exactly the surjectivity and injectivity of the coordinate encoding in Theorem 1.3.58.
Structural Commentary: Columns are basis-image coordinates, read by inner products.
Key Insight: Each entry measures $T\mathbf{v}_j$ against one output direction. □

Corollary 2.1.40. *Inner Products Through an Operator Matrix*
Let $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be an orthonormal basis of a finite-dimensional inner-product space $\mathcal{V}$, and let $T \in \mathcal{L}(\mathcal{V})$.
Then, for all $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, $\langle T\mathbf{v}, \mathbf{w} \rangle = \langle [T]_{\beta \leftarrow \beta} [\mathbf{v}]_\beta, [\mathbf{w}]_\beta \rangle_{\mathbb{F}^n}$.

Proof. By Proposition 1.3.53, $[T\mathbf{v}]_\beta = [T]_{\beta \leftarrow \beta} [\mathbf{v}]_\beta$. Applying Theorem 2.1.38 to $T\mathbf{v}$ and $\mathbf{w}$ gives the result.
Structural Commentary: Orthonormal coordinates turn $\langle T\mathbf{v}, \mathbf{w} \rangle$ into matrix computation.
Key Insight: Orthonormal coordinates make operator geometry computable by matrices. □

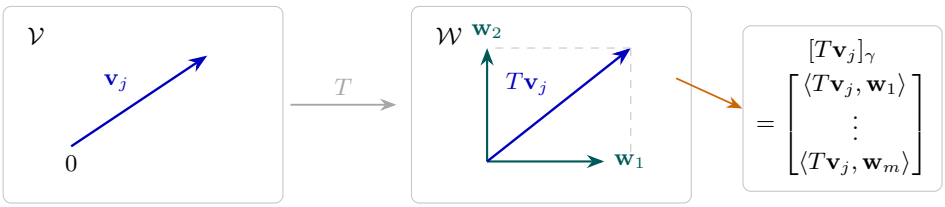


Figure 14: The jth column of $[T]_{\gamma \leftarrow \beta}$ is obtained by resolving $T\mathbf{v}_j$ against the orthonormal basis of $\mathcal{W}$.

Theorem 2.1.41. *Riesz Representation Theorem*
Let $\mathcal{V}$ be a finite-dimensional inner-product space and let $\varphi : \mathcal{V} \to \mathbb{F}$ be a linear functional. Then there is a unique vector $\mathbf{u} \in \mathcal{V}$ such that $\varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u} \rangle$ for every $\mathbf{v} \in \mathcal{V}$.

Proof. Let $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be an orthonormal basis of $\mathcal{V}$.
By Theorem 2.1.33, $\varphi(\mathbf{v}) = \varphi\left(\sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j\right) = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \varphi(\mathbf{e}_j)$.

Set $\mathbf{u} = \sum_{j=1}^{n} \overline{\varphi(\mathbf{e}_j)}\,\mathbf{e}_j$. Since inner product is conjugate linear in second slot, $\langle \mathbf{v}, \mathbf{u} \rangle = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle\, \varphi(\mathbf{e}_j) = \varphi(\mathbf{v})$.
For uniqueness, suppose $\langle \mathbf{v}, \mathbf{u}_1 \rangle = \langle \mathbf{v}, \mathbf{u}_2 \rangle \ \ \forall \mathbf{v} \in \mathcal{V}$. Taking $\mathbf{v} = \mathbf{u}_1 - \mathbf{u}_2$ gives $\|\mathbf{u}_1 - \mathbf{u}_2\|^2 = 0$, hence $\mathbf{u}_1 = \mathbf{u}_2$.
Structural Commentary: An orthonormal basis packages a functional as one representing vector.
Key Insight: Finite-dimensional functionals are inner products with fixed vectors. □

Definition 2.1.42. *Riesz Vector*
Let $\mathcal{V}$ be finite-dimensional and let $\varphi : \mathcal{V} \to \mathbb{F}$ be a linear functional. The unique vector $\mathbf{u} \in \mathcal{V}$ satisfying $\varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u} \rangle$ for every $\mathbf{v} \in \mathcal{V}$ is the *Riesz vector* representing φ.

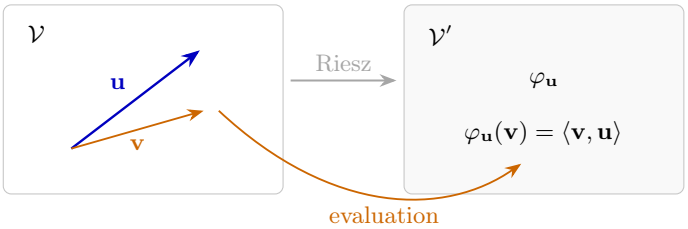


Figure 15: Riesz identifies a functional with its representing vector.

Perspective 2.1.43. *Riesz as Coordinate Duality*
The vector $\sum_{j=1}^{n} \overline{\varphi(\mathbf{e}_j)}\mathbf{e}_j$ seems to depend on the orthonormal basis, but Theorem 2.1.41 shows that the resulting representing vector is intrinsic.

Definition 2.1.44. *Gram Matrix*
For vectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ in an inner-product space, the *Gram matrix* is

$$\operatorname{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n) := \begin{bmatrix} \langle \mathbf{u}_1, \mathbf{u}_1 \rangle & \langle \mathbf{u}_2, \mathbf{u}_1 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_1 \rangle \\ \langle \mathbf{u}_1, \mathbf{u}_2 \rangle & \langle \mathbf{u}_2, \mathbf{u}_2 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{u}_1, \mathbf{u}_n \rangle & \langle \mathbf{u}_2, \mathbf{u}_n \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_n \rangle \end{bmatrix}.$$

Proposition 2.1.45. *Gram Matrix Quadratic Form*
For any vectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$, Gram matrix $G = \operatorname{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$ is Hermitian and satisfies $\mathbf{c}^* G \mathbf{c} = \left\| \sum_{j=1}^{n} c_j \mathbf{u}_j \right\|^2$ for every $\mathbf{c} = [c_1 \ \cdots \ c_n]^t \in \mathbb{F}^n$. Hence the Gram quadratic form is nonnegative. If $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are linearly independent, then it is positive for every nonzero $\mathbf{c}$.

Proof. Hermitian symmetry follows from conjugate symmetry of the inner product. Expanding the squared norm gives $\left\| \sum_{j=1}^{n} c_j \mathbf{u}_j \right\|^2 = \sum_{i,j} \overline{c_i}\, \langle \mathbf{u}_j, \mathbf{u}_i \rangle\, c_j = \mathbf{c}^* G \mathbf{c}$. The right-hand side is a squared norm, so it is nonnegative.
If $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are linearly independent, the squared norm is 0 only when all $c_j = 0$.
Structural Commentary: A Gram matrix records every pairwise inner product in a list of vectors.
Key Insight: The quadratic form of a Gram matrix is exactly the squared norm of a linear combination. □

2.2 Orthogonal Complements and Projections

Orthogonal complements turn orthogonality into a subspace operation; projections turn it into canonical decompositions and best approximations.

Definition 2.2.1. *Orthogonal Complement*
If $\mathcal{U} \subseteq \mathcal{V}$, then the *orthogonal complement* of $\mathcal{U}$ is $\mathcal{U}^{\perp} := \{\mathbf{v} \in \mathcal{V} : \langle \mathbf{v}, \mathbf{u} \rangle = 0 \text{ for every } \mathbf{u} \in \mathcal{U}\}$.

Proposition 2.2.2. *Basic Properties of Orthogonal Complements*
For subsets $\mathcal{U}, \mathcal{W} \subseteq \mathcal{V}$,

i. $\mathcal{U}^{\perp}$ is a subspace of $\mathcal{V}$;
ii. $\{\mathbf{0}\}^{\perp} = \mathcal{V}$;
iii. $\mathcal{V}^{\perp} = \{\mathbf{0}\}$;
iv. $\mathcal{U} \cap \mathcal{U}^{\perp} \subseteq \{\mathbf{0}\}$;
v. if $\mathcal{U} \subseteq \mathcal{W}$, then $\mathcal{W}^{\perp} \subseteq \mathcal{U}^{\perp}$.

Proof.

(i) We have $\mathbf{0} \in \mathcal{U}^\perp$. If $\mathbf{v}, \mathbf{w} \in \mathcal{U}^\perp$ and $a \in \mathbb{F}$, then for every $\mathbf{u} \in \mathcal{U}$, $\langle a\mathbf{v} + \mathbf{w}, \mathbf{u} \rangle = a \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{u} \rangle = 0$.
(ii) For every $\mathbf{v} \in \mathcal{V}$, $\langle \mathbf{v}, \mathbf{0} \rangle = 0$, so $\mathbf{v} \in \{\mathbf{0}\}^\perp$.
(iii) If $\mathbf{v} \in \mathcal{V}^\perp$, then $\langle \mathbf{v}, \mathbf{v} \rangle = 0$, so $\mathbf{v} = \mathbf{0}$.
(iv) If $\mathbf{v} \in \mathcal{U} \cap \mathcal{U}^\perp$, then $\langle \mathbf{v}, \mathbf{v} \rangle = 0$, so $\mathbf{v} = \mathbf{0}$.
(v) If $\mathbf{v} \in \mathcal{W}^\perp$, then $\mathbf{v}$ is orthogonal to every vector in $\mathcal{W}$, hence to every vector in $\mathcal{U}$.

Structural Commentary: Orthogonal complements are solution spaces of homogeneous inner-product equations. Larger input sets impose more equations, so their complements become smaller.
Key Insight: Orthogonality constraints are linear constraints. □

Theorem 2.2.3. *Direct Sum with the Orthogonal Complement*
If $\mathcal{U}$ is a finite-dimensional subspace of $\mathcal{V}$, then $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^\perp$.

Proof. Let $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ be an orthonormal basis of $\mathcal{U}$. For $\mathbf{v} \in \mathcal{V}$, define $\mathbf{u} = \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$ and set $\mathbf{w} = \mathbf{v} - \mathbf{u}$.

Clearly $\mathbf{u} \in \mathcal{U}$. For each k, $\langle \mathbf{w}, \mathbf{e}_k \rangle = \langle \mathbf{v}, \mathbf{e}_k \rangle - \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \langle \mathbf{e}_j, \mathbf{e}_k \rangle = 0$.
If $\mathbf{x} \in \mathcal{U}$, write $\mathbf{x} = \sum_{k=1}^{m} a_k \mathbf{e}_k$. Then $\langle \mathbf{w}, \mathbf{x} \rangle = \sum_{k=1}^{m} \overline{a_k} \langle \mathbf{w}, \mathbf{e}_k \rangle = 0$, so $\mathbf{w} \in \mathcal{U}^\perp$ and $\mathbf{v} = \mathbf{u} + \mathbf{w}$.
Hence $\mathcal{V} = \mathcal{U} + \mathcal{U}^\perp$. The intersection is $\{\mathbf{0}\}$ by Proposition 2.2.2, so the sum is direct.
Structural Commentary: Each vector splits into a $\mathcal{U}$-component and an orthogonal residual.
Key Insight: Projection plus residual gives the canonical direct-sum decomposition. □

Corollary 2.2.4. *Dimension of the Orthogonal Complement*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U}$ is a subspace of $\mathcal{V}$, then $\dim(\mathcal{U}^\perp) = \dim(\mathcal{V}) - \dim(\mathcal{U})$.

Proof. Since $\mathcal{V}$ is finite-dimensional, $\mathcal{U}$ is finite-dimensional by Corollary 1.1.26. Theorem 2.2.3 gives $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^\perp$. Thus Corollary 1.2.11 gives $\dim(\mathcal{V}) = \dim(\mathcal{U}) + \dim(\mathcal{U}^\perp)$.
Structural Commentary: Dimensions add across the orthogonal split.
Key Insight: Orthogonal complements contain exactly the dimensions not used by $\mathcal{U}$. □

Corollary 2.2.5. *Double Orthogonal Complement*
If $\mathcal{U}$ is a finite-dimensional subspace of $\mathcal{V}$, then $\mathcal{U} = (\mathcal{U}^\perp)^\perp$.

Proof. If $\mathbf{u} \in \mathcal{U}$, then $\mathbf{u}$ is orthogonal to every vector in $\mathcal{U}^\perp$, so $\mathcal{U} \subseteq (\mathcal{U}^\perp)^\perp$. Conversely, let $\mathbf{v} \in (\mathcal{U}^\perp)^\perp$. By Theorem 2.2.3, write $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{U}^\perp$. Since $\mathcal{U} \subseteq (\mathcal{U}^\perp)^\perp$, we have $\mathbf{w} = \mathbf{v} - \mathbf{u} \in (\mathcal{U}^\perp)^\perp$.
Hence $\mathbf{w} \in \mathcal{U}^\perp \cap (\mathcal{U}^\perp)^\perp$, so $\mathbf{w} = \mathbf{0}$ and $\mathbf{v} = \mathbf{u} \in \mathcal{U}$.
Structural Commentary: The orthogonal split leaves no extra direction beyond $\mathcal{U}$.
Key Insight: Taking the orthogonal complement twice recovers the original finite-dimensional subspace. □

Perspective 2.2.6. *Orthogonal Versus Complementary*
Orthogonality gives $\mathcal{W} \subseteq \mathcal{U}^\perp$ and $\mathcal{U} \subseteq \mathcal{W}^\perp$, not necessarily equality. In finite dimensions, equality follows once $\dim(\mathcal{U}) + \dim(\mathcal{W}) = \dim(\mathcal{V})$ by Corollary 2.2.4.

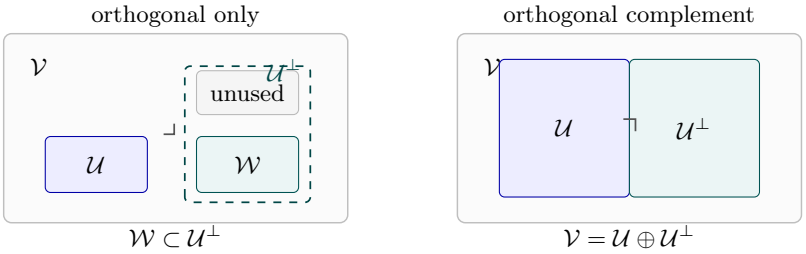


Figure 16: Orthogonal subspaces are complements only when their sum fills $\mathcal{V}$.

Proposition 2.2.7. *Orthogonal Projection onto a Line*
Let $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ with $\mathbf{v} \neq \mathbf{0}$. Define $c := \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}$, $\mathbf{w} := \mathbf{u} - c\mathbf{v}$. Then $\mathbf{w} \perp \mathbf{v}$ and $\mathbf{u} = c\mathbf{v} + \mathbf{w}$.
The vector $\operatorname{proj}_{\mathbf{v}}(\mathbf{u}) := c\mathbf{v} = \left\langle \mathbf{u}, \frac{\mathbf{v}}{\|\mathbf{v}\|} \right\rangle \frac{\mathbf{v}}{\|\mathbf{v}\|}$ is the orthogonal projection of $\mathbf{u}$ onto $\operatorname{span}(\mathbf{v})$.

Proof. The decomposition $\mathbf{u} = c\mathbf{v} + \mathbf{w}$ follows from the definition of $\mathbf{w}$.
Moreover, $\langle \mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{u} - c\mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - c \langle \mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \|\mathbf{v}\|^2 = 0$. Thus $\mathbf{w} \perp \mathbf{v}$.
Structural Commentary: Choose c so the remainder $\mathbf{u} - c\mathbf{v}$ has zero inner product with $\mathbf{v}$.
Key Insight: Projection is the part detected by pairing with the direction vector. □

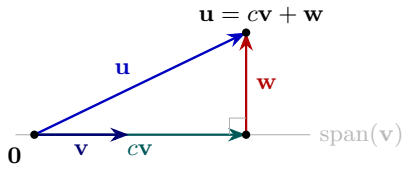


Figure 17: Writing $\mathbf{u}$ as a component parallel to $\mathbf{v}$ plus a component orthogonal to $\mathbf{v}$.

Definition 2.2.8. *Orthogonal Projection onto a Subspace*
Let $\mathcal{U}$ be a finite-dimensional subspace of $\mathcal{V}$. For $\mathbf{v} \in \mathcal{V}$, write the unique decomposition $\mathbf{v} = \mathbf{u} + \mathbf{w}$, $\mathbf{u} \in \mathcal{U}$, $\mathbf{w} \in \mathcal{U}^\perp$, from Theorem 2.2.3. The *orthogonal projection* of $\mathcal{V}$ onto $\mathcal{U}$ is the map $P_\mathcal{U} : \mathcal{V} \to \mathcal{V}$ defined by $P_\mathcal{U}\mathbf{v} = \mathbf{u}$.

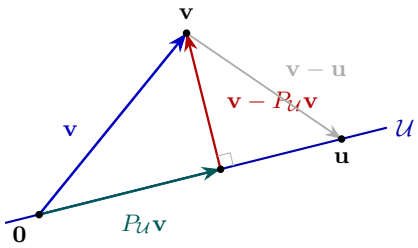


Figure 18: Orthogonal projection decomposes $\mathbf{v}$ into $P_\mathcal{U}\mathbf{v}$ plus an orthogonal residual.

Proposition 2.2.9. *Properties of Orthogonal Projection*
Let $\mathcal{U}$ be a finite-dimensional subspace of $\mathcal{V}$. Then:

i. $P_\mathcal{U} \in \mathcal{L}(\mathcal{V})$;
ii. $P_\mathcal{U}\mathbf{u} = \mathbf{u}$ *for every* $\mathbf{u} \in \mathcal{U}$;
iii. $P_\mathcal{U}\mathbf{w} = \mathbf{0}$ *for every* $\mathbf{w} \in \mathcal{U}^\perp$;
iv. $\operatorname{range}(P_\mathcal{U}) = \mathcal{U}$;
v. $\ker(P_\mathcal{U}) = \mathcal{U}^\perp$, *i.e. the kernel/nullspace of* $P_\mathcal{U}$ *is* $\mathcal{U}^\perp$;
vi. $\mathbf{v} - P_\mathcal{U}\mathbf{v} \in \mathcal{U}^\perp$ *for every* $\mathbf{v} \in \mathcal{V}$;
vii. $P_\mathcal{U}^2 = P_\mathcal{U}$;
viii. $\|P_\mathcal{U}\mathbf{v}\| \leq \|\mathbf{v}\|$ *for every* $\mathbf{v} \in \mathcal{V}$;
ix. if $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ *is an orthonormal basis of* $\mathcal{U}$, *then* $P_\mathcal{U}\mathbf{v} = \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$, $\mathbf{v} \in \mathcal{V}$.

Proof.

(i) For linearity, let $a \in \mathbb{F}$ and write $\mathbf{v}_r = \mathbf{u}_r + \mathbf{w}_r$ with $\mathbf{u}_r \in \mathcal{U}$ and $\mathbf{w}_r \in \mathcal{U}^\perp$ for $r = 1, 2$.
Then $a\mathbf{v}_1 + \mathbf{v}_2 = (a\mathbf{u}_1 + \mathbf{u}_2) + (a\mathbf{w}_1 + \mathbf{w}_2)$, where first summand lies in $\mathcal{U}$ and second in $\mathcal{U}^\perp$.
By uniqueness of the orthogonal decomposition, $P_\mathcal{U}(a\mathbf{v}_1 + \mathbf{v}_2) = aP_\mathcal{U}\mathbf{v}_1 + P_\mathcal{U}\mathbf{v}_2$.
(ii) This follows from $\mathbf{u} = \mathbf{u} + \mathbf{0}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{0} \in \mathcal{U}^\perp$.
(iii) This follows from $\mathbf{w} = \mathbf{0} + \mathbf{w}$ with $\mathbf{0} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{U}^\perp$.
(iv) The definition gives $\operatorname{range}(P_\mathcal{U}) \subseteq \mathcal{U}$, and (ii) gives $\mathcal{U} \subseteq \operatorname{range}(P_\mathcal{U})$.
(v) Part (iii) gives $\mathcal{U}^\perp \subseteq \ker(P_\mathcal{U})$.
Conversely, if $P_\mathcal{U}\mathbf{v} = \mathbf{0}$, then $\mathbf{v} = \mathbf{0} + (\mathbf{v} - P_\mathcal{U}\mathbf{v})$ with $\mathbf{v} - P_\mathcal{U}\mathbf{v} \in \mathcal{U}^\perp$, so $\mathbf{v} \in \mathcal{U}^\perp$.
(vi) This is exactly the residual term in the decomposition $\mathbf{v} = P_\mathcal{U}\mathbf{v} + (\mathbf{v} - P_\mathcal{U}\mathbf{v})$.
(vii) By (ii), $P_\mathcal{U}(P_\mathcal{U}\mathbf{v}) = P_\mathcal{U}\mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$.
(viii) Write $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$, $\mathbf{w} \in \mathcal{U}^\perp$. Then $\mathbf{u} \perp \mathbf{w}$ and $\|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{w}\|^2 \geq \|\mathbf{u}\|^2 = \|P_\mathcal{U}\mathbf{v}\|^2$.
Since norms are nonnegative, $\|P_\mathcal{U}\mathbf{v}\| \leq \|\mathbf{v}\|$.
(ix) This is the explicit $\mathcal{U}$-component constructed in the proof of Theorem 2.2.3.

Structural Commentary: Orthogonal projection keeps the $\mathcal{U}$-component and discards the $\mathcal{U}^\perp$-component.
Key Insight: $P_\mathcal{U}$ is the identity on $\mathcal{U}$ and zero on $\mathcal{U}^\perp$. □

Definition 2.2.10. *Rank-One Projection Matrix*
Let $\mathbf{u} \in \mathbb{F}^n$ be a unit vector. The *rank-one projection matrix* onto $\operatorname{span}(\mathbf{u})$ is $P_{\mathbf{u}} := \mathbf{u}\mathbf{u}^* \in \operatorname{Mat}(n, n, \mathbb{F})$, where * denotes conjugate transpose.
For every $\mathbf{x} \in \mathbb{F}^n$, $P_{\mathbf{u}}\mathbf{x} = \langle \mathbf{x}, \mathbf{u} \rangle_{\mathbb{F}^n} \mathbf{u}$.

Proposition 2.2.11. *Properties of Rank-One Projection Matrices*
Let $\mathbf{u} \in \mathbb{F}^n$ be a unit vector. Then:

i. $\operatorname{range}(P_{\mathbf{u}}) = \operatorname{span}(\mathbf{u})$;
ii. $\ker(P_{\mathbf{u}}) = \operatorname{span}(\mathbf{u})^{\perp}$;
iii. $P_{\mathbf{u}}^* = P_{\mathbf{u}}$;
iv. $P_{\mathbf{u}}^2 = P_{\mathbf{u}}$.

Proof.

(i) The formula $P_{\mathbf{u}}\mathbf{x} = \langle \mathbf{x}, \mathbf{u} \rangle \mathbf{u}$ gives $\operatorname{range}(P_{\mathbf{u}}) \subseteq \operatorname{span}(\mathbf{u})$, and $P_{\mathbf{u}}\mathbf{u} = \mathbf{u}$ gives the reverse inclusion.
(ii) The same formula gives $P_{\mathbf{u}}\mathbf{x} = 0$ exactly when $\langle \mathbf{x}, \mathbf{u} \rangle \mathbf{u} = \mathbf{0}$.
Since $\mathbf{u} \neq \mathbf{0}$, this is equivalent to $\langle \mathbf{x}, \mathbf{u} \rangle = 0$, i.e. $\mathbf{x} \in \operatorname{span}(\mathbf{u})^{\perp}$.
(iii) Since $P_{\mathbf{u}} = \mathbf{u}\mathbf{u}^*$, $P_{\mathbf{u}}^* = (\mathbf{u}\mathbf{u}^*)^* = \mathbf{u}\mathbf{u}^* = P_{\mathbf{u}}$.
(iv) Because $\mathbf{u}^*\mathbf{u} = \|\mathbf{u}\|_2^2 = 1$, $P_{\mathbf{u}}^2 = \mathbf{u}(\mathbf{u}^*\mathbf{u})\mathbf{u}^* = \mathbf{u}\mathbf{u}^* = P_{\mathbf{u}}$.

Structural Commentary: A rank-one projection keeps the $\mathbf{u}$ direction and kills its orthogonal complement.
Key Insight: The outer product $\mathbf{u}\mathbf{u}^*$ encodes projection onto one unit direction. □

Theorem 2.2.12. *Orthogonal Projection Minimizes Distance*
Let $\mathcal{U}$ be a finite-dimensional subspace of $\mathcal{V}$. If $\mathbf{v} \in \mathcal{V}$ and $\mathbf{u} \in \mathcal{U}$, then $\|\mathbf{v} - P_{\mathcal{U}}\mathbf{v}\| \leq \|\mathbf{v} - \mathbf{u}\|$.
Equality holds if and only if $\mathbf{u} = P_{\mathcal{U}}\mathbf{v}$.

Proof. Since $\mathbf{v} - P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}^{\perp}$ and $P_{\mathcal{U}}\mathbf{v} - \mathbf{u} \in \mathcal{U}$, the two vectors are orthogonal.
Thus $\|\mathbf{v} - \mathbf{u}\|^2 = \|(\mathbf{v} - P_{\mathcal{U}}\mathbf{v}) + (P_{\mathcal{U}}\mathbf{v} - \mathbf{u})\|^2 = \|\mathbf{v} - P_{\mathcal{U}}\mathbf{v}\|^2 + \|P_{\mathcal{U}}\mathbf{v} - \mathbf{u}\|^2$.
Taking square roots gives the inequality. Equality holds exactly when $\|P_{\mathcal{U}}\mathbf{v} - \mathbf{u}\| = 0$, i.e. when $\mathbf{u} = P_{\mathcal{U}}\mathbf{v}$.
Structural Commentary: Projection error is orthogonal to every in-subspace correction.
Key Insight: Moving away from the projection adds a perpendicular error term and can only increase distance.
□

Definition 2.2.13. *Distance to a Subspace*
Let $\mathcal{U}$ be a nonempty subset of an inner-product space $\mathcal{V}$.
The *distance* from $\mathbf{v} \in \mathcal{V}$ to $\mathcal{U}$ is $d(\mathbf{v}, \mathcal{U}) := \inf_{\mathbf{u} \in \mathcal{U}} \|\mathbf{v} - \mathbf{u}\|$.
If $\mathcal{U}$ is finite-dimensional, Theorem 2.2.12 gives $d(\mathbf{v}, \mathcal{U}) = \|\mathbf{v} - P_{\mathcal{U}}\mathbf{v}\|$.

2.3 Least Squares and Metric Optimization

Least squares is projection theory as optimization: normal equations, full-rank formulas, and minimum-norm solutions.

Principle 2.3.1. *Least Squares as Orthogonal Projection*
In an inner-product space, the best approximation to $\mathbf{v}$ from a finite-dimensional subspace $\mathcal{U}$ is $P_{\mathcal{U}}\mathbf{v}$. The residual $\mathbf{v} - P_{\mathcal{U}}\mathbf{v}$ is orthogonal to every allowable correction inside $\mathcal{U}$.

Theorem 2.3.2. *Normal Equations for Projection*
Let $\mathcal{U}$ be a finite-dimensional subspace of an inner-product space $\mathcal{V}$, and suppose $\mathcal{U} = \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$.
For $\mathbf{v} \in \mathcal{V}$, the projection has the form $P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{u}_j$, where $(c_1, \ldots, c_n)$ solves the normal equations

$$\begin{bmatrix} \langle \mathbf{u}_1, \mathbf{u}_1 \rangle & \langle \mathbf{u}_2, \mathbf{u}_1 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_1 \rangle \\ \langle \mathbf{u}_1, \mathbf{u}_2 \rangle & \langle \mathbf{u}_2, \mathbf{u}_2 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{u}_1, \mathbf{u}_n \rangle & \langle \mathbf{u}_2, \mathbf{u}_n \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_n \rangle \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} \langle \mathbf{v}, \mathbf{u}_1 \rangle \\ \langle \mathbf{v}, \mathbf{u}_2 \rangle \\ \vdots \\ \langle \mathbf{v}, \mathbf{u}_n \rangle \end{bmatrix}.$$

This system is consistent. If $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are linearly independent, then the solution is unique.

Proof. Since $P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}$, write $P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{u}_j$.
The residual $\mathbf{v} - P_{\mathcal{U}}\mathbf{v}$ lies in $\mathcal{U}^{\perp}$, so for each i, $0 = \left\langle \mathbf{v} - \sum_{j=1}^{n} c_j \mathbf{u}_j, \mathbf{u}_i \right\rangle = \langle \mathbf{v}, \mathbf{u}_i \rangle - \sum_{j=1}^{n} c_j \langle \mathbf{u}_j, \mathbf{u}_i \rangle$.
These equations are exactly the displayed system, so consistency follows from the existence of $P_{\mathcal{U}}\mathbf{v}$.
Conversely, if $(d_1, \ldots, d_n)$ solves displayed system and $\mathbf{q} := \sum_{j=1}^{n} d_j \mathbf{u}_j$, then $\langle \mathbf{v} - \mathbf{q}, \mathbf{u}_i \rangle = 0$ for every i.

Since $\mathbf{u}_i$ span $\mathcal{U}$, it follows that $\mathbf{v} - \mathbf{q} \in \mathcal{U}^\perp$, hence $\mathbf{q} = P_{\mathcal{U}}\mathbf{v}$ by uniqueness of the decomposition. If $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are linearly independent, they form a basis of $\mathcal{U}$, so the coordinates of this projection in that basis are unique.
Structural Commentary: Normal equations express residual orthogonality.
Key Insight: The projection error is perpendicular to $\mathcal{U}$. □

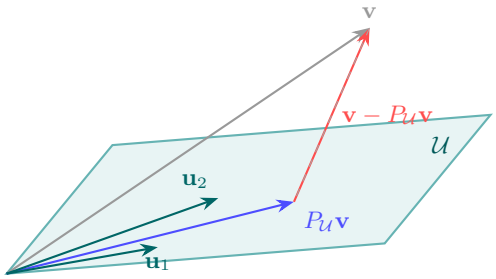


Figure 19: Normal equations impose $\langle \mathbf{v} - P_{\mathcal{U}}\mathbf{v}, \mathbf{u}_i \rangle = 0$ for each spanning vector.

Theorem 2.3.3. *Matrix Least Squares and Normal Equations*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $\mathbf{y} \in \mathbb{F}^m$, where A^* denotes the conjugate transpose of A.
A vector $\mathbf{x}_0 \in \mathbb{F}^n$ satisfies $\|\mathbf{y} - A\mathbf{x}_0\|_2 = \min_{\mathbf{x}\in\mathbb{F}^n} \|\mathbf{y} - A\mathbf{x}\|_2$ if and only if $A^*A\mathbf{x}_0 = A^*\mathbf{y}$.
The normal equations are always consistent, and their solution is unique if and only if $\mathrm{rank}(A) = n$.

Proof. Let $\mathcal{U} = \mathrm{col}(A)$, and write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$. Since $\{A\mathbf{x} : \mathbf{x} \in \mathbb{F}^n\} = \mathcal{U}$, Theorem 2.2.12 shows that $\mathbf{x}_0$ is least-squares optimal exactly when $A\mathbf{x}_0 = P_{\mathcal{U}}\mathbf{y}$, or equivalently when $\mathbf{y} - A\mathbf{x}_0 \in \mathcal{U}^\perp$.

($\Rightarrow$) If $\mathbf{x}_0$ is least-squares optimal, then $\mathbf{y} - A\mathbf{x}_0 \in \mathcal{U}^\perp$. Hence $\langle \mathbf{y} - A\mathbf{x}_0, \mathbf{a}_j \rangle = 0$, $j = 1, \ldots, n$.
These are the entries of $A^*(\mathbf{y} - A\mathbf{x}_0)$, so $A^*(\mathbf{y} - A\mathbf{x}_0) = \mathbf{0}$ and $A^*A\mathbf{x}_0 = A^*\mathbf{y}$.

($\Leftarrow$) If $A^*A\mathbf{x}_0 = A^*\mathbf{y}$, then $A^*(\mathbf{y} - A\mathbf{x}_0) = \mathbf{0}$. Thus $\langle \mathbf{y} - A\mathbf{x}_0, \mathbf{a}_j \rangle = 0$ for every j, so $\mathbf{y} - A\mathbf{x}_0$ is orthogonal to every linear combination of the columns of A. Hence $\mathbf{y} - A\mathbf{x}_0 \in \mathcal{U}^\perp$.
Hence $A\mathbf{x}_0 = P_{\mathcal{U}}\mathbf{y}$, so $\mathbf{x}_0$ is least-squares optimal.

Consistency follows by choosing $\mathbf{x}_0$ such that $A\mathbf{x}_0 = P_{\mathcal{U}}\mathbf{y}$.
If $\mathbf{x}_0, \mathbf{x}_1$ solve the normal equations, then $\|A(\mathbf{x}_0 - \mathbf{x}_1)\|_2^2 = \langle \mathbf{x}_0 - \mathbf{x}_1, A^*A(\mathbf{x}_0 - \mathbf{x}_1) \rangle = 0$, so $\mathbf{x}_0 - \mathbf{x}_1 \in \ker(A)$.
Conversely, if $\mathbf{z} \in \ker(A)$ and $\mathbf{x}_0$ solves the normal equations, then $\mathbf{x}_0 + \mathbf{z}$ also solves them. Thus uniqueness is equivalent to $\ker(A) = \{\mathbf{0}\}$, equivalently $\mathrm{rank}(A) = n$.
Structural Commentary: Least squares is orthogonal projection onto the column space of A.
Key Insight: The normal equations say exactly that the residual is orthogonal to every column of A. □

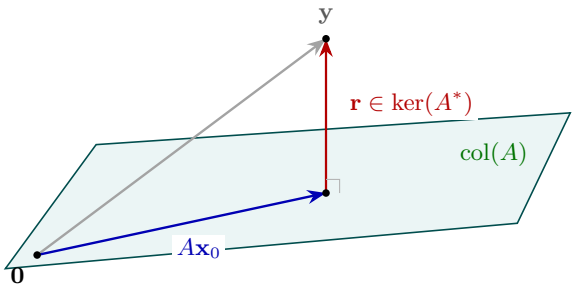


Figure 20: Fitted vector and orthogonal residual: $A\mathbf{x}_0 = P_{\mathrm{col}(A)}\mathbf{y}$ and $\mathbf{r} = \mathbf{y} - A\mathbf{x}_0 \in \mathrm{col}(A)^\perp = \ker(A^*)$.

Perspective 2.3.4. *Incidence Data and Least Squares*
For an incidence matrix B as in Example 4.1.51, $B\mathbf{x} = \mathbf{b}$ seeks node potentials matching edge differences. If loop consistency fails, least squares projects $\mathbf{b}$ onto $\mathrm{col}(B)$, where $B^*(B\mathbf{x}_0 - \mathbf{b}) = \mathbf{0}$ (Theorem 2.3.3 on B). The constant-vector kernel means potentials are unique only after fixing one node or choosing minimum norm.

Corollary 2.3.5. *Projection Formula for Full Column Rank*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$ have rank n.
Then A^*A is invertible and the orthogonal projection onto $\mathrm{col}(A)$ *is* $P_{\mathrm{col}(A)} = A(A^*A)^{-1}A^*$.

Proof. If $A^*A\mathbf{x} = \mathbf{0}$, then $\|A\mathbf{x}\|_2^2 = \langle \mathbf{x}, A^*A\mathbf{x} \rangle = 0$, so $A\mathbf{x} = \mathbf{0}$.
Since $\mathrm{rank}(A) = n$, we get $\mathbf{x} = \mathbf{0}$. Hence A^*A is invertible. By Theorem 2.3.3, the least-squares solution is

$\mathbf{x}_0 = (A^*A)^{-1}A^*\mathbf{y}$. The fitted vector is $P_{\mathrm{col}(A)}\mathbf{y} = A\mathbf{x}_0 = A(A^*A)^{-1}A^*\mathbf{y}$.
Since this holds for every $\mathbf{y}$, the displayed operator formula follows.
Structural Commentary: Full column rank makes the normal-equation coordinates unique.
Key Insight: Solve for coefficients, then map back by A. □

Definition 2.3.6. *Minimum Norm Solution*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $\mathbf{y} \in \mathrm{col}(A)$.
A solution $\mathbf{s}$ of $A\mathbf{x} = \mathbf{y}$ is a *minimum norm solution* if $\|\mathbf{s}\|_2 \leq \|\mathbf{x}\|_2$ for every solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{y}$.

Theorem 2.3.7. *Minimum Norm Solution of a Consistent Linear System*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$, $\mathbf{y} \in \mathbb{F}^m$, and suppose $A\mathbf{x} = \mathbf{y}$ is consistent.

i. There is a unique minimum norm solution $\mathbf{s} \in \mathbb{F}^n$, and $\mathbf{s} \in \mathrm{col}(A^*)$.
ii. $\mathbf{s}$ is the only solution of $A\mathbf{x} = \mathbf{y}$ that lies in $\mathrm{col}(A^*)$.
iii. The system $AA^*\mathbf{u} = \mathbf{y}$ is consistent. If $AA^*\mathbf{u}_0 = \mathbf{y}$, then $\mathbf{s} = A^*\mathbf{u}_0$.

Proof. First note that $\mathrm{col}(A^*) = (\ker A)^\perp$. Indeed, if $\mathbf{z} = A^*\mathbf{u}$ and $\mathbf{x} \in \ker(A)$, then $\langle \mathbf{x}, \mathbf{z}\rangle = \langle A\mathbf{x}, \mathbf{u}\rangle = 0$, so $\mathbf{z}$ is orthogonal to every vector in $\ker(A)$. Thus $\mathrm{col}(A^*) \subseteq (\ker A)^\perp$. Also,

$$\dim(\mathrm{col}(A^*)) = \mathrm{rank}(A^*) = \mathrm{rank}(A) = n - \dim(\ker(A)) = \dim((\ker A)^\perp),$$

so the two subspaces are equal. Hence $\mathbb{F}^n = \mathrm{col}(A^*) \oplus \ker(A)$ orthogonally.

(i) Choose a solution $\mathbf{x}_0$ of $A\mathbf{x} = \mathbf{y}$ and write $\mathbf{x}_0 = \mathbf{s} + \mathbf{v}$, with $\mathbf{s} \in \mathrm{col}(A^*)$ and $\mathbf{v} \in \ker(A)$. Then $A\mathbf{s} = A\mathbf{x}_0 = \mathbf{y}$. If $\mathbf{x}$ is any solution, then $\mathbf{x} - \mathbf{s} \in \ker(A)$, so every solution has the form $\mathbf{s} + \mathbf{w}$ with $\mathbf{w} \in \ker(A)$. Since $\mathbf{s} \perp \mathbf{w}$, $\|\mathbf{s} + \mathbf{w}\|_2^2 = \|\mathbf{s}\|_2^2 + \|\mathbf{w}\|_2^2 \geq \|\mathbf{s}\|_2^2$, with equality only when $\mathbf{w} = \mathbf{0}$. Thus $\mathbf{s}$ is the unique solution of minimum norm.
(ii) If $\mathbf{s}' \in \mathrm{col}(A^*)$ is another solution, then $\mathbf{s}' - \mathbf{s} \in \ker(A) \cap \mathrm{col}(A^*) = \{\mathbf{0}\}$, so $\mathbf{s}' = \mathbf{s}$.
(iii) Since $\mathbf{s} \in \mathrm{col}(A^*)$, write $\mathbf{s} = A^*\mathbf{z}$. Then $AA^*\mathbf{z} = A\mathbf{s} = \mathbf{y}$, so $AA^*\mathbf{u} = \mathbf{y}$ is consistent. Conversely, if $AA^*\mathbf{u}_0 = \mathbf{y}$, then $A^*\mathbf{u}_0$ is a solution in $\mathrm{col}(A^*)$, so $A^*\mathbf{u}_0 = \mathbf{s}$ by (ii).

Structural Commentary: Solutions are $\mathbf{s} + \ker(A)$; $\mathrm{col}(A^*)$ meets this set orthogonally.
Key Insight: The minimum norm solution has no nullspace component. □

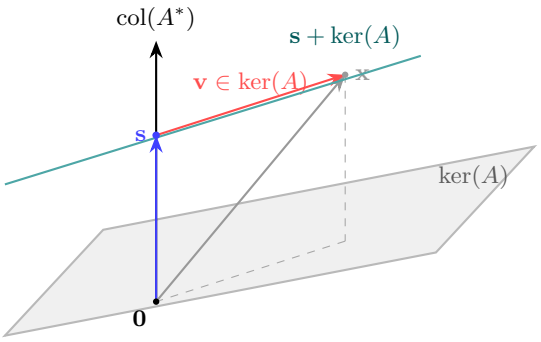


Figure 21: Among all solutions $\mathbf{s} + \ker(A)$, the unique one in $\mathrm{col}(A^*) = (\ker A)^\perp$ has minimum norm.

Perspective 2.3.8. *Roadmap to the Pseudoinverse*
Theorem 2.3.3 projects $\mathbf{y}$ onto $\mathrm{col}(A)$; Theorem 2.3.7 selects the minimum-norm preimage in $\mathrm{col}(A^*)$. Definition 4.3.16 and Theorem 4.3.19 package both choices as $A^\dagger\mathbf{y}$.

2.4 Adjoints and Orthogonal Operators

Adjoints express dual action in inner-product language. They become conjugate transposes in orthonormal coordinates and lead to self-adjoint maps, projections, isometries, and unitary or orthogonal operators.

Definition 2.4.1. *Adjoint of a Linear Map*
Let $\mathcal{V}$ and $\mathcal{W}$ be inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
A function $S : \mathcal{W} \to \mathcal{V}$ is an *adjoint* of T if $\langle T\mathbf{v}, \mathbf{w}\rangle_{\mathcal{W}} = \langle \mathbf{v}, S\mathbf{w}\rangle_{\mathcal{V}}$, $\mathbf{v} \in \mathcal{V}$, $\mathbf{w} \in \mathcal{W}$.

Lemma 2.4.2. *Uniqueness of Adjoints*
A linear map has at most one adjoint.

Proof. Suppose $S_1, S_2 : \mathcal{W} \to \mathcal{V}$ are adjoints of T. For each $\mathbf{w} \in \mathcal{W}$ and every $\mathbf{v} \in \mathcal{V}$, $\langle \mathbf{v}, S_1\mathbf{w} - S_2\mathbf{w}\rangle_{\mathcal{V}} = 0$. Taking $\mathbf{v} = S_1\mathbf{w} - S_2\mathbf{w}$ gives $S_1\mathbf{w} = S_2\mathbf{w}$. Hence $S_1 = S_2$.
Structural Commentary: Same tests force candidate adjoints to agree.
Key Insight: Inner products separate vectors. □

Theorem 2.4.3. *Existence of the Adjoint*
Let $\mathcal{V}$ be a finite-dimensional inner-product space, let $\mathcal{W}$ be an inner-product space, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then T has a unique adjoint $T^* : \mathcal{W} \to \mathcal{V}$, and T^* is linear.

Proof. For each $\mathbf{w} \in \mathcal{W}$, the map $\varphi_{\mathbf{w}} : \mathcal{V} \to \mathbb{F}$ defined by $\varphi_{\mathbf{w}}(\mathbf{v}) = \langle T\mathbf{v}, \mathbf{w}\rangle_{\mathcal{W}}$ is a linear functional. By Theorem 2.1.41, there is a unique vector $T^*\mathbf{w} \in \mathcal{V}$ such that $\langle T\mathbf{v}, \mathbf{w}\rangle_{\mathcal{W}} = \langle \mathbf{v}, T^*\mathbf{w}\rangle_{\mathcal{V}}$ for every $\mathbf{v} \in \mathcal{V}$.
Thus T^* is an adjoint, and it is unique by Lemma 2.4.2. For $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W}$ and $a \in \mathbb{F}$,

$$\langle \mathbf{v}, T^*(a\mathbf{w}_1 + \mathbf{w}_2)\rangle_{\mathcal{V}} = \langle T\mathbf{v}, a\mathbf{w}_1 + \mathbf{w}_2\rangle_{\mathcal{W}} = \overline{a}\,\langle T\mathbf{v}, \mathbf{w}_1\rangle_{\mathcal{W}} + \langle T\mathbf{v}, \mathbf{w}_2\rangle_{\mathcal{W}} = \langle \mathbf{v}, aT^*\mathbf{w}_1 + T^*\mathbf{w}_2\rangle_{\mathcal{V}}\,.$$

for every $\mathbf{v} \in \mathcal{V}$. Taking the difference of the two second-slot vectors and then setting $\mathbf{v}$ equal to that difference gives $T^*(a\mathbf{w}_1 + \mathbf{w}_2) = aT^*\mathbf{w}_1 + T^*\mathbf{w}_2$.
Structural Commentary: Each $\mathbf{w}$ gives a Riesz-represented functional on $\mathcal{V}$.
Key Insight: The adjoint is pointwise Riesz representation for T. □

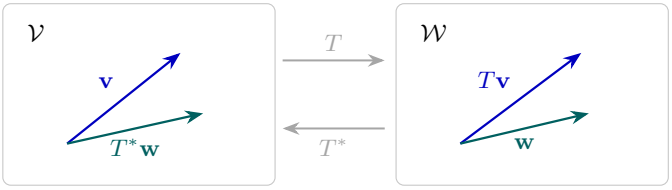


Figure 22: Adjoint relation: $\langle T\mathbf{v}, \mathbf{w}\rangle_{\mathcal{W}} = \langle \mathbf{v}, T^*\mathbf{w}\rangle_{\mathcal{V}}$.

Definition 2.4.4. *Conjugate Transpose*
For $A = [a_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$, the *conjugate transpose* of A is $A^* = [\overline{a_{ji}}] \in \mathrm{Mat}(n, m, \mathbb{F})$.
Over $\mathbb{R}$, this is the ordinary transpose.

Theorem 2.4.5. *Matrix of the Adjoint*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner-product spaces. Let $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be an orthonormal basis of $\mathcal{V}$ and let $\gamma = (\mathbf{f}_1, \ldots, \mathbf{f}_m)$ be an orthonormal basis of $\mathcal{W}$. If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $[T^*]_{\beta\leftarrow\gamma} = \big([T]_{\gamma\leftarrow\beta}\big)^*$.

Proof. Let $A = [T]_{\gamma\leftarrow\beta}$ and $B = [T^*]_{\beta\leftarrow\gamma}$.
By Theorem 2.1.39, $B_{i,j} = \langle T^*\mathbf{f}_j, \mathbf{e}_i\rangle_{\mathcal{V}} = \overline{\langle \mathbf{e}_i, T^*\mathbf{f}_j\rangle_{\mathcal{V}}} = \overline{\langle T\mathbf{e}_i, \mathbf{f}_j\rangle_{\mathcal{W}}} = \overline{A_{j,i}}$. Thus $B = A^*$.
Structural Commentary: Entries swap by moving T across the inner product.
Key Insight: In orthonormal coordinates, taking adjoints is exactly taking conjugate transposes. □

Proposition 2.4.6. *Basic Adjoint Identities*
For finite-dimensional inner-product spaces and composable linear maps:

i. $(aS + T)^* = \overline{a}S^* + T^*$ *for* $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ *and* $a \in \mathbb{F}$;
ii. $(SR)^* = R^*S^*$ *for* $R \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ *and* $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$;
iii. $(T^*)^* = T$;
iv. $I_{\mathcal{V}}^* = I_{\mathcal{V}}$.

Proof. Verify the defining identity; uniqueness gives equality.

(i) For $\mathbf{v} \in \mathcal{V}$, $\mathbf{w} \in \mathcal{W}$, $\langle (aS + T)\mathbf{v}, \mathbf{w}\rangle = a\langle S\mathbf{v}, \mathbf{w}\rangle + \langle T\mathbf{v}, \mathbf{w}\rangle = \langle \mathbf{v}, \overline{a}S^*\mathbf{w} + T^*\mathbf{w}\rangle$, so $(aS+T)^* = \overline{a}S^* + T^*$.
(ii) For $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$, $\langle SR\mathbf{u}, \mathbf{w}\rangle = \langle R\mathbf{u}, S^*\mathbf{w}\rangle = \langle \mathbf{u}, R^*S^*\mathbf{w}\rangle$, so $(SR)^* = R^*S^*$.
(iii) Conjugating $\langle T\mathbf{v}, \mathbf{w}\rangle = \langle \mathbf{v}, T^*\mathbf{w}\rangle$ gives $\langle T^*\mathbf{w}, \mathbf{v}\rangle = \langle \mathbf{w}, T\mathbf{v}\rangle$, so T is the adjoint of T^*.
(iv) Since $\langle I\mathbf{v}, \mathbf{w}\rangle = \langle \mathbf{v}, I\mathbf{w}\rangle$, the identity map is its own adjoint.

Structural Commentary: Put both sides in one inner-product slot, then use uniqueness.
Key Insight: Like matrix transposes, adjoints reverse products and conjugate scalars. □

Proposition 2.4.7. *Adjoint Kernel-Range Orthogonality*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
Then $\ker(T^*) = (\mathrm{range}(T))^\perp$, $\mathrm{range}(T^*) = (\ker(T))^\perp$, *and* $\ker(T) = (\mathrm{range}(T^*))^\perp$, $\mathrm{range}(T) = (\ker(T^*))^\perp$.
Equivalently, the nullspace of an adjoint is the orthogonal complement of the range of the original map.

Proof. For $\mathbf{w} \in \mathcal{W}$,

$$\begin{aligned} \mathbf{w} \in \ker(T^*) &\iff \langle \mathbf{v}, T^*\mathbf{w} \rangle_{\mathcal{V}} = 0 \quad \text{for every } \mathbf{v} \in \mathcal{V} \\ &\iff \langle T\mathbf{v}, \mathbf{w} \rangle_{\mathcal{W}} = 0 \quad \text{for every } \mathbf{v} \in \mathcal{V} \\ &\iff \mathbf{w} \in (\operatorname{range}(T))^{\perp}. \end{aligned}$$

Applying this identity to T^* gives $\ker(T) = (\operatorname{range}(T^*))^{\perp}$.
Since all spaces are finite-dimensional, the ranges and kernels are finite-dimensional subspaces.
Taking orthogonal complements and using Corollary 2.2.5 gives

$$(\ker(T^*))^{\perp} = ((\operatorname{range}(T))^{\perp})^{\perp} = \operatorname{range}(T), \qquad (\ker(T))^{\perp} = ((\operatorname{range}(T^*))^{\perp})^{\perp} = \operatorname{range}(T^*)$$

Structural Commentary: Adjoints trade ranges for kernels, and orthogonal complements reverse that trade.
Key Insight: A vector is killed by T^* exactly when it is orthogonal to everything T can produce. □

Corollary 2.4.8. *Matrix Fundamental Subspaces as Orthogonal Complements*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$ act as $A : \mathbb{F}^n \to \mathbb{F}^m$ with standard inner products.
Then $\ker(A) = \operatorname{col}(A^*)^{\perp}$ and $\operatorname{col}(A^*) = \ker(A)^{\perp}$; $\ker(A^*) = \operatorname{col}(A)^{\perp}$ and $\operatorname{col}(A) = \ker(A^*)^{\perp}$.
Over $\mathbb{R}$, this says $\ker(A) \perp \operatorname{row}(A)$ in $\mathbb{R}^n$ and $\ker(A^t) \perp \operatorname{col}(A)$ in $\mathbb{R}^m$, with both pairs complementary.

Proof. By Theorem 2.4.5, the adjoint of A is represented by A^*.
Proposition 2.4.7 gives $\ker(A^*) = \operatorname{col}(A)^{\perp}$ and $\ker(A) = \operatorname{col}(A^*)^{\perp}$. Taking orthogonal complements and using Corollary 2.2.5 gives the two column-space identities. Over $\mathbb{R}$, $A^* = A^t$ and $\operatorname{col}(A^t) = \operatorname{row}(A)$.
Structural Commentary: The four fundamental subspaces form two orthogonal-complement pairs.
Key Insight: Adjoints turn missing directions into kernels. □

Definition 2.4.9. *Self-Adjoint and Hermitian*
An operator $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space is *self-adjoint* if $T^* = T$. A square matrix A is *Hermitian* if $A^* = A$. In orthonormal coordinates, these are the same condition by Theorem 2.4.5.

Lemma 2.4.10. *Zero Complex Quadratic Form*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$.
If $\langle T\mathbf{v}, \mathbf{v} \rangle = 0$ for every $\mathbf{v} \in \mathcal{V}$, then $T = 0$.

Proof. Set $Q(\mathbf{x}) = \langle T\mathbf{x}, \mathbf{x} \rangle$.
For $\mathbf{u}, \mathbf{w} \in \mathcal{V}$, expansion gives the polarization identity for the sesquilinear form $(\mathbf{u}, \mathbf{w}) \mapsto \langle T\mathbf{u}, \mathbf{w} \rangle$:

$$\langle T\mathbf{u}, \mathbf{w} \rangle = \frac{Q(\mathbf{u}+\mathbf{w}) - Q(\mathbf{u}-\mathbf{w})}{4} + i\,\frac{Q(\mathbf{u}+i\mathbf{w}) - Q(\mathbf{u}-i\mathbf{w})}{4}.$$

The hypothesis makes every term on the right zero, so $\langle T\mathbf{u}, \mathbf{w} \rangle = 0$ for all $\mathbf{u}, \mathbf{w}$.
Taking $\mathbf{w} = T\mathbf{u}$ gives $\|T\mathbf{u}\|^2 = 0$, hence $T\mathbf{u} = \mathbf{0}$ for every $\mathbf{u}$.
Structural Commentary: Over $\mathbb{C}$, diagonal values of a sesquilinear form determine all mixed values.
Key Insight: Polarization turns the one-vector condition into a two-vector condition. □

Proposition 2.4.11. *Self-Adjointness from Real Quadratic Values*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$.
Then T is self-adjoint if and only if $\langle T\mathbf{v}, \mathbf{v} \rangle \in \mathbb{R}$ for every $\mathbf{v} \in \mathcal{V}$.

Proof. For each $\mathbf{v}$, $\langle (T - T^*)\mathbf{v}, \mathbf{v} \rangle = \langle T\mathbf{v}, \mathbf{v} \rangle - \langle T^*\mathbf{v}, \mathbf{v} \rangle = \langle T\mathbf{v}, \mathbf{v} \rangle - \overline{\langle T\mathbf{v}, \mathbf{v} \rangle}$.
If all values $\langle T\mathbf{v}, \mathbf{v} \rangle$ are real, this expression is zero for every $\mathbf{v}$, and Lemma 2.4.10 gives $T - T^* = 0$.
Conversely, if $T = T^*$, then $\langle T\mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{v}, T\mathbf{v} \rangle = \overline{\langle T\mathbf{v}, \mathbf{v} \rangle}$, so $\langle T\mathbf{v}, \mathbf{v} \rangle$ is real.
Structural Commentary: The imaginary part of $\langle T\mathbf{v}, \mathbf{v} \rangle$ is exactly the diagonal trace of $T - T^*$.
Key Insight: Over $\mathbb{C}$, real quadratic values force the operator to equal its adjoint. □

Proposition 2.4.12. *Zero Quadratic Form for Self-Adjoint Operators*
Let T be a self-adjoint operator on a finite-dimensional inner-product space $\mathcal{V}$.
If $\langle T\mathbf{v}, \mathbf{v} \rangle = 0$ for every $\mathbf{v} \in \mathcal{V}$, then $T = 0$.

Proof. If $\mathbb{F} = \mathbb{C}$, this is Lemma 2.4.10. Assume $\mathbb{F} = \mathbb{R}$.
For $\mathbf{u}, \mathbf{w} \in \mathcal{V}$, self-adjointness gives $\langle T\mathbf{w}, \mathbf{u} \rangle = \langle \mathbf{w}, T\mathbf{u} \rangle = \langle T\mathbf{u}, \mathbf{w} \rangle$, so the real polarization identity gives

$$\langle T\mathbf{u}, \mathbf{w} \rangle = \frac{\langle T(\mathbf{u}+\mathbf{w}), \mathbf{u}+\mathbf{w} \rangle - \langle T(\mathbf{u}-\mathbf{w}), \mathbf{u}-\mathbf{w} \rangle}{4} = 0.$$

Taking $\mathbf{w} = T\mathbf{u}$ gives $T\mathbf{u} = \mathbf{0}$ for every $\mathbf{u}$.
Structural Commentary: Self-adjointness supplies the missing symmetry needed for real polarization.
Key Insight: A self-adjoint operator is determined by its quadratic form. □

Definition 2.4.13. *Orthogonal Projection Matrix*
A square matrix $P \in \mathrm{Mat}(n, n, \mathbb{F})$ is an *orthogonal projection matrix* if $P^2 = P$ and $P^* = P$.

Theorem 2.4.14. *Orthogonal Projections as Self-Adjoint Idempotents*
Let $\mathcal{V}$ be finite-dimensional and let $P \in \mathcal{L}(\mathcal{V})$.
Then P is the orthogonal projection onto $\mathrm{range}(P)$ *if and only if $P^2 = P$ and $P^* = P$.*

Proof.
($\Rightarrow$) If $P = P_{\mathcal{U}}$ for $\mathcal{U} = \mathrm{range}(P)$, then Proposition 2.2.9 gives $P^2 = P$. For $\mathbf{v} = \mathbf{u} + \mathbf{w}$ and $\mathbf{x} = \mathbf{a} + \mathbf{b}$ with $\mathbf{u}, \mathbf{a} \in \mathcal{U}$ and $\mathbf{w}, \mathbf{b} \in \mathcal{U}^\perp$, $\langle P\mathbf{v}, \mathbf{x}\rangle = \langle \mathbf{u}, \mathbf{a} + \mathbf{b}\rangle = \langle \mathbf{u}, \mathbf{a}\rangle = \langle \mathbf{u} + \mathbf{w}, \mathbf{a}\rangle = \langle \mathbf{v}, P\mathbf{x}\rangle$, so $P^* = P$.
($\Leftarrow$) Suppose $P^2 = P$ and $P^* = P$. For each $\mathbf{v}$, $\mathbf{v} = P\mathbf{v} + (\mathbf{v} - P\mathbf{v})$ with $P\mathbf{v} \in \mathrm{range}(P)$ and $P(\mathbf{v} - P\mathbf{v}) = \mathbf{0}$. Thus the residual lies in $\ker(P)$. Since $P = P^*$, Proposition 2.4.7 gives $\ker(P) = (\mathrm{range}(P))^\perp$.
Hence every $\mathbf{v}$ has the orthogonal decomposition $\mathbf{v} = P\mathbf{v} + (\mathbf{v} - P\mathbf{v})$ with first part in $\mathrm{range}(P)$, so P is exactly the orthogonal projection onto $\mathrm{range}(P)$.

Structural Commentary: Idempotence splits by range and kernel; self-adjointness makes the split orthogonal.
Key Insight: An idempotent is orthogonal exactly when $\ker(P) = (\mathrm{range}(P))^\perp$. □

Theorem 2.4.15. *Unitary Coordinate Form of an Orthogonal Projection*
Let $P \in \mathrm{Mat}(n, n, \mathbb{F})$ be an orthogonal projection matrix and let $r = \mathrm{rank}(P)$. Then P is unitarily similar to $I_r \oplus 0_{n-r}$. Equivalently, there is a unitary matrix U such that

$$U^*PU = \begin{bmatrix} I_r & 0 \\ 0 & 0_{n-r} \end{bmatrix}.$$

Proof. Let $\mathcal{U} = \mathrm{col}(P)$. Identifying P with its induced operator on $\mathbb{F}^n$, Theorem 2.4.14 says P is the orthogonal projection onto $\mathcal{U}$, so P is the identity on $\mathcal{U}$ and zero on $\mathcal{U}^\perp$. Choose an orthonormal basis $\mathbf{u}_1, \ldots, \mathbf{u}_r$ of $\mathcal{U}$ and extend it to an orthonormal basis $\mathbf{u}_1, \ldots, \mathbf{u}_n$ of $\mathbb{F}^n$. Let $U = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n]$.
Then U is unitary. For $j \le r$, $P\mathbf{u}_j = \mathbf{u}_j$, and for $j > r$, $\mathbf{u}_j \in \mathcal{U}^\perp$ so $P\mathbf{u}_j = \mathbf{0}$. Therefore the jth column of U^*PU is the coordinate column of $P\mathbf{u}_j$ in this orthonormal basis, giving the displayed block diagonal form.
Structural Commentary: Orthogonal projection splits the space into range and orthogonal kernel.
Key Insight: In a basis adapted to $\mathrm{col}(P) \oplus \mathrm{col}(P)^\perp$, projection keeps the first coordinates and kills the rest. □

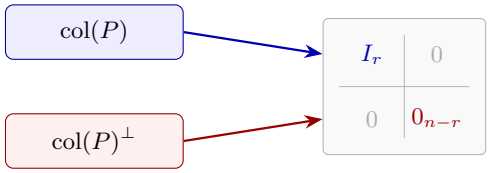


Figure 23: Adapted basis: $U^*PU = I_r \oplus 0_{n-r}$, identity on $\mathrm{col}(P)$ and zero on $\mathrm{col}(P)^\perp$.

Definition 2.4.16. *Isometry*
Let $\mathcal{V}$ be an inner-product space. An operator $T \in \mathcal{L}(\mathcal{V})$ is an *isometry* if $\|T\mathbf{v}\| = \|\mathbf{v}\|$ for every $\mathbf{v} \in \mathcal{V}$.

Proposition 2.4.17. *Basic Properties of Isometries*
Let $\mathcal{V}$ be an inner-product space and let $S, T \in \mathcal{L}(\mathcal{V})$ be isometries.

i. ST is an isometry;
ii. T is injective;
iii. if $\mathcal{V}$ is finite-dimensional, then T is invertible and T^{-1} is an isometry.

Proof.
(i) For every $\mathbf{v}$, $\|ST\mathbf{v}\| = \|T\mathbf{v}\| = \|\mathbf{v}\|$.
(ii) If $T\mathbf{v} = \mathbf{0}$, then $\|\mathbf{v}\| = \|T\mathbf{v}\| = 0$, so $\mathbf{v} = \mathbf{0}$. Thus $\ker(T) = \{\mathbf{0}\}$.
(iii) In finite dimensions, injectivity implies invertibility by Theorem 1.3.27.
If $\mathbf{u} = T\mathbf{v}$, then $\left\|T^{-1}\mathbf{u}\right\| = \|\mathbf{v}\| = \|T\mathbf{v}\| = \|\mathbf{u}\|$.

Structural Commentary: Isometries preserve length, so they cannot collapse nonzero vectors.
Key Insight: In finite dimensions, length preservation already forces an invertible geometry-preserving map. □

Theorem 2.4.18. *Isometries Preserve Inner Products*
Let $\mathcal{V}$ be an inner-product space and let $T \in \mathcal{L}(\mathcal{V})$.
Then T is an isometry if and only if $\langle T\mathbf{u}, T\mathbf{v}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle$ for every $\mathbf{u}, \mathbf{v} \in \mathcal{V}$.

Proof.

($\Rightarrow$) Suppose T is an isometry. Then every norm appearing in the polarization identities of Theorem 2.1.17 is unchanged after applying T. Therefore $\langle T\mathbf{u}, T\mathbf{v}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle$.

($\Leftarrow$) Suppose T preserves inner products. Then $\|T\mathbf{u}\|^2 = \langle T\mathbf{u}, T\mathbf{u}\rangle = \langle \mathbf{u}, \mathbf{u}\rangle = \|\mathbf{u}\|^2$, so T is an isometry.

Structural Commentary: Polarization converts preservation of norms into preservation of inner products.
Key Insight: Length preservation for all vectors also preserves angle information. □

Theorem 2.4.19. *Isometries and the Adjoint Product*
Let $\mathcal{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. Then T is an isometry if and only if $T^*T = I_{\mathcal{V}}$.

Proof.

($\Rightarrow$) Suppose T is an isometry. By Theorem 2.4.18, $\langle T^*T\mathbf{u}, \mathbf{v}\rangle = \langle T\mathbf{u}, T\mathbf{v}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle$ for all $\mathbf{u}, \mathbf{v}$. Thus $\langle (T^*T - I_{\mathcal{V}})\mathbf{u}, \mathbf{v}\rangle = 0$ for every $\mathbf{v}$, so $T^*T\mathbf{u} = \mathbf{u}$ for every $\mathbf{u}$.

($\Leftarrow$) Suppose $T^*T = I_{\mathcal{V}}$. Then $\|T\mathbf{u}\|^2 = \langle T\mathbf{u}, T\mathbf{u}\rangle = \langle T^*T\mathbf{u}, \mathbf{u}\rangle = \|\mathbf{u}\|^2$ for every $\mathbf{u}$, so T is an isometry.

Structural Commentary: The operator T^*T records inner products after applying T.
Key Insight: T is an isometry exactly when this Gram operator is the identity. □

Definition 2.4.20. *Unitary and Orthogonal*
An operator $U \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space is *unitary* if $U^*U = UU^* = I_{\mathcal{V}}$.
A matrix $U \in \mathrm{Mat}(n, n, \mathbb{F})$ is *unitary* if $U^*U = I_n$. If $\mathbb{F} = \mathbb{R}$, a unitary matrix is called *orthogonal*.

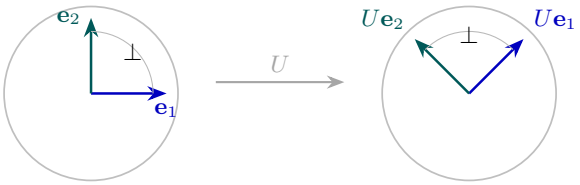


Figure 24: Unitary invariance: $\|U\mathbf{x}\| = \|\mathbf{x}\|$ and $\langle U\mathbf{x}, U\mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle$.

Theorem 2.4.21. *Equivalent Conditions for a Unitary Matrix*
Let $U = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n] \in \mathrm{Mat}(n, n, \mathbb{F})$. The following are equivalent:

i. U is unitary, i.e. $U^*U = I_n$;
ii. the columns $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are orthonormal in $\mathbb{F}^n$;
iii. $UU^* = I_n$;
iv. U^* is unitary;
v. the rows of U are orthonormal;
vi. U is invertible and $U^{-1} = U^*$;
vii. $\|U\mathbf{x}\|_2 = \|\mathbf{x}\|_2$ for every $\mathbf{x} \in \mathbb{F}^n$;
viii. $\langle U\mathbf{x}, U\mathbf{y}\rangle_{\mathbb{F}^n} = \langle \mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n}$ for every $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$.

Proof.

(i) $\Rightarrow$ (ii). Since $(U^*U)_{ij} = \mathbf{u}_i^*\mathbf{u}_j = \langle \mathbf{u}_j, \mathbf{u}_i\rangle_{\mathbb{F}^n}$, equation $U^*U = I_n$ states that columns of U are orthonormal.

(ii) $\Rightarrow$ (iii). Orthonormal columns give $U^*U = I_n$. A left inverse of a square matrix is also a right inverse by Theorem 1.3.56, so $UU^* = I_n$.

(iii) $\Rightarrow$ (iv). The equation $UU^* = I_n$ is $(U^*)^*U^* = I_n$, so U^* is unitary.

(iv) $\Rightarrow$ (v). Apply (i) $\Rightarrow$ (ii) to U^*; the columns of U^* are the conjugates of the rows of U.

(v) $\Rightarrow$ (vi). Orthonormal rows give $UU^* = I_n$. Thus U^* is the inverse of U, so $U^{-1} = U^*$.

(vi) $\Rightarrow$ (vii). If $U^{-1} = U^*$, then $U^*U = I_n$. Therefore $\|U\mathbf{x}\|_2^2 = \langle U\mathbf{x}, U\mathbf{x}\rangle_{\mathbb{F}^n} = \langle U^*U\mathbf{x}, \mathbf{x}\rangle_{\mathbb{F}^n} = \|\mathbf{x}\|_2^2$ for every $\mathbf{x} \in \mathbb{F}^n$. Since both norms are nonnegative, $\|U\mathbf{x}\|_2 = \|\mathbf{x}\|_2$.

(vii) $\Rightarrow$ ($viii$). Apply Theorem 2.4.18 to $\mathbf{x} \mapsto U\mathbf{x}$.

($viii$) $\Rightarrow$ (i). For every $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, $\langle (U^*U - I_n)\mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n} = \langle U\mathbf{x}, U\mathbf{y}\rangle_{\mathbb{F}^n} - \langle \mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n} = 0$. Taking $\mathbf{y} = (U^*U - I_n)\mathbf{x}$ shows $(U^*U - I_n)\mathbf{x} = \mathbf{0}$ for every $\mathbf{x}$, so $U^*U = I_n$.

Structural Commentary: A unitary matrix is an isometry written in orthonormal coordinates.
Key Insight: Columns, rows, inverses, norms, and inner products are different faces of the same condition. □

Proposition 2.4.22. *Algebra of Unitary Matrices*
Let $U, V \in \mathrm{Mat}(n, n, \mathbb{F})$ and $W \in \mathrm{Mat}(m, m, \mathbb{F})$, where $V \oplus W$ denotes the block diagonal matrix with blocks V and W.

i. If U and V are unitary, then UV is unitary.

ii. $V \oplus W$ is unitary if and only if V and W are unitary.

Proof.

(i) Compute $(UV)^*(UV) = V^*U^*UV = V^*V = I_n$.
(ii) Since $(V \oplus W)^*(V \oplus W) = V^*V \oplus W^*W$, this product equals I_{n+m} exactly when $V^*V = I_n$ and $W^*W = I_m$.

Structural Commentary: Unitary structure is stable under composition and blockwise orthogonal sums.
Key Insight: The equation $U^*U = I$ makes every verification a short adjoint calculation. □

Theorem 2.4.23. *Extending Isometric Columns to a Unitary Matrix*
If $X \in \mathrm{Mat}(m, n, \mathbb{F})$ and $X^*X = I_n$, then there exists $X' \in \mathrm{Mat}(m, m-n, \mathbb{F})$ such that $[X\ X']$ is unitary.

Proof. Write $X = [\mathbf{x}_1\ \cdots\ \mathbf{x}_n]$. Since $(X^*X)_{ij} = \langle \mathbf{x}_j, \mathbf{x}_i \rangle_{\mathbb{F}^m}$, the condition $X^*X = I_n$ says $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are orthonormal in $\mathbb{F}^m$. By Corollary 2.1.32, extend this list to an orthonormal basis $\mathbf{x}_1, \ldots, \mathbf{x}_m$ of $\mathbb{F}^m$.
Let $X' = [\mathbf{x}_{n+1}\ \cdots\ \mathbf{x}_m]$. Then the columns of $[X\ X']$ are orthonormal, so $[X\ X']$ is unitary.
Structural Commentary: Isometric columns are a partial orthonormal basis.
Key Insight: Complete the columns by extending the orthonormal list. □

Definition 2.4.24. *Unitary and Permutation Similarity*
Square matrices $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ are *unitarily similar* if $A = UBU^*$ for some unitary U. They are *orthogonally similar* if $\mathbb{F} = \mathbb{R}$ and U can be chosen real orthogonal.
A *permutation matrix* is a square matrix with exactly one entry equal to 1 in each row and each column, and all other entries equal to 0. Matrices are *permutation similar* if $A = PBP^*$ for some permutation matrix P.

Proposition 2.4.25. *Permutation Matrices Are Orthogonal*
If $P = [\mathbf{e}_{\sigma(1)}\ \cdots\ \mathbf{e}_{\sigma(n)}]$ is a permutation matrix, then $P^*P = PP^* = I_n$. Hence P is real orthogonal.

Proof. The columns of P are a rearrangement of the standard orthonormal basis of $\mathbb{R}^n$, hence they are orthonormal. Theorem 2.4.21 gives $P^*P = PP^* = I_n$.
Structural Commentary: A permutation matrix only reorders the standard basis.
Key Insight: Reordering orthonormal vectors preserves orthonormality. □

Definition 2.4.26. *Householder Matrix*
Let $\mathbf{w} \in \mathbb{F}^n$ be nonzero and set $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|_2$.
The *Householder matrix* associated to $\mathbf{w}$ is $U_{\mathbf{w}} := I_n - 2P_{\mathbf{u}} = I_n - 2\frac{\mathbf{w}\mathbf{w}^*}{\|\mathbf{w}\|_2^2}$.
It acts by $\mathbf{x} \mapsto U_{\mathbf{w}}\mathbf{x} = \mathbf{x} - 2\langle \mathbf{x}, \mathbf{u} \rangle_{\mathbb{F}^n} \mathbf{u}$.

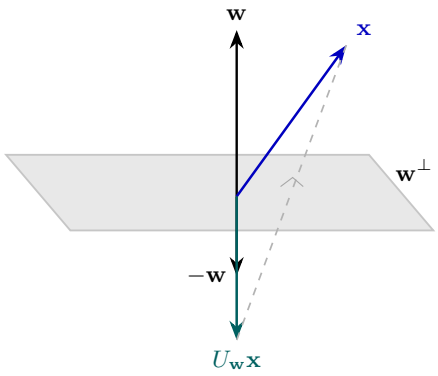


Figure 25: A Householder matrix fixes $\mathbf{w}^\perp$ and flips the span($\mathbf{w}$) component.

Theorem 2.4.27. *Householder Matrices Are Unitary Reflections*
Every Householder matrix $U_{\mathbf{w}}$ is unitary, Hermitian, and involutive: $U_{\mathbf{w}}^*U_{\mathbf{w}} = I_n$, $U_{\mathbf{w}}^* = U_{\mathbf{w}}$, $U_{\mathbf{w}}^2 = I_n$.
If $\mathbf{w} \in \mathbb{R}^n$, then $U_{\mathbf{w}}$ is real orthogonal and symmetric.

Proof. Let $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|_2$. By Proposition 2.2.11, $P_{\mathbf{u}}^* = P_{\mathbf{u}}$ and $P_{\mathbf{u}}^2 = P_{\mathbf{u}}$.
Therefore $U_{\mathbf{w}}^* = I_n - 2P_{\mathbf{u}} = U_{\mathbf{w}}$, $U_{\mathbf{w}}^2 = (I_n - 2P_{\mathbf{u}})^2 = I_n - 4P_{\mathbf{u}} + 4P_{\mathbf{u}}^2 = I_n$.
Thus $U_{\mathbf{w}}^*U_{\mathbf{w}} = U_{\mathbf{w}}^2 = I_n$, so $U_{\mathbf{w}}$ is unitary. The real case is the same statement with conjugation removed.
Structural Commentary: A Householder matrix reflects across an orthogonal hyperplane.
Key Insight: The identity $P_{\mathbf{u}}^2 = P_{\mathbf{u}}$ makes $(I - 2P_{\mathbf{u}})^2 = I$. □

Theorem 2.4.28. *Householder Map Between Equal-Norm Vectors*
Let $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$ *satisfy* $\|\mathbf{x}\|_2 = \|\mathbf{y}\|_2 \neq 0$. *Define* $\sigma = \begin{cases} 1, & \langle \mathbf{x}, \mathbf{y} \rangle_{\mathbb{F}^n} = 0, \\ -\dfrac{\overline{\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbb{F}^n}}}{|\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbb{F}^n}|}, & \langle \mathbf{x}, \mathbf{y} \rangle_{\mathbb{F}^n} \neq 0, \end{cases}$ *where* $\mathbf{w} = \mathbf{y} - \sigma \mathbf{x}$.
Then $\mathbf{w} \neq \mathbf{0}$, $|\sigma| = 1$, *and* $\sigma U_{\mathbf{w}} \mathbf{x} = \mathbf{y}$.

Proof. Clearly $|\sigma| = 1$. Let $z = \langle \mathbf{x}, \mathbf{y} \rangle$ and $r = \|\mathbf{x}\|_2 = \|\mathbf{y}\|_2$. By construction, $\sigma z = -|z|$, with the convention that this equals 0 when $z = 0$. Hence $\|\mathbf{w}\|_2^2 = \|\mathbf{y} - \sigma \mathbf{x}\|_2^2 = 2r^2 - \sigma z - \overline{\sigma z} = 2\left(r^2 + |z|\right) > 0$, so $\mathbf{w} \neq \mathbf{0}$.
Since $\mathbf{y} = \mathbf{w} + \sigma \mathbf{x}$, $\sigma U_{\mathbf{w}} \mathbf{x} = \sigma \mathbf{x} - 2\sigma \frac{\langle \mathbf{x}, \mathbf{w} \rangle}{\|\mathbf{w}\|_2^2} \mathbf{w} = \mathbf{y} - \left(1 + 2\sigma \frac{\langle \mathbf{x}, \mathbf{w} \rangle}{\|\mathbf{w}\|_2^2}\right) \mathbf{w}$.
But $\langle \mathbf{x}, \mathbf{w} \rangle = z - \overline{\sigma} r^2$, so $2\sigma \langle \mathbf{x}, \mathbf{w} \rangle = 2\sigma z - 2r^2 = -2(r^2 + |z|) = -\|\mathbf{w}\|_2^2$.
Coefficient of $\mathbf{w}$ is hence zero, proving $\sigma U_{\mathbf{w}} \mathbf{x} = \mathbf{y}$.
Structural Commentary: The scalar σ chooses the phase so that $\mathbf{y} - \sigma \mathbf{x}$ gives a stable reflection direction.
Key Insight: Householder reflections connect equal-norm vectors up to a unit scalar. □

Corollary 2.4.29. *Householder Unitary Annihilator*
Let $\mathbf{x} = [x_1 \ \cdots \ x_n]^t \in \mathbb{F}^n$ *be nonzero. Define* $\sigma = \begin{cases} 1, & x_1 = 0, \\ -\dfrac{\overline{x_1}}{|x_1|}, & x_1 \neq 0, \end{cases}$ *where* $\mathbf{w} = \|\mathbf{x}\|_2 \mathbf{e}_1 - \sigma \mathbf{x}$.
Then $\sigma U_{\mathbf{w}} \mathbf{x} = \|\mathbf{x}\|_2 \mathbf{e}_1$. *If* $\mathbf{x}$ *is a unit vector, then* $\overline{\sigma} U_{\mathbf{w}}$ *is unitary and has* $\mathbf{x}$ *as its first column.*

Proof. Apply Theorem 2.4.28 with $\mathbf{y} = \|\mathbf{x}\|_2 \mathbf{e}_1$, noting that $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 \|\mathbf{x}\|_2$. If $\|\mathbf{x}\|_2 = 1$, then $(\sigma U_{\mathbf{w}})\mathbf{x} = \mathbf{e}_1$, so $(\sigma U_{\mathbf{w}})^* \mathbf{e}_1 = \mathbf{x}$. Since $(\sigma U_{\mathbf{w}})^* = \overline{\sigma} U_{\mathbf{w}}$, this says the first column of $\overline{\sigma} U_{\mathbf{w}}$ is $\mathbf{x}$.
Structural Commentary: The reflector concentrates all mass of $\mathbf{x}$ into the first coordinate.
Key Insight: Householder matrices produce explicit unitary matrices with a prescribed first column. □

3 Spectral and Operator Structure

Spectral and operator structure studies linear maps through invariant subspaces, polynomial relations, canonical forms, spectral theorems, and operator functions.

3.1 Eigenstructure and Invariant Subspaces

Spectral linear algebra understands an operator through subspaces it preserves. Eigenstructure, triangularization, diagonalization, and commuting families all refine this invariant-subspace viewpoint.

Definition 3.1.1. *Eigenvalue and Eigenvector*
Let $T \in \mathcal{L}(\mathcal{V})$. A scalar $\lambda \in \mathbb{F}$ is an *eigenvalue* of T if there exists $\mathbf{v} \in \mathcal{V}$ with $\mathbf{v} \neq \mathbf{0}$ and $T\mathbf{v} = \lambda \mathbf{v}$. Such a vector $\mathbf{v}$ is an *eigenvector* of T corresponding to λ.

Definition 3.1.2. *Eigenspace*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\lambda \in \mathbb{F}$. The *eigenspace* of T corresponding to λ is $E_\lambda(T) := \ker(T - \lambda I_{\mathcal{V}})$. Thus λ is an eigenvalue of T if and only if $E_\lambda(T) \neq \{\mathbf{0}\}$. Equivalently, $E_\lambda(T)$ is the nullspace of the shifted operator $T - \lambda I_{\mathcal{V}}$.

Definition 3.1.3. *Spectrum and Geometric Multiplicity*
Let $T \in \mathcal{L}(\mathcal{V})$. The *spectrum* of T is $\operatorname{spec}(T) := \{\lambda \in \mathbb{F} : \lambda \text{ is an eigenvalue of } T\}$.
The *geometric multiplicity* of $\lambda \in \mathbb{F}$ is $\dim E_\lambda(T)$. For a square matrix $A \in \operatorname{Mat}(n, n, \mathbb{F})$, $E_\lambda(A) := \ker(A - \lambda I_n)$ and $\operatorname{spec}(A)$ denotes the spectrum of the induced operator on $\mathbb{F}^n$.

Proposition 3.1.4. *Eigenvectors as Kernel Vectors*
Let $T \in \mathcal{L}(\mathcal{V})$, $\lambda \in \mathbb{F}$. *Vector* $\mathbf{v}$ *is eigenvector of* T *corresponding to* λ *if and only if* $\mathbf{v} \in \ker(T - \lambda I_{\mathcal{V}})$, $\mathbf{v} \neq \mathbf{0}$.

Proof.
($\Rightarrow$) If $\mathbf{v}$ is an eigenvector corresponding to λ, then $\mathbf{v} \neq \mathbf{0}$ and $T\mathbf{v} = \lambda \mathbf{v}$, so $(T - \lambda I_{\mathcal{V}})\mathbf{v} = \mathbf{0}$.
($\Leftarrow$) If $\mathbf{v} \in \ker(T - \lambda I_{\mathcal{V}})$ and $\mathbf{v} \neq \mathbf{0}$, then $(T - \lambda I_{\mathcal{V}})\mathbf{v} = \mathbf{0}$, so $T\mathbf{v} = \lambda \mathbf{v}$.
Structural Commentary: Eigenvectors are kernels of shifted operators.
Key Insight: Subtract $\lambda I_{\mathcal{V}}$ to convert $T\mathbf{v} = \lambda \mathbf{v}$ into homogeneous form. □

Theorem 3.1.5. *Equivalent Conditions for an Eigenvalue*
Let $\mathcal{V}$ *be finite-dimensional, let* $T \in \mathcal{L}(\mathcal{V})$, *and let* $\lambda \in \mathbb{F}$. *The following are equivalent:*

(i) λ *is an eigenvalue of* T;
(ii) $T - \lambda I_{\mathcal{V}}$ *is not injective;*

(iii) $T - \lambda I_{\mathcal{V}}$ *is not surjective;*
(iv) $T - \lambda I_{\mathcal{V}}$ *is not invertible.*

Proof.

$(i) \Rightarrow (ii)$ If λ is an eigenvalue, Proposition 3.1.4 gives nonzero vector in $\ker(T-\lambda I_{\mathcal{V}})$, so $T-\lambda I_{\mathcal{V}}$ not injective.
$(ii) \Rightarrow (i)$ If $T - \lambda I_{\mathcal{V}}$ not injective, then $\ker(T - \lambda I_{\mathcal{V}})$ contains nonzero vector, so λ is eigenvalue (Prop 3.1.4).
$(ii) \Rightarrow (iii)$ If $T - \lambda I_{\mathcal{V}}$ were surjective, Theorem 1.3.27 would make it injective, a contradiction.
$(iii) \Rightarrow (iv)$ If $T - \lambda I_{\mathcal{V}}$ were invertible, then it would be surjective, a contradiction.
$(iv) \Rightarrow (ii)$ If $T - \lambda I_{\mathcal{V}}$ were injective, Theorem 1.3.27 would make it invertible, a contradiction.

Structural Commentary: In finite dimensions, noninvertibility has equivalent forms.
Key Insight: A scalar λ is spectral precisely when $T - \lambda I_{\mathcal{V}}$ has a nontrivial kernel. □

Theorem 3.1.6. *Eigenvectors for Distinct Eigenvalues Are Independent*
Let $T \in \mathcal{L}(\mathcal{V})$. Suppose $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T, and let $\mathbf{v}_1, \ldots, \mathbf{v}_m$ be corresponding eigenvectors. Then $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent.

Proof. Suppose the list is linearly dependent. Let k be smallest positive integer s.t. $\mathbf{v}_k \in \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$. Then $k \geq 2$, $(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$ is linearly independent, and $\exists$ scalars $a_1, \ldots, a_{k-1}$ s.t. $\mathbf{v}_k = a_1\mathbf{v}_1 + \cdots + a_{k-1}\mathbf{v}_{k-1}$. Applying T gives $\lambda_k\mathbf{v}_k = a_1\lambda_1\mathbf{v}_1 + \cdots + a_{k-1}\lambda_{k-1}\mathbf{v}_{k-1}$. Multiplying first displayed equation by λ_k and subtracting gives $0 = a_1(\lambda_k - \lambda_1)\mathbf{v}_1 + \cdots + a_{k-1}(\lambda_k - \lambda_{k-1})\mathbf{v}_{k-1}$. Since $\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}$ are linearly independent and $\lambda_k \neq \lambda_i$ for $i < k$, all $a_i = 0$. Hence $\mathbf{v}_k = \mathbf{0}$, contradicting that $\mathbf{v}_k$ is an eigenvector.
Structural Commentary: A smallest dependence isolates the first new eigenvector that depends on earlier ones. Applying T and subtracting separates the distinct eigenvalue factors.
Key Insight: Distinct eigenvalues make the same dependence relation impossible after applying T. □

Corollary 3.1.7. *Number of Distinct Eigenvalues*
If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$, then T has at most $\dim(\mathcal{V})$ distinct eigenvalues.

Proof. If T had more than $\dim(\mathcal{V})$ distinct eigenvalues, choose $m > \dim(\mathcal{V})$ of them, one eigenvector for each. By Theorem 3.1.6, these eigenvectors form linearly independent list of length m, contradicting Theorem 1.1.21.
Structural Commentary: Distinct eigenvalues give independent directions.
Key Insight: Count one eigenvector per eigenspace. □

Corollary 3.1.8. *Invertibility and the Zero Eigenvalue*
Let $\mathcal{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. Then T is invertible if and only if $0 \notin \operatorname{spec}(T)$.

Proof. Apply Theorem 3.1.5 with $\lambda = 0$. Then 0 is an eigenvalue exactly when $T - 0I_{\mathcal{V}} = T$ is not invertible.
Structural Commentary: The zero eigenvalue is the spectral form of singularity.
Key Insight: Invertibility is equivalent to having trivial kernel, and $E_0(T) = \ker T$. □

Corollary 3.1.9. *Matrix Invertibility and the Spectrum*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$. Then A is invertible if and only if $0 \notin \operatorname{spec}(A)$.

Proof.

$(\Rightarrow)$ If A is invertible, then $\ker(A) = \{\mathbf{0}\}$, so 0 is not an eigenvalue of A.
$(\Leftarrow)$ If $0 \notin \operatorname{spec}(A)$, then $\ker(A) = \{\mathbf{0}\}$. Corollary 1.3.48 gives full rank, hence A is invertible.

Structural Commentary: Matrix invertibility is the coordinate form of trivial kernel.
Key Insight: A zero eigenvalue is exactly a nonzero solution of $A\mathbf{x} = \mathbf{0}$. □

Definition 3.1.10. *Invariant Subspace*
Let $T \in \mathcal{L}(\mathcal{V})$. A subspace $\mathcal{U}$ of $\mathcal{V}$ is *invariant under T* if $T\mathbf{u} \in \mathcal{U}$ for every $\mathbf{u} \in \mathcal{U}$.

Equivalently, $\mathcal{U}$ is invariant under T exactly when the restriction $T|_{\mathcal{U}}$ is an operator on $\mathcal{U}$.

Proposition 3.1.11. *One-Dimensional Invariant Subspaces and Eigenvectors*
Let $T \in \mathcal{L}(\mathcal{V})$, $\mathbf{v} \neq \mathbf{0}$. Then $\operatorname{span}(\mathbf{v})$ is invariant under T if and only if there exists $\lambda \in \mathbb{F}$ such that $T\mathbf{v} = \lambda\mathbf{v}$.

Proof.

$(\Rightarrow)$ If $\operatorname{span}(\mathbf{v})$ is invariant under T, then $T\mathbf{v} \in \operatorname{span}(\mathbf{v})$, so $T\mathbf{v} = \lambda\mathbf{v}$ for some $\lambda \in \mathbb{F}$.
$(\Leftarrow)$ If $T\mathbf{v} = \lambda\mathbf{v}$ and $\mathbf{u} = a\mathbf{v} \in \operatorname{span}(\mathbf{v})$, then $T\mathbf{u} = aT\mathbf{v} = a\lambda\mathbf{v} \in \operatorname{span}(\mathbf{v})$.

Structural Commentary: One-dimensional invariance means rescaling one direction.
Key Insight: In dimension one, staying inside the subspace means being a scalar multiple of the same vector. □

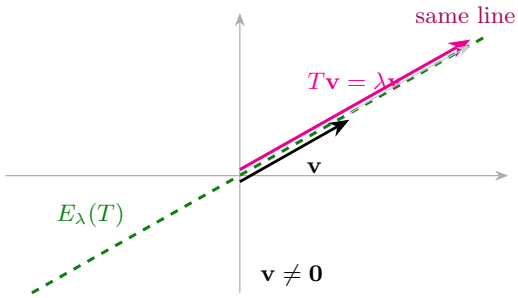


Figure 26: An eigendirection is a one-dimensional invariant subspace.

Proposition 3.1.12. *Invariant Subspaces and Block Form*
Let $T \in \mathcal{L}(\mathcal{V})$ and suppose $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.

(i) If $\mathcal{U}$ is invariant under T, then in a basis adapted to $\mathcal{U} \oplus \mathcal{W}$, matrix of T has upper block-triangular form

$$\begin{bmatrix} A & B \\ 0 & D \end{bmatrix}.$$

(ii) If both $\mathcal{U}$ and $\mathcal{W}$ are invariant under T, then the matrix of T has the block-diagonal form

$$\begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix}.$$

Proof. Choose basis $(\mathbf{u}_1, \ldots, \mathbf{u}_p)$ of $\mathcal{U}$ and basis $(\mathbf{w}_1, \ldots, \mathbf{w}_q)$ of $\mathcal{W}$, and use concatenation as basis of $\mathcal{V}$.

(i) If $\mathcal{U}$ is invariant, then each $T\mathbf{u}_j$ lies in $\mathcal{U}$, so its coordinates along $\mathbf{w}_1, \ldots, \mathbf{w}_q$ are zero. These entries form the lower-left block.

(ii) If $\mathcal{W}$ is also invariant, then each $T\mathbf{w}_j$ lies in $\mathcal{W}$, so its coordinates along $\mathbf{u}_1, \ldots, \mathbf{u}_p$ are zero. These entries form the upper-right block.

Structural Commentary: Block form records how an operator moves vectors between chosen summands. Invariant summands force the corresponding off-diagonal movement to vanish.
Key Insight: Zero blocks are coordinate evidence of invariant subspaces. □

Perspective 3.1.13. *Matrix Viewpoint on Structure*
Basic block matrices already capture important geometry. Upper block-triangular form records the presence of an invariant subspace; block diagonal form records a genuine decomposition into independent invariant pieces.

Definition 3.1.14. *Diagonal and Upper-Triangular Matrices*
The *diagonal* of a square matrix is its upper-left to lower-right entries. A square matrix is *diagonal* if all off-diagonal entries are zero, and *upper triangular* if all entries below the diagonal are zero.

Theorem 3.1.15. *Invariant Subspaces, Blocks, and Projections*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$, let $\mathcal{U}$ be an r-dimensional subspace of $\mathbb{F}^n$ with $1 \leq r \leq n-1$, and let $Q = [Q_1\ Q_2]$ be unitary, where the columns of Q_1 and Q_2 are orthonormal bases of $\mathcal{U}$ and $\mathcal{U}^\perp$, respectively. Let $P = Q_1 Q_1^*$ be the orthogonal projection onto $\mathcal{U}$. The following are equivalent:

(i) $\mathcal{U}$ is invariant under A;

(ii) $Q^*AQ = \begin{bmatrix} B & X \\ 0 & C \end{bmatrix}$ for some compatible matrices B, X, C;

(iii) $PAP = AP$.

Proof. Let $M = Q^*AQ = \begin{bmatrix} B & X \\ Y & C \end{bmatrix}$ and $E = Q^*PQ = I_r \oplus 0_{n-r}$. Since Q is unitary and $P = Q_1 Q_1^*$, the matrix of P in the Q-coordinates is E. Also $Y = Q_2^* A Q_1$, and $\ker(Q_2^*) = \mathrm{col}(Q_1) = \mathcal{U}$.

(i) $\Rightarrow$ *(ii)* If $A\mathcal{U} \subseteq \mathcal{U}$, then each column of AQ_1 lies in $\mathcal{U} = \ker(Q_2^*)$, so $Y = Q_2^* A Q_1 = 0$.

(ii) $\Rightarrow$ *(iii)* If $Y = 0$, then $EME = \begin{bmatrix} B & 0 \\ 0 & 0 \end{bmatrix}$ and $ME = \begin{bmatrix} B & 0 \\ 0 & 0 \end{bmatrix}$.
Thus $Q^*(PAP)Q = EME = ME = Q^*(AP)Q$, so $PAP = AP$.

(iii) $\Rightarrow$ *(i)* If $PAP = AP$, then $EME = ME$. Since $EME = \begin{bmatrix} B & 0 \\ 0 & 0 \end{bmatrix}$ and $ME = \begin{bmatrix} B & 0 \\ Y & 0 \end{bmatrix}$, we get $Y = 0$.
Hence the columns of AQ_1 lie in $\ker(Q_2^*) = \mathcal{U}$. Since the columns of Q_1 span $\mathcal{U}$, linearity gives $A\mathcal{U} \subseteq \mathcal{U}$.

Structural Commentary: In adapted coordinates, the lower-left block is the component of $A\mathcal{U}$ in $\mathcal{U}^\perp$.
Key Insight: $PAP = AP$ exactly kills this component. □

Corollary 3.1.16. *Reducing Subspaces and Commuting Projections*
Maintain the notation of Theorem 3.1.15. The following are equivalent:

(i) $\mathcal{U}$ *is invariant under both* A *and* A^**;*
(ii) $Q^*AQ = B \oplus C$ *for some* $B \in \mathrm{Mat}(r, r, \mathbb{F})$ *and* $C \in \mathrm{Mat}(n-r, n-r, \mathbb{F})$*;*
(iii) $PA = AP$*.*

Proof. Let $M = Q^*AQ = \begin{bmatrix} B & X \\ Y & C \end{bmatrix}$ and $E = Q^*PQ = I_r \oplus 0_{n-r}$.
Then $Q^*A^*Q = M^*$, $Q^*(PA)Q = EM$, and $Q^*(AP)Q = ME$.

(i) $\Rightarrow$ *(ii)* By Theorem 3.1.15, A-invariance gives $Y = 0$. Applying the same theorem to A^* gives that the lower-left block of M^* is zero, namely $X^* = 0$, so $X = 0$. Thus $M = B \oplus C$.

(ii) $\Rightarrow$ *(iii)* If $M = B \oplus C$, then $EM = ME$, hence $PA = AP$.

(iii) $\Rightarrow$ *(i)* If $PA = AP$, then $EM = ME$. Since $EM = \begin{bmatrix} B & X \\ 0 & 0 \end{bmatrix}$ and $ME = \begin{bmatrix} B & 0 \\ Y & 0 \end{bmatrix}$, we get $X = Y = 0$. The equation $Y = 0$ gives A-invariance by Theorem 3.1.15. Since the lower-left block of M^* is $X^* = 0$, applying the same theorem to A^* gives A^*-invariance.

Structural Commentary: Reducing subspaces are exactly block diagonal decompositions in adapted coordinates.
Key Insight: Commutation with P kills both off-diagonal blocks. □

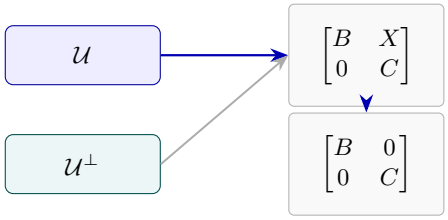


Figure 27: Adapted coordinates: invariance gives upper block form; $PA = AP$ gives block diagonal form.

Definition 3.1.17. *Restriction and Quotient Operators*
Let $T \in \mathcal{L}(\mathcal{V})$ and let $\mathcal{U}$ be invariant under T.

(i) The *restriction operator* $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ is defined by $T|_{\mathcal{U}}(\mathbf{u}) = T\mathbf{u}$ for $\mathbf{u} \in \mathcal{U}$.
(ii) The *quotient operator* $T/\mathcal{U} \in \mathcal{L}(\mathcal{V}/\mathcal{U})$ is defined by $(T/\mathcal{U})(\mathbf{v} + \mathcal{U}) = T\mathbf{v} + \mathcal{U}$, $\mathbf{v} \in \mathcal{V}$.

Proposition 3.1.18. *Restriction and Quotient Operators Are Well-Defined*
Let $T \in \mathcal{L}(\mathcal{V})$ *and let* $\mathcal{U}$ *be invariant under* T*. Then* $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ *and* $T/\mathcal{U} \in \mathcal{L}(\mathcal{V}/\mathcal{U})$*.*

Proof. Since $\mathcal{U}$ is invariant under T, $T\mathbf{u} \in \mathcal{U}$ whenever $\mathbf{u} \in \mathcal{U}$, so $T|_{\mathcal{U}}$ maps $\mathcal{U}$ into itself. Its linearity is inherited from T. For the quotient operator, suppose $\mathbf{v} + \mathcal{U} = \mathbf{w} + \mathcal{U}$. Then $\mathbf{v} - \mathbf{w} \in \mathcal{U}$.
Invariance gives $T(\mathbf{v} - \mathbf{w}) \in \mathcal{U}$, so $T\mathbf{v} - T\mathbf{w} \in \mathcal{U}$ and hence $T\mathbf{v} + \mathcal{U} = T\mathbf{w} + \mathcal{U}$. Thus $T/\mathcal{U}$ is well-defined.
For $a \in \mathbb{F}$, $(T/\mathcal{U})(a(\mathbf{v} + \mathcal{U}) + (\mathbf{w} + \mathcal{U})) = T(a\mathbf{v} + \mathbf{w}) + \mathcal{U} = a(T\mathbf{v} + \mathcal{U}) + (T\mathbf{w} + \mathcal{U})$, so $T/\mathcal{U}$ is linear.
Structural Commentary: Restriction stays inside $\mathcal{U}$; quotient collapses $\mathcal{U}$.
Key Insight: Invariance makes both operators well-defined. □

Perspective 3.1.19. *Two Smaller Views of an Operator*
When $\mathcal{U}$ is invariant under T, the operators $T|_{\mathcal{U}}$ and $T/\mathcal{U}$ split attention between what happens inside $\mathcal{U}$ and what remains after $\mathcal{U}$ is collapsed. These two views often simplify an operator, even though they need not determine the whole operator by themselves.

Theorem 3.1.20. *Complex Operators Have Eigenvalues*
Every operator on a finite-dimensional, nonzero complex vector space has an eigenvalue.

Proof. Let $T \in \mathcal{L}(\mathcal{V})$. By Proposition 3.2.6, T has a nonconstant annihilating polynomial. Since $\mathcal{V}$ is complex and nonzero, Theorem 3.2.40 implies that one root of this polynomial is an eigenvalue of T.
Structural Commentary: Finite dimension gives an annihilating polynomial; over $\mathbb{C}$, it factors linearly.
Key Insight: One noninjective factor produces an eigenvector. □

Definition 3.1.21. *Commutant*
For $\emptyset \neq \mathcal{F} \subseteq \mathrm{Mat}(n, n, \mathbb{F})$, the *commutant* of $\mathcal{F}$ is $\mathrm{Comm}(\mathcal{F}) := \{X \in \mathrm{Mat}(n, n, \mathbb{F}) : XA = AX \ \forall A \in \mathcal{F}\}$.
For a single matrix A, write $\mathrm{Comm}(A)$.

Lemma 3.1.22. *Commuting with a Block Scalar Diagonal Matrix*
Let $\lambda_1, \dots, \lambda_k \in \mathbb{F}$ and let $\Lambda = \lambda_1 I_{n_1} \oplus \cdots \oplus \lambda_k I_{n_k}$.
Partition $A = [A_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$ conformally with Λ, so $A_{ij} \in \mathrm{Mat}(n_i, n_j, \mathbb{F})$.

(i) If A is block diagonal, then $A\Lambda = \Lambda A$.
(ii) If $A\Lambda = \Lambda A$ and $\lambda_1, \dots, \lambda_k$ are distinct, then A is block diagonal.

Proof. Block multiplication gives $(A\Lambda)_{ij} = \lambda_j A_{ij}$ and $(\Lambda A)_{ij} = \lambda_i A_{ij}$. If A is block diagonal, then $A_{ij} = 0$ for $i \neq j$, and these two expressions agree for every block. Conversely, if $A\Lambda = \Lambda A$, then $(\lambda_j - \lambda_i)A_{ij} = 0$ for every i, j. Distinctness of the λ_i forces $A_{ij} = 0$ whenever $i \neq j$, so A is block diagonal.
Structural Commentary: Distinct scalar blocks force commuting matrices to respect the block decomposition.
Key Insight: Distinct block scalars make every off-diagonal block visible and force it to vanish. □

Example 3.1.23. *Distinctness Is Necessary*
The distinctness hypothesis in Lemma 3.1.22 cannot be dropped. If $\lambda_1 = \cdots = \lambda_k = 1$, then $\Lambda = I_n$ commutes with every matrix in $\mathrm{Mat}(n, n, \mathbb{F})$, including matrices that are not block diagonal for the chosen partition.

Theorem 3.1.24. *Common Eigenvector for a Commuting Family*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, and let $\mathcal{F} \subseteq \mathcal{L}(\mathcal{V})$ be a nonempty commuting family: $ST = TS$ for all $S, T \in \mathcal{F}$. Then there exists $\mathbf{v} \neq \mathbf{0}$ that is an eigenvector of every operator in $\mathcal{F}$.

Proof. First suppose $\mathcal{F} = \{T_1, \dots, T_m\}$ is finite, and induct on m. The case $m = 1$ is Theorem 3.1.20.
For $m > 1$, choose $\lambda \in \mathrm{spec}(T_1)$ and set $\mathcal{U} = E_\lambda(T_1)$. Then $\mathcal{U} \neq \{\mathbf{0}\}$.
If $\mathbf{u} \in \mathcal{U}$ and $j \geq 2$, commutation gives $(T_1 - \lambda I_\mathcal{V})T_j\mathbf{u} = T_j(T_1 - \lambda I_\mathcal{V})\mathbf{u} = \mathbf{0}$, so $\mathcal{U}$ is invariant under $T_2, \dots, T_m$.
Their restrictions to $\mathcal{U}$ still commute, and by induction have a common eigenvector $\mathbf{v} \in \mathcal{U}$, $\mathbf{v} \neq \mathbf{0}$.
Since $\mathbf{v} \in E_\lambda(T_1)$, it is also an eigenvector of T_1. For general $\mathcal{F}$, the subspace $\mathrm{span}(\mathcal{F})$ of $\mathcal{L}(\mathcal{V})$ is finite-dimensional. Choose a basis $T_1, \dots, T_m$ of this span from the spanning set $\mathcal{F}$.
The finite case gives a nonzero $\mathbf{v}$ and scalars μ_j with $T_j\mathbf{v} = \mu_j\mathbf{v}$. If $S \in \mathcal{F}$, write $S = \sum_j a_j T_j$.
Then $S\mathbf{v} = (\sum_j a_j\mu_j)\mathbf{v}$, so $\mathbf{v}$ is an eigenvector of every member of $\mathcal{F}$.
Structural Commentary: Commutation makes one eigenspace invariant, reducing the family by one.
Key Insight: Find the common eigenvector inside an eigenspace. □

Theorem 3.1.25. *Criteria for an Upper-Triangular Operator Matrix*
Let $T \in \mathcal{L}(\mathcal{V})$ and let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$. The following are equivalent:

(i) $[T]_{\beta \leftarrow \beta}$ is upper triangular;
(ii) $T\mathbf{v}_j \in \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ for each $j = 1, \dots, n$;
(iii) $\mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ is invariant under T for each $j = 1, \dots, n$.

Proof.

(i) ⇒ *(ii)* The jth column of $[T]_{\beta \leftarrow \beta}$ is the coordinate column of $T\mathbf{v}_j$ in the basis β. Since the matrix is upper triangular, this column has zero entries below position j, so $T\mathbf{v}_j \in \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$.
(ii) ⇒ *(i)* If $T\mathbf{v}_j \in \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ for every j, then the jth coordinate column of $T\mathbf{v}_j$ has zero entries below position j. Hence $[T]_{\beta \leftarrow \beta}$ is upper triangular.
(ii) ⇒ *(iii)* If $\mathbf{u} \in \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$, then linearity gives $T\mathbf{u}$ as a linear combination of $T\mathbf{v}_1, \dots, T\mathbf{v}_j$, all of which lie in $\mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$. Hence this span is invariant.
(iii) ⇒ *(ii)* Apply invariance of $\mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ to $\mathbf{v}_j$.

Structural Commentary: Upper triangularity records an invariant flag.
Key Insight: Columns below the diagonal vanish exactly when $T\mathbf{v}_j$ uses only earlier basis vectors and itself. □

Corollary 3.1.26. *Upper-Triangular Form in an Orthonormal Basis*
Let $\mathcal{V}$ be a finite-dimensional inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. If T has an upper-triangular matrix with respect to some basis of $\mathcal{V}$, then T has an upper-triangular matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof. Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis for which $[T]_{\beta \leftarrow \beta}$ is upper triangular. By Theorem 3.1.25, each subspace

$$\mathcal{U}_j = \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j), \qquad j = 1, \dots, n,$$

is invariant under T. Apply Gram–Schmidt to β and obtain an orthonormal basis $\gamma = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ with

$$\mathrm{span}(\mathbf{e}_1, \dots, \mathbf{e}_j) = \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_j) = \mathcal{U}_j.$$

Thus each $\mathrm{span}(\mathbf{e}_1, \dots, \mathbf{e}_j)$ is invariant under T, and Theorem 3.1.25 shows that $[T]_{\gamma \leftarrow \gamma}$ is upper triangular.
Structural Commentary: Upper triangularity is preservation of a nested flag of subspaces. Gram–Schmidt preserves that flag while making the basis orthonormal.
Key Insight: Triangular form depends on the flag, not on the particular basis used inside the flag. □

Theorem 3.1.27. *Complex Upper Triangularisation*
Let $\mathcal{V}$ be a finite-dimensional complex vector space and let $T \in \mathcal{L}(\mathcal{V})$. Then T has an upper-triangular matrix with respect to some basis of $\mathcal{V}$.

Proof. Induct on $n = \dim(\mathcal{V})$. If $n \leq 1$, the result is immediate.
Suppose $n > 1$ and the result holds in smaller dimensions. Choose an eigenvector $\mathbf{v}_1$ of T by Theorem 3.1.20, and set $\mathcal{U} = \operatorname{span}(\mathbf{v}_1)$. Then $\mathcal{U}$ is T-invariant, so the quotient operator $T/\mathcal{U}$ is defined on $\mathcal{V}/\mathcal{U}$, which has dimension $n-1$. By induction, choose a basis $(\mathbf{v}_2 + \mathcal{U}, \ldots, \mathbf{v}_n + \mathcal{U})$ of $\mathcal{V}/\mathcal{U}$ with respect to which $T/\mathcal{U}$ is upper triangular, and choose representatives $\mathbf{v}_2, \ldots, \mathbf{v}_n \in \mathcal{V}$.
The lifted list $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$: any relation descends to the quotient basis, forcing all coefficients of $\mathbf{v}_2, \ldots, \mathbf{v}_n$ to be zero, and then the coefficient of $\mathbf{v}_1$ is zero. Now $T\mathbf{v}_1 \in \operatorname{span}(\mathbf{v}_1)$.
For $j \geq 2$, upper triangularity of $T/\mathcal{U}$ gives $T\mathbf{v}_j + \mathcal{U} = (T/\mathcal{U})(\mathbf{v}_j + \mathcal{U}) \in \operatorname{span}(\mathbf{v}_2 + \mathcal{U}, \ldots, \mathbf{v}_j + \mathcal{U})$, so $T\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_j)$. Hence Theorem 3.1.25 gives an upper-triangular matrix for T in the basis $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$.
Structural Commentary: Lift an upper-triangular quotient basis above an eigenline.
Key Insight: The quotient builds the remaining invariant flag. □

Proposition 3.1.28. *Invertibility from an Upper-Triangular Matrix*
Suppose $T \in \mathcal{L}(\mathcal{V})$ has an upper-triangular matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$ with respect to some basis of $\mathcal{V}$. Then T is invertible if and only if $\lambda_j \neq 0$ for every j.

Proof. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be such a basis.

($\Leftarrow$) If every $\lambda_j \neq 0$, then upper-triangular matrix $[T]_{\beta \leftarrow \beta}$ is invertible by Theorem 1.3.82, so T is invertible.
($\Rightarrow$) Suppose T is invertible. If $\lambda_j = 0$ for some j, set $\mathcal{U}_j = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_j)$ and $\mathcal{U}_0 = \{\mathbf{0}\}$. Upper triangularity gives $T(\mathcal{U}_j) \subseteq \mathcal{U}_{j-1}$ because the coefficient of $\mathbf{v}_j$ in $T\mathbf{v}_j$ is $\lambda_j = 0$ and $T\mathbf{v}_i \in \mathcal{U}_{j-1}$ for $i < j$.
Since $\dim(\mathcal{U}_j) > \dim(\mathcal{U}_{j-1})$, the map $\mathcal{U}_j \to \mathcal{U}_{j-1}$ induced by T is not injective. Thus there is a nonzero $\mathbf{u} \in \mathcal{U}_j$ with $T\mathbf{u} = \mathbf{0}$, contradicting invertibility. Hence every $\lambda_j \neq 0$.

Structural Commentary: Diagonal entries control each flag layer.
Key Insight: A zero diagonal entry forces a kernel vector. □

Proposition 3.1.29. *Eigenvalues from an Upper-Triangular Matrix*
Suppose $T \in \mathcal{L}(\mathcal{V})$ has an upper-triangular matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$ with respect to some basis of $\mathcal{V}$. Then the eigenvalues of T are precisely $\lambda_1, \ldots, \lambda_n$.

Proof. For $\lambda \in \mathbb{F}$, the matrix of $T - \lambda I_{\mathcal{V}}$ in the same basis is upper triangular with diagonal entries $\lambda_1 - \lambda, \ldots, \lambda_n - \lambda$. By Proposition 3.1.28, $T - \lambda I_{\mathcal{V}}$ is not invertible if and only if $\lambda = \lambda_j$ for some j. By Theorem 3.1.5, this is equivalent to λ being an eigenvalue of T.
Structural Commentary: Scalar shifts preserve triangular form.
Key Insight: Eigenvalues occur exactly when one shifted diagonal entry becomes zero. □

Perspective 3.1.30. *Why Triangular Form Matters*
Upper-triangular form is the first useful simplification beyond finding one eigenvector. It preserves enough of the operator to read eigenvalues from the diagonal while still allowing nontrivial coupling above the diagonal.

Definition 3.1.31. *Diagonalizable Operator*
An operator $T \in \mathcal{L}(\mathcal{V})$ is *diagonalizable* if T has a diagonal matrix with respect to some basis of $\mathcal{V}$.

Proposition 3.1.32. *Eigenspaces for Distinct Eigenvalues Form a Direct Sum*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and suppose $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T. Then $E_{\lambda_1}(T) + \cdots + E_{\lambda_m}(T)$ is a direct sum. Consequently, $\dim E_{\lambda_1}(T) + \cdots + \dim E_{\lambda_m}(T) \leq \dim(\mathcal{V})$.

Proof. Suppose $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ with $\mathbf{u}_j \in E_{\lambda_j}(T)$. If some $\mathbf{u}_j$ were nonzero, then the nonzero vectors among $\mathbf{u}_1, \ldots, \mathbf{u}_m$ would be eigenvectors corresponding to distinct eigenvalues, and Theorem 3.1.6 would make them linearly independent. This contradicts their sum being $\mathbf{0}$. Hence every $\mathbf{u}_j = \mathbf{0}$, so the sum is direct.
By Theorem 1.2.12, $\dim(E_{\lambda_1}(T) + \cdots + E_{\lambda_m}(T)) = \dim E_{\lambda_1}(T) + \cdots + \dim E_{\lambda_m}(T)$.
The left side is the dimension of a subspace of $\mathcal{V}$, so Proposition 1.1.29 gives the stated inequality.
Structural Commentary: Distinct eigenspaces form independent pieces.
Key Insight: A zero sum across them would contradict eigenvector independence. □

Theorem 3.1.33. *Conditions Equivalent to Diagonalizability*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of T. Set $n = \dim(\mathcal{V})$. The following are equivalent:

(i) T is diagonalizable;
(ii) $\mathcal{V}$ has a basis consisting of eigenvectors of T;

(iii) $\exists$ *one-dimensional subspaces* $\mathcal{U}_1, \dots, \mathcal{U}_n$ *of* $\mathcal{V}$*, each invariant under* T*, such that* $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_n$*;*
(iv) $\mathcal{V} = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_m}(T)$*;*
(v) $\dim(\mathcal{V}) = \dim E_{\lambda_1}(T) + \cdots + \dim E_{\lambda_m}(T)$.

Proof.

(i) $\Rightarrow$ *(ii)* If T is diagonalizable, then T has a diagonal matrix with respect to some basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$. Hence $T\mathbf{v}_j = a_j\mathbf{v}_j$ for each j, so this basis consists of eigenvectors of T.
(ii) $\Rightarrow$ *(i)* If $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of eigenvectors of T, then $T\mathbf{v}_j = a_j\mathbf{v}_j$ for each j. Thus the matrix of T with respect to this basis is diagonal.
(ii) $\Rightarrow$ *(iii)* Let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ be an eigenbasis, and put $\mathcal{U}_j = \operatorname{span}(\mathbf{v}_j)$. Each $\mathcal{U}_j$ is one-dimensional and invariant under T by Proposition 3.1.11; the basis decomposition gives $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_n$.
(iii) $\Rightarrow$ *(ii)* Choose nonzero $\mathbf{v}_j \in \mathcal{U}_j$. Proposition 3.1.11 makes each $\mathbf{v}_j$ an eigenvector of T, and the direct-sum decomposition makes $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ a basis of $\mathcal{V}$.
(ii) $\Rightarrow$ *(iv)* Every vector in an eigenbasis lies in one of $E_{\lambda_1}(T), \dots, E_{\lambda_m}(T)$, so these eigenspaces span $\mathcal{V}$. Proposition 3.1.32 shows their sum is direct.
(iv) $\Rightarrow$ *(v)* If $\mathcal{V} = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_m}(T)$, then dimensions add by Theorem 1.2.12.
(v) $\Rightarrow$ *(ii)* For each j, choose a basis β_j of $E_{\lambda_j}(T)$, and concatenate these bases into a list β. The length of β is $\dim(\mathcal{V})$. If a linear combination of vectors in β is zero, grouping terms by eigenspace gives $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ with $\mathbf{u}_j \in E_{\lambda_j}(T)$. Proposition 3.1.32 gives $\mathbf{u}_j = \mathbf{0}$ for every j; since each β_j is independent, all coefficients are zero. Thus β is independent, Proposition 1.1.29 makes β a basis of $\mathcal{V}$ consisting of eigenvectors of T.

Structural Commentary: Diagonalizability means the eigenspaces exhaust $\mathcal{V}$. Directness is automatic for distinct eigenspaces, so the real test is spanning, or equivalently the dimension count.
Key Insight: A diagonal basis is exactly an eigenbasis. □

Perspective 3.1.34. *Diagonalization as Exhaustion by Eigenspaces*
Distinct eigenspaces are already separated by Proposition 3.1.32. Thus diagonalizability is the stronger statement that these separated pieces account for every direction in $\mathcal{V}$.

Corollary 3.1.35. *Enough Eigenvalues Imply Diagonalizability*
If $\mathcal{V}$ *is finite-dimensional and* $T \in \mathcal{L}(\mathcal{V})$ *has* $\dim(\mathcal{V})$ *distinct eigenvalues, then* T *is diagonalizable.*

Proof. Let $n = \dim(\mathcal{V})$, and let $\lambda_1, \dots, \lambda_n$ be distinct eigenvalues of T. For each j, choose an eigenvector $\mathbf{v}_j$ corresponding to λ_j. By Theorem 3.1.6, $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is linearly independent. Since this list has length n, Proposition 1.1.29 makes it a basis of $\mathcal{V}$. Thus $\mathcal{V}$ has an eigenbasis, so T is diagonalizable by Theorem 3.1.33.
Structural Commentary: One eigenvector from each distinct eigenvalue already gives a full-size independent list.
Key Insight: Once the independent eigenvectors reach $\dim(\mathcal{V})$, they must form a basis. □

Theorem 3.1.36. *Commutant of a Diagonalizable Matrix*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ *be diagonalizable. Suppose its distinct eigenvalues are* $\mu_1, \dots, \mu_d$*, and set* $n_i = \dim E_{\mu_i}(A)$*. Write* $A = S(\mu_1 I_{n_1} \oplus \cdots \oplus \mu_d I_{n_d})S^{-1}$*. Then* $\operatorname{Comm}(A) = \{S(X_1 \oplus \cdots \oplus X_d)S^{-1} : X_i \in \operatorname{Mat}(n_i, n_i, \mathbb{C})\}$*. In particular,* $\dim \operatorname{Comm}(A) = n_1^2 + \cdots + n_d^2$.

Proof. Let $\Lambda = \mu_1 I_{n_1} \oplus \cdots \oplus \mu_d I_{n_d}$. Matrix B commutes with A if and only if $S^{-1}BS$ commutes with Λ.
By Lemma 3.1.22, this happens exactly when $S^{-1}BS = X_1 \oplus \cdots \oplus X_d$ with $X_i \in \operatorname{Mat}(n_i, n_i, \mathbb{C})$.
Dimension count follows because conjugation by S is a vector-space isomorphism and $\dim \operatorname{Mat}(n_i, n_i, \mathbb{C}) = n_i^2$.
Structural Commentary: Diagonalizing A reduces commutation to respecting equal-eigenvalue blocks.
Key Insight: Distinct eigenvalue blocks cannot mix under a commuting matrix. □

Theorem 3.1.37. *Simultaneous Schur Triangularisation*
Let $\emptyset \neq \mathcal{F} \subseteq \operatorname{Mat}(n, n, \mathbb{C})$ *be a commuting family.*
Then there is a unitary $U \in \operatorname{Mat}(n, n, \mathbb{C})$ *such that* U^*AU *is upper triangular for every* $A \in \mathcal{F}$*.*
If every matrix in $\mathcal{F}$ *is real and has only real eigenvalues, then* U *can be chosen real orthogonal.*

Proof. Induct on n. The case $n = 1$ is immediate.
For $n > 1$, Theorem 3.1.24 gives a common eigenvector; normalize it to a unit vector $\mathbf{x}$ and extend to an orthonormal basis. Let $V = [\mathbf{x}\ V_2]$ be the corresponding unitary matrix.
For every $A \in \mathcal{F}$, since $A\mathbf{x} = \lambda_A\mathbf{x}$, we have $V^*AV = \begin{bmatrix} \lambda_A & * \\ 0 & A' \end{bmatrix}$.
If $A, B \in \mathcal{F}$, then $AB = BA$. The lower-right block of $V^*ABV = (V^*AV)(V^*BV)$ is $A'B'$, while the lower-right block of V^*BAV is $B'A'$. Hence $A'B' = B'A'$, so the compressed family is still commuting.
By induction, choose a unitary W that simultaneously upper triangularises all compressed blocks.
Then $U = V(1 \oplus W)$ is unitary and U^*AU is upper triangular for every $A \in \mathcal{F}$.
For the real case, run the same induction over $\mathbb{R}$. A common real eigenvector exists as follows. If all matrices

are scalar, any real unit vector works. Otherwise choose a nonscalar $A_0 \in \mathcal{F}$.
Since A_0 has only real eigenvalues, some real eigenspace $\mathcal{U} = E_\lambda(A_0)$ is nonzero; since A_0 is not scalar, $\mathcal{U} \neq \mathbb{R}^n$. Commutation gives $(A_0 - \lambda I)B\mathbf{u} = B(A_0 - \lambda I)\mathbf{u} = \mathbf{0}$ for $B \in \mathcal{F}$ and $\mathbf{u} \in \mathcal{U}$, so $\mathcal{U}$ is invariant under every B. The restricted family on $\mathcal{U}$ is commuting and still has only real eigenvalues: in a basis adapted to $\mathcal{U}$ each B has block form $\begin{bmatrix} B|_{\mathcal{U}} & * \\ 0 & * \end{bmatrix}$, so the characteristic polynomial of $B|_{\mathcal{U}}$ divides that of B. Induction on $\dim(\mathcal{U})$ gives a common real eigenvector. Extending by real orthonormal bases gives a real orthogonal U.
Structural Commentary: A common eigenline creates the first block; commutation descends to the compression.
Key Insight: Simultaneous triangularisation is Schur induction with a common eigenvector at every step. □

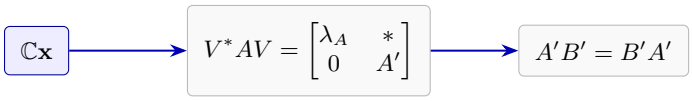


Figure 28: A shared eigenline gives a simultaneous first block; lower-right compressions still commute.

Corollary 3.1.38. *Eigenvalues of Sums and Products of Commuting Matrices*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$ and $AB = BA$. Their eigenvalues can be ordered as $\lambda_1, \ldots, \lambda_n$ and $\mu_1, \ldots, \mu_n$ so that $A + B$ has eigenvalues $\lambda_i + \mu_i$ and AB has eigenvalues $\lambda_i\mu_i$, counted with multiplicity.

Proof. Choose simultaneous Schur forms $T = U^*AU$ and $R = U^*BU$. Set $\lambda_i = t_{ii}$ and $\mu_i = r_{ii}$. Then $T + R$ and TR are upper triangular with diagonals $\lambda_i + \mu_i$ and $\lambda_i\mu_i$, and are similar to $A + B$ and AB.
Structural Commentary: Simultaneous Schur form aligns both spectra on one diagonal.
Key Insight: In common triangular coordinates, diagonals add and multiply. □

3.2 Polynomial Structure of Operators

Polynomial structure records the equations an operator satisfies and turns them into invariant pieces through characteristic and minimal polynomials, Cayley–Hamilton, generalized eigenspaces, and nilpotent parts.

Definition 3.2.1. *Powers of an Operator*
Let $T \in \mathcal{L}(\mathcal{V})$. For a positive integer m, $T^m := \underbrace{T \cdots T}_{m \text{ times}}$ and $T^0 := I_{\mathcal{V}}$.
If T is invertible, then $T^{-m} := (T^{-1})^m$ for positive integers m.

Definition 3.2.2. *Polynomial Applied to an Operator*
Let $T \in \mathcal{L}(\mathcal{V})$ and let $p \in \mathbb{F}[z]$ be $p(z) = a_0 + a_1 z + \cdots + a_m z^m$. Then $p(T) := a_0 I_{\mathcal{V}} + a_1 T + \cdots + a_m T^m$.

Definition 3.2.3. *Annihilating Polynomial*
Let $T \in \mathcal{L}(\mathcal{V})$. A nonzero polynomial $p \in \mathbb{F}[z]$ *annihilates* T if $p(T) = 0$.

Definition 3.2.4. *Product of Polynomials*
If $p, q \in \mathbb{F}[z]$, their product $pq \in \mathbb{F}[z]$ is defined by $(pq)(z) = p(z)q(z)$ for $z \in \mathbb{F}$.

Proposition 3.2.5. *Multiplicative Properties of Operator Polynomials*
Let $p, q \in \mathbb{F}[z]$ and $T \in \mathcal{L}(\mathcal{V})$. Then
(i) $(pq)(T) = p(T)q(T)$;
(ii) $p(T)q(T) = q(T)p(T)$.

Proof.
(i) Write $p(z) = \sum_{j=0}^{m} a_j z^j$ and $q(z) = \sum_{k=0}^{n} b_k z^k$. Then $(pq)(z) = \sum_{j=0}^{m} \sum_{k=0}^{n} a_j b_k z^{j+k}$.
Substituting T gives $(pq)(T) = \sum_{j=0}^{m} \sum_{k=0}^{n} a_j b_k T^{j+k} = \left(\sum_{j=0}^{m} a_j T^j \right) \left(\sum_{k=0}^{n} b_k T^k \right) = p(T)q(T)$.
(ii) Since polynomial multiplication is commutative, $pq = qp$.
By part (i), $p(T)q(T) = (pq)(T) = (qp)(T) = q(T)p(T)$.
Structural Commentary: Polynomial evaluation preserves products.
Key Insight: Expanding a polynomial product works the same with T as with z. □

Proposition 3.2.6. *Existence of Annihilating Polynomials*
Let $\mathcal{V}$ be finite-dimensional with $n = \dim(\mathcal{V}) > 0$, and let $T \in \mathcal{L}(\mathcal{V})$.
Then T has a nonconstant annihilating polynomial of degree at most n^2.

Proof. By Theorem 1.3.58, the vector space $\mathcal{L}(\mathcal{V})$ has dimension n^2, so the n^2+1 operators $I_{\mathcal{V}}, T, T^2, \ldots, T^{n^2}$ are linearly dependent. Hence there are scalars $a_0, \ldots, a_{n^2}$, not all zero, such that $\sum_{j=0}^{n^2} a_j T^j = 0$.

If $a_1 = \cdots = a_{n^2} = 0$, then $a_0 I_{\mathcal{V}} = 0$, forcing $a_0 = 0$ because $\mathcal{V} \neq \{\mathbf{0}\}$, a contradiction. Thus $p(z) = \sum_{j=0}^{n^2} a_j z^j$ is a nonconstant annihilating polynomial.

Structural Commentary: Finite-dimensionality makes the powers of one operator linearly dependent inside $\mathcal{L}(\mathcal{V})$.

Key Insight: A dependence among $I_{\mathcal{V}}, T, \ldots, T^{n^2}$ is exactly a polynomial equation satisfied by T. □

Proposition 3.2.7. *Polynomial Evaluation on Eigenvectors*
Let $T \in \mathcal{L}(\mathcal{V})$, let $p \in \mathbb{F}[z]$, and suppose $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Then $p(T)\mathbf{v} = p(\lambda)\mathbf{v}$.

Proof. Induct on j to get $T^j\mathbf{v} = \lambda^j\mathbf{v}$ for every $j \geq 0$.

If $p(z) = \sum_{j=0}^{m} a_j z^j$, then $p(T)\mathbf{v} = \sum_{j=0}^{m} a_j T^j\mathbf{v} = \sum_{j=0}^{m} a_j \lambda^j\mathbf{v} = p(\lambda)\mathbf{v}$.

Structural Commentary: On a one-dimensional eigenspace, T acts like the scalar λ.

Key Insight: Polynomial expressions in T reduce to the same polynomial evaluated at λ. □

Motivation 3.2.8. *Two Spectral Polynomials*
The characteristic polynomial records the whole primary decomposition with dimensions.
The minimal polynomial records the shortest polynomial relation satisfied by the operator.

Definition 3.2.9. *Characteristic Polynomial*
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, the *characteristic polynomial* of A is $p_A(z) := \det(zI_n - A)$.

Theorem 3.2.10. *Basic Form of the Characteristic Polynomial*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$. Then p_A is monic of degree n and $p_A(z) = z^n - (\operatorname{tr} A)z^{n-1} + \cdots + (-1)^n \det A$. Moreover, for $\lambda \in \mathbb{F}$, $\lambda \in \operatorname{spec}(A)$ if and only if $p_A(\lambda) = 0$.

Proof. The only determinant term of degree greater than $n-2$ comes from the identity permutation:

$$(z - a_{11})(z - a_{22}) \cdots (z - a_{nn}) = z^n - (a_{11} + \cdots + a_{nn})z^{n-1} + \cdots .$$

Thus p_A is monic and the coefficient of z^{n-1} is $-\operatorname{tr} A$; also $p_A(0) = \det(-A) = (-1)^n \det A$. Finally, $p_A(\lambda) = 0$ if and only if $\lambda I_n - A$ is singular, equivalently $\ker(A - \lambda I_n) \neq \{\mathbf{0}\}$.

Structural Commentary: The diagonal product determines the top two coefficients.

Key Insight: Eigenvalues are precisely the roots of $\det(zI_n - A)$. □

Definition 3.2.11. *Algebraic Multiplicity*
For $\lambda \in \mathbb{F}$, let $m_A(\lambda)$ be the multiplicity of λ as a root of p_A, with $m_A(\lambda) = 0$ if $\lambda \notin \operatorname{spec}(A)$.
For $\lambda \in \operatorname{spec}(A)$, this is the *algebraic multiplicity* of λ. Unqualified multiplicity means algebraic multiplicity.

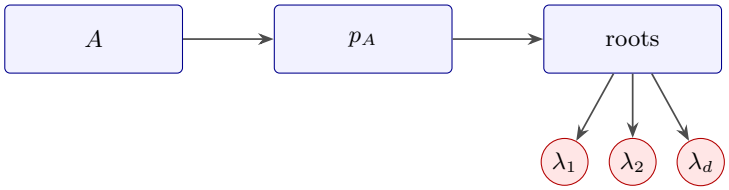


Figure 29: Form $p_A(z) = \det(zI_n - A)$ and factor it; each root λ is counted by $m_A(\lambda)$.

Method 3.2.12. *Characteristic Equation and Eigenspace Computation*
To compute eigenvalues and eigenspaces of $A \in \mathrm{Mat}(n, n, \mathbb{F})$:

(i) Form the characteristic polynomial $p_A(z) = \det(zI_n - A)$ from Definition 3.2.9.
(ii) Solve the characteristic equation $p_A(\lambda) = 0$. By Theorem 3.2.10, the solutions are exactly the eigenvalues in $\mathbb{F}$.
(iii) For each eigenvalue λ, row-reduce $A - \lambda I_n$ and compute

$$E_\lambda(A) = \ker(A - \lambda I_n).$$

(iv) Choose a basis of each eigenspace. The dimension of this basis is the geometric multiplicity of λ.

Algebraic multiplicity comes from the factorisation of p_A, while geometric multiplicity comes from the nullspace computation.

Proposition 3.2.13. *Upper-Triangular Eigenvalues*
If $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$ *is upper triangular, then* $p_A(z) = \prod_{j=1}^{n} (z - a_{jj})$.
Thus $\mathrm{spec}(A) = \{a_{jj} : 1 \le j \le n\}$, *and for* $\lambda \in \mathrm{spec}(A)$, $m_A(\lambda) = \#\{j : a_{jj} = \lambda\}$.

Proof. Since $zI_n - A$ is upper triangular with diagonal entries $z - a_{jj}$, $p_A(z) = \det(zI_n - A) = \prod_{j=1}^{n} (z - a_{jj})$.
Thus the roots of p_A are exactly the diagonal entries, and the exponent of $(z - \lambda)$ is $\#\{j : a_{jj} = \lambda\}$.
Structural Commentary: Triangular form removes all off-diagonal determinant contributions.
Key Insight: Once a matrix is triangular, the spectrum is visible on the diagonal. □

Theorem 3.2.14. *Similarity Preserves Characteristic Data*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ *and suppose* $B = SAS^{-1}$ *for some invertible* S. *Then* $p_B = p_A$. *In particular, for each* $\lambda \in \mathrm{spec}(A)$, $m_B(\lambda) = m_A(\lambda)$ *and* $\dim E_\lambda(B) = \dim E_\lambda(A)$.

Proof. For every z, $p_B(z) = \det(zI_n - SAS^{-1}) = \det(S(zI_n - A)S^{-1}) = \det(S)\det(zI_n - A)\det(S)^{-1} = p_A(z)$.
Thus roots and root multiplicities agree, so $m_B(\lambda) = m_A(\lambda)$. Also $B - \lambda I_n = S(A - \lambda I_n)S^{-1}$.
Hence $\mathbf{x} \in \ker(A - \lambda I_n)$ if and only if $S\mathbf{x} \in \ker(B - \lambda I_n)$, S restricts to isomorphism between two eigenspaces.
Structural Commentary: Similarity changes coordinates, not the shifted determinant or shifted kernel dimension.
Key Insight: Conjugation inserts S and S^{-1}, which cancel in determinants and identify kernels. □

Proposition 3.2.15. *Block Upper-Triangular Characteristic Polynomial*
Let $A = \begin{bmatrix} B & C \\ 0 & D \end{bmatrix}$, B *and* D *are square. Then* $p_A(z) = p_B(z)p_D(z)$, *so* $m_A(\lambda) = m_B(\lambda) + m_D(\lambda)$ $\forall \lambda \in \mathbb{F}$.

Proof. The shifted matrix is block upper triangular, $zI - A = \begin{bmatrix} zI - B & -C \\ 0 & zI - D \end{bmatrix}$.
Theorem 4.1.8 gives $p_A(z) = \det(zI - B)\det(zI - D) = p_B(z)p_D(z)$.
Roots of a product are the roots of the factors, with multiplicity added.
Structural Commentary: The off-diagonal block couples vectors but not determinant factors.
Key Insight: In block upper-triangular form, spectral data lives on the diagonal blocks. □

Theorem 3.2.16. *Geometric Multiplicity Is at Most Algebraic Multiplicity*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ *and let* $\lambda \in \mathrm{spec}(A)$. *Then* $\dim E_\lambda(A) \le m_A(\lambda)$.

Proof. Let $k = \dim E_\lambda(A)$ and choose a basis $\mathbf{x}_1, \ldots, \mathbf{x}_k$ of $E_\lambda(A)$. Extend it to a basis of $\mathbb{F}^n$, and let S be the matrix whose columns are this basis. In the new basis,

$$S^{-1}AS = \begin{bmatrix} \lambda I_k & * \\ 0 & C \end{bmatrix}.$$

By Theorem 3.2.14, Proposition 3.2.15, $p_A(z) = p_{S^{-1}AS}(z) = (z - \lambda)^k p_C(z)$. Thus $m_A(\lambda) \ge k = \dim E_\lambda(A)$.
Structural Commentary: An eigenspace-first basis exposes a λI_k block.
Key Insight: Each independent eigenvector forces one visible factor $z - \lambda$ in the characteristic polynomial. □

Theorem 3.2.17. *Diagonalizability and Multiplicity*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. *Then* A *is diagonalizable if and only if* $\dim E_\lambda(A) = m_A(\lambda)$ *for every* $\lambda \in \mathrm{spec}(A)$.

Proof.
(⇒) If A is diagonalizable, then A is similar to a diagonal matrix D. Theorem 3.2.14 gives $m_A(\lambda) = m_D(\lambda)$ and $\dim E_\lambda(A) = \dim E_\lambda(D)$. For diagonal D, both numbers count diagonal entries equal to λ.
(⇐) Assume $\dim E_\lambda(A) = m_A(\lambda)$ for every eigenvalue.
If $\mu_1, \ldots, \mu_d$ are distinct eigenvalues, then $\sum_{i=1}^{d} m_A(\mu_i) = n$. Hence $\dim E_{\mu_1}(A) + \cdots + \dim E_{\mu_d}(A) = n$.
Theorem 3.1.33 gives an eigenbasis, so A is diagonalizable.
Structural Commentary: Diagonalizability means root counts match eigenspace dimensions.
Key Insight: No missing eigendirections means an eigenbasis. □

Definition 3.2.18. *Characteristic Polynomial of an Operator*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, and let $T \in \mathcal{L}(\mathcal{V})$. If $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of T and $d_j = \dim \mathcal{G}_{\lambda_j}(T)$, then the *characteristic polynomial* of T is $q_T(z) := \prod_{j=1}^{m} (z - \lambda_j)^{d_j}$.

Proposition 3.2.19. *Degree and Roots of q_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, let $T \in \mathcal{L}(\mathcal{V})$, and let q_T be the characteristic polynomial of T.

(i) $\deg q_T = \dim(\mathcal{V})$.
(ii) The roots of q_T are precisely the eigenvalues of T.

Proof. Write $q_T(z) = \prod_{j=1}^{m} (z - \lambda_j)^{d_j}$, where $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues and $d_j = \dim \mathcal{G}_{\lambda_j}(T)$.

(i) Proposition 3.2.64 gives $\deg q_T = \sum_j d_j = \dim(\mathcal{V})$.
(ii) The displayed factorization has exactly the roots $\lambda_1, \ldots, \lambda_m$.

Structural Commentary: Each primary summand contributes one root factor.
Key Insight: Multiplicity is dimension. □

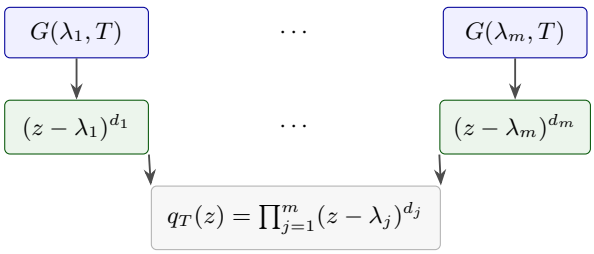


Figure 30: Each primary summand contributes $(z - \lambda_j)^{d_j}$, where $d_j = \dim \mathcal{G}_{\lambda_j}(T)$.

Definition 3.2.20. *Monic Polynomial*
A nonzero polynomial is *monic* if its highest-degree coefficient is 1.

Theorem 3.2.21. *Existence of the Minimal Polynomial*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dim over $\mathbb{F}$ and let $T \in \mathcal{L}(\mathcal{V})$. Then T has a unique least-degree monic annihilator μ_T.

Proof. Proposition 3.2.6 gives a nonzero annihilator; scaling makes it monic, so a least-degree monic annihilator exists. If p and q are two of them, then $(p - q)(T) = 0$ and $\deg(p - q) < \deg p$.
If $p \neq q$, scaling $p - q$ gives a smaller monic annihilator, impossible. Hence $p = q$.
Structural Commentary: Existence comes from finite-dimensional dependence; uniqueness from minimality.
Key Insight: Two least monic annihilators cannot differ. □

Definition 3.2.22. *Minimal Polynomial of an Operator*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. The *minimal polynomial* of T, denoted μ_T, is the unique monic polynomial of smallest degree such that $\mu_T(T) = 0$.

Definition 3.2.23. *Minimal Polynomial of a Matrix*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. The *minimal polynomial* of A, denoted μ_A, is the unique monic polynomial of smallest positive degree such that $\mu_A(A) = 0$.

Theorem 3.2.24. *Cayley–Hamilton Theorem for Operators*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, q_T be characteristic polynomial of $T \in \mathcal{L}(\mathcal{V})$. Then $q_T(T) = 0$.

Proof. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues and set $d_j = \dim \mathcal{G}_{\lambda_j}(T)$.
It suffices, by primary decomposition, to show $q_T(T)$ kills each $\mathcal{G}_{\lambda_j}(T)$.
On $\mathcal{G}_{\lambda_j}(T)$, the operator $T - \lambda_j I_{\mathcal{V}}$ is nilpotent, so Corollary 3.2.56 gives $(T - \lambda_j I_{\mathcal{V}})^{d_j} = 0$ on $\mathcal{G}_{\lambda_j}(T)$.
Since the factors of $q_T(T)$ commute, this factor kills the whole product on $\mathcal{G}_{\lambda_j}(T)$. Hence $q_T(T) = 0$.
Structural Commentary: Check the polynomial on each primary summand.
Key Insight: One factor of $q_T(T)$ kills each summand. □

Theorem 3.2.25. *Cayley–Hamilton Theorem*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have characteristic polynomial $p_A(z) = \det(zI_n - A)$. Then $p_A(A) = 0$.

Proof. Choose a Schur form $A = UTU^*$ with diagonal $\lambda_1, \ldots, \lambda_n$.
Since $p_A(A) = U p_A(T) U^*$ and $p_A(z) = \prod_{i=1}^{n} (z - \lambda_i)$, it suffices to prove $(T - \lambda_1 I_n)(T - \lambda_2 I_n) \cdots (T - \lambda_n I_n) = 0$.
Let $Q_j := \prod_{i=1}^{j} (T - \lambda_i I_n)$. We show that the first j columns of Q_j are zero. For $j = 1$, the first column of $T - \lambda_1 I_n$ is zero as T is upper triangular with first diagonal entry λ_1. Assume the first j columns of Q_j are zero.
Since $T - \lambda_{j+1} I_n$ is upper triangular and its $(j+1, j+1)$ entry is zero, its first $j+1$ columns lie in $\mathrm{span}(\mathbf{e}_1, \ldots, \mathbf{e}_j)$.

By the induction hypothesis, Q_j kills this span, so $Q_{j+1} = Q_j(T - \lambda_{j+1} I_n)$ has first $j+1$ zero columns.
At $j = n$, all columns of Q_n are zero, so $Q_n = 0$.
Structural Commentary: Schur form reduces the theorem to a triangular column-killing induction.
Key Insight: Each factor $T - \lambda_i I_n$ kills the next leading column. □

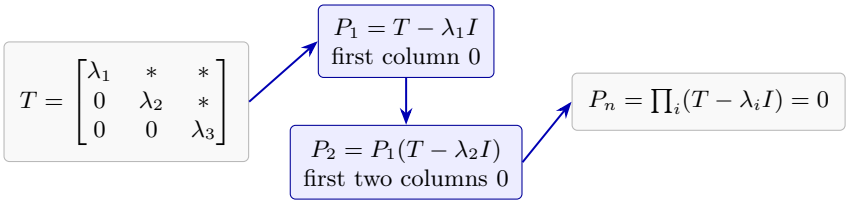


Figure 31: In Schur form, the factors $T - \lambda_i I$ kill the leading columns one layer at a time.

Theorem 3.2.26. *Basic Properties of the Minimal Polynomial*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$, and let $\lambda_1, \ldots, \lambda_d$ be the distinct eigenvalues of A.

(i) μ_A exists, is unique, and $\deg \mu_A \leq n$.
(ii) If a nonzero polynomial f annihilates A, then μ_A divides f. In particular, μ_A divides p_A.
(iii) There are integers q_i satisfying $1 \leq q_i \leq m_A(\lambda_i)$ such that $\mu_A(z) = \prod_{i=1}^{d} (z - \lambda_i)^{q_i}$.

Proof.

(i) Cayley–Hamilton supplies the monic annihilator p_A, so a least-degree monic annihilator μ_A exists and has degree at most n. If g is another least-degree monic annihilator, write $g = h\mu_A + r$, where $r = 0$ or $\deg r < \deg \mu_A$. Evaluating at A gives $r(A) = 0$. If $r \neq 0$, scaling r to be monic would give a smaller-degree monic annihilator, a contradiction. Hence $r = 0$, so $g = h\mu_A$.
Since g and μ_A are monic of the same degree, $h = 1$ and $g = \mu_A$.
(ii) If $f(A) = 0$, divide $f = q\mu_A + r$ with $\deg r < \deg \mu_A$. Evaluating at A gives $r(A) = 0$, hence $r = 0$ and $\mu_A \mid f$. Taking $f = p_A$ gives $\mu_A \mid p_A$.
(iii) Since $\mu_A \mid p_A = \prod_{i=1}^{d} (z - \lambda_i)^{m_A(\lambda_i)}$, its roots are eigenvalues with exponents at most the algebraic multiplicities. If $A\mathbf{v} = \lambda_i \mathbf{v}$ and $\mathbf{v} \neq \mathbf{0}$, then $\mu_A(\lambda_i)\mathbf{v} = \mu_A(A)\mathbf{v} = 0$, so every λ_i occurs.

Structural Commentary: Polynomial division makes μ_A the least monic generator of annihilators.
Key Insight: Cayley–Hamilton gives existence; eigenvectors force the roots. □

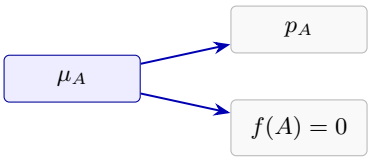


Figure 32: μ_A is the least monic annihilator; arrows indicate that μ_A divides p_A and every annihilator f.

Theorem 3.2.27. *Minimal Polynomial of a Spectrally Disjoint Direct Sum*
Let $A = A_1 \oplus \cdots \oplus A_d$, where the spectra of $A_1, \ldots, A_d$ are pairwise disjoint. Then $\mu_A = \mu_{A_1} \mu_{A_2} \cdots \mu_{A_d}$.

Proof. Let $f = \prod_i \mu_{A_i}$. Since $f(A_i) = 0$ for all i, blockwise evaluation gives $f(A) = 0$, and therefore $\mu_A \mid f$.
Conversely, $\mu_A(A_i) = 0$ for each i, so $\mu_{A_i} \mid \mu_A$ for each i. The roots of μ_{A_i} are exactly $\mathrm{spec}(A_i)$ by Theorem 3.2.35, so the disjointness of the spectra makes the polynomials μ_{A_i} pairwise relatively prime.
Hence their product f divides μ_A. Since $\mu_A \mid f$ and $f \mid \mu_A$, and both are monic, $\mu_A = f$.
Structural Commentary: Direct sums are annihilated blockwise; disjoint spectra make the block factors coprime.
Key Insight: Each spectral component contributes its own minimal-polynomial factor. □

Theorem 3.2.28. *Similarity Preserves the Minimal Polynomial*
If $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$ are similar, then $p_A = p_B$ and $\mu_A = \mu_B$.

Proof. The identity $p_A = p_B$ is Theorem 3.2.14. Let $A = SBS^{-1}$, then $f(A) = Sf(B)S^{-1}$ for every polynomial f, so A and B have the same annihilating polynomials. Hence $\mu_A = \mu_B$.
Structural Commentary: Similarity preserves polynomial equations.
Key Insight: Conjugation commutes with polynomial evaluation. □

Theorem 3.2.29. *Diagonalizability and the Minimal Polynomial*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have distinct eigenvalues $\lambda_1, \ldots, \lambda_d$.
Then A is diagonalizable if and only if $\mu_A(z) = (z - \lambda_1) \cdots (z - \lambda_d)$.
Equivalently, A is diagonalizable if and only if μ_A is square-free.

Proof. Set $p(z) := \prod_{i=1}^{d}(z - \lambda_i)$.
($\Rightarrow$) If $A = S\Lambda S^{-1}$ is diagonalizable, then $p(\Lambda) = 0$, so $p(A) = 0$.
Since μ_A contains every factor $z - \lambda_i$ and divides p, we have $\mu_A = p$.
($\Leftarrow$) Suppose $\mu_A = p$. On the primary summand $\mathcal{G}_{\lambda_i}(A)$, the restriction of $A - \lambda_j I_n$ for $j \neq i$ is $(A - \lambda_i I_n) + (\lambda_i - \lambda_j)I$, a nonzero scalar plus a nilpotent operator. By Lemma 3.2.57, these restricted factors are invertible. Since $0 = p(A)|_{\mathcal{G}_{\lambda_i}(A)} = \left(\prod_{j \neq i}(A - \lambda_j I_n)\right)|_{\mathcal{G}_{\lambda_i}(A)}(A - \lambda_i I_n)|_{\mathcal{G}_{\lambda_i}(A)}$, the invertible product of the $j \neq i$ factors can be cancelled, giving $(A - \lambda_i I_n) = 0$ on $\mathcal{G}_{\lambda_i}(A)$.
Thus every primary summand is an eigenspace, so A is diagonalizable.
Structural Commentary: Repeated factors of μ_A record nilpotent parts in primary blocks.
Key Insight: Square-free μ_A kills each primary block in one step. □

Corollary 3.2.30. *Square-Free Annihilator Criterion*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. If a polynomial f with no repeated roots satisfies $f(A) = 0$, then A is diagonalizable.

Proof. Since $\mu_A \mid f$, the minimal polynomial is square-free; Theorem 3.2.29 applies.
Structural Commentary: A square-free annihilator forces square-free μ_A.
Key Insight: Square-free μ_A means diagonalizable. □

Method 3.2.31. *Computing μ_T from a Matrix*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ be the matrix of T in some basis.
(i) For $m = 1, 2, \ldots$, test the monic candidate $p_m(z) = a_0 + a_1 z + \cdots + a_{m-1}z^{m-1} + z^m$.
(ii) Solve linear system by comparing matrix entries in $a_0 I + a_1 A + \cdots + a_{m-1}A^{m-1} = -A^m$ for $a_0, \ldots, a_{m-1}$.
(iii) If no solution exists, increase m. The first solvable candidate is μ_T.

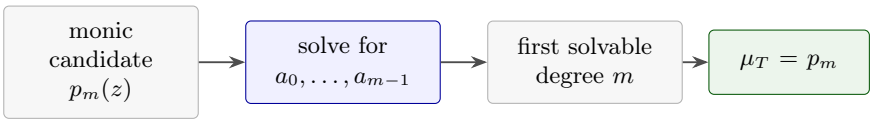


Figure 33: Solve $a_0 I + \cdots + a_{m-1}A^{m-1} = -A^m$; the first solvable monic candidate is μ_T.

Method 3.2.32. *Testing a Minimal Polynomial from p_A*
Suppose $p_A(z) = \prod_i (z - \lambda_i)^{a_i}$ is known.
i. Write each candidate as $m_{\mathbf{q}}(z) = \prod_i (z - \lambda_i)^{q_i}$, where $1 \leq q_i \leq a_i$.
ii. Test exponent vectors $\mathbf{q} = (q_i)$ in increasing total degree $\sum_i q_i$.
iii. For each $\mathbf{q}$, compute $m_{\mathbf{q}}(A) = \prod_i (A - \lambda_i I)^{q_i}$.
iv. The first candidate with $m_{\mathbf{q}}(A) = 0$ is μ_A, by Theorem 3.2.26.

Theorem 3.2.33. *Annihilators Are Multiples of μ_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dim, $T \in \mathcal{L}(\mathcal{V})$, and $f \in \mathbb{F}[z]$ be nonzero. Then $f(T) = 0$ if and only if μ_T divides f.

Proof.
($\Rightarrow$) Suppose $f(T) = 0$. Divide $f = s\mu_T + r$, where $r = 0$ or $\deg r < \deg \mu_T$.
Then $r(T) = 0$, so minimality forces $r = 0$. Hence μ_T divides f.
($\Leftarrow$) If $f = \mu_T s$, then $f(T) = \mu_T(T)s(T) = 0$.
Structural Commentary: Polynomial division reduces every annihilator to its remainder modulo μ_T.
Key Insight: Minimality forces that remainder to be zero. □

Corollary 3.2.34. *q_T Is a Multiple of μ_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$.
Then the characteristic polynomial q_T is a polynomial multiple of the minimal polynomial μ_T.

Proof. Theorem 3.2.24 gives $q_T(T) = 0$. Theorem 3.2.33 then implies that μ_T divides q_T.
Structural Commentary: Cayley–Hamilton turns the characteristic polynomial into an annihilator.
Key Insight: The minimal polynomial divides every annihilator, hence divides q_T. □

Theorem 3.2.35. *Roots of μ_T Are Eigenvalues*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. Then the roots of μ_T are precisely the eigenvalues of T.

Proof.

($\Rightarrow$) If $\mu_T(\lambda) = 0$, write $\mu_T(z) = (z - \lambda)g(z)$. Since $\deg g < \deg \mu_T$, minimality gives $g(T) \neq 0$; choose $\mathbf{v}$ with $\mathbf{u} = g(T)\mathbf{v} \neq \mathbf{0}$. Then $(T - \lambda I_\mathcal{V})\mathbf{u} = \mu_T(T)\mathbf{v} = \mathbf{0}$.
($\Leftarrow$) If $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then $0 = \mu_T(T)\mathbf{v} = \mu_T(\lambda)\mathbf{v}$, so $\mu_T(\lambda) = 0$.

Structural Commentary: A root splits off one shifted operator.
Key Insight: The quotient cannot already annihilate T. □

Perspective 3.2.36. *Characteristic vs. Minimal Polynomial*
The characteristic polynomial records every primary block with its dimension. The minimal polynomial records the least polynomial needed to annihilate all blocks at once.

Proposition 3.2.37. *Minimal Polynomial Survives Complexification*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{R}$ and let $T \in \mathcal{L}(\mathcal{V})$. Then $\mu_{T_\mathbb{C}} = \mu_T$ in $\mathbb{C}[z]$.

Proof.

($\Rightarrow$) Since $\mu_T(T) = 0$, Proposition 3.3.29 gives $\mu_T(T_\mathbb{C}) = (\mu_T(T))_\mathbb{C} = 0$. Hence $\mu_{T_\mathbb{C}}$ divides μ_T in $\mathbb{C}[z]$.
($\Leftarrow$) Let $q = \mu_{T_\mathbb{C}} = r + is$, with $r, s \in \mathbb{R}[z]$. Since q is monic, r is monic and $\deg r = \deg q$.
For every $\mathbf{u} \in \mathcal{V}$, $0 = q(T_\mathbb{C})(\mathbf{u} + i\mathbf{0}) = r(T)\mathbf{u} + is(T)\mathbf{u}$. Thus $r(T) = 0$, so $\deg \mu_T \leq \deg r = \deg q$. The first part gives opposite degree inequality. As q divides μ_T and both are monic with equal degree, $q = \mu_T$.

Structural Commentary: Annihilation passes to complexification and back by real-part extraction.
Key Insight: Evaluate $q(T_\mathbb{C})$ on $\mathbf{u} + i\mathbf{0}$. □

Definition 3.2.38. *Companion Matrix*
Let $f(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0$ be monic. For $n \geq 2$, the *companion matrix* of f is

$$C_f := \begin{bmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -c_{n-1} \end{bmatrix}.$$

For $n = 1$, set $C_f = [-c_0]$.

Theorem 3.2.39. *Companion Matrix Realises a Polynomial*
For every monic polynomial f, the companion matrix C_f satisfies $p_{C_f} = f$ and $\mu_{C_f} = f$.

Proof. Let $\mathbf{e}_1, \ldots, \mathbf{e}_n$ be the standard basis. For $0 \leq j < n$, $C_f^j \mathbf{e}_1 = \mathbf{e}_{j+1}$.
The last-column relation gives $C_f^n \mathbf{e}_1 = -c_0\mathbf{e}_1 - c_1 C_f \mathbf{e}_1 - \cdots - c_{n-1}C_f^{n-1}\mathbf{e}_1$. Thus $f(C_f)\mathbf{e}_1 = 0$.
Since $\mathbf{e}_j = C_f^{j-1}\mathbf{e}_1$ and $f(C_f)$ commutes with C_f, we get $f(C_f)\mathbf{e}_j = C_f^{j-1} f(C_f)\mathbf{e}_1 = \mathbf{0}$ $\forall j$, so $f(C_f) = 0$.
If $g(z) = \sum_{i=0}^{m} b_i z^i$ has $m < n$, then $g(C_f)\mathbf{e}_1 = b_0\mathbf{e}_1 + b_1\mathbf{e}_2 + \cdots + b_m\mathbf{e}_{m+1}$, which is zero only when all $b_i = 0$.
Hence no nonzero polynomial of degree $< n$ annihilates C_f. Therefore $\deg \mu_{C_f} = n$.
Since f is a monic degree-n annihilator and $\mu_{C_f} \mid f$, we have $\mu_{C_f} = f$.
Cayley–Hamilton makes p_{C_f} a monic degree-n annihilator, so $\mu_{C_f} \mid p_{C_f}$. Both p_{C_f} and f are monic of degree n, and both are multiples of the monic degree-n polynomial $\mu_{C_f} = f$, hence $p_{C_f} = f$.
Structural Commentary: The cyclic vector $\mathbf{e}_1$ generates the basis, and last column imposes relation $f(C_f) = 0$.
Key Insight: The companion matrix makes f the first possible annihilating relation. □

Theorem 3.2.40. *Annihilating Polynomials Control the Spectrum*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let p be a nonconstant annihilating polynomial for T.

(i) Every eigenvalue of T is a root of p.
(ii) If $\mathbb{F} = \mathbb{C}$, then at least one root of p is an eigenvalue of T.

Proof.

(i) If $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then Proposition 3.2.7 gives $0 = p(T)\mathbf{v} = p(\lambda)\mathbf{v}$. Since $\mathbf{v} \neq \mathbf{0}$, $p(\lambda) = 0$.
(ii) Factor $p(z) = c(z - \lambda_1)\cdots(z - \lambda_m)$ over $\mathbb{C}$, with $c \neq 0$. Then $0 = p(T) = c(T - \lambda_1 I_\mathcal{V})\cdots(T - \lambda_m I_\mathcal{V})$. At least one factor $T - \lambda_j I_\mathcal{V}$ is not injective; otherwise product would be injective. Hence λ_j is an eigenvalue.

Structural Commentary: Annihilating polynomial restricts spectrum to its roots; over $\mathbb{C}$, one root must appear.
Key Insight: Eigenvectors turn $p(T)$ into $p(\lambda)$. □

Proposition 3.2.41. *Kernel and Range of a Polynomial in an Operator Are Invariant*
Let $T \in \mathcal{L}(\mathcal{V})$ and $p \in \mathbb{F}[z]$. Then $\ker(p(T))$ and $\operatorname{range}(p(T))$ are invariant under T.

Proof. If $\mathbf{v} \in \ker(p(T))$, then $p(T)\mathbf{v} = \mathbf{0}$.
Since $p(T)$ commutes with T by Proposition 3.2.5, $p(T)(T\mathbf{v}) = T(p(T)\mathbf{v}) = T\mathbf{0} = \mathbf{0}$.
If $\mathbf{v} \in \operatorname{range}(p(T))$, write $\mathbf{v} = p(T)\mathbf{u}$. Then $T\mathbf{v} = Tp(T)\mathbf{u} = p(T)(T\mathbf{u})$, so $T\mathbf{v} \in \operatorname{range}(p(T))$.
Structural Commentary: Polynomial kernels and ranges are stable under the original operator.
Key Insight: Since T commutes with $p(T)$, it preserves both $\ker p(T)$ and $\operatorname{range} p(T)$. □

Theorem 3.2.42. *Sylvester Linear Matrix Equation*
Let $A \in \operatorname{Mat}(m, m, \mathbb{C})$ and $B \in \operatorname{Mat}(n, n, \mathbb{C})$. If $\operatorname{spec}(A) \cap \operatorname{spec}(B) = \emptyset$, then for every $C \in \operatorname{Mat}(m, n, \mathbb{C})$ the equation $AX - XB = C$ has a unique solution $X \in \operatorname{Mat}(m, n, \mathbb{C})$.

Proof. Let $L(X) = AX - XB$. Since L has equal finite-dimensional domain and codomain, it suffices to show $\ker L = \{0\}$. If $AX = XB$, then $p(A)X = Xp(B)$ for every polynomial p.
With $p = p_B$, Cayley–Hamilton gives $p_B(A)X = Xp_B(B) = 0$. For every $\lambda \in \operatorname{spec}(A)$, disjointness gives $p_B(\lambda) \neq 0$; hence $p_B(A)$ is invertible by Corollary 3.5.9. Thus $X = 0$, so L is bijective.
Structural Commentary: Polynomial calculus pass through homogeneous solutions; disjoint spectra invert $p_B(A)$.
Key Insight: Spectral separation turns an intertwining relation into the zero relation. □

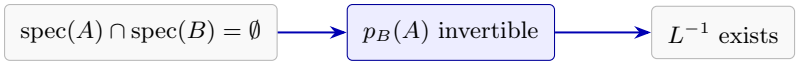


Figure 34: Disjoint spectra make $p_B(A)$ invertible, so the Sylvester map $L(X) = AX - XB$ is invertible.

Corollary 3.2.43. *Removing a Spectrally Separated Off-Diagonal Block*
Let $M = \begin{bmatrix} A & C \\ 0 & B \end{bmatrix}$ and $\operatorname{spec}(A) \cap \operatorname{spec}(B) = \emptyset$. Then M is similar to $A \oplus B$.

Proof. By Theorem 3.2.42, choose X such that $AX - XB = -C$. Then

$$\begin{bmatrix} I & -X \\ 0 & I \end{bmatrix} \begin{bmatrix} A & C \\ 0 & B \end{bmatrix} \begin{bmatrix} I & X \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & AX + C - XB \\ 0 & B \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}.$$

Structural Commentary: A block shear cancels the off-diagonal block via Sylvester's equation.
Key Insight: Spectrally separated diagonal blocks cannot carry essential upper-right coupling. □

Theorem 3.2.44. *Schur Form Grouped by Eigenvalues*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ have distinct eigenvalues $\lambda_1, \ldots, \lambda_d$ with algebraic multiplicities $n_1, \ldots, n_d$. Then A is unitarily similar to a block upper-triangular matrix

$$T = \begin{bmatrix} T_{11} & T_{12} & \cdots & T_{1d} \\ 0 & T_{22} & \cdots & T_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & T_{dd} \end{bmatrix}, \qquad T_{ii} \in \operatorname{Mat}(n_i, n_i, \mathbb{C}),$$

where each T_{ii} is upper triangular with all diagonal entries equal to λ_i. Moreover, A is similar to $T_{11} \oplus \cdots \oplus T_{dd}$.

Proof. Order the Schur diagonal as $\lambda_1, \ldots, \lambda_1, \lambda_2, \ldots, \lambda_2, \ldots, \lambda_d, \ldots, \lambda_d$.
Theorem 3.4.10 gives the displayed block Schur form. Since $\operatorname{spec}(T_{11})$ is disjoint from the trailing block spectrum, Corollary 3.2.43 splits off T_{11}. Iterating splits off every diagonal block.
Structural Commentary: Grouped Schur form isolates spectral blocks; Sylvester shears split disjoint groups.
Key Insight: Cross-group upper blocks are coordinate artefacts. □

Corollary 3.2.45. *Commuting with Disjoint Blocks*
Let $A = A_{11} \oplus \cdots \oplus A_{dd}$, where the spectra of $A_{11}, \ldots, A_{dd}$ are pairwise disjoint. Partition $B = [B_{ij}]$ conformally with A. If $AB = BA$, then $B_{ij} = 0$ for all $i \neq j$.

Proof. Equating (i, j) blocks in $AB = BA$ gives $A_{ii}B_{ij} - B_{ij}A_{jj} = 0$.
For $i \neq j$, the spectra of A_{ii} and A_{jj} are disjoint, so Theorem 3.2.42 with $C = 0$ gives $B_{ij} = 0$.
Structural Commentary: A commuting matrix cannot mix direct-sum blocks with disjoint spectra.
Key Insight: The off-diagonal block solves a homogeneous Sylvester equation, whose only solution is zero. □

Generalized eigenvectors repair the main defect of ordinary eigenvectors: an operator may not have enough eigendirections to form a basis. The replacement is to study vectors killed by powers of shifted operators.

Definition 3.2.46. *Generalized Eigenvectors and Generalized Eigenspaces*
Let $T \in \mathcal{L}(\mathcal{V})$, $\lambda \in \mathbb{F}$. Nonzero vector $\mathbf{v} \in \mathcal{V}$ is a *generalized eigenvector* of T to λ if $(T - \lambda I_{\mathcal{V}})^j \mathbf{v} = \mathbf{0}$ for some positive integer j. The *generalized eigenspace* corresponding to λ is

$$\mathcal{G}_\lambda(T) := \{\mathbf{0}\} \cup \{\mathbf{v} \neq \mathbf{0} : \mathbf{v} \text{ is a generalized eigenvector of } T \text{ corresponding to } \lambda\}.$$

Every eigenvector is a generalized eigenvector by taking $j = 1$. The point is that higher powers of $T - \lambda I_{\mathcal{V}}$ may reveal directions that are not eigenvectors but are eventually pushed into the eigenspace.

Proposition 3.2.47. *Kernels of Powers Increase*
Let $T \in \mathcal{L}(\mathcal{V})$. Then $\{\mathbf{0}\} = \ker(T^0) \subseteq \ker(T) \subseteq \ker(T^2) \subseteq \cdots \subseteq \ker(T^k) \subseteq \ker(T^{k+1}) \subseteq \cdots$.

Proof. If $\mathbf{v} \in \ker(T^k)$, then $T^k \mathbf{v} = \mathbf{0}$, hence $T^{k+1}\mathbf{v} = T(T^k \mathbf{v}) = \mathbf{0}$. Thus $\ker(T^k) \subseteq \ker(T^{k+1})$ for every $k \geq 0$.
Structural Commentary: Powers of T form an increasing chain of kernels.
Key Insight: Once a vector is killed by T^k, applying one more T cannot revive it. □

Proposition 3.2.48. *Equality Forces Stabilization*
Let $T \in \mathcal{L}(\mathcal{V})$ and $m \geq 0$. If $\ker(T^m) = \ker(T^{m+1})$, then $\ker(T^m) = \ker(T^{m+1}) = \ker(T^{m+2}) = \cdots$.

Proof. By Proposition 3.2.47, it suffices to prove $\ker(T^{m+k+1}) \subseteq \ker(T^{m+k})$ for every $k \geq 0$.
Take $\mathbf{v} \in \ker(T^{m+k+1})$. Then $T^{m+1}(T^k \mathbf{v}) = T^{m+k+1}\mathbf{v} = \mathbf{0}$, so $T^k \mathbf{v} \in \ker(T^{m+1}) = \ker(T^m)$.
Hence $T^{m+k}\mathbf{v} = T^m(T^k \mathbf{v}) = \mathbf{0}$, proving the reverse inclusion.
Structural Commentary: Stabilization at one step propagates through the whole tail.
Key Insight: A later kernel vector can be shifted down into the already-stable kernel. □

Theorem 3.2.49. *Kernels Stop Growing by $\dim(\mathcal{V})$*
If $\mathcal{V}$ is finite-dimensional, $T \in \mathcal{L}(\mathcal{V})$, and $n = \dim(\mathcal{V})$, then $\ker(T^n) = \ker(T^{n+1}) = \ker(T^{n+2}) = \cdots$.

Proof. If $\ker(T^n) \neq \ker(T^{n+1})$, Prop 3.2.48 implies all inclusions $\{\mathbf{0}\} = \ker(T^0) \subsetneq \ker(T) \subsetneq \cdots \subsetneq \ker(T^{n+1})$ are strict. Each strict inclusion raises dimension by at least 1, so $\dim \ker(T^{n+1}) \geq n + 1$, impossible because $\ker(T^{n+1}) \subseteq \mathcal{V}$. Thus $\ker(T^n) = \ker(T^{n+1})$, and Proposition 3.2.48 gives the tail equality.
Structural Commentary: Strict kernel growth is dimension-limited.
Key Insight: Dimension bounds force the kernel chain to stabilize no later than step $\dim(\mathcal{V})$. □

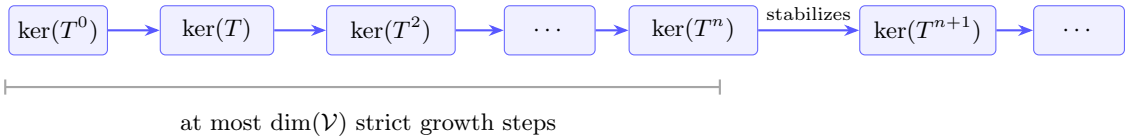


Figure 35: The kernel filtration grows until it stabilizes; after equality appears once, the tail is constant.

Theorem 3.2.50. *Ranks of Matrix Powers Stabilize*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$. Then:

(i) $\mathrm{rank}(A^k) \geq \mathrm{rank}(A^{k+1})$ for $k = 1, 2, \ldots$;
(ii) if $\mathrm{rank}(A^k) = \mathrm{rank}(A^{k+1})$, then $\mathrm{col}(A^k) = \mathrm{col}(A^{k+p})$ and $\mathrm{rank}(A^k) = \mathrm{rank}(A^{k+p})$ for every $p = 1, 2, \ldots$;
(iii) if A is not invertible, then there is a least positive integer $q \leq n$ such that $\mathrm{rank}(A^q) = \mathrm{rank}(A^{q+1})$.

Proof.

(i) Since $A^{k+1} = A^k A$, each column of A^{k+1} is a linear combination of the columns of A^k. Hence $\mathrm{col}(A^{k+1}) \subseteq \mathrm{col}(A^k)$, so $\mathrm{rank}(A^{k+1}) \leq \mathrm{rank}(A^k)$.

(ii) The inclusion $\mathrm{col}(A^{k+1}) \subseteq \mathrm{col}(A^k)$ and equality of dimensions give $\mathrm{col}(A^{k+1}) = \mathrm{col}(A^k)$. Let $U = \mathrm{col}(A^k)$. Then multiplication by A maps U onto U, so induction gives $\mathrm{col}(A^{k+p}) = U$ for every $p \geq 1$.

(iii) If A is not invertible, then Corollary 1.3.64 gives $\mathrm{rank}(A) \leq n-1$. If all inequalities $\mathrm{rank}(A^k) > \mathrm{rank}(A^{k+1})$ held for $k = 1, \ldots, n$, then $\mathrm{rank}(A^{n+1}) \leq \mathrm{rank}(A) - n < 0$, impossible. Hence equality occurs for some $q \leq n$; choose the least such q.

Structural Commentary: The column spaces of powers form a decreasing chain. In finite dimensions, a decreasing chain cannot keep losing dimension forever.
Key Insight: Once two consecutive ranks agree, the image has stabilized under multiplication by A. □

Definition 3.2.51. *Index of a Matrix*
If $A \in \mathrm{Mat}(n, n, \mathbb{F})$ is invertible, define its *index* to be 0. If A is not invertible, its index is the least positive integer q such that $\mathrm{rank}(A^q) = \mathrm{rank}(A^{q+1})$.

Theorem 3.2.52. *Stable Kernel-Range Decomposition*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and set $n = \dim(\mathcal{V})$. Then $\mathcal{V} = \ker(T^n) \oplus \operatorname{range}(T^n)$.

Proof. First show the intersection is trivial. If $\mathbf{v} \in \ker(T^n) \cap \operatorname{range}(T^n)$, then $\mathbf{v} = T^n\mathbf{u}$ and $T^n\mathbf{v} = \mathbf{0}$. Hence $T^{2n}\mathbf{u} = \mathbf{0}$, so $\mathbf{u} \in \ker(T^{2n}) = \ker(T^n)$ by Theorem 3.2.49. Thus $\mathbf{v} = T^n\mathbf{u} = \mathbf{0}$.
By Rank-Nullity Theorem 1.3.19, $\dim\ker(T^n) + \dim\operatorname{range}(T^n) = \dim(\mathcal{V})$.
The zero intersection and the dimension count imply the sum is direct and equals $\mathcal{V}$.
Structural Commentary: At the stable power, killed directions and surviving image directions separate cleanly.
Key Insight: Stabilization turns the usual rank-nullity count into an actual direct-sum decomposition. □

Theorem 3.2.53. *Description of Generalized Eigenspaces*
Let $\mathcal{V}$ be finite-dimensional, $T \in \mathcal{L}(\mathcal{V})$, $\lambda \in \mathbb{F}$, and $n = \dim(\mathcal{V})$. Then $\mathcal{G}_\lambda(T) = \ker\big((T - \lambda I_\mathcal{V})^n\big)$.
In particular, $\mathcal{G}_\lambda(T)$ is a subspace of $\mathcal{V}$.

Proof. Set $S = T - \lambda I_\mathcal{V}$.
($\Rightarrow$) If $\mathbf{v} \in \mathcal{G}_\lambda(T)$, then $S^j\mathbf{v} = \mathbf{0}$ for some $j \geq 1$. If $j \leq n$, Proposition 3.2.47 gives $\mathbf{v} \in \ker(S^n)$; if $j > n$, Theorem 3.2.49 gives $\ker(S^j) = \ker(S^n)$.
($\Leftarrow$) If $\mathbf{v} \in \ker(S^n)$, then either $\mathbf{v} = \mathbf{0}$ or $\mathbf{v}$ is a generalized eigenvector of T corresponding to λ.
Thus $\mathcal{G}_\lambda(T) = \ker(S^n)$, which is a subspace.
Structural Commentary: Generalized eigenspaces are stable shifted kernels.
Key Insight: Finite dimension makes exponent $\dim(\mathcal{V})$ enough. □

Proposition 3.2.54. *Generalized Eigenspaces Are Invariant*
Let $\mathcal{V}$ be finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{G}_\lambda(T)$ is invariant under T for every $\lambda \in \mathbb{F}$.

Proof. By Theorem 3.2.53, $\mathcal{G}_\lambda(T) = \ker((T - \lambda I_\mathcal{V})^n)$, where $n = \dim(\mathcal{V})$.
This is the kernel of a polynomial in T, so it is invariant under T by Proposition 3.2.41.
Structural Commentary: A generalized eigenspace is cut out by a polynomial equation in T.
Key Insight: Polynomial kernels commute with the operator that defines them. □

Definition 3.2.55. *Nilpotent Operator*
An operator $N \in \mathcal{L}(\mathcal{V})$ is *nilpotent* if $N^m = 0$ for some positive integer m.

Corollary 3.2.56. *Nilpotent Power Bound*
If $\mathcal{V}$ is finite-dimensional, $N \in \mathcal{L}(\mathcal{V})$ is nilpotent, and $n = \dim(\mathcal{V})$, then $N^n = 0$.

Proof. Since N is nilpotent, $\ker(N^m) = \mathcal{V}$ for some m.
The kernel chain has stabilized by the nth step by Theorem 3.2.49, so $\ker(N^n) = \mathcal{V}$. Hence $N^n = 0$.
Structural Commentary: Nilpotence says some kernel in the chain is already the whole space.
Key Insight: The stabilization bound pulls that full-kernel conclusion back to exponent $\dim(\mathcal{V})$. □

Lemma 3.2.57. *Nonzero Scalar Plus Nilpotent Is Invertible*
Let $N \in \mathcal{L}(\mathcal{V})$ be nilpotent and let $a \in \mathbb{F}$ with $a \neq 0$. Then $aI_\mathcal{V} + N$ is invertible.

Proof. Choose m such that $N^m = 0$. Then $(aI_\mathcal{V} + N)\left(a^{-1}\sum_{r=0}^{m-1}(-a^{-1}N)^r\right) = I_\mathcal{V} - (-a^{-1}N)^m = I_\mathcal{V}$, and the same computation in the reverse order gives a two-sided inverse.
Structural Commentary: A nilpotent perturbation has a finite geometric-series inverse.
Key Insight: Nilpotence truncates the Neumann series exactly. □

Theorem 3.2.58. *Generalized Eigenvectors for Distinct Eigenvalues Are Independent*
Let $\mathcal{V}$ be finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$. Suppose $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T, and let $\mathbf{v}_1, \ldots, \mathbf{v}_m$ be corresponding generalized eigenvectors. Then $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent.

Proof. Let $n = \dim(\mathcal{V})$ and suppose $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$.
Fix $r \in \{1, \ldots, m\}$ and apply $Q_r(T) := \prod_{i \neq r}(T - \lambda_i I_\mathcal{V})^n$ to the relation.
For $i \neq r$, Theorem 3.2.53 gives $Q_r(T)\mathbf{v}_i = \mathbf{0}$, so $a_r Q_r(T)\mathbf{v}_r = \mathbf{0}$.
On $\mathcal{G}_{\lambda_r}(T)$, the operator $T - \lambda_i I_\mathcal{V}$ equals $(T - \lambda_r I_\mathcal{V}) + (\lambda_r - \lambda_i)I_\mathcal{V}$. The first term is nilpotent on $\mathcal{G}_{\lambda_r}(T)$ and $\lambda_r - \lambda_i \neq 0$, so Lemma 3.2.57 makes each factor invertible on $\mathcal{G}_{\lambda_r}(T)$. Thus $Q_r(T)\mathbf{v}_r \neq \mathbf{0}$, forcing $a_r = 0$. Since r was arbitrary, all coefficients vanish.
Structural Commentary: Polynomial filters isolate one generalized eigenspace at a time.
Key Insight: Distinct eigenvalues let polynomial filters kill the other components and preserve the target one. □

Theorem 3.2.59. *Nilpotent Operators and Strictly Upper-Triangular Matrices*
Let $\mathcal{V}$ be finite-dimensional and $N \in \mathcal{L}(\mathcal{V})$. Then N is nilpotent if and only if N has a strictly upper-triangular matrix in some basis.

Proof. Let $n = \dim(\mathcal{V})$. By Corollary 3.2.56, $\{\mathbf{0}\} \subseteq \ker(N) \subseteq \ker(N^2) \subseteq \cdots \subseteq \ker(N^n) = \mathcal{V}$.

($\Rightarrow$) Choose a basis of $\ker(N)$, extend it to a basis of $\ker(N^2)$, and continue until obtaining a basis of $\mathcal{V}$. If $\mathbf{v}$ first appears in $\ker(N^k)$, then $N\mathbf{v} \in \ker(N^{k-1})$, which is spanned by earlier basis vectors. Hence the matrix of N is strictly upper triangular.

($\Leftarrow$) If the matrix of N is $n \times n$ strictly upper triangular, then N sends each basis vector into the span of earlier basis vectors. After n applications every basis vector is sent to $\mathbf{0}$, so $N^n = 0$.

Structural Commentary: Kernel filtration gives adapted basis; triangular form records motion to earlier layers.
Key Insight: Nilpotence is strict upper-triangularity in a suitable basis. □

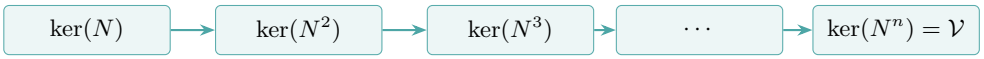


Figure 36: Extending bases through the kernel filtration makes N move vectors to earlier layers.

Perspective 3.2.60. *Eigenvalue Plus Nilpotent Part*
On $\mathcal{G}_\lambda(T)$, the operator T behaves as $\lambda I_\mathcal{V}$ plus the nilpotent operator $T - \lambda I_\mathcal{V}$. Generalized eigenvectors are therefore ordinary eigenvectors enriched by nilpotent chain data.

Theorem 3.2.61. *Primary Decomposition over $\mathbb{C}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dim over $\mathbb{C}$, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T. Then

(i) $\mathcal{V} = \mathcal{G}_{\lambda_1}(T) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T)$;
(ii) each $(T - \lambda_j I_\mathcal{V})|_{\mathcal{G}_{\lambda_j}(T)}$ *is nilpotent.*

Proof. Let $n = \dim(\mathcal{V})$.

(i) Proceed by induction on n. If $n = 1$, then T has single eigenvalue λ_1 and $\mathcal{V} = \mathcal{G}_{\lambda_1}(T)$.
Assume $n > 1$. Let $S = T - \lambda_1 I_\mathcal{V}$, $\mathcal{U} := \operatorname{range}(S^n)$.
By Theorems 3.2.52 and 3.2.53, $\mathcal{V} = \ker(S^n) \oplus \operatorname{range}(S^n) = \mathcal{G}_{\lambda_1}(T) \oplus \mathcal{U}$. The subspace $\mathcal{U}$ is T-invariant by Proposition 3.2.41, and $\dim(\mathcal{U}) < n$ because $\mathcal{G}_{\lambda_1}(T) \neq \{\mathbf{0}\}$. If $\mathcal{U} = \{\mathbf{0}\}$, then (i) follows.
Otherwise distinct eigenvalues of $T|_\mathcal{U}$ are $\lambda_2, \ldots, \lambda_m$: every eigenvalue of $T|_\mathcal{U}$ is an eigenvalue of T but not λ_1, while an eigenvector $\mathbf{v}$ of T for λ_j, $j \geq 2$, satisfies $S^n\mathbf{v} = (\lambda_j - \lambda_1)^n\mathbf{v}$ and hence lies in $\mathcal{U}$.
Induction gives $\mathcal{U} = \mathcal{G}_{\lambda_2}(T|_\mathcal{U}) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T|_\mathcal{U})$. For $j \geq 2$, $\mathcal{G}_{\lambda_j}(T|_\mathcal{U}) = \mathcal{G}_{\lambda_j}(T)$.
The inclusion $\subseteq$ follows from the definition. Conversely, if $\mathbf{v} \in \mathcal{G}_{\lambda_j}(T)$, write $\mathbf{v} = \mathbf{g} + \mathbf{u}$ with $\mathbf{g} \in \mathcal{G}_{\lambda_1}(T)$ and $\mathbf{u} \in \mathcal{U}$. For some r, $(T - \lambda_j I_\mathcal{V})^r\mathbf{v} = \mathbf{0}$. Since both summands are T-invariant, the two terms $(T - \lambda_j I_\mathcal{V})^r\mathbf{g}$ and $(T - \lambda_j I_\mathcal{V})^r\mathbf{u}$ remain in the two direct summands and have sum zero, so both are zero. Thus $\mathbf{g} \in \mathcal{G}_{\lambda_1}(T) \cap \mathcal{G}_{\lambda_j}(T) = \{\mathbf{0}\}$ by Theorem 3.2.58, and $\mathbf{v} = \mathbf{u} \in \mathcal{G}_{\lambda_j}(T|_\mathcal{U})$. Hence (i) follows.

(ii) By Proposition 3.2.54, the restriction is an operator on $\mathcal{G}_{\lambda_j}(T)$. By Theorem 3.2.53, $(T - \lambda_j I_\mathcal{V})^n\mathbf{v} = \mathbf{0}$ $\forall \mathbf{v} \in \mathcal{G}_{\lambda_j}(T)$. Hence nth power of $(T - \lambda_j I_\mathcal{V})|_{\mathcal{G}_{\lambda_j}(T)}$ is zero.

Structural Commentary: Peel off one generalized eigenspace, then induct on the invariant range.
Key Insight: Stable kernel-range splitting isolates one eigenvalue at a time. □

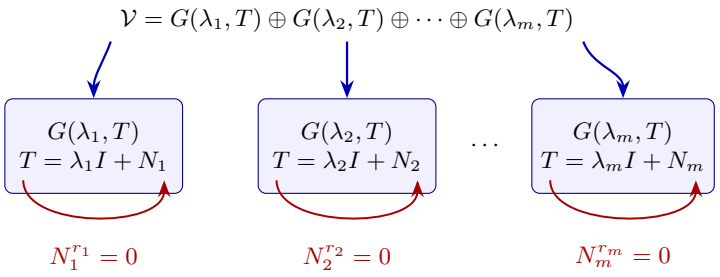


Figure 37: Primary decomposition: on $\mathcal{G}_{\lambda_j}(T)$, $T = \lambda_j I$ plus nilpotent.

Corollary 3.2.62. *Basis of Generalized Eigenvectors*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and $T \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{V}$ has a generalized-eigenvector basis for T.

Proof. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T. By Theorem 3.2.61, $\mathcal{V} = \mathcal{G}_{\lambda_1}(T) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T)$. Choose a basis of each generalized eigenspace and concatenate the bases. Directness makes the concatenated list a basis of $\mathcal{V}$, and every nonzero vector in each list is a generalized eigenvector.
Structural Commentary: A direct sum basis is obtained by choosing bases in each summand.
Key Insight: Bases of the direct-sum pieces concatenate. □

Definition 3.2.63. *Multiplicity of an Eigenvalue*
Let $\mathcal{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$.
The *multiplicity* of an eigenvalue λ of T is $\dim \mathcal{G}_\lambda(T) = \dim \ker\big((T - \lambda I_\mathcal{V})^{\dim(\mathcal{V})}\big)$.
This is the algebraic multiplicity in generalized-eigenspace language.

Proposition 3.2.64. *Sum of Multiplicities*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional complex vector space and let $T \in \mathcal{L}(\mathcal{V})$. Then the sum of the multiplicities of all eigenvalues of T equals $\dim(\mathcal{V})$.

Proof. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T. By Theorem 3.2.61, $\mathcal{V} = \mathcal{G}_{\lambda_1}(T) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T)$.
Taking dimensions, by Corollary 1.2.11, $\dim(\mathcal{V}) = \sum_{j=1}^{m} \dim \mathcal{G}_{\lambda_j}(T)$, exactly the sum of the multiplicities.
Structural Commentary: Multiplicity is dimension of the generalized-eigenspace summand.
Key Insight: Multiplicities are dimensions of the direct-sum pieces. □

Theorem 3.2.65. *Block Diagonal Form with Upper-Triangular Primary Blocks*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, and let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of $T \in \mathcal{L}(\mathcal{V})$, with multiplicities $d_1, \dots, d_m$. Then T has a matrix of the form

$$\begin{bmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{bmatrix},$$

where each $A_j \in \mathrm{Mat}(d_j, d_j, \mathbb{C})$ is upper triangular with diagonal $\lambda_j, \dots, \lambda_j$.

Proof. Set $\mathcal{U}_j = \mathcal{G}_{\lambda_j}(T)$ and $N_j = (T - \lambda_j I_\mathcal{V})|_{\mathcal{U}_j}$.
By Theorem 3.2.61, the $\mathcal{U}_j$ form a T-invariant direct-sum decomposition and each N_j is nilpotent.
Choose a basis of each $\mathcal{U}_j$ in which N_j is strictly upper triangular. Then $T|_{\mathcal{U}_j} = \lambda_j I_{\mathcal{U}_j} + N_j$ has an upper triangular matrix with diagonal $\lambda_j, \dots, \lambda_j$. Concatenating these bases gives the desired block diagonal matrix.
Structural Commentary: Primary decomposition gives the blocks; nilpotent triangularisation shapes them.
Key Insight: On each generalized eigenspace, T is eigenvalue plus nilpotent. □

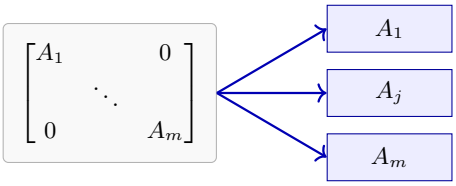


Figure 38: Primary blocks: $A_j = \lambda_j I + N_j$, with N_j strictly upper triangular.

3.3 Jordan and Canonical Forms

Jordan and canonical forms turn primary decomposition into explicit blocks. Jordan chains, complexification, and real block forms organize the nilpotent and real spectral structure.

Definition 3.3.1. *Jordan Chains, Blocks, and Matrices*
For $k \geq 1$, the *nilpotent Jordan block* $J_k(0)$ is the $k \times k$ matrix with 1 on the first superdiagonal and 0 elsewhere:

$$J_k(0) = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 & 1 \\ 0 & \cdots & \cdots & 0 & 0 \end{bmatrix}.$$

For $\lambda \in \mathbb{C}$, set $J_k(\lambda) := \lambda I_k + J_k(0)$. A *Jordan matrix* is a direct sum of Jordan blocks. A basis is a *Jordan basis* for an operator T if the matrix of T in that basis is a Jordan matrix.

Perspective 3.3.2. *Jordan Strings as Columns*
A length-k Jordan string for A at λ is $(\mathbf{x}_1, \dots, \mathbf{x}_k)$ with $(A - \lambda I)\mathbf{x}_1 = \mathbf{0}$; $(A - \lambda I)\mathbf{x}_j = \mathbf{x}_{j-1}$, $2 \le j \le k$.
For $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_k]$, $AX = XJ_k(\lambda)$.
If all strings form the columns of a basis matrix M, then $AM = MJ$ and $M^{-1}AM = J$.

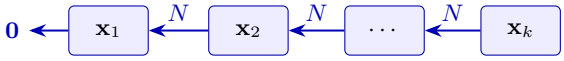


Figure 39: Jordan chain for $N = A - \lambda I$; columns give $AX = XJ_k(\lambda)$.

Method 3.3.3. *Filippov Column-Space Jordan Strings*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$, let $\lambda \in \mathrm{spec}(A)$, and set $N = A - \lambda I_n$ on the generalized eigenspace $\mathcal{G}_\lambda(A) = \ker(N^n)$.

(i) Work first inside $\mathrm{range}(N) \cap \mathcal{G}_\lambda(A)$, where N is again nilpotent.
(ii) Build Jordan strings for this smaller nilpotent operator.
(iii) For each string ending in $\mathbf{u}$, solve $N\mathbf{v} = \mathbf{u}$ in $\mathcal{G}_\lambda(A)$ and lift the string by appending $\mathbf{v}$.
(iv) Add enough vectors from $\ker(N)$ to complete a basis of $\mathcal{G}_\lambda(A)$; these added vectors are length-one strings.
(v) Repeat the process recursively until all of $\mathcal{G}_\lambda(A)$ is covered, then concatenate the strings for all eigenvalues.

Theorem 3.3.5 justifies the lifts; Perspective 3.3.2 turns the resulting columns into $AM = MJ$.

Proposition 3.3.4. *Powers of a Jordan Block*
For $p \ge 0$, $J_k(\lambda)^p = \sum_{r=0}^{\min(p,k-1)} \binom{p}{r} \lambda^{p-r} J_k(0)^r$. Thus $J_k(\lambda)^p \to 0$ if and only if $|\lambda| < 1$.

Proof.
($\Rightarrow$) If $J_k(\lambda)^p \to 0$, then its diagonal entries λ^p tend to 0, so $|\lambda| < 1$.
($\Leftarrow$) If $|\lambda| < 1$, then for each fixed $0 \le r < k$ the scalar $\binom{p}{r}\lambda^{p-r}$ tends to 0, because a polynomial in p is dominated by the exponential decay of $|\lambda|^p$.

Structural Commentary: Jordan powers are exponential-polynomial terms.
Key Insight: Polynomial growth cannot defeat exponential decay when $|\lambda| < 1$. □

Theorem 3.3.5. *Nilpotent Jordan Chain Basis*
Let $\mathcal{V}$ be finite-dim, $N \in \mathcal{L}(\mathcal{V})$ be nilpotent. Then $\exists$ vectors $\mathbf{v}_1, \dots, \mathbf{v}_r \in \mathcal{V}$ and nonnegative integers $m_1, \dots, m_r$ such that $\mathcal{B}_N = \left(N^{m_1}\mathbf{v}_1, \dots, N\mathbf{v}_1, \mathbf{v}_1, \dots, N^{m_r}\mathbf{v}_r, \dots, N\mathbf{v}_r, \mathbf{v}_r\right)$ is a basis of $\mathcal{V}$ and $N^{m_j+1}\mathbf{v}_j = \mathbf{0}$ for every j.

Proof. Induct on $n = \dim(\mathcal{V})$. If $N = 0$, any basis works with all $m_j = 0$.
Now assume $N \neq 0$. Then N is not injective, so $\mathrm{range}(N)$ is a proper subspace, and $N|_{\mathrm{range}(N)}$ is nilpotent.
By induction, $\mathrm{range}(N)$ has a chain basis $\left(N^{m_1}\mathbf{u}_1, \dots, N\mathbf{u}_1, \mathbf{u}_1, \dots, N^{m_r}\mathbf{u}_r, \dots, N\mathbf{u}_r, \mathbf{u}_r\right)$ with $N^{m_j+1}\mathbf{u}_j = \mathbf{0}$.
Choose $\mathbf{v}_j \in \mathcal{V}$ such that $N\mathbf{v}_j = \mathbf{u}_j$, and lift the chains to $\mathcal{L} = \left(N^{m_1+1}\mathbf{v}_1, \dots, N\mathbf{v}_1, \mathbf{v}_1, \dots, N^{m_r+1}\mathbf{v}_r, \dots, N\mathbf{v}_r, \mathbf{v}_r\right)$.
The list $\mathcal{L}$ is linearly independent. Indeed, applying N to a dependence among vectors in $\mathcal{L}$ gives a dependence among the range-chain basis, so all coefficients except possibly those on the leading vectors $N^{m_j+1}\mathbf{v}_j$ vanish.
The remaining relation is a relation among $N^{m_j+1}\mathbf{v}_j = N^{m_j}\mathbf{u}_j$, which are members of the range-chain basis, so those coefficients vanish too. Also $N(\mathrm{span}(\mathcal{L})) = \mathrm{range}(N)$: each range-chain basis vector $N^q\mathbf{u}_j$ is $N(N^q\mathbf{v}_j)$ for the corresponding lifted vector. Extend $\mathcal{L}$ to a basis by adding $\mathbf{w}_1, \dots, \mathbf{w}_s$. For each i, choose $\mathbf{x}_i \in \mathrm{span}(\mathcal{L})$ such that $N\mathbf{x}_i = N\mathbf{w}_i$, and set $\mathbf{z}_i = \mathbf{w}_i - \mathbf{x}_i$. Then $N\mathbf{z}_i = \mathbf{0}$. Since each $\mathbf{x}_i$ lies in $\mathrm{span}(\mathcal{L})$, replacing $\mathbf{w}_i$ by $\mathbf{z}_i$ preserves the span and linear independence of the full basis. Thus $\mathcal{L}, \mathbf{z}_1, \dots, \mathbf{z}_s$ is the desired chain basis.
Structural Commentary: Build chains in $\mathrm{range}(N)$, lift them once, then add kernel chains.
Key Insight: Applying N turns lifted-chain dependences into known range-chain dependences. □

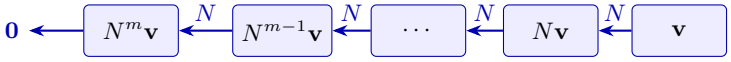


Figure 40: A nilpotent chain shifts left under N; in this basis order it becomes one Jordan block.

Theorem 3.3.6. *Jordan Form over $\mathbb{C}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{V}$ has a Jordan basis for T.

Proof. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T. By Theorem 3.2.61, $\mathcal{V} = \mathcal{G}_{\lambda_1}(T) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T)$, and each $N_j := (T - \lambda_j I_\mathcal{V})|_{\mathcal{G}_{\lambda_j}(T)}$ is nilpotent. Apply Theorem 3.3.5 to N_j on each generalized eigenspace.
The resulting chain basis of $\mathcal{G}_{\lambda_j}(T)$ is a basis of that summand, and the direct-sum decomposition makes the

concatenation of these bases a basis of $\mathcal{V}$. In a chain of length r for N_j, the matrix of N_j is $J_r(0)$, so the matrix of $T|_{\mathcal{G}_{\lambda_j}(T)} = \lambda_j I + N_j$ on the same chain is $J_r(\lambda_j)$. Therefore the concatenated basis is a Jordan basis for T.
Structural Commentary: Primary pieces decompose into nilpotent chains.
Key Insight: Jordan form is "eigenvalue plus nilpotent chains" on every generalized eigenspace. □

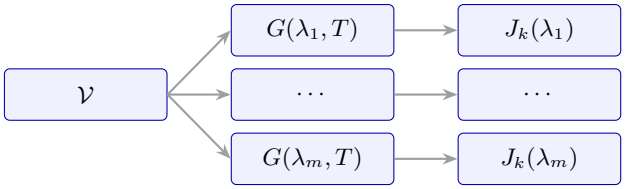


Figure 41: Primary pieces $\mathcal{G}_{\lambda_j}(T)$ give nilpotent chains, hence Jordan blocks.

Corollary 3.3.7. *Diagonalizability in Jordan Form*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have Jordan form J. For each eigenvalue λ, $\dim E_\lambda(A)$ equals the number of λ-blocks in J. Hence A is diagonalizable if and only if every Jordan block has size 1.

Proof. Write $A = SJS^{-1}$. Then $S : E_\lambda(J) \to E_\lambda(A)$ is an isomorphism. A block $J_k(\mu)$ contributes to $E_\lambda(J)$ only when $\mu = \lambda$, and then $\dim \ker(J_k(\lambda) - \lambda I_k) = \dim \ker J_k(0) = 1$.
Thus $\dim E_\lambda(A)$ is the number of λ-blocks. By Theorem 3.2.17, diagonalizability means geometric and algebraic multiplicities agree for every λ; here this holds exactly when every block has size 1.
Structural Commentary: Each Jordan block contributes one eigenvector direction.
Key Insight: Missing eigenvectors are measured by block sizes greater than 1. □

Lemma 3.3.8. *Powers of a Nilpotent Jordan Block*
For $k \geq 1$ and $p \geq 0$, $\mathrm{rank}\big(J_k(0)^p\big) = \max(k - p, 0)$ and $\dim \ker\big(J_k(0)^p\big) = \min(p, k)$.

Proof. Let $N = J_k(0)$. In the standard basis, $N\mathbf{e}_1 = \mathbf{0}$ and $N\mathbf{e}_j = \mathbf{e}_{j-1}$ for $2 \leq j \leq k$. Iterating gives

$$N^p \mathbf{e}_j = \begin{cases} \mathbf{0}, & j \leq p, \\ \mathbf{e}_{j-p}, & j > p. \end{cases}$$

If $p < k$, the image is $\mathrm{span}(\mathbf{e}_1, \ldots, \mathbf{e}_{k-p})$, so the rank is $k - p$; if $p \geq k$, the image is $\{\mathbf{0}\}$, so the rank is 0.
Hence $\mathrm{rank}(N^p) = \max(k - p, 0)$. Rank-nullity gives $\dim \ker(N^p) = k - \max(k - p, 0) = \min(p, k)$.
Structural Commentary: A nilpotent Jordan block is a left-shift on the standard basis.
Key Insight: Each power deletes one more leading layer of the chain. □

Definition 3.3.9. *Weyr Characteristic*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ and $\lambda \in \mathrm{spec}(A)$.
For $p \geq 1$, define $w_p(\lambda, A) := \mathrm{rank}\big((A - \lambda I_n)^{p-1}\big) - \mathrm{rank}\big((A - \lambda I_n)^p\big)$.
The positive part of the sequence $w_1, w_2, \ldots$ is the *Weyr characteristic* of A associated with λ.

Theorem 3.3.10. *Jordan Block Sizes from Rank Drops*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$, $\lambda \in \mathrm{spec}(A)$, and $p \geq 1$. Then $w_p(\lambda, A)$ is the number of λ-Jordan blocks of size at least p; hence $w_p(\lambda, A) - w_{p+1}(\lambda, A)$ is the number of λ-blocks of size exactly p.

Proof. Rank is similarity-invariant, so assume A is in Jordan form.
Blocks $J_k(\mu)$ with $\mu \neq \lambda$ contribute 0 to the rank drop because $J_k(\mu) - \lambda I_k$ is invertible.
A block $J_k(\lambda)$ contributes $\mathrm{rank}\big(J_k(0)^{p-1}\big) - \mathrm{rank}\big(J_k(0)^p\big)$, which is 1 exactly when $p \leq k$ by Lemma 3.3.8.
Summing over blocks proves the first claim; subtracting the $p + 1$ count gives the exact-size count.
Structural Commentary: Rank drops count surviving chain lengths.
Key Insight: A length-k block contributes to drops $1, \ldots, k$. □

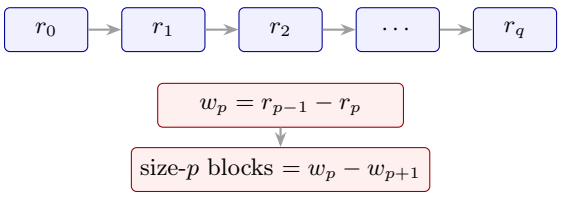


Figure 42: For $r_p = \mathrm{rank}((A - \lambda I)^p)$, drops count surviving chains; $w_p - w_{p+1}$ counts size-p blocks.

Method 3.3.11. *Computing Jordan Blocks by Rank Drops*
For each eigenvalue λ of $A \in \mathrm{Mat}(n, n, \mathbb{C})$:

(i) Compute $r_p = \operatorname{rank}\big((A - \lambda I_n)^p\big)$ for $p = 0, 1, 2, \ldots$ until the differences stabilize at 0.
(ii) Set $w_p = r_{p-1} - r_p$. Then w_p is the number of λ-blocks of size at least p.
(iii) The number of blocks $J_p(\lambda)$ is $w_p - w_{p+1}$.
(iv) Assemble the Jordan form by taking the direct sum over all eigenvalues.

Theorem 3.3.12. *Nonzero Eigenvalues of AB and BA*
Let $A \in \mathrm{Mat}(m, n, \mathbb{C})$, $B \in \mathrm{Mat}(n, m, \mathbb{C})$, and suppose $n \geq m$. Then $p_{BA}(z) = z^{n-m} p_{AB}(z)$. Consequently, for $\lambda \neq 0$, $m_{AB}(\lambda) = m_{BA}(\lambda)$. If $m_{AB}(0) = k$, then $m_{BA}(0) = k + n - m$.

Proof. For $z \neq 0$, Sylvester's determinant identity gives

$$\det(zI_n - BA) = z^n \det(I_n - z^{-1}BA) = z^n \det(I_m - z^{-1}AB) = z^{n-m} \det(zI_m - AB).$$

Both sides are polynomials in z, so the identity holds for all z. The claimed multiplicities follow by factoring p_{AB} and comparing with $p_{BA} = z^{n-m} p_{AB}$.
Structural Commentary: Sylvester's determinant identity moves the rectangular product across the determinant.
Key Insight: Passing from AB to BA only adds the necessary extra zero roots when the sizes differ. □

Theorem 3.3.13. *Nonzero Jordan Blocks of AB and BA*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$. For every $\lambda \neq 0$, the Jordan blocks of AB and BA at λ coincide.

Proof. Set $X = \begin{bmatrix} AB & A \\ 0 & 0 \end{bmatrix}$ and $Y = \begin{bmatrix} 0 & A \\ 0 & BA \end{bmatrix}$. Then $\begin{bmatrix} I_n & 0 \\ B & I_n \end{bmatrix} X \begin{bmatrix} I_n & 0 \\ -B & I_n \end{bmatrix} = Y$, so X and Y are similar.
For $\lambda \neq 0$ and $p \geq 1$, the shifted powers have block triangular forms

$$(X - \lambda I_{2n})^p = \begin{bmatrix} (AB - \lambda I_n)^p & * \\ 0 & (-\lambda)^p I_n \end{bmatrix}, \qquad (Y - \lambda I_{2n})^p = \begin{bmatrix} (-\lambda)^p I_n & * \\ 0 & (BA - \lambda I_n)^p \end{bmatrix}$$

The diagonal block $(-\lambda)^p I_n$ is invertible.
Block Gaussian elimination by invertible row and column operations removes the starred block, so

$$\operatorname{rank}\big((X - \lambda I_{2n})^p\big) = n + \operatorname{rank}\big((AB - \lambda I_n)^p\big),$$
$$\operatorname{rank}\big((Y - \lambda I_{2n})^p\big) = n + \operatorname{rank}\big((BA - \lambda I_n)^p\big).$$

Since X and Y are similar, these left sides agree for all p.
Hence the shifted rank drops of AB and BA at λ agree, so Theorem 3.3.10 gives the same Jordan blocks.
Structural Commentary: Similar block embeddings make the nonzero shifted ranks match.
Key Insight: At $\lambda \neq 0$, the extra block is invertible and splits off. □

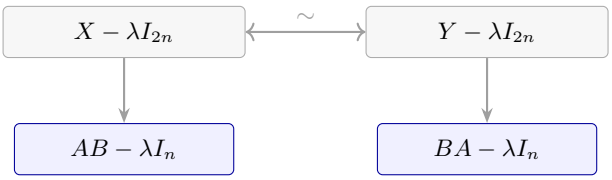


Figure 43: For $\lambda \neq 0$, the auxiliary similarity reduces the comparison to shifted ranks of AB and BA.

Corollary 3.3.14. *Uniqueness of Jordan Form*
If J and J' are Jordan matrices similar to $A \in \mathrm{Mat}(n, n, \mathbb{C})$, they differ only by a permutation of Jordan blocks.

Proof. Similarity preserves spectra and the ranks of $(X - \lambda I_n)^p$. Thus J and J' have the same Weyr characteristic for each eigenvalue. Theorem 3.3.10 then gives the same block multiplicities, so only block order may differ.
Structural Commentary: Weyr data is similarity-invariant and fixes block multiplicities.
Key Insight: Jordan form is unique modulo block order. □

Corollary 3.3.15. *Similarity Criterion for AB and BA*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$. AB is similar to BA if and only if $\operatorname{rank}\big((AB)^p\big) = \operatorname{rank}\big((BA)^p\big)$, $p = 1, \ldots, n$.

Proof. If AB and BA are similar, then their powers are similar and have equal rank.
Conversely, Theorem 3.3.13 gives equality of all nonzero Jordan blocks. The displayed rank equalities, together with the trivial equality at $p = 0$, give the same Weyr characteristic at $\lambda = 0$ by Theorem 3.3.10. Hence the nilpotent Jordan blocks also agree, so the two Jordan forms are identical up to block order.
Structural Commentary: Nonzero blocks agree automatically; only the nilpotent blocks must be tested.
Key Insight: Powers of AB and BA measure exactly the zero-Jordan-block data. □

Method 3.3.16. *Checking Whether AB and BA Are Similar*
For $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$:

(i) The nonzero Jordan blocks of AB and BA already match.
(ii) Compute $\operatorname{rank}\big((AB)^p\big)$ and $\operatorname{rank}\big((BA)^p\big)$ for $p = 1, \dots, n$.
(iii) If all ranks agree, then $AB \sim BA$; if some rank differs, then only the nilpotent Jordan structure differs.

Definition 3.3.17. *Nonderogatory Matrix*
A matrix $A \in \mathrm{Mat}(n, n, \mathbb{C})$ is *nonderogatory* if its Jordan form has exactly one Jordan block for each distinct eigenvalue. Equivalently, $\dim E_\lambda(A) = 1$ for every $\lambda \in \operatorname{spec}(A)$.

Lemma 3.3.18. *Commutant of One Jordan Block*
If $J = J_m(\lambda)$ and $XJ = JX$, then $X = p(J)$ for some polynomial p with $\deg p < m$.

Proof. It suffices to treat $N = J_m(0)$, since $J_m(\lambda) = \lambda I_m + N$.
The vector $\mathbf{e}_m$ is cyclic for N: $(\mathbf{e}_m, N\mathbf{e}_m, \dots, N^{m-1}\mathbf{e}_m)$ is the standard basis in reverse order.
Choose the unique polynomial p with $\deg p < m$ such that $p(N)\mathbf{e}_m = X\mathbf{e}_m$.
For every $r = 0, \dots, m-1$, $p(N)N^r\mathbf{e}_m = N^r p(N)\mathbf{e}_m = N^r X\mathbf{e}_m = XN^r\mathbf{e}_m$.
Thus $p(N)$ and X agree on a basis, so $X = p(N) = q(J)$ for $q(z) = p(z - \lambda)$.
Structural Commentary: A single Jordan block is generated by one cyclic vector.
Key Insight: The cyclic vector determines the commutant. □

Theorem 3.3.19. *Nonderogatory Commutants Are Polynomials*
If $A \in \mathrm{Mat}(n, n, \mathbb{C})$ is nonderogatory, then $\operatorname{Comm}(A) = \{p(A) : p \in \mathbb{C}[z]\}$.

Proof. The inclusion $\{p(A) : p \in \mathbb{C}[z]\} \subseteq \operatorname{Comm}(A)$ follows from Proposition 3.2.5.
For reverse inclusion, first replace A by its Jordan form $J = J_{m_1}(\lambda_1) \oplus \cdots \oplus J_{m_d}(\lambda_d)$, where λ_i are distinct.
If $XJ = JX$ and $X = [X_{ij}]$ is partitioned conformally with J, the block equation is $X_{ij}J_{m_j}(\lambda_j) = J_{m_i}(\lambda_i)X_{ij}$.
Corollary 3.2.45 gives $X_{ij} = 0$ for $i \neq j$. Each diagonal block X_{ii} commutes with $J_{m_i}(\lambda_i)$, so Lemma 3.3.18 gives $X_{ii} = p_i(J_{m_i}(\lambda_i))$ for some polynomial p_i with $\deg p_i < m_i$. The ideals generated by $(z - \lambda_i)^{m_i}$ are pairwise coprime, so Hermite interpolation gives a polynomial p with $p \equiv p_i \pmod{(z - \lambda_i)^{m_i}}$ for every i.
On the ith block, $p - p_i = (z - \lambda_i)^{m_i} q_i$ for some q_i, and $(J_{m_i}(\lambda_i) - \lambda_i I)^{m_i} = 0$. Hence $p(J_{m_i}(\lambda_i)) = p_i(J_{m_i}(\lambda_i)) = X_{ii}$ for every i, so $X = p(J)$. Conjugating back to A proves the claim.
Structural Commentary: Disjoint eigenvalue blocks force block diagonality; one Jordan block per eigenvalue makes each diagonal commutant polynomial.
Key Insight: Hermite interpolation glues the separate block polynomials into one polynomial in A. □

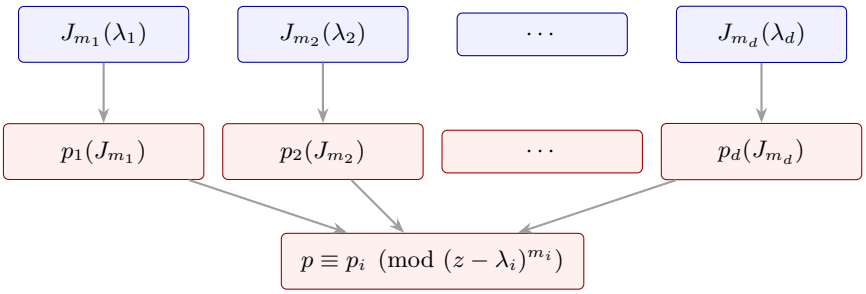


Figure 44: Blockwise commutant polynomials glue by interpolation.

Theorem 3.3.20. *Conjugate Similarity Criterion*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. Then A is similar to $\overline{A}$ if and only if each $J_k(\lambda)$ occurs in the Jordan form of A with the same multiplicity as $J_k(\overline{\lambda})$.

Proof. Let J be a Jordan form of A. Then $\overline{J}$ is a Jordan form of $\overline{A}$, obtained by replacing each $J_k(\lambda)$ by $J_k(\overline{\lambda})$.
($\Rightarrow$) If A is similar to $\overline{A}$, then J and $\overline{J}$ are similar.
By Corollary 3.3.14, their Jordan blocks have same multiplicities.
($\Leftarrow$) If the multiplicities agree, then J and $\overline{J}$ differ only by a permutation of blocks. Hence A is similar to $\overline{A}$.
Structural Commentary: Conjugation changes eigenvalue labels but not block sizes.
Key Insight: Non-real Jordan blocks must occur in conjugate pairs. □

Corollary 3.3.21. *Hermitian Products and Conjugate Products*
Let $A, H, K \in \mathrm{Mat}(n, n, \mathbb{C})$, and suppose H and K are Hermitian.
Then HK is similar to KH, and $A\overline{A}$ is similar to $\overline{A}A$.

Proof. For each $p = 1, \dots, n$, adjoints preserves rank, $((HK)^p)^* = (KH)^p$. Hence $\operatorname{rank}\big((HK)^p\big) = \operatorname{rank}\big((KH)^p\big)$. Similarly, entrywise conjugation preserves rank and gives $\overline{(A\overline{A})^p} = (\overline{A}A)^p$, so $\operatorname{rank}\big((A\overline{A})^p\big) = \operatorname{rank}\big((\overline{A}A)^p\big)$. Apply Corollary 3.3.15 in both cases.
Structural Commentary: Adjoint and conjugation preserve rank and reverse product order.
Key Insight: The rank-power test turns order reversal into similarity. □

Definition 3.3.22. *Reversal Matrix*
For $k \geq 1$, *reversal matrix* $R_k \in \operatorname{Mat}(k, k, \mathbb{C})$ defined $(R_k)_{ij} = \begin{cases} 1, & i+j = k+1, \\ 0, & i+j \neq k+1 \end{cases}$, thus $R_k^t = R_k$, $R_k^2 = I_k$.

Lemma 3.3.23. *Reversal Conjugates a Jordan Block to Its Transpose*
For every $k \geq 1$ and $\lambda \in \mathbb{C}$, $J_k(\lambda)^t = R_k J_k(\lambda) R_k$. Moreover, $R_k J_k(\lambda)$ and $J_k(\lambda) R_k$ are symmetric.

Proof. Let $N = J_k(0)$. Since R_k reverses standard basis order, $R_k N R_k = N^t$.
Adding scalar gives $R_k J_k(\lambda) R_k = J_k(\lambda)^t$. As $R_k = R_k^t = R_k^{-1}$, this identity gives $J_k(\lambda)^t R_k = R_k J_k(\lambda)$ and $R_k J_k(\lambda)^t = J_k(\lambda) R_k$, so $R_k J_k(\lambda)$ and $J_k(\lambda) R_k$ are symmetric.
Structural Commentary: Reversal changes the superdiagonal shift into the subdiagonal shift.
Key Insight: Transposition reads the same Jordan chain backwards. □

Theorem 3.3.24. *Transpose Similarity and Symmetric Factorisations*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$. Then:

(i) A is similar to A^t via a symmetric similarity matrix.
(ii) $A = BC = DE$, where B, C, D, E are symmetric and B, E are invertible.

Proof. Choose a Jordan form $A = SJS^{-1}$, and let R be the direct sum of the reversal matrices matching the Jordan blocks of J. By Lemma 3.3.23, $J^t = RJR$, $R^t = R$, $R^2 = I_n$, and both RJ and JR are symmetric.

(i) Set $P = (S^{-1})^t R S^{-1}$. Then $P^t = P$, $P^{-1} = SRS^t$, and $PAP^{-1} = (S^{-1})^t RJRS^t = (S^{-1})^t J^t S^t = A^t$.
(ii) Using $R^2 = I_n$, $A = (SRS^t)\big((S^{-1})^t RJS^{-1}\big)$ and $A = (SJRS^t)\big((S^{-1})^t RS^{-1}\big)$.
The products equal SJS^{-1} because $S^t(S^{-1})^t = I_n$ and $R^2 = I_n$. All four factors are symmetric because R, RJ, JR are symmetric; SRS^t and $(S^{-1})^t RS^{-1}$ are invertible.

Structural Commentary: Reduce to Jordan blocks, reverse chains, then conjugate back.
Key Insight: Symmetric involutions produce both transpose similarity and symmetric factors. □

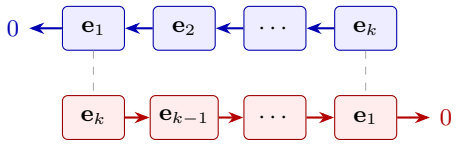


Figure 45: Reversal gives $R_k J_k(0) R_k = J_k(0)^t$: the same chain read backwards.

Theorem 3.3.25. *Conjugate Similarity and Hermitian Products*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$. The following are equivalent:

(i) A is similar to $\overline{A}$.
(ii) A is similar to A^*.
(iii) A is similar to A^* via a Hermitian similarity matrix.
(iv) $A = HK = LM$, where H, K, L, M are Hermitian, and H, M are invertible.
(v) $A = HK$ for some Hermitian matrices H, K.

Proof.

(i) $\Rightarrow$ *(ii)* Theorem 3.3.24 applied to $\overline{A}$ gives $\overline{A} \sim (\overline{A})^t = A^*$. Thus $A \sim A^*$.
(ii) $\Rightarrow$ *(iii)* Choose invertible S such that $A = SA^*S^{-1}$, so $AS = SA^*$. For $\theta \in \mathbb{R}$, put $S_\theta = e^{i\theta} S$.
Then $AS_\theta = S_\theta A^*$ and, by taking adjoints, $AS_\theta^* = S_\theta^* A^*$. Hence $A(S_\theta + S_\theta^*) = (S_\theta + S_\theta^*)A^*$.
The matrix $H_\theta = S_\theta + S_\theta^*$ is Hermitian. Also $S_\theta^{-1} H_\theta = I_n + e^{-2i\theta} S^{-1} S^*$. Thus H_θ is singular exactly when $-e^{2i\theta}$ is an eigenvalue of $S^{-1}S^*$. There are only finitely many such values, so choose θ for which H_θ is invertible. Then $A = H_\theta A^* H_\theta^{-1}$.
(iii) $\Rightarrow$ *(iv)* Let R be invertible Hermitian and satisfy $A = RA^*R^{-1}$. Then $AR = RA^*$ and $R^{-1}A = A^*R^{-1}$.
Thus A^*R^{-1} and RA^* are Hermitian, and $A = R(A^*R^{-1}) = (RA^*)R^{-1}$.
This gives the two asserted Hermitian factorizations, with the first factor in the first product and the second factor in the second product invertible.

(*iv*) $\Rightarrow$ (*v*) The first factorization in (iv) already has the form $A = HK$ with H and K Hermitian.
(*v*) $\Rightarrow$ (*i*) If $A = HK$ with H, K Hermitian, then Corollary 3.3.21 gives $HK \sim KH$.
By Theorem 3.3.24, $KH \sim (KH)^t = H^t K^t = \overline{H}\,\overline{K} = \overline{A}$. Hence $A \sim \overline{A}$.

Structural Commentary: The equivalences move between conjugation, adjoints, and Hermitian products by combining transpose similarity with a Hermitianized similarity matrix.
Key Insight: A phase-rotated average $e^{i\theta}S + e^{-i\theta}S^*$ turns an arbitrary similarity to A^* into a Hermitian one. □

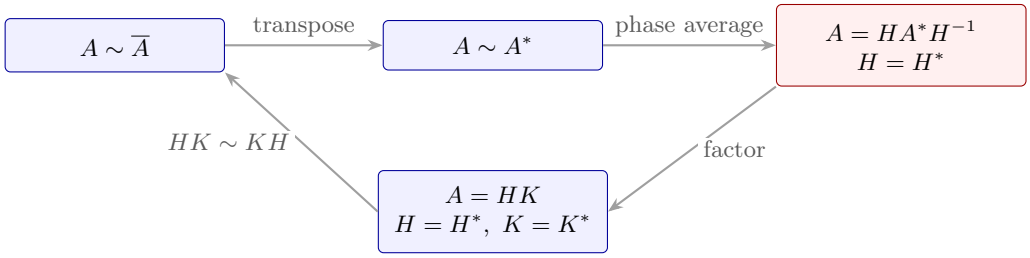


Figure 46: Conjugate similarity, adjoint similarity, and Hermitian product factorisations are equivalent.

Definition 3.3.26. *Complexification of a Real Vector Space*
Let $\mathcal{V}$ be a real vector space. The *complexification* of $\mathcal{V}$ is $\mathcal{V}_{\mathbb{C}} := \mathcal{V} \times \mathcal{V}$, where $(\mathbf{u}, \mathbf{v})$ is written as $\mathbf{u} + i\mathbf{v}$. Addition and complex scalar multiplication are defined by

$$(\mathbf{u}_1 + i\mathbf{v}_1) + (\mathbf{u}_2 + i\mathbf{v}_2) = (\mathbf{u}_1 + \mathbf{u}_2) + i(\mathbf{v}_1 + \mathbf{v}_2),$$
$$(a + bi)(\mathbf{u} + i\mathbf{v}) = (a\mathbf{u} - b\mathbf{v}) + i(b\mathbf{u} + a\mathbf{v}), \qquad a, b \in \mathbb{R}.$$

We identify $\mathcal{V}$ with the real subspace $\{\mathbf{u} + i\mathbf{0} : \mathbf{u} \in \mathcal{V}\}$.

Proposition 3.3.27. *Complexification Preserves Bases*
Let $\mathcal{V}$ be real. Then $\mathcal{V}_{\mathbb{C}}$ is complex, and every real basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ of $\mathcal{V}$ is a complex basis of $\mathcal{V}_{\mathbb{C}}$. Thus $\dim_{\mathbb{C}}(\mathcal{V}_{\mathbb{C}}) = \dim_{\mathbb{R}}(\mathcal{V})$.

Proof. The vector-space axioms follow componentwise from those of $\mathcal{V}$.
For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, write $\mathbf{u} = \sum_{j=1}^{n} a_j \mathbf{v}_j$ and $\mathbf{v} = \sum_{j=1}^{n} b_j \mathbf{v}_j$, $a_j, b_j \in \mathbb{R}$. Then $\mathbf{u} + i\mathbf{v} = \sum_{j=1}^{n} (a_j + ib_j)\mathbf{v}_j$, so the list spans.
If $\sum_{j=1}^{n} \lambda_j \mathbf{v}_j = \mathbf{0}$ and $\lambda_j = a_j + ib_j$, then $\sum_{j=1}^{n} a_j \mathbf{v}_j = \mathbf{0}$ and $\sum_{j=1}^{n} b_j \mathbf{v}_j = \mathbf{0}$.
Real linear independence gives $a_j = b_j = 0$ for all j, hence $\lambda_j = 0$ for all j.
Structural Commentary: Complexification keeps the basis length unchanged.
Key Insight: Separate real and imaginary parts. □

Definition 3.3.28. *Complexification of a Real Operator*
Let $\mathcal{V}$ be a real vector space and let $T \in \mathcal{L}(\mathcal{V})$.
The *complexification* of T is the operator $T_{\mathbb{C}} \in \mathcal{L}(\mathcal{V}_{\mathbb{C}})$ defined by $T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}) := T\mathbf{u} + iT\mathbf{v}$.

Proposition 3.3.29. *Complexification Preserves Operator Data*
Let $\mathcal{V}$ be a real vector space and let $T \in \mathcal{L}(\mathcal{V})$.

(i) $T_{\mathbb{C}} \in \mathcal{L}(\mathcal{V}_{\mathbb{C}})$.
(ii) $(T_{\mathbb{C}})^k(\mathbf{u} + i\mathbf{v}) = T^k\mathbf{u} + iT^k\mathbf{v}$ *for every integer* $k \geq 0$.
(iii) If β is a real basis of $\mathcal{V}$, then $[T_{\mathbb{C}}]_{\beta \leftarrow \beta} = [T]_{\beta \leftarrow \beta}$, *now regarded as a complex matrix.*

Proof.

(i) Additivity is immediate. For $\alpha = a + bi \in \mathbb{C}$,
$$T_{\mathbb{C}}\big(\alpha(\mathbf{u} + i\mathbf{v})\big) = T(a\mathbf{u} - b\mathbf{v}) + iT(b\mathbf{u} + a\mathbf{v}) = (aT\mathbf{u} - bT\mathbf{v}) + i(bT\mathbf{u} + aT\mathbf{v}) = \alpha T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}),$$
so $T_{\mathbb{C}}$ is complex-linear.

(ii) For $k = 0$, the claim is the identity. If it holds for k, then
$$(T_{\mathbb{C}})^{k+1}(\mathbf{u} + i\mathbf{v}) = T_{\mathbb{C}}(T^k\mathbf{u} + iT^k\mathbf{v}) = T^{k+1}\mathbf{u} + iT^{k+1}\mathbf{v}.$$
Thus the formula holds for all $k \geq 0$.

(iii) If $T\mathbf{v}_j = \sum_i a_{ij}\mathbf{v}_i$ with $a_{ij} \in \mathbb{R}$, then $T_{\mathbb{C}}\mathbf{v}_j = \sum_i a_{ij}\mathbf{v}_i$. Hence the coordinate matrix is unchanged.

Structural Commentary: Complexification keeps the coordinate matrix unchanged.
Key Insight: Same real entries; larger scalar field. □



Figure 47: Complexification packages two real vectors: $T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}) = T\mathbf{u} + iT\mathbf{v}$.

Method 3.3.30. *Complexifying a Real Matrix*
To use complex eigenvalues for a real matrix $A \in \mathrm{Mat}(n, n, \mathbb{R})$:

(i) Regard the same real matrix A as a complex matrix on $\mathbb{C}^n$.
(ii) Find complex eigenpairs or invariant data for this complexified operator.
(iii) If $A(\mathbf{u} + i\mathbf{v}) = (a + bi)(\mathbf{u} + i\mathbf{v})$, separate parts: $A\mathbf{u} = a\mathbf{u} - b\mathbf{v}$, $A\mathbf{v} = b\mathbf{u} + a\mathbf{v}$
(iv) Conclude that $\mathrm{span}(\mathbf{u}, \mathbf{v}) \subseteq \mathbb{R}^n$ is an invariant line or plane.

Theorem 3.3.31. *Invariant Line or Plane over $\mathbb{R}$*
Every operator on a nonzero finite-dimensional real vector space has an invariant subspace of dimension 1 *or* 2.

Proof. Let $\mathcal{V}$ be a nonzero finite-dimensional real vector space and let $T \in \mathcal{L}(\mathcal{V})$.
By Theorem 3.1.20, $T_{\mathbb{C}}$ has an eigenvalue $\lambda = a + bi$ with eigenvector $\mathbf{u} + i\mathbf{v} \neq \mathbf{0}$.
The equation $T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}) = (a + bi)(\mathbf{u} + i\mathbf{v})$ gives $T\mathbf{u} = a\mathbf{u} - b\mathbf{v}$ and $T\mathbf{v} = b\mathbf{u} + a\mathbf{v}$.
Thus $\mathcal{U} = \mathrm{span}(\mathbf{u}, \mathbf{v})$ is nonzero, has dimension 1 or 2, and is invariant under T.
Structural Commentary: A complex eigenvector decomposes into two real directions that are closed under T.
Key Insight: Nonreal eigenvalues become real invariant planes; real eigenvalues become invariant lines. □

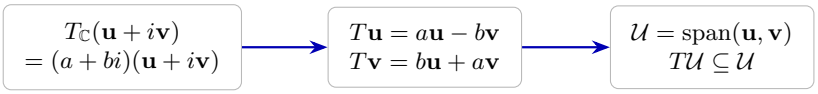


Figure 48: Separate a complex eigenpair into a real invariant plane.

Perspective 3.3.32. *Real Normal Operators Use Planes*
Over $\mathbb{R}$, normal operators need not have enough eigenvectors. The replacement for a missing conjugate eigenpair is a two-dimensional invariant plane carrying a rotation-dilation block.

Proposition 3.3.33. *Two-Dimensional Real Normal Blocks*
Let $\dim(\mathcal{V}) = 2$ *over* $\mathbb{R}$ *and let* $T \in \mathcal{L}(\mathcal{V})$.
Then T *is normal but not self-adjoint if and only if, in every orthonormal basis,* $[T] = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, $b \neq 0$.
Moreover, some orthonormal basis makes $b > 0$.

Proof.

(⇒) In an orthonormal basis, write $A = [T] = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$.
Normality gives $A^tA = AA^t$. Comparing the $(1,1)$ and $(2,2)$ entries gives $b^2 = c^2$. If $c = b$, then $A = A^t$, impossible because T is not self-adjoint. Thus $c = -b$ and $b \neq 0$. Comparing off-diagonal entries gives $-ab + bd = ab - bd$, so $d = a$. Since the original orthonormal basis was arbitrary, every orthonormal basis has the displayed form.

(⇐) A matrix $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ with $b \neq 0$ satisfies $A^tA = AA^t$ and is not symmetric. Hence T is normal but not self-adjoint. If $b < 0$, replace the second basis vector by its negative to make $b > 0$.

Structural Commentary: In real dimension 2, non-self-adjoint normality is exactly a scalar plus a rotation block.
Key Insight: Normality forces $c = -b$ and $d = a$. □

Theorem 3.3.34. *Real Normal Operator Block Form*
Let $\mathcal{V}$ *be finite-dimensional over* $\mathbb{R}$ *and let* $T \in \mathcal{L}(\mathcal{V})$. *Then* T *is normal if and only if some orthonormal basis makes* $[T]$ *block diagonal, with each diagonal block either* 1*-by-*1 *or of the form* $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, $b > 0$.

Proof.

(⇒) Induct on $n = \dim(\mathcal{V})$. The cases $n \leq 1$ are immediate.
For $n \geq 2$, Theorem 3.3.31 gives an invariant subspace $\mathcal{U}$ with $\dim(\mathcal{U}) \in \{1, 2\}$.
By Corollary 3.4.26, $\mathcal{U}^{\perp}$ is invariant under T and both restrictions are normal.
If $\dim(\mathcal{U}) = 1$, take a unit vector in $\mathcal{U}$. If $\dim(\mathcal{U}) = 2$, use the Real Spectral Theorem 3.4.13 when $T|_{\mathcal{U}}$ is

self-adjoint, giving scalar 1-by-1 blocks on $\mathcal{U}$, and use Proposition 3.3.33 otherwise.
By induction, choose the required orthonormal block basis on $\mathcal{U}^\perp$. Concatenating it with the basis chosen on $\mathcal{U}$ gives an orthonormal basis of $\mathcal{V}$ because $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^\perp$.

($\Leftarrow$) Each allowed block is normal. If $[T]$ is the block diagonal sum of these blocks, then $[T]^*[T]$ and $[T][T]^*$ are the corresponding block diagonal sums of B^*B and BB^*, hence are equal block by block.

Structural Commentary: Normality lets invariant lines or planes split orthogonally.
Key Insight: Real normal operators split into scalar lines and rotation-dilation planes. □

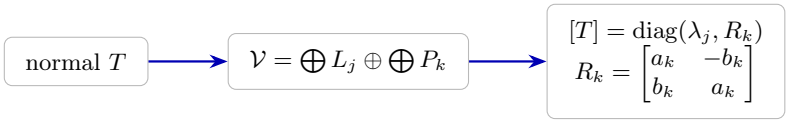


Figure 49: Orthogonal line/plane split into scalar and rotation blocks.

Example 3.3.35. *Plane Rotations*
For $\theta \in \mathbb{R}$, the real matrix $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is an isometry of $\mathbb{R}^2$. If θ is not an integer multiple of π, then R_θ has no real eigenvalue, which is why real isometries require two-dimensional rotation blocks.

Theorem 3.3.36. *Real Isometry Block Form*
Let $\mathcal{V}$ be finite-dimensional over $\mathbb{R}$ and let $S \in \mathcal{L}(\mathcal{V})$. Then S is an isometry if and only if some orthonormal basis makes $[S]$ block diagonal, with each diagonal block one of $[1]$, $[-1]$, $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ with $0 < \theta < \pi$.

Proof.

($\Rightarrow$) Since S is an isometry, $S^*S = I_\mathcal{V}$ by Theorem 2.4.19. Finite-dimensional invertibility gives $SS^* = I_\mathcal{V}$, so S is normal. Apply Theorem 3.3.34. A 1-by-1 block $[\lambda]$ has $\lambda^2 = 1$, so $\lambda = \pm 1$.
A 2-by-2 block $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ has $a^2 + b^2 = 1$ and $b > 0$, so $a = \cos\theta$ and $b = \sin\theta$ for a unique $\theta \in (0, \pi)$.

($\Leftarrow$) Each displayed block B satisfies $B^tB = I$ on its coordinate subspace. Therefore the block diagonal matrix satisfies $[S]^t[S] = I$, so S preserves norms and is an isometry.

Structural Commentary: Isometries are the unit-length special case of real normal blocks.
Key Insight: Scalar blocks reduce to ± 1, and planar normal blocks reduce to rotations. □

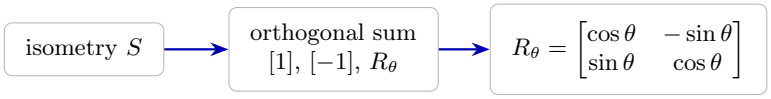


Figure 50: Normal form: fixed/flipped lines and rotation planes.

Proposition 3.3.37. *Real Eigenvalues of a Complexification*
Let $\mathcal{V}$ be real, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda \in \mathbb{R}$. Then λ is an eigenvalue of T if and only if it is an eigenvalue of $T_\mathbb{C}$.

Proof.

($\Rightarrow$) If $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then $T_\mathbb{C}(\mathbf{v} + i\mathbf{0}) = \lambda(\mathbf{v} + i\mathbf{0})$.

($\Leftarrow$) If $T_\mathbb{C}(\mathbf{u} + i\mathbf{v}) = \lambda(\mathbf{u} + i\mathbf{v})$ and $\mathbf{u} + i\mathbf{v} \neq \mathbf{0}$, then $T\mathbf{u} = \lambda\mathbf{u}$ and $T\mathbf{v} = \lambda\mathbf{v}$.
Since $\mathbf{u}$ or $\mathbf{v}$ is nonzero, T has eigenvalue λ.

Structural Commentary: Real eigenvalues survive complexification exactly.
Key Insight: For real λ, real and imaginary parts decouple. □

Lemma 3.3.38. *Conjugation Transfers Generalized Eigenspaces*
Let $\mathcal{V}$ be real, let $T \in \mathcal{L}(\mathcal{V})$, let $\lambda \in \mathbb{C}$, and let $k \geq 0$. Then $(T_\mathbb{C} - \lambda I)^k(\mathbf{u} + i\mathbf{v}) = \mathbf{0} \iff (T_\mathbb{C} - \overline{\lambda} I)^k(\mathbf{u} - i\mathbf{v}) = \mathbf{0}$.

Proof. Let $\Gamma(\mathbf{u} + i\mathbf{v}) = \mathbf{u} - i\mathbf{v}$.
Then Γ is a conjugate-linear bijection, $\Gamma T_\mathbb{C} = T_\mathbb{C}\Gamma$, and $\Gamma\big((T_\mathbb{C} - \lambda I)^k\mathbf{x}\big) = (T_\mathbb{C} - \overline{\lambda} I)^k\Gamma\mathbf{x}$.

($\Rightarrow$) Apply the displayed identity to $\mathbf{x} = \mathbf{u} + i\mathbf{v}$.

($\Leftarrow$) Apply the first direction with $\overline{\lambda}$ in place of λ and $\mathbf{u} - i\mathbf{v}$ in place of $\mathbf{u} + i\mathbf{v}$.

Structural Commentary: Conjugation commutes with $T_\mathbb{C}$ and conjugates scalars.
Key Insight: Conjugation pairs generalized eigenspaces. □

Corollary 3.3.39. *Conjugate Eigenvalues Have Equal Multiplicity*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{R}$ and let $T \in \mathcal{L}(\mathcal{V})$.
If λ is an eigenvalue of $T_{\mathbb{C}}$, then $\overline{\lambda}$ is an eigenvalue of $T_{\mathbb{C}}$, and $\dim_{\mathbb{C}} \mathcal{G}_{\lambda}(T_{\mathbb{C}}) = \dim_{\mathbb{C}} \mathcal{G}_{\overline{\lambda}}(T_{\mathbb{C}})$.

Proof. Lemma 3.3.38 with $k = 1$ sends a nonzero eigenvector for λ to a nonzero eigenvector for $\overline{\lambda}$. Thus $\overline{\lambda}$ is an eigenvalue. With $k = \dim_{\mathbb{C}} \mathcal{V}_{\mathbb{C}}$, Theorem 3.2.53 identifies generalized eigenspaces with the corresponding stabilized kernels. By Lemma 3.3.38, $\Gamma(\mathbf{u}+i\mathbf{v}) = \mathbf{u}-i\mathbf{v}$ restricts to conjugate-linear bijection $\mathcal{G}_{\lambda}(T_{\mathbb{C}}) \longrightarrow \mathcal{G}_{\overline{\lambda}}(T_{\mathbb{C}})$. Conjugate-linear bijections preserve complex dimension, so the multiplicities agree.
Structural Commentary: Conjugation mirrors generalized eigenspaces.
Key Insight: Nonreal spectral data of a real operator arrives in equal-sized conjugate pairs. □

Corollary 3.3.40. *Odd-Dimensional Real Operators Have Eigenvalues*
Every operator on a nonzero odd-dimensional real vector space has a real eigenvalue.

Proof. Let $T \in \mathcal{L}(\mathcal{V})$, where $\dim_{\mathbb{R}} \mathcal{V}$ is odd. By Proposition 3.3.27, $\dim_{\mathbb{C}} \mathcal{V}_{\mathbb{C}}$ is odd.
The sum of the multiplicities of all eigenvalues of $T_{\mathbb{C}}$ equals this odd dimension by Proposition 3.2.64. Nonreal eigenvalues contribute in conjugate pairs of equal multiplicity by Corollary 3.3.39, hence contribute an even total. Therefore $T_{\mathbb{C}}$ has at least one real eigenvalue, which is an eigenvalue of T by Proposition 3.3.37.
Structural Commentary: Odd dimension forces a real eigenvalue.
Key Insight: Parity forces at least one real spectral direction. □

Definition 3.3.41. *Characteristic Polynomial of a Real Operator*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real vector space and let $T \in \mathcal{L}(\mathcal{V})$.
The *characteristic polynomial* of T is $q_T := q_{T_{\mathbb{C}}}$, the characteristic polynomial of its complexification.

Theorem 3.3.42. *Characteristic Polynomial over $\mathbb{R}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real vector space, let $T \in \mathcal{L}(\mathcal{V})$, and let q_T be its characteristic polynomial.

(i) $q_T \in \mathbb{R}[z]$.
(ii) $\deg q_T = \dim_{\mathbb{R}} \mathcal{V}$.
(iii) The real zeros of q_T are precisely the eigenvalues of T.
(iv) $q_T(T) = 0$.

Proof.

(i) Factor $q_{T_{\mathbb{C}}}$ over $\mathbb{C}$. Real eigenvalues give real linear factors. If $\lambda \notin \mathbb{R}$ has multiplicity m, then $\overline{\lambda}$ has multiplicity m by Corollary 3.3.39, and $(z-\lambda)^m(z-\overline{\lambda})^m = \left(z^2 - 2\operatorname{Re}(\lambda)z + |\lambda|^2\right)^m$, which is real.
(ii) Proposition 3.2.19 and Proposition 3.3.27 give $\deg q_T = \dim_{\mathbb{R}} \mathcal{V}$.
(iii) Real zeros of $q_T = q_{T_{\mathbb{C}}}$ are real eigenvalues of $T_{\mathbb{C}}$ (Prop 3.2.19), hence eigenvalues of T (Prop 3.3.37).
(iv) Cayley–Hamilton over $\mathbb{C}$ gives $q_T(T_{\mathbb{C}}) = 0$. By (i) and Proposition 3.3.29, $q_T(T_{\mathbb{C}}) = (q_T(T))_{\mathbb{C}}$. If $q_T(T)$ were nonzero on some $\mathbf{u} \in \mathcal{V}$, then $(q_T(T))_{\mathbb{C}}(\mathbf{u} + i\mathbf{0}) = q_T(T)\mathbf{u} + i\mathbf{0}$ would be nonzero, a contradiction. Hence $q_T(T) = 0$.

Structural Commentary: Nonreal factors pair into real quadratics.
Key Insight: Complexify, then pair conjugates. □

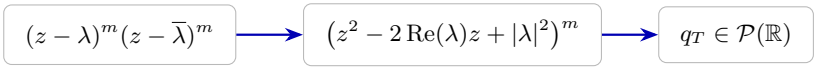


Figure 51: Conjugate factors pair into real quadratics.

Corollary 3.3.43. *q_T Is a Multiple of μ_T over $\mathbb{R}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real vector space and let $T \in \mathcal{L}(\mathcal{V})$.
Then $\deg \mu_T \leq \dim_{\mathbb{R}} \mathcal{V}$, and q_T is a polynomial multiple of μ_T in $\mathbb{R}[z]$.

Proof. Theorem 3.3.42 gives $q_T(T) = 0$.
By Theorem 3.2.33, μ_T divides q_T in $\mathbb{R}[z]$. The degree bound follows from $\deg q_T = \dim_{\mathbb{R}} \mathcal{V}$.
Structural Commentary: Cayley–Hamilton makes q_T a real annihilator of T.
Key Insight: The minimal polynomial divides every real annihilator. □

3.4 Spectral Theory of Structured Operators

Structured spectral theory studies operators whose geometry and spectrum align. Normal, self-adjoint, positive, and unitary maps admit orthogonal decompositions and spectral resolutions.

Definition 3.4.1. *Normal Operator*
An operator $T \in \mathcal{L}(\mathcal{V})$ on a finite-dim inner-product space is *normal* if it commutes with its adjoint: $TT^* = T^*T$.
A square matrix A is *normal* if $A^*A = AA^*$.
Every self-adjoint operator, every Hermitian matrix, and every unitary matrix is normal.

Theorem 3.4.2. *Normal Operators and Equal Adjoint Norms*
An operator $T \in \mathcal{L}(\mathcal{V})$ is normal if and only if $\|T\mathbf{v}\| = \|T^*\mathbf{v}\|$ for every $\mathbf{v} \in \mathcal{V}$.

Proof. The operator $A = T^*T - TT^*$ is self-adjoint.
For every $\mathbf{v} \in \mathcal{V}$, $\langle A\mathbf{v}, \mathbf{v}\rangle = \langle T^*T\mathbf{v}, \mathbf{v}\rangle - \langle TT^*\mathbf{v}, \mathbf{v}\rangle = \|T\mathbf{v}\|^2 - \|T^*\mathbf{v}\|^2$.
Thus T is normal exactly when $A = 0$, which by Proposition 2.4.12 is equivalent to the displayed quadratic form being zero for every $\mathbf{v}$. This is equivalent to $\|T\mathbf{v}\| = \|T^*\mathbf{v}\|$ for every $\mathbf{v}$.
Structural Commentary: Normality is measured by the self-adjoint commutator $T^*T - TT^*$.
Key Insight: A normal operator and its adjoint move every vector the same distance. □

Theorem 3.4.3. *Normal Operators Share Eigenvectors with Their Adjoints*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal. If $T\mathbf{v} = \lambda\mathbf{v}$ for some $\mathbf{v} \neq \mathbf{0}$, then $T^*\mathbf{v} = \overline{\lambda}\mathbf{v}$.

Proof. The operator $T - \lambda I_{\mathcal{V}}$ is normal by Proposition 2.4.6 and scalar multiples of $I_{\mathcal{V}}$ commute with T and T^*. Applying Theorem 3.4.2 to $T - \lambda I_{\mathcal{V}}$ gives $0 = \|(T - \lambda I_{\mathcal{V}})\mathbf{v}\| = \|(T - \lambda I_{\mathcal{V}})^*\mathbf{v}\| = \left\|(T^* - \overline{\lambda} I_{\mathcal{V}})\mathbf{v}\right\|$.
Structural Commentary: For normal operators, kernels of $T - \lambda I$ and $T^* - \overline{\lambda} I$ coincide.
Key Insight: Normality makes eigenvector information visible from both T and T^*. □

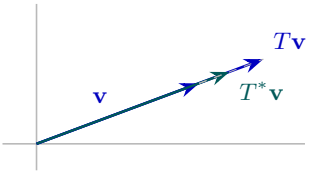


Figure 52: For normal T, $T\mathbf{v} = \lambda\mathbf{v}$ implies $T^*\mathbf{v} = \overline{\lambda}\mathbf{v}$.

Theorem 3.4.4. *Orthogonal Eigenvectors for Normal Operators*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal. Eigenvectors of T corresponding to distinct eigenvalues are orthogonal.

Proof. Suppose $T\mathbf{u} = \alpha\mathbf{u}$ and $T\mathbf{v} = \beta\mathbf{v}$ with $\alpha \neq \beta$. By Theorem 3.4.3, $T^*\mathbf{v} = \overline{\beta}\mathbf{v}$.
Hence $\alpha\langle \mathbf{u}, \mathbf{v}\rangle = \langle T\mathbf{u}, \mathbf{v}\rangle = \langle \mathbf{u}, T^*\mathbf{v}\rangle = \langle \mathbf{u}, \overline{\beta}\mathbf{v}\rangle = \beta\langle \mathbf{u}, \mathbf{v}\rangle$. Since $\alpha - \beta \neq 0$, $\langle \mathbf{u}, \mathbf{v}\rangle = 0$.
Structural Commentary: Normality lets the operator move from one eigenvector to the other through the adjoint.
Key Insight: Distinct eigenvalue scalings are incompatible unless the inner product is zero. □

Theorem 3.4.5. *Eigenvalues of Self-Adjoint Operators Are Real*
Every eigenvalue of a self-adjoint operator on a finite-dimensional inner-product space is real.

Proof. Let $T = T^*$ and $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Then $\lambda\|\mathbf{v}\|^2 = \langle \lambda\mathbf{v}, \mathbf{v}\rangle = \langle T\mathbf{v}, \mathbf{v}\rangle = \langle \mathbf{v}, T\mathbf{v}\rangle = \langle \mathbf{v}, \lambda\mathbf{v}\rangle = \overline{\lambda}\|\mathbf{v}\|^2$.
Since $\|\mathbf{v}\|^2 > 0$, $\lambda = \overline{\lambda}$, so $\lambda \in \mathbb{R}$.
Structural Commentary: Self-adjointness moves T between inner-product slots without changing the value.
Key Insight: An eigenvalue must equal its conjugate because the eigenvector has positive squared norm. □

Proposition 3.4.6. *Invertible Quadratic Expressions*
Let $T \in \mathcal{L}(\mathcal{V})$ be self-adjoint on a finite-dimensional inner-product space.
If $b, c \in \mathbb{R}$ and $b^2 < 4c$, then $T^2 + bT + cI_{\mathcal{V}}$ is invertible.

Proof. Let $S = T^2 + bT + cI_{\mathcal{V}}$. For $\mathbf{v} \neq \mathbf{0}$,

$$\begin{aligned}\langle S\mathbf{v}, \mathbf{v}\rangle &= \|T\mathbf{v}\|^2 + b\langle T\mathbf{v}, \mathbf{v}\rangle + c\|\mathbf{v}\|^2 \geq \|T\mathbf{v}\|^2 - |b|\,\|T\mathbf{v}\|\,\|\mathbf{v}\| + c\|\mathbf{v}\|^2 \\ &= \left(\|T\mathbf{v}\| - \frac{|b|}{2}\|\mathbf{v}\|\right)^2 + \left(c - \frac{b^2}{4}\right)\|\mathbf{v}\|^2 > 0\end{aligned}$$

Thus $S\mathbf{v} \neq \mathbf{0}$ for every nonzero $\mathbf{v}$, so $\ker(S) = \{\mathbf{0}\}$. In finite dimensions, injectivity implies invertibility.
Structural Commentary: Negative discriminant keeps this self-adjoint quadratic positive.
Key Insight: Self-adjointness converts $\langle T^2\mathbf{v}, \mathbf{v}\rangle$ into $\|T\mathbf{v}\|^2$, so completing the square controls the kernel. □

Lemma 3.4.7. *Self-Adjoint Operators Have Eigenvalues*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real inner-product space. Every self-adjoint $T \in \mathcal{L}(\mathcal{V})$ has an eigenvalue.

Proof. Choose $\mathbf{v} \neq \mathbf{0}$ and set $n = \dim(\mathcal{V})$.
The list $\mathbf{v}, T\mathbf{v}, \dots, T^n\mathbf{v}$ is linearly dependent, so some nonzero real polynomial p satisfies $p(T)\mathbf{v} = \mathbf{0}$.
Factor p over $\mathbb{R}$ as $p(x) = a \prod_{j=1}^{M} (x^2 + b_j x + c_j) \prod_{k=1}^{m} (x - \lambda_k)$, where $a \neq 0$, $b_j^2 < 4c_j$, $\lambda_k \in \mathbb{R}$.
By Proposition 3.4.6, each quadratic factor $Q_j(T) := T^2 + b_j T + c_j I_{\mathcal{V}}$ is invertible. Let $R(T) = \prod_j Q_j(T)$ and $L(T) = \prod_k (T - \lambda_k I_{\mathcal{V}})$. The factors commute because they are polynomials in T, and $R(T)$ is invertible.
If $m = 0$, then $p(T) = aR(T)$ would be invertible, contradicting $p(T)\mathbf{v} = \mathbf{0}$ with $\mathbf{v} \neq \mathbf{0}$. Hence $m \geq 1$.
Since $\mathbf{0} = p(T)\mathbf{v} = aR(T)L(T)\mathbf{v}$ and $aR(T)$ is invertible, $L(T)\mathbf{v} = \mathbf{0}$. Thus $L(T)$ has nontrivial kernel.
If every $T - \lambda_k I_{\mathcal{V}}$ were injective, their product $L(T)$ would be injective, a contradiction. Therefore some $T - \lambda_k I_{\mathcal{V}}$ is not injective, and λ_k is an eigenvalue of T.
Structural Commentary: An annihilating polynomial remains; quadratic factors are invertible.
Key Insight: The nontrivial kernel must come from a real linear factor, hence from an eigenvalue. □

Proposition 3.4.8. *Self-Adjoint Operators and Invariant Subspaces*
Let $T \in \mathcal{L}(\mathcal{V})$ be self-adjoint, and let $\mathcal{U}$ be a subspace of $\mathcal{V}$ invariant under T. Then:

i. $\mathcal{U}^\perp$ is invariant under T;
ii. $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ is self-adjoint;
iii. $T|_{\mathcal{U}^\perp} \in \mathcal{L}(\mathcal{U}^\perp)$ is self-adjoint.

Proof.

(i) If $\mathbf{v} \in \mathcal{U}^\perp$ and $\mathbf{u} \in \mathcal{U}$, then $\langle T\mathbf{v}, \mathbf{u}\rangle = \langle \mathbf{v}, T\mathbf{u}\rangle = 0$ because $T\mathbf{u} \in \mathcal{U}$. Hence $T\mathbf{v} \in \mathcal{U}^\perp$.
(ii) For $\mathbf{u}, \mathbf{v} \in \mathcal{U}$, $\langle (T|_{\mathcal{U}})\mathbf{u}, \mathbf{v}\rangle = \langle T\mathbf{u}, \mathbf{v}\rangle = \langle \mathbf{u}, T\mathbf{v}\rangle = \langle \mathbf{u}, (T|_{\mathcal{U}})\mathbf{v}\rangle$. Thus $T|_{\mathcal{U}}$ is self-adjoint.
(iii) By (i), $\mathcal{U}^\perp$ is invariant, so (ii)'s argument applies to $T|_{\mathcal{U}^\perp}$.

Structural Commentary: Self-adjointness preserves orthogonal complements of invariant subspaces.
Key Insight: Move T across the inner product; invariance of $\mathcal{U}$ then forces orthogonality to persist. □

Theorem 3.4.9. *Schur Triangularisation*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. Then T has an upper-triangular matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof. Proceed by induction on $n = \dim(\mathcal{V})$. The cases $n = 0$ and $n = 1$ are immediate.
Suppose $n \geq 2$ and the result holds for smaller dimensions. By Theorem 3.1.20, choose a unit eigenvector $\mathbf{e}_1$ of T, say $T\mathbf{e}_1 = \lambda\mathbf{e}_1$. Let $\mathcal{U} = \operatorname{span}(\mathbf{e}_1)$ and define $S \in \mathcal{L}(\mathcal{U}^\perp)$ by $S\mathbf{w} = P_{\mathcal{U}^\perp}(T\mathbf{w})$.
By induction, choose an orthonormal basis $(\mathbf{e}_2, \dots, \mathbf{e}_n)$ of $\mathcal{U}^\perp$ with respect to which S has an upper-triangular matrix. Then $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ is an orthonormal basis of $\mathcal{V}$.
For $j \geq 2$, the coordinates of $T\mathbf{e}_j$ along $\mathbf{e}_2, \dots, \mathbf{e}_n$ are exactly the coordinates of $S\mathbf{e}_j$, hence have no entries below row j. Also $T\mathbf{e}_1 = \lambda\mathbf{e}_1$. Therefore $[T]_{\gamma \leftarrow \gamma}$ is upper triangular for $\gamma = (\mathbf{e}_1, \dots, \mathbf{e}_n)$.
Structural Commentary: Schur triangularisation builds an eigenline first, then triangularises the projected action on its orthogonal complement.
Key Insight: The lower-right block is controlled by the compression $P_{\mathcal{U}^\perp} T|_{\mathcal{U}^\perp}$. □

Theorem 3.4.10. *Matrix Schur Triangularisation with Ordered Diagonal*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \dots, \lambda_n$ in a prescribed order. If $A\mathbf{x} = \lambda_1\mathbf{x}$ and $\|\mathbf{x}\|_2 = 1$, then there is a unitary $U = [\mathbf{x}\ U_2]$ such that U^*AU is upper triangular with diagonal $\lambda_1, \dots, \lambda_n$.
If A is real, all $\lambda_i \in \mathbb{R}$, and $\mathbf{x} \in \mathbb{R}^n$, then U may be chosen real orthogonal.

Proof. Induct on n. The case $n = 1$ is immediate. Extend $\mathbf{x}$ to an orthonormal basis and set $V = [\mathbf{x}\ V_2]$.
Since $A\mathbf{x} = \lambda_1\mathbf{x}$, the first column of V^*AV is $\lambda_1\mathbf{e}_1$, so $V^*AV = \begin{bmatrix} \lambda_1 & * \\ 0 & A' \end{bmatrix}$.
Block triangularity gives $p_{V^*AV}(z) = (z - \lambda_1)p_{A'}(z)$, so A' has eigenvalues $\lambda_2, \dots, \lambda_n$, counted with algebraic multiplicity. By induction, choose unitary W with $W^*A'W = T'$ upper triangular and diagonal $\lambda_2, \dots, \lambda_n$.
Then $U := V(I_1 \oplus W)$ has first column $\mathbf{x}$ and $U^*AU = \begin{bmatrix} \lambda_1 & * \\ 0 & T' \end{bmatrix}$.
The real case uses real orthonormal extensions; the block A' is real, and the same characteristic-polynomial factorization leaves the real eigenvalue list $\lambda_2, \dots, \lambda_n$ for the induction step.
Structural Commentary: Isolate the chosen eigenline, then repeat on the lower-right block.
Key Insight: The diagonal order is controlled by which eigenline is peeled off first. □

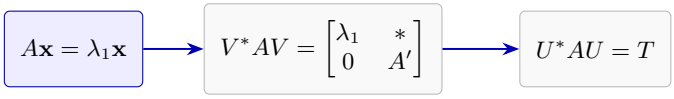


Figure 53: Schur induction: isolate an eigenline, then triangularise the compressed block.

Corollary 3.4.11. *Eigenvalues of the Inverse*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be invertible with eigenvalues $\lambda_1, \ldots, \lambda_n$, counted with algebraic multiplicity. Then A^{-1} has eigenvalues $\lambda_1^{-1}, \ldots, \lambda_n^{-1}$, counted with algebraic multiplicity.

Proof. Take a Schur form $A = UTU^*$ with diagonal $\lambda_1, \ldots, \lambda_n$.
Then $A^{-1} = UT^{-1}U^*$, and T^{-1} is upper triangular with diagonal $\lambda_1^{-1}, \ldots, \lambda_n^{-1}$.
Structural Commentary: Invert the Schur form and read the diagonal.
Key Insight: The inverse spectrum is the reciprocal spectrum. □

Theorem 3.4.12. *Complex Spectral Theorem*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. The following are equivalent:

i. T is normal;
ii. $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of T;
iii. T has a diagonal matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof.

(i) $\Rightarrow$ (ii) By Schur triangularisation (Theorem 3.4.9), choose an orthonormal basis $\gamma = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ in which $A = [T]_{\gamma \leftarrow \gamma} = [a_{ij}]$ is upper triangular. Show by induction on k that the kth row has no entries to the right of the diagonal. Assume this has been proved for rows $1, \ldots, k-1$.
Upper triangularity gives $T\mathbf{e}_k = \sum_{i \le k} a_{ik}\mathbf{e}_i$ and $T^*\mathbf{e}_k = \sum_{j \ge k} \overline{a_{kj}}\mathbf{e}_j$.
The induction hypothesis gives $a_{ik} = 0$ for $i < k$, so $\|T\mathbf{e}_k\|^2 = |a_{kk}|^2$.
Since T is normal, Theorem 3.4.2 gives $|a_{kk}|^2 = \|T\mathbf{e}_k\|^2 = \|T^*\mathbf{e}_k\|^2 = |a_{kk}|^2 + \sum_{j>k} |a_{kj}|^2$.
Thus each a_{kj} with $j > k$ is zero. Induction makes A diagonal, so γ is an orthonormal basis of eigenvectors.

(ii) $\Rightarrow$ (iii) In an orthonormal eigenbasis, the matrix of T is diagonal.

(iii) $\Rightarrow$ (i) If $[T]$ is diagonal in an orthonormal basis, then $[T^*] = [T]^*$ is diagonal as well.
Diagonal matrices commute, so $TT^* = T^*T$; hence T is normal.

Structural Commentary: Schur gives triangular form; normality kills the upper entries.
Key Insight: Equal norms of $T\mathbf{e}_j$ and $T^*\mathbf{e}_j$ force each row tail to vanish. □

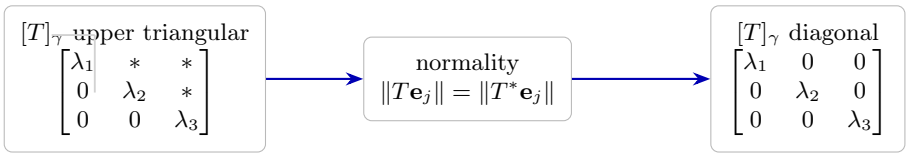


Figure 54: Normality turns Schur triangular form diagonal by forcing row tails to vanish.

Theorem 3.4.13. *Real Spectral Theorem*
Let $\mathcal{V}$ be a finite-dimensional real inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. The following are equivalent:

i. T is self-adjoint;
ii. $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of T;
iii. T has a diagonal matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof.

(i) $\Rightarrow$ (ii) Assume $T = T^*$ and induct on $n = \dim(\mathcal{V})$. For $n = 0$ there is nothing to prove, and for $n = 1$ every nonzero vector is an eigenvector after normalization.
Let $n \ge 2$. Lemma 3.4.7 gives a unit eigenvector $\mathbf{u}$, and set $\mathcal{U} = \mathrm{span}(\mathbf{u})$. The subspace $\mathcal{U}$ is T-invariant.
By Proposition 3.4.8, $\mathcal{U}^\perp$ is T-invariant and the restriction $T|_{\mathcal{U}^\perp}$ is self-adjoint.
Since $\dim(\mathcal{U}^\perp) = n - 1$, the induction hypothesis gives an orthonormal eigenbasis of $\mathcal{U}^\perp$.
Adjoining $\mathbf{u}$ gives an orthonormal eigenbasis of $\mathcal{V}$ by Theorem 2.2.3.

(ii) $\Rightarrow$ (iii) In an orthonormal eigenbasis, the matrix of T is diagonal.

(iii) $\Rightarrow$ (i) In an orthonormal basis where $[T]$ is real diagonal, $[T^*] = [T]^t = [T]$. Hence $T = T^*$.

Structural Commentary: Split off one eigendirection, then restrict to its orthogonal complement.
Key Insight: One eigenvector reduces the dimension. □

Perspective 3.4.14. *Normal Means Diagonal in Orthogonal Coordinates*
Complex spectral theorem states that normal operators are precisely the operators whose full structure is visible in an orthonormal eigenbasis. This is why normality is the natural coordinate-free version of being diagonal.

Proposition 3.4.15. *Basic Spectral Properties of Normal Operators*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space, let $T \in \mathcal{L}(\mathcal{V})$ be normal, and let $p \in \mathbb{C}[z]$.

(i) *$p(T)$ is normal and commutes with T.*
(ii) *T^* is normal and $E_\lambda(T) = E_{\overline{\lambda}}(T^*)$ for every $\lambda \in \mathbb{C}$.*
(iii) *$\ker T = \ker T^*$ and $\operatorname{range}(T) = \operatorname{range}(T^*)$.*

Proof. Choose an orthonormal eigenbasis $\mathbf{e}_1, \dots, \mathbf{e}_n$ with $T\mathbf{e}_j = \mu_j \mathbf{e}_j$.

(i) $p(T)\mathbf{e}_j = p(\mu_j)\mathbf{e}_j$, $p(T)$ is diagonal in this basis, hence normal. It commutes with T by Proposition 3.2.5.
(ii) Since $T^*\mathbf{e}_j = \overline{\mu_j}\mathbf{e}_j$, T^* is diagonal in the same basis, hence normal. Also $T\mathbf{v} = \lambda\mathbf{v}$ iff $T^*\mathbf{v} = \overline{\lambda}\mathbf{v}$.
(iii) Part (ii) with $\lambda = 0$ gives $\ker T = \ker T^*$. Both ranges are $\operatorname{span}\{\mathbf{e}_j : \mu_j \neq 0\}$.

Structural Commentary: Normality gives one orthonormal basis that diagonalises both T and T^*.
Key Insight: Kernel and range are read from the nonzero diagonal entries in that basis. □

Theorem 3.4.16. *Spectral Resolution of a Normal Operator*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $\mu_1, \dots, \mu_d$ be distinct scalars. For $T \in \mathcal{L}(\mathcal{V})$, consider:

(i) *T is normal and $\operatorname{spec}(T) \subseteq \{\mu_1, \dots, \mu_d\}$.*
(ii) *$\exists$ pairwise orthogonal projections $P_1, \dots, P_d \in \mathcal{L}(\mathcal{V})$ such that $P_iP_j = 0$ for $i \neq j$, $P_1 + \cdots + P_d = I_\mathcal{V}$, and $T = \mu_1 P_1 + \cdots + \mu_d P_d$*
(iii) *Any projections satisfying (ii) obey $\operatorname{range}(P_j) = E_{\mu_j}(T)$ whenever $\mu_j \in \operatorname{spec}(T)$.*

Then (i) and (ii) are equivalent, and any projections satisfying (ii) satisfy (iii).

Proof.

(i) $\Rightarrow$ (ii). By Theorem 3.4.12, $\mathcal{V} = \bigoplus_{\lambda \in \operatorname{spec}(T)} E_\lambda(T)$ as an orthogonal direct sum.

Let P_j be the orthogonal projection onto $E_{\mu_j}(T)$, and set $P_j = 0$ when $\mu_j \notin \operatorname{spec}(T)$. Since $\operatorname{spec}(T) \subseteq \{\mu_1, \dots, \mu_d\}$, these projections are pairwise orthogonal, sum to $I_\mathcal{V}$, and satisfy $T = \sum_j \mu_j P_j$.

(ii) $\Rightarrow$ (i). Since $P_j^* = P_j$ and $P_iP_j = 0$ for $i \neq j$, $T^* = \sum_{j=1}^{d} \overline{\mu_j} P_j$ and $T^*T = \sum_{j=1}^{d} |\mu_j|^2 P_j = TT^*$.

Thus T is normal. If $\lambda \in \operatorname{spec}(T)$ and $\mathbf{0} \neq \mathbf{x} \in E_\lambda(T)$, then $0 = (T - \lambda I_\mathcal{V})\mathbf{x} = \sum_{j=1}^{d} (\mu_j - \lambda) P_j \mathbf{x}$.

The summands lie in the pairwise orthogonal ranges $\operatorname{range}(P_j)$, hence $0 = \sum_{j=1}^{d} |\mu_j - \lambda|^2 \, \|P_j\mathbf{x}\|^2$.

Because $\sum_j P_j = I_\mathcal{V}$, at least one $P_k\mathbf{x}$ is nonzero, so $\mu_k = \lambda$. Thus $\operatorname{spec}(T) \subseteq \{\mu_1, \dots, \mu_d\}$.

(ii) $\Rightarrow$ (iii). For $\mathbf{x} \in \mathcal{V}$, $\mathbf{x} = \sum_{i=1}^{d} P_i\mathbf{x}$ and $(T - \mu_j I_\mathcal{V})\mathbf{x} = \sum_{i \neq j} (\mu_i - \mu_j) P_i \mathbf{x}$.

The summands are orthogonal, so $\|(T - \mu_j I_\mathcal{V})\mathbf{x}\|^2 = \sum_{i \neq j} |\mu_i - \mu_j|^2 \, \|P_i\mathbf{x}\|^2$, which vanishes exactly when $P_i\mathbf{x} = 0$ for all $i \neq j$, equivalently $\mathbf{x} = P_j\mathbf{x}$. Thus $E_{\mu_j}(T) = \operatorname{range}(P_j)$.

Structural Commentary: A normal operator decomposes into scalar actions on orthogonal eigenspaces.
Key Insight: Spectral projections isolate eigenspaces; the operator is the weighted sum of those projections. □



Figure 55: Spectral resolution: $\mathcal{V} = \bigoplus_j \operatorname{range}(P_j)$, $P_iP_j = 0$ for $i \neq j$, $\sum_j P_j = I_\mathcal{V}$, and $T = \sum_j \mu_j P_j$.

Corollary 3.4.17. *Functional Calculus from Spectral Resolution*
Let $T = \sum_{j=1}^{d} \mu_j P_j$ be the spectral resolution of a normal operator and let $f \in \mathbb{C}[z]$. Then $f(T) = \sum_{j=1}^{d} f(\mu_j) P_j$. In particular, for each j there is a polynomial p_j such that $p_j(T) = P_j$.

Proof. Since $P_iP_j = 0$ for $i \neq j$ and $P_j^2 = P_j$, induction gives $T^m = \sum_j \mu_j^m P_j$ for every $m \geq 0$.
Linearity then gives the displayed formula for every polynomial f.
For the final assertion, choose the Lagrange polynomial p_j satisfying $p_j(\mu_i) = \delta_{ij}$. Then $p_j(T) = P_j$.
Structural Commentary: Orthogonal spectral projections multiply like coordinate idempotents.
Key Insight: Polynomial functions of a normal operator act by scalar functions on each eigenspace. □

Lemma 3.4.18. *Adjoints of Normal Matrices as One Polynomial*
Let $A_1, \dots, A_k$ be normal complex matrices. Then $\exists$ a polynomial $p \in \mathbb{C}[z]$ s.t. $p(A_j) = A_j^*$ $\forall j = 1, \dots, k$.

Proof. Let $\Omega = \bigcup_{j=1}^{k} \operatorname{spec}(A_j)$. Since Ω is finite, choose a Lagrange interpolation polynomial p satisfying $p(\lambda) = \overline{\lambda}$ for every $\lambda \in \Omega$. For each A_j, Corollary 3.4.17 gives $p(A_j) = A_j^*$.
Structural Commentary: One polynomial interpolates conjugation on all spectra.
Key Insight: For normal matrices, A^* is $p(A)$ with $p(\lambda) = \overline{\lambda}$. □

Theorem 3.4.19. *Fuglede–Putnam Theorem*
Let $A \in \operatorname{Mat}(m, m, \mathbb{C})$ and $B \in \operatorname{Mat}(n, n, \mathbb{C})$ be normal, and let $X \in \operatorname{Mat}(m, n, \mathbb{C})$.
Then $AX = XB \iff A^*X = XB^*$.

Proof.

(⇒) Assume $AX = XB$. Then $A^rX = XB^r$ for every $r \geq 0$, so $p(A)X = Xp(B)$ for every polynomial p. By Lemma 3.4.18, choose p with $p(A) = A^*$ and $p(B) = B^*$. Thus $A^*X = XB^*$.
(⇐) Assume $A^*X = XB^*$. Apply the same argument to the normal matrices A^* and B^*, choosing q with $q(A^*) = A$ and $q(B^*) = B$. Hence $AX = XB$.

Structural Commentary: An intertwining relation passes through every polynomial in the two normal matrices.
Key Insight: Normality makes adjoints polynomial, so intertwining transfers to adjoints. □

Method 3.4.20. *Finding a Spectral Decomposition of a Normal Matrix*
For a normal matrix $A \in \operatorname{Mat}(n, n, \mathbb{C})$:

i. Find the distinct eigenvalues $\mu_1, \dots, \mu_d$.
ii. For each j, solve $(A - \mu_j I_n)\mathbf{x} = 0$ to get $E_{\mu_j}(A)$.
iii. Orthonormalise a basis of each eigenspace.
iv. Concatenate these orthonormal bases as the columns of U; then U is unitary and $A = U\Lambda U^*$, where Λ repeats each μ_j according to $\dim E_{\mu_j}(A)$.

Theorem 3.4.21. *Spectral Classification of Normal Operators*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$ be normal.

(i) T is unitary if and only if $|\lambda| = 1$ for every $\lambda \in \operatorname{spec}(T)$.
(ii) T is self-adjoint if and only if $\operatorname{spec}(T) \subseteq \mathbb{R}$.
(iii) T is an orthogonal projection if and only if $\operatorname{spec}(T) \subseteq \{0, 1\}$.
(iv) T is skew-adjoint, meaning $T^* = -T$, if and only if every $\lambda \in \operatorname{spec}(T)$ is purely imaginary.

Proof. Write the spectral resolution $T = \sum_j \mu_j P_j$, where the μ_j are the distinct eigenvalues and $P_j \neq 0$.
Then $I_{\mathcal{V}} = \sum_j P_j$, $T^* = \sum_j \overline{\mu_j} P_j$, $T^*T = \sum_j |\mu_j|^2 P_j$, and $T^2 = \sum_j \mu_j^2 P_j$.
Since ranges of P_j are nonzero and independent, an identity $\sum_j \alpha_j P_j = 0$ holds only when every $\alpha_j = 0$.

(i) T is unitary iff $T^*T = I_{\mathcal{V}} = \sum_j P_j$, iff $|\mu_j| = 1$ for every j.
(ii) T is self-adjoint iff $T = T^*$, iff $\mu_j = \overline{\mu_j}$ for every j.
(iii) T is an orthogonal projection iff $T = T^*$ and $T^2 = T$, iff $\mu_j = \overline{\mu_j}$ and $\mu_j^2 = \mu_j$ $\forall j$, iff $\mu_j \in \{0, 1\}$ $\forall j$.
(iv) $T^* = -T$ iff $\overline{\mu_j} = -\mu_j$ for every j, iff every μ_j is purely imaginary.

Structural Commentary: For normal operators, structural classes are scalar conditions on spectral blocks.
Key Insight: For normal T, these properties reduce to scalar tests. □

Proposition 3.4.22. *Cartesian Decomposition and Normality*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space.
For every $T \in \mathcal{L}(\mathcal{V})$, there are unique self-adjoint operators $H, K \in \mathcal{L}(\mathcal{V})$ such that $T = H + iK$.
They are $H = \frac{1}{2}(T + T^*)$ and $K = \frac{1}{2i}(T - T^*)$. Moreover, T is normal if and only if $HK = KH$.

Proof. The displayed formulas give self-adjoint H, K and $T = H + iK$. If $T = X + iY$ with X, Y self-adjoint, then $T^* = X - iY$, so $X = \frac{1}{2}(T + T^*)$ and $Y = \frac{1}{2i}(T - T^*)$; uniqueness follows.
For $T = H + iK$, $T^*T = (H - iK)(H + iK) = H^2 + K^2 + i(HK - KH)$,
whereas $TT^* = (H + iK)(H - iK) = H^2 + K^2 - i(HK - KH)$. Thus $T^*T = TT^*$ if and only if $HK = KH$.
Structural Commentary: Every complex operator splits into Hermitian and skew-Hermitian parts.
Key Insight: Normality says the two Hermitian coordinates of T commute. □

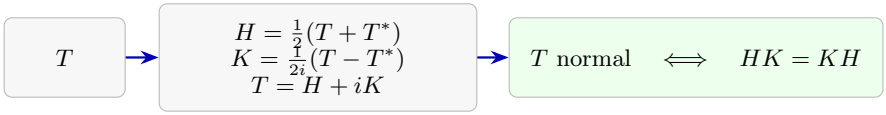


Figure 56: Cartesian decomposition: T is normal exactly when $HK = KH$.

Lemma 3.4.23. *Upper-Triangular Normal Matrices Are Diagonal*
If $A \in \mathrm{Mat}(n, n, \mathbb{C})$ is upper triangular and normal, then A is diagonal.

Proof. Induct on n. The case $n = 1$ is immediate.
Since A is upper triangular, $A\mathbf{e}_1 = a_{11}\mathbf{e}_1$, while $A^*\mathbf{e}_1 = \overline{a_{11}}\mathbf{e}_1 + \overline{a_{12}}\mathbf{e}_2 + \cdots + \overline{a_{1n}}\mathbf{e}_n$.
Normality gives $\|A\mathbf{e}_1\|_2 = \|A^*\mathbf{e}_1\|_2$, hence $a_{12} = \cdots = a_{1n} = 0$. Thus $A = a_{11} \oplus B$ with B upper triangular.
The block diagonal normality equation gives $B^*B = BB^*$, so B is normal. By induction, B is diagonal.
Structural Commentary: Normality forbids upper-triangular row tails.
Key Insight: Compare the lengths of $A\mathbf{e}_1$ and $A^*\mathbf{e}_1$ to kill the first row, then induct. □

Proposition 3.4.24. *Normal Block Upper-Triangular Matrices Collapse*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Let $A = \begin{bmatrix} B & X \\ 0 & C \end{bmatrix} \in \mathrm{Mat}(n, n, \mathbb{F})$, where B and C are square.
Then A is normal if and only if $X = 0$ and both B and C are normal.

Proof.
($\Rightarrow$) Suppose A is normal. Comparing the $(1,1)$ blocks of A^*A and AA^* gives $B^*B = BB^* + XX^*$.
Taking traces and using cyclicity, $\mathrm{tr}(XX^*) = \mathrm{tr}(B^*B) - \mathrm{tr}(BB^*) = 0$. Since $\mathrm{tr}(XX^*) = \|X\|_F^2$, we get $X = 0$. Thus $A = B \oplus C$, and $A^*A = AA^*$ reduces blockwise to $B^*B = BB^*$ and $C^*C = CC^*$.
($\Leftarrow$) If $X = 0$ and B, C are normal, then $A = B \oplus C$, so blockwise multiplication gives $A^*A = AA^*$.
Structural Commentary: A normal operator cannot have one-way coupling between orthogonal blocks.
Key Insight: The off-diagonal block contributes the nonnegative trace $\|X\|_F^2$, so normality forces it to vanish. □

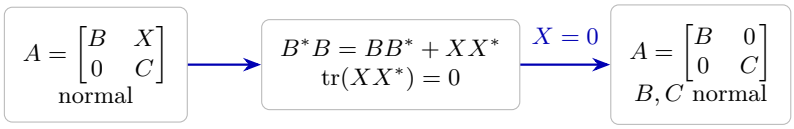


Figure 57: Normality removes the upper-right block in a block upper-triangular matrix.

Theorem 3.4.25. *Invariant Subspaces of Normal Matrices Reduce*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ be normal, and let $\mathcal{U} \subseteq \mathbb{F}^n$ be invariant under A. Then $\mathcal{U}$ is invariant under A^*. If P is the orthogonal projection onto $\mathcal{U}$, then $AP = PA$.

Proof. The cases $\mathcal{U} = \{0\}$ and $\mathcal{U} = \mathbb{F}^n$ are immediate. Otherwise choose a unitary $Q = [Q_1\ Q_2]$ whose columns form orthonormal bases of $\mathcal{U}$ and $\mathcal{U}^\perp$. Invariance of $\mathcal{U}$ says AQ_1 has columns in $\mathcal{U} = \mathrm{col}(Q_1)$, hence $Q_2^*AQ_1 = 0$.
Therefore $Q^*AQ = \begin{bmatrix} B & X \\ 0 & C \end{bmatrix}$. This matrix is normal because it is unitarily similar to A.
Proposition 3.4.24 gives $X = 0$, so $Q^*AQ = B \oplus C$. Hence $Q^*A^*Q = B^* \oplus C^*$, which shows $\mathcal{U}$ is A^*-invariant.
Also $Q^*PQ = I_{\dim(\mathcal{U})} \oplus 0$, so Q^*AQ commutes with Q^*PQ; conjugating back gives $AP = PA$.
Structural Commentary: In adapted coordinates, normal invariance becomes block diagonality.
Key Insight: Normal invariant subspaces are automatically reducing subspaces. □

Corollary 3.4.26. *Invariant Subspaces of Normal Operators Reduce*
Let $\mathcal{V}$ be a finite-dimensional inner-product space, let $T \in \mathcal{L}(\mathcal{V})$ be normal, let $\mathcal{U}$ be invariant under T. Then:
(i) $\mathcal{U}^\perp$ is invariant under T;
(ii) $\mathcal{U}$ is invariant under T^*;

(iii) $(T|_{\mathcal{U}})^* = (T^*)|_{\mathcal{U}}$;
(iv) $T|_{\mathcal{U}}$ *and* $T|_{\mathcal{U}^\perp}$ *are normal.*

Proof. Choose an orthonormal basis adapted to $\mathcal{U} \oplus \mathcal{U}^\perp$.
In these coordinates, Theorem 3.4.25 makes the matrix of T block diagonal, say $A = B \oplus C$.
The block diagonal form shows $\mathcal{U}^\perp$ is invariant under T and $\mathcal{U}$ is invariant under T^*. It also gives $(T|_{\mathcal{U}})^* = (T^*)|_{\mathcal{U}}$ because the adjoint block on $\mathcal{U}$ is B^*. Finally, normality of A reduces blockwise to normality of B and C.
Structural Commentary: Normal operator turns every invariant subspace into an orthogonal reducing summand.
Key Insight: Normality kills the one-way coupling between $\mathcal{U}$ and $\mathcal{U}^\perp$. □

Theorem 3.4.27. *Simultaneous Unitary Diagonalisation of Commuting Normal Operators*
Let $\emptyset \neq \mathcal{F} \subseteq \mathcal{L}(\mathcal{V})$ be a normal family on a finite-dimensional complex inner-product space. Then $\mathcal{F}$ is pairwise commuting if and only if $\mathcal{V}$ has an orthonormal basis of common eigenvectors for $\mathcal{F}$.

Proof.
($\Leftarrow$) In a common orthonormal eigenbasis, every $A \in \mathcal{F}$ is diagonal. Hence all operators in $\mathcal{F}$ commute.
($\Rightarrow$) Induct on $\dim(\mathcal{V})$. The zero-dimensional case is vacuous. If every $A \in \mathcal{F}$ is scalar, any orthonormal basis works. Otherwise choose a nonscalar $A_0 \in \mathcal{F}$ and write its spectral resolution as $A_0 = \sum_j \mu_j P_j$.
Since A_0 is nonscalar, at least two nonzero ranges $\operatorname{range}(P_j)$ occur, and each has dimension strictly smaller than $\dim(\mathcal{V})$. By Corollary 3.4.17, each P_j is a polynomial in A_0.
If $B \in \mathcal{F}$, then $BA_0 = A_0B$, so B commutes with every polynomial in A_0, in particular with P_j. Hence $B(\operatorname{range}(P_j)) \subseteq \operatorname{range}(P_j)$ because $P_jB\mathbf{x} = BP_j\mathbf{x} = B\mathbf{x}$ whenever $P_j\mathbf{x} = \mathbf{x}$.
The same commutation preserves $\ker(P_j) = \operatorname{range}(P_j)^\perp$.
In the decomposition $\mathcal{V} = \operatorname{range}(P_j) \oplus \ker(P_j)$, the matrix of B is therefore block diagonal; since B is normal, its diagonal restrictions are normal block by block.
Thus, on each nonzero $\operatorname{range}(P_j)$, the restricted family is again pairwise commuting and normal.
By induction it has a common orthonormal eigenbasis. Concatenating these bases over all j gives a common orthonormal eigenbasis of $\mathcal{V}$.

Structural Commentary: Spectral projections reduce the whole commuting family.
Key Insight: Diagonalise one nonscalar member, then induct on its eigenspaces. □

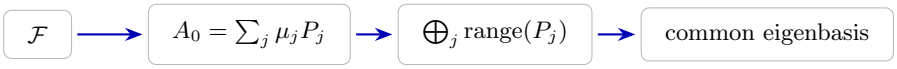


Figure 58: For $A_0 = \sum_j \mu_j P_j$, commutativity gives $BP_j = P_jB$; the ranges reduce $\mathcal{F}$, and induction gives a common orthonormal eigenbasis.

Corollary 3.4.28. *Sums and Products of Commuting Normal Operators*
Let $S, T \in \mathcal{L}(\mathcal{V})$ be commuting normal operators on a finite-dimensional complex inner-product space. Then $S + T$ and ST are normal.

Proof. By Theorem 3.4.27, S and T are diagonal in a common orthonormal basis.
Then $S + T$ and ST are also diagonal in that basis, hence normal.
Structural Commentary: Commuting normal operators share orthonormal coordinates.
Key Insight: In common diagonal form, sums and products remain diagonal. □

Theorem 3.4.29. *Normality Transfers from AB to BA*
Let $A, B \in \operatorname{Mat}(n, n, \mathbb{C})$. If A, B, and AB are normal, then BA is normal.

Proof. By Theorem 3.3.12, AB and BA have the same eigenvalues counted with algebraic multiplicity. Also,

$$\begin{aligned}\operatorname{tr}\big((BA)^*(BA)\big) &= \operatorname{tr}(A^*B^*BA) = \operatorname{tr}(AA^*B^*B)\\ &= \operatorname{tr}(A^*AB^*B) = \operatorname{tr}(A^*ABB^*) = \operatorname{tr}(B^*A^*AB) = \operatorname{tr}\big((AB)^*(AB)\big),\end{aligned}$$

using cyclicity of trace and normality of A and B. Thus $\|BA\|_F^2 = \|AB\|_F^2$, so $\Delta(BA) = \Delta(AB)$.
Since AB is normal, $\Delta(AB) = 0$; hence $\Delta(BA) = 0$, and Theorem 4.3.9 gives that BA is normal.
Structural Commentary: Same spectrum and Frobenius norm give the same Schur defect.
Key Insight: Schur defect turns normality of AB into normality of BA. □

Definition 3.4.30. *Circulant Matrix*
On the 0-indexed basis of $\mathbb{C}^n$, define the *cyclic shift* $P_n\mathbf{e}_j = \mathbf{e}_{(j-1) \bmod n}$.
A matrix $C \in \mathrm{Mat}(n, n, \mathbb{C})$ is *circulant* if, for some $a_0, \dots, a_{n-1} \in \mathbb{C}$,

$$C = a_0 I_n + a_1 P_n + \cdots + a_{n-1} P_n^{n-1} = p(P_n),$$
$$p(z) = a_0 + a_1 z + \cdots + a_{n-1} z^{n-1}$$

Theorem 3.4.31. *Fourier Basis for Cyclic Shifts and Circulants*
Let $n \geq 1$, let $\omega_n := e^{2\pi i/n}$, and let P_n be the cyclic shift.
For $0 \leq k < n$, set $\mathbf{f}_k := (\omega_n^{jk})_{j=0}^{n-1}$; $F_n := [\mathbf{f}_0 \ \cdots \ \mathbf{f}_{n-1}]$; and $\Omega_n := \mathrm{diag}(1, \omega_n, \dots, \omega_n^{n-1})$.
Then $F_n^* F_n = nI_n$, so $Q_n := n^{-1/2} F_n$ is unitary, and $P_n F_n = F_n \Omega_n$.
If $C = p(P_n)$ is circulant, then $Q_n^* C Q_n = p(\Omega_n) = \mathrm{diag}\big(p(1), p(\omega_n), \dots, p(\omega_n^{n-1})\big)$.

Proof. For $0 \leq k, \ell \leq n-1$, $\langle \mathbf{f}_k, \mathbf{f}_\ell \rangle = \sum_{r=0}^{n-1} (\omega_n^{k-\ell})^r = n\delta_{k\ell}$. For $k = \ell$ the sum is n; for $k \neq \ell$, $W = \omega_n^{k-\ell}$ has $W \neq 1$ and $W^n = 1$, so $(1-W)\sum_{r=0}^{n-1} W^r = 0$. Hence $F_n^* F_n = nI_n$, and Theorem 2.4.21 makes Q_n unitary.
Also $P_n \mathbf{f}_k = \omega_n^k \mathbf{f}_k$, so $Q_n^* P_n Q_n = \Omega_n$; applying p gives the formula.
Structural Commentary: Fourier coordinates diagonalize every circulant.
Key Insight: Orthogonality of roots of unity turns a cyclic matrix problem into scalar evaluation. □

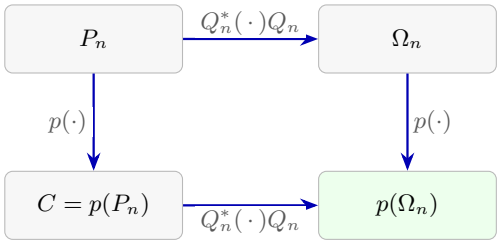


Figure 59: Fourier diagonalizes P_n; polynomial evaluation diagonalizes $C = p(P_n)$.

Proposition 3.4.32. *Permutation Similarity Reorders Diagonal Entries*
Let $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ and let $P = [\mathbf{e}_{\sigma(1)} \ \cdots \ \mathbf{e}_{\sigma(n)}]$ be a permutation matrix. Then $P^* \Lambda P = \mathrm{diag}(\lambda_{\sigma(1)}, \dots, \lambda_{\sigma(n)})$.

Proof. The (i, j) entry of $P^*\Lambda P$ is $\mathbf{e}_{\sigma(i)}^* \Lambda \mathbf{e}_{\sigma(j)} = \lambda_{\sigma(j)} \mathbf{e}_{\sigma(i)}^* \mathbf{e}_{\sigma(j)} = \lambda_{\sigma(j)} \delta_{ij}$.
Thus all off-diagonal entries are zero, and the jth diagonal entry is $\lambda_{\sigma(j)}$.
Structural Commentary: Permutation similarity applies the same basis reordering to rows and columns.
Key Insight: Diagonal entries move exactly as the standard basis vectors are rearranged. □

Corollary 3.4.33. *Normal Matrices with the Same Spectrum Are Unitarily Similar*
If $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$ are normal and have the same eigenvalues counted with algebraic multiplicity, then A and B are unitarily similar.

Proof. By Theorem 3.4.12, write $A = U\Lambda U^*$ and $B = V\Gamma V^*$ with U, V unitary and Λ, Γ diagonal.
Since eigenvalues agree with multiplicity, Proposition 3.4.32 gives a permutation matrix P such that $\Gamma = P^*\Lambda P$.
Hence $B = (VP^*U^*)A(UPV^*)$, and UPV^* is unitary.
Structural Commentary: Normal matrices are classified up to unitary similarity by their eigenvalue multiset.
Key Insight: After diagonalising both matrices, only a permutation of diagonal entries remains. □

Theorem 3.4.34. *Description of Complex Isometries*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $S \in \mathcal{L}(\mathcal{V})$. Then S is an isometry if and only if $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of S whose eigenvalues all have absolute value 1.

Proof. If $S\mathbf{e}_j = \lambda_j \mathbf{e}_j$ in an orthonormal basis with $|\lambda_j| = 1$, then for $\mathbf{v} = \sum_j c_j \mathbf{e}_j$,
$\|S\mathbf{v}\|^2 = \sum_j |\lambda_j c_j|^2 = \sum_j |c_j|^2 = \|\mathbf{v}\|^2$, so S is an isometry. Conversely, if S is an isometry, then $S^*S = I_{\mathcal{V}}$ by Theorem 2.4.19. Finite-dimensional invertibility gives $SS^* = I_{\mathcal{V}}$, so S is normal. By Complex Spectral Theorem, choose orthonormal eigenbasis $S\mathbf{e}_j = \lambda_j \mathbf{e}_j$. Since $\mathbf{e}_j$ is unit and S preserves norms, $|\lambda_j| = \|S\mathbf{e}_j\| = 1$.
Structural Commentary: Complex isometries are normal with unit-circle spectrum.
Key Insight: In an eigenbasis, isometry means $|\lambda_j| = 1$. □

Definition 3.4.35. *Positive Operator*
$T \in \mathcal{L}(\mathcal{V})$ is *positive* if $T = T^*$ and $\langle T\mathbf{v}, \mathbf{v}\rangle \geq 0$ for all $\mathbf{v} \in \mathcal{V}$.

Definition 3.4.36. *Square Root of an Operator*
An operator $R \in \mathcal{L}(\mathcal{V})$ is a *square root* of $T \in \mathcal{L}(\mathcal{V})$ if $R^2 = T$.

Theorem 3.4.37. *Characterization of Positive Operators*
For $T \in \mathcal{L}(\mathcal{V})$, the following are equivalent:

i. T *is positive;*
ii. T *is self-adjoint and every eigenvalue of* T *is nonnegative;*
iii. T *has a positive square root;*
iv. T *has a self-adjoint square root;*
v. there exists $R \in \mathcal{L}(\mathcal{V})$ *such that* $T = R^*R$.

Proof.

(*i*) $\Rightarrow$ (*ii*) If $T\mathbf{v} = \lambda\mathbf{v}$ and $\mathbf{v} \neq \mathbf{0}$, then $\lambda \in \mathbb{R}$ by Theorem 3.4.5, and $0 \leq \langle T\mathbf{v}, \mathbf{v}\rangle = \lambda \|\mathbf{v}\|^2$, so $\lambda \geq 0$.
(*ii*) $\Rightarrow$ (*iii*) By the real or complex spectral theorem, choose an orthonormal eigenbasis $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ with $T\mathbf{e}_j = \lambda_j \mathbf{e}_j$ and $\lambda_j \geq 0$. Define $S\mathbf{e}_j = \sqrt{\lambda_j}\mathbf{e}_j$ and extend linearly.
Then S is diagonal with nonnegative real diagonal entries in an orthonormal basis, hence self-adjoint and positive, and $S^2\mathbf{e}_j = \lambda_j \mathbf{e}_j = T\mathbf{e}_j$ for every j. Thus $S^2 = T$.
(*iii*) $\Rightarrow$ (*iv*) A positive square root is self-adjoint.
(*iv*) $\Rightarrow$ (*v*) If $T = R^2$ with $R = R^*$, then $T = R^*R$.
(*v*) $\Rightarrow$ (*i*) If $T = R^*R$, then $T^* = T$ and $\langle T\mathbf{v}, \mathbf{v}\rangle = \langle R\mathbf{v}, R\mathbf{v}\rangle = \|R\mathbf{v}\|^2 \geq 0$ for every $\mathbf{v}$, so T is positive.

Structural Commentary: Positivity equals nonnegative spectrum and square-root factorization.
Key Insight: In an eigenbasis, the proof is scalar: $\lambda \geq 0$ iff $\sqrt{\lambda}$ exists. □

Proposition 3.4.38. *Unique Positive Square Root*
Every positive operator on $\mathcal{V}$ has a unique positive square root.

Proof. Existence follows from Theorem 3.4.37. For uniqueness, decompose $\mathcal{V} = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_m}(T)$.
Let R be a positive square root of T. Since $R^2 = T$, we have $RT = R^3 = TR$, so each $E_{\lambda_j}(T)$ is invariant under R. On $E_{\lambda_j}(T)$, the restriction $R_j := R|_{E_{\lambda_j}(T)}$ is positive and satisfies $R_j^2 = \lambda_j I$.
By the spectral theorem for the positive operator R_j, every eigenvalue ρ of R_j satisfies $\rho \geq 0$ and $\rho^2 = \lambda_j$, hence $\rho = \sqrt{\lambda_j}$. Therefore $R_j = \sqrt{\lambda_j} I$ on $E_{\lambda_j}(T)$.
Thus every positive square root agrees with the same scalar operator on each eigenspace $E_{\lambda_j}(T)$, so it is unique.
Structural Commentary: Each eigenspace determines its own scalar root.
Key Insight: Positivity selects the nonnegative root. □

Definition 3.4.39. *Positive Square Root Notation*
If T is positive, then $\sqrt{T}$ denotes the unique positive square root of T.

3.5 Operator Functions

Operator functions build new operators from old ones. Functional calculus, spectral mapping, exponentials, and square roots use the spectral and nilpotent structure developed earlier.

Theorem 3.5.1. *Polynomial Functional Calculus for Diagonalizable Matrices*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ be diagonalizable and write $A = S\Lambda S^{-1}$, where $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$.
If $p \in \mathbb{F}[z]$, then $p(A) = S\,\mathrm{diag}(p(\lambda_1), \ldots, p(\lambda_n))S^{-1}$.

Proof. For every $j \geq 0$, $A^j = S\Lambda^j S^{-1}$.

If $p(z) = \sum_j c_j z^j$, then $p(A) = \sum_j c_j A^j = S\left(\sum_j c_j \Lambda^j\right) S^{-1} = Sp(\Lambda)S^{-1}$.

Since Λ is diagonal, $p(\Lambda) = \mathrm{diag}(p(\lambda_1), \ldots, p(\lambda_n))$.
Structural Commentary: Diagonalization reduces $p(A)$ to scalar evaluation on eigenvalues.
Key Insight: Compute functions of A by applying the function to the eigenvalues. □

Definition 3.5.2. *Matrix Exponential*
For $A \in \mathrm{Mat}(n, n, \mathbb{C})$ and $t \in \mathbb{C}$, define $e^{tA} := \sum_{k=0}^{\infty} \frac{t^k A^k}{k!}$.

Proposition 3.5.3. *Pure Modes for Powers and Exponentials*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ be diagonalizable with $A = S\Lambda S^{-1}$, where $\Lambda = \mathrm{diag}(\lambda_1,\dots,\lambda_n)$ and the columns of S are $\mathbf{v}_1,\dots,\mathbf{v}_n$. If $\mathbf{x}_0 = \sum_{j=1}^{n} c_j\mathbf{v}_j$, then $A^k\mathbf{x}_0 = \sum_{j=1}^{n} c_j\lambda_j^k\mathbf{v}_j$ for $k \geq 0$; and $e^{tA}\mathbf{x}_0 = \sum_{j=1}^{n} c_j e^{\lambda_j t}\mathbf{v}_j$ for $t \in \mathbb{C}$.

Proof. Let $c = (c_1,\dots,c_n)^t$, so $\mathbf{x}_0 = Sc$.
Then $A^k = S\Lambda^k S^{-1}$, and Definition 3.5.2 gives $e^{tA} = S\left(\sum_{r=0}^{\infty} \frac{t^r\Lambda^r}{r!}\right)S^{-1} = S\,\mathrm{diag}(e^{\lambda_1 t},\dots,e^{\lambda_n t})S^{-1}$.
Applying these two formulas to Sc gives the stated identities.
Structural Commentary: Eigenbasis coordinates decouple powers and exponentials.
Key Insight: Diagonalization turns coupled evolution into independent scalar modes. □

Corollary 3.5.4. *Simple Spectrum Commutants Are Polynomials*
If $A \in \mathrm{Mat}(n,n,\mathbb{C})$ has n distinct eigenvalues, then $\mathrm{Comm}(A) = \{p(A) : p \in \mathbb{C}[z]\}$, $\dim \mathrm{Comm}(A) = n$.

Proof. Write $A = S\,\mathrm{diag}(\lambda_1,\dots,\lambda_n)S^{-1}$ with distinct λ_i. Theorem 3.1.36 says every $B \in \mathrm{Comm}(A)$ has the form $S\,\mathrm{diag}(\xi_1,\dots,\xi_n)S^{-1}$. Choose the Lagrange interpolation polynomial $p(z) = \sum_{i=1}^{n} \xi_i \prod_{j\neq i} \frac{z-\lambda_j}{\lambda_i-\lambda_j}$.
Then $p(\lambda_i) = \xi_i$ for every i, so Theorem 3.5.1 gives $B = p(A)$.
The dimension statement is the case $n_1 = \cdots = n_n = 1$ of Theorem 3.1.36.
Structural Commentary: Simple spectrum leaves one scalar choice per eigendirection.
Key Insight: Interpolation converts arbitrary diagonal data into a polynomial in A. □

Corollary 3.5.5. *Reduction Modulo the Characteristic Polynomial*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ and let $f \in \mathbb{C}[z]$. If $f = p_A q + r$ with $\deg r < n$, then $f(A) = r(A)$.

Proof. Evaluate division identity at A: $f(A) = p_A(A)q(A) + r(A) = 0q(A) + r(A) = r(A)$.
Theorem 3.2.25 gives $p_A(A) = 0$.
Structural Commentary: Cayley–Hamilton makes polynomial expressions in A reducible modulo p_A.
Key Insight: Every high power of A can be rewritten using powers below n. □

Corollary 3.5.6. *Powers Span at Most n Dimensions*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$. Then $\mathrm{span}(I_n, A, A^2, \dots) = \mathrm{span}(I_n, A, \dots, A^{n-1})$, so this has dimension at most n.

Proof. For each $k \geq 0$, Corollary 3.5.5 applied to $f(z) = z^{n+k}$ gives a polynomial r_k of degree at most $n-1$ such that $A^{n+k} = r_k(A)$. Thus every power of A lies in $\mathrm{span}(I_n, A, \dots, A^{n-1})$.
Structural Commentary: Polynomial reduction collapses the infinite power list to its first n terms.
Key Insight: Cayley–Hamilton gives the recurrence relation for powers of A. □

Corollary 3.5.7. *Inverse from the Characteristic Polynomial*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ have $p_A(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0$.
If A is invertible, then $A^{-1} = -c_0^{-1}(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_2 A + c_1 I_n)$.

Proof. Since A is invertible, $0 \notin \mathrm{spec}(A)$, so $c_0 = p_A(0) = (-1)^n \det A \neq 0$.
Cayley–Hamilton gives $A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I_n = 0$.
Move $c_0 I_n$ to the other side and multiply by $-c_0^{-1}A^{-1}$.
Structural Commentary: The characteristic equation contains an identity term exactly when A is invertible.
Key Insight: Solve Cayley–Hamilton for A^{-1} after factoring out one A. □

Proposition 3.5.8. *Eigenvalues of a Polynomial in a Matrix*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, let $p \in \mathbb{C}[z]$, and let $\lambda_1,\dots,\lambda_n$ be the eigenvalues of A, counted with algebraic multiplicity. Then the eigenvalues of $p(A)$ are $p(\lambda_1),\dots,p(\lambda_n)$, counted with algebraic multiplicity.

Proof. Choose a Schur form $U^*AU = T$ with diagonal $\lambda_1,\dots,\lambda_n$. Then $p(A) = Up(T)U^*$, and $p(T)$ is upper triangular with diagonal $p(\lambda_1),\dots,p(\lambda_n)$. Hence $p(A)$ has precisely these eigenvalues.
Structural Commentary: Schur form reduces the claim to the diagonal.
Key Insight: A polynomial preserves triangular form and applies p to each diagonal entry. □

Corollary 3.5.9. *Polynomial Invertibility Test*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ and $p \in \mathbb{C}[z]$. If $p(\lambda) \neq 0$ for every $\lambda \in \mathrm{spec}(A)$, then $p(A)$ is invertible.

Proof. By Proposition 3.5.8, $0 \notin \mathrm{spec}(p(A))$. Corollary 3.1.9 gives invertibility.
Structural Commentary: Invertibility of $p(A)$ is checked on spectral values of A.
Key Insight: Avoiding zero on the spectrum prevents $p(A)$ from having a kernel. □

Proposition 3.5.10. *Exponential of a Jordan Block*
For $k \geq 1$, $\lambda \in \mathbb{C}$, and $t \in \mathbb{C}$, $e^{tJ_k(\lambda)} = e^{\lambda t} \sum_{r=0}^{k-1} \frac{t^r}{r!} J_k(0)^r$.

Proof. Since $J_k(\lambda) = \lambda I_k + J_k(0)$ and the two summands commute, $e^{tJ_k(\lambda)} = e^{\lambda t I_k} e^{tJ_k(0)}$.
The first factor is $e^{\lambda t} I_k$, and the second exponential truncates because $J_k(0)^k = 0$.
Structural Commentary: The scalar part supplies $e^{\lambda t}$; the nilpotent part supplies a finite polynomial in t.
Key Insight: Nilpotence turns the exponential series into a finite sum. □

Method 3.5.11. *Solving $\mathbf{x}'(t) = A\mathbf{x}(t)$ by Jordan Form*
To solve $\mathbf{x}'(t) = A\mathbf{x}(t)$ with $\mathbf{x}(0) = \mathbf{x}_0$:

(i) Put $A = SJS^{-1}$, where $J = \bigoplus_{i=1}^{r} J_{k_i}(\lambda_i)$.
(ii) Set $\mathbf{y}(t) = S^{-1}\mathbf{x}(t)$ and $\mathbf{y}_0 = S^{-1}\mathbf{x}_0$. Then $\mathbf{y}'(t) = J\mathbf{y}(t)$ and $\mathbf{x}(t) = S\mathbf{y}(t)$.
(iii) Split $\mathbf{y} = (\mathbf{y}_1, \ldots, \mathbf{y}_r)$ according to the Jordan blocks. Then $\mathbf{y}_i'(t) = J_{k_i}(\lambda_i)\mathbf{y}_i(t)$.
(iv) For each block, use Proposition 3.5.10: $\mathbf{y}_i(t) = e^{tJ_{k_i}(\lambda_i)}\mathbf{y}_i(0) = e^{\lambda_i t} \sum_{q=0}^{k_i-1} \frac{t^q}{q!} J_{k_i}(0)^q \mathbf{y}_i(0)$.
(v) Reassemble $\mathbf{y}(t)$ and transform back: $\mathbf{x}(t) = S\mathbf{y}(t)$.

Thus a size-k block with eigenvalue λ contributes entries of the form $e^{\lambda t}p(t)$, where $\deg p < k$.

Perspective 3.5.12. *Jordan Functions and Polynomial Factors*
For a Jordan block $J_k(\lambda) = \lambda I_k + N$ with $N^k = 0$, any polynomial f satisfies $f(J_k(\lambda)) = \sum_{r=0}^{k-1} \frac{f^{(r)}(\lambda)}{r!} N^r$.
Nilpotent part adds derivative terms; exponentials gives polynomial factors (Proposition 3.5.10, Method 3.5.11).

Example 3.5.13. *Nilpotent Jordan Block Evolution*
$N = J_3(0) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. Since $N^3 = 0$, $e^{tN} = I_3 + tN + \frac{t^2}{2}N^2 = \begin{bmatrix} 1 & t & \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix}$.
Nilpotent Jordan blocks create polynomial, not exponential, growth along chains.

Lemma 3.5.14. *Identity Plus Nilpotent Has a Square Root*
Let $\mathcal{V}$ be a complex vector space and let $N \in \mathcal{L}(\mathcal{V})$ be nilpotent. Then $I_\mathcal{V} + N$ has a square root.

Proof. Choose $m \geq 1$ such that $N^m = 0$. If $m = 1$, then $N = 0$ and $I_\mathcal{V}$ is a square root.
Otherwise seek $R = I_\mathcal{V} + a_1 N + \cdots + a_{m-1}N^{m-1}$. With $a_0 = 1$, choose $a_1, \ldots, a_{m-1}$ recursively so the coefficients of R^2 match those of $I_\mathcal{V} + N$: $2a_1 = 1$, and $2a_k + \sum_{r=1}^{k-1} a_r a_{k-r} = 0 \quad (2 \leq k \leq m-1)$
Terms of degree $\geq m$ vanish because $N^m = 0$, so $R^2 = I_\mathcal{V} + N$.
Structural Commentary: Nilpotence reduces the square-root series to a finite polynomial.
Key Insight: Match coefficients recursively. □



Figure 60: Nilpotence truncates the square-root series; coefficients are chosen so $R^2 = I + N$.

Theorem 3.5.15. *Invertible Complex Operators Have Square Roots*
Let $\mathcal{V}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$ be invertible. Then T has a square root.

Proof. Let $\mathcal{U}_j = \mathcal{G}_{\lambda_j}(T)$ for the distinct eigenvalues of T. Primary decomposition gives $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$, and invertibility gives $\lambda_j \neq 0$. Put $N_j = (T - \lambda_j I_\mathcal{V})|_{\mathcal{U}_j}$.
Then $T|_{\mathcal{U}_j} = \lambda_j \left(I_{\mathcal{U}_j} + \lambda_j^{-1} N_j\right)$ with $\lambda_j^{-1} N_j$ nilpotent. By Lemma 3.5.14, choose $B_j^2 = I_{\mathcal{U}_j} + \lambda_j^{-1} N_j$; choose $\alpha_j^2 = \lambda_j$ and set $R_j = \alpha_j B_j$. Then $R_j^2 = T|_{\mathcal{U}_j}$. Define $R = R_1 \oplus \cdots \oplus R_m$ on the direct sum. Then $R^2 = T$.
Structural Commentary: Take square roots on primary blocks, then recombine.
Key Insight: Invertibility makes every scalar block nonzero. □

4 Matrix Analysis and Numerical Structure

Matrix analysis turns linear structure into quantitative matrix tools: invariants, decompositions, norms, conditioning, perturbation bounds, and numerical algorithms.

4.1 Matrix Classes and Invariants

Matrix classes package operator structure in coordinates. Trace, determinant, rank, positivity, idempotents, nilpotents, spectral radius, and structured products give computable invariants.

Definition 4.1.1. *Trace and Determinant of a Matrix*
For $A = [A_{j,k}] \in \mathrm{Mat}(n, n, \mathbb{F})$, the *trace* of A is $\mathrm{tr}(A) := A_{1,1} + \cdots + A_{n,n}$.
Let $\mathfrak{S}_n$ denote the set of permutations of $\{1, \ldots, n\}$. The *sign* of $\sigma \in \mathfrak{S}_n$ is $\mathrm{sign}(\sigma) = 1$ if the number of pairs $j < k$ with $\sigma(j) > \sigma(k)$ is even, and $\mathrm{sign}(\sigma) = -1$ otherwise.
The *determinant* of A is $\det A := \sum_{\sigma \in \mathfrak{S}_n} \mathrm{sign}(\sigma) A_{\sigma(1),1} \cdots A_{\sigma(n),n}$.

Definition 4.1.2. *Gram Determinant*
Vectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ in inner-product space has *Gram determinant* $\mathrm{GramDet}(\mathbf{u}_1, \ldots, \mathbf{u}_n) := \det \mathrm{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$.

Example 4.1.3. *Trace Functional and the Frobenius Inner Product*
On $\mathrm{Mat}(n, n, \mathbb{F})$ with the Frobenius inner product from Example 2.1.6, the Riesz vector of the trace functional $A \mapsto \mathrm{tr}(A)$ is I_n, since $\langle A, I_n \rangle_F = \sum_{j=1}^{n} a_{jj} = \mathrm{tr}(A)$.

Proposition 4.1.4. *Trace Is Cyclic for Two Factors*
If $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $B \in \mathrm{Mat}(n, m, \mathbb{F})$, then $\mathrm{tr}(AB) = \mathrm{tr}(BA)$.

Proof. Expand both traces by entries: $\mathrm{tr}(AB) = \sum_{j=1}^{m} \sum_{k=1}^{n} A_{j,k} B_{k,j} = \sum_{k=1}^{n} \sum_{j=1}^{m} B_{k,j} A_{j,k} = \mathrm{tr}(BA)$.
Structural Commentary: Cyclicity is just reindexing the same double sum.
Key Insight: The two double sums contain exactly the same scalar terms. □

Theorem 4.1.5. *Basic Determinant Laws*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) $\det I_n = 1$.
(ii) $\det(A)$ *is linear as a function of each column of A when the other columns are fixed.*
(iii) Swapping two columns negates the determinant; equal columns force $\det A = 0$.
(iv) If A is triangular, then $\det A = A_{1,1} \cdots A_{n,n}$.
(v) $\det(AB) = \det(A) \det(B)$.
(vi) A is invertible if and only if $\det A \neq 0$. *In that case* $\det(A^{-1}) = (\det A)^{-1}$.
(vii) $\det(A^t) = \det(A)$.

Proof.

(i) $\det I_n = \sum_{\sigma \in \mathfrak{S}_n} \mathrm{sign}(\sigma) (I_n)_{\sigma(1),1} \cdots (I_n)_{\sigma(n),n} = 1$ as product is nonzero only when $\sigma(k) = k \ \forall k$.

(ii) For fixed k, $\det(A_{\cdot,1}, \ldots, \mathbf{x}, \ldots, A_{\cdot,n}) = \sum_{\sigma \in \mathfrak{S}_n} \mathrm{sign}(\sigma) x_{\sigma(k)} \prod_{\ell \neq k} A_{\sigma(\ell),\ell}$, which is linear in $\mathbf{x}$.

(iii) If $B_{\cdot,k} = A_{\cdot,\tau(k)}$ for transposition τ, then $\det B = \sum_{\rho \in \mathfrak{S}_n} \mathrm{sign}(\rho\tau) \prod_{k=1}^{n} A_{\rho(k),k} = -\det A$.
If the swapped columns are equal, then $B = A$, so $\det A = -\det A$ and $\det A = 0$.

(iv) If A is upper triangular and $\prod_{k=1}^{n} A_{\sigma(k),k} \neq 0$, then $\sigma(k) \leq k$ for all k, hence $\sigma(k) = k$ for all k.
Thus $\det A = A_{1,1} \cdots A_{n,n}$. If A is lower triangular, nonzero products require $\sigma(k) \geq k$ for all k, again forcing $\sigma(k) = k$ for all k, so the same diagonal product formula holds.

(v) If F is alternating and multilinear in columns, then expanding each column in the standard basis gives $F(A_{\cdot,1}, \ldots, A_{\cdot,n}) = \sum_{i_1, \ldots, i_n} A_{i_1,1} \cdots A_{i_n,n} F(\mathbf{e}_{i_1}, \ldots, \mathbf{e}_{i_n}) = \sum_{\sigma \in \mathfrak{S}_n} \mathrm{sign}(\sigma) F(I_n) A_{\sigma(1),1} \cdots A_{\sigma(n),n}$.
The terms with repeated i_j vanish by alternation, and the surviving ordered bases differ from $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ by sign of permutation. For $H(B) = \det(AB)$, map H is alternating multilinear and $H(I_n) = \det A$.
Hence $\det(AB) = H(B) = H(I_n) \det B = \det(A) \det(B)$.

(vi) If A is invertible, then $1 = \det I_n = \det(AA^{-1}) = \det(A) \det(A^{-1})$.
If $A_{\cdot,k} = \sum_{\ell \neq k} c_\ell A_{\cdot,\ell}$, then $\det A = \sum_{\ell \neq k} c_\ell \det(A_{\cdot,1}, \ldots, A_{\cdot,\ell}, \ldots, A_{\cdot,\ell}, \ldots, A_{\cdot,n}) = 0$.
Thus $\det A \neq 0$ iff the columns of A are independent, iff A is invertible.

(vii) Using the permutation definition and replacing σ by σ^{-1},
$\det(A^t) = \sum_{\sigma \in \mathfrak{S}_n} \mathrm{sign}(\sigma) A_{1,\sigma(1)} \cdots A_{n,\sigma(n)} = \sum_{\tau \in \mathfrak{S}_n} \mathrm{sign}(\tau) A_{\tau(1),1} \cdots A_{\tau(n),n} = \det(A)$; $\mathrm{sign}(\tau^{-1}) = \mathrm{sign}(\tau)$.

Structural Commentary: The determinant is the normalized alternating multilinear measure of columns.
Key Insight: Multilinearity expands columns; alternation kills repeated directions. □

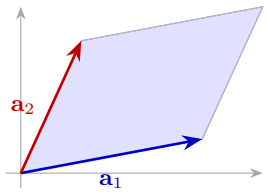


Figure 61: Determinant is signed column-volume; swaps reverse orientation and dependence gives zero volume.

Method 4.1.6. *Determinant by Structure*
To compute or simplify a determinant:

(i) first look for triangular or block triangular form;
(ii) use column linearity to split complicated columns only when it reduces structure;
(iii) use column swaps to move useful pivots into position, tracking a factor of -1 for each swap;
(iv) detect zero determinants from dependent columns or repeated columns;
(v) for products or changes of basis, use multiplicativity instead of expanding permutations.

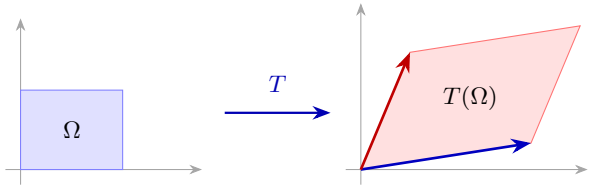


Figure 62: A linear map scales oriented volume by its determinant; singular maps collapse volume to zero.

Lemma 4.1.7. *Identity Blocks Do Not Change Determinant*
Let $A \in \mathrm{Mat}(m, m, \mathbb{F})$ *and let* $n \geq 0$. *Then* $\det(I_n \oplus A) = \det(A) = \det(A \oplus I_n)$.

Proof. In the permutation formula for $I_n \oplus A$, a nonzero term must satisfy $\sigma(k) = k$ for $1 \leq k \leq n$. Remaining choices are exactly permutations of $\{n{+}1, \ldots, n{+}m\}$, after relabeling, gives $\det(A)$. Hence $\det(I_n \oplus A) = \det(A)$.
The same argument applied to the last n coordinates gives $\det(A \oplus I_n) = \det(A)$.
The case $n = 0$ is the empty identity block convention.
Structural Commentary: Identity blocks add directions without changing volume.
Key Insight: In the determinant expansion, the identity block forces the matching coordinates to be fixed. □

Theorem 4.1.8. *Determinant of an Upper Block-Triangular Matrix*
Let $A \in \mathrm{Mat}(r, r, \mathbb{F})$, $B \in \mathrm{Mat}(r, s, \mathbb{F})$, *and* $D \in \mathrm{Mat}(s, s, \mathbb{F})$. *Then* $\det \begin{bmatrix} A & B \\ 0 & D \end{bmatrix} = \det(A)\det(D)$.

Proof. Factor $\begin{bmatrix} A & B \\ 0 & D \end{bmatrix} = \begin{bmatrix} I_r & 0 \\ 0 & D \end{bmatrix} \begin{bmatrix} I_r & B \\ 0 & I_s \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & I_s \end{bmatrix}$.
The first and last factors have determinants $\det(D)$ and $\det(A)$ by Lemma 4.1.7.
The middle factor is upper triangular with every diagonal entry equal to 1, so its determinant is 1.
Multiplicativity of determinant gives the formula.
Structural Commentary: Block upper triangular matrices separate determinant contribution into diagonal blocks. The off-diagonal block does not affect the determinant.
Key Insight: Factor the matrix into two block diagonal pieces and one unit upper triangular piece. □

Proposition 4.1.9. *Schur Complement Determinant Formula*
Let $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, $A \in \mathrm{Mat}(r, r, \mathbb{F})$, $D \in \mathrm{Mat}(s, s, \mathbb{F})$.
If A *is invertible, then the Schur complement of* A *in* M *is* $M/A := D - CA^{-1}B$, *and* $\det(M) = \det(A)\det(M/A)$.

Proof. Block Gaussian elimination gives $\begin{bmatrix} I_r & 0 \\ -CA^{-1} & I_s \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & D - CA^{-1}B \end{bmatrix}$.
The elimination matrix is lower block-triangular with identity diagonal blocks, so its determinant is 1. Take determinants, apply Theorem 4.1.8 to right-hand block triangular matrix, $\det(M) = \det(A)\det(D - CA^{-1}B)$.
Structural Commentary: Schur complementation is block Gaussian elimination.
Key Insight: Eliminating C splits the determinant into $\det(A)$ times the remaining block determinant. □

Proposition 4.1.10. *Schur Complement of the Lower-Right Block*
Let $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, $A \in \mathrm{Mat}(r, r, \mathbb{F})$, $D \in \mathrm{Mat}(s, s, \mathbb{F})$.
If D *is invertible, then the Schur complement of* D *in* M *is* $M/D := A - BD^{-1}C$, *and* $\det(M) = \det(D)\det(M/D)$.

Proof. Block Gaussian elimination gives $\begin{bmatrix} I_r & -BD^{-1} \\ 0 & I_s \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A - BD^{-1}C & 0 \\ C & D \end{bmatrix}$.
The elimination matrix has determinant 1. The transpose of the matrix on the right is upper block-triangular, and $\det(X^t) = \det(X)$ by Theorem 4.1.5. Therefore Theorem 4.1.8 gives $\det(M) = \det(A - BD^{-1}C)\det(D)$.
Structural Commentary: This is the lower-right version of Schur complementation. Eliminating the upper-right block isolates the determinant contribution of D.
Key Insight: Eliminate by blocks, then read the determinant from the diagonal blocks. □

Theorem 4.1.11. *Sylvester Determinant Identity*
Let $A \in \mathrm{Mat}(n, m, \mathbb{F})$ *and* $B \in \mathrm{Mat}(m, n, \mathbb{F})$. *Then* $\det(I_n + AB) = \det(I_m + BA)$.

Proof. Let $M = \begin{bmatrix} I_n & -A \\ B & I_m \end{bmatrix}$.
Using Proposition 4.1.10 with lower-right block I_m gives $\det(M) = \det(I_n + AB)$. Using Proposition 4.1.9 with upper-left block I_n gives $\det(M) = \det(I_m + BA)$. Hence the two determinants are equal.
Structural Commentary: Two Schur complements of one block matrix give Sylvester's identity.
Key Insight: Compare the same determinant after eliminating different identity blocks. □

Theorem 4.1.12. *Cauchy–Binet Formula*
Let $A \in \mathrm{Mat}(k, n, \mathbb{F})$, $B \in \mathrm{Mat}(n, k, \mathbb{F})$, *and* $k \le n$. *For* $J = \{j_1 < \cdots < j_k\} \subseteq \{1, \ldots, n\}$, *let* A^J *use the columns of* A *in* J *and* B_J *use the rows of* B *in* J.
Then $\det(AB) = \sum_J \det(A^J)\det(B_J)$, *where* J *ranges over all* k*-subsets.*

Proof. Put $S(A, B) := \sum_J \det(A^J)\det(B_J)$. For an elementary $n \times n$ matrix E, determinant multilinearity and row/column operation rules give $S(A, B) = S(AE, E^{-1}B)$, and $AB = (AE)(E^{-1}B)$.
If $\operatorname{rank} A < k$, both sides vanish. Otherwise reduce A by elementary column operations to $A' = [C\ 0]$ with C invertible, and transform B oppositely to B'.
Then $AB = A'B'$ and $\det(A'B') = \det(C)\det(B'_{J_0}) = S(A', B')$, $J_0 = \{1, \ldots, k\}$, as every other k-minor of A' is zero. Hence $\det(AB) = S(A, B)$.
Structural Commentary: Cauchy–Binet is determinant multiplicativity for rectangular middle dimensions.
Key Insight: The middle space contributes one k-coordinate shadow at a time. □

Proposition 4.1.13. *Bordered Determinant Formulas*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$, *let* $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, *and let* $c \in \mathbb{F}$.

(i) If $c \neq 0$, *then* $\det \begin{bmatrix} c & \mathbf{x}^t \\ \mathbf{y} & A \end{bmatrix} = c \det\left(A - \frac{1}{c}\mathbf{y}\mathbf{x}^t\right)$.

(ii) If A *is invertible, then* $\det \begin{bmatrix} A & \mathbf{x} \\ \mathbf{y}^t & c \end{bmatrix} = \left(c - \mathbf{y}^t A^{-1}\mathbf{x}\right)\det(A)$

Proof.

(i) This is Proposition 4.1.9 with the scalar block c in the upper-left corner.
(ii) This is Proposition 4.1.9 with A as the upper-left block and $D = c$.

Structural Commentary: Bordered determinants are Schur complements with scalar borders.
Key Insight: Treat the border as a 1×1 block and apply the Schur complement formula. □

Definition 4.1.14. *Minors, Cofactors, and Adjugate*
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, let M_{ij} delete row i and column j from A; set $\det(\emptyset_{0\times 0}) = 1$. Define $C_{ij} := (-1)^{i+j}\det(M_{ij})$. Then $C = [C_{ij}]$ is the *cofactor matrix*, and $\operatorname{adj}(A) := C^t$ is the *adjugate*.

Lemma 4.1.15. *Cofactor Expansion*
For $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$, $\det A = \sum_{k=1}^{n} a_{ik}C_{ik}$ *and* $\det A = \sum_{k=1}^{n} a_{kj}C_{kj}$.

Proof. Using $\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{r=1}^{n} a_{r,\sigma(r)}$, group terms by $\sigma(i) = k$.
For fixed k, the remaining permutation sum equals $(-1)^{i+k}\det(M_{ik}) = C_{ik}$, giving the row formula. Applying this to A^t and using Theorem 4.1.5 gives the column formula.

Structural Commentary: Cofactor expansion is the big determinant formula collected around one row or column.
Key Insight: Fix one entry; the rest is its minor. □

Remark 4.1.16. *Adjugate Form of the Bordered Formula*
Proposition 4.1.13 gives the adjugate form $\det \begin{bmatrix} A & \mathbf{x} \\ \mathbf{y}^t & c \end{bmatrix} = c\det(A) - \mathbf{y}^t \operatorname{adj}(A)\mathbf{x}$. No invertibility of A is required.

Theorem 4.1.17. *Adjugate Identity and Cramer's Rule*
Let $A \in \operatorname{Mat}(n,n,\mathbb{F})$ and let $\mathbf{b} \in \mathbb{F}^n$. Then $A\operatorname{adj}(A) = \operatorname{adj}(A)A = (\det A)I_n$. If $\det A \neq 0$, then, for $A\mathbf{x} = \mathbf{b}$, $A^{-1} = \frac{1}{\det A}\operatorname{adj}(A)$ and $x_j = \frac{\det A_j(\mathbf{b})}{\det A}$ for $1 \leq j \leq n$, where $A_j(\mathbf{b})$ replaces column j of A by $\mathbf{b}$.

Proof. Since $(\operatorname{adj} A)_{kj} = C_{jk}$, $(A \operatorname{adj} A)_{ij} = \sum_k a_{ik}C_{jk}$.
For $i = j$ this is $\det A$ by Lemma 4.1.15; for $i \neq j$ it is the cofactor expansion of a matrix with rows i and j equal, hence 0. Thus $A\operatorname{adj}(A) = (\det A)I_n$. The same argument applied to A^t gives $\operatorname{adj}(A)A = (\det A)I_n$. If $\det A \neq 0$, division gives the inverse formula. Finally, $(\operatorname{adj} A\,\mathbf{b})_j = \sum_i C_{ij}b_i = \det A_j(\mathbf{b})$, by expansion down column j, so $x = A^{-1}\mathbf{b}$ gives Cramer's rule.
Structural Commentary: Cofactors assemble all one-column replacements of A into a single inverse formula.
Key Insight: Off-diagonal products vanish because they are determinants with two equal rows or columns. □

Theorem 4.1.18. *Trace and Determinant Are Similarity Invariants*
If $A, B \in \operatorname{Mat}(n,n,\mathbb{F})$ are similar, then $\operatorname{tr}(A) = \operatorname{tr}(B)$, $\det(A) = \det(B)$.

Proof. Write $B = PAP^{-1}$ with P invertible.
Proposition 4.1.4 gives $\operatorname{tr}(B) = \operatorname{tr}\big(P(AP^{-1})\big) = \operatorname{tr}\big((AP^{-1})P\big) = \operatorname{tr}(A)$.
Theorem 4.1.5 gives $\det(B) = \det(P)\det(A)\det(P^{-1}) = \det(A)$.
Structural Commentary: Trace and determinant ignore the coordinate system used to write an operator.
Key Insight: Conjugating by P inserts P and P^{-1}, which cancel inside trace and determinant. □

Principle 4.1.19. *Trace and Determinant Are Coordinate-Free*
Trace and determinant are first computed from a matrix, but their real structural meaning is operator-level: changing basis conjugates the matrix, and conjugation preserves both quantities.

Theorem 4.1.20. *Trace and Determinant of an Operator Are Basis-Independent*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let β, γ be bases of $\mathcal{V}$.
Then $\operatorname{tr}\big([T]_{\beta\leftarrow\beta}\big) = \operatorname{tr}\big([T]_{\gamma\leftarrow\gamma}\big)$, $\det\big([T]_{\beta\leftarrow\beta}\big) = \det\big([T]_{\gamma\leftarrow\gamma}\big)$.

Proof. By Theorem 1.3.66, the two matrices are similar: $[T]_{\gamma\leftarrow\gamma} = C_{\gamma\leftarrow\beta}[T]_{\beta\leftarrow\beta}C_{\beta\leftarrow\gamma}$.
Theorem 4.1.18 now gives equality of traces and determinants.
Structural Commentary: Basis changes conjugate matrices; trace and determinant remain operator invariants.
Key Insight: Similar matrices have the same trace and determinant. □

Definition 4.1.21. *Trace and Determinant of an Operator*
Let $\mathcal{V}$ be finite-dim, let $T \in \mathcal{L}(\mathcal{V})$. Choose any basis β of $\mathcal{V}$. Define $\operatorname{tr} T := \operatorname{tr}\big([T]_{\beta\leftarrow\beta}\big)$, $\det T := \det\big([T]_{\beta\leftarrow\beta}\big)$.
Theorem 4.1.20 shows these definitions do not depend on chosen basis.

Proposition 4.1.22. *Unitary Matrices and Isometries Have Determinant of Modulus One*
Let $U \in \operatorname{Mat}(n,n,\mathbb{F})$ be unitary. Then $|\det U| = 1$. More generally, if $S \in \mathcal{L}(\mathcal{V})$ is an isometry on a finite-dimensional inner-product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, then $|\det S| = 1$.

Proof. The permutation definition gives $\det(U^*) = \overline{\det U}$. By Theorem 4.1.5, taking determinants in $U^*U = I_n$ gives $1 = \det(U^*)\det(U) = \overline{\det U}\det U = |\det U|^2$. Hence $|\det U| = 1$.
For an isometry S, choose an orthonormal basis and let U be the matrix of S in that basis. Then U is unitary by Theorem 2.4.19, and $\det S = \det U$.
Structural Commentary: In orthonormal coordinates, isometries are unitary or orthogonal, so the determinant calculation is the matrix one.
Key Insight: Determinants turn $U^*U = I$ into $|\det U|^2 = 1$. □

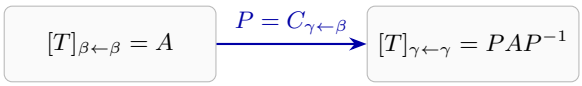


Figure 63: Basis changes conjugate matrices; trace and determinant are unchanged.

Proposition 4.1.23. *Trace and Determinant from q_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dim over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $T \in \mathcal{L}(\mathcal{V})$, $q_T(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0$, $n = \dim \mathcal{V}$. Then $\operatorname{tr} T = -c_{n-1}$ and $\det T = (-1)^n c_0$. If $\mathbb{F} = \mathbb{C}$ and $\lambda_1, \ldots, \lambda_n$ list the eigenvalues of T with algebraic multiplicity, then $\operatorname{tr} T = \lambda_1 + \cdots + \lambda_n$, $\det T = \lambda_1 \cdots \lambda_n$.

Proof. Choose a basis β and set $A = [T]_{\beta \leftarrow \beta}$.
Then $q_T = p_A$, and by Definition 4.1.21, $\operatorname{tr} T = \operatorname{tr} A$ and $\det T = \det A$.
Theorem 3.2.10 gives $\operatorname{tr} T = -c_{n-1}$ and $\det T = (-1)^n c_0$.
If $\mathbb{F} = \mathbb{C}$, then comparing $q_T(z) = (z - \lambda_1) \cdots (z - \lambda_n)$ with the displayed form of q_T gives the formulas.
Structural Commentary: Characteristic polynomial stores trace and determinant in first and last coefficients.
Key Insight: Compare coefficients of $\det(zI - T)$. □

Proposition 4.1.24. *Trace and Determinant from Eigenvalues*
Let $\lambda_1, \ldots, \lambda_n$ list the eigenvalues of $A \in \operatorname{Mat}(n, n, \mathbb{C})$ according to m_A. Then $\operatorname{tr} A = \sum_{j=1}^{n} \lambda_j$ and $\det A = \prod_{j=1}^{n} \lambda_j$.

Proof. By Fundamental Theorem of Algebra, $p_A(z) = \prod_{j=1}^{n} (z - \lambda_j) = z^n - \left(\sum_{j=1}^{n} \lambda_j \right) z^{n-1} + \cdots + (-1)^n \prod_{j=1}^{n} \lambda_j$.
Theorem 3.2.10 gives $p_A(z) = z^n - (\operatorname{tr} A) z^{n-1} + \cdots + (-1)^n \det A$. Uniqueness of coefficients gives the identities.
Structural Commentary: Coefficients of p_A encode symmetric functions of the eigenvalues.
Key Insight: Trace is the first root sum and determinant is the full root product. □

Proposition 4.1.25. *Transpose, Conjugate, and Adjoint Spectra*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$. Then $p_{A^t}(z) = p_A(z)$, $p_{\overline{A}}(z) = \overline{p_A(\overline{z})}$, and $p_{A^*}(z) = \overline{p_A(\overline{z})}$.
Consequently, if $\lambda \in \operatorname{spec}(A)$, then $m_{A^t}(\lambda) = m_A(\lambda)$ and $m_{\overline{A}}(\overline{\lambda}) = m_{A^*}(\overline{\lambda}) = m_A(\lambda)$.

Proof. Use determinant identities:

$$p_{A^t}(z) = \det(zI_n - A^t) = \det((zI_n - A)^t) = p_A(z),$$
$$p_{\overline{A}}(z) = \det(zI_n - \overline{A}) = \det(\overline{\overline{z} I_n - A}) = \overline{\det(\overline{z} I_n - A)} = \overline{p_A(\overline{z})}.$$

Since $A^* = \overline{A}^t$, the first identity applied to $\overline{A}$ gives the formula for A^*.
Multiplicity is preserved because conjugating a polynomial sends a factor $(z - \lambda)^k$ to $(z - \overline{\lambda})^k$.
Structural Commentary: Transpose preserves p_A; conjugation conjugates its roots.
Key Insight: The adjoint combines conjugation with transposition, so its spectrum is conjugated. □

Proposition 4.1.26. *Transpose Preserves Spectrum and Geometric Multiplicity*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$ and $\lambda \in \mathbb{F}$. Then $\lambda \in \operatorname{spec}(A) \iff \lambda \in \operatorname{spec}(A^t)$, and $\dim E_\lambda(A) = \dim E_\lambda(A^t)$.

Proof. Since $(A - \lambda I_n)^t = A^t - \lambda I_n$, Theorem 1.3.49 gives $\operatorname{rank}(A - \lambda I_n) = \operatorname{rank}(A^t - \lambda I_n)$.
Matrix rank-nullity then gives $\dim E_\lambda(A) = n - \operatorname{rank}(A - \lambda I_n) = n - \operatorname{rank}(A^t - \lambda I_n) = \dim E_\lambda(A^t)$.
($\Rightarrow$) If $\lambda \in \operatorname{spec}(A)$, then $E_\lambda(A) \neq \{\mathbf{0}\}$, so $E_\lambda(A^t) \neq \{\mathbf{0}\}$ and $\lambda \in \operatorname{spec}(A^t)$.
($\Leftarrow$) The same argument with A and A^t interchanged gives the converse.
Structural Commentary: Transposition preserves rank, so shifted kernels have the same dimension.
Key Insight: Spectrum and geometric multiplicity are kernel facts for $A - \lambda I_n$. □

Corollary 4.1.27. *Square Matrix Algebraic Invertibility Dictionary*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \operatorname{Mat}(n, n, \mathbb{F})$. The following are equivalent:

(i) A is invertible;
(ii) $\mathbf{a}_1, \ldots, \mathbf{a}_n$ form a basis of $\mathbb{F}^n$;
(iii) $\mathbf{a}_1, \ldots, \mathbf{a}_n$ are linearly independent;
(iv) $\mathbf{a}_1, \ldots, \mathbf{a}_n$ span $\mathbb{F}^n$;
(v) for every $\mathbf{b} \in \mathbb{F}^n$, the system $A\mathbf{x} = \mathbf{b}$ has a unique solution;
(vi) the homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the zero solution.

Proof.
(*i*) $\Rightarrow$ (*ii*) By Theorem 1.3.54, the columns of an invertible matrix form a basis.
(*ii*) $\Rightarrow$ (*iii*) A basis is linearly independent.
(*iii*) $\Rightarrow$ (*iv*) In $\mathbb{F}^n$, n linearly independent vectors form a basis, hence span.
(*iv*) $\Rightarrow$ (*v*) In $\mathbb{F}^n$, n spanning vectors form a basis, so every $\mathbf{b}$ has unique coordinates $\mathbf{x}$ in $A\mathbf{x} = \mathbf{b}$.
(*v*) $\Rightarrow$ (*vi*) Apply (v) to $\mathbf{b} = \mathbf{0}$; since $\mathbf{x} = \mathbf{0}$ is a solution, it is the only one.
(*vi*) $\Rightarrow$ (*i*) Condition (vi) says the columns are independent.
Hence they form a basis of $\mathbb{F}^n$, and Corollary 1.3.62 gives invertibility.

Structural Commentary: Square invertibility is column basishood.
Key Insight: Invertible columns give unique coordinates. □

Corollary 4.1.28. *Nonsingularity by Rank, Determinant, and Spectrum*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$. The following are equivalent:

(i) *A is invertible;*
(ii) $\operatorname{rank}(A) = n$;
(iii) $\det A \neq 0$;
(iv) $0 \notin \operatorname{spec}(A)$.

Over $\mathbb{C}$, this is also equivalent to all eigenvalues of A being nonzero.

Proof.

(i) $\Rightarrow$ *(ii)* This is Corollary 1.3.64.
(ii) $\Rightarrow$ *(iii)* Full rank gives invertibility by Corollary 1.3.64, hence $\det A \neq 0$ by Theorem 4.1.5.
(iii) $\Rightarrow$ *(iv)* If $\det A \neq 0$, then A is invertible by Theorem 4.1.5; hence $0 \notin \operatorname{spec}(A)$ by Corollary 3.1.9.
(iv) $\Rightarrow$ *(i)* This is Corollary 3.1.9.

Over $\mathbb{C}$, Theorem 3.2.10 identifies $\lambda_1, \dots, \lambda_n$ as the roots of p_A, so (iv) is equivalent to $\lambda_j \neq 0$ for all j.
Structural Commentary: Rank, determinant, and spectrum detect singularity.
Key Insight: Singularity can be seen as rank loss, zero determinant, or the zero eigenvalue. □

Method 4.1.29. *Trace–Determinant Invariant Method*
To get eigenvalue data without solving for eigenvalues:

(i) choose a basis in which $[T]_{\beta \leftarrow \beta}$ is easiest to compute;
(ii) compute $\operatorname{tr} T$ and $\det T$ from this matrix;
(iii) if the eigenvalues are $\lambda_1, \dots, \lambda_n$, conclude $\lambda_1 + \cdots + \lambda_n = \operatorname{tr} T$, and $\lambda_1 \cdots \lambda_n = \det T$;
(iv) for equations $ST - TS = A$, first test $\operatorname{tr} A = 0$; otherwise no solution exists.

Theorem 4.1.30. *Full-Rank Products Preserve Rank*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$. If $X \in \mathrm{Mat}(p,m,\mathbb{F})$ has full column rank and $Y \in \mathrm{Mat}(n,q,\mathbb{F})$ has full row rank, then $\operatorname{rank}(A) = \operatorname{rank}(XAY)$. This holds when X and Y are square invertible matrices of appropriate sizes.

Proof. Because X has full column rank, the induced map $L_X : \mathbb{F}^m \to \mathbb{F}^p$, $L_X(\mathbf{u}) = X\mathbf{u}$, has kernel $\{\mathbf{0}\}$.
Thus $\mathbf{z} \in \ker(XA) \iff X(A\mathbf{z}) = \mathbf{0} \iff A\mathbf{z} = \mathbf{0} \iff \mathbf{z} \in \ker(A)$.
Since XA and A have the same number of columns, Corollary 1.3.48 gives $\operatorname{rank}(XA) = \operatorname{rank}(A)$.
Because Y has full row rank, Y^t has full column rank. Applying the left-multiplication result to A^t and Y^t, then using Theorem 1.3.49, gives $\operatorname{rank}(AY) = \operatorname{rank}((AY)^t) = \operatorname{rank}(Y^tA^t) = \operatorname{rank}(A^t) = \operatorname{rank}(A)$.
Applying the left-multiplication result once more to AY gives $\operatorname{rank}(XAY) = \operatorname{rank}(AY) = \operatorname{rank}(A)$.
Structural Commentary: Multiplication by a full-column-rank matrix on the left loses no output information, while full row rank on the right loses no input information.
Key Insight: Rank changes only when multiplication introduces a new kernel or collapses the row side. □

Theorem 4.1.31. *Rank Bounds for Products and Concatenations*
Let $A \in \mathrm{Mat}(m,k,\mathbb{F})$, $B \in \mathrm{Mat}(k,n,\mathbb{F})$, and $C \in \mathrm{Mat}(m,p,\mathbb{F})$. Then $\operatorname{rank}(AB) \leq \min(\operatorname{rank}(A), \operatorname{rank}(B))$ and $\operatorname{rank}(AB) = \operatorname{rank}(A)$ if and only if $\operatorname{col}(AB) = \operatorname{col}(A)$.
Also, $\max(\operatorname{rank}(A), \operatorname{rank}(C)) \leq \operatorname{rank}\begin{bmatrix} A & C \end{bmatrix}$ and $\operatorname{rank}\begin{bmatrix} A & C \end{bmatrix} = \operatorname{rank}(A)$ if and only if $\operatorname{col}(A) = \operatorname{col}(A) + \operatorname{col}(C)$.

Proof. Each column of AB is a linear combination of the columns of A, so $\operatorname{col}(AB) \subseteq \operatorname{col}(A)$. Therefore $\operatorname{rank}(AB) \leq \operatorname{rank}(A)$; since both spaces are finite-dimensional, equality holds exactly when $\operatorname{col}(AB) = \operatorname{col}(A)$.
Applying this same argument to $(AB)^t = B^tA^t$ and using Theorem 1.3.49 gives $\operatorname{rank}(AB) \leq \operatorname{rank}(B)$.
For the block matrix, its columns are precisely the columns of A and C, hence $\operatorname{col}\begin{bmatrix} A & C \end{bmatrix} = \operatorname{col}(A) + \operatorname{col}(C)$.
This space contains both $\operatorname{col}(A)$ and $\operatorname{col}(C)$, giving the lower bound. Equality with $\operatorname{rank}(A)$ occurs exactly when the containing subspace has the same dimension as $\operatorname{col}(A)$, equivalently when $\operatorname{col}(A) + \operatorname{col}(C) = \operatorname{col}(A)$.
Structural Commentary: Products can only reduce accessible column directions, while horizontal concatenation can only add column directions.
Key Insight: Rank tracks whether a matrix operation preserves, loses, or adds column-space dimension. □

Theorem 4.1.32. *Sylvester Rank Inequality*
Let $A \in \mathrm{Mat}(m,k,\mathbb{F})$ and $B \in \mathrm{Mat}(k,n,\mathbb{F})$. Then $\operatorname{rank}(A) + \operatorname{rank}(B) - k \leq \operatorname{rank}(AB)$.

Proof. Let $R = \mathrm{col}(B) \subseteq \mathbb{F}^k$. The columns of AB are A applied to the columns of B, so $\mathrm{col}(AB) = A(R)$.
Apply rank-nullity to the restriction $T_A|_R : R \to \mathbb{F}^m$. Its range is $A(R) = \mathrm{col}(AB)$ and its kernel is $R \cap \ker(A)$.
Hence $\mathrm{rank}(AB) = \dim(A(R)) = \dim(R) - \dim(R \cap \ker(A)) \geq \mathrm{rank}(B) - \dim(\ker(A))$.
Since A has k columns, Corollary 1.3.48 gives $\dim(\ker(A)) = k - \mathrm{rank}(A)$. Substitution gives desired inequality.
Structural Commentary: Multiplication by A can lose only the part of $\mathrm{col}(B)$ lying in $\ker(A)$. Sylvester's inequality is this loss estimate in dimensions.
Key Insight: The shared middle dimension k bounds how much rank can disappear in the product. □

Theorem 4.1.33. *Frobenius Rank Inequality*
Let $A \in \mathrm{Mat}(m, k, \mathbb{F})$, $B \in \mathrm{Mat}(k, p, \mathbb{F})$, and $C \in \mathrm{Mat}(p, n, \mathbb{F})$.
Then $\mathrm{rank}(AB) + \mathrm{rank}(BC) \leq \mathrm{rank}(B) + \mathrm{rank}(ABC)$.

Proof. Let $R = \mathrm{col}(B)$ and $S = \mathrm{col}(BC)$. Since every column of BC is a linear combination of columns of B, we have $S \subseteq R$. The induced map T_A maps R onto $\mathrm{col}(AB)$ and maps S onto $\mathrm{col}(ABC)$.
Rank-nullity on the restrictions $T_A|_R$ and $T_A|_S$ gives

$$\mathrm{rank}(AB) = \dim R - \dim(R \cap \ker(A)), \quad \mathrm{rank}(ABC) = \dim S - \dim(S \cap \ker(A)).$$

Since $S \cap \ker(A) \subseteq R \cap \ker(A)$, we have $\dim(S \cap \ker(A)) \leq \dim(R \cap \ker(A))$.
Thus $\dim R - \dim(R \cap \ker(A)) + \dim S \leq \dim R + \dim S - \dim(S \cap \ker(A))$.
Substituting $\dim R = \mathrm{rank}(B)$ and $\dim S = \mathrm{rank}(BC)$ gives the stated inequality.
Structural Commentary: Frobenius tracks A on nested column spaces.
Key Insight: The smaller column space cannot lose more dimensions to $\ker(A)$ than the larger one. □

Corollary 4.1.34. *Rank Inequalities from Frobenius*
The following are special cases of Theorem 4.1.33:

(i) $\mathrm{rank}(AB) \leq \min(\mathrm{rank}(A), \mathrm{rank}(B))$;
(ii) $\mathrm{rank}(A) + \mathrm{rank}(B) - k \leq \mathrm{rank}(AB)$.

Proof.

(i) Apply Frobenius to conformable factors X, Y, Z. Taking $Z = 0$ gives $\mathrm{rank}(XY) \leq \mathrm{rank}(Y)$, because $\mathrm{rank}(Y0) = \mathrm{rank}(XY0) = 0$. Taking $X = 0$ gives $\mathrm{rank}(YZ) \leq \mathrm{rank}(Y)$.
Renaming the factors gives $\mathrm{rank}(AB) \leq \mathrm{rank}(B)$ and $\mathrm{rank}(AB) \leq \mathrm{rank}(A)$.

(ii) Taking the middle matrix to be I_k gives $\mathrm{rank}(A) + \mathrm{rank}(B) \leq k + \mathrm{rank}(AB)$, which is Sylvester's inequality.

Structural Commentary: Frobenius is the three-factor parent inequality. The usual upper and lower product-rank bounds appear by making one factor zero or identity.
Key Insight: Zero factors isolate upper bounds; an identity middle factor yields the Sylvester lower bound. □

Proposition 4.1.35. *Sylvester's Law of Nullity*
For a matrix X, let $\mathrm{nullity}(X) = \dim(\ker(X))$; this is the dimension of its nullspace.

(i) If $A \in \mathrm{Mat}(m, k, \mathbb{F})$ and $B \in \mathrm{Mat}(k, n, \mathbb{F})$, then $\mathrm{nullity}(AB) \leq \mathrm{nullity}(A) + \mathrm{nullity}(B)$.
(ii) If $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$, then $\max(\mathrm{nullity}(A), \mathrm{nullity}(B)) \leq \mathrm{nullity}(AB)$.

Proof.

(i) By rank-nullity and Theorem 4.1.32,

$$\begin{aligned}\mathrm{nullity}(AB) &= n - \mathrm{rank}(AB) \leq n - \mathrm{rank}(A) - \mathrm{rank}(B) + k \\ &= (k - \mathrm{rank}(A)) + (n - \mathrm{rank}(B)) = \mathrm{nullity}(A) + \mathrm{nullity}(B).\end{aligned}$$

(ii) For square A, B, $\mathrm{rank}(AB) \leq \mathrm{rank}(A)$, $\mathrm{rank}(AB) \leq \mathrm{rank}(B)$. Subtract from n gives desired lower bounds.

Structural Commentary: Nullity is rank loss from the domain.
Key Insight: Use rank-nullity to translate rank bounds into kernel bounds. □

Theorem 4.1.36. *Rank Triangle Inequality*
If $A, B \in \mathrm{Mat}(m, n, \mathbb{F})$, then $|\mathrm{rank}(A) - \mathrm{rank}(B)| \leq \mathrm{rank}(A + B) \leq \mathrm{rank}(A) + \mathrm{rank}(B)$.

Proof. Since $\mathrm{col}(A + B) \subseteq \mathrm{col}(A) + \mathrm{col}(B)$, Theorem 1.2.10 gives $\mathrm{rank}(A + B) \leq \dim(\mathrm{col}(A) + \mathrm{col}(B)) \leq \mathrm{rank}(A) + \mathrm{rank}(B)$. Applying this upper bound to $A = (A + B) - B$ gives $\mathrm{rank}(A) \leq \mathrm{rank}(A + B) + \mathrm{rank}(B)$, so $\mathrm{rank}(A) - \mathrm{rank}(B) \leq \mathrm{rank}(A + B)$. Applying it to $B = (A + B) - A$ gives $\mathrm{rank}(B) - \mathrm{rank}(A) \leq \mathrm{rank}(A + B)$.
Together these are exactly $|\mathrm{rank}(A) - \mathrm{rank}(B)| \leq \mathrm{rank}(A + B)$.
Structural Commentary: The rank of a sum is controlled by the sum of the two column spaces. The reverse bounds come by solving for either summand.
Key Insight: Rank behaves like a dimension-valued size under addition. □

Definition 4.1.37. *Commutator*
For $S, T \in \mathcal{L}(\mathcal{V})$, the *commutator* of S and T is $[S,T] := ST - TS$.
For $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$, the *matrix commutator* is $[A,B] := AB - BA$.

Proposition 4.1.38. *Trace Obstruction for Commutators*
If $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$, then $\mathrm{tr}(AB - BA) = 0$. Hence any matrix similar to a commutator has trace zero.

Proof. By linearity of trace and Proposition 4.1.4, $\mathrm{tr}(AB - BA) = \mathrm{tr}(AB) - \mathrm{tr}(BA) = 0$.
If C is similar to $AB - BA$, then Theorem 4.1.18 gives $\mathrm{tr}(C) = \mathrm{tr}(AB - BA) = 0$.
Structural Commentary: Trace kills commutators because it is cyclic.
Key Insight: AB and BA have the same trace, so their difference has trace zero. □

Lemma 4.1.39. *Scalar Matrices and One-Step Orbits*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$ with $n \geq 2$.
Then $(\mathbf{x}, A\mathbf{x})$ is linearly dependent for every $\mathbf{x} \in \mathbb{F}^n$ if and only if A is a scalar matrix.

Proof.
($\Rightarrow$) Suppose $A = cI_n$. Then $A\mathbf{x} = c\mathbf{x}$ for every $\mathbf{x}$, so $(\mathbf{x}, A\mathbf{x})$ is linearly dependent.
($\Leftarrow$) Suppose $(\mathbf{x}, A\mathbf{x})$ is linearly dependent for every $\mathbf{x}$.
For each standard basis vector $\mathbf{e}_i$, $\exists a_i \in \mathbb{F}$ such that $A\mathbf{e}_i = a_i\mathbf{e}_i$. For $i \geq 2$, dependence of $\mathbf{e}_1 + \mathbf{e}_i$ and $A(\mathbf{e}_1 + \mathbf{e}_i) = a_1\mathbf{e}_1 + a_i\mathbf{e}_i$ means $a_1\mathbf{e}_1 + a_i\mathbf{e}_i = b(\mathbf{e}_1 + \mathbf{e}_i)$ for some $b \in \mathbb{F}$. Comparing the independent vectors $\mathbf{e}_1, \mathbf{e}_i$ gives $a_1 = b = a_i$. Hence every a_i equals a_1, so $A = a_1 I_n$.

Structural Commentary: A scalar matrix preserves every one-dimensional coordinate direction and every diagonal line between two coordinates. These diagonal lines force all coordinate scalars to agree.
Key Insight: If every vector is an eigenvector, then there can be only one eigenvalue. □

Lemma 4.1.40. *Nonscalar Matrices Have a Zero Diagonal Entry After Similarity*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$ with $n \geq 2$.
If A is not a scalar matrix, then A is similar over $\mathbb{F}$ to a matrix whose $(1,1)$ entry is 0.

Proof. By Lemma 4.1.39, there is $\mathbf{x} \in \mathbb{F}^n$ such that $(\mathbf{x}, A\mathbf{x})$ is linearly independent. Extend $(\mathbf{x}, A\mathbf{x})$ to a basis β of $\mathbb{F}^n$. In the basis β, the first column of the matrix of A is the coordinate vector of $A\mathbf{x}$, namely $(0,1,0,\ldots,0)^t$. Thus the $(1,1)$ entry is 0, and this matrix is similar to A.
Structural Commentary: Nonscalarity produces a vector whose first iterate points in a genuinely new direction. Using these two vectors as the start of a basis forces a zero in the first diagonal position.
Key Insight: Put $\mathbf{x}$ first and $A\mathbf{x}$ second in the basis. □

Theorem 4.1.41. *Similarity to a Constant-Diagonal Matrix*
Let $n \geq 1$, let $A \in \mathrm{Mat}(n,n,\mathbb{F})$, and assume $k \cdot 1_{\mathbb{F}} \neq 0$ for every $1 \leq k \leq n$. Then A is similar over $\mathbb{F}$ to a matrix whose diagonal entries are all equal to $\frac{1}{n}\mathrm{tr}(A)$.

Proof. Set $\tau = \frac{1}{n}\mathrm{tr}(A)$ and $B = A - \tau I_n$. Then $\mathrm{tr}(B) = 0$. It is enough to show that every trace-zero matrix is similar to a matrix with zero diagonal, because Theorem 1.3.71 then adds τI_n back to the diagonal.
To prove the trace-zero claim by induction on n. If $n = 1$, a trace-zero matrix is $[0]$. Let $n \geq 2$ and assume the claim holds for smaller sizes. Let $\mathrm{tr}(B) = 0$. If B is scalar, then $B = cI_n$ and $0 = \mathrm{tr}(B) = nc$.
Since $n \cdot 1_{\mathbb{F}} \neq 0$, we get $c = 0$, so B already has zero diagonal.
If B is not scalar, Lemma 4.1.40 makes B similar to $\begin{bmatrix} 0 & * \\ * & B_1 \end{bmatrix}$, and $\mathrm{tr}(B_1) = 0$.
By induction, choose an invertible S_1 such that $S_1^{-1}B_1S_1$ has zero diagonal. Conjugating the displayed block matrix by $1 \oplus S_1$ leaves the first diagonal entry equal to 0 and replaces the lower-right block by $S_1^{-1}B_1S_1$.
Hence the resulting similar matrix has every diagonal entry equal to 0.
Structural Commentary: The proof removes one zero diagonal entry at a time.
Key Insight: Similarity creates the first zero; trace zero handles the smaller block. □

Theorem 4.1.42. *Shoda's Theorem in the Distinct-Diagonal Case*
Let $n \geq 1$ and assume $k \cdot 1_{\mathbb{F}} \neq 0$ for every $1 \leq k \leq n$. For $A \in \mathrm{Mat}(n,n,\mathbb{F})$, the matrix A is a commutator in $\mathrm{Mat}(n,n,\mathbb{F})$ if and only if $\mathrm{tr}(A) = 0$. In particular, no nonzero scalar matrix is a commutator.

Proof.
($\Rightarrow$) Proposition 4.1.38 gives $\mathrm{tr}(A) = 0$.
Thus cI_n can be a commutator only when $nc = \mathrm{tr}(cI_n) = 0$, hence only when $c = 0$.

($\Leftarrow$) Suppose $\operatorname{tr}(A) = 0$. By Theorem 4.1.41, there is an invertible S such that $B = S^{-1}AS = [b_{ij}]$ has zero diagonal. Let $X = \operatorname{diag}(1, 2, \ldots, n)$, viewing the integers as scalars in $\mathbb{F}$, and define $Y = [y_{ij}]$ by

$$y_{ij} = \begin{cases} (i-j)^{-1}b_{ij}, & i \neq j, \\ 0, & i = j. \end{cases}$$

The hypothesis ensures that $i - j \neq 0$ in $\mathbb{F}$ whenever $i \neq j$. Thus $XY - YX = [(i-j)y_{ij}] = B$. Hence $A = (SXS^{-1})(SYS^{-1}) - (SYS^{-1})(SXS^{-1})$, so A is a commutator.

Structural Commentary: Trace is the obstruction; similarity exposes zero diagonal form.
Key Insight: Cyclicity forces trace zero; distinct diagonals realize it. □

Remark 4.1.43. *Field Hypothesis in the Diagonal Proof*
The proof above uses that the integers $1, \ldots, n$ are distinct and that each smaller block size remains nonzero in $\mathbb{F}$. In small characteristic, this diagonal proof must be replaced by a more delicate argument.

Corollary 4.1.44. *The Identity Is Not a Commutator*
Assume $\dim \mathcal{V} \cdot 1_{\mathbb{F}} \neq 0$. *There do not exist* $S, T \in \mathcal{L}(\mathcal{V})$ *such that* $ST - TS = I_{\mathcal{V}}$.

Proof. If $ST - TS = I_{\mathcal{V}}$, choose a basis and let $A = [S]$ and $B = [T]$ in that basis. Then $AB - BA = I_n$. Taking traces and using Proposition 4.1.4, $0 = \operatorname{tr}(AB) - \operatorname{tr}(BA) = \operatorname{tr}(I_n) = n \cdot 1_{\mathbb{F}}$, contradicting the hypothesis.
Structural Commentary: Commutators have zero trace, while the identity has dimension as its trace.
Key Insight: Trace detects a scalar obstruction invisible from multiplication alone. □

Definition 4.1.45. *Idempotent and Nilpotent Matrices*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$.

(i) A is *idempotent* if $A^2 = A$.
(ii) A is *nilpotent* if $A^k = 0$ for some positive integer k.

An idempotent matrix is an algebraic projection: it fixes its range and kills its kernel.

Proposition 4.1.46. *Basic Invariants of Idempotents*
If $P \in \operatorname{Mat}(n, n, \mathbb{F})$ *satisfies* $P^2 = P$, *then* $\mathbb{F}^n = \operatorname{range}(P) \oplus \ker(P)$, P *is diagonalizable, and its only possible eigenvalues are* 0 *and* 1. *Moreover* $\operatorname{tr}(P) = \operatorname{rank}(P)$.
If $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, *then* P *is the orthogonal projection onto* $\operatorname{range}(P)$ *if and only if* $P = P^*$.

Proof. For every $\mathbf{x}$, write $\mathbf{x} = P\mathbf{x} + (\mathbf{x} - P\mathbf{x})$. The first term lies in $\operatorname{range}(P)$, while $P(\mathbf{x} - P\mathbf{x}) = P\mathbf{x} - P^2\mathbf{x} = 0$, so the second term lies in $\ker(P)$. The intersection is zero because $\mathbf{x} = P\mathbf{y}$ and $P\mathbf{x} = 0$ imply $\mathbf{x} = P^2\mathbf{y} = P\mathbf{x} = 0$. In a basis adapted to $\operatorname{range}(P) \oplus \ker(P)$, the matrix of P is $I_r \oplus 0$, where $r = \operatorname{rank}(P)$.
Hence P is diagonalizable, its eigenvalues lie in $\{0, 1\}$, and $\operatorname{tr}(P) = r$.
For the final claim, self-adjoint idempotents are exactly orthogonal projections by Theorem 2.4.14.
Structural Commentary: Idempotents split the space into fixed directions and killed directions.
Key Insight: In adapted coordinates, every idempotent is just $I_r \oplus 0$. □

Proposition 4.1.47. *Basic Invariants of Nilpotents*
For $N \in \operatorname{Mat}(n, n, \mathbb{F})$, *the following are equivalent:*

(i) N *is nilpotent;*
(ii) the characteristic polynomial of N *is* t^n;
(iii) $N^n = 0$.

In this case, $\operatorname{tr}(N) = 0$, $\det(N) = 0$ *for* $n \geq 1$, *and the spectral radius of* N *over* $\mathbb{C}$ *is* 0.

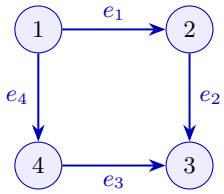
Proof. If $N^k = 0$, then every eigenvalue λ of N over an algebraic closure satisfies $\lambda^k = 0$, hence $\lambda = 0$. Thus the characteristic polynomial is t^n. If the characteristic polynomial is t^n, Cayley–Hamilton gives $N^n = 0$. The implication $N^n = 0 \Rightarrow$ nilpotent is immediate. The trace, determinant, and spectral-radius statements follow from the eigenvalue data and from Definitions 4.1.1 and 4.1.88.
Structural Commentary: Nilpotent matrices have no spectral mass away from zero.
Key Insight: Cayley–Hamilton turns the characteristic polynomial t^n into the equation $N^n = 0$. □

Definition 4.1.48. *Positive Semidefinite and Positive Definite Matrices*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let $A \in \operatorname{Mat}(n, n, \mathbb{F})$.
The matrix A is *positive semidefinite*, written $A \succeq 0$, if $A = A^*$ and $\mathbf{x}^* A\mathbf{x} \geq 0$ for every $\mathbf{x} \in \mathbb{F}^n$.
It is *positive definite*, written $A \succ 0$, if $A = A^*$ and $\mathbf{x}^* A\mathbf{x} > 0$ for every nonzero $\mathbf{x} \in \mathbb{F}^n$.
Negative semidefinite and negative definite mean $-A \succeq 0$ and $-A \succ 0$, respectively.

Proposition 4.1.49. *Rectangular Gram Products Are Positive Semidefinite*
Let $R \in \mathrm{Mat}(m,n,\mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$. Then $R^*R \succeq 0$, $RR^* \succeq 0$, and $(R^*R)_{j,j} = \|R_{\cdot,j}\|_2^2$; $(RR^*)_{i,i} = \|R_{i,\cdot}\|_2^2$. Full column rank gives $R^*R \succ 0$; full row rank gives $RR^* \succ 0$.

Proof. The matrices are self-adjoint by Proposition 2.4.6.
For $\mathbf{x} \in \mathbb{F}^n$ and $\mathbf{y} \in \mathbb{F}^m$, $\mathbf{x}^*R^*R\mathbf{x} = \|R\mathbf{x}\|_2^2 \geq 0$ and $\mathbf{y}^*RR^*\mathbf{y} = \|R^*\mathbf{y}\|_2^2 \geq 0$. Hence both products are positive semidefinite. Taking $\mathbf{x} = \mathbf{e}_j$ and $\mathbf{y} = \mathbf{e}_i$ gives the diagonal formulas. If R has full column rank, then $R\mathbf{x} \neq 0$ for $\mathbf{x} \neq 0$, so $R^*R \succ 0$. If R has full row rank, apply the same argument to R^*.
Structural Commentary: Multiplying a matrix by its adjoint produces a Gram matrix of columns or rows.
Key Insight: The quadratic form is a squared norm, so no cancellation can make it negative. □

Definition 4.1.50. *Directed Incidence Matrix*
For edges $e = u \to v$ on vertices $1,\dots,n$, $B \in \mathrm{Mat}(m,n,\mathbb{R})$ has row e equal to $-\mathbf{e}_u^t + \mathbf{e}_v^t$.



$$B = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 \end{bmatrix}$$

Figure 64: A directed incidence matrix records oriented edge differences from vertices to edges.

Example 4.1.51. *A Directed Incidence Matrix*
For Figure 64, $B = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 \end{bmatrix}$. For node potentials $\mathbf{x} = (x_1,x_2,x_3,x_4)^t$, $B\mathbf{x} = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_3 - x_4 \\ x_4 - x_1 \end{bmatrix}$.
Hence $\ker(B) = \mathrm{span}\{(1,1,1,1)^t\}$, and $\mathrm{rank}(B) = 3$. The loop vector $\mathbf{y} = (1,1,-1,-1)^t$ spans $\ker(B^t)$.
Thus $\mathbf{b} \in \mathrm{col}(B)$ if and only if $\mathbf{y}^t\mathbf{b} = b_1 + b_2 - b_3 - b_4 = 0$. This is Kirchhoff loop compatibility.
Structural Commentary: Incidence subspaces encode graph constraints.
Key Insight: A sparse matrix can encode both the algebra of a linear map and the combinatorics of a graph.

Proposition 4.1.52. *Incidence Laplacians Are Positive Semidefinite*
Let B be a directed incidence matrix and set $L = B^tB$. Then $L \succeq 0$ and $\mathbf{x}^tL\mathbf{x} = \sum_{e=(u\to v)} (x_v - x_u)^2$, with $\ker(L)$ equal to the vectors constant on each connected component.

Proof. Since $L = B^tB$, Proposition 4.1.49 gives $L \succeq 0$.
For $e = (u \to v)$, $(B\mathbf{x})_e = x_v - x_u$, so $\mathbf{x}^tL\mathbf{x} = \|B\mathbf{x}\|_2^2 = \sum_{e=(u\to v)} (x_v - x_u)^2$. Thus $\mathbf{x} \in \ker(L)$ if and only if $B\mathbf{x} = \mathbf{0}$, equivalently $x_u = x_v$ on every edge, which means $\mathbf{x}$ is constant on each connected component.
Structural Commentary: Laplacians sum squared edge differences.
Key Insight: A graph Laplacian is a Gram product, so positivity is automatic. □

Theorem 4.1.53. *Equivalent Forms of Positive Semidefiniteness*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$. The following are equivalent:

(i) $A \succeq 0$.
(ii) *A is Hermitian and all eigenvalues of A are nonnegative.*
(iii) $A = B^*B$ *for some* $B \in \mathrm{Mat}(m,n,\mathbb{F})$.
(iv) *A has a positive semidefinite square root.*

In this case the positive semidefinite square root is unique.
If $A = U\Lambda U^*$ with $\Lambda = \mathrm{diag}(\lambda_1,\dots,\lambda_n)$ and $\lambda_i \geq 0$, then $A^{1/2} = U\,\mathrm{diag}(\sqrt{\lambda_1},\dots,\sqrt{\lambda_n})U^*$.

Proof.

(i) $\Rightarrow$ *(ii)* This is the matrix form of Theorem 3.4.37.
(ii) $\Rightarrow$ *(iv)* By the spectral theorem, write $A = U\Lambda U^*$, where U is unitary and $\Lambda = \mathrm{diag}(\lambda_1,\dots,\lambda_n)$ with $\lambda_i \geq 0$. Then $A^{1/2} = U\,\mathrm{diag}(\sqrt{\lambda_1},\dots,\sqrt{\lambda_n})U^*$ is positive semidefinite and satisfies $(A^{1/2})^2 = A$.
(iv) $\Rightarrow$ *(iii)* If $C \succeq 0$ and $C^2 = A$, then $C = C^*$ and $A = C^*C$, so (iii) holds with $m = n$.
(iii) $\Rightarrow$ *(i)* If $A = B^*B$, then $A^* = A$ and $\mathbf{x}^*A\mathbf{x} = \mathbf{x}^*B^*B\mathbf{x} = \|B\mathbf{x}\|_2^2 \geq 0$, so $A \succeq 0$.

The positive semidefinite square root is unique by Proposition 3.4.38.
Structural Commentary: Positive semidefinite matrices are Gram-type matrices with nonnegative spectrum.
Key Insight: Positivity is best checked either on eigenvalues or by finding a factor B with $A = B^*B$. □

Method 4.1.54. *PSD Spectral Factorisation Method*
To analyse a Hermitian matrix A:

(i) diagonalise $A = U\Lambda U^*$;
(ii) check the signs of the diagonal entries of Λ;
(iii) if $\Lambda \geq 0$, set $A^{1/2} = U\Lambda^{1/2}U^*$ and use $A = (A^{1/2})^*A^{1/2}$;
(iv) translate kernel, rank, and zero-quadratic-form questions into the factor $A^{1/2}$.

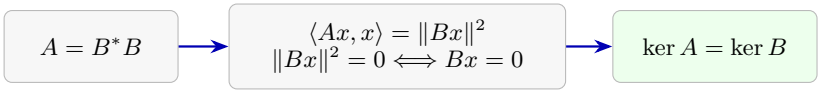


Figure 65: For $A = B^*B$, the squared-norm identity gives $\ker A = \ker B$.

Theorem 4.1.55. *Equivalent Forms of Positive Definiteness*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. The following are equivalent:

(i) $A \succ 0$.
(ii) *A is Hermitian and all eigenvalues of A are positive.*
(iii) $A \succeq 0$ *and A is invertible.*
(iv) $A = B^*B$ *for some invertible* $B \in \mathrm{Mat}(n, n, \mathbb{F})$.
(v) *A is invertible and* $A^{-1} \succ 0$.

Proof.

(i) $\Rightarrow$ (ii) By definition A is Hermitian. If $(\lambda, \mathbf{v})$ is an eigenpair with $\mathbf{v} \neq \mathbf{0}$, then $\lambda \|\mathbf{v}\|_2^2 = \mathbf{v}^*A\mathbf{v} > 0$, so every eigenvalue is positive.
(ii) $\Rightarrow$ (iii) Positive eigenvalues imply $A \succeq 0$ by Theorem 4.1.53; they also exclude 0, so A is invertible.
(iii) $\Rightarrow$ (iv) By Theorem 4.1.53, A has a positive semidefinite square root $A^{1/2}$.
Since A is invertible, $A^{1/2}$ is invertible, and $A = (A^{1/2})^*A^{1/2}$.
(iv) $\Rightarrow$ (i) If $A = B^*B$ and B is invertible, then for every nonzero $\mathbf{x}$, $\mathbf{x}^*A\mathbf{x} = \|B\mathbf{x}\|_2^2 > 0$.
(ii) $\Leftrightarrow$ (v) Hermitian-ness is preserved by inversion, and eigenvalues of A^{-1} are reciprocal eigenvalues of A. Thus positive eigenvalues of A are equivalent to positive eigenvalues of A^{-1}.

Structural Commentary: Positive definite matrices are the invertible part of the positive semidefinite cone.
Key Insight: Positive definite means every eigenvalue is strictly on the positive side of zero. □

Proposition 4.1.56. *Trace and Determinant of Positive Semidefinite Matrices*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ be positive semidefinite, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Then $\operatorname{tr} A \geq 0$, with equality if and only if $A = 0$, and $\det A \geq 0$, with strict inequality if and only if A is positive definite.

Proof. List the eigenvalues of A as $\lambda_1, \ldots, \lambda_n$. By Theorem 4.1.53, each $\lambda_j \geq 0$.
Hence $\operatorname{tr} A = \sum_{j=1}^{n} \lambda_j \geq 0$ and $\det A = \prod_{j=1}^{n} \lambda_j \geq 0$.
The trace is zero exactly when every $\lambda_j = 0$, which for a Hermitian matrix means $A = 0$. The determinant is positive exactly when every $\lambda_j > 0$, which is positive definiteness by Theorem 4.1.55.
Structural Commentary: Trace and determinant aggregate the nonnegative eigenvalues by sum and product.
Key Insight: Equality is controlled by which eigenvalues vanish. □

Theorem 4.1.57. *Kernel, Rank, and Columns of B^*B*
Let $B \in \mathrm{Mat}(m, n, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and set $A = B^*B$. Then:

(i) $A \succeq 0$;
(ii) $\ker(A) = \ker(B)$;
(iii) $\operatorname{rank}(A) = \operatorname{rank}(B)$;
(iv) $\operatorname{col}(A) = \operatorname{col}(B^*)$;
(v) $A = 0$ *if and only if* $B = 0$.

Proof.

(i) Positive semidefiniteness follows from Theorem 4.1.53.
(ii) If $B\mathbf{x} = 0$, then $A\mathbf{x} = 0$. Conversely, if $A\mathbf{x} = 0$, then $0 = \mathbf{x}^*A\mathbf{x} = \|B\mathbf{x}\|_2^2$ so $B\mathbf{x} = 0$.

(iii) Since A and B both have n columns, rank-nullity and (ii) give $\operatorname{rank}(A) = \operatorname{rank}(B)$.
(iv) Since $A = A^*$, Proposition 2.4.7 and (ii) give $\operatorname{col}(A) = \ker(A)^\perp = \ker(B)^\perp = \operatorname{col}(B^*)$.
(v) By (iii), $\operatorname{rank}(A) = 0$ if and only if $\operatorname{rank}(B) = 0$.
A matrix has rank zero exactly when it is the zero matrix, so $A = 0$ if and only if $B = 0$.

Structural Commentary: The Gram product B^*B preserves the kernel of B.
Key Insight: The identity $\mathbf{x}^*B^*B\mathbf{x} = \|B\mathbf{x}\|_2^2$ is the whole mechanism. □

Corollary 4.1.58. *Principal Blocks and Column Inclusion*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$ *be PSD, where* $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, *and partition* $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{12}^* & A_{22} \end{bmatrix}$, $A_{11} \in \operatorname{Mat}(k, k, \mathbb{F})$.
Then $A_{11} \succeq 0$, $\operatorname{col}(A_{12}) \subseteq \operatorname{col}(A_{11})$, *and* $\operatorname{rank}(A_{12}) \leq \operatorname{rank}(A_{11})$. *If* $A \succ 0$, *then* $A_{11} \succ 0$.

Proof. Write $A = B^*B$ and partition $B = [B_1 \; B_2]$ conformally. Then $A_{11} = B_1^*B_1$ and $A_{12} = B_1^*B_2$.
Hence $A_{11} \succeq 0$ and $\operatorname{col}(A_{12}) = \operatorname{col}(B_1^*B_2) \subseteq \operatorname{col}(B_1^*) = \operatorname{col}(B_1^*B_1) = \operatorname{col}(A_{11})$, equality $\operatorname{col}(B_1^*) = \operatorname{col}(B_1^*B_1)$ follows from Theorem 4.1.57. The rank inequality follows by taking dimensions
If $A \succ 0$ and $\mathbf{x} \neq 0$ in $\mathbb{F}^k$, then $\mathbf{x}^*A_{11}\mathbf{x} = \begin{bmatrix} \mathbf{x} \\ 0 \end{bmatrix}^* A \begin{bmatrix} \mathbf{x} \\ 0 \end{bmatrix} > 0$, so $A_{11} \succ 0$.
Structural Commentary: In a positive block matrix, cross-block columns lie in principal block's column space.
Key Insight: Factor first; then column inclusion is immediate from $A_{12} = B_1^*B_2$. □

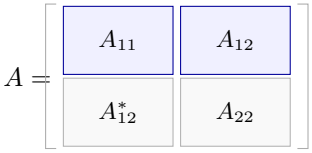


Figure 66: For $A \succeq 0$, factorisation forces $\operatorname{col}(A_{12}) \subseteq \operatorname{col}(A_{11})$.

Definition 4.1.59. *Principal and Leading Principal Minors*
For $A \in \operatorname{Mat}(n, n, \mathbb{C})$, a *principal minor* is the determinant of a principal submatrix of A.
A *leading principal minor* is the determinant of the leading $k \times k$ principal submatrix for some $k = 1, \ldots, n$.

Theorem 4.1.60. *Sylvester's Criterion*
A Hermitian matrix $A \in \operatorname{Mat}(n, n, \mathbb{C})$ *is positive definite if and only if all leading principal minors are positive.*

Proof.
(⇒) If $A \succ 0$, then each leading principal submatrix is positive definite by Corollary 4.1.58. Its eigenvalues are positive, so its determinant is positive.
(⇐) Induct on n. If $n = 1$, the single leading principal minor is the only diagonal entry, so positivity of that minor is exactly positive definiteness. Assume $n \geq 2$ and write $A = \begin{bmatrix} B & \mathbf{y} \\ \mathbf{y}^* & a \end{bmatrix}$, $B \in \operatorname{Mat}(n-1, n-1, \mathbb{C})$
The leading principal minors of B are positive, so $B \succ 0$.
Since $\det(B) > 0$ and $\det(A) > 0$, Proposition 4.1.9 gives $a - \mathbf{y}^*B^{-1}\mathbf{y} = \frac{\det(A)}{\det(B)} > 0$.
For $\mathbf{u} \in \mathbb{C}^{n-1}$ and $t \in \mathbb{C}$, $\begin{bmatrix} \mathbf{u} \\ t \end{bmatrix}^* A \begin{bmatrix} \mathbf{u} \\ t \end{bmatrix} = (\mathbf{u} + B^{-1}\mathbf{y}t)^*B(\mathbf{u} + B^{-1}\mathbf{y}t) + \left(a - \mathbf{y}^*B^{-1}\mathbf{y}\right)|t|^2$.
First term is nonnegative as $B \succ 0$, and second coefficient is positive. If displayed vector is nonzero, then either $t \neq 0$ and second term is positive, or $t = 0$ and then $\mathbf{u} \neq 0$, making first term positive. Thus $A \succ 0$.

Structural Commentary: Leading minors make each leading block positive and final Schur complement positive.
Key Insight: Complete the square with the leading block. □

Method 4.1.61. 2×2 *Quadratic Form Test*
Let $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ be real symmetric, $\Delta = ac - b^2$, and $q(x, y) = \begin{bmatrix} x & y \end{bmatrix} A \begin{bmatrix} x \\ y \end{bmatrix} = ax^2 + 2bxy + cy^2$.
For $a \neq 0$, $q(x, y) = a\left(x + \frac{b}{a}y\right)^2 + \frac{\Delta}{a}y^2$.
If $a = 0$, swap variables when $c \neq 0$; if $a = c = 0$, use $q = 2bxy$. Thus:
(i) $q \succ 0$ if and only if $a > 0$ and $\Delta > 0$;
(ii) $q \prec 0$ if and only if $a < 0$ and $\Delta > 0$;
(iii) q is indefinite if and only if $\Delta < 0$;
(iv) if $\Delta = 0$, then $q \succeq 0$ iff $a, c \geq 0$, and $q \preceq 0$ iff $a, c \leq 0$.

This is the 2×2 form of Sylvester's criterion and inertia.
Structural Commentary: Completing the square is the 2×2 shadow of the LDL^* and Schur-complement tests.
Key Insight: Δ and the leading sign classify the form.

Corollary 4.1.62. *Zero Quadratic Form Detects the Kernel*
If $A \in \mathrm{Mat}(n, n, \mathbb{F})$ is positive semidefinite, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, then $\mathbf{x}^* A\mathbf{x} = 0$ if and only if $A\mathbf{x} = 0$.

Proof. Write $A = B^*B$.
($\Rightarrow$) If $\mathbf{x}^* A\mathbf{x} = 0$, then $\|B\mathbf{x}\|_2^2 = 0$, so $B\mathbf{x} = 0$ and hence $A\mathbf{x} = B^*B\mathbf{x} = 0$.
($\Leftarrow$) If $A\mathbf{x} = 0$, then $\mathbf{x}^* A\mathbf{x} = 0$.
Structural Commentary: A zero PSD quadratic value has no hidden cancellation.
Key Insight: Zero length-square means the factored vector itself is zero. □

Theorem 4.1.63. *Stability Rules for Positive Semidefinite Matrices*
Let $A, B \succeq 0$ over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Whenever the expressions are defined:

(i) $aA + bB \succeq 0$ for $a, b \geq 0$; if $a, b > 0$ and one of A, B is positive definite, then $aA + bB \succ 0$.
(ii) $S^*AS \succeq 0$; moreover, $S^*AS \succ 0$ if and only if $S\mathbf{x} \notin \ker(A)$ for all $\mathbf{x} \neq \mathbf{0}$.
(iii) $A \oplus B \succeq 0$, with $A \oplus B \succ 0$ if and only if $A \succ 0$ and $B \succ 0$.

Proof.
(i) $\mathbf{x}^*(aA + bB)\mathbf{x} = a\mathbf{x}^*A\mathbf{x} + b\mathbf{x}^*B\mathbf{x} \geq 0$. If, for example, $A \succ 0$ and $a > 0$, then for every nonzero $\mathbf{x}$ the first summand is positive and the second is nonnegative, so $aA + bB \succ 0$; the case $B \succ 0$ is identical.
(ii) $\mathbf{x}^*S^*AS\mathbf{x} = (S\mathbf{x})^*A(S\mathbf{x}) \geq 0$. By Corollary 4.1.62, equality occurs exactly when $S\mathbf{x} \in \ker(A)$.
(iii) $(\mathbf{x} \oplus \mathbf{y})^*(A \oplus B)(\mathbf{x} \oplus \mathbf{y}) = \mathbf{x}^*A\mathbf{x} + \mathbf{y}^*B\mathbf{y} \geq 0$. This expression is positive for every nonzero $\mathbf{x} \oplus \mathbf{y}$ exactly when each summand is positive on its own nonzero vectors, namely when $A \succ 0$ and $B \succ 0$.

Structural Commentary: The PSD cone is stable under sums, pullbacks, and direct sums.
Key Insight: Test the quadratic form after each construction. □

Proposition 4.1.64. *Explicit Square Root for a 2×2 Positive Semidefinite Matrix*
Let $A \in \mathrm{Mat}(2, 2, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, be positive semidefinite and nonzero. Set $\tau = \left(\operatorname{tr} A + 2\sqrt{\det A}\right)^{1/2}$. Then $\tau > 0$ and $A^{1/2} = \frac{1}{\tau}\left(A + \sqrt{\det A}\, I_2\right)$.

Proof. Let $C = \frac{1}{\tau}\left(A + \sqrt{\det A}\, I_2\right)$. By Proposition 4.1.56, $\det A \geq 0$ and $\operatorname{tr} A > 0$, hence $\tau > 0$; by Theorem 4.1.63, $C \succeq 0$. Cayley–Hamilton gives $A^2 - (\operatorname{tr} A)A + (\det A)I_2 = 0$, so

$$C^2 = \frac{1}{\tau^2}\left(A^2 + 2\sqrt{\det A}\, A + (\det A)I_2\right) = \frac{1}{\tau^2}\left((\operatorname{tr} A)A + 2\sqrt{\det A}\, A\right) = A.$$

Thus $C = A^{1/2}$ by uniqueness of the positive semidefinite square root.
Structural Commentary: In dimension 2, Cayley–Hamilton makes $A^{1/2}$ linear in A.
Key Insight: Square the linear candidate and use Cayley–Hamilton. □

Theorem 4.1.65. *Simultaneous Congruence with One Positive Definite Form*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ be Hermitian, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $A \succ 0$.
Then, for some invertible S and real diagonal Λ, $A = SS^*$ and $B = S\Lambda S^*$.
The diagonal entries of Λ are the eigenvalues of $A^{-1}B$.

Proof. Let $C = A^{-1/2}BA^{-1/2} = U\Lambda U^*$ be spectral decomposition, $S = A^{1/2}U$. Then $SS^* = A$ and $S\Lambda S^* = B$. Also $A^{-1}B = A^{-1/2}CA^{1/2}$, so $A^{-1}B$ is similar to C and has eigenvalues given by the diagonal entries of Λ.
Structural Commentary: The positive definite form A normalizes B to a Hermitian matrix.
Key Insight: Normalize, diagonalise, then transform back. □

Corollary 4.1.66. *Products with One Positive Definite Factor*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $A \succ 0$.

(i) If B is Hermitian, then AB is diagonalizable with real spectrum.
(ii) If $B \succeq 0$, then AB is diagonalizable with nonnegative real spectrum; if $B \neq 0$, one eigenvalue is positive.
(iii) If $B \succ 0$, then AB is diagonalizable with positive real spectrum.

Proof. Let $C = A^{1/2}BA^{1/2}$. Since $A^{-1/2}(AB)A^{1/2} = C$, the matrices AB and C are similar.
(i) If B is Hermitian, then C is Hermitian, hence diagonalizable with real spectrum.
(ii) If $B \succeq 0$, then $C \succeq 0$ by Theorem 4.1.63, so its spectrum is nonnegative.
If $B \neq 0$, then $C \neq 0$, hence some eigenvalue is positive.
(iii) If $B \succ 0$, then $C \succ 0$ by Theorem 4.1.63, so all eigenvalues are positive.

Structural Commentary: A positive definite factor makes AB similar to a Hermitian congruence.
Key Insight: Use $AB \sim A^{1/2}BA^{1/2}$. □

Theorem 4.1.67. *Principal Axes for Real Quadratic Forms*
Let $A \in \mathrm{Mat}(n,n,\mathbb{R})$ be symmetric. Then:

(i) there are a real orthogonal Q and real $\lambda_1,\dots,\lambda_n$ such that, with $\mathbf{y} = Q^t\mathbf{x}$, $\mathbf{x}^tA\mathbf{x} = \sum_{j=1}^{n} \lambda_j y_j^2$.
(ii) $A \succeq 0$ if and only if every $\lambda_j \geq 0$;
(iii) $A \succ 0$ if and only if every $\lambda_j > 0$;
(iv) A is indefinite if and only if some $\lambda_j > 0$ and some $\lambda_k < 0$.

Proof.

(i) By the real spectral theorem, write $A = Q\,\mathrm{diag}(\lambda_1,\dots,\lambda_n)Q^t$.
Then $\mathbf{x}^tA\mathbf{x} = \mathbf{x}^tQ\,\mathrm{diag}(\lambda_j)Q^t\mathbf{x} = \mathbf{y}^t\,\mathrm{diag}(\lambda_j)\mathbf{y} = \sum_{j=1}^{n} \lambda_j y_j^2$.
(ii) The displayed sum is nonnegative for all $\mathbf{y}$ exactly when each coefficient λ_j is nonnegative.
(iii) The displayed sum is positive for all nonzero $\mathbf{y}$ exactly when each coefficient λ_j is positive.
(iv) The diagonal form takes both signs exactly when its coefficients include both signs.

Structural Commentary: Principal axes turn the form into signed squares.
Key Insight: Eigenvalue signs classify the quadratic form. □

Method 4.1.68. *Quadratic Form Principal-Axes Method*
To analyse a real quadratic form:

(i) write it as $\mathbf{x}^tA\mathbf{x}$ with $A = A^t$; split each cross coefficient equally between a_{ij} and a_{ji};
(ii) diagonalise $A = Q\Lambda Q^t$;
(iii) change variables by $\mathbf{y} = Q^t\mathbf{x}$;
(iv) classify the form from the signs of the eigenvalues in Λ.

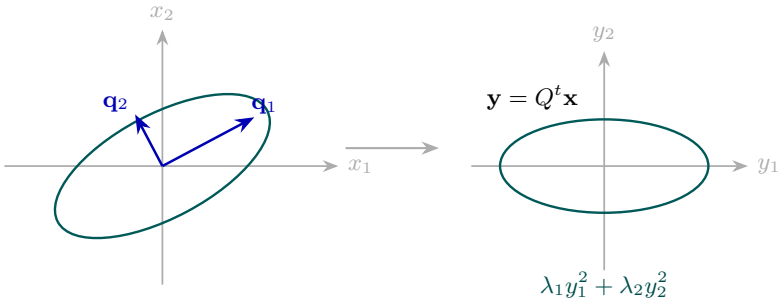


Figure 67: Orthogonal coordinates reveal the principal axes of a quadratic form.

Definition 4.1.69. *$*$-Congruence and Hermitian Inertia*
Two matrices $A, B \in \mathrm{Mat}(n,n,\mathbb{C})$ are *$*$-congruent* if $A = SBS^*$ for some invertible $S \in \mathrm{Mat}(n,n,\mathbb{C})$.
For Hermitian A, its *inertia* is $\mathrm{in}(A) = (n_+(A), n_-(A), n_0(A))$, where these are the numbers of positive, negative, and zero eigenvalues of A, counted with multiplicity.

Lemma 4.1.70. *Positive Subspaces and Inertia*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ be Hermitian. The maximal dimensions of subspaces on which $\mathbf{x}^*A\mathbf{x}$ is positive or negative for every nonzero $\mathbf{x}$ are $n_+(A)$ and $n_-(A)$, respectively.

Proof. Let P_+, $P_{\leq 0}$ be positive, nonpositive spectral subspaces. Then $\dim P_+ = n_+(A)$, P_+ is positive for A. If $\dim \mathcal{U} > n_+(A)$, then $\dim \mathcal{U} + \dim P_{\leq 0} > n_+(A) + n_-(A) + n_0(A) = n$, so $\mathcal{U} \cap P_{\leq 0}$ contains a nonzero vector. On this vector quadratic form is nonpositive, so $\mathcal{U}$ cannot be positive. Apply this to $-A$ for negative case.
Structural Commentary: The positive spectral subspace is maximal; the nonpositive part blocks enlargement.
Key Insight: Inertia counts maximal sign-definite dimensions. □

Theorem 4.1.71. *Sylvester's Inertia Theorem*
Hermitian matrices A, B of the same size are $*$-congruent if and only if $n_+(A) = n_+(B)$ and $n_-(A) = n_-(B)$.

Proof.

($\Rightarrow$) Suppose $A = SBS^*$ with S invertible. Then $\mathbf{x}^*A\mathbf{x} = (S^*\mathbf{x})^*B(S^*\mathbf{x})$.
Thus S^* transports positive and negative subspaces, so Lemma 4.1.70 gives $n_\pm(A) = n_\pm(B)$.

($\Leftarrow$) Let the common inertia be (p, q, r) and set $D = I_p \oplus (-I_q) \oplus 0_r$.
By the spectral theorem, each of A and B is $*$-congruent to D, hence they are $*$-congruent to each other.

Structural Commentary: $*$-congruence is a coordinate change for Hermitian quadratic forms.
Key Insight: Spectral decomposition reduces the form to signed squares and zero directions. □

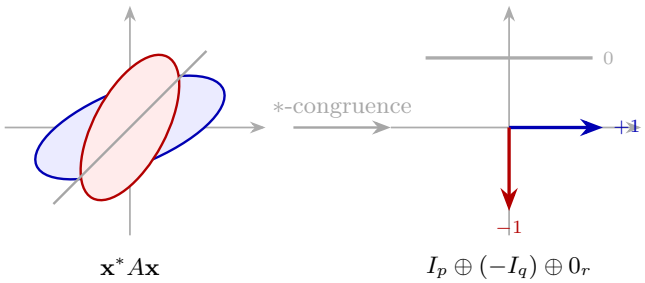


Figure 68: $*$-congruence reduces a Hermitian form to signed coordinate axes.

Definition 4.1.72. *Phase Inertia Matrix*
A *phase inertia matrix* is a diagonal complex matrix whose nonzero diagonal entries have modulus 1.

Proposition 4.1.73. *Phase Inertia Form for Normal Matrices*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be normal.
Then A is $*$-congruent to a phase inertia matrix D, whose entries record the spectral rays of A.

Proof. Write $A = U\,\mathrm{diag}(\lambda_j)U^*$. Write $\lambda_j = r_j d_j$, where $r_j > 0$ and $|d_j| = 1$ if $\lambda_j \neq 0$, while $r_j = 1$ and $d_j = 0$ if $\lambda_j = 0$. Set $D = \mathrm{diag}(d_j)$, $S = U\,\mathrm{diag}(\sqrt{r_j})$. Then S is invertible and $A = SDS^*$.
The entries $d_j = e^{i\theta}$ count exactly the eigenvalues on the ray $\{\rho e^{i\theta} : \rho > 0\}$.
Structural Commentary: Normal diagonalization separates modulus from phase; $*$-congruence absorbs moduli.
Key Insight: Only spectral phases survive. □

Perspective 4.1.74. *Hermitian Inertia as Real Phase Inertia*
For Hermitian matrices, the only nonzero spectral rays are the positive and negative real axes.
Sylvester's inertia theorem is therefore the real-ray version of phase inertia.

Definition 4.1.75. *Kronecker Product*
Let $A = [a_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$ and $B \in \mathrm{Mat}(p, q, \mathbb{F})$. The *Kronecker product* of A and B is the block matrix

$$A \otimes B := \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{bmatrix} \in \mathrm{Mat}(mp, nq, \mathbb{F}).$$

The Kronecker product is also called the tensor product of matrices.

Proposition 4.1.76. *Basic Kronecker Identities*
Whenever the expressions are defined,

(i) $c(A \otimes B) = (cA) \otimes B = A \otimes (cB)$;
(ii) $(A + B) \otimes C = A \otimes C + B \otimes C$ *and* $A \otimes (B + C) = A \otimes B + A \otimes C$;
(iii) $(A \otimes B) \otimes C = A \otimes (B \otimes C)$;
(iv) $(A \otimes B)^t = A^t \otimes B^t$;
(v) $I_m \otimes I_n = I_{mn}$.

If $\mathbb{F} = \mathbb{C}$, then also $\overline{A \otimes B} = \overline{A} \otimes \overline{B}$ and $(A \otimes B)^* = A^* \otimes B^*$.

Proof.

(i) The (i, j) block of $c(A \otimes B)$ is $c(a_{ij}B) = (ca_{ij})B = a_{ij}(cB)$, which is (i, j) block of $(cA) \otimes B$ and $A \otimes (cB)$.
(ii) The (i, j) block of $(A + B) \otimes C$ is $(a_{ij} + b_{ij})C = a_{ij}C + b_{ij}C$, so $(A + B) \otimes C = A \otimes C + B \otimes C$.
The proof of $A \otimes (B + C) = A \otimes B + A \otimes C$ is identical inside each block.
(iii) If $B = [b_{rs}]$, then both $(A \otimes B) \otimes C$ and $A \otimes (B \otimes C)$ have blocks $a_{ij}b_{rs}C$ in the same relative positions.
(iv) The (j, i) block of $(A \otimes B)^t$ is $(a_{ij}B)^t = a_{ij}B^t$, which is the (j, i) block of $A^t \otimes B^t$.

(v) The block matrix $I_m \otimes I_n$ has I_n on its diagonal blocks and zero elsewhere, hence equals I_{mn}.

If $\mathbb{F} = \mathbb{C}$, conjugation is checked blockwise, adjoint identity follows from conjugation together with part (iv).
Structural Commentary: Kronecker products replace scalar entries by whole scaled blocks. Standard algebraic rules survive because each block is still controlled by scalar distributivity and transposition.
Key Insight: Check identities block-by-block; each block reduces to an ordinary scalar-matrix identity. □

Theorem 4.1.77. *Mixed Product Property*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $B \in \mathrm{Mat}(p,q,\mathbb{F})$, $C \in \mathrm{Mat}(n,r,\mathbb{F})$, $D \in \mathrm{Mat}(q,s,\mathbb{F})$. $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$.

Proof. Write $A = [a_{ik}]$ and $C = [c_{kj}]$. The (i,j) block of $(A \otimes B)(C \otimes D)$ is
$\sum_{k=1}^{n} (a_{ik}B)(c_{kj}D) = \left(\sum_{k=1}^{n} a_{ik}c_{kj}\right) BD = (AC)_{ij}BD$, which is exactly the (i,j) block of $(AC) \otimes (BD)$.
Structural Commentary: The outer indices multiply as AC, while the inner block action multiplies as BD.
Key Insight: Kronecker multiplication separates into ordinary multiplication in each factor. □

Corollary 4.1.78. *Inverse of a Kronecker Product*
If $A \in \mathrm{Mat}(m,m,\mathbb{F})$ and $B \in \mathrm{Mat}(n,n,\mathbb{F})$ are invertible, then $A \otimes B$ is invertible and $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.

Proof. By Theorem 4.1.77, $(A \otimes B)(A^{-1} \otimes B^{-1}) = I_m \otimes I_n = I_{mn}$ and the reverse product is identical.
Structural Commentary: Inverting a Kronecker product inverts each independent factor.
Key Insight: The mixed product property turns the inverse check into two ordinary inverse checks. □

Proposition 4.1.79. *Kronecker Product and Sum Spectra in the Diagonalizable Case*
Let $A = P\Lambda P^{-1}$, $B = QMQ^{-1}$, where $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ and $M = \mathrm{diag}(\mu_1, \ldots, \mu_q)$.
Then $A \otimes B \sim \Lambda \otimes M$; and $A \otimes I_q + I_n \otimes B \sim \Lambda \otimes I_q + I_n \otimes M$.
Their diagonal entries are respectively $\lambda_i\mu_j$ and $\lambda_i + \mu_j$ for $1 \le i \le n$, $1 \le j \le q$.

Proof. By Theorem 4.1.77 and $(P \otimes Q)^{-1} = P^{-1} \otimes Q^{-1}$, $A \otimes B = (P \otimes Q)(\Lambda \otimes M)(P^{-1} \otimes Q^{-1})$.
Similarly, $A \otimes I_q + I_n \otimes B = (P \otimes Q)(\Lambda \otimes I_q + I_n \otimes M)(P^{-1} \otimes Q^{-1})$.
The middle matrices are diagonal with the stated (i,j)-indexed entries.
Structural Commentary: Eigenvectors tensor together, and Kronecker sums add the two scalar eigenvalues.
Key Insight: Diagonalize each factor first; the Kronecker operation becomes entrywise on eigenvalue pairs. □

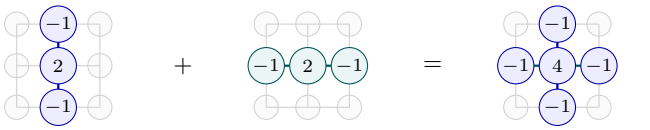


Figure 69: Vertical plus horizontal second differences give the five-point stencil.

Definition 4.1.80. *Vectorisation*
Let $X = [\mathbf{x}_1\ \mathbf{x}_2\ \cdots\ \mathbf{x}_n] \in \mathrm{Mat}(m,n,\mathbb{F})$. The *vectorisation* of X is

$$\mathrm{vec}(X) := \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{bmatrix} \in \mathbb{F}^{mn}.$$

Thus vec stacks the columns of X vertically.

Theorem 4.1.81. *Vectorisation of a Triple Product*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $X \in \mathrm{Mat}(n,p,\mathbb{F})$, and $B \in \mathrm{Mat}(p,q,\mathbb{F})$. Then $\mathrm{vec}(AXB) = (B^t \otimes A)\,\mathrm{vec}(X)$.

Proof. Write $B = [\mathbf{b}_1\ \cdots\ \mathbf{b}_q]$ and $X = [\mathbf{x}_1\ \cdots\ \mathbf{x}_p]$.
The kth column of AXB is $AX\mathbf{b}_k = A\sum_{i=1}^{p} b_{ik}\mathbf{x}_i = \sum_{i=1}^{p} b_{ik}A\mathbf{x}_i = (\mathbf{b}_k^t \otimes A)\,\mathrm{vec}(X)$.
Stacking these q columns gives $(B^t \otimes A)\,\mathrm{vec}(X)$.
Structural Commentary: Vectorisation rewrites two-sided multiplication as one matrix action.
Key Insight: Left multiplication contributes A; right multiplication contributes B^t. □

Perspective 4.1.82. *Vectorising a Minimal-Polynomial Test*
For $A \in \mathrm{Mat}(n,n,\mathbb{F})$ and a monic candidate $p_m(z) = a_0 + a_1 z + \cdots + a_{m-1}z^{m-1} + z^m$, the matrix equation $a_0 I + a_1 A + \cdots + a_{m-1}A^{m-1} = -A^m$ can be written as the ordinary linear system

$$\begin{bmatrix} \mathrm{vec}(I) & \mathrm{vec}(A) & \cdots & \mathrm{vec}(A^{m-1}) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{m-1} \end{bmatrix} = -\,\mathrm{vec}(A^m)$$

Structural Commentary: Vectorisation turns matrix identities into linear systems.
Key Insight: Polynomial membership becomes a coefficient system in vectorized powers.

Definition 4.1.83. *Hadamard Product*
Let $A = [a_{ij}], B = [b_{ij}] \in \mathrm{Mat}(m,n,\mathbb{F})$.
The *Hadamard product*, or *entrywise product*, of A and B is $A \circ B := [a_{ij}b_{ij}] \in \mathrm{Mat}(m,n,\mathbb{F})$.

Proposition 4.1.84. *Basic Hadamard Identities*
For matrices of compatible sizes:

(i) $A \circ B = B \circ A$;
(ii) $A \circ (\gamma B + C) = \gamma(A \circ B) + A \circ C$;
(iii) $(A \circ B)^* = A^* \circ B^*$;
(iv) *if E is the all-ones matrix, then* $A \circ E = A$;
(v) *if $D = \mathrm{diag}(d_1, \ldots, d_n)$, then* $A \circ D = \mathrm{diag}(a_{11}d_1, \ldots, a_{nn}d_n)$.

Proof.

(i) $(A \circ B)_{ij} = a_{ij}b_{ij} = b_{ij}a_{ij} = (B \circ A)_{ij}$.
(ii) $\big(A \circ (\gamma B + C)\big)_{ij} = a_{ij}(\gamma b_{ij} + c_{ij}) = \gamma a_{ij}b_{ij} + a_{ij}c_{ij}$.
(iii) $\big((A \circ B)^*\big)_{ij} = \overline{a_{ji}b_{ji}} = \overline{a_{ji}}\,\overline{b_{ji}} = (A^* \circ B^*)_{ij}$.
(iv) $(A \circ E)_{ij} = a_{ij}$.
(v) Since D has zero off-diagonal entries, $A \circ D$ is diagonal with entries $a_{ii}d_i$.

Structural Commentary: Hadamard multiplication is entrywise, with no row-column summation.
Key Insight: Prove Hadamard identities entry by entry. □

Theorem 4.1.85. *Schur Product Theorem*
If $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$ are positive semidefinite, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, then $A \circ B$ is positive semidefinite.

Proof. By Theorem 4.1.53, regard A and B as Gram matrices: $a_{ij} = \langle \mathbf{u}_j, \mathbf{u}_i \rangle$, $b_{ij} = \langle \mathbf{v}_j, \mathbf{v}_i \rangle$ for suitable vectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ and $\mathbf{v}_1, \ldots, \mathbf{v}_n$. Then $(A \circ B)_{ij} = \langle \mathbf{u}_j, \mathbf{u}_i \rangle \langle \mathbf{v}_j, \mathbf{v}_i \rangle = \langle \mathbf{u}_j \otimes \mathbf{v}_j, \mathbf{u}_i \otimes \mathbf{v}_i \rangle$.
Thus $A \circ B$ is the Gram matrix of the vectors $\mathbf{u}_j \otimes \mathbf{v}_j$, hence is positive semidefinite by Proposition 2.1.45.
Structural Commentary: The Schur product is the Gram matrix of paired tensor vectors.
Key Insight: Pair the jth vector from each Gram representation and take tensor products. □

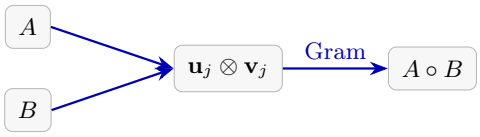


Figure 70: For Gram matrices, entrywise products become tensor inner products, so $A \circ B$ is Gram.

Corollary 4.1.86. *Positive Definite Hadamard Products*
Let $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$ be positive semidefinite, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

(i) If A is positive definite and every diagonal entry of B is positive, then $A \circ B$ is positive definite.
(ii) If A and B are positive definite, then $A \circ B$ is positive definite.

Proof.

(i) Let $\lambda_{\min} > 0$ be the smallest eigenvalue of A. Then $A - \lambda_{\min} I_n \succeq 0$, so $(A - \lambda_{\min} I_n) \circ B \succeq 0$ by Theorem 4.1.85. Thus, for $\mathbf{x} \neq \mathbf{0}$, $\mathbf{x}^*(A \circ B)\mathbf{x} = \mathbf{x}^*((A - \lambda_{\min} I_n) \circ B)\mathbf{x} \lambda_{\min} \sum_{j=1}^{n} b_{jj} |x_j|^2 > 0$.
(ii) If $B \succ 0$, then $b_{jj} = \mathbf{e}_j^* B \mathbf{e}_j > 0$ for every j, so (i) applies.

Structural Commentary: Subtract the spectral floor of A; the diagonal of B gives strictness.
Key Insight: Use $I_n \circ B = \mathrm{diag}(b_{11}, \ldots, b_{nn})$. □

Method 4.1.87. *Schur Product Positivity Method*
To prove an entrywise product is positive semidefinite:

(i) factor or interpret each matrix as a Gram matrix;
(ii) pair the corresponding vectors;
(iii) take tensor products of the paired vectors;
(iv) identify the Hadamard product as the Gram matrix of those tensor products.

Definition 4.1.88. *Spectral Radius and Convergent Matrix*
For $A \in \mathrm{Mat}(n, n, \mathbb{C})$, the *spectral radius* is $\rho(A) := \max\{|\lambda| : \lambda \in \mathrm{spec}(A)\}$.
We say $A^p \to 0$ if every entry of A^p converges to 0 as $p \to \infty$.

Theorem 4.1.89. *Matrix Powers Converge to Zero*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. Then $A^p \to 0$ if and only if $\rho(A) < 1$.

Proof. Choose a Jordan form $A = SJS^{-1}$.
Since S and S^{-1} are fixed, $A^p \to 0$ if and only if every Jordan block power in J^p converges to 0.
($\Rightarrow$) Each block power tends to 0, so Proposition 3.3.4 gives $|\lambda| < 1$ for every eigenvalue. Thus $\rho(A) < 1$.
($\Leftarrow$) If $\rho(A) < 1$, then every eigenvalue satisfies $|\lambda| < 1$. Proposition 3.3.4 makes each block power tend to 0.
Structural Commentary: Similarity reduces matrix-power convergence to Jordan block-power convergence.
Key Insight: The largest eigenvalue modulus controls the asymptotic size of all blocks. □

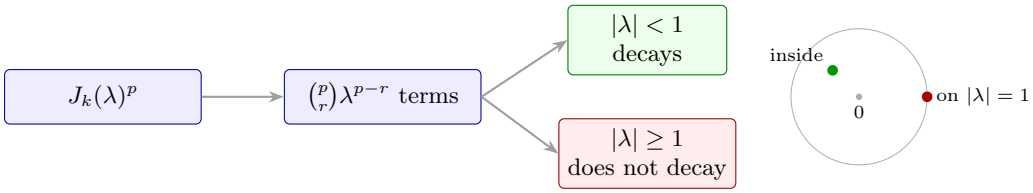


Figure 71: The gray circle is $|\lambda| = 1$: inside it, powers decay; on or outside it, they do not.

Definition 4.1.90. *Power Bounded Matrix*
A matrix $A \in \mathrm{Mat}(n, n, \mathbb{C})$ is *power bounded* if for each entry position (i, j), the sequence of entries of A^p in that position is bounded for $p = 1, 2, \ldots$.

Theorem 4.1.91. *Power Bounded Jordan Criterion*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. The following are equivalent:

(i) A is power bounded.
(ii) $\rho(A) \leq 1$, and every eigenvalue λ with $|\lambda| = 1$ has only 1×1 Jordan blocks.
(iii) $\rho(A) \leq 1$, and $\dim E_\lambda(A) = m_A(\lambda)$ for every $\lambda \in \mathrm{spec}(A)$ with $|\lambda| = 1$.

Proof. Work in Jordan form.

(i) $\Rightarrow$ (ii): If $|\lambda| > 1$, the diagonal entries λ^p are unbounded.
If $|\lambda| = 1$ and $J_k(\lambda)$ has $k \geq 2$, then the first superdiagonal contains $p\lambda^{p-1}$, also unbounded.
(ii) $\Rightarrow$ (i): Blocks with $|\lambda| < 1$ converge to 0 by Proposition 3.3.4; 1×1 blocks with $|\lambda| = 1$ are bounded.
Finite direct sums and fixed similarities preserve boundedness.
(ii) $\Leftrightarrow$ (iii): For fixed λ, $\dim E_\lambda(A)$ counts Jordan blocks, while $m_A(\lambda)$ counts their total size.
Equality holds exactly when every block has size 1.

Structural Commentary: Unit-circle eigenvalues are harmless only when their nilpotent parts vanish.
Key Insight: Nontrivial Jordan blocks on the unit circle create polynomial growth. □

Definition 4.1.92. *Probability Vectors and Markov Matrices*
A vector $\mathbf{x} \in \mathbb{R}^n$ is a *probability vector* if its entries are nonnegative and sum to 1. Let $\mathbf{e} := (1, \ldots, 1)^t$. A matrix $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{R})$ is a *Markov matrix* if $a_{ij} \geq 0$ for all i, j and $A\mathbf{e} = \mathbf{e}$; equivalently, every row sum is 1.

Lemma 4.1.93. *Basic Properties of Markov Matrices*
If $A \in \mathrm{Mat}(n, n, \mathbb{R})$ is Markov, then $1 \in \mathrm{spec}(A)$, and each A^p is Markov with entries in $[0, 1]$.

Proof. Since $A\mathbf{e} = \mathbf{e}$, 1 is an eigenvalue. Inductively, A^p has nonnegative entries and $A^p\mathbf{e} = \mathbf{e}$, so A^p is Markov.
Each nonnegative row sums to 1, so all entries lie in $[0, 1]$.
Structural Commentary: Markov structure is closed under powers.
Key Insight: Nonnegative row sums of 1 bound every entry. □

Theorem 4.1.94. *Unit-Circle Spectrum of Markov Matrices*
If $A \in \mathrm{Mat}(n,n,\mathbb{R})$ is Markov, then $\rho(A) = 1$, and $\dim E_\lambda(A) = m_A(\lambda)$ for every $\lambda \in \mathrm{spec}(A)$ with $|\lambda| = 1$.

Proof. Lemma 4.1.93 makes A power bounded. Theorem 4.1.91 gives $\rho(A) \leq 1$ and $\dim E_\lambda(A) = m_A(\lambda)$ on $|\lambda| = 1$. Since $A\mathbf{e} = \mathbf{e}$, $1 \in \mathrm{spec}(A)$, so $\rho(A) \geq 1$. Thus $\rho(A) = 1$.
Structural Commentary: Bounded Markov powers rule out unit-circle Jordan growth.
Key Insight: $A\mathbf{e} = \mathbf{e}$ forces the eigenvalue 1. □

Theorem 4.1.95. *Positive Markov Matrices Converge to a Stationary Distribution*
Let $n \geq 2$ and let $A \in \mathrm{Mat}(n,n,\mathbb{R})$ be a Markov matrix with positive entries. Then there is a unique probability vector $\mathbf{x}$ with $A^t\mathbf{x} = \mathbf{x}$; its entries are positive and $\lim_{p\to\infty} A^p = \mathbf{e}\mathbf{x}^t$.

Proof. Let $A\mathbf{y} = \lambda\mathbf{y}$ with $|\lambda| = 1$, and choose k with $|y_k| = \|\mathbf{y}\|_\infty$.
Positivity and the kth row sum give $\|\mathbf{y}\|_\infty = |\lambda y_k| = \left|\sum_{j=1}^n a_{kj} y_j\right| \leq \sum_{j=1}^n a_{kj}\,|y_j| \leq \|\mathbf{y}\|_\infty$.
Because every a_{kj} is positive, equality in the last inequality forces $|y_j| = \|\mathbf{y}\|_\infty$ for all j. Equality in the triangle inequality then forces all nonzero y_j to have the same argument. Hence $\mathbf{y} = c\mathbf{e}$, and $A\mathbf{e} = \mathbf{e}$ gives $\lambda = 1$. Thus $E_1(A) = \mathrm{span}(\mathbf{e})$ and no other eigenvalue lies on the unit circle.
By Theorem 4.1.94, the Jordan form is $[1] \oplus J$ with $\rho(J) < 1$. Choose $S = [\mathbf{e}\ Y]$ such that $A = S([1] \oplus J)S^{-1}$, and let $\mathbf{x}^t$ be the first row of S^{-1}. Then $\mathbf{x}^t A = \mathbf{x}^t$ and $\mathbf{x}^t\mathbf{e} = 1$. Since $J^p \to 0$, $A^p = S([1] \oplus J^p)S^{-1} \to \mathbf{e}\mathbf{x}^t$.
The nonnegative powers A^p imply $\mathbf{e}\mathbf{x}^t \geq 0$, so $\mathbf{x} \geq 0$. Since $A^t\mathbf{x} = \mathbf{x}$, $\mathbf{x} \neq 0$, and A^t has positive entries, in fact $\mathbf{x} > 0$. Finally, $\dim E_1(A^t) = \dim E_1(A) = 1$ because $A - I_n$ and $(A - I_n)^t$ have the same rank, so $\mathbf{x}^t\mathbf{e} = 1$ gives uniqueness among probability vectors.
Structural Commentary: Positivity leaves one simple unit-circle block; all other blocks decay.
Key Insight: The limit is the rank-one projection $\mathbf{e}\mathbf{x}^t$. □

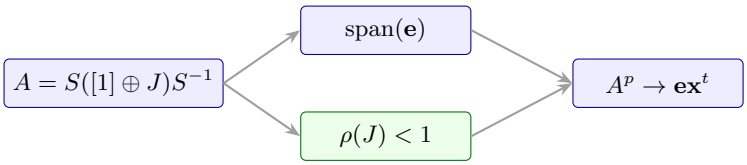


Figure 72: The line span($\mathbf{e}$) remains; all other Jordan directions decay, so $A^p \to \mathbf{e}\mathbf{x}^t$.

4.2 Matrix Decompositions

Decompositions expose usable structure in a matrix. LU, QR, Cholesky, SVD, compact SVD, and polar decomposition separate elimination, geometry, positivity, rank, and stretching.

Perspective 4.2.1. *Matrix Factorisation Roadmap*
Factorisations expose structure by task: elimination gives LU and LDL^t (Theorem 4.2.5, Corollary 4.2.6); positivity gives Cholesky (Theorem 4.2.10); geometry gives QR (Theorem 4.2.8); singular directions give SVD, compact SVD, and the pseudoinverse (Theorems 4.2.16, 4.2.20, Definition 4.3.16); polar decomposition separates motion from stretch (Theorem 4.2.25). Spectral factorisations appeared earlier: diagonalisation, spectral theorems, Jordan form, and Schur form (Theorems 3.1.33, 3.4.12, 3.4.13, 3.3.6, 3.4.10).
Key Insight: Choose the factorisation that exposes the needed structure.

Definition 4.2.2. *Full-Rank Factorization*
A factorization $A = XY$ with X full column rank and Y full row rank is called a *full-rank factorization* of A.

Theorem 4.2.3. *Full-Rank Factorization*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have rank $r \geq 1$, and let the columns of $X \in \mathrm{Mat}(m,r,\mathbb{F})$ be a basis of $\mathrm{col}(A)$. Then there is a unique $Y \in \mathrm{Mat}(r,n,\mathbb{F})$ such that $A = XY$, and $\mathrm{rank}(X) = \mathrm{rank}(Y) = r$.

Proof. Write $A = [\mathbf{a}_1\ \cdots\ \mathbf{a}_n]$. Since the columns of X form a basis of $\mathrm{col}(A)$, for each j there is a unique $\mathbf{y}_j \in \mathbb{F}^r$ such that $\mathbf{a}_j = X\mathbf{y}_j$. Set $Y = [\mathbf{y}_1\ \cdots\ \mathbf{y}_n]$. Then $A = [X\mathbf{y}_1\ \cdots\ X\mathbf{y}_n] = XY$ and uniqueness of the coordinates $\mathbf{y}_j$ gives uniqueness of Y. The columns of X are independent, so $\mathrm{rank}(X) = r$.
By Theorem 4.1.31, $r = \mathrm{rank}(A) = \mathrm{rank}(XY) \leq \mathrm{rank}(Y) \leq r$ because Y has only r rows. Hence $\mathrm{rank}(Y) = r$.
Structural Commentary: A matrix factors through a basis of its column space.
Key Insight: Full-rank factorization separates column directions from their coordinates. □

Definition 4.2.4. *LU Factorisation*
For $A \in \mathrm{Mat}(n,n,\mathbb{F})$, an *LU factorisation* is a product $A = LU$, where L is lower triangular and U is upper triangular. It is *unit lower triangular* if every diagonal entry of L is 1.

Theorem 4.2.5. *LU Factorisation from Nonzero Leading Principal Minors*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$. If every leading principal minor of A is nonzero, then $A = LU$ for a unique unit lower triangular L and a unique upper triangular U.

Proof. Induct on n. The case $n = 1$ is immediate. Write $A = \begin{bmatrix} a & \mathbf{r} \\ \mathbf{c} & B \end{bmatrix}$, $a \neq 0$.
Set $\boldsymbol{\ell} = a^{-1}\mathbf{c}$ and $S = B - a^{-1}\mathbf{cr}$. For each leading $k \times k$ principal block of A, the block determinant formula gives determinant a times the corresponding leading $(k-1)\times(k-1)$ principal minor of S. Hence all leading principal minors of S are nonzero. By induction, $S = L_1U_1$ with L_1 unit lower triangular and U_1 upper triangular.
Then $A = \begin{bmatrix} 1 & 0 \\ \boldsymbol{\ell} & L_1 \end{bmatrix} \begin{bmatrix} a & \mathbf{r} \\ 0 & U_1 \end{bmatrix}$. For uniqueness, comparing the first row and first column of LU determines a, $\mathbf{r}$, and $\boldsymbol{\ell}$, then reduces uniqueness to the lower-right Schur complement.
Structural Commentary: Gaussian elimination peels off one pivot and passes to a Schur complement.
Key Insight: Nonzero leading principal minors are exactly the no-row-swap pivot condition. □

Corollary 4.2.6. *Symmetric LU Gives an LDL^t Factorisation*
Let $A = A^t \in \mathrm{Mat}(n,n,\mathbb{R})$ have nonzero leading principal minors.
Then $A = LDL^t$, with L unit lower triangular and D diagonal invertible.

Proof. By Theorem 4.2.5, write $A = LU$ uniquely with L unit lower triangular and U upper triangular.
Since the leading principal minors are nonzero, each $u_{ii} \neq 0$. Put $D = \mathrm{diag}(u_{11}, \ldots, u_{nn})$ and $U_0 = D^{-1}U$; then U_0 is unit upper triangular and $A = LDU_0$. Since $A = A^t$, also $A = U_0^t D L^t$. This is another LU factorisation with unit lower factor U_0^t, so uniqueness gives $L = U_0^t$. Hence $A = LDL^t$.
Structural Commentary: Symmetry makes the upper triangular unit factor the transpose of the lower one.
Key Insight: Once the pivots are separated into D, the two triangular halves mirror each other. □

Definition 4.2.7. *Wide and Narrow QR Factorisation*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$ with $m \geq n$, factorisation $A = V \begin{bmatrix} R \\ 0 \end{bmatrix}$, $V \in \mathrm{Mat}(m,m,\mathbb{F})$ unitary, $R \in \mathrm{Mat}(n,n,\mathbb{F})$ upper triangular is called a *wide QR factorisation*. Writing $V = [Q\ Q']$ gives the *narrow QR factorisation* $A = QR$.

Theorem 4.2.8. *QR Factorisation*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and suppose $m \geq n$.

(i) $\exists$ a unitary V and an upper triangular R with real nonnegative diagonal entries such that $A = V \begin{bmatrix} R \\ 0 \end{bmatrix}$.
Writing $V = [Q\ Q']$ with Q the first n columns gives $A = QR$, with Q orthonormal columns.
(ii) If $\mathrm{rank}(A) = n$, then narrow $A = QR$ is unique when R has positive diagonal entries.

Proof.

(i) At step k, after the first $k-1$ columns have been triangularized, apply Corollary 2.4.29 to the tail of the kth column. This gives a unitary $I_{k-1} \oplus U_k$ that sends that tail to $r_{kk}\mathbf{e}_1$ with $r_{kk} \geq 0$.
Since the first $k-1$ entries of every earlier column already contain all possible nonzero entries, acting only on rows $k, \ldots, m$ preserves the earlier triangular columns.
After n steps, $U_n \cdots U_1 A = \begin{bmatrix} R \\ 0 \end{bmatrix}$, where R is upper triangular with nonnegative diagonal entries.
Let $V = (U_n \cdots U_1)^*$. Then $A = V \begin{bmatrix} R \\ 0 \end{bmatrix}$, and $V = [Q\ Q']$ gives $A = QR$ with Q orthonormal columns.
If $\mathrm{rank}(A) = n$, then $n = \mathrm{rank}(A) = \mathrm{rank}(R)$, so the triangular matrix R is invertible and its nonnegative diagonal entries are positive.

(ii) Suppose $A = Q_1R_1 = Q_2R_2$, where $Q_i^*Q_i = I_n$ and the R_i are upper triangular with positive diagonal entries. Then $A^*A = R_1^*R_1 = R_2^*R_2$. Let $D = R_1R_2^{-1}$. Substituting $R_1 = DR_2$ gives $D^*D = I_n$.
Thus the columns of D are orthonormal. Since D is upper triangular, the first column has only one possible nonzero entry; orthogonality with it forces the rest of the first row to vanish.
Inducting on the lower-right principal submatrix shows D is diagonal, and $D^*D = I_n$ gives diagonal entries of modulus 1. These diagonal entries are ratios of positive diagonal entries of R_1 and R_2, so they are positive. Hence $D = I_n$. Therefore $R_1 = R_2$, and then $Q_1 = Q_2$.

Structural Commentary: Householder steps triangularise columns; full rank makes the diagonal positive.
Key Insight: Positive diagonals remove the sign/phase ambiguity. □

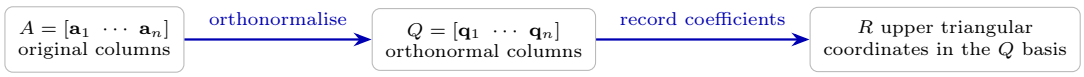


Figure 73: QR writes $A = QR$ while preserving prefix column spaces: $\operatorname{col}(A_k) = \operatorname{col}(Q_k)$.

Corollary 4.2.9. *Prefix Column Spaces in QR Factorisation*
Let $A = QR$ be the narrow QR factorisation of $A \in \operatorname{Mat}(m, n, \mathbb{F})$ with $\operatorname{rank}(A) = n$ and positive diagonal entries in R. For $k = 1, \ldots, n$, let A_k and Q_k be the first k columns of A and Q, respectively. Then $\operatorname{col}(A_k) = \operatorname{col}(Q_k)$. Thus the columns of Q_k form an orthonormal basis of $\operatorname{col}(A_k)$.

Proof. Let R_k be the leading $k \times k$ principal submatrix of R. Since R is upper triangular with positive diagonal entries, R_k is invertible. The first k columns of $A = QR$ satisfy $A_k = Q_k R_k$, because upper triangularity prevents later columns of Q from contributing. Hence $\operatorname{col}(A_k) \subseteq \operatorname{col}(Q_k)$ and $Q_k = A_k R_k^{-1}$ gives the reverse inclusion. The columns of Q_k are orthonormal because Q has orthonormal columns.
Structural Commentary: QR preserves prefix column spaces while orthonormalising them.
Key Insight: Invertible R_k only changes coordinates within that space. □

Theorem 4.2.10. *Cholesky Factorisation*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. If $A \succeq 0$, then $A = LL^*$ for some lower triangular L with nonnegative diagonal entries. If $A \succ 0$, then L is unique and has positive diagonal entries.

Proof. Let $B = A^{1/2}$ and take a QR factorisation $B = QR$ with R upper triangular and diagonal entries nonnegative. Then $A = B^*B = R^*R$, so $L = R^*$ works.
If $A \succ 0$, then L is invertible, so its diagonal entries are positive. For uniqueness, suppose $A = L_1 L_1^* = L_2 L_2^*$ with L_i lower triangular and positive diagonal. Set $R_i = L_i^*$. Then R_i is upper triangular with positive diagonal and $B^2 = A = R_i^* R_i$. Define $Q_i = B R_i^{-1}$. Then $Q_i^* Q_i = (R_i^{-1})^* B^2 R_i^{-1} = (R_i^{-1})^* R_i^* R_i R_i^{-1} = I_n$, so $B = Q_i R_i$ are QR factorisations with positive diagonal. Uniqueness of QR gives $R_1 = R_2$, hence $L_1 = L_2$.
Structural Commentary: Cholesky is QR applied to $A^{1/2}$.
Key Insight: Triangularise the square root, then set $L = R^*$. □

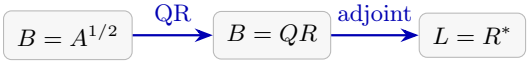


Figure 74: Apply QR to $B = A^{1/2}$; then $A = R^*R = LL^*$, with uniqueness in the positive definite case.

Theorem 4.2.11. *Polar Decomposition*
For every $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space, $\exists$ an isometry $S \in \mathcal{L}(\mathcal{V})$ s.t. $T = S\sqrt{T^*T}$.

Proof. Let $P = \sqrt{T^*T}$. Since T^*T is positive, P is well defined by Proposition 3.4.38.
For every $\mathbf{v} \in \mathcal{V}$, $\|T\mathbf{v}\|^2 = \langle T^*T\mathbf{v}, \mathbf{v}\rangle = \langle P^2\mathbf{v}, \mathbf{v}\rangle = \|P\mathbf{v}\|^2$.
Define $S_1 : \operatorname{range}(P) \to \operatorname{range}(T)$ by $S_1(P\mathbf{v}) = T\mathbf{v}$. The norm identity shows this is well defined: if $P\mathbf{v}_1 = P\mathbf{v}_2$, then $\|T(\mathbf{v}_1 - \mathbf{v}_2)\| = \|P(\mathbf{v}_1 - \mathbf{v}_2)\| = 0$. It also shows S_1 is an isometry onto $\operatorname{range}(T)$.
The same norm identity gives $\ker(P) = \ker(T)$, so $\operatorname{rank}(P) = \operatorname{rank}(T)$ by rank-nullity. Hence $\operatorname{range}(P)^\perp$ and $\operatorname{range}(T)^\perp$ have the same dimension. Choose any isometry between these complements and combine it orthogonally with S_1 to obtain an isometry S on all of $\mathcal{V}$.
Then $SP\mathbf{v} = S_1(P\mathbf{v}) = T\mathbf{v}$ for every $\mathbf{v}$, so $T = S\sqrt{T^*T}$.
Structural Commentary: The positive factor $\sqrt{T^*T}$ reproduces exactly the lengths produced by T.
Key Insight: Once the metric stretching has been extracted, the remaining map is forced to be an isometry. □

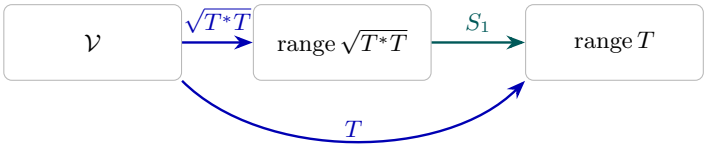


Figure 75: Polar decomposition builds S_1 using $\|T\mathbf{v}\| = \left\|\sqrt{T^*T}\mathbf{v}\right\|$.

Definition 4.2.12. *Singular Values*
For $T \in \mathcal{L}(\mathcal{V})$, *singular values* of T are eigenvalues of $\sqrt{T^*T}$, where each eigenvalue λ is repeated $\dim E_\lambda(\sqrt{T^*T})$.

Theorem 4.2.13. *Singular Value Decomposition*
Let $T \in \mathcal{L}(\mathcal{V})$ have singular values $s_1, \dots, s_n$. Then there exist orthonormal bases $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ and $(\mathbf{f}_1, \dots, \mathbf{f}_n)$ of $\mathcal{V}$ such that $T\mathbf{v} = s_1 \langle \mathbf{v}, \mathbf{e}_1 \rangle \mathbf{f}_1 + \cdots + s_n \langle \mathbf{v}, \mathbf{e}_n \rangle \mathbf{f}_n$ for every $\mathbf{v} \in \mathcal{V}$.

Proof. Let $|T| = \sqrt{T^*T}$. Since $|T|$ is positive, the real or complex spectral theorem gives an orthonormal basis $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ with $|T|\,\mathbf{e}_j = s_j \mathbf{e}_j$. Thus $|T|\,\mathbf{v} = \sum_{j=1}^{n} s_j \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$.
By Theorem 4.2.11, $T = S\,|T|$ for some isometry S. Put $\mathbf{f}_j = S\mathbf{e}_j$. The vectors $\mathbf{f}_1, \dots, \mathbf{f}_n$ form an orthonormal basis, and applying S to the displayed expansion gives the desired formula.
Structural Commentary: SVD diagonalises $|T|$, then transports its eigenvectors by S.
Key Insight: The s_j are stretches; the $\mathbf{f}_j$ are their landing directions. □

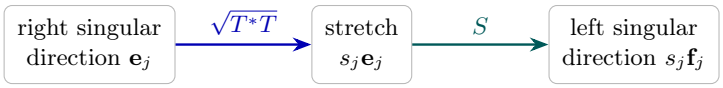


Figure 76: Since $T = S\sqrt{T^*T}$, each $\mathbf{e}_j$ maps to $s_j\mathbf{f}_j$.

Proposition 4.2.14. *Singular Values from T^*T*
The singular values of $T \in \mathcal{L}(\mathcal{V})$ are the nonnegative square roots of the eigenvalues of T^*T, with the same eigenspace multiplicities.

Proof. By the spectral theorem, choose an orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_n$ and nonnegative scalars $\lambda_1, \dots, \lambda_n$ such that $T^*T\mathbf{e}_j = \lambda_j \mathbf{e}_j$. The unique positive square root of T^*T sends $\mathbf{e}_j$ to $\sqrt{\lambda_j}\mathbf{e}_j$.
Thus the singular values are $\sqrt{\lambda_j}$, with the same eigenspace multiplicities.
Structural Commentary: The positive square root keeps eigenspaces and sends λ to $\sqrt{\lambda}$.
Key Insight: Singular values can be read from T^*T directly. □

Definition 4.2.15. *Matrix Singular Values*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$, let $q = \min(m, n)$, and let $r = \mathrm{rank}(A)$. The *positive singular values* of A are the positive eigenvalues of $|A| = (A^*A)^{1/2}$, written $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$.
The full list of singular values is $\sigma_1, \dots, \sigma_r, \sigma_{r+1}, \dots, \sigma_q$, where $\sigma_{r+1} = \cdots = \sigma_q = 0$.

Theorem 4.2.16. *Matrix Singular Value Decomposition*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$, where $q = \min(m, n)$. Then $\exists$ unitary matrices $V \in \mathrm{Mat}(m, m, \mathbb{F})$, $W \in \mathrm{Mat}(n, n, \mathbb{F})$ such that $A = V\Sigma W^*$, $\Sigma = \begin{bmatrix} \mathrm{diag}(\sigma_1, \dots, \sigma_q) & 0 \\ 0 & 0 \end{bmatrix} \in \mathrm{Mat}(m, n, \mathbb{R})$ with the evident deletion of empty blocks.

Proof. If $A = 0$, take $\Sigma = 0$ and choose any unitary V, W. Assume $r = \mathrm{rank}(A) \geq 1$.
Diagonalise A^*A as $W^*A^*AW = \mathrm{diag}(\sigma_1^2, \dots, \sigma_r^2, 0, \dots, 0)$, where $\mathbf{w}_1, \dots, \mathbf{w}_n$ are the columns of W.
For $1 \leq j \leq r$, set $\mathbf{v}_j = \sigma_j^{-1} A\mathbf{w}_j$. Then $\mathbf{v}_i^*\mathbf{v}_j = \frac{1}{\sigma_i\sigma_j}\mathbf{w}_i^*A^*A\mathbf{w}_j = \delta_{ij}$ so $\mathbf{v}_1, \dots, \mathbf{v}_r$ are orthonormal.
Complete them to an orthonormal basis of $\mathbb{F}^m$ and let V be the matrix with these columns.
For $j > r$, $\|A\mathbf{w}_j\|_2^2 = \mathbf{w}_j^*A^*A\mathbf{w}_j = 0$, hence $A\mathbf{w}_j = 0$. Thus $AW = V\Sigma$, equivalently $A = V\Sigma W^*$.
Structural Commentary: A^*A determines the right singular directions and singular values.
Key Insight: Apply A and normalize to get the left singular directions. □

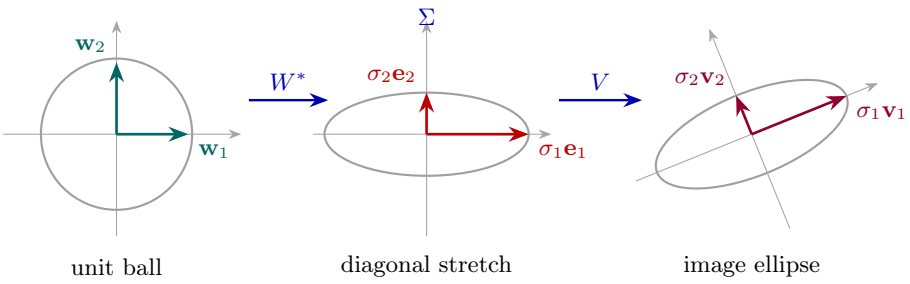


Figure 77: SVD: right-unitary motion, diagonal stretch, left-unitary motion.

Proposition 4.2.17. *Basic Singular Value Identities*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$, where $q = \min(m,n)$ and $r = \mathrm{rank}(A)$. Let $c \in \mathbb{F}$.

(i) The positive eigenvalues of A^*A and AA^* are $\sigma_1^2, \ldots, \sigma_r^2$.
(ii) $\sum_{j=1}^{q} \sigma_j^2 = \mathrm{tr}(A^*A) = \mathrm{tr}(AA^*) = \|A\|_F^2$, where $\|A\|_F^2 := \mathrm{tr}(A^*A)$.
(iii) A and A^* have the same singular values. Over $\mathbb{C}$, so do A^t and $\overline{A}$.
(iv) The singular values of cA are $|c|\,\sigma_1, \ldots, |c|\,\sigma_q$.

Proof. Let $A = V\Sigma W^*$ be an SVD. Then $A^*A = W(\Sigma^*\Sigma)W^*$ and $AA^* = V(\Sigma\Sigma^*)V^*$.

(i) The nonzero diagonal entries of $\Sigma^*\Sigma$ and $\Sigma\Sigma^*$ are $\sigma_1^2, \ldots, \sigma_r^2$.
(ii) Since A^*A and $\Sigma^*\Sigma$ are unitarily similar, and AA^* and $\Sigma\Sigma^*$ are unitarily similar,
$\mathrm{tr}(A^*A) = \mathrm{tr}(\Sigma^*\Sigma) = \sum_{j=1}^{q} \sigma_j^2 = \mathrm{tr}(\Sigma\Sigma^*) = \mathrm{tr}(AA^*)$. The final equality is the definition of $\|A\|_F$.
(iii) For A^*, the two Gram matrices are interchanged. Over $\mathbb{C}$, $(A^t)^*A^t = (AA^*)^t$ and $(\overline{A})^*\overline{A} = (A^*A)^t$.
Since a matrix and its transpose have the same characteristic polynomial, the spectra agree.
(iv) Since $(cA)^*(cA) = |c|^2 A^*A$, singular values scale by $|c|$.

Structural Commentary: Unitary factors do not change Gram spectra; Σ contains all singular data.
Key Insight: Reduce each claim to $\Sigma^*\Sigma$ and $\Sigma\Sigma^*$. □

Corollary 4.2.18. *Square Invertibility and Singular Values*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$, $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$, have singular values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. The following are equivalent:

(i) A is invertible;
(ii) $\mathrm{rank}(A) = n$;
(iii) $\sigma_n > 0$;
(iv) $A^*A \succ 0$.

If $r = \mathrm{rank}(A) < n$, then $\sigma_1, \ldots, \sigma_r > 0$ and $\sigma_{r+1} = \cdots = \sigma_n = 0$.

Proof.

(i) ⇒ *(ii)* By Corollary 1.3.64, invertibility gives full rank.
(ii) ⇒ *(iii)* By Definition 4.2.15, the number of positive singular values is $\mathrm{rank}(A)$.
(iii) ⇒ *(iv)* If $\sigma_n > 0$, then A has full column rank, so Proposition 4.1.49 gives $A^*A \succ 0$.
(iv) ⇒ *(i)* If $A\mathbf{x} = \mathbf{0}$, then $\mathbf{x}^*A^*A\mathbf{x} = 0$; PSD gives $\mathbf{x} = \mathbf{0}$. Thus $\ker(A) = \{\mathbf{0}\}$, so A is invertible.

The rank-defect statement follows again from Definition 4.2.15.
Structural Commentary: SVD sees nonsingularity as the absence of zero stretch directions.
Key Insight: A zero singular value is exactly a nonzero vector sent to zero. □

Definition 4.2.19. *Compact Singular Value Decomposition*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have rank $r \geq 1$.
A factorisation $A = X\Sigma_r Y^*$ is a *compact singular value decomposition* if $X \in \mathrm{Mat}(m,r,\mathbb{F})$ and $Y \in \mathrm{Mat}(n,r,\mathbb{F})$ have orthonormal columns and $\Sigma_r = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$ with $\sigma_1 \geq \cdots \geq \sigma_r > 0$.

Theorem 4.2.20. *Compact Singular Value Decomposition*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have rank $r \geq 1$.

(i) If $A = V\Sigma W^*$ is an SVD and X, Y are the first r columns of V, W, then $A = X\Sigma_r Y^*$.
(ii) Conversely, if X, Y have orthonormal columns and $\Sigma_r = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$ with $\sigma_1 \geq \cdots \geq \sigma_r > 0$, then $X\Sigma_r Y^*$ has positive singular values $\sigma_1, \ldots, \sigma_r$.

Proof.

(i) Partition $V = [X\ X_0]$, $W = [Y\ Y_0]$, and $\Sigma = \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix}$. Multiplication gives $A = V\Sigma W^* = X\Sigma_r Y^*$.
(ii) Let $B = X\Sigma_r Y^*$. Since $X^*X = Y^*Y = I_r$, $B^*B = Y\Sigma_r^2 Y^*$.
Completing Y to a unitary matrix $\widetilde{Y} = [Y\ Y_0]$ gives $\widetilde{Y}^*B^*B\widetilde{Y} = \Sigma_r^2 \oplus 0$.
Thus the positive eigenvalues of B^*B are $\sigma_1^2, \ldots, \sigma_r^2$, so the positive singular values of B are $\sigma_1, \ldots, \sigma_r$.

Structural Commentary: Compact SVD keeps only the nonzero singular directions.
Key Insight: The rank-r data X, Σ_r, Y reconstructs A. □

Theorem 4.2.21. *Column and Kernel Spaces from Compact SVD*
Let $A = X\Sigma_r Y^*$ be a compact SVD. Then $\mathrm{col}(A) = \mathrm{col}(X) = (\ker A^*)^\perp$, and $\ker(A) = \mathrm{col}(Y)^\perp = (\mathrm{col}(A^*))^\perp$.

Proof. Since $A = X\Sigma_r Y^*$ and $AY\Sigma_r^{-1} = X$, $\operatorname{col}(A) = \operatorname{col}(X)$.
For $\mathbf{z} \in \mathbb{F}^n$, $A\mathbf{z} = 0 \iff X\Sigma_r Y^*\mathbf{z} = 0 \iff Y^*\mathbf{z} = 0 \iff \mathbf{z} \perp \operatorname{col}(Y)$, because X has independent columns and Σ_r is invertible. The orthogonal-complement identities follow from Proposition 2.4.7.
Structural Commentary: Compact SVD exposes image directions X and null directions $\operatorname{col}(Y)^\perp$.
Key Insight: A acts only on the right singular subspace $\operatorname{col}(Y)$. □

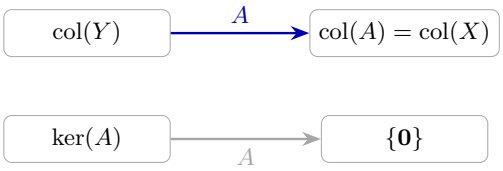


Figure 78: Compact SVD: A maps $\operatorname{col}(Y)$ onto $\operatorname{col}(X)$ and kills $\ker(A) = \operatorname{col}(Y)^\perp$.

Theorem 4.2.22. *Uniqueness of Compact SVD up to Singular Subspaces*
Let $A = X_1\Sigma_r Y_1^* = X_2\Sigma_r Y_2^*$ be compact SVDs with the same Σ_r. Then $X_1 = X_2U$, $Y_1 = Y_2U$ for some unitary U satisfying $U\Sigma_r = \Sigma_r U$. Hence $U = U_1 \oplus \cdots \oplus U_d$ along the equal singular-value blocks of Σ_r.

Proof. Theorem 4.2.21 gives $\operatorname{col}(X_1) = \operatorname{col}(X_2)$ and $\operatorname{col}(Y_1) = \operatorname{col}(Y_2)$. Since each X_i and Y_i has orthonormal columns, there are unitary $U, V \in \operatorname{Mat}(r, r, \mathbb{F})$ such that $X_1 = X_2U$ and $Y_1 = Y_2V$. Substitution and multiplication on the left by X_2^* and on the right by Y_2 give $\Sigma_r = U\Sigma_r V^*$, hence $U\Sigma_r = \Sigma_r V$.
Therefore $U\Sigma_r^2 U^* = \Sigma_r^2$, so U commutes with Σ_r^2. Lemma 3.1.22 makes U block diagonal on equal singular-value blocks. Hence $U\Sigma_r = \Sigma_r U$, and $U\Sigma_r = \Sigma_r V$ forces $V = U$ because Σ_r is invertible.
Structural Commentary: Compact SVD is unique up to unitary changes inside equal singular-value blocks.
Key Insight: Distinct singular values fix subspaces; repeated ones only fix the subspace, not its basis. □

Definition 4.2.23. *Matrix Modulus*
For $A \in \operatorname{Mat}(m, n, \mathbb{F})$, the *modulus* of A is $|A| := (A^*A)^{1/2} \in \operatorname{Mat}(n, n, \mathbb{F})$.
Thus $|A^*| = (AA^*)^{1/2} \in \operatorname{Mat}(m, m, \mathbb{F})$.

Proposition 4.2.24. *Modulus from SVD*
Let $A = V\Sigma W^*$ be an SVD. Then $|A| = W(\Sigma^*\Sigma)^{1/2}W^*$ and $|A^*| = V(\Sigma\Sigma^*)^{1/2}V^*$.

Proof. Since $A^*A = W(\Sigma^*\Sigma)W^*$ and $AA^* = V(\Sigma\Sigma^*)V^*$, the displayed matrices are positive semidefinite square roots of A^*A and AA^*, respectively. Uniqueness of the positive semidefinite square root gives the claim.
Structural Commentary: The modulus keeps only the singular-value stretch.
Key Insight: In an SVD, $|A|$ is the right singular-vector version of the diagonal stretch. □

Theorem 4.2.25. *Rectangular Polar Decomposition*
Let $A = V\Sigma W^* \in \operatorname{Mat}(m, n, \mathbb{F})$ be an SVD.

(i) If $m \geq n$, let V_n be the first n columns of V and set $U = V_nW^*$. Then $U^*U = I_n$ and $A = U\,|A|$.
(ii) If $m \leq n$, let W_m be the first m columns of W and set $U = VW_m^*$. Then $UU^* = I_m$ and $A = |A^*|\,U$.
(iii) If $m = n$, then $U = VW^*$ is unitary and $A = U\,|A| = |A^*|\,U$.

Proof.

(i) Proposition 4.2.24 gives $|A| = W(\Sigma^*\Sigma)^{1/2}W^*$.
Since $V_n(\Sigma^*\Sigma)^{1/2} = V\Sigma$, $U\,|A| = V_nW^*W(\Sigma^*\Sigma)^{1/2}W^* = V\Sigma W^* = A$. Also $U^*U = WV_n^*V_nW^* = I_n$.
(ii) Apply (i) to A^* with the SVD $A^* = W\Sigma^*V^*$, obtaining $A^* = U_0\,|A^*|$ with $U_0 = W_mV^*$ and $U_0^*U_0 = I_m$.
Taking adjoints gives $A = |A^*|\,U_0^* = |A^*|\,VW_m^*$, and $U_0^*(U_0^*)^* = U_0^*U_0 = I_m$. Thus $U = VW_m^*$ works.
(iii) This is the common case of (i) and (ii), with $U = VW^*$ unitary.

Structural Commentary: Polar decomposition compresses the two SVD unitary factors into one partial isometry.
Key Insight: SVD records every singular direction; polar form records stretch plus motion. □

Method 4.2.26. *SVD and Polar Computation Method*
To compute an SVD and the associated polar factors of $A \in \operatorname{Mat}(m, n, \mathbb{F})$:

(i) Diagonalise A^*A; the square roots of its positive eigenvalues are the positive singular values.
(ii) Choose orthonormal right singular vectors $\mathbf{w}_j$ for the positive eigenvalues and set $\mathbf{v}_j = \sigma_j^{-1}A\mathbf{w}_j$.
(iii) Complete the $\mathbf{v}_j$ and $\mathbf{w}_j$ to unitary matrices V, W to obtain $A = V\Sigma W^*$.
Omit zero-singular directions for compact SVD.
(iv) Use $|A| = W(\Sigma^*\Sigma)^{1/2}W^*$ and merge V, W as in Theorem 4.2.25 to obtain the polar factor.

Theorem 4.2.27. *Uniqueness of the Positive Polar Factor*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$.

(i) If $m \geq n$ and $A = UB$ with $U^*U = I_n$ and $B \succeq 0$, then $B = |A|$.
If $\mathrm{rank}(A) = n$, then $U = A\,|A|^{-1}$ is unique.
(ii) If $m \leq n$ and $A = CU$ with $UU^* = I_m$ and $C \succeq 0$, then $C = |A^*|$.
If $\mathrm{rank}(A) = m$, then $U = |A^*|^{-1} A$ is unique.

Proof.

(i) $A^*A = BU^*UB = B^2$, $B = |A|$ by uniqueness of positive square root. If $\mathrm{rank}(A) = n$, then $\ker(A) = \{\mathbf{0}\}$. Since $\mathbf{x}^*A^*A\mathbf{x} = \|A\mathbf{x}\|_2^2$, this gives $\ker(A^*A) = \{\mathbf{0}\}$ and $|A|$ is invertible. Thus $U = A\,|A|^{-1}$.
(ii) $AA^* = CUU^*C = C^2$, so $C = |A^*|$. If $\mathrm{rank}(A) = m$, then A^* has full column rank, so $\ker(A^*) = \{\mathbf{0}\}$. Since $\mathbf{y}^*AA^*\mathbf{y} = \|A^*\mathbf{y}\|_2^2$, this gives $\ker(AA^*) = \{\mathbf{0}\}$ and $|A^*|$ is invertible. Thus $U = |A^*|^{-1} A$.

Structural Commentary: The positive factor is fixed by the Gram matrix.
Key Insight: Full rank makes the partial isometry unique. □

Theorem 4.2.28. *Normality and Polar Commutation*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$ and suppose $A = U\,|A|$ with U unitary. Then A is normal if and only if $U\,|A| = |A|\,U$.

Proof.

($\Rightarrow$) If A is normal, then $|A|^2 = A^*A = AA^* = U\,|A|^2\,U^*$. Thus $U\,|A|^2\,U^* = |A|^2$. The matrix $U\,|A|\,U^*$ is PSD and squares to $|A|^2$, so uniqueness of positive square root gives $U\,|A|\,U^* = |A|$. Hence $U\,|A| = |A|\,U$.
($\Leftarrow$) If $U\,|A| = |A|\,U$, then $AA^* = U\,|A|^2\,U^* = |A|^2 = A^*A$.

Structural Commentary: Normality says that the polar motion preserves the stretching eigenspaces.
Key Insight: Normality is commutation of polar motion and stretch. □

Proposition 4.2.29. *Absolute Determinant Comes from the Positive Polar Factor*
For $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space, $|\det T| = \det\sqrt{T^*T}$.

Proof. By Theorem 4.2.11, $T = S\sqrt{T^*T}$ for some isometry S. Hence $|\det T| = |\det S|\left|\det\sqrt{T^*T}\right| = \det\sqrt{T^*T}$, where Proposition 4.1.22 gives $|\det S| = 1$, and positivity of $\sqrt{T^*T}$ gives nonnegative determinant.
Structural Commentary: Polar decomposition isolates stretch; only the positive factor changes volume.
Key Insight: Determinant sign or phase belongs to the isometric part; volume belongs to the positive part. □

4.3 Matrix Norms and Conditioning

Matrix norms measure stretch, energy, approximation quality, and sensitivity. Singular values organize norms, low-rank approximation, pseudoinverses, conditioning, and error bounds.

Definition 4.3.1. *Spectral Norm and Extreme Singular Values*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$, define the *spectral norm* $\|A\|_2 := \max_{\|\mathbf{x}\|_2=1} \|A\mathbf{x}\|_2$.
If the singular values of A are $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$, $q = \min(m,n)$, set $\sigma_{\max}(A) := \sigma_1$.
When A is square, $\sigma_{\min}(A) := \sigma_n$.

Theorem 4.3.2. *Spectral Norm is the Largest Singular Value*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$. Then $\|A\|_2 = \sigma_{\max}(A)$.
Moreover, if $A = V\Sigma W^*$ is an SVD and $\mathbf{w}_1$ is the first column of W, then $\|A\mathbf{w}_1\|_2 = \sigma_1$.

Proof. Let $\mathbf{w}_1, \ldots, \mathbf{w}_n$ be the columns of W. For $\mathbf{x} = \sum_{j=1}^n c_j\mathbf{w}_j$ with $\|\mathbf{x}\|_2 = 1$, unitary invariance of the Euclidean norm gives $\|A\mathbf{x}\|_2^2 = \|\Sigma W^*\mathbf{x}\|_2^2 = \sum_{j=1}^q \sigma_j^2\,|c_j|^2 \leq \sigma_1^2 \sum_{j=1}^n |c_j|^2 = \sigma_1^2$.
Thus $\|A\|_2 \leq \sigma_1$. Taking $\mathbf{x} = \mathbf{w}_1$ gives equality.
Structural Commentary: The right singular basis diagonalises the stretching computation.
Key Insight: The largest stretch occurs in a top right singular direction. □

Proposition 4.3.3. *Basic Properties of the Spectral Norm*
For compatible matrices A, B and scalar c:

(i) $\|A\|_2 \geq 0$, and $\|A\|_2 = 0$ if and only if $A = 0$;
(ii) $\|cA\|_2 = |c|\,\|A\|_2$;
(iii) $\|A + B\|_2 \leq \|A\|_2 + \|B\|_2$;
(iv) $\|AB\|_2 \leq \|A\|_2\,\|B\|_2$.

Proof.

(i) Since $\|A\mathbf{x}\|_2 \geq 0$, $\|A\|_2 \geq 0$. Also $\|A\|_2 = 0$ iff $A\mathbf{x} = 0$ for every unit $\mathbf{x}$, iff $A = 0$.
(ii) $\|cA\|_2 = \max_{\|\mathbf{x}\|_2=1} \|cA\mathbf{x}\|_2 = |c|\,\|A\|_2$.
(iii) For $\|\mathbf{x}\|_2 = 1$, $\|(A+B)\mathbf{x}\|_2 \leq \|A\mathbf{x}\|_2 + \|B\mathbf{x}\|_2 \leq \|A\|_2 + \|B\|_2$. Maximize over unit $\mathbf{x}$.
(iv) For $\|\mathbf{x}\|_2 = 1$, $\|AB\mathbf{x}\|_2 \leq \|A\|_2\,\|B\mathbf{x}\|_2 \leq \|A\|_2\,\|B\|_2$. Maximize over unit $\mathbf{x}$.

Structural Commentary: The spectral norm is the operator norm induced by Euclidean geometry.
Key Insight: Bound each output vector first, then take the worst case. □

Definition 4.3.4. *Operator Norm on $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$*
For $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$, define $\|A\|_{\text{op}} := \sup_{\|\mathbf{h}\|_2 \leq 1} \|A\mathbf{h}\|_2$.
Equivalently, after standard bases identify A with its matrix, $\|A\|_{\text{op}}$ is the spectral norm from Definition 4.3.1.
Thus $\|A\mathbf{h}\|_2 \leq \|A\|_{\text{op}}\,\|\mathbf{h}\|_2$, and sums and products obey the norm estimates in Proposition 4.3.3.
The induced metric on $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ is $d(A, B) := \|A - B\|_{\text{op}}$; this is the metric used for continuity of derivative maps in Definition 15.2.12.

Proposition 4.3.5. *Frobenius and Spectral Norm Comparison*
Let $A \in \text{Mat}(m, n, \mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$. Then $\|A\|_F^2 = \sum_{j=1}^q \sigma_j^2$ and $\|A\|_2 \leq \|A\|_F$. If $A \neq 0$, equality $\|A\|_2 = \|A\|_F$ holds if and only if $\text{rank}(A) = 1$.

Proof. The Frobenius identity is Proposition 4.2.17. Hence $\|A\|_2^2 = \sigma_1^2 \leq \sum_{j=1}^q \sigma_j^2 = \|A\|_F^2$. Assume $A \neq 0$.

($\Rightarrow$) If $\|A\|_2 = \|A\|_F$, then $\sum_{j=2}^q \sigma_j^2 = 0$, so only σ_1 is nonzero and $\text{rank}(A) = 1$.
($\Leftarrow$) If $\text{rank}(A) = 1$, then $\sigma_2 = \cdots = \sigma_q = 0$, so $\|A\|_2 = \|A\|_F$.

Structural Commentary: Frobenius counts all singular energy; spectral keeps the largest part.
Key Insight: Equality means exactly one nonzero singular value. □

Theorem 4.3.6. *Submultiplicativity of Frobenius and Spectral Norms*
Let $A \in \text{Mat}(m, n, \mathbb{F})$, $B \in \text{Mat}(n, k, \mathbb{F})$. Then $\|AB\|_F \leq \|A\|_2\,\|B\|_F \leq \|A\|_F\,\|B\|_F$ and $\|AB\|_2 \leq \|A\|_2\,\|B\|_2$.

Proof. Write $B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_k]$. Then $\|AB\|_F^2 = \sum_{j=1}^k \|A\mathbf{b}_j\|_2^2 \leq \|A\|_2^2 \sum_{j=1}^k \|\mathbf{b}_j\|_2^2 = \|A\|_2^2\,\|B\|_F^2$.
Proposition 4.3.5 gives $\|A\|_2 \leq \|A\|_F$. The spectral-norm inequality is Proposition 4.3.3.
Structural Commentary: Frobenius norm is column energy; each column of AB is A applied to a column of B.
Key Insight: Control every column by the spectral norm of A. □

Proposition 4.3.7. *Unitary Invariance of Singular Values*
Let $A \in \text{Mat}(m, n, \mathbb{F})$, and let $U \in \text{Mat}(m, m, \mathbb{F})$ and $V \in \text{Mat}(n, n, \mathbb{F})$ be unitary. Then A and UAV have the same singular values. In particular, $\|UAV\|_2 = \|A\|_2$ and $\|UAV\|_F = \|A\|_F$.

Proof. Since $(UAV)^*(UAV) = V^*A^*AV$, the Gram matrices of A and UAV are unitarily similar, hence have the same eigenvalues. Singular values are the nonnegative square roots of these eigenvalues. The two norm identities follow from Theorem 4.3.2 and Proposition 4.3.5.
Structural Commentary: Left and right unitary factors change coordinates, not singular-value data.
Key Insight: Singular values are spectral data of A^*A, and unitary conjugation preserves that spectrum. □

Example 4.3.8. *Similarity Need Not Preserve Singular Values*
Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$. Then $B = DAD^{-1}$ with $D = \text{diag}(2, 1)$, so A and B are similar.
However $A^*A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$, $B^*B = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$ have different eigenvalues, so A and B have different singular values.
Singular values are preserved by unitary left-right equivalence, not by arbitrary similarity.

Theorem 4.3.9. *Schur's Inequality and Normality*
Let $A \in \text{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \ldots, \lambda_n$, counted algebraically.
Then $\sum_{j=1}^n |\lambda_j|^2 \leq \|A\|_F^2$ with equality if and only if A is normal.

Proof. Let $A = UTU^*$ be a Schur form, where $T = [t_{ij}]$ is upper triangular and $t_{jj} = \lambda_j$.
Then $\|A\|_F^2 = \text{tr}(T^*T) = \sum_{j=1}^n |\lambda_j|^2 + \sum_{i<j} |t_{ij}|^2$. Inequality holds. Equality holds exactly when T is diagonal.
($\Rightarrow$) If equality holds, then T is diagonal, so A is unitarily diagonalizable and hence normal.
($\Leftarrow$) If A is normal, then T is normal and upper triangular, hence diagonal by Lemma 3.4.23.

Structural Commentary: Schur form splits Frobenius norm into eigenvalue mass plus off-diagonal mass.
Key Insight: Normality means the Schur off-diagonal part vanishes. □

Definition 4.3.10. *Defect from Normality*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \dots, \lambda_n$, counted with algebraic multiplicity.
The *defect from normality* of A is $\Delta(A) := \|A\|_F^2 - \sum_{j=1}^{n} |\lambda_j|^2$.

By Theorem 4.3.9, $\Delta(A) \geq 0$, with equality if and only if A is normal.

Theorem 4.3.11. *Eigenvalues Versus Singular Values*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \dots, \lambda_n$ ordered so that $|\lambda_1| \geq \cdots \geq |\lambda_n|$, and singular values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. Then:

(i) $|\det A| = \prod_{j=1}^{n} |\lambda_j| = \prod_{j=1}^{n} \sigma_j$;
(ii) $\rho(A) \leq \|A\|_2$;
(iii) $\sigma_n \leq |\lambda_n|$;
(iv) *A is normal if and only if $|\lambda_j| = \sigma_j$ for every j;*
(v) *A is positive semidefinite if and only if $\lambda_j = \sigma_j$ for every j.*

Proof.

(i) The eigenvalue product gives $|\det A| = \prod_j |\lambda_j|$.
If $A = V\Sigma W^*$ is an SVD, then $|\det V| = |\det W| = 1$, so $|\det A| = |\det \Sigma| = \prod_j \sigma_j$.
(ii) If $A\mathbf{x} = \lambda\mathbf{x}$ with $\|\mathbf{x}\|_2 = 1$, then $|\lambda| = \|A\mathbf{x}\|_2 \leq \|A\|_2$. Maximize over eigenvalues.
(iii) If A is singular, both sides are 0 or the inequality is immediate.
If A is invertible, apply (ii) to A^{-1}: $\frac{1}{|\lambda_n|} = \rho(A^{-1}) \leq \|A^{-1}\|_2 = \frac{1}{\sigma_n}$.
(iv) If A is normal, the spectral theorem gives $\sigma_j = |\lambda_j|$. Conversely, equality gives $\sum_j |\lambda_j|^2 = \sum_j \sigma_j^2 = \|A\|_F^2$, so Theorem 4.3.9 gives normality.
(v) If $A \succeq 0$, then A is normal and $\lambda_j \geq 0$, so (iv) gives $\lambda_j = \sigma_j$. Conversely, equality gives normality with nonnegative real eigenvalues, hence $A \succeq 0$ by Theorem 4.1.53.

Structural Commentary: Normality aligns eigenvalue moduli with singular values; positivity removes the moduli.
Key Insight: Normality is the bridge that turns spectral data into singular-value data. □

Theorem 4.3.12. *Trace Bound by Singular Values*
If $A \in \mathrm{Mat}(n, n, \mathbb{C})$, then $|\operatorname{tr} A| \leq \sum_{j=1}^{n} \sigma_j(A)$. Consequently, $\sum_{j=1}^{n} |\lambda_j(A)| \leq \sum_{j=1}^{n} \sigma_j(A)$ and $\sum_{j=1}^{n} |a_{jj}| \leq \sum_{j=1}^{n} \sigma_j(A)$.

Proof. Let $A = V\Sigma W^*$ be an SVD.

Writing $\mathbf{v}_k, \mathbf{w}_k$ for the columns of V, W, $\operatorname{tr} A = \sum_{k=1}^{n} \sigma_k \mathbf{w}_k^* \mathbf{v}_k$, $|\mathbf{w}_k^* \mathbf{v}_k| \leq 1$, so $|\operatorname{tr} A| \leq \sum_k \sigma_k$.
For eigenvalues, take a Schur form T of A and a diagonal unitary D with $\operatorname{tr}(DT) = \sum_j |\lambda_j|$.

For diagonal entries, take diagonal unitary D with $\operatorname{tr}(DA) = \sum_j |a_{jj}|$.

In both cases, unitary invariance of singular values and the trace bound give the result.
Structural Commentary: Phase rotations turn trace estimates into eigenvalue and diagonal estimates.
Key Insight: Prove $|\operatorname{tr} A| \leq \sum_j \sigma_j$ once, then rotate phases. □

Proposition 4.3.13. *Rank-One SVD Terms and Truncation Error*
Let $A = X\Sigma_r Y^* = \sum_{j=1}^{r} \sigma_j \mathbf{x}_j \mathbf{y}_j^*$ be a compact SVD, with $\sigma_1 \geq \cdots \geq \sigma_r > 0$.
For $0 \leq k \leq r$, set $A_k := \sum_{j=1}^{k} \sigma_j \mathbf{x}_j \mathbf{y}_j^*$, $A_0 := 0$.
Then the rank-one summands are mutually orthogonal in the Frobenius inner product, and

$$\|A\|_F^2 = \sum_{j=1}^{r} \sigma_j^2, \qquad \|A - A_k\|_F^2 = \sum_{j=k+1}^{r} \sigma_j^2, \qquad \|A - A_k\|_2 = \begin{cases} \sigma_{k+1}, & k < r, \\ 0, & k = r. \end{cases}$$

Proof. For $i, j \in \{1, \dots, r\}$, $\langle \sigma_i \mathbf{x}_i \mathbf{y}_i^*, \sigma_j \mathbf{x}_j \mathbf{y}_j^* \rangle_F = \sigma_i \sigma_j \operatorname{tr}(\mathbf{y}_i \mathbf{x}_i^* \mathbf{x}_j \mathbf{y}_j^*) = \sigma_i \sigma_j \delta_{ij}$.
Pythagoras in the Frobenius inner product gives the two Frobenius formulas. The residual $A - A_k$ has compact SVD $\sum_{j=k+1}^{r} \sigma_j \mathbf{x}_j \mathbf{y}_j^*$, so Theorem 4.3.2 gives the spectral-norm formula.
Structural Commentary: Compact SVD decomposes a matrix into orthogonal rank-one energy packets.
Key Insight: Truncation error is exactly the energy left in the discarded singular directions. □

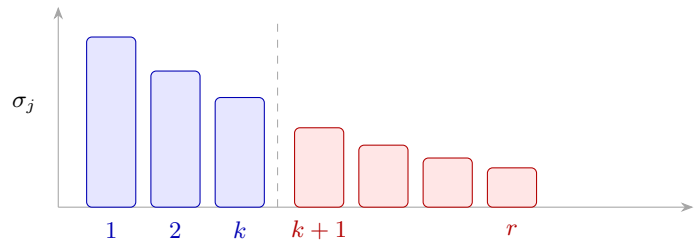


Figure 79: Blue bars form A_k; red bars are discarded, with $\|A - A_k\|_F^2 = \sum_{j>k} \sigma_j^2$.

Theorem 4.3.14. *Eckart–Young Low-Rank Approximation*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$, where $q = \min(m, n)$ and $\sigma_j = 0$ for $j > q$. If A_k is the rank-k truncated SVD, $0 \leq k < q$, and $B \in \mathrm{Mat}(m, n, \mathbb{F})$ has $\mathrm{rank}(B) \leq k$, then

$$\|A - A_k\|_2 \leq \|A - B\|_2, \qquad \|A - A_k\|_F \leq \|A - B\|_F.$$

Proof. Let $\mathbf{w}_1, \ldots, \mathbf{w}_n$ be right singular vectors for A.

i. Since $\mathrm{rank}(B) \leq k$, rank-nullity gives $\dim \ker(B) \geq n - k$. The subspaces $\ker(B)$ and $\mathrm{span}(\mathbf{w}_1, \ldots, \mathbf{w}_{k+1})$ in $\mathbb{F}^n$ have dimensions summing at least $n + 1$, so their intersection contains a nonzero vector.
Choose unit $\mathbf{x}$ in this intersection. Let $\mathbf{x} = \sum_{i=1}^{k+1} c_i \mathbf{w}_i$ gives $\sum_i |c_i|^2 = 1$ and $\|A\mathbf{x}\|_2^2 = \sum_{i=1}^{k+1} \sigma_i^2 |c_i|^2 \geq \sigma_{k+1}^2$.
Thus $\|A - B\|_2 \geq \|(A - B)\mathbf{x}\|_2 = \|A\mathbf{x}\|_2 \geq \sigma_{k+1} = \|A - A_k\|_2$.

ii. Choose an $(n-k)$-dimensional subspace $\mathcal{U} \subseteq \ker(B)$, with projection P and orthonormal basis $\mathbf{z}_1, \ldots, \mathbf{z}_{n-k}$.
Since $B\mathbf{z}_j = 0$, $\|A - B\|_F^2 \geq \sum_{j=1}^{n-k} \|(A - B)\mathbf{z}_j\|_2^2 = \sum_{j=1}^{n-k} \|A\mathbf{z}_j\|_2^2 = \sum_{i=1}^{n} \sigma_i^2 \|P\mathbf{w}_i\|_2^2$.
Put $\alpha_i = \|P\mathbf{w}_i\|_2^2$. Then $0 \leq \alpha_i \leq 1$ and $\sum_i \alpha_i = \mathrm{tr}(P) = n - k$.
Among all such weights, the sum $\sum_i \sigma_i^2 \alpha_i$ is minimized by placing weight 1 on the $n - k$ smallest values of the decreasing sequence σ_i^2, hence $\sum_i \sigma_i^2 \alpha_i \geq \sum_{i=k+1}^{n} \sigma_i^2 = \|A - A_k\|_F^2$.

Structural Commentary: Rank-k competitors vanish on at least $n - k$ directions.
Key Insight: The optimal loss is precisely the singular-value tail. □

Method 4.3.15. *Truncated SVD Approximation Method*
To compress or approximate A by rank k:

(i) compute or estimate the leading singular triples $(\sigma_j, \mathbf{x}_j, \mathbf{y}_j)$;
(ii) keep $A_k = \sum_{j=1}^{k} \sigma_j \mathbf{x}_j \mathbf{y}_j^*$;
(iii) use $\|A - A_k\|_F^2 = \sum_{j>k} \sigma_j^2$ and $\|A - A_k\|_2 = \sigma_{k+1}$ to quantify the lost information.

Definition 4.3.16. *Moore–Penrose Pseudoinverse*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$. If $A = 0$, define $A^\dagger = 0_{n,m}$.
If $\mathrm{rank}(A) = r \geq 1$ and $A = X\Sigma_r Y^*$ is a compact SVD, define $A^\dagger := Y\Sigma_r^{-1} X^* \in \mathrm{Mat}(n, m, \mathbb{F})$.
This is the *pseudoinverse* or *Moore–Penrose pseudoinverse* of A.

Proposition 4.3.17. *Pseudoinverse is Well Defined*
The definition of $A^\dagger$ does not depend on the chosen compact SVD of A.

Proof. Let $A = X_1\Sigma_r Y_1^* = X_2\Sigma_r Y_2^*$. By Theorem 4.2.22, $X_1 = X_2 U$, $Y_1 = Y_2 U$, and $U\Sigma_r^{-1}U^* = \Sigma_r^{-1}$.
Hence $Y_1\Sigma_r^{-1}X_1^* = Y_2 U\Sigma_r^{-1}U^* X_2^* = Y_2\Sigma_r^{-1}X_2^*$.
Structural Commentary: Compact SVD ambiguity cancels in $Y\Sigma_r^{-1}X^*$.
Key Insight: Inverting the singular values preserves the same block symmetries. □

Theorem 4.3.18. *Basic Properties of the Pseudoinverse*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$. Then:

(i) $\mathrm{col}(A^\dagger) = \mathrm{col}(A^*)$ *and* $\ker(A^\dagger) = \ker(A^*)$;
(ii) $AA^\dagger$ *is the orthogonal projection of* $\mathbb{F}^m$ *onto* $\mathrm{col}(A)$;
(iii) $A^\dagger A$ *is the orthogonal projection of* $\mathbb{F}^n$ *onto* $\mathrm{col}(A^*)$;
(iv) $AA^\dagger A = A$ *and* $A^\dagger AA^\dagger = A^\dagger$;
(v) $(A^\dagger)^\dagger = A$;
(vi) if $m = n$ and A is invertible, then $A^\dagger = A^{-1}$.

Proof. The zero case is immediate. Let $A = X\Sigma_r Y^*$ be a compact SVD; then $A^\dagger = Y\Sigma_r^{-1}X^*$.

(i) $\operatorname{col}(A^\dagger) = \operatorname{col}(Y) = \operatorname{col}(A^*)$ and $\ker(A^\dagger) = \ker(X^*) = \ker(A^*)$.
(ii) $AA^\dagger = XX^*$, the orthogonal projection onto $\operatorname{col}(X) = \operatorname{col}(A)$.
(iii) $A^\dagger A = YY^*$, the orthogonal projection onto $\operatorname{col}(Y) = \operatorname{col}(A^*)$.
(iv) These follow from $XX^*X = X$ and $YY^*Y = Y$.
(v) The compact SVD of $A^\dagger$ is $Y\Sigma_r^{-1}X^*$, so $(A^\dagger)^\dagger = A$.
(vi) If A is square and invertible, then $AA^\dagger = A^\dagger A = I_n$, hence $A^\dagger = A^{-1}$.

Structural Commentary: The pseudoinverse inverts nonzero singular directions and projects away null directions.
Key Insight: Replace each $\sigma_j > 0$ by σ_j^{-1} on the same singular subspaces. □

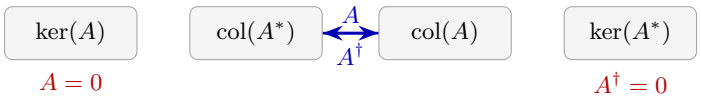


Figure 80: A and $A^\dagger$ invert the nonzero singular subspaces; they vanish on $\ker(A)$ and $\ker(A^*)$, respectively.

Theorem 4.3.19. *Pseudoinverse and Minimum-Norm Least Squares*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$, let $\mathbf{y} \in \mathbb{F}^m$, and set $\mathbf{x}_0 = A^\dagger\mathbf{y}$.

(i) If $A\mathbf{x} = \mathbf{y}$ is consistent, then $A\mathbf{x}_0 = \mathbf{y}$ and $\mathbf{x}_0$ is the unique solution of smallest Euclidean norm.
(ii) For arbitrary $\mathbf{y}$, $\mathbf{x}_0$ is a least-squares solution of $A\mathbf{x} \approx \mathbf{y}$, and it is the unique least-squares solution of smallest Euclidean norm.

Proof. Let $P = AA^\dagger = P_{\operatorname{col}(A)}$ and note that $\mathbf{x}_0 \in \operatorname{col}(A^\dagger) = \operatorname{col}(A^*) = (\ker A)^\perp$.

(i) If $\mathbf{y} \in \operatorname{col}(A)$, then $A\mathbf{x}_0 = P\mathbf{y} = \mathbf{y}$. For any solution $\mathbf{x}$, $\mathbf{x} - \mathbf{x}_0 \in \ker(A)$, so $\|\mathbf{x}\|_2^2 = \|\mathbf{x}_0 + (\mathbf{x} - \mathbf{x}_0)\|_2^2 = \|\mathbf{x}_0\|_2^2 + \|\mathbf{x} - \mathbf{x}_0\|_2^2$ with equality only when $\mathbf{x} = \mathbf{x}_0$.
(ii) For arbitrary $\mathbf{y}$, $A\mathbf{x}_0 = P\mathbf{y}$, so $\mathbf{y} - A\mathbf{x}_0 \perp \operatorname{col}(A)$ and $\mathbf{x}_0$ is least-squares optimal by Theorem 2.3.3. Any least-squares solution $\mathbf{x}$ satisfies $A\mathbf{x} = P\mathbf{y} = A\mathbf{x}_0$, hence the same orthogonal decomposition gives $\|\mathbf{x}\|_2 \geq \|\mathbf{x}_0\|_2$, with equality only for $\mathbf{x} = \mathbf{x}_0$.

Structural Commentary: Least squares projects onto $\operatorname{col}(A)$, then removes the $\ker(A)$ component.
Key Insight: $A^\dagger\mathbf{y}$ is the fitted solution orthogonal to $\ker(A)$. □

Method 4.3.20. *Pseudoinverse Solution Method*
For a system $A\mathbf{x} \approx \mathbf{y}$:

(i) compute a compact SVD $A = X\Sigma_r Y^*$;
(ii) form $A^\dagger = Y\Sigma_r^{-1}X^*$;
(iii) set $\mathbf{x}_0 = A^\dagger\mathbf{y}$;
(iv) interpret $A\mathbf{x}_0 = P_{\operatorname{col}(A)}\mathbf{y}$ and $\mathbf{x}_0 \in (\ker A)^\perp$.

Definition 4.3.21. *Spectral Condition Number*
If $A \in \mathrm{Mat}(n,n,\mathbb{F})$ is invertible, its *spectral condition number* is $\kappa_2(A) := \|A\|_2 \left\|A^{-1}\right\|_2$.
Equivalently, $\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$. If A is singular, $\kappa_2(A)$ is undefined.

Proposition 4.3.22. *Properties of the Spectral Condition Number*
Let $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$ be invertible.

(i) $\kappa_2(AB) \leq \kappa_2(A)\kappa_2(B)$;
(ii) $\kappa_2(A) \geq 1$, with equality if and only if A is a nonzero scalar multiple of a unitary matrix;
(iii) $\kappa_2(cA) = \kappa_2(A)$ for every nonzero scalar c;
(iv) $\kappa_2(A) = \kappa_2(A^*) = \kappa_2(\overline{A}) = \kappa_2(A^t) = \kappa_2(A^{-1})$ over $\mathbb{C}$, with the evident real special cases;
(v) $\kappa_2(A^*A) = \kappa_2(A)^2$;
(vi) if U, V are unitary, then $\kappa_2(UAV) = \kappa_2(A)$.

Proof.

(i) Use submultiplicativity: $\kappa_2(AB) = \|AB\|_2 \left\|B^{-1}A^{-1}\right\|_2 \leq \|A\|_2 \|B\|_2 \left\|B^{-1}\right\|_2 \left\|A^{-1}\right\|_2$.
(ii) Since $I_n = AA^{-1}$, $1 = \|I_n\|_2 \leq \kappa_2(A)$. Equality means $\sigma_{\max}(A) = \sigma_{\min}(A)$, equivalently every singular value is the same positive number s. If $A = V\Sigma W^*$, then $\Sigma = sI_n$, so $A = sVW^*$ is a nonzero scalar multiple of a unitary matrix. The converse is immediate.
(iii) This follows from $\|cA\|_2 = |c| \|A\|_2$ and $(cA)^{-1} = c^{-1}A^{-1}$.
(iv) These matrices have the same singular values as A, except A^{-1} has singular values $\sigma_n^{-1} \geq \cdots \geq \sigma_1^{-1}$.

(v) The singular values of A^*A are $\sigma_1(A)^2, \dots, \sigma_n(A)^2$.
(vi) This follows from Proposition 4.3.7.

Structural Commentary: Condition number compares largest and smallest singular stretches.
Key Insight: A well-conditioned matrix stretches every direction by nearly the same factor. □

Corollary 4.3.23. *Positive Definite Condition Number by Eigenvalues*
Let $A \succ 0$ have eigenvalues $0 < \lambda_1 \le \cdots \le \lambda_n$. Then $\|A\|_2 = \lambda_n$; $\|A^{-1}\|_2 = \frac{1}{\lambda_1}$; and $\kappa_2(A) = \frac{\lambda_n}{\lambda_1}$.

Proof. By Theorems 4.1.55 and 4.3.11, the singular values of A are $\lambda_1, \dots, \lambda_n$.
Hence Theorem 4.3.2 gives $\|A\|_2 = \lambda_n$. Since $\operatorname{spec}(A^{-1}) = \{\lambda_1^{-1}, \dots, \lambda_n^{-1}\}$, same argument gives $\|A^{-1}\|_2 = \lambda_1^{-1}$, and multiplying yields $\kappa_2(A) = \lambda_n/\lambda_1$.
Structural Commentary: For $A \succ 0$, singular and eigenvalue stretches coincide.
Key Insight: Conditioning compares the largest and smallest positive eigendirections. □

Perspective 4.3.24. *Determinant is Not Conditioning*
Determinants measure volume, not sensitivity.
$\kappa_2(cA) = \kappa_2(A)$ for $c \neq 0$ (Proposition 4.3.22), but $\det(cA) = c^n \det(A)$.

Theorem 4.3.25. *Condition Number and Linear-System Error*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$ be invertible, let $\mathbf{y} \neq \mathbf{0}$, and let $A\mathbf{x} = \mathbf{y}$.
If $A(\Delta\mathbf{x}) = \Delta\mathbf{y}$, then $\frac{\|\Delta\mathbf{x}\|_2}{\|\mathbf{x}\|_2} \le \kappa_2(A) \frac{\|\Delta\mathbf{y}\|_2}{\|\mathbf{y}\|_2}$

Proof. From $\mathbf{y} = A\mathbf{x}$, we have $\|\mathbf{y}\|_2 \le \|A\|_2 \|\mathbf{x}\|_2$, hence $1/\|\mathbf{x}\|_2 \le \|A\|_2 / \|\mathbf{y}\|_2$. Also $\Delta\mathbf{x} = A^{-1}\Delta\mathbf{y}$, so $\|\Delta\mathbf{x}\|_2 \le \|A^{-1}\|_2 \|\Delta\mathbf{y}\|_2$. Multiplying the two estimates gives the displayed bound.
Structural Commentary: Solution error is controlled by the ratio of extreme singular stretches.
Key Insight: $\kappa_2(A)$ measures worst-case error magnification. □

Theorem 4.3.26. *Condition Number and Matrix Perturbation*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$ be invertible. Suppose $A\mathbf{x} = \mathbf{y}$; $(A + \Delta A)(\mathbf{x} + \Delta\mathbf{x}) = \mathbf{y}$, and $\mathbf{x} + \Delta\mathbf{x} \neq \mathbf{0}$.
Then $\frac{\|\Delta\mathbf{x}\|_2}{\|\mathbf{x}+\Delta\mathbf{x}\|_2} \le \kappa_2(A) \frac{\|\Delta A\|_2}{\|A\|_2}$.

Proof. Subtracting gives $A\Delta\mathbf{x} + \Delta A(\mathbf{x} + \Delta\mathbf{x}) = \mathbf{0}$, so $\Delta\mathbf{x} = -A^{-1}\Delta A(\mathbf{x} + \Delta\mathbf{x})$.
Thus $\|\Delta\mathbf{x}\|_2 \le \|A^{-1}\|_2 \|\Delta A\|_2 \|\mathbf{x} + \Delta\mathbf{x}\|_2 = \kappa_2(A) \frac{\|\Delta A\|_2}{\|A\|_2} \|\mathbf{x} + \Delta\mathbf{x}\|_2$. Divide by $\|\mathbf{x} + \Delta\mathbf{x}\|_2 \neq 0$.
Structural Commentary: Perturbing the matrix changes the operator applied to the perturbed solution.
Key Insight: The same condition number controls matrix errors and right-hand-side errors. □

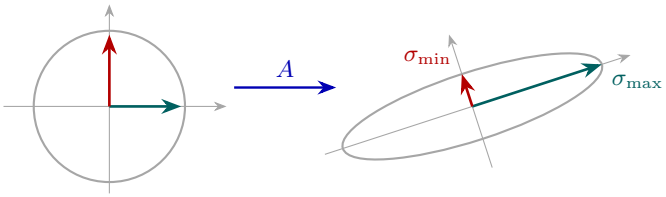


Figure 81: A maps a unit error ball to an ellipse; $\kappa_2(A) = \sigma_{\max}/\sigma_{\min}$ is the long/short semiaxis ratio.

4.4 Perturbation and Stability

Perturbation theory studies how matrix information changes under small errors. Bauer–Fike, Gersgorin, interlacing, Weyl inequalities, and pseudospectra measure robustness.

Lemma 4.4.1. *Small Spectral-Norm Perturbations of the Identity*
If $E \in \operatorname{Mat}(n, n, \mathbb{C})$ and $\|E\|_2 < 1$, then $I_n + E$ is invertible.

Proof. Theorem 4.3.11 gives $\rho(E) \le \|E\|_2 < 1$, so $-1 \notin \operatorname{spec}(E)$.
Hence $0 \notin \operatorname{spec}(I_n + E)$, and $I_n + E$ is invertible.
Structural Commentary: Spectral-norm smallness traps every eigenvalue inside the unit disk.
Key Insight: If -1 is not an eigenvalue of E, then $I + E$ has no kernel. □

Theorem 4.4.2. *Bauer–Fike Eigenvalue Perturbation Bound*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ be diagonalizable, with $A = S\Lambda S^{-1}$ and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$.
If $\Delta A \in \operatorname{Mat}(n, n, \mathbb{C})$ and $\mu \in \operatorname{spec}(A + \Delta A)$, then $\min_{1 \le j \le n} |\mu - \lambda_j| \le \kappa_2(S) \|\Delta A\|_2$.

Proof. If $\mu = \lambda_j$ for some j, the claim is immediate. Otherwise $\Lambda - \mu I_n$ is invertible.
Since μ is an eigenvalue of $A + \Delta A$, the matrix $A + \Delta A - \mu I_n$ is singular.
Conjugating gives $S^{-1}(A+\Delta A-\mu I_n)S = (\Lambda-\mu I_n)\left(I_n + (\Lambda - \mu I_n)^{-1}S^{-1}\Delta AS\right)$ so the second factor is singular.
Lemma 4.4.1 implies $1 \le \left\|(\Lambda - \mu I_n)^{-1}S^{-1}\Delta AS\right\|_2 \le \frac{\left\|S^{-1}\right\|_2\|\Delta A\|_2\|S\|_2}{\min_j|\mu-\lambda_j|}$. Rearranging gives the bound.
Structural Commentary: Diagonalize, then test how perturbation breaks shifted invertibility.
Key Insight: Eigenvalue sensitivity is controlled by how ill-conditioned the eigenvector matrix S is. □

Method 4.4.3. *Bauer–Fike Perturbation Method*
To bound eigenvalue movement for a diagonalizable matrix:

(i) diagonalize $A = S\Lambda S^{-1}$;
(ii) estimate $\|\Delta A\|_2$;
(iii) compute or bound $\kappa_2(S)$;
(iv) place every eigenvalue of $A + \Delta A$ inside the union of disks $|z - \lambda_j| \le \kappa_2(S)\,\|\Delta A\|_2$.

Perspective 4.4.4. *Pseudospectral Intuition*
For a matrix A, the ε-pseudospectrum records complex numbers z for which $A - zI$ is singular or nearly singular. Equivalently, it asks where a perturbation of size at most ε can move the spectrum. Bauer–Fike gives a clean enclosure for diagonalizable matrices, while Gersgorin and norm bounds give cruder but often cheaper enclosures. The main lesson is that eigenvalues of nonnormal matrices can be much more sensitive than their locations alone suggest.

Definition 4.4.5. *Gersgorin Disks*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$. The *deleted absolute row sum* in row k is $R_k^{\mathrm{off}}(A) := \sum_{j\neq k} |a_{kj}|$.

The *kth Gersgorin disk* of A is $\mathrm{Ger}_k(A) := \{z \in \mathbb{C} : |z - a_{kk}| \le R_k^{\mathrm{off}}(A)\}$, where $\mathrm{Ger}(A) := \bigcup_{k=1}^{n} \mathrm{Ger}_k(A)$.

Theorem 4.4.6. *Gersgorin Disk Theorem*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$. *Then* $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A) = \bigcup_{k=1}^{n} \mathrm{Ger}_k(A)$.

Proof. Let $\lambda \in \mathrm{spec}(A)$ and choose an eigenvector $\mathbf{x} = [x_1 \ \cdots \ x_n]^t$.
Pick k such that $|x_k| = \max_j |x_j|$; then $x_k \neq 0$. The kth row of $A\mathbf{x} = \lambda\mathbf{x}$ gives $(\lambda - a_{kk})x_k = \sum_{j\neq k} a_{kj}x_j$.
Divide by x_k and use $|x_j/x_k| \le 1$: $|\lambda - a_{kk}| = \left|\sum_{j\neq k} a_{kj}\frac{x_j}{x_k}\right| \le \sum_{j\neq k} |a_{kj}|\left|\frac{x_j}{x_k}\right| \le R_k^{\mathrm{off}}(A)$.
Thus $\lambda \in \mathrm{Ger}_k(A) \subseteq \mathrm{Ger}(A)$.
Structural Commentary: Normalize by a largest component; all row ratios have modulus at most one.
Key Insight: Row k traps λ in the disk centered at a_{kk} with radius $R_k^{\mathrm{off}}(A)$. □

Method 4.4.7. *Gersgorin Localisation Method*
For $A = [a_{ij}]$, Gersgorin localisation gives a disk enclosure for the spectrum.

(i) For each row k, place a disk at the diagonal entry a_{kk} with radius $R_k^{\mathrm{off}}(A) = \sum_{j\neq k} |a_{kj}|$.
The row enclosure is $\mathrm{Ger}(A) = \bigcup_{k=1}^{n} \mathrm{Ger}_k(A)$.
(ii) Theorem 4.4.6 guarantees that every eigenvalue of A lies in $\mathrm{Ger}(A)$.
(iii) For a sharper enclosure, repeat the same construction for A^t. The centers are still a_{kk}, while the radii becomes deleted column sums of A. Then use $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A) \cap \mathrm{Ger}(A^t)$. This is the intersection of two unions of disks; an eigenvalue need not lie in the overlap of each individual row-column disk pair.

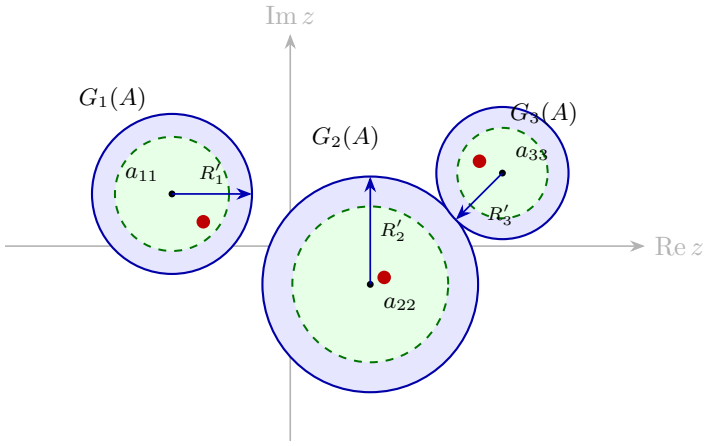


Figure 82: Blue row disks contain the spectrum; green dashed column disks give the sharper row-column enclosure $\mathrm{Ger}(A) \cap \mathrm{Ger}(A^t)$. Red points illustrate possible eigenvalue locations.

Corollary 4.4.8. *Row-Column Gersgorin Refinement*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. *Then* $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A) \cap \mathrm{Ger}(A^t)$.
If $R_k(A) = \sum_{j=1}^{n} |a_{kj}|$ *and* $R_{\max}(A) = \max_{1 \le k \le n} R_k(A)$, *then* $\mathrm{spec}(A) \subseteq \{z \in \mathbb{C} : |z| \le \min(R_{\max}(A), R_{\max}(A^t))\}$.

Proof. Theorem 4.4.6 gives $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A)$.
Proposition 4.1.26 gives $\mathrm{spec}(A) = \mathrm{spec}(A^t)$, so applying Gersgorin to A^t gives $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A^t)$.
Also $\mathrm{Ger}_k(A) \subseteq \{z : |z| \le |a_{kk}| + R_k^{\mathrm{off}}(A) = R_k(A)\}$, and similarly for A^t.
Structural Commentary: Row disks and column disks are both valid because A and A^t have the same spectrum.
Key Insight: Intersecting independent enclosures usually sharpens the eigenvalue search region. □

Definition 4.4.9. *Diagonal Dominance*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$. Matrix A is *diagonally dominant* if $|a_{kk}| \ge R_k^{\mathrm{off}}(A)$ for every k, and *strictly diagonally dominant* if $|a_{kk}| > R_k^{\mathrm{off}}(A)$ for every k.

Corollary 4.4.10. *Strict Diagonal Dominance Implies Invertibility*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$.

(i) If A is strictly diagonally dominant, then A is invertible.
(ii) If A has real spectrum and nonnegative diagonal entries, then diagonal dominance gives $\mathrm{spec}(A) \subseteq [0, \infty)$*; strict diagonal dominance gives* $\mathrm{spec}(A) \subseteq (0, \infty)$.

Proof.

(i) Strict dominance gives $0 \notin \mathrm{Ger}_k(A)$ for every k, since $|a_{kk}| > R_k^{\mathrm{off}}(A)$.
Thus $0 \notin \mathrm{spec}(A)$ by Gersgorin, so A is invertible by Corollary 3.1.9.
(ii) If $a_{kk} \ge 0$ and $a_{kk} \ge R_k^{\mathrm{off}}(A)$, then each $\mathrm{Ger}_k(A)$ lies in the closed right half-plane. In the strict case, each disk lies in the open right half-plane. Gersgorin and the real-spectrum hypothesis give the claim.

Structural Commentary: Gersgorin converts diagonal dominance into disk placement.
Key Insight: The disks control whether eigenvalues can reach 0 or the left half-plane. □

Corollary 4.4.11. *Diagonal Dominance and Positive Semidefiniteness*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$ *be Hermitian with nonnegative diagonal entries.*

(i) If A is diagonally dominant, then A is positive semidefinite.
(ii) If A is diagonally dominant and invertible, then A is positive definite.
(iii) If A is strictly diagonally dominant, then A is positive definite.

Proof. Hermitian matrices have real spectrum.

(i) Corollary 4.4.10 gives $\mathrm{spec}(A) \subseteq [0, \infty)$; hence $A \succeq 0$ by Theorem 4.1.53.
(ii) If A is invertible, then $0 \notin \mathrm{spec}(A)$, so every eigenvalue is positive; hence $A \succ 0$.
(iii) Strict diagonal dominance implies invertibility by Corollary 4.4.10; apply (ii).

Structural Commentary: Gersgorin traps the real spectrum in $[0, \infty)$; invertibility removes 0.
Key Insight: For Hermitian matrices, positivity is spectral location. □

Theorem 4.4.12. *Weak Diagonal Dominance Criterion*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$ be diagonally dominant with no zero entries.
If $|a_{kk}| > R_k^{\mathrm{off}}(A)$ for some k, then A is invertible.

Proof. Assume $A\mathbf{x} = 0$ with $\mathbf{x} \neq \mathbf{0}$, and choose k with $|x_k| = \|\mathbf{x}\|_\infty$.
Row k gives $|a_{kk}| \|\mathbf{x}\|_\infty = \left| -\sum_{j \neq k} a_{kj} x_j \right| \leq \sum_{j \neq k} |a_{kj}| |x_j| \leq R_k^{\mathrm{off}}(A) \|\mathbf{x}\|_\infty \leq |a_{kk}| \|\mathbf{x}\|_\infty$.
Hence all inequalities are equalities. Since no a_{kj} vanishes, $|x_j| = \|\mathbf{x}\|_\infty$ for every j. Applying the same chain to any row i gives $|a_{ii}| = R_i'(A)$ for every i, contradicting the strict row. Thus $\ker(A) = \{\mathbf{0}\}$, so A is invertible.
Structural Commentary: A zero vector in the kernel would force equality in every Gersgorin row.
Key Insight: One strict row breaks that equality pattern. □

Definition 4.4.13. *Increasing Hermitian Eigenvalue Order*
For a Hermitian matrix $A \in \mathrm{Mat}(n, n, \mathbb{C})$, write $\lambda_1(A) \leq \lambda_2(A) \leq \cdots \leq \lambda_n(A)$ for its eigenvalues counted with algebraic multiplicity. If only one Hermitian matrix is under discussion, we write λ_i instead of $\lambda_i(A)$.

Theorem 4.4.14. *Rayleigh Quotient on a Spectral Interval*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian with increasing eigenvalues $\lambda_1 \leq \cdots \leq \lambda_n$ and corresponding orthonormal eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$. For a unit vector, write $\mathrm{Ray}_A(\mathbf{x}) := \mathbf{x}^* A \mathbf{x}$. If $1 \leq p \leq q \leq n$ and $\mathbf{x} \in \mathrm{span}(\mathbf{u}_p, \ldots, \mathbf{u}_q)$ is a unit vector, then $\lambda_p \leq \mathrm{Ray}_A(\mathbf{x}) \leq \lambda_q$.

Proof. Write $\mathbf{x} = \sum_{i=p}^{q} c_i \mathbf{u}_i$. Orthonormality gives $\sum_{i=p}^{q} |c_i|^2 = 1$, and $\mathbf{x}^* A \mathbf{x} = \sum_{i=p}^{q} \lambda_i |c_i|^2$.
This is a convex combination of $\lambda_p, \ldots, \lambda_q$, so it lies between the smallest and largest of them.
Structural Commentary: In an eigenbasis, the Rayleigh quotient is an eigenvalue average.
Key Insight: Spectral coordinates turn a quadratic form into weighted scalar data. □

Corollary 4.4.15. *Rayleigh Extrema and Diagonal Entry Bounds*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian.

(i) If $\mathcal{U}_{p,q} = \mathrm{span}(\mathbf{u}_p, \ldots, \mathbf{u}_q)$, then $\min_{\substack{\mathbf{x} \in \mathcal{U}_{p,q} \\ \|\mathbf{x}\|_2 = 1}} \mathbf{x}^* A \mathbf{x} = \lambda_p(A)$, and $\max_{\substack{\mathbf{x} \in \mathcal{U}_{p,q} \\ \|\mathbf{x}\|_2 = 1}} \mathbf{x}^* A \mathbf{x} = \lambda_q(A)$.
(ii) For every unit vector $\mathbf{x}$, $\lambda_1(A) \leq \mathbf{x}^* A \mathbf{x} \leq \lambda_n(A)$.
(iii) If $A = [a_{ij}]$, then $\lambda_1(A) \leq a_{ii} \leq \lambda_n(A)$ for $i = 1, \ldots, n$.

Proof.
(i) Theorem 4.4.14 gives the inequalities on $\mathcal{U}_{p,q}$, with equality at $\mathbf{u}_p$ and $\mathbf{u}_q$.
(ii) This is (i) with $p = 1$ and $q = n$.
(iii) Apply (ii) to $\mathbf{x} = \mathbf{e}_i$, since $\mathbf{e}_i^* A \mathbf{e}_i = a_{ii}$.
Structural Commentary: Rayleigh extrema on extreme eigendirections; diagonal entries are coordinate quotients.
Key Insight: The whole numerical range of a Hermitian matrix sits inside its spectral interval. □

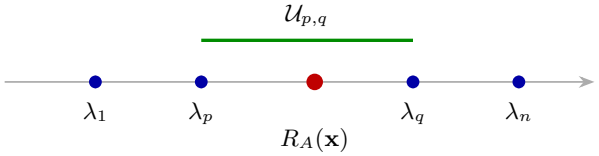


Figure 83: For $\mathbf{x} \in \mathcal{U}_{p,q}$, $\mathrm{Ray}_A(\mathbf{x})$ is a convex combination of $\lambda_p, \ldots, \lambda_q$.

Theorem 4.4.16. *Rayleigh Minimax Principle*
Let $A = A^* \in \mathrm{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1 \leq \cdots \leq \lambda_n$, and set $R_A(\mathbf{x}) = \mathbf{x}^* A \mathbf{x} / (\mathbf{x}^* \mathbf{x})$ for $\mathbf{x} \neq \mathbf{0}$.
For $1 \leq k \leq n$, $\lambda_k = \min_{\substack{\mathcal{S} \subseteq \mathbb{C}^n \\ \dim \mathcal{S} = k}} \max_{\mathbf{0} \neq \mathbf{x} \in \mathcal{S}} R_A(\mathbf{x}) = \max_{\substack{\mathcal{S} \subseteq \mathbb{C}^n \\ \dim \mathcal{S} = n-k+1}} \min_{\mathbf{0} \neq \mathbf{x} \in \mathcal{S}} R_A(\mathbf{x})$.

Proof. Let $\mathbf{u}_1, \ldots, \mathbf{u}_n$ be an orthonormal eigenbasis in increasing order.
Set $E_k = \mathrm{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$; $F_k = \mathrm{span}(\mathbf{u}_k, \ldots, \mathbf{u}_n)$.
By Theorem 4.4.14, $\max_{\mathbf{0} \neq \mathbf{x} \in E_k} R_A(\mathbf{x}) = \lambda_k$ and $\min_{\mathbf{0} \neq \mathbf{x} \in F_k} R_A(\mathbf{x}) = \lambda_k$.
If $\dim \mathcal{S} = k$, then $\mathcal{S} \cap F_k \neq \{\mathbf{0}\}$, so $\max_{\mathbf{0} \neq \mathbf{x} \in \mathcal{S}} R_A(\mathbf{x}) \geq \lambda_k$; equality is attained by E_k.
If $\dim \mathcal{T} = n - k + 1$, then $\mathcal{T} \cap E_k \neq \{\mathbf{0}\}$, so $\min_{\mathbf{0} \neq \mathbf{x} \in \mathcal{T}} R_A(\mathbf{x}) \leq \lambda_k$; equality is attained by F_k.
Structural Commentary: Dimension forces intersections with spectral flags.
Key Insight: Minimax is Rayleigh's quotient plus dimension counting. □

Method 4.4.17. *Rayleigh Eigenbasis Method*
To bound a Hermitian quadratic form:

(i) diagonalise A with an orthonormal eigenbasis;
(ii) expand the unit vector in that basis;
(iii) rewrite $\mathbf{x}^*A\mathbf{x}$ as $\sum_i \lambda_i |c_i|^2$;
(iv) use the eigenvalue range allowed by the support of $\mathbf{x}$.

Theorem 4.4.18. *Rank-One Positive Perturbation Interlacing*
Let $A, E \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian, with $E \succeq 0$ and $\mathrm{rank}(E) \le 1$.
Then $\lambda_i(A) \le \lambda_i(A+E) \le \lambda_{i+1}(A) \quad (1 \le i < n)$, $\lambda_n(A) \le \lambda_n(A+E)$
Equivalently, $\lambda_i(A-E) \le \lambda_i(A) \le \lambda_{i+1}(A-E) \quad (1 \le i < n)$, $\lambda_n(A-E) \le \lambda_n(A)$.

Proof. Let $\mathbf{u}_j$ and $\mathbf{v}_j$ be orthonormal eigenvectors of A and $A+E$, respectively, in increasing eigenvalue order. For the lower bound, set $\mathcal{U} = \mathrm{span}(\mathbf{u}_i, \dots, \mathbf{u}_n)$, $\mathcal{V} = \mathrm{span}(\mathbf{v}_1, \dots, \mathbf{v}_i)$. Since $\dim\mathcal{U} + \dim\mathcal{V} = n+1$, choose a unit $\mathbf{x} \in \mathcal{U} \cap \mathcal{V}$. Then $\lambda_i(A) \le \mathbf{x}^*A\mathbf{x} \le \mathbf{x}^*(A+E)\mathbf{x} \le \lambda_i(A+E)$ by Theorem 4.4.14 and $E \succeq 0$.
For the upper bound, set $\mathcal{U} = \mathrm{span}(\mathbf{u}_1, \dots, \mathbf{u}_{i+1})$, $\mathcal{V} = \mathrm{span}(\mathbf{v}_i, \dots, \mathbf{v}_n)$, $\mathcal{W} = \ker(E)$. Here $\dim(\mathcal{U} \cap \mathcal{V}) \ge 2$ and $\dim\mathcal{W} \ge n-1$, so choose a unit $\mathbf{x} \in \mathcal{U} \cap \mathcal{V} \cap \mathcal{W}$. Thus $\lambda_i(A+E) \le \mathbf{x}^*(A+E)\mathbf{x} = \mathbf{x}^*A\mathbf{x} \le \lambda_{i+1}(A)$.
The statements for $A-E$ follow by applying the proved result to $A-E$ and E.
Structural Commentary: Interlacing comes from test vectors in overlapping spectral subspaces.
Key Insight: Rank one leaves a large kernel, so the upper-bound vector can avoid the perturbation. □

Corollary 4.4.19. *Rank-Two Indefinite Interlacing*
Let $A, H \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian, $n \ge 2$. Suppose $\mathrm{rank}(H) \le 2$ and H has at most one positive and one negative eigenvalue. Then

$$\lambda_1(A+H) \le \lambda_2(A), \qquad \lambda_{i-1}(A) \le \lambda_i(A+H) \le \lambda_{i+1}(A) \quad (2 \le i < n), \qquad \lambda_{n-1}(A) \le \lambda_n(A+H).$$

Proof. Write $H = E_+ - E_-$ with $E_\pm \succeq 0$ and $\mathrm{rank}(E_\pm) \le 1$. Set $B = A - E_-$; then $A = B + E_-$ and $A+H = B+E_+$. For $2 \le i < n$, rank-one interlacing gives $\lambda_{i-1}(A) \le \lambda_i(B) \le \lambda_i(A+H) \le \lambda_{i+1}(B) \le \lambda_{i+1}(A)$.
The endpoints follow similarly: $\lambda_1(A+H) \le \lambda_2(B) \le \lambda_2(A)$, and $\lambda_{n-1}(A) \le \lambda_n(B) \le \lambda_n(A+H)$.
Structural Commentary: Such an H is a difference of two rank-one PSD perturbations.
Key Insight: Interlace once with E_-, then once with E_+. □

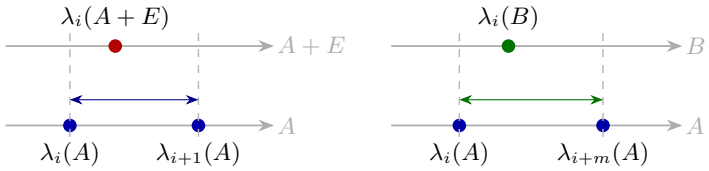


Figure 84: Rank-one and principal-submatrix interlacing place new eigenvalues between old ones.

Theorem 4.4.20. *Cauchy Interlacing: One Principal Deletion*
Let $A = \begin{bmatrix} B & \mathbf{y} \\ \mathbf{y}^* & c \end{bmatrix} \in \mathrm{Mat}(n+1, n+1, \mathbb{C})$ be Hermitian, with $B \in \mathrm{Mat}(n, n, \mathbb{C})$.
Then, for $1 \le i \le n$, $\lambda_i(A) \le \lambda_i(B) \le \lambda_{i+1}(A)$.

Proof. Let $\mathbf{u}_j$ and $\mathbf{v}_j$ be orthonormal eigenvectors of A and B, respectively, in increasing eigenvalue order. Identify $\mathbb{C}^n$ with $\{(\mathbf{z}, 0)\} \subseteq \mathbb{C}^{n+1}$, so $(\mathbf{z},0)^*A(\mathbf{z},0) = \mathbf{z}^*B\mathbf{z}$.
For the lower bound, set $\mathcal{S} = \mathrm{span}((\mathbf{v}_1,0), \dots, (\mathbf{v}_i,0))$ and $\mathcal{T} = \mathrm{span}(\mathbf{u}_i, \dots, \mathbf{u}_{n+1})$. Their dimensions are i and $n+2-i$, so $\dim\mathcal{S} + \dim\mathcal{T} > n+1$ and $\mathcal{S} \cap \mathcal{T}$ contains a nonzero vector.
Choose a unit $\mathbf{x} = (\mathbf{z}, 0)$ in the intersection. The spectral expansion in the $\mathbf{u}_j$ gives $\mathbf{x}^*A\mathbf{x} \ge \lambda_i(A)$, while the expansion in the $\mathbf{v}_j$ gives $\mathbf{z}^*B\mathbf{z} \le \lambda_i(B)$. Hence $\lambda_i(A) \le \lambda_i(B)$.
For the upper bound, take $\mathcal{S}' = \mathrm{span}((\mathbf{v}_i,0), \dots, (\mathbf{v}_n,0))$ and $\mathcal{T}' = \mathrm{span}(\mathbf{u}_1, \dots, \mathbf{u}_{i+1})$. Again the dimensions sum to $n+2$, so choose a unit $\mathbf{x} = (\mathbf{z},0) \in \mathcal{S}' \cap \mathcal{T}'$.
Now $\mathbf{z}^*B\mathbf{z} \ge \lambda_i(B)$ and $\mathbf{x}^*A\mathbf{x} \le \lambda_{i+1}(A)$, giving $\lambda_i(B) \le \lambda_{i+1}(A)$.
Structural Commentary: A principal submatrix is a coordinate slice of the larger quadratic form.
Key Insight: Deleting one coordinate costs one dimension, so eigenvalues shift by at most one index. □

Corollary 4.4.21. *Singular Values Interlace After Deleting a Row or Column*
Let B be obtained from $A \in \mathrm{Mat}(n, n, \mathbb{C})$ by deleting one row or column.
With singular values ordered decreasingly, $\sigma_i(A) \ge \sigma_i(B) \ge \sigma_{i+1}(A) \quad (1 \le i < n)$.

Proof. For a row deletion, permute rows so that $PA = \begin{bmatrix} B \\ \mathbf{y}^* \end{bmatrix}$.
Then PA has the same singular values as A, and BB^* is a principal submatrix of $(PA)(PA)^*$.
Cauchy interlacing applied to these Gram matrices gives the result after taking square roots and reversing the increasing eigenvalue order. A column deletion follows by applying the row case to A^*.
Structural Commentary: Row deletion makes BB^* a principal submatrix of a permuted AA^*.
Key Insight: Singular-value interlacing is Cauchy interlacing applied to Gram matrices. □

Corollary 4.4.22. *Cauchy Interlacing for Principal Submatrices*
Let $B \in \mathrm{Mat}(n, n, \mathbb{C})$ be a principal submatrix of Hermitian $A \in \mathrm{Mat}(n+m, n+m, \mathbb{C})$.
Then, for $1 \le i \le n$, $\lambda_i(A) \le \lambda_i(B) \le \lambda_{i+m}(A)$.

Proof. After a permutation similarity, B is leading.
Delete the other m coordinates one at a time and apply Theorem 4.4.20 at each step.
Structural Commentary: Principal submatrices come from repeated coordinate deletion.
Key Insight: Each deletion shifts eigenvalue indices by at most one. □

Theorem 4.4.23. *Schur Majorization of the Diagonal*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian, with increasing eigenvalues $\lambda_i(A)$ and increasing diagonal entries d_i.
Then $\sum_{i=1}^{k} \lambda_i(A) \le \sum_{i=1}^{k} d_i$, $(1 \le k \le n)$ with equality for $k = n$.

Proof. Permute coordinates so that $a_{ii} = d_i$, and let A_k be the leading $k \times k$ principal submatrix.
Cauchy interlacing gives $\lambda_i(A) \le \lambda_i(A_k)$ for $1 \le i \le k$, hence $\sum_{i=1}^{k} \lambda_i(A) \le \sum_{i=1}^{k} \lambda_i(A_k) = \mathrm{tr}(A_k) = \sum_{i=1}^{k} d_i$.
For $k = n$, this is $\mathrm{tr}(A) = \mathrm{tr}(A)$.
Structural Commentary: Ordered diagonal sums are traces of leading principal blocks.
Key Insight: Diagonal majorization is Cauchy interlacing plus trace. □

Method 4.4.24. *Eigenvalue Interlacing Method*
To prove or use an interlacing estimate:

(i) order Hermitian eigenvalues increasingly;
(ii) identify dimension lost: rank-one perturbation, one deleted coordinate, m-coordinate principal submatrix;
(iii) choose test vectors by intersecting spectral subspaces with the unaffected subspace;
(iv) convert the Rayleigh quotient bounds into eigenvalue inequalities.

Theorem 4.4.25. *Weyl Eigenvalue Inequalities*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian. For $1 \le i \le n$,

$$\lambda_i(A+B) \le \lambda_{i+j}(A) + \lambda_{n-j}(B), \qquad 0 \le j \le n-i,$$
$$\lambda_{i-j+1}(A) + \lambda_j(B) \le \lambda_i(A+B), \qquad 1 \le j \le i.$$

Proof. Let $\mathbf{u}_k, \mathbf{v}_k, \mathbf{w}_k$ be increasing eigenbases for $A, A+B, B$.
Fix i and $0 \le j \le n-i$, and set $\mathcal{U} = \mathrm{span}(\mathbf{u}_1, \dots, \mathbf{u}_{i+j})$, $\mathcal{V} = \mathrm{span}(\mathbf{v}_i, \dots, \mathbf{v}_n)$, $\mathcal{W} = \mathrm{span}(\mathbf{w}_1, \dots, \mathbf{w}_{n-j})$.
Their dimensions sum to $2n+1$, so choose a unit $\mathbf{x} \in \mathcal{U} \cap \mathcal{V} \cap \mathcal{W}$.
Then $\lambda_i(A+B) \le \mathbf{x}^*(A+B)\mathbf{x} = \mathbf{x}^*A\mathbf{x} + \mathbf{x}^*B\mathbf{x} \le \lambda_{i+j}(A) + \lambda_{n-j}(B)$.
The lower bound follows by applying this upper bound to $-A$ and $-B$ and using $\lambda_k(-C) = -\lambda_{n-k+1}(C)$.
Structural Commentary: Weyl bounds come from intersecting the spectral flags of A, B, and $A+B$.
Key Insight: One shared test vector controls all three Rayleigh quotients. □

Method 4.4.26. *Weyl Bound Method*
To bound eigenvalues of $A + B$:

(i) choose the target index i;
(ii) use the upper Weyl family with $j = 0, \dots, n-i$ to get upper bounds;
(iii) use the lower Weyl family with $j = 1, \dots, i$ to get lower bounds;
(iv) for positive semidefinite or low-rank B, insert the known zero or nonnegative eigenvalues to recover monotonicity or interlacing.

Theorem 4.4.27. *Brauer Rank-One Eigenvalue Shift*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$, let $A\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$, and let $\mathbf{y} \in \mathbb{C}^n$. If $\lambda, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A with multiplicity, then the eigenvalues of $A + \mathbf{x}\mathbf{y}^*$ are $\lambda + \mathbf{y}^*\mathbf{x}, \lambda_2, \dots, \lambda_n$.
Moreover, $\mathbf{x}$ is an eigenvector of $A + \mathbf{x}\mathbf{y}^*$ corresponding to $\lambda + \mathbf{y}^*\mathbf{x}$.

Proof. Put $\mathbf{u} = \mathbf{x}/\|\mathbf{x}\|_2$. By Schur triangularisation, choose a unitary $U = [\mathbf{u}\ U_2]$ such that $U^*AU = \begin{bmatrix} \lambda & * \\ 0 & T' \end{bmatrix}$, where the diagonal entries of T' are $\lambda_2, \dots, \lambda_n$. Also $U^*(\mathbf{x}\mathbf{y}^*)U = \begin{bmatrix} \mathbf{y}^*\mathbf{x} & * \\ 0 & 0 \end{bmatrix}$. Therefore $U^*(A + \mathbf{x}\mathbf{y}^*)U$ is upper triangular with diagonal entries $\lambda + \mathbf{y}^*\mathbf{x}, \lambda_2, \dots, \lambda_n$. Finally, $(A + \mathbf{x}\mathbf{y}^*)\mathbf{x} = A\mathbf{x} + \mathbf{x}(\mathbf{y}^*\mathbf{x}) = (\lambda + \mathbf{y}^*\mathbf{x})\mathbf{x}$.
Structural Commentary: A rank-one eigenvector update changes only the first Schur diagonal entry.
Key Insight: Align the update with the first coordinate; only one diagonal term shifts. □

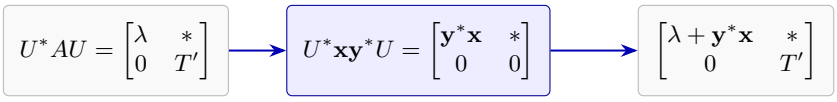


Figure 85: Brauer's update shifts one Schur diagonal entry.

Corollary 4.4.28. *Damped Rank-One Eigenvalue Reset*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have eigenpair $(\lambda, \mathbf{x})$, let $\mathbf{y}^*\mathbf{x} = 1$, and let $\tau \in \mathbb{C}$. If $\lambda, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A, then $A_\tau := \tau A + (1 - \tau)\lambda\mathbf{x}\mathbf{y}^*$ has eigenvalues $\lambda, \tau\lambda_2, \dots, \tau\lambda_n$.

Proof. The eigenvalues of τA are $\tau\lambda, \tau\lambda_2, \dots, \tau\lambda_n$. Since $A_\tau = \tau A + \mathbf{x}((1 - \overline{\tau})\overline{\lambda}\mathbf{y})^*$, Theorem 4.4.27 shifts only $\tau\lambda$ by $(1 - \tau)\lambda\mathbf{y}^*\mathbf{x} = (1 - \tau)\lambda$. Thus $\tau\lambda$ becomes λ, while the other eigenvalues stay $\tau\lambda_i$.
Structural Commentary: Damping scales the spectrum; Brauer's update restores one eigenvalue.
Key Insight: Brauer's theorem changes one spectral coordinate without moving the rest. □

4.5 Numerical Linear Algebra

Numerical linear algebra turns matrix structure into algorithms for direct solves, least squares, eigenvalues, SVD, structured transforms, Krylov methods, iteration, and preconditioning.

Perspective 4.5.1. *Solve with Factors Before Forming Inverses*
For $A\mathbf{x} = \mathbf{b}$, use available factors such as LU, QR, or Cholesky to solve triangular or orthogonal subproblems. This avoids forming A^{-1} and preserves sparse or triangular structure.

Perspective 4.5.2. *Iterative Refinement by Residual Correction*
Given an approximate solution $\mathbf{x}_k$ of $A\mathbf{x} = \mathbf{b}$, set $\mathbf{r}_k = \mathbf{b} - A\mathbf{x}_k$.
Solve $A\mathbf{d}_k = \mathbf{r}_k$ with the same factors or preconditioner, then update $\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{d}_k$.
Exact arithmetic corrects in one step; floating point helps when $\mathbf{r}_k$ is accurate and conditioning is moderate.

Method 4.5.3. *Augmented Inverse Computation Preview*
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$:

(i) Form $[\,A\ I_n\,]$.
(ii) Apply Gauss–Jordan reduction to the left block.
(iii) If $[\,A\ I_n\,] \sim [\,I_n\ B\,]$, then $B = A^{-1}$.
(iv) Otherwise A is not invertible, by Corollary 1.3.64.

Proposition 4.5.4. *QR Route for Least Squares*
Let $A = QR \in \mathrm{Mat}(m, n, \mathbb{F})$ be a full-column-rank narrow QR factorisation. For $\mathbf{y} \in \mathbb{F}^m$, the least-squares solution solves $R\mathbf{x} = Q^*\mathbf{y}$. Moreover $P_{\mathrm{col}(A)} = QQ^*$.

Proof. The normal equations reduce by $A = QR$ and $Q^*Q = I_n$ to $R^*R\mathbf{x} = R^*Q^*\mathbf{y}$. Since R is invertible, this is equivalent to $R\mathbf{x} = Q^*\mathbf{y}$. Also $\mathrm{col}(A) = \mathrm{col}(Q)$, and the columns of Q are orthonormal, so $P_{\mathrm{col}(A)} = QQ^*$.
Structural Commentary: QR replaces $\mathrm{col}(A)$ by an orthonormal basis Q.
Key Insight: Project onto Q, then solve triangularly. □

Perspective 4.5.5. *Why Normal Equations Can Be Numerically Fragile*
For full-column-rank least squares, solving $A^*A\mathbf{x} = A^*\mathbf{y}$ squares the spectral condition number because $\kappa_2(A^*A) = \kappa_2(A)^2$. QR-based methods avoid this squaring and therefore preserve conditioning better.

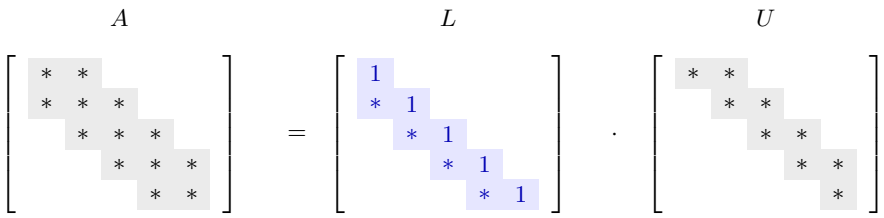


Figure 86: A tridiagonal matrix can factor as bidiagonal L times bidiagonal U, retaining local band structure.

Definition 4.5.6. *Band and Tridiagonal Matrices*
For $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$ and $1 \leq w \leq n$, say that A has *half-bandwidth* w if $a_{ij} = 0$ whenever $|i - j| \geq w$. Thus $w = 1$ gives a diagonal matrix, while $w = 2$ gives a *tridiagonal* matrix.

Method 4.5.7. *Banded Elimination Preview*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ have half-bandwidth w.

(i) Eliminate without row exchanges that leave the band.
(ii) At pivot k, update only entries with $|i - k| < w$ and $|j - k| < w$.
(iii) Obtain $A = LU$ with L and U of comparable bandwidth.

Cost: $O(nw^2)$ operations; for tridiagonal matrices ($w = 2$), $O(n)$.

Perspective 4.5.8. *Five-Point Stencils Are Banded but Less Local*
The 1D second-difference pattern $-1, 2, -1$ is tridiagonal; the row-ordered $N \times N$ five-point stencil is still sparse, but vertical neighbors create diagonals offset by about N. Thus bandwidth grows with grid width, and incomplete LU keeps a prescribed sparse pattern for a cheap approximate solve or preconditioner.

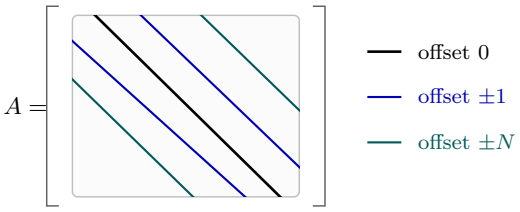


Figure 87: Row ordering gives matrix bands at offsets $0, \pm 1, \pm N$.

Definition 4.5.9. *Upper Hessenberg Matrix*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$. Then A is *upper Hessenberg* if $a_{ij} = 0$ whenever $i > j + 1$.
Equivalently, all entries below the first subdiagonal are zero.

Numerical linear algebra also replaces a matrix by a unitarily similar one with many forced zero entries. Upper Hessenberg form is the standard preliminary reduction before QR-type eigenvalue algorithms.

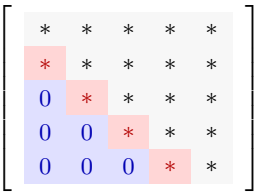


Figure 88: Upper Hessenberg pattern: red is the first subdiagonal; blue entries below it are forced zero.

Theorem 4.5.10. *Unitary Upper Hessenberg Reduction*
Every $A \in \mathrm{Mat}(n, n, \mathbb{F})$ is unitarily similar to upper Hessenberg matrix with real nonnegative subdiagonal entries.

Proof. Set $A_0 = A$. For $k = 1, \ldots, n - 2$, suppose columns $1, \ldots, k - 1$ of A_{k-1} already have zeros below the first subdiagonal. Let $\mathbf{w}$ be the tail of column k in rows $k + 1, \ldots, n$.
By Corollary 2.4.29, choose a unitary V_k with $V_k^* \mathbf{w} = \alpha \mathbf{e}_1$, $\alpha \geq 0$.
Put $U_k = I_k \oplus V_k$ and $A_k = U_k^* A_{k-1} U_k$. Right multiplication by U_k changes only columns $k+1, \ldots, n$, so columns $1, \ldots, k-1$ keep their existing Hessenberg zeros. Left multiplication by U_k^* acts only on rows $k+1, \ldots, n$; those rows are already zero in columns $1, \ldots, k - 1$, so the zeros remain. In column k, the tail becomes $V_k^* \mathbf{w} = \alpha \mathbf{e}_1$, so entries below row $k + 1$ are zero and the new subdiagonal entry is real nonnegative.

After $n-2$ steps, $A_{n-2} = U^*AU$ is upper Hessenberg, where $U = U_1 \cdots U_{n-2}$ is unitary.
Structural Commentary: Each trailing unitary creates one Hessenberg column and preserves earlier ones.
Key Insight: Kill one column tail at a time. □

Perspective 4.5.11. *Why Hessenberg Form Matters*
Hessenberg form preserves spectral data by unitary similarity, making QR-type eigenvalue algorithms cheaper.

Method 4.5.12. *QR Eigenvalue Algorithm Preview*
For a square matrix A_0, the basic QR eigenvalue iteration is:

(i) In practice, first reduce A_0 to Hessenberg form by Theorem 4.5.10.
(ii) Factor $A_k = Q_k R_k$, with Q_k unitary and R_k upper triangular.
(iii) Set $A_{k+1} = R_k Q_k = Q_k^* A_k Q_k$, using shifts to speed convergence.

Each step is a unitary similarity, so the spectrum is preserved.

Method 4.5.13. *Hessenberg QR Step by Plane Rotations*
Let $H \in \mathrm{Mat}(n, n, \mathbb{C})$ be upper Hessenberg and let $\mu \in \mathbb{C}$.

(i) Set $B_0 = H - \mu I_n$.
(ii) For $j = 1, \ldots, n-1$, choose a unitary plane rotation G_j acting on coordinates $j, j+1$, so that $(G_j^* B_{j-1})_{j+1,j} = 0$ and $B_j = G_j^* B_{j-1}$.
(iii) Put $Q = G_1 \cdots G_{n-1}$ and $R = B_{n-1}$; then $H - \mu I_n = QR$.
(iv) Define the shifted QR update $H_+ = RQ + \mu I_n = Q^* H Q$

Adjacent rotations implement the Hessenberg step by local bulge chasing.

Theorem 4.5.14. *Power Method Dominant-Eigenvector Preview*
Let $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$, where $\mathbf{v}_1, \ldots, \mathbf{v}_n$ is a basis of $\mathbb{C}^n$, and assume $|\lambda_1| > \max_{j \geq 2} |\lambda_j|$.
If $\mathbf{u}_0 = \sum_{j=1}^{n} c_j \mathbf{v}_j$ with $c_1 \neq 0$, then $\lambda_1^{-k} A^k \mathbf{u}_0 \longrightarrow c_1 \mathbf{v}_1$. Thus the directions of $A^k \mathbf{u}_0$ converge to $\mathrm{span}\{\mathbf{v}_1\}$.

Proof. Set $\rho = \max_{j \geq 2} |\lambda_j / \lambda_1| < 1$. Since $A^k \mathbf{v}_j = \lambda_j^k \mathbf{v}_j$, $\lambda_1^{-k} A^k \mathbf{u}_0 = c_1 \mathbf{v}_1 + \sum_{j=2}^{n} c_j \left(\frac{\lambda_j}{\lambda_1}\right)^k \mathbf{v}_j$.

The remainder has norm at most $\rho^k \sum_{j=2}^{n} |c_j| \, \|\mathbf{v}_j\|$, hence tends to 0.
Structural Commentary: The power method is spectral filtering by repeated multiplication.
Key Insight: A spectral gap makes every non-dominant component decay relative to the dominant one. □

Method 4.5.15. *Inverse and Shifted Power Iteration Preview*
Let $\mathcal{N}(\mathbf{z}) := \mathbf{z} / \|\mathbf{z}\|_2$ for $\mathbf{z} \neq \mathbf{0}$.

(i) *Inverse*: if $0 \notin \mathrm{spec}(A)$, solve $A\mathbf{y}_{k+1} = \mathbf{x}_k$ and set $\mathbf{x}_{k+1} = \mathcal{N}(\mathbf{y}_{k+1})$. This is power iteration on A^{-1} and targets minimal $|\lambda|$.
(ii) *Shift-invert*: for $\sigma \notin \mathrm{spec}(A)$, $(A - \sigma I)\mathbf{y}_{k+1} = \mathbf{x}_k$, $\mathbf{x}_{k+1} = \mathcal{N}(\mathbf{y}_{k+1})$. Eigenvalues become $(\lambda - \sigma)^{-1}$, targeting λ nearest σ.
(iii) *Shifted power*: $\mathbf{x}_{k+1} = \mathcal{N}\big((A - \sigma I)\mathbf{x}_k\big)$; eigenvalues become $\lambda - \sigma$, emphasizing large $|\lambda - \sigma|$.

Use factors of A or $A - \sigma I$ for solves; do not form explicit inverses.

Method 4.5.16. *SVD Numerics Preview*
Stable numerical SVD computation proceeds as:

(i) Avoid forming A^*A when possible, since it can square the condition number.
(ii) Use orthogonal or unitary transformations to reduce A to bidiagonal form.
(iii) Compute singular values from the bidiagonal problem, then use them for low-rank approximation, pseudoinverse truncation, and least-squares diagnostics through Theorem 4.3.14 and Definition 4.3.21.

Definition 4.5.17. *Discrete Fourier Matrix*
For $n \geq 1$, set $\omega_n = e^{2\pi i/n}$ and $F_n := [\omega_n^{jk}]_{j,k=0}^{n-1}$.
Then $F_n \mathbf{c}$ is the unnormalised DFT of $\mathbf{c} \in \mathbb{C}^n$; $Q_n := n^{-1/2} F_n$ is the normalized Fourier matrix.

Proposition 4.5.18. *Inverse of the Fourier Matrix*
The Fourier matrix satisfies $F_n^* F_n = n I_n$, and therefore $F_n^{-1} = \frac{1}{n} F_n^*$.

Proof. Theorem 3.4.31 gives $Q_n^* Q_n = I_n$ for $Q_n = n^{-1/2} F_n$. Multiplying by n gives $F_n^* F_n = n I_n$, so $n^{-1} F_n^*$ is a left inverse of F_n. Since F_n is square, Theorem 1.3.56 makes it the inverse.
Structural Commentary: Fourier inversion is conjugate transpose plus one scalar normalization.
Key Insight: The inverse is simple because the Fourier columns are orthogonal. □

Method 4.5.19. *Radix-Two FFT Split*
Let $n = 2m$ and $\mathbf{y} = F_n\mathbf{c}$.

(i) Split $\mathbf{c}' := (c_{2r})_{r=0}^{m-1}$ and $\mathbf{c}'' := (c_{2r+1})_{r=0}^{m-1}$.
(ii) Compute $\mathbf{y}' := F_m\mathbf{c}'$ and $\mathbf{y}'' := F_m\mathbf{c}''$.
(iii) With $d_j := \omega_n^j$, and using $\omega_n^2 = \omega_m$ and $\omega_n^m = -1$, set $y_j = y'_j + d_j y''_j$; and $y_{j+m} = y'_j - d_j y''_j$ for $0 \le j < m$.
(iv) For $n = 2^\ell$, repeat (i)–(iii) for ℓ stages.
n^2 dense products become about $\frac{1}{2}n\log_2 n$ twiddle multiplications plus additions.

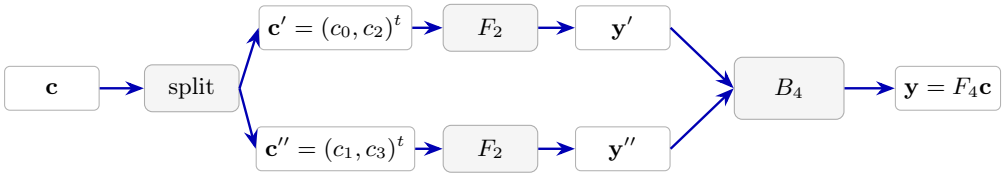


Figure 89: For $n = 4$: split, apply two F_2 transforms, then $B_4\binom{\mathbf{y}'}{\mathbf{y}''} = \binom{\mathbf{y}'+D\mathbf{y}''}{\mathbf{y}'-D\mathbf{y}''}$ with $D = \mathrm{diag}(1, i)$.

Definition 4.5.20. *Krylov Subspace*
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$ and a vector $\mathbf{b} \in \mathbb{F}^n$, the mth *Krylov subspace* generated by $(A, \mathbf{b})$ is $\mathcal{K}_m(A, \mathbf{b}) := \mathrm{span}\{\mathbf{b}, A\mathbf{b}, \ldots, A^{m-1}\mathbf{b}\}$.

Perspective 4.5.21. *Krylov Methods Preview*
Krylov methods work in $\mathcal{K}_m(A, \mathbf{b})$: Arnoldi and Lanczos build bases, while CG and GMRES impose projection or residual-minimization conditions using only matrix-vector products.

Method 4.5.22. *Arnoldi, Lanczos, and CG Preview*
Let $\mathcal{K}_m = \mathcal{K}_m(A, \mathbf{b})$ and let $Q_m = [\mathbf{q}_1 \ \cdots \ \mathbf{q}_m]$ be an orthonormal basis.

(i) *Arnoldi*: $AQ_m = Q_m H_m + h_{m+1,m}\mathbf{q}_{m+1}\mathbf{e}_m^t$, with H_m upper Hessenberg.
(ii) *Lanczos*: if $A = A^*$, then H_m is tridiagonal and $A\mathbf{q}_j = \beta_{j-1}\mathbf{q}_{j-1} + \alpha_j\mathbf{q}_j + \beta_j\mathbf{q}_{j+1}$.
(iii) *CG*: if $A \succ 0$, choose $\mathbf{x}_k \in \mathbf{x}_0 + \mathcal{K}_k(A, \mathbf{r}_0)$ with $\mathbf{r}_k \perp \mathcal{K}_k(A, \mathbf{r}_0)$; search directions satisfy $\mathbf{p}_i^* A\mathbf{p}_j = 0$, $i \ne j$.

Each step uses one matrix-vector product.

Definition 4.5.23. *Matrix Splitting Iteration*
Let $A = S - T$ with S invertible, and set $B := S^{-1}T$.
The *splitting iteration* for $A\mathbf{x} = \mathbf{b}$ is $S\mathbf{x}_{k+1} = T\mathbf{x}_k + \mathbf{b}$, and $\mathbf{x}_{k+1} = B\mathbf{x}_k + S^{-1}\mathbf{b}$.
For a solution $\mathbf{x}_*$, the error $\mathbf{e}_k := \mathbf{x}_k - \mathbf{x}_*$ satisfies $\mathbf{e}_{k+1} = B\mathbf{e}_k$.

Theorem 4.5.24. *Splitting Convergence Criterion*
Let $A = S - T$, with S invertible, and set $B = S^{-1}T$. For $\mathbf{x}_{k+1} = B\mathbf{x}_k + S^{-1}\mathbf{b}$:

(i) if A is invertible, then $\mathbf{x}_k \to A^{-1}\mathbf{b}$ for all $\mathbf{x}_0, \mathbf{b}$ if and only if $\rho(B) < 1$;
(ii) $\rho(B) < 1$ implies A is invertible.

Proof.

(i) Let $\mathbf{x}_* = A^{-1}\mathbf{b}$ and $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}_*$. Then $\mathbf{e}_k = B^k\mathbf{e}_0$, so convergence for all $\mathbf{x}_0, \mathbf{b}$ is equivalent to $B^k \to 0$, equivalently $\rho(B) < 1$ by Theorem 4.1.89.
(ii) If $\rho(B) < 1$, then $1 \notin \mathrm{spec}(B)$, so $I_n - B$ is invertible. Since $A = S - T = S(I_n - B)$, A is invertible.

Structural Commentary: A stationary method is a matrix-power problem in disguise.
Key Insight: The iteration matrix $B = S^{-1}T$ controls the errors. □

Corollary 4.5.25. *Singular Matrices Obstruct Splitting Convergence*
Let $A = S - T$, with S invertible. If A is singular, then $1 \in \mathrm{spec}(S^{-1}T)$.
Hence $\rho(S^{-1}T) \ge 1$, so $\rho(S^{-1}T) < 1$ cannot hold.

Proof. Choose $\mathbf{0} \ne \mathbf{x} \in \ker A$. Then $(S - T)\mathbf{x} = \mathbf{0}$, so $S\mathbf{x} = T\mathbf{x}$ and $S^{-1}T\mathbf{x} = \mathbf{x}$. Thus 1 is eigenvalue of $S^{-1}T$.
Structural Commentary: Singular A forces an undamped error mode.
Key Insight: A null vector of A becomes a fixed error mode. □

Method 4.5.26. *Stationary Iterative Method*
For $A\mathbf{x} = \mathbf{b}$, a stationary iteration is:

(i) Choose fixed M and $\mathbf{c}$.
(ii) Update $\mathbf{x}_{k+1} = M\mathbf{x}_k + \mathbf{c}$.

(iii) If $\mathbf{x}_*$ exists, use $\mathbf{x}_{k+1} - \mathbf{x}_* = M(\mathbf{x}_k - \mathbf{x}_*)$ and Theorem 4.1.89 for convergence.

Method 4.5.27. *Jacobi, Gauss–Seidel, and SOR Splittings*
Let $A = D+L+U$, with D diagonal, L strictly lower triangular, and U strictly upper triangular. For a splitting $A = S - T$, use $S\mathbf{x}_{k+1} = T\mathbf{x}_k + \mathbf{b}$ and $B = S^{-1}T$.

(i) *Jacobi*: $S = D$, $T = -(L+U)$, $B_J = -D^{-1}(L+U)$.
(ii) *Gauss–Seidel*: $S = D + L$, $T = -U$, $B_{\mathrm{GS}} = -(D+L)^{-1}U$.
(iii) *SOR*: for $\omega \neq 0$, $S_\omega = \frac{1}{\omega}D + L$; $T_\omega = \left(\frac{1}{\omega} - 1\right) D - U$; and $B_\omega = S_\omega^{-1}T_\omega$.

Convergence is controlled by $\rho(B)$; $\omega = 1$ gives Gauss–Seidel.

Corollary 4.5.28. *Strict Diagonal Dominance Gives Jacobi Convergence*
If $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$ is strictly row diagonally dominant, then Jacobi converges for every right-hand side.

Proof. Here $a_{ii} \neq 0$ and $B_J = -D^{-1}(L+U) = I_n - D^{-1}A$. Thus $(B_J)_{ii} = 0$ and $\sum_{j \neq i} |(B_J)_{ij}| = \sum_{j \neq i} |a_{ij}| \,/\, |a_{ii}| < 1$.
Gersgorin gives $\mathrm{spec}(B_J) \subset \{z : |z| < 1\}$, so $\rho(B_J) < 1$; Theorem 4.5.24 applies.
Structural Commentary: Diagonal dominance puts the Jacobi spectrum in the unit disk.
Key Insight: Small off-diagonal row sums force the error matrix to have spectral radius below 1. □

Example 4.5.29. *Two-by-Two Jacobi and Gauss–Seidel*
For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $\mathbf{b} \in \mathbb{R}^2$, write $A = D + L + U$.
The Jacobi iteration matrix is $B_J = -D^{-1}(L+U) = \begin{bmatrix} 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 \end{bmatrix}$, so $\rho(B_J) = \frac{1}{2}$.
The Gauss–Seidel iteration matrix is $B_{\mathrm{GS}} = -(D+L)^{-1}U = \begin{bmatrix} 0 & -\frac{1}{2} \\ 0 & \frac{1}{4} \end{bmatrix}$, so $\rho(B_{\mathrm{GS}}) = \frac{1}{4}$.
Both converge by Theorem 4.5.24, and smaller spectral radius predicts faster asymptotic decay for Gauss–Seidel.

Example 4.5.30. *Poisson Tridiagonal Stationary Rates*
For $A_n = \mathrm{tridiag}(-1, 2, -1)$, Jacobi gives $B_J = I_n - \frac{1}{2}A_n$; $\lambda_k(B_J) = \cos\frac{k\pi}{n+1}$; and $\rho(B_J) = \cos\frac{\pi}{n+1}$.
For the same ordering, Gauss–Seidel satisfies $\rho(B_{\mathrm{GS}}) = \rho(B_J)^2 = \cos^2\frac{\pi}{n+1}$. As $n \to \infty$, both radii tend to 1; fine grids need acceleration or preconditioning.

Perspective 4.5.31. *Preconditioning Preview*
A preconditioner replaces $A\mathbf{x} = \mathbf{b}$ by an equivalent system such as $M^{-1}A\mathbf{x} = M^{-1}\mathbf{b}$ or $AM^{-1}\mathbf{y} = \mathbf{b}$, with $\mathbf{x} = M^{-1}\mathbf{y}$. It should improve conditioning or convergence while being much cheaper to apply than A^{-1}.

Part II
Algebraic Structures

5 Algebraic Language and Categories

5.1 Sets, Maps, Relations, and Universal Properties

5.2 Categories and Functors

5.3 Algebraic Structures as Objects

5.4 Exactness and Diagrams

6 Group Theory

6.1 Groups, Subgroups, and Homomorphisms

6.2 Group Actions

6.3 Normality and Structure Theorems

6.4 Sylow Theory and Finite Groups

6.5 Representation-Theoretic Preview

7 Rings, Ideals, and Modules

7.1 Rings and Ideals

7.2 Polynomial Rings and Factorization

7.3 Modules

7.4 Structure Theorems

7.5 Linear Operators as Modules

8 Fields and Galois Theory

8.1 Field Extensions

8.2 Splitting Fields and Normality

8.3 Galois Groups

8.4 Classical Applications

8.5 Scalar Extension in Linear Algebra

9 Tensor, Exterior, and Homological Algebra

9.1 Tensor Products and Universal Properties

9.2 Symmetric and Exterior Algebras

9.3 Multilinear Algebra Revisited

9.4 Chain Complexes and Homology

9.5 Derived Functors Preview

Part III
Algebraic Geometry and Commutative Algebra

10 Commutative Algebra

10.1 Prime Ideals and Spectra
10.2 Localization and Local Rings
10.3 Noetherian Rings and Modules
10.4 Integral Dependence and Dimension
10.5 Regularity and Singularities Preview

11 Algebraic Geometry

11.1 Affine Algebraic Sets
11.2 Morphisms and Coordinate Rings
11.3 Projective Geometry
11.4 Schemes
11.5 Sheaves and Cohomology Preview
11.6 Smoothness, Tangent Spaces, and Dimension
11.7 Divisors, Line Bundles, and Curves

Part IV
Analysis and Continuous Structure

12 Foundations of Analysis

Foundations of Analysis fixes the language of proofs, sets, number systems, cardinality, and real sequences before topology begins.

12.1 Mathematical Language, Sets, and Functions

This subsection fixes the basic language of Analysis: membership, containment, functions, images, preimages, and definition-level proof patterns.

Principle 12.1.1. *Language Carries Structure*
Objects, hypotheses, conclusions, and quantifier order matter. For instance,

$$\forall x \in X\ \exists y \in Y\ P(x,y), \qquad \exists y \in Y\ \forall x \in X\ P(x,y)$$

are different: in the first y may depend on x; in the second one y works for all x.

Principle 12.1.2. *Well-Formed Statements and Bound Variables*
A mathematical statement is well formed only after every symbol has a declared role and every expression has a defined meaning. A variable introduced by a quantifier, summation sign, integral, set-builder condition, or similar construction is *bound* inside that construction; changing its name does not change the statement as long as its scope is unchanged and no free variable is captured.

Definition 12.1.3. *Finite Sums Indexed by Integers*
Let $a_i \in \mathbb{C}$ be assigned for integers $m \le i \le n$. Define $\sum_{i=m}^{n} a_i := a_m + a_{m+1} + \cdots + a_n$.
If $m > n$, the sum over the empty index set is 0.

Proposition 12.1.4. *Basic Laws for Finite Sums*
Let $a_i, b_i \in \mathbb{C}$ and $c \in \mathbb{C}$ be assigned for $m \le i \le n$.

(i) If $m \le r < n$, then $\sum_{i=m}^{n} a_i = \sum_{i=m}^{r} a_i + \sum_{i=r+1}^{n} a_i$.

(ii) $\sum_{i=m}^{n} (a_i + b_i) = \sum_{i=m}^{n} a_i + \sum_{i=m}^{n} b_i$ and $\sum_{i=m}^{n} ca_i = c \sum_{i=m}^{n} a_i$.

(iii) $\left|\sum_{i=m}^{n} a_i\right| \le \sum_{i=m}^{n} |a_i|$.

(iv) If $a_i, b_i \in \mathbb{R}$ and $a_i \le b_i$ for every i, then $\sum_{i=m}^{n} a_i \le \sum_{i=m}^{n} b_i$.

Proof.

(i) By Definition 12.1.3, associativity of addition in $\mathbb{C}$, $\sum_{i=m}^{n} a_i = \sum_{i=m}^{r} a_i + \sum_{i=r+1}^{n} a_i$.

(ii) By associativity, commutativity, and distributivity in the field $\mathbb{C}$ (Theorem 12.2.32), $\sum_{i=m}^{n} (a_i + b_i) = \sum_{i=m}^{n} a_i + \sum_{i=m}^{n} b_i$ and $\sum_{i=m}^{n} ca_i = c \sum_{i=m}^{n} a_i$.

(iii) Induct on $n - m$. The case $n = m$ is immediate.
If the claim holds up to n, then by Theorem 12.2.38, $\left|\sum_{i=m}^{n+1} a_i\right| \le \left|\sum_{i=m}^{n} a_i\right| + |a_{n+1}| \le \sum_{i=m}^{n} |a_i| + |a_{n+1}| = \sum_{i=m}^{n+1} |a_i|$.

(iv) Induct on $n - m$. The case $n = m$ is the hypothesis $a_m \le b_m$.
If the claim holds up to n, then Proposition 12.2.13 gives $\sum_{i=m}^{n+1} a_i = \sum_{i=m}^{n} a_i + a_{n+1} \le \sum_{i=m}^{n} b_i + b_{n+1} = \sum_{i=m}^{n+1} b_i$.

Structural Commentary: Finite sums are iterated addition, so their laws are inherited from the scalar field.
Key Insight: No convergence hypothesis is needed for a finite sum. □

Principle 12.1.5. *Mathematical Induction*
Let $P_1, P_2, \ldots$ be statements indexed by positive integers. To prove P_n for every $n \in \mathbb{N}$, it is enough to prove:

(i) *basis*: P_1 is true;

(ii) *induction step*: for every $n \in \mathbb{N}$, P_n implies P_{n+1}.

The induction step is an implication; it does not prove P_{n+1} in isolation.
The basis starts the chain, and the implication carries truth from one statement to the next.

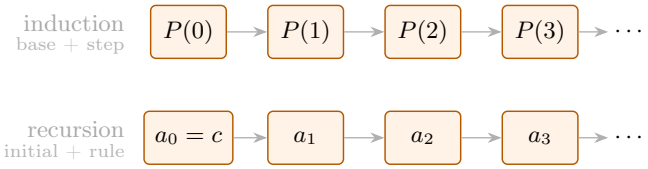


Figure 90: Induction propagates truth, while recursion propagates values along the same index chain.

Method 12.1.6. *Proof Grammar and Negation*
In theorem statements, put the hypotheses before the conclusion and keep quantifiers in the order meant.
In particular, switching $\forall$ and $\exists$ usually changes the statement; see Principle 12.1.1.
Before applying a rule, also check that the displayed expressions are well formed; an undefined expression makes the sentence ill formed, not merely false.
The word "or" is inclusive in mathematics, and the basic negation rules are

$$\neg(a \wedge b) \iff \neg a \vee \neg b, \qquad \neg(a \vee b) \iff \neg a \wedge \neg b, \qquad \neg(a \Rightarrow b) \iff a \wedge \neg b$$

Thus a proof by contradiction of $a \Rightarrow b$ assumes both a and $\neg b$ and derives an impossibility.
Key Insight: Before proving or negating a statement, parse its quantifiers and logical connectives.

Definition 12.1.7. *Sets and Operations*
A *set* is a collection. Write $x \in E$, $x \notin E$, $\emptyset$, and $A \subseteq B$; $A = B$ iff $A \subseteq B$ and $B \subseteq A$. For $A, B \subseteq X$,

$$\begin{aligned} A \cup B &:= \{x \in X : x \in A \text{ or } x \in B\}, \\ A \cap B &:= \{x \in X : x \in A \text{ and } x \in B\}, \\ A \setminus B &:= \{x \in X : x \in A \text{ and } x \notin B\}, \\ A^c &:= X \setminus A. \end{aligned}$$

The *power set* is $\mathcal{P}(X) := \{A : A \subseteq X\}$.

Remark 12.1.8. *Set-Builder Notation*
$\{x \in X : P(x)\}$ means all $x \in X$ satisfying $P(x)$.
For sets (A_n), $\bigcup_n A_n$ means membership in some A_n; $\bigcap_n A_n$, membership in every A_n.
Key Insight: Expand membership conditions.

Definition 12.1.9. *Indexed Families of Sets*
An *indexed family* $\{A_i\}_{i \in I}$ is one set $A_i \subseteq X$ for each index $i \in I$. Define

$$\begin{aligned} \bigcup_{i \in I} A_i &:= \{x \in X : x \in A_i \text{ for some } i \in I\}, \\ \bigcap_{i \in I} A_i &:= \{x \in X : x \in A_i \text{ for every } i \in I\}. \end{aligned}$$

Key Insight: Union means "some index"; intersection means "every index".

Method 12.1.10. *Proof by Two Containments*
To prove $A = B$, prove both inclusions.

(i) To prove $A \subseteq B$, let $x \in A$ be arbitrary and use the definitions and hypotheses to show $x \in B$.
(ii) To prove $B \subseteq A$, let $x \in B$ be arbitrary and use the definitions and hypotheses to show $x \in A$.
(iii) Once both containments are proved, conclude $A = B$ by extensionality of sets.

If both sides are described by membership conditions, this is a chain, $x \in A \iff \cdots \iff x \in B$.

Proposition 12.1.11. *Basic Set Algebra*
Let $A, B, C \subseteq X$. Then

(i) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
(ii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$;
(iii) $(A \cup B)^c = A^c \cap B^c$;
(iv) $(A \cap B)^c = A^c \cup B^c$.

Proof. It suffices to prove (i) and (iii); the others are similar. Let $x \in X$. Then

(i) $x \in A \cap (B \cup C) \iff x \in A$ and $(x \in B$ or $x \in C) \iff (x \in A \cap B)$ or $(x \in A \cap C) \iff x \in (A \cap B) \cup (A \cap C)$
(iii) $x \in (A \cup B)^c \iff x \notin A \cup B \iff x \notin A$ and $x \notin B \iff x \in A^c \cap B^c$

Structural Commentary: Set algebra is membership logic.
Key Insight: Translate sets into conditions on x. □

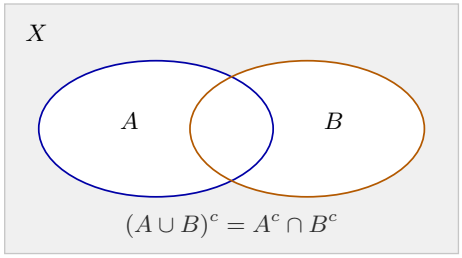


Figure 91: De Morgan's law says that being outside a union is the same as being outside each set.

Definition 12.1.12. *Cartesian Product*
For sets X and Y, the *Cartesian product* is $X \times Y := \{(x, y) : x \in X,\ y \in Y\}$.
The order matters: the first coordinate comes from X, and the second comes from Y.

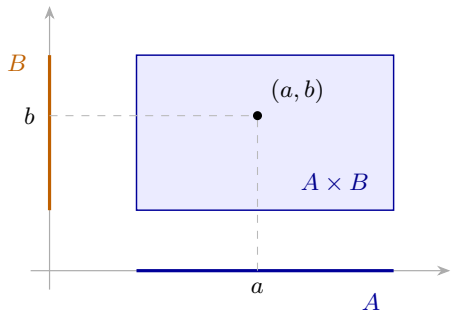


Figure 92: $A \times B$: first coordinate in A, second in B.

Definition 12.1.13. *Relations and Equivalence*
A *relation* from X to Y is a subset $R \subseteq X \times Y$. If $(x, y) \in R$, one often writes xRy.
A relation $\sim$ on X is an *equivalence relation* if, for all $x, y, z \in X$,

i. $x \sim x$;
ii. $x \sim y$ implies $y \sim x$;
iii. $x \sim y$ and $y \sim z$ imply $x \sim z$.

The equivalence class of x is $[x] := \{y \in X : y \sim x\}$.

Proposition 12.1.14. *Equivalence Classes Partition the Set*
Let $\sim$ be an equivalence relation on X. Then the equivalence classes form a partition of X.

Proof. Every $x \in X$ lies in $[x]$ since $x \sim x$. If $[x] \cap [y] \neq \emptyset$, choose z in the intersection.
Then $x \sim z \sim y$, so $u \in [x]$ implies $u \sim y$ and $u \in [y]$. Hence $[x] \subseteq [y]$, and symmetry gives equality.
Structural Commentary: Equivalence means partition.
Key Insight: Touching classes coincide. □

Definition 12.1.15. *Functions, Images, and Preimages*
A *function* $f : X \to Y$ assigns to each $x \in X$ a unique element $f(x) \in Y$. The set X is the domain and Y is the codomain. The *range* of f is $\text{range}(f) := f(X) := \{f(x) : x \in X\}$.
If $A \subseteq X$, the *image* of A under f is $f(A) := \{f(x) : x \in A\}$.
If $B \subseteq Y$, the *preimage* of B under f is $f^{-1}(B) := \{x \in X : f(x) \in B\}$.
This does not require f to have an inverse function.
If $A \subseteq X$, the *restriction* of f to A is denoted $f|_A : A \to Y$.
If $g : Y \to Z$, the *composition* $g \circ f : X \to Z$ is defined by $(g \circ f)(x) = g(f(x))$.
The *identity function* on X is denoted id_X.

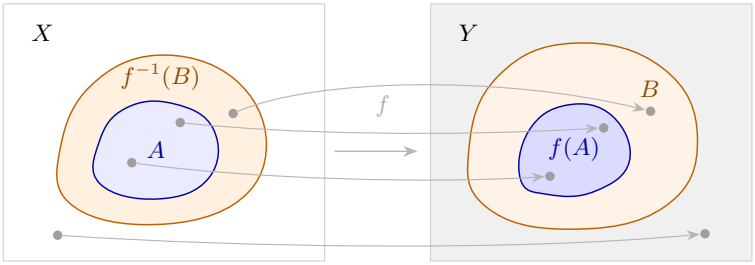


Figure 93: Image–preimage containment: $f(A) \subseteq B \iff A \subseteq f^{-1}(B)$.

Proposition 12.1.16. *Image and Preimage Calculus*
Let $f : X \to Y$, let $\{A_i\}_{i\in I}$ be a nonempty family of subsets of X, and let $\{B_i\}_{i\in I}$ be a nonempty family of subsets of Y. Then

(i) $f\left(\bigcup_{i\in I} A_i\right) = \bigcup_{i\in I} f(A_i)$;
(ii) $f\left(\bigcap_{i\in I} A_i\right) \subseteq \bigcap_{i\in I} f(A_i)$;
(iii) $f^{-1}\left(\bigcup_{i\in I} B_i\right) = \bigcup_{i\in I} f^{-1}(B_i)$;
(iv) $f^{-1}\left(\bigcap_{i\in I} B_i\right) = \bigcap_{i\in I} f^{-1}(B_i)$;
(v) $f^{-1}(Y \setminus B) = X \setminus f^{-1}(B)$ *for every* $B \subseteq Y$;
(vi) $f(f^{-1}(B)) = B \cap \operatorname{range}(f)$ *for every* $B \subseteq Y$.

Proof.

(i) $y \in f(\bigcup_i A_i)$ iff $y = f(x)$ for some $x \in A_i$ and some i, iff $y \in \bigcup_i f(A_i)$.
(ii) If $y = f(x)$ with $x \in \bigcap_i A_i$, then $y \in f(A_i)$ for every i.
(iii) $x \in f^{-1}(\bigcup_i B_i)$ iff $f(x) \in B_i$ for some i, iff $x \in \bigcup_i f^{-1}(B_i)$.
(iv) $x \in f^{-1}(\bigcap_i B_i)$ iff $f(x) \in B_i$ for every i, iff $x \in \bigcap_i f^{-1}(B_i)$.
(v) For $x \in X$, $x \in f^{-1}(Y \setminus B) \iff f(x) \notin B \iff x \notin f^{-1}(B) \iff x \in X \setminus f^{-1}(B)$
(vi) $y \in f(f^{-1}(B))$ iff $y = f(x)$ for some x with $f(x) \in B$, iff $y \in B \cap \operatorname{range}(f)$.

Structural Commentary: Preimages are Boolean-exact; images need not be.
Key Insight: Pull back conditions through f. □

Definition 12.1.17. *Injective, Surjective, and Bijective Maps*
Let $f : X \to Y$.

i. f is *injective* if $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
ii. f is *surjective* if $\operatorname{range}(f) = Y$.
iii. f is *bijective* if it is both injective and surjective.

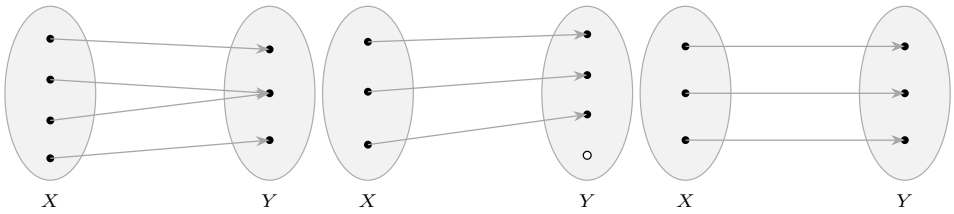


Figure 94: Surjective: covers codomain; injective: no collisions; bijective: both.

Theorem 12.1.18. *Bijective Means Invertible*
A function $f : X \to Y$ is bijective iff there exists a function $g : Y \to X$ s.t. $g \circ f = \mathrm{id}_X$, $f \circ g = \mathrm{id}_Y$. When such a function exists, it is unique and is denoted $f^{-1} : Y \to X$.

Proof.

($\Rightarrow$) If f is bijective, define $g(y)$ as the unique x with $f(x) = y$. Then $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$.
($\Leftarrow$) If such a g exists, applying g to $f(x_1) = f(x_2)$ gives $x_1 = x_2$, and $y = f(g(y))$ gives surjectivity.

If g and h are two-sided inverses, then $g(y) = g(f(h(y))) = h(y)$, so the inverse is unique.
Structural Commentary: A true inverse undoes f on both sides.
Key Insight: Bijective means exactly one preimage per output. □

Example 12.1.19. *Image and Preimage for the Square Map*
Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x^2$. Then $f([-2,1]) = [0,4]$, $f^{-1}([1,4]) = [-2,-1] \cup [1,2]$. The map is not injective because $f(1) = f(-1)$, and it is not surjective as a map $\mathbb{R} \to \mathbb{R}$ because no real number maps to -1.

12.2 Number Systems and Completeness

Number systems supply the ordered and algebraic background for Analysis: $\mathbb{Q}$ has gaps, $\mathbb{R}$ repairs them by completeness, and $\mathbb{C}$ keeps field algebra without order.

Perspective 12.2.1. *Number Systems from Representatives*
Build by representatives, then pass to classes:

$$\mathbb{Z}: \ (a,b) \mapsto [(a,b)], \qquad \mathbb{Q}: \ (a,b),\ b \neq 0 \mapsto [(a,b)], \qquad \mathbb{R}: \ \text{cuts or Cauchy data} \mapsto [\cdot].$$

The pattern is: choose $\sim$, prove $+, \cdot, <, \ldots$ are well defined on $X/\!\sim$, then use the classes as the number system; compare Definition 12.1.13 and Proposition 1.2.18.
Key Insight: Representatives are scaffolding; the quotient laws are the interface.

Proposition 12.2.2. *The Rational Gap at $\sqrt{2}$*
There is no rational p such that $p^2 = 2$.

Proof. Let $p = m/n$. Then $m^2 = 2n^2$, so m is even; $m = 2k$. Hence $n^2 = 2k^2$, so n is even, a contradiction.
Structural Commentary: This exhibits a gap in $\mathbb{Q}$.
Key Insight: $\sqrt{2}$ is not rational. □

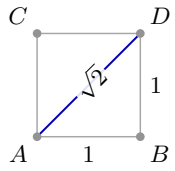


Figure 95: The diagonal of a unit square has length $\sqrt{2}$, so the rational gap is geometric as well as algebraic.

Example 12.2.3. *No Largest Lower Side and No Smallest Upper Side*
Let $A := \{p \in \mathbb{Q} : p > 0,\ p^2 < 2\}$, and $B := \{p \in \mathbb{Q} : p > 0,\ p^2 > 2\}$.
For every $p > 0$ define $q := p - \frac{p^2-2}{p+2} = \frac{2p+2}{p+2}$. Then $q^2 - 2 = \frac{2(p^2-2)}{(p+2)^2}$. Hence if $p \in A$, then $q \in A$ and $q > p$; if $p \in B$, then $q \in B$ and $0 < q < p$. Thus A has no largest element in $\mathbb{Q}$, and B has no smallest element in $\mathbb{Q}$.

Remark 12.2.4. *Density Does Not Mean Completeness*
Between two rationals $r < s$ there is another rational, namely $(r+s)/2$. Nevertheless, $\mathbb{Q}$ is not complete.

Definition 12.2.5. *Ordered Sets and Bounds*
An *order* on S is a relation $<$ such that exactly one of $x < y$, $x = y$, or $y < x$ holds, and $x < y < z$ implies $x < z$. An *ordered set* is a set with an order. For $E \subseteq S$, an *upper bound* is a $\beta \in S$ with $x \leq \beta$ for all $x \in E$; then E is *bounded above*. Lower bounds reverse the inequality.

Definition 12.2.6. *Supremum, Infimum, and Least-Upper-Bound Property*
For $E \subseteq S$, $\sup E$ is least upper bound of E, and $\inf E$ is greatest lower bound of E, when these exist. Ordered set S has *least-upper-bound property* if every nonempty bounded-above subset of S has a supremum in S.

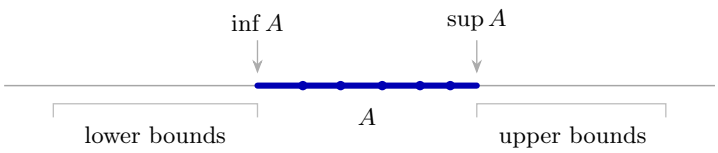


Figure 96: The supremum is the least upper bound, while the infimum is the greatest lower bound.

Example 12.2.7. *Suprema Need Not Be Elements*
In $\mathbb{Q}$, the set $E_1 = \{r \in \mathbb{Q} : r < 0\}$ and the set $E_2 = \{r \in \mathbb{Q} : r \leq 0\}$ both have supremum 0, but $0 \notin E_1$ and $0 \in E_2$. If $E = \{1/n : n = 1, 2, 3, \ldots\}$, then $\sup E = 1$ while $\inf E = 0$, and the infimum is not a member of E.

Theorem 12.2.8. *Least Upper Bounds Give Greatest Lower Bounds*
If S has the least-upper-bound property, then every nonempty bounded-below set $B \subseteq S$ has an infimum.

Proof. Let L be the set of lower bounds of B. Then L is nonempty and every $x \in B$ is an upper bound of L, so $\alpha = \sup L$ exists. Since every $x \in B$ bounds L above, $\alpha \le x$ for all $x \in B$; hence α is a lower bound of B. If $\beta > \alpha$, then $\beta \notin L$, so β is not a lower bound. Thus $\alpha = \inf B$.
Structural Commentary: Infima are suprema of lower bounds.
Key Insight: Completeness above gives completeness below. □

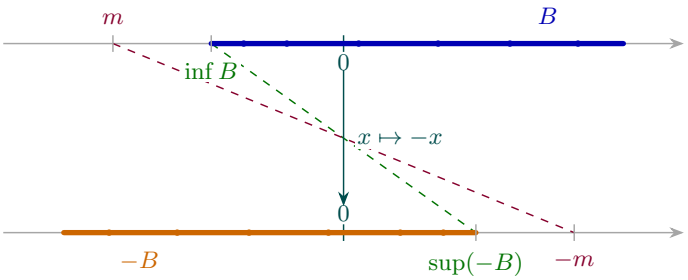


Figure 97: Reflecting B through 0 turns lower bounds of B into upper bounds of $-B$, so $\inf B = -\sup(-B)$.

Corollary 12.2.9. *Infimum by Reflection*
If $B \subseteq \mathbb{R}$ is nonempty and bounded below, then $\inf B = -\sup(-B)$ and $-B := \{-b : b \in B\}$.

Proof. Since B is nonempty and any lower bound m of B gives an upper bound $-m$ of $-B$, let $s = \sup(-B)$. For $b \in B$, $-b \le s$, so $-s \le b$; hence $-s$ is a lower bound of B. If t is any lower bound of B, then $-t$ is an upper bound of $-B$, so $s \le -t$ and $t \le -s$. Thus $-s = \inf B$.
Structural Commentary: Reflection swaps lower bounds of B with upper bounds of $-B$.
Key Insight: The infimum of B is the negative supremum of its reflection. □

Definition 12.2.10. *Fields and Ordered Fields*
A *field* is a set F with addition and multiplication satisfying closure, commutativity, associativity, identities $0, 1$ with $1 \neq 0$, additive inverses, multiplicative inverses for nonzero elements, and distributivity.
An *ordered field* is a field such that $y < z$ implies $x + y < x + z$, and $x > 0$ and $y > 0$ imply $xy > 0$.

Remark 12.2.11. *Fields as Objects and Scalars*
Unlike their scalar role in Definition 1.1.1, fields are now Analysis objects: $\mathbb{Q}$ is ordered but incomplete, while $\mathbb{R}$ has the least-upper-bound property.

Proposition 12.2.12. *Basic Consequences of the Field Axioms*
Every field has cancellation, unique identities and inverses, $0x = 0$, no zero divisors, and the sign rules

$$(-x)y = -(xy) = x(-y), \qquad (-x)(-y) = xy.$$

Proof. Cancellation comes from adding or multiplying by the relevant inverse; uniqueness follows from cancellation. Also $0x + 0x = (0 + 0)x = 0x$, so $0x = 0$. If $x, y \neq 0$ and $xy = 0$, multiplying by $x^{-1}y^{-1}$ gives $1 = 0$, impossible. The sign rules follow because the displayed negatives are additive inverses of xy.
Structural Commentary: The field axioms license ordinary algebra.
Key Insight: Inverses force cancellation and signs. □

Proposition 12.2.13. *Basic Inequality Rules in Ordered Fields*
In every ordered field,

(i) $x > 0$ *iff* $-x < 0$;
(ii) multiplying an inequality by a positive element preserves the inequality;
(iii) multiplying an inequality by a negative element reverses the inequality;
(iv) if $x \neq 0$, *then* $x^2 > 0$, *so in particular* $1 > 0$;
(v) if $0 < x < y$, *then* $0 < 1/y < 1/x$.

Proof.

(i) If $x > 0$, then $0 = -x + x > -x + 0$, so $-x < 0$; the converse is the same with $-x$.
(ii) If $y < z$ and $x > 0$, then $z - y > 0$, so $x(z - y) > 0$ and hence $xy < xz$.
(iii) Apply (ii) to $-x > 0$.
(iv) Either $x > 0$ or $-x > 0$, so $x^2 > 0$; take $x = 1$ for $1 > 0$.

(v) Since $x, y > 0$, multiply $x < y$ by $(1/x)(1/y) > 0$ to get $1/y < 1/x$.

Structural Commentary: Ordered fields make algebraic signs compatible with order.
Key Insight: Positivity decides whether multiplication preserves order. □

Definition 12.2.14. *Real Absolute Value and Distance*
For $x \in \mathbb{R}$, define $|x| := \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$.
For $a, b \in \mathbb{R}$, the distance between a and b on the real line is $|a - b|$.

Proposition 12.2.15. *Real Absolute-Value Laws*
For $x, y, a, b, c \in \mathbb{R}$,

(i) $|x| \geq 0$, $|x| = 0$ *iff* $x = 0$, *and* $-|x| \leq x \leq |x|$;
(ii) $|xy| = |x|\,|y|$;
(iii) $|x + y| \leq |x| + |y|$;
(iv) $|a - b| \leq |a - c| + |c - b|$.

Proof.

(i) Immediate from Definition 12.2.14.
(ii) Check the four sign cases for x and y.
(iii) Add $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$, then $-(|x| + |y|) \leq x + y \leq |x| + |y|$. Since $|x| + |y| \geq 0$, this gives $|x + y| \leq |x| + |y|$.
(iv) Apply (iii) to $a - b = (a - c) + (c - b)$.

Structural Commentary: Absolute value turns order bounds into distance estimates.
Key Insight: The triangle inequality on $\mathbb{R}$ is the inequality $|(a - c) + (c - b)| \leq |a - c| + |c - b|$. □

Perspective 12.2.16. *ε-Closeness Becomes Metric Language*
On $\mathbb{R}$, ε-close means $|x - y| < \varepsilon$, i.e. $d(x, y) < \varepsilon$ with $d(x, y) = |x - y|$; may be abstracted to metric balls.
Key Insight: Absolute value is distance on $\mathbb{R}$.

Perspective 12.2.17. *Completeness Becomes a Theorem*
Dedekind cuts build completeness from $\mathbb{Q}$: real numbers are rational cuts, rational cuts embed $\mathbb{Q}$, and least upper bounds arise by taking unions of cuts.

Definition 12.2.18. *Dedekind Cuts and Their Order*
A *cut* is a set $\alpha \subseteq \mathbb{Q}$ such that

(i) α is nonempty and $\alpha \neq \mathbb{Q}$;
(ii) if $p \in \alpha$, $q \in \mathbb{Q}$, and $q < p$, then $q \in \alpha$;
(iii) if $p \in \alpha$, then some $r \in \alpha$ satisfies $p < r$.

Let $\mathscr{D}$ be the set of all cuts. For $\alpha, \beta \in \mathscr{D}$, define $\alpha < \beta$ to mean $\alpha \subsetneq \beta$.
Rational letters $p, q, r, \ldots$ denote elements of $\mathbb{Q}$; Greek letters $\alpha, \beta, \gamma, \ldots$ denote cuts.
A cut is the rational left side of a real number: a proper initial segment with no last point.

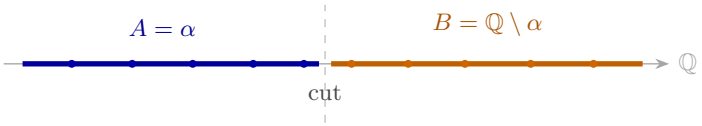


Figure 98: A Dedekind cut splits $\mathbb{Q}$ into α and $\mathbb{Q} \setminus \alpha$, with no last rational in α.

Remark 12.2.19. *Left Cuts Versus Two-Sided Cuts*
Two-sided notation writes a cut as $A|B$, with $A = \alpha$ and $B = \mathbb{Q} \setminus \alpha$; Definition 12.2.18 records only A.
Thus "no largest element" is the same left-side condition.

Definition 12.2.20. *Arithmetic of Cuts*
For cuts α, β, define $\alpha + \beta := \{r + s : r \in \alpha,\ s \in \beta\}$ and $0^* := \{q \in \mathbb{Q} : q < 0\}$.
Let candidate additive inverse of α be $-\alpha := \{p \in \mathbb{Q} : \exists r \in \mathbb{Q},\ r > 0,\ -p - r \notin \alpha\}$. Let $\mathscr{D}^+ := \{\alpha \in \mathscr{D} : \alpha > 0^*\}$.
For $\alpha, \beta \in \mathscr{D}^+$, define $\alpha\beta := \{p \in \mathbb{Q} : \exists r \in \alpha,\ \exists s \in \beta,\ r > 0,\ s > 0,\ p \leq rs\}$ and $1^* := \{q \in \mathbb{Q} : q < 1\}$.
Extend multiplication to all cuts by setting $\alpha 0^* = 0^* \alpha = 0^*$ and, for nonzero signs,

$$\alpha\beta := \begin{cases} (-\alpha)(-\beta), & \alpha < 0^*,\ \beta < 0^*, \\ -\big((-\alpha)\beta\big), & \alpha < 0^*,\ \beta > 0^*, \\ -\big(\alpha(-\beta)\big), & \alpha > 0^*,\ \beta < 0^*. \end{cases}$$

Motivation: arithmetic uses rational approximations inside cuts; signs come after positive products.

Definition 12.2.21. *Rational Cuts*
For $r \in \mathbb{Q}$, define the *rational cut*, $r^* := \{p \in \mathbb{Q} : p < r\}$.
The map $r \mapsto r^*$ is intended to identify $\mathbb{Q}$ with a subfield of the cut field.

Theorem 12.2.22. *Existence of the Real Field*
There exists an ordered field $\mathbb{R}$ with the least-upper-bound property. Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.

Proof. Construct $\mathbb{R}$ from $\mathbb{Q}$ using Definitions 12.2.18–12.2.21.

(i) *Order.* The relation $\alpha < \beta$ is transitive, and at most one of $\alpha < \beta$, $\alpha = \beta$, and $\beta < \alpha$ can hold. For trichotomy, suppose $\alpha \not\subseteq \beta$ and $\alpha \neq \beta$. Choose $p \in \alpha$ with $p \notin \beta$. If $q \in \beta$, then $q < p$, since $p \notin \beta$ and β is downward closed. Hence $q \in \alpha$, so $\beta \subseteq \alpha$. Since $\alpha \neq \beta$, $\beta < \alpha$. Thus $\mathscr{D}$ is ordered.
Commentary: One missing rational decides containment.

(ii) *Least Upper Bounds.* Let $A \subseteq \mathscr{D}$ be nonempty and bounded above by a cut β. Define $\gamma := \bigcup_{\alpha \in A} \alpha$.
Since some $\alpha_0 \in A$ is nonempty, γ is nonempty. Since every $\alpha \in A$ lies in β, $\gamma \subseteq \beta$, so $\gamma \neq \mathbb{Q}$. If $p \in \gamma$ and $q < p$, then $p \in \alpha_1$ for some $\alpha_1 \in A$, hence $q \in \alpha_1 \subseteq \gamma$. If $p \in \gamma$, choose $\alpha_1 \in A$ with $p \in \alpha_1$ and then choose $r \in \alpha_1$ with $p < r$; then $r \in \gamma$. Thus γ is a cut.
Every $\alpha \in A$ satisfies $\alpha \leq \gamma$. If $\delta < \gamma$, choose $s \in \gamma \setminus \delta$. Then $s \in \alpha$ for some $\alpha \in A$, so $\delta < \alpha$ and δ is not an upper bound of A. Hence $\gamma = \sup A$.
Motivation: Supremum is union.
Commentary: Bounded unions stay proper and give the smallest upper bound.

(iii) *Addition.* The set $\alpha + \beta$ is a cut. It is nonempty by choosing one element from each summand.
If $r' \notin \alpha$ and $s' \notin \beta$, then every $r \in \alpha$ and $s \in \beta$ satisfies $r < r'$ and $s < s'$, so $r + s < r' + s'$; hence $r' + s' \notin \alpha + \beta$, and $\alpha + \beta$ is proper. If $p = r + s \in \alpha + \beta$ and $q < p$, then $q = r + (s + q - p)$ with $s+q-p < s$, so $s+q-p \in \beta$ and $q \in \alpha+\beta$. If $p = r+s$, choose $r_1 \in \alpha$ with $r < r_1$; then $p < r_1+s \in \alpha+\beta$. Thus $\alpha + \beta$ is a cut. Commutativity and associativity follow by rewriting representatives in $\mathbb{Q}$.
The cut 0^* is the additive identity: $\alpha + 0^* \subseteq \alpha$ by downward closure, while if $p \in \alpha$, choose $r \in \alpha$ with $p < r$ and write $p = r + (p - r)$ with $p - r < 0$.
The inverse $-\alpha$ is a cut, and $\alpha + (-\alpha) = 0^*$. The inclusion $\alpha + (-\alpha) \subseteq 0^*$ follows because $s \in -\alpha$ forces $-s \notin \alpha$, so any $r \in \alpha$ satisfies $r < -s$. Conversely, if $v < 0$, put $w = -v/2 > 0$ and choose an integer n with $nw \in \alpha$ but $(n+1)w \notin \alpha$; then $p = -(n+2)w$ lies in $-\alpha$ and $v = nw + p \in \alpha + (-\alpha)$.
Commentary: Addition inherits its laws from $\mathbb{Q}$ once closure and inverses are checked.

(iv) *Order Compatibility with Addition.* The additive laws make $\mathscr{D}$ an additive group.
If $\beta < \gamma$, then $\alpha+\beta \subseteq \alpha+\gamma$. Equality would imply $\beta = \gamma$ after adding $-\alpha$ to both sides, so $\alpha+\beta < \alpha+\gamma$. Consequently, $\alpha > 0^*$ if and only if $-\alpha < 0^*$.
Commentary: Cancellation turns inclusion into strict inclusion.

(v) *Positive Multiplication.* If $\alpha, \beta \in \mathscr{D}^+$, then $\alpha\beta$ is a cut. Since each positive cut contains some positive rational, $\alpha\beta$ is nonempty. Choose positive $r' \notin \alpha$ and $s' \notin \beta$; then every positive $r \in \alpha$ and $s \in \beta$ satisfies $rs < r's'$, so $r's' \notin \alpha\beta$, and the product is proper. Downward closure is built into the condition $p \leq rs$. If $p \leq rs$ with $r, s > 0$, then either choose q with $p < q < rs$, or, when needed, choose $r_1 \in \alpha$ with $r < r_1$ and take q with $p < q \leq r_1 s$; hence the product has no largest element.
Commutativity follows from $rs = sr$. Associativity and distributivity are proved by unfolding membership and using the rational identities $(rs)t = r(st)$ and $r(s+t) = rs+rt$, with a positive rational slack inserted whenever a representative must be made strictly larger inside a cut. The identity is 1^*: if $p \leq rs$ with $r \in \alpha$, $s < 1$, and $r, s > 0$, then $p < r$, so $p \in \alpha$; conversely, if $p \in \alpha$, choose $r \in \alpha$ with $p < r$ and then choose $s \in \mathbb{Q}$ with $0 < s < 1$ and $p \leq rs$. In particular, if $\alpha > 0^*$ and $\beta > 0^*$, then $\alpha\beta > 0^*$.
Commentary: Positivity avoids sign reversals.

(vi) *Signed Multiplication.* The definitions extend multiplication to all cuts by reducing to positive factors and additive inverses. Since $\delta = -(-\delta)$ and the sign rules for addition have already been proved, commutativity, associativity, and inverses reduce to the positive case.
For distributivity, split into signs of $\alpha, \beta, \gamma, \beta + \gamma$. For example, if $\alpha > 0^*$, $\beta < 0^*$, and $\beta + \gamma > 0^*$, then $\gamma = (\beta + \gamma) + (-\beta)$, so positive distributivity gives $\alpha\gamma = \alpha(\beta + \gamma) + \alpha(-\beta)$. Because $\alpha(-\beta) = -(\alpha\beta)$, adding $\alpha\beta$ to both sides gives $\alpha\beta + \alpha\gamma = \alpha(\beta + \gamma)$. The other sign cases are identical reductions after replacing negative terms by their additive inverses.
Commentary: Distributivity reduces to positive cases plus signs.

(vii) *Embedding $\mathbb{Q}$.* Each r^* is a cut. For $r, s \in \mathbb{Q}$, $r^* + s^* = (r + s)^*$, $r^*s^* = (rs)^*$, and $r^* < s^* \iff r < s$.
For addition, if $p = u + v$ with $u < r$ and $v < s$, then $p < r + s$. Conversely, if $p < r + s$, choose $t > 0$ with $2t = r + s - p$ and put $r' = r - t$, $s' = s - t$; then $r' < r$, $s' < s$, and $p = r' + s'$. For positive products, if $p \leq uv$ with $0 < u < r$ and $0 < v < s$, then $p < rs$. Conversely, if $0 < p < rs$, choose positive $u < r$ and $v < s$ close enough to r and s that $p \leq uv$; if $p \leq 0$, use any positive $u < r$, $v < s$. The signed product

cases follow from the definitions by additive inverses. The order equivalence follows because $r < s$ puts r in s^* but not in r^*, while $r^* < s^*$ gives some $p < s$ with $p \notin r^*$, hence $r \le p < s$.
Commentary: The map $r \mapsto r^*$ preserves arithmetic and order.

Identify $\mathbb{Q}$ with the subfield $\mathbb{Q}^* := \{r^* : r \in \mathbb{Q}\} \subseteq \mathscr{D}$, and write $\mathbb{R}$ for $\mathscr{D}$. Items (i), (iii)–(vi) give an ordered field; item (ii) gives the least-upper-bound property; item (vii) embeds $\mathbb{Q}$ as a subfield.
Any two ordered fields with the least-upper-bound property are isomorphic, so this construction characterizes the real field up to ordered-field structure. □

Perspective 12.2.23. *Cauchy Construction as a Parallel Model*
Cuts build $\mathbb{R}$ here; Cauchy representatives give parallel model:

$$(a_n) \sim (b_n) \iff |a_n - b_n| \to 0, \qquad \mathbb{R} \simeq \{\text{rational Cauchy sequences}\}/\sim$$

This is the representative pattern from Perspective 12.2.1; compare the completion skeleton in Perspective 13.2.17. Definition 12.1.13 is the first well-definedness checkpoint.

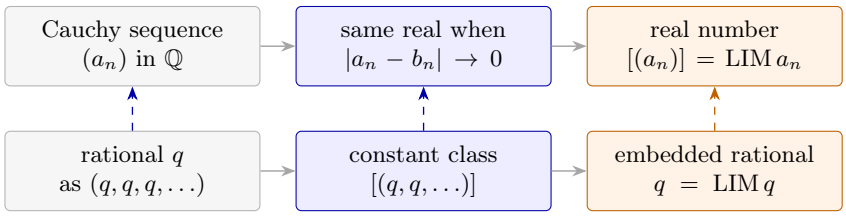


Figure 99: Real numbers as equivalence classes of rational Cauchy representatives.

Theorem 12.2.24. *Archimedean Property and Density of $\mathbb{Q}$*

(i) If $x, y \in \mathbb{R}$ and $x > 0$, then there is a positive integer n such that $nx > y$.
(ii) If $x, y \in \mathbb{R}$ and $x < y$, then there exists $p \in \mathbb{Q}$ such that $x < p < y$.

Proof.

(i) Let $A = \{nx : n \in \mathbb{N}\}$. If no nx exceeded y, then A would be bounded above. By Theorem 12.2.22, let $\alpha = \sup A$. Since $x > 0$, $\alpha - x < \alpha$, so $\alpha - x$ is not an upper bound.
Hence $\alpha - x < mx$ for some $m \in \mathbb{N}$, and $\alpha < (m+1)x \in A$, a contradiction.
(ii) Since $y - x > 0$, part (i) gives $n \in \mathbb{N}$ with $n(y - x) > 1$. Apply part (i) again to get $m_1, m_2 \in \mathbb{N}$ with $m_1 > nx$ and $m_2 > -nx$. Then some $m \in \mathbb{Z}$ satisfies $m - 1 \le nx < m$.
Thus $nx < m \le nx + 1 < ny$, so $x < m/n < y$ and $m/n \in \mathbb{Q}$.

Structural Commentary: Completeness makes $\{nx : n \in \mathbb{N}\}$ unbounded.
Key Insight: Archimedean unboundedness gives rational density. □

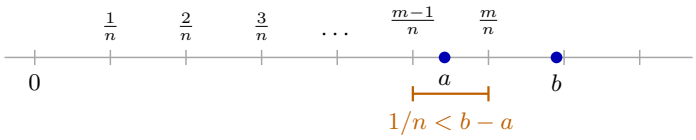


Figure 100: Choose n so the spacing $1/n$ is smaller than $b - a$; then one grid point m/n falls inside (a, b).

Theorem 12.2.25. *Existence and Uniqueness of Positive nth Roots*
For every $x \in \mathbb{R}$ with $x > 0$ and every positive integer n, $\exists$ a unique positive real number y such that $y^n = x$.

Proof.

(i) *Uniqueness.* If $0 < y_1 < y_2$, then $y_1^n < y_2^n$.
(ii) *Existence.* Let $E := \{t \in \mathbb{R} : t > 0,\ t^n < x\}$. $E \neq \varnothing$: for $t = x/(1+x)$, $0 < t < 1$ and $t^n \le t < x$.
Also E is bounded above by $1 + x$, since $t > 1 + x$ implies $t^n \ge t > x$. By Theorem 12.2.22, let $y = \sup E$.
For $0 < a < b$, $b^n - a^n = (b-a)(b^{n-1} + b^{n-2}a + \cdots + a^{n-1}) < (b-a)nb^{n-1}$.
If $y^n < x$, choose h with $0 < h < 1$ and $h < \frac{x - y^n}{n(y+1)^{n-1}}$.
Then $(y+h)^n - y^n < hn(y+h)^{n-1} < hn(y+1)^{n-1} < x - y^n$.
Thus $y + h \in E$, contradicting that y is an upper bound. If $y^n > x$, set $k := \frac{y^n - x}{ny^{n-1}}$. Then $0 < k < y$.
For $t \ge y - k$, $y^n - t^n \le y^n - (y-k)^n < kny^{n-1} = y^n - x$, so $t^n > x$ and $t \notin E$.
Hence $y - k$ is an upper bound for E, contradicting $y = \sup E$. Therefore $y^n = x$.

Structural Commentary: The root is $\sup\{t > 0 : t^n < x\}$.
Key Insight: Completeness supplies the missing root. □

Corollary 12.2.26. *Density of Rational and Irrational Points*
If $x < y$ in $\mathbb{R}$, then there exist $r \in \mathbb{Q}$ and $s \in \mathbb{R} \setminus \mathbb{Q}$ such that $x < r < y$ and $x < s < y$.

Proof. The rational point r is supplied by Theorem 12.2.24. Let $\eta = \sqrt{2}$.
Choose $n \in \mathbb{N}$ with $0 < \eta/n < y - x$, and choose $q \in \mathbb{Q}$ with $x - \eta/n < q < y - \eta/n$.
Then $s := q + \eta/n$ lies in (x, y) and is irrational, since $s \in \mathbb{Q}$ would imply $\eta = n(s - q) \in \mathbb{Q}$.
Structural Commentary: Irrational density is rational density shifted by a small irrational amount.
Key Insight: Once one irrational is known, every interval contains irrational translates of rational points. □

Corollary 12.2.27. *Product Rule for Positive Roots*
If $a, b > 0$ and n is a positive integer, then $(ab)^{1/n} = a^{1/n} b^{1/n}$.

Proof. Let $\alpha = a^{1/n}$ and $\beta = b^{1/n}$. Then $ab = \alpha^n \beta^n = (\alpha\beta)^n$ by commutativity of multiplication.
The uniqueness part of Theorem 12.2.25 gives the identity.
Structural Commentary: Use uniqueness after checking nth powers.
Key Insight: Root identities are uniqueness arguments. □

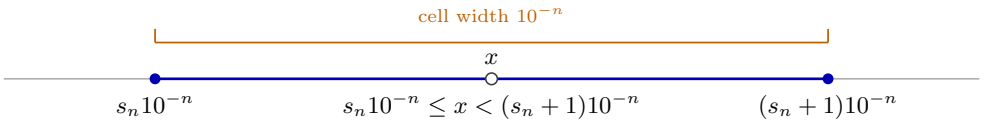


Figure 101: At precision 10^{-n}, the truncation $s_n 10^{-n}$ traps x in a cell of width 10^{-n}.

Proposition 12.2.28. *Decimal Truncations Squeeze a Real*
Let $x \ge 0$. For $n \ge 0$, set $s_n := \max\{k \in \mathbb{Z}_{\ge 0} : k10^{-n} \le x\}$.
Then $s_n 10^{-n} \le x < (s_n + 1)10^{-n}$; and $0 \le x - s_n 10^{-n} < 10^{-n}$; so $s_n 10^{-n} \to x$.
Moreover $s_{n+1} = 10s_n + a_{n+1}$ for some $a_{n+1} \in \{0, \dots, 9\}$.

Proof. The maximum exists by the Archimedean property, Theorem 12.2.24, and well-ordering, Principle 12.3.3.
Maximality gives the displayed inequalities. The squeeze and Theorem 12.4.25 give $s_n 10^{-n} \to x$.
At levels n and $n + 1$, $10s_n \le s_{n+1} < 10s_n + 10$, so $a_{n+1} := s_{n+1} - 10s_n \in \{0, \dots, 9\}$.
Structural Commentary: Decimal existence is an Archimedean grid argument followed by a squeeze.
Key Insight: Refining the mesh by a factor of 10 adds exactly one digit. □

Definition 12.2.29. *Extended Real Number System*
The *extended real number system* is $\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}$.
The order on $\mathbb{R}$ is preserved, and $-\infty < x < +\infty$, $(x \in \mathbb{R})$.
Every nonempty subset of $\overline{\mathbb{R}}$ has least upper bound; if $E \subseteq \mathbb{R}$ is nonempty and unbounded above in $\mathbb{R}$, then $\sup E = +\infty$. Same convention applies to greatest lower bounds.

Remark 12.2.30. *Extended Arithmetic Conventions*
$\overline{\mathbb{R}}$ is not a field. For real x, $x + \infty = +\infty$, $x - \infty = -\infty$, and $\frac{x}{+\infty} = \frac{x}{-\infty} = 0$.
If $x > 0$, then $x(+\infty) = +\infty$ and $x(-\infty) = -\infty$; if $x < 0$, the signs reverse.
Real numbers are called *finite* when one wants to distinguish them from $\pm\infty$.

Definition 12.2.31. *Complex Numbers*
A *complex number* is an ordered pair (a, b) of real numbers. For $x = (a, b)$ and $y = (c, d)$, define equality componentwise and set $x + y := (a + c, b + d)$ and $xy := (ac - bd, ad + bc)$

Theorem 12.2.32. *The Complex Numbers Form a Field*
With the operations in Definition 12.2.31, the set of complex numbers is a field.
Its additive and multiplicative identities are $(0, 0)$ and $(1, 0)$.

Proof. The field axioms are checked directly from the two coordinate formulas and the field structure of $\mathbb{R}$.
If $x = (a, b) \neq (0, 0)$, then $a^2 + b^2 > 0$ by Proposition 12.2.13, and $x^{-1} = \left(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2}\right)$ satisfies $xx^{-1} = (1, 0)$.
Structural Commentary: Inverses use conjugate-over-square.
Key Insight: Positivity of $a^2 + b^2$ supplies the denominator for every nonzero complex number. □

Proposition 12.2.33. *Real Numbers Inside Complex Numbers*
For all real a, b, $(a,0)+(b,0)=(a+b,0)$, and $(a,0)(b,0)=(ab,0)$.
Thus complex numbers of form $(a,0)$ have same arithmetic as real numbers, and identify $(a,0)$ with a.
The complex unit is then $i := (0,1)$.

Proposition 12.2.34. *Coordinate Form and i^2*
We have $i^2=-1$. Moreover, if $a,b\in\mathbb{R}$, then $(a,b)=a+bi$.

Proof. First, $i^2=(0,1)(0,1)=(-1,0)=-1$. Also, $a+bi=(a,0)+(b,0)(0,1)=(a,0)+(0,b)=(a,b)$.
Structural Commentary: The ordered-pair construction recovers the usual notation after $i=(0,1)$ is named.
Key Insight: The symbol i is not assumed; it is produced by the pair model. □

Definition 12.2.35. *Conjugate, Real Part, and Imaginary Part*
If $z=a+bi$ with $a,b\in\mathbb{R}$, then the *complex conjugate* of z is $\overline{z} := a-bi$.
The numbers a and b are the *real part* and *imaginary part*, written $a=\operatorname{Re}(z)$ and $b=\operatorname{Im}(z)$.

Proposition 12.2.36. *Basic Conjugation Identities*
If $z,w\in\mathbb{C}$, then

(i) $\overline{z+w}=\overline{z}+\overline{w}$;
(ii) $\overline{zw}=\overline{z}\,\overline{w}$;
(iii) $z+\overline{z}=2\operatorname{Re}(z)$ *and* $z-\overline{z}=2i\operatorname{Im}(z)$;
(iv) $z\overline{z}$ *is real and positive unless* $z=0$.

Proof. Write $z=a+bi$ and $w=c+di$.

(i) $\overline{z+w}=(a+c)-(b+d)i=\overline{z}+\overline{w}$.
(ii) $\overline{zw}=(ac-bd)-(ad+bc)i=(a-bi)(c-di)=\overline{z}\,\overline{w}$.
(iii) $z+\overline{z}=2a=2\operatorname{Re}(z)$ and $z-\overline{z}=2bi=2i\operatorname{Im}(z)$.
(iv) $z\overline{z}=a^2+b^2$, positive unless $a=b=0$, i.e. unless $z=0$.

Structural Commentary: Conjugation fixes Re and negates Im.
Key Insight: $z\overline{z}=a^2+b^2\geq 0$. □

Definition 12.2.37. *Absolute Value of a Complex Number*
For $z\in\mathbb{C}$, define $|z| := (z\overline{z})^{1/2}$, the nonnegative square root supplied by Theorem 12.2.25 and Proposition 12.2.36.

Theorem 12.2.38. *Absolute Value Identities*
For $z,w\in\mathbb{C}$,

(i) $|z|>0$ *unless* $z=0$, *and* $|0|=0$;
(ii) $|\overline{z}|=|z|$;
(iii) $|zw|=|z|\,|w|$;
(iv) $|\operatorname{Re}(z)|\leq|z|$;
(v) $|z+w|\leq|z|+|w|$.

Proof.

(i) If $z=a+bi$, then $|z|=(a^2+b^2)^{1/2}$; this is positive unless $a=b=0$, and $|0|=0$.
(ii) $|\overline{z}|^2=\overline{z}z=z\overline{z}=|z|^2$, so nonnegative square roots give $|\overline{z}|=|z|$.
(iii) $|zw|^2=zw\overline{zw}=z\overline{z}w\overline{w}=|z|^2\,|w|^2$; take nonnegative square roots.
(iv) If $z=a+bi$, then $|\operatorname{Re}(z)|=|a|\leq(a^2+b^2)^{1/2}=|z|$.
(v) $z\overline{w}$ and $\overline{z}w$ are conjugates, $|z+w|^2=|z|^2+2\operatorname{Re}(z\overline{w})+|w|^2\leq|z|^2+2\,|z\overline{w}|+|w|^2=(|z|+|w|)^2$, using (iii) and (iv). Take nonnegative square roots.

Structural Commentary: The complex absolute value behaves like length because $z\overline{z}$ is a squared length.
Key Insight: Triangle inequality is a real-part estimate plus multiplicativity. □

Theorem 12.2.39. *Finite Schwarz Inequality for Complex Scalars*
If $a_1,\dots,a_n$ and $b_1,\dots,b_n$ are complex numbers, then $\left|\sum_{j=1}^{n} a_j\overline{b_j}\right|^2\leq\left(\sum_{j=1}^{n}|a_j|^2\right)\left(\sum_{j=1}^{n}|b_j|^2\right)$.

Proof. This is Theorem 2.1.11 in $\mathbb{C}^n$ with the standard inner product.
Equivalently, set $A=\sum|a_j|^2$, $B=\sum|b_j|^2$, and $C=\sum a_j\overline{b_j}$; when $B>0$, $\sum|Ba_j-Cb_j|^2=B(AB-|C|^2)\geq 0$.
Structural Commentary: This scalar inequality is the coordinate model for geometric Cauchy–Schwarz inequality.
Key Insight: A nonnegative square sum controls the mixed term. □

12.3 Cardinality and Infinite Sets

Cardinality counts sets by matching them with other sets, using the earlier language of functions, power sets, and bijections.

Definition 12.3.1. *Same Cardinality*
Two sets A and B have *same cardinality*, or are *equivalent*, if $\exists$ a bijection $f : A \to B$. Then $A \sim_{\#} B$.
The relation $\sim_{\#}$ is an equivalence relation: $A \sim_{\#} A$ by identity map, symmetry follows from Theorem 12.1.18, and transitivity follows by composing bijections.

Definition 12.3.2. *Finite, Countable, and Uncountable*
Set $J_0 := \emptyset$, and for $n \geq 1$ set $J_n := \{1, 2, \ldots, n\}$. A set A is

(i) *finite* if $A \sim_{\#} J_n$ for some $n \in \mathbb{N}$;
(ii) *infinite* if it is not finite;
(iii) *countable* if $A \sim_{\#} \mathbb{N}$;
(iv) *uncountable* if it is neither finite nor countable;
(v) *at most countable* if it is finite or countable.

Countable sets are also called enumerable or denumerable.

Principle 12.3.3. *Well-Ordering of $\mathbb{N}$*
Every nonempty subset of $\mathbb{N}$ has a least element, justifying first unused indices, first repeated values, and increasing listings of infinite subsets of $\mathbb{N}$.

Proposition 12.3.4. *Finite Cardinality is Well Defined*
If $A \sim_{\#} J_m$ and $A \sim_{\#} J_n$, then $m = n$. Thus $\#A = n$ is well defined for finite sets.

Proof. By transitivity of $\sim_{\#}$, it suffices to prove $J_m \sim_{\#} J_n \Rightarrow m = n$. Induct on m.
For $m = 0$, only $J_0 = \emptyset$ is bijective to J_0.
If $m, n \geq 1$ and $f : J_m \to J_n$ is bijective, remove m and its partner $f(m)$. The remaining sets are bijective and relabel as J_{m-1} and J_{n-1}, so induction gives $m - 1 = n - 1$, hence $m = n$.
Structural Commentary: Finite counting is invariant under bijection.
Key Insight: Remove one matched element and count down. □

Definition 12.3.5. *Summation over a Finite Set*
Let X be a finite set with $\#X = n$, and let $f : X \to \mathbb{C}$.
Choose any bijection $g : J_n \to X$, where $J_n = \{1, \ldots, n\}$, and set $\sum_{x \in X} f(x) := \sum_{i=1}^{n} f(g(i))$.
When $X = \emptyset$, this is the empty sum 0.
Key Insight: Proposition 12.3.6 shows that the value is independent of the chosen bijection.

Proposition 12.3.6. *Finite Set Summation Laws*
Let X, Y be finite sets.

(i) Well-definedness: different bijections $J_{\#X} \to X$ give the same value in Definition 12.3.5.
(ii) Substitution: if $h : Y \to X$ is a bijection and $f : X \to \mathbb{C}$, then $\sum_{x \in X} f(x) = \sum_{y \in Y} f(h(y))$.
(iii) Linearity: for $f, g : X \to \mathbb{C}$ and $c \in \mathbb{C}$, $\sum_{x \in X} (f(x)+g(x)) = \sum_{x \in X} f(x) + \sum_{x \in X} g(x)$ and $\sum_{x \in X} cf(x) = c \sum_{x \in X} f(x)$.
(iv) Disjoint unions: if $X \cap Y = \emptyset$ and $f : X \cup Y \to \mathbb{C}$, then $\sum_{z \in X \cup Y} f(z) = \sum_{x \in X} f(x) + \sum_{y \in Y} f(y)$.
(v) Finite Fubini: if $f : X \times Y \to \mathbb{C}$, then $\sum_{x \in X} \sum_{y \in Y} f(x, y) = \sum_{(x,y) \in X \times Y} f(x, y) = \sum_{y \in Y} \sum_{x \in X} f(x, y)$.

Proof.

(i) For $f : X \to \mathbb{C}$, let $n = \#X$ and let $g, k : J_n \to X$ be bijections. Induct on n. The case $n = 0$ is empty. If $n \geq 1$, let $j \in J_n$ satisfy $k(j) = g(n)$. Removing n from g and j from k gives bijections onto $X \setminus \{g(n)\}$; after relabeling by J_{n-1}, the induction hypothesis makes the remaining sums equal. Adding the common last term $f(g(n)) = f(k(j))$ gives equality of the two sums.

(ii) Let $g : J_{\#Y} \to Y$ be a bijection. Then $h \circ g : J_{\#Y} \to X$ is a bijection, so $\#Y = \#X$ by Proposition 12.3.4. Hence, by (i) and Definition 12.3.5, $\sum_{x \in X} f(x) = \sum_{i=1}^{\#Y} f(h(g(i))) = \sum_{y \in Y} f(h(y))$.

(iii) Choose a bijection $\gamma : J_{\#X} \to X$.
Proposition 12.1.4 gives $\sum_{x \in X} (f(x) + g(x)) = \sum_{x \in X} f(x) + \sum_{x \in X} g(x)$ and $\sum_{x \in X} cf(x) = c \sum_{x \in X} f(x)$

(iv) Let $\alpha : J_{\#X} \to X$ and $\beta : J_{\#Y} \to Y$ be bijections, and enumerate $X \cup Y$ by listing α first and then β. Since $X \cap Y = \emptyset$, this is a bijection, and Proposition 12.1.4 gives $\sum_{z \in X \cup Y} f(z) = \sum_{x \in X} f(x) + \sum_{y \in Y} f(y)$.

(v) Let $\alpha : J_{\#X} \to X$ and $\beta : J_{\#Y} \to Y$ be bijections.
The maps $\ell((i-1)\#Y + j) = (\alpha(i), \beta(j))$ and $r((j-1)\#X + i) = (\alpha(i), \beta(j))$ for $i \in J_{\#X}$ and $j \in J_{\#Y}$ give the two lexicographic bijections $J_{(\#X)(\#Y)} \to X \times Y$.
Applying Definition 12.3.5, Proposition 12.1.4, and (i) yields $\sum_{x\in X}\sum_{y\in Y} f(x,y) = \sum_{(x,y)\in X\times Y} f(x,y) = \sum_{y\in Y}\sum_{x\in X} f(x,y)$.

Structural Commentary: A finite set sum is an order-free finite sum.
Key Insight: Bijections are bookkeeping devices; the value does not depend on the bookkeeping order. □

Definition 12.3.7. *Sequences as Enumerations*
A *sequence* in set A is $x : \mathbb{N} \to A$, written (x_n), where $x_n := x(n)$. Terms of sequence need not be distinct.
If A is countable, then its elements can be arranged as a sequence with distinct terms. Conversely, a sequence whose terms are distinct and whose range is A gives a bijection $\mathbb{N} \to A$.

Example 12.3.8. *The Integers Are Countable*
The set $\mathbb{Z}$ is countable. One explicit enumeration is given by

$$f(n) = \begin{cases} \frac{n}{2}, & n \text{ even}, \\ -\frac{n-1}{2}, & n \text{ odd}, \end{cases} \qquad n \in \mathbb{N}.$$

This lists $\mathbb{Z}$ as $0, 1, -1, 2, -2, 3, -3, \ldots$.

Theorem 12.3.9. *Infinite Subsets of Countable Sets*
Every infinite subset of a countable set is countable.

Proof. Let $E \subseteq A$ be infinite, and enumerate A by distinct terms (x_n). Let $n_1 < n_2 < \cdots$ be exactly the indices for which $x_{n_k} \in E$, listed in increasing order. Then $k \mapsto x_{n_k}$ is a bijection from $\mathbb{N}$ onto E.
Structural Commentary: The proof keeps only the entries of the original enumeration that land in E.
Key Insight: An infinite subset of an enumeration inherits an enumeration by first appearances. □

Proposition 12.3.10. *Subsets and Images of At Most Countable Sets*
Let A be at most countable.

(i) If $B \subseteq A$, then B is at most countable.
(ii) If $f : A \to Y$ is a function, then $f(A)$ is at most countable.

Proof. If A is finite, both claims are immediate. Otherwise write $A = \{a_n : n \in \mathbb{N}\}$ with the a_n distinct.

(i) If B is infinite, let $I_B := \{n : a_n \in B\} = \{n_1 < n_2 < \cdots\}$.
Then $k \mapsto a_{n_k}$ bijects $\mathbb{N}$ onto B; the finite case is trivial.
(ii) Let $I := \{n : f(a_n) \notin \{f(a_m) : m < n\}\}$. By (i), I is at most countable, $n \mapsto f(a_n)$ bijects I onto $f(A)$.

Structural Commentary: Subsets select indices; images select first appearances.
Key Insight: Subsets and images preserve at-most-countability. □

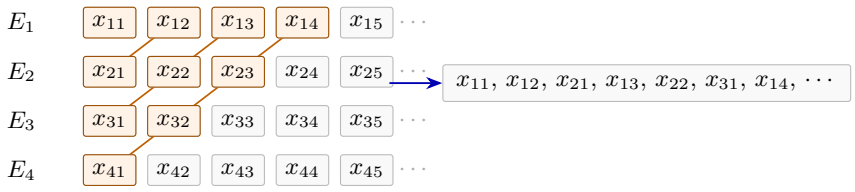


Figure 102: Diagonal scan of countable rows into one sequence, skipping repeats.

Theorem 12.3.11. *Countable Union of Countable Sets*
Let (E_n) be a sequence of countable sets and put $S := \bigcup_{n=1}^{\infty} E_n$. Then S is countable.

Proof. Enumerate each E_n as $(x_{nk})_{k\in\mathbb{N}}$.
Reading the array (x_{nk}) by diagonals, as in Figure 102, gives a sequence whose image is S.
Thus S is at most countable by Proposition 12.3.10.
Since $E_1 \subseteq S$ and E_1 is countable, S is infinite; hence S is countable.
Structural Commentary: The diagonal order turns a two-indexed list into a one-indexed list.
Key Insight: Countably many countable rows can still be scanned by one sequence. □

Corollary 12.3.12. *At Most Countable Unions*
If A is at most countable and B_α is at most countable for each $\alpha \in A$, then $\bigcup_{\alpha\in A} B_\alpha$ is at most countable.

Proof. Let $T := \bigcup_{\alpha\in A} B_\alpha$. If $T = \emptyset$, we are done. Otherwise enumerate the nonempty sets among the B_α as $B_{\alpha_1}, B_{\alpha_2}, \ldots$, repeating indices if needed, and list each B_{α_j} as $(b_{jk})_{k\in\mathbb{N}}$.
A diagonal scan of (b_{jk}) has image T, so Proposition 12.3.10 makes T at most countable.
Structural Commentary: Finite pieces cause no difficulty; they can be embedded into countable rows.
Key Insight: At most countable indexing + at most countable indexed sets gives at most countable total set. □

Theorem 12.3.13. *Finite Tuples over a Countable Set*
Let A be countable. For each fixed $n \in \mathbb{N}$, the set of all n-tuples $(a_1, \ldots, a_n)$ with $a_k \in A$ is countable.

Proof. Let $B_n := \{(a_1, \ldots, a_n) : a_k \in A\}$, $B_1 = A$. If B_{n-1} is countable, $B_n = \bigcup_{b\in B_{n-1}} \{(b, a) : a \in A\}$, each set $\{(b, a) : a \in A\}$ is equivalent to A. Hence B_n is countable by Theorem 12.3.11.
Structural Commentary: Build an n-tuple by appending one coordinate to an $(n-1)$-tuple.
Key Insight: Induction plus countable unions controls finite products. □

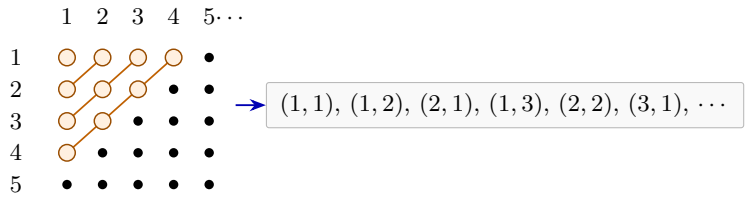


Figure 103: Diagonal enumeration of $\mathbb{N} \times \mathbb{N}$ by finite $i + j$ diagonals.

Corollary 12.3.14. *Countable Cartesian Products*
If A, B are at most countable, then $A \times B$ is at most countable.
If A, B are countable and nonempty, then $A \times B$ is countable. Hence $\mathbb{N} \times \mathbb{N}$ is countable.

Proof. For $a \in A$, $\{a\} \times B \sim B$, and $A \times B = \bigcup_{a\in A} \big(\{a\} \times B\big)$, thus $A \times B$ at most countable (Corollary 12.3.12). If A, B are countable and $b_0 \in B$, then $a \mapsto (a, b_0)$ injects A into $A \times B$, so $A \times B$ is infinite; hence it is countable.
Structural Commentary: A product is a countable union of countable slices.
Key Insight: Two countable coordinates can still be scanned by one sequence. □

Corollary 12.3.15. *The Rationals Are Countable*
The set $\mathbb{Q}$ is countable.

Proof. Let $D := \{(a, b) \in \mathbb{Z} \times \mathbb{Z} : a \neq 0\}$ and $\pi : D \to \mathbb{Q}$ by $\pi(a, b) = b/a$.
Since $\mathbb{Z}$ is countable, Corollary 12.3.14 and Proposition 12.3.10 make D at most countable. Since $\pi(D) = \mathbb{Q}$, Proposition 12.3.10 makes $\mathbb{Q}$ at most countable. As $\mathbb{Z} \subseteq \mathbb{Q}$, it is countable.
Structural Commentary: Repeated fractions only add repetitions to the list.
Key Insight: Countable pairs produce countably many rational values. □

Theorem 12.3.16. *Cantor's Theorem*
For every set X, no function $f : X \to \mathcal{P}(X)$ is surjective. Hence X and $\mathcal{P}(X)$ cannot have the same cardinality.

Proof. Suppose $f : X \to \mathcal{P}(X)$ were surjective, and set $D := \{x \in X : x \notin f(x)\}$.
Since $D \subseteq X$, we have $D \in \mathcal{P}(X)$, so surjectivity gives $D = f(a)$ for some $a \in X$.
But then $a \in D \iff a \notin f(a) \iff a \notin D$, a contradiction.
Structural Commentary: The diagonal set is defined to disagree with each proposed row at its own index.
Key Insight: Power sets are too large to be exhausted by their base set. □

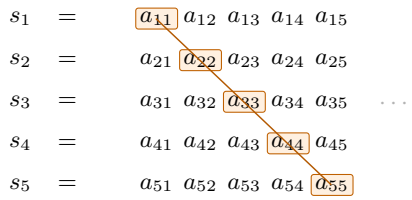


Figure 104: Cantor diagonalization: $s(n) = 1 - a_{nn}$ differs from row s_n at position n.

Theorem 12.3.17. *Binary Sequences Are Uncountable*
Let A be the set of all sequences whose terms are the digits 0 and 1. Then A is uncountable.

Proof. Suppose A were countable, $A = \{s_1, s_2, \ldots\}$ with $s_n = (s_n(k))_{k\in\mathbb{N}}$. Define $s \in A$ by $s(k) := 1 - s_k(k)$. Then $s(n) \neq s_n(n)$ for every n, so $s \neq s_n$ for all n. Hence $s \notin A$, a contradiction. Thus A is uncountable.
Structural Commentary: The construction attacks the diagonal of a proposed enumeration.
Key Insight: A sequence can be built to escape every row of a countable list. □

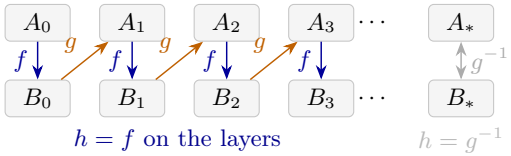


Figure 105: Schroeder–Bernstein: alternating images leave a remainder paired by g^{-1}.

Theorem 12.3.18. *Schroeder–Bernstein Theorem*
If there are injections $f : A \to B$ and $g : B \to A$, then there is a bijection $h : A \to B$.

Proof. Let $A_0 := A \setminus g(B)$, and recursively define $B_n := f(A_n)$ and $A_{n+1} := g(B_n)$.
The sets A_n are pairwise disjoint: if $A_i \cap A_j \neq \emptyset$ with $i < j$, then $i = 0$ already contradicts $A_j \subseteq g(B)$; if $i > 0$, injectivity of g and then f steps backward to $A_{i-1} \cap A_{j-1} \neq \emptyset$, and iteration gives the same contradiction.
Since f is injective, the sets $B_n = f(A_n)$ are pairwise disjoint as well.
Let $A_* := A \setminus \bigcup_{n\geq 0} A_n$ and $B_* := B \setminus \bigcup_{n\geq 0} B_n$. The map g restricts to a bijection $B_* \to A_*$.
Indeed, $g(B_*)$ avoids every A_n, while if $a \in A_*$ then $a \notin A_0$, so $a = g(b)$ for a unique $b \in B$; this b cannot lie in any B_n, otherwise $a \in A_{n+1}$.
Define $h(x) := \begin{cases} f(x), & x \in \bigcup_{n\geq 0} A_n, \\ g^{-1}(x), & x \in A_*, \end{cases}$ where g^{-1} denotes the inverse of $g : B_* \to A_*$.
This maps each A_n bijectively onto B_n and maps A_* bijectively onto B_*. Hence h is a bijection $A \to B$.
Structural Commentary: Two injections carve both sets into matched layers plus a common remainder.
Key Insight: Use f on the escaping layers and g^{-1} on the stable remainder. □

Principle 12.3.19. *Choice Principle*
For nonempty sets $\{X_i\}_{i\in I}$, the axiom of choice asserts $\exists c : I \to \bigcup_{i\in I} X_i$ such that $c(i) \in X_i\ \forall i \in I$.
Use direct construction for finite or explicit choices; use choice only for arbitrary simultaneous choices.

Definition 12.3.20. *Ordered Set Vocabulary*

(i) $\leq_X$ is a *partial order* if it is reflexive, anti-symmetric, and transitive.
(ii) $C \subseteq X$ is a *chain* if any two points of C are comparable.
(iii) X is *well-ordered* if every nonempty $Y \subseteq X$ has a least element.
(iv) For $Y \subseteq X$, $u \in X$ is an *upper bound* if $y \leq_X u$ for all $y \in Y$.
(v) $m \in Y$ is *maximal* if no $y \in Y$ satisfies $m <_X y$; minimal is dual.

Principle 12.3.21. *Zorn's Lemma*
If a nonempty partially ordered set X has the property that every chain in X has an upper bound in X, then X contains a maximal element. This is equivalent in strength to the axiom of choice, Principle 12.3.19.

Method 12.3.22. *Cardinality Proof Patterns*

(i) Same cardinality: construct a bijection.
(ii) Countable: construct a sequence whose range contains every element.

(iii) Uncountable: show every proposed sequence misses some element.
(iv) Choice/maximality: cite Principle 12.3.19 for arbitrary simultaneous choices and Principle 12.3.21 for maximal elements in partially ordered sets.

Key Idea: Use maps: bijections prove equal cardinality, enumerations $\mathbb{N} \to A$ prove countability, and diagonal arguments show every proposed enumeration misses an element.

12.4 Real Sequences and Completeness Methods

Sequences are the first testing ground for limits, order, and completeness. The main tools are subsequences, Cauchy criteria, monotone convergence, limsup/liminf, and Bolzano–Weierstrass.

Definition 12.4.1. *Convergent Real and Complex Sequences*
A real or complex sequence (s_n) *converges* to s if, $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $n \geq N$ implies $|s_n - s| < \varepsilon$.
Notation is $s_n \to s$ or $\lim_{n\to\infty} s_n = s$. If no such s exists, the sequence *diverges*.

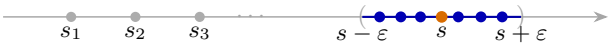


Figure 106: For all $n \geq N$, the terms s_n stay inside $(s-\varepsilon, s+\varepsilon)$.

Definition 12.4.2. *Range and Boundedness of a Real Sequence*
The *range* of (s_n) is set $\{s_n : n \in \mathbb{N}\}$. Real sequence is *bounded* if its range is bounded above and below in $\mathbb{R}$.

Theorem 12.4.3. *Limit Algebra for Complex Sequences*
Suppose (s_n) and (t_n) are complex sequences with $s_n \to s$ and $t_n \to t$. Then, for every $c \in \mathbb{C}$,

(i) $s_n + t_n \to s + t$;
(ii) $cs_n \to cs$ *and* $c + s_n \to c + s$;
(iii) $s_n t_n \to st$;
(iv) if $s \neq 0$ *and* $s_n \neq 0$ *for all* n, *then* $1/s_n \to 1/s$.

Proof.

(i) Given $\varepsilon > 0$, choose N such that $|s_n - s| < \varepsilon/2$, $|t_n - t| < \varepsilon/2$ for $n \geq N$.
Then $|(s_n + t_n) - (s + t)| \leq |s_n - s| + |t_n - t| < \varepsilon$.
(ii) Scalar multiplication uses $|cs_n - cs| = |c|\,|s_n - s|$; translation is (i) with constant sequence c.
(iii) Since $s_n \to s$, choose $N \geq 2$ with $|s_n - s| < 1$ for $n \geq N$ and set $M := \max\{|s_1|, \ldots, |s_{N-1}|, |s| + 1\}$.
Then $|s_n t_n - st| \leq |s_n|\,|t_n - t| + |t|\,|s_n - s| \leq M\,|t_n - t| + |t|\,|s_n - s| \to 0$.
(iv) If $s \neq 0$, then for all sufficiently large n, $|s_n| > |s|/2$. Hence $\left|\frac{1}{s_n} - \frac{1}{s}\right| = \frac{|s_n - s|}{|s_n||s|} \leq \frac{2}{|s|^2}\,|s_n - s| \to 0$.

Structural Commentary: Boundedness controls products; nonzero limits control reciprocals.
Key Insight: Product and reciprocal rules are estimates after the easy sum rule. □

Theorem 12.4.4. *Coordinatewise Convergence in $\mathbb{R}^k$*
Let $\mathbf{x}_n = (\alpha_{j,n})_{j=1}^k$ and $\mathbf{x} = (\alpha_j)_{j=1}^k$ in $\mathbb{R}^k$. Then $\mathbf{x}_n \to \mathbf{x} \iff \alpha_{j,n} \to \alpha_j$ $(1 \leq j \leq k)$.
Moreover, if $\mathbf{x}_n \to \mathbf{x}$, $\mathbf{y}_n \to \mathbf{y}$, and $\beta_n \to \beta$ in $\mathbb{R}$, then $\mathbf{x}_n + \mathbf{y}_n \to \mathbf{x} + \mathbf{y}$; $\mathbf{x}_n \cdot \mathbf{y}_n \to \mathbf{x} \cdot \mathbf{y}$; $\beta_n \mathbf{x}_n \to \beta \mathbf{x}$.

Proof.

($\Rightarrow$) Since $|\alpha_{j,n} - \alpha_j| \leq \|\mathbf{x}_n - \mathbf{x}\|_2$ for each j, each coordinate converges.
($\Leftarrow$) Given $\varepsilon > 0$, choose N so that $|\alpha_{j,n} - \alpha_j| < \varepsilon/\sqrt{k}$ for all j and $n \geq N$. Then $\|\mathbf{x}_n - \mathbf{x}\|_2 < \varepsilon$.

The algebraic conclusions follow coordinatewise from Theorem 12.4.3.
Structural Commentary: Euclidean convergence is exactly simultaneous coordinate convergence.
Key Insight: The norm packages finitely many coordinate estimates into one estimate. □

Remark 12.4.5. *Coordinate Metrics Give the Same Limits*
For $k \in \mathbb{N}$ and $x, y \in \mathbb{R}^k$, set $d_1(x,y) = \sum_{j=1}^k |x_j - y_j|$, $d_2(x,y) = \left(\sum_{j=1}^k |x_j - y_j|^2\right)^{1/2}$, $d_\infty(x,y) = \max_{1\leq j\leq k} |x_j - y_j|$.
Since $d_\infty \leq d_2 \leq d_1 \leq k d_\infty$, these have same convergent sequences, equivalently same coordinatewise criterion.

Definition 12.4.6. *Subsequences and Subsequential Limits*
Let (p_n) be a sequence. If $n_1 < n_2 < n_3 < \cdots$ is a strictly increasing sequence of positive integers, then (p_{n_k}) is a *subsequence* of (p_n). If $p_{n_k} \to p$, then p is a *subsequential limit* of (p_n).

Remark 12.4.7. *Subsequences as Selection Maps*
Equivalently, a subsequence is the composite of the original sequence, viewed as a function on $\mathbb{N}$ as in Definition 12.3.7, with a strictly increasing map $\sigma : \mathbb{N} \to \mathbb{N}$. In this notation the subsequence is $k \mapsto p_{\sigma(k)}$.
Key Insight: A subsequence keeps the original order but is allowed to skip terms.

Remark 12.4.8. *Subsequences of Convergent Sequences*
If $p_n \to p$, then every subsequence of (p_n) also converges to p. Conversely, if every subsequence of (p_n) converges to p, then (p_n) itself converges to p, since (p_n) is one of its subsequences.

Definition 12.4.9. *Monotone Real Sequence*
A real sequence (s_n) is *monotonically increasing* if $s_n \le s_{n+1}$ for all n, and *monotonically decreasing* if $s_n \ge s_{n+1}$ for all n. A *monotonic* sequence is either increasing or decreasing.

Definition 12.4.10. *Dominant Index*
For a real sequence (s_n), an index n is *dominant* if $s_n \ge s_m$ for every $m > n$.

Theorem 12.4.11. *Monotone Subsequence Theorem*
Every real sequence has a monotone subsequence.

Proof. If the dominant indices $n_1 < n_2 < \cdots$ are infinite, then $s_{n_1} \ge s_{n_2} \ge \cdots$, giving a decreasing subsequence. Otherwise choose N beyond all dominant indices and set $n_1 > N$.
Given $n_k > N$, nondominance gives $n_{k+1} > n_k$ with $s_{n_{k+1}} > s_{n_k}$; hence (s_{n_k}) is increasing.
Structural Commentary: Dominant indices give decreasing case; nondominance recursively gives increasing case.
Key Insight: The proof organizes arbitrary real behavior into one ordered thread. □

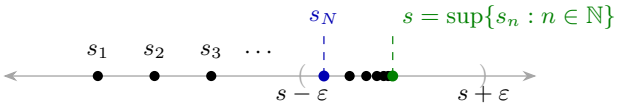


Figure 107: For an increasing bounded sequence, $s = \sup\{s_n\}$ is the limit.

Theorem 12.4.12. *Monotone Convergence Theorem for Sequences*
A monotonic real sequence converges if and only if it is bounded.

Proof.
($\Rightarrow$) If $s_n \to s$, choose N with $|s_n - s| < 1$ for $n \ge N$. The tail lies in $(s-1, s+1)$ and the initial set $\{s_1, \ldots, s_{N-1}\}$ is finite, so (s_n) is bounded.
($\Leftarrow$) Let $E := \{s_n : n \in \mathbb{N}\}$. If (s_n) is increasing, set $s := \sup E$. Given $\varepsilon > 0$, choose N with $s - \varepsilon < s_N \le s$; then $s - \varepsilon < s_n \le s$ for $n \ge N$, so $s_n \to s$. If (s_n) is decreasing, set $s := \inf E$. Given $\varepsilon > 0$, choose N with $s \le s_N < s + \varepsilon$; then $s \le s_n \le s_N < s + \varepsilon$ for $n \ge N$, so $s_n \to s$.
Structural Commentary: Monotonicity traps bounded tails at $\sup E$ or $\inf E$.
Key Insight: The least upper bound is forced to be the limit. □

Theorem 12.4.13. *Bolzano–Weierstrass for Real Sequences*
Every bounded real sequence has a convergent subsequence.

Proof. By Theorem 12.4.11, choose a monotone subsequence. It is bounded because the original sequence is bounded, so Theorem 12.4.12 gives convergence.
Structural Commentary: Boundedness prevents escape; monotone subsequences force one ordered thread.
Key Insight: Monotone subsequence plus monotone convergence is the real-line compactness mechanism. □

Definition 12.4.14. *Cauchy Real Sequence*
A real sequence (s_n) is *Cauchy* if, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that $m, n \ge N$ implies $|s_n - s_m| < \varepsilon$.

Lemma 12.4.15. *Cauchy Plus One Convergent Subsequence*
If a real Cauchy sequence has a subsequence converging to $s \in \mathbb{R}$, then the whole sequence converges to s.

Proof. Given $\varepsilon > 0$, choose N_0 so that $m, n \ge N_0$ implies $|s_m - s_n| < \varepsilon/2$. Choose j with $n_j \ge N_0$ and $|s_{n_j} - s| < \varepsilon/2$. Then for every $n \ge N_0$, $|s_n - s| \le |s_n - s_{n_j}| + |s_{n_j} - s| < \varepsilon$.
Structural Commentary: Cauchy tails follow any convergent thread.
Key Insight: Mutual tail closeness turns one subsequential limit into the full limit. □

Theorem 12.4.16. *Real Cauchy Criterion*
A real sequence converges in $\mathbb{R}$ if and only if it is Cauchy.

Proof. If $s_n \to s$, choose N such that $|s_n - s| < \varepsilon/2$ for $n \geq N$; then $|s_n - s_m| < \varepsilon$ for $m, n \geq N$. Conversely, suppose (s_n) is Cauchy. Choose N so that $m, n \geq N$ implies $|s_m - s_n| < 1$. Then the tail lies in $(s_N - 1, s_N + 1)$, and the finitely many initial terms are bounded, so (s_n) is bounded. By Theorem 12.4.13, some subsequence converges; Lemma 12.4.15 makes the whole sequence converge.
Structural Commentary: Completeness of $\mathbb{R}$ means Cauchy behavior has a real endpoint.
Key Insight: Cauchy gives boundedness, Bolzano–Weierstrass gives a subsequential endpoint, and the Cauchy property pulls the whole sequence there. □

Definition 12.4.17. *Divergence to $\pm\infty$*
For a real sequence (s_n),

$$s_n \to +\infty \iff \forall M \in \mathbb{R},\ \exists N \in \mathbb{N} :\ n \geq N \Rightarrow s_n \geq M,$$
$$s_n \to -\infty \iff \forall M \in \mathbb{R},\ \exists N \in \mathbb{N} :\ n \geq N \Rightarrow s_n \leq M.$$

This extends $\to$ notation for divergent real sequences; Definition 12.4.1 is unchanged.

Proposition 12.4.18. *Unbounded Real Sequences Have Infinite Subsequences*
A real sequence is unbounded iff some subsequence tends to $+\infty$ or $-\infty$.

Proof. Set $n_0 = 0$. If (s_n) is unbounded above, every tail is unbounded above, so choose $n_k > n_{k-1}$ with $s_{n_k} \geq k$; then $s_{n_k} \to +\infty$. The unbounded-below case similarly gives $s_{n_k} \leq -k$, hence $s_{n_k} \to -\infty$.
Conversely, a subsequence tending to $\pm\infty$ is unbounded, so (s_n) is unbounded.
Structural Commentary: Unbounded tails supply the thresholds $\pm k$.
Key Insight: Large late terms form a divergent subsequence. □

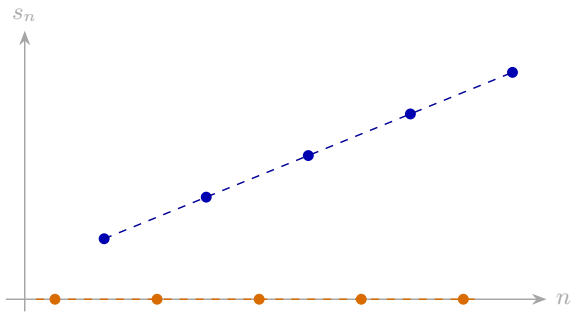


Figure 108: For $s_n = [(-1)^n + 1]n$, odd terms (orange) stay at 0, while even terms (blue) diverge to $+\infty$.

Definition 12.4.19. *Upper and Lower Limits*
For real sequence (s_n), let $E := \{x \in \overline{\mathbb{R}} : \exists\, n_1 < n_2 < \cdots,\ s_{n_k} \to x\}$, allowing $x = \pm\infty$.
Set $s^* := \sup E$ and $s_* := \inf E$; equivalently, $\limsup_{n\to\infty} s_n = s^*$ and $\liminf_{n\to\infty} s_n = s_*$.

Theorem 12.4.20. *Characterization of the Upper Limit*
Let (s_n) be a real sequence, and let E, s^*, s_* be as in Definition 12.4.19. Then:

(i) $s^* \in E$.
(ii) If $x > s^*$, then $s_n < x$ eventually.
(iii) These two properties characterize s^* uniquely.

Dually, s_* is uniquely characterized by $s_* \in E$ and $x < s_*$ implies $s_n > x$ eventually.

Proof.

(i) Since $s^* = \sup E$: if $s^* = +\infty$, use Proposition 12.4.18; if $s^* \in \mathbb{R}$, use Theorem 13.2.4. If $s^* = -\infty$ and $s_n \not\to -\infty$, infinitely many terms are bounded below; they are either unbounded above or have a bounded subsequence, giving an element of E above $-\infty$, a contradiction.
(ii) If $s_n \geq x > s^*$ infinitely often, that subsequence is unbounded above or has a finite subsequential limit $y \geq x$; both contradict $s^* = \sup E < x$. Hence $s_n < x$ eventually.
(iii) If $p < q$ both satisfy (i)–(ii), choose $p < x < q$. Then $s_n < x$ eventually, contradicting $q \in E$.

Apply the same argument to $(-s_n)$ for s_*.
Structural Commentary: Limsup is the maximal subsequential limit and eventual ceiling.
Key Insight: Limsup combines subsequences with eventual upper bounds. □

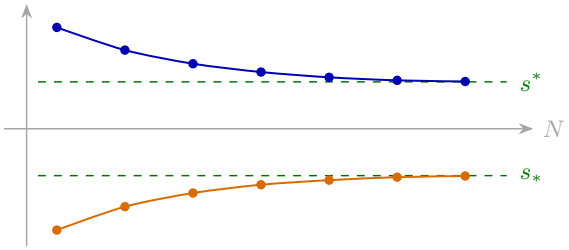


Figure 109: Blue tail suprema u_N decrease to $s^* = \limsup s_n$; orange tail infima ℓ_N increase to $s_* = \liminf s_n$.

Proposition 12.4.21. *Tail Suprema Formula*
For a real sequence (s_n), put $T_N := \{s_n : n \geq N\}$ and, in $\overline{\mathbb{R}}$, $u_N := \sup T_N$, $\ell_N := \inf T_N$.
Then $u_N \downarrow$, $\ell_N \uparrow$, and $\limsup_{n\to\infty} s_n = \inf_N u_N$ and $\liminf_{n\to\infty} s_n = \sup_N \ell_N$.

Proof. Since $T_{N+1} \subseteq T_N$, $u_N \downarrow$ and $\ell_N \uparrow$. Let $s^* = \limsup s_n$ and $u = \inf_N u_N$.
If $x > s^*$, then $s_n < x$ eventually, so $u \leq u_N \leq x$ for some N. If $y < s^*$, then $s^* \in E$ gives a subsequence tending to s^*; hence every tail meets (y, ∞), so $u_N \geq y$ for all N and $u \geq y$. Thus $u = s^*$.
Apply the same formula to $(-s_n)$ and use Corollary 12.2.9.
Structural Commentary: Tail envelopes converge to lim sup and lim inf.
Key Insight: Shrinking tails compute the extremal subsequential limits. □

Theorem 12.4.22. *Convergence and Equal Upper/Lower Limits*
A real sequence (s_n) converges to a finite number $s \in \mathbb{R}$ if and only if $\limsup_{n\to\infty} s_n = \liminf_{n\to\infty} s_n = s$.

Proof.
($\Rightarrow$) If $s_n \to s$, every subsequence converges to s, and the tail is bounded, so no subsequence tends to $\pm\infty$. Thus $E = \{s\}$, hence $\limsup s_n = \liminf s_n = s$.
($\Leftarrow$) Suppose $\limsup s_n = \liminf s_n = s$. Given $\varepsilon > 0$, Theorem 12.4.20 gives $s_n < s + \varepsilon$ and $s_n > s - \varepsilon$ eventually. Hence $|s_n - s| < \varepsilon$ eventually, so $s_n \to s$.

Structural Commentary: Equality of the two extremal subsequential limits traps the whole tail.
Key Insight: Limsup supplies an eventual ceiling; liminf supplies an eventual floor. □

Theorem 12.4.23. *Order Comparison for Upper and Lower Limits*
If $s_n \leq t_n$ for all $n \geq N$, where N is fixed, then:
(i) $\limsup_{n\to\infty} s_n \leq \limsup_{n\to\infty} t_n$.
(ii) $\liminf_{n\to\infty} s_n \leq \liminf_{n\to\infty} t_n$.

Proof. Discard finitely many terms, so $s_n \leq t_n$ for all n.
(i) Put $a = \limsup s_n$ and $b = \limsup t_n$. If $x < a$, then $s_n > x$ infinitely often, hence $t_n > x$ infinitely often; Theorem 12.4.20 gives $x \leq b$. Thus $a \leq b$.
(ii) Apply (i) to $-t_n \leq -s_n$ to get $\limsup(-t_n) \leq \limsup(-s_n)$, equivalently $\liminf s_n \leq \liminf t_n$.

Structural Commentary: Eventual order is inherited by the extremal subsequential limits.
Key Insight: Limsup and liminf ignore finitely many initial terms. □

Theorem 12.4.24. *Pinching Theorem for Real Sequences*
Let (a_n), (s_n), and (b_n) be real sequences with $a_n \leq s_n \leq b_n$ for all sufficiently large n.
If $a_n \to s$ and $b_n \to s$, then $s_n \to s$.

Proof. Let $\varepsilon > 0$. Choose N_0 s.t. $a_n \leq s_n \leq b_n$ for $n \geq N_0$; N_1 s.t. $s - \varepsilon < a_n$ for $n \geq N_1$; and N_2 s.t. $b_n < s + \varepsilon$ for $n \geq N_2$. If $N = \max\{N_0, N_1, N_2\}$, then $s - \varepsilon < s_n < s + \varepsilon$ for all $n \geq N$. Hence $s_n \to s$.
Structural Commentary: Two convergent bounds squeeze the middle sequence to the same limit.
Key Insight: Once both walls are inside the ε-neighborhood, the middle term has no room to escape. □

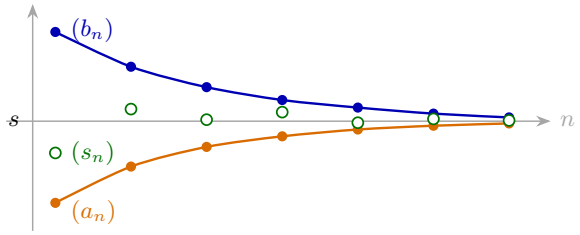


Figure 110: With $a_n \le s_n \le b_n$ and $a_n, b_n \to s$, the middle sequence also tends to s.

Theorem 12.4.25. *Special Sequence Limits*

(i) If $p > 0$, then $\lim_{n\to\infty} n^{-p} = 0$.
(ii) If $p > 0$, then $\lim_{n\to\infty} \sqrt[n]{p} = 1$.
(iii) $\lim_{n\to\infty} \sqrt[n]{n} = 1$.
(iv) If $p > 0$ and $\alpha \in \mathbb{R}$, then $\lim_{n\to\infty} \frac{n^\alpha}{(1+p)^n} = 0$.
(v) If $|x| < 1$, then $\lim_{n\to\infty} x^n = 0$.

Proof.

(i) Given $\varepsilon > 0$, choose $N > (1/\varepsilon)^{1/p}$ by Theorem 12.2.24. Then $n \ge N$ gives $n^{-p} < \varepsilon$.
(ii) If $p > 1$, set $x_n := \sqrt[n]{p} - 1$. Then $x_n > 0$ and $1 + nx_n \le (1 + x_n)^n = p$, so $0 < x_n \le (p-1)/n \to 0$. By Theorem 12.4.24, $x_n \to 0$. If $p = 1$, then $\sqrt[n]{p} = 1$ for every n. If $0 < p < 1$, then $1/p > 1$, so $\sqrt[n]{1/p} \to 1$ by the previous case. Since $\sqrt[n]{p} = 1/\sqrt[n]{1/p}$ and limits preserve reciprocals away from 0, $\sqrt[n]{p} \to 1$.
(iii) Put $x_n := \sqrt[n]{n} - 1$. Then $x_n \ge 0$, and for $n \ge 2$ the binomial theorem gives $n = (1 + x_n)^n \ge \frac{n(n-1)}{2} x_n^2$. Thus $0 \le x_n \le \sqrt{2/(n-1)}$, so Theorem 12.4.24 gives $x_n \to 0$.
(iv) Choose an integer $k > \alpha$ with $k > 0$. For $n > 2k$, $(1+p)^n \ge \binom{n}{k} p^k = \frac{n(n-1)\cdots(n-k+1)}{k!} p^k > \frac{n^k p^k}{2^k k!}$. Hence $0 < \frac{n^\alpha}{(1+p)^n} < \frac{2^k k!}{p^k} n^{\alpha-k}$, and $n^{\alpha-k} \to 0$ by (i). Apply Theorem 12.4.24.
(v) If $x \ne 0$, write $|x| = (1+p)^{-1}$ with $p > 0$ and apply (iv) with $\alpha = 0$ to get $|x|^n \to 0$. Since $-|x|^n \le x^n \le |x|^n$, Theorem 12.4.24 gives $x^n \to 0$; $x = 0$ is immediate.

Structural Commentary: Each limit is reduced to an elementary comparison sequence.
Key Insight: The binomial theorem supplies the right upper bounds for roots and exponential domination. □

Method 12.4.26. *Sequence Proof Patterns*

(i) *Convergence*: fix $\varepsilon > 0$, then find N so that all later terms lie in the relevant ε-ball.
(ii) *Uniqueness*: compare two proposed limits with $\varepsilon/2$ and use the triangle inequality.
(iii) *Subsequences*: choose $n_k \uparrow$ recursively with distances $< 1/k$.
(iv) *Cauchy arguments*: estimate the diameter of tails rather than guessing the limit first.
(v) *Monotone sequences*: use sup or inf and then prove eventual trapping.

Key Insight: Most sequence proofs are bookkeeping for quantifiers, indices, and one controlling inequality.

13 Metric and Topological Structure

Metric and Topological Structure abstracts limits, continuity, compactness, completeness, and connectedness, ending with only the general topology needed later.

13.1 Metric Spaces and Euclidean Topology

Metric topology studies sets through balls: openness, closure, and limit points come from distance. Euclidean estimates rely on norms, Cauchy–Schwarz, and the triangle inequality.

Definition 13.1.1. *Metric Space*
A *metric space* is a set X equipped with a function $d : X \times X \to \mathbb{R}$ such that, for all $p, q, r \in X$,

i. $d(p, q) > 0$ if $p \ne q$, and $d(p, p) = 0$;
ii. $d(p, q) = d(q, p)$;
iii. $d(p, q) \le d(p, r) + d(r, q)$.

The function d is called a *distance function* or *metric*.

Example 13.1.2. *Euclidean Metric and Subspaces*
The Euclidean space $\mathbb{R}^k$ is a metric space with $d(\mathbf{x},\mathbf{y}) := \|\mathbf{x}-\mathbf{y}\|_2$, $\mathbf{x},\mathbf{y}\in\mathbb{R}^k$.
Every subset Y of a metric space X is a metric space by restricting the same distance function to points of Y.

Example 13.1.3. *Discrete Metric*
On any set X, define $d(x,y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$.
This is the *discrete metric*: distinct points are all distance 1 apart.

Example 13.1.4. *Coordinate Norm Metrics on $\mathbb{R}^k$*
By Proposition 2.1.15 and Theorem 2.1.11, $d_1(\mathbf{x},\mathbf{y}) := \|\mathbf{x}-\mathbf{y}\|_1$; $d_2(\mathbf{x},\mathbf{y}) := \|\mathbf{x}-\mathbf{y}\|_2$; $d_\infty(\mathbf{x},\mathbf{y}) := \|\mathbf{x}-\mathbf{y}\|_\infty$ are metrics on $\mathbb{R}^k$, and $\forall \mathbf{x},\mathbf{y}\in\mathbb{R}^k$, $d_\infty(\mathbf{x},\mathbf{y}) \le d_2(\mathbf{x},\mathbf{y}) \le d_1(\mathbf{x},\mathbf{y}) \le \sqrt{k}\, d_2(\mathbf{x},\mathbf{y})$; $d_2(\mathbf{x},\mathbf{y}) \le \sqrt{k}\, d_\infty(\mathbf{x},\mathbf{y})$.
Hence d_1, d_2, d_∞ have the same convergent sequences.

Definition 13.1.5. *Euclidean Intervals, Cells, Balls, and Convexity*
In $\mathbb{R}$, $(a,b) := \{x\in\mathbb{R} : a < x < b\}$ and $[a,b] := \{x\in\mathbb{R} : a \le x \le b\}$; $[a,b)$ and $(a,b]$ use the mixed inequalities.
If $a_i < b_i$ for $i = 1,\dots,k$, the *k-cell* is $\{\mathbf{x} = (x_1,\dots,x_k)\in\mathbb{R}^k : a_i \le x_i \le b_i,\ 1 \le i \le k\}$.
For $\mathbf{x}\in\mathbb{R}^k$ and $r > 0$, the *open ball* is $B_r(\mathbf{x}) := \{\mathbf{y}\in\mathbb{R}^k : \|\mathbf{y}-\mathbf{x}\|_2 < r\}$,
and *closed ball* is $\overline{B}_r(\mathbf{x}) := \{\mathbf{y}\in\mathbb{R}^k : \|\mathbf{y}-\mathbf{x}\|_2 \le r\}$.
A set $E\subseteq\mathbb{R}^k$ is *convex* if $\lambda\mathbf{x} + (1-\lambda)\mathbf{y}\in E$ whenever $\mathbf{x},\mathbf{y}\in E$ and $0 < \lambda < 1$.

Remark 13.1.6. *Euclidean Vocabulary*
Integer lattice is $\mathbb{Z}^k$. *First orthant* are points with nonnegative coordinates.
Unit ball is $B^k := \{\mathbf{x}\in\mathbb{R}^k : \|\mathbf{x}\|_2 \le 1\}$; and *unit sphere* is $S^{k-1} := \{\mathbf{x}\in\mathbb{R}^k : \|\mathbf{x}\|_2 = 1\}$.
$\lambda\mathbf{x} + (1-\lambda)\mathbf{y}$, $0 \le \lambda \le 1$, is a *convex combination*; all such points form the line segment from $\mathbf{x}$ to $\mathbf{y}$.

Proposition 13.1.7. *Balls and Cells Are Convex*
Open balls, closed balls, and k-cells in $\mathbb{R}^k$ are convex.

Proof. Let $\mathbf{y},\mathbf{z}\in B_r(\mathbf{x})$ and $0 < \lambda < 1$.
Then $\|\lambda\mathbf{y} + (1-\lambda)\mathbf{z} - \mathbf{x}\|_2 = \|\lambda(\mathbf{y}-\mathbf{x}) + (1-\lambda)(\mathbf{z}-\mathbf{x})\|_2 \le \lambda\|\mathbf{y}-\mathbf{x}\|_2 + (1-\lambda)\|\mathbf{z}-\mathbf{x}\|_2 < r$.
The same proof with $\le r$ gives closed balls. For a k-cell, each coordinate of $\lambda\mathbf{y} + (1-\lambda)\mathbf{z}$ lies between the corresponding coordinates of $\mathbf{y}$ and $\mathbf{z}$, hence remains in the same interval $[a_i, b_i]$.
Structural Commentary: Convexity is checked by following the line segment between two points.
Key Insight: Balls use the triangle inequality; cells use coordinate inequalities. □

Definition 13.1.8. *Product Metrics*
For metric spaces (X, d_X) and (Y, d_Y), and $p = (x,y)$, $p' = (x',y')$ in $X\times Y$, define

$$d_E(p,p') := \sqrt{d_X(x,x')^2 + d_Y(y,y')^2},$$
$$d_{\max}(p,p') := \max\{d_X(x,x'), d_Y(y,y')\},$$
$$d_{\text{sum}}(p,p') := d_X(x,x') + d_Y(y,y')$$

These are the Euclidean, maximum, and sum product metrics; the sum metric is often simplest for estimates.

Proposition 13.1.9. *Comparing Product Metrics*
On $X\times Y$, $d_{\max} \le d_E \le d_{\text{sum}} \le 2d_{\max}$. Hence the three product metrics have same convergent sequences.

Proof. Let $a = d_X(x,x')$ and $b = d_Y(y,y')$.
Then $\max\{a,b\} \le \sqrt{a^2+b^2} \le a+b \le 2\max\{a,b\}$; convergence follows by squeezing.
Structural Commentary: The three product metrics differ by controlled constants.
Key Insight: Equivalent metric estimates preserve convergence. □

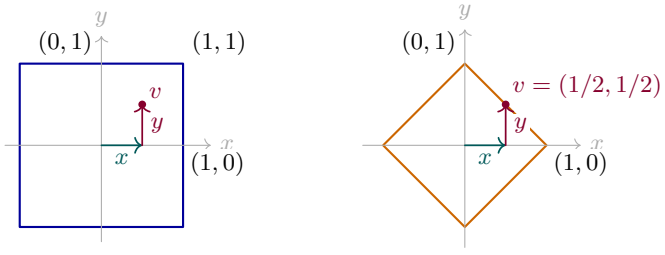
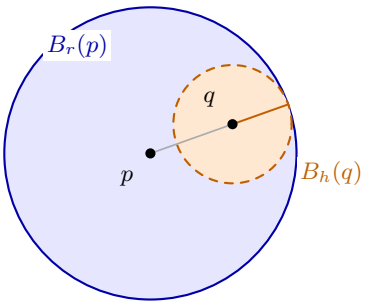


Figure 111: Unit balls in $\mathbb{R}^2$: $\max\{|x|, |y|\} < 1$ and $|x| + |y| < 1$.



$h = r - d(p,q) > 0$

Figure 112: Every point inside a metric ball has a smaller ball still contained in the original ball.

Definition 13.1.10. *Metric Topology Terminology*
Let X be a metric space, and let all points and sets below be understood as points of X and subsets of X.

(i) A *neighborhood* of p with radius $r > 0$ is $B_r(p) := \{q \in X : d(p,q) < r\}$.
The corresponding *closed ball* is $\overline{B}_r(p) := \{q \in X : d(p,q) \leq r\}$.
(ii) A point p is a *limit point* of E if every neighborhood of p contains a point $q \neq p$ with $q \in E$.
(iii) If $p \in E$ and p is not a limit point of E, then p is an *isolated point* of E.
(iv) A set E is *closed* if every limit point of E is a point of E.
(v) A point p is an *interior point* of E if some neighborhood of p is contained in E.
(vi) A set E is *open* if every point of E is an interior point of E.
(vii) The *complement* of E is $E^c := X \setminus E$.
(viii) A set E is *dense* in X if every point of X is either a limit point of E, a point of E, or both.

For $X = \mathbb{R}$, this gives $B_\varepsilon(a) = (a - \varepsilon, a + \varepsilon)$.

Remark 13.1.11. *Ambient Space Matters*
Openness and closedness depend on the ambient metric space; they are not opposites.

Theorem 13.1.12. *Every Neighborhood Is Open*
Every neighborhood in a metric space is an open set.

Proof. Let $E = B_r(p)$, and choose $q \in E$. Put $h := r - d(p,q) > 0$.
If $s \in B_h(q)$, then $d(p,s) \leq d(p,q) + d(q,s) < d(p,q) + h = r$.
Thus $B_h(q) \subseteq E$, so q is an interior point of E. Since q was arbitrary, E is open.
Structural Commentary: Around any point inside a ball, the leftover distance to boundary gives a smaller ball.
Key Insight: The triangle inequality turns radius slack into openness. □

Theorem 13.1.13. *Limit Points Force Infinitely Many Nearby Points*
If p is a limit point of E, then every neighborhood of p contains infinitely many points of E.

Proof. Suppose some neighborhood N of p contains only finitely many points of E distinct from p, say $q_1, \ldots, q_n$.
If there are none, then N itself contradicts that p is a limit point. Otherwise put $\rho := \min_{1 \leq m \leq n} d(p, q_m) > 0$.
The neighborhood $B_\rho(p)$ contains no point $q \in E$ with $q \neq p$, contradicting the definition of limit point.
Structural Commentary: A finite list of nearby exceptions has a smallest positive distance from p.
Key Insight: Limit points cannot be witnessed by only finitely many neighboring points. □

Corollary 13.1.14. *Finite Point Sets Have No Limit Points*
A finite point set in a metric space has no limit points.

Proof. Let $E \subseteq X$ be finite. If p were a limit point of E, then by Theorem 13.1.13, every neighborhood of p would contain infinitely many distinct points of E, impossible as E is finite. Hence E has no limit points.
Structural Commentary: This corollary is the finite obstruction to accumulation.
Key Insight: A limit point requires infinitely many nearby witnesses. □

Theorem 13.1.15. *Arbitrary De Morgan Laws*
Let X be a set. For any collection $\{E_\alpha\}$ of subsets of X, $\left(\bigcup_\alpha E_\alpha\right)^c = \bigcap_\alpha E_\alpha^c$ *and* $\left(\bigcap_\alpha E_\alpha\right)^c = \bigcup_\alpha E_\alpha^c$.

Proof. The first identity follows from membership: $x \in \left(\bigcup_\alpha E_\alpha\right)^c \iff x \notin E_\alpha \ \forall \alpha \iff x \in \bigcap_\alpha E_\alpha^c$.
The second follows by applying the first identity to the complements E_α^c, or by the same membership argument.
Structural Commentary: Complements exchange unions and intersections.
Key Insight: Negating "for some" gives "for every", and conversely. □

Theorem 13.1.16. *Open Sets and Closed Complements*
A set E is open if and only if its complement E^c is closed.

Proof.
($\Rightarrow$) Suppose E is open. If x is a limit point of E^c and $x \in E$, then some neighborhood N of x satisfies $N \subseteq E$, contradicting $N \cap (E^c \setminus \{x\}) \neq \emptyset$. Hence $x \in E^c$, so E^c is closed.
($\Leftarrow$) Suppose E^c is closed. If $x \in E$, then $x \notin E^c$, hence x is not a limit point of E^c. Thus some neighborhood N of x has $N \cap E^c = \emptyset$, i.e. $N \subseteq E$. Hence E is open.

Structural Commentary: Openness of a set is equivalent to closure of the outside.
Key Insight: Interior points of E cannot be limit points of E^c. □

Corollary 13.1.17. *Closed Sets and Open Complements*
A set F is closed if and only if F^c is open.

Proof. Apply Theorem 13.1.16 with $E = F^c$. Then F^c is open if and only if $(F^c)^c$ is closed. Since $(F^c)^c = F$, this is exactly the assertion that F^c is open if and only if F is closed.
Structural Commentary: This is the same open–closed duality after taking complements.
Key Insight: Closedness of F is openness of the outside F^c. □

Theorem 13.1.18. *Basic Topological Calculus*
In a metric space:

(i) any union of open sets is open;
(ii) any intersection of closed sets is closed;
(iii) any finite intersection of open sets is open;
(iv) any finite union of closed sets is closed.

Proof.
(i) Let $G := \bigcup_\alpha G_\alpha$. If $x \in G$, then $x \in G_\alpha$ for some α.
Since G_α is open, some neighborhood of x lies in $G_\alpha \subseteq G$. Hence G is open.
(ii) Let $F := \bigcap_\alpha F_\alpha$. By Theorem 13.1.15, $F^c = \bigcup_\alpha F_\alpha^c$.
Each F_α^c is open by Corollary 13.1.17, so F^c is open by (i). Hence F is closed.
(iii) Let $H := \bigcap_{i=1}^n G_i$. If $x \in H$, let $r_i > 0$, $B_{r_i}(x) \subseteq G_i$, $r := \min_{1 \le i \le n} r_i$. Then $B_r(x) \subseteq H$, so H is open.
(iv) Let $F := \bigcup_{i=1}^n F_i$. By Theorem 13.1.15, $F^c = \bigcap_{i=1}^n F_i^c$.
Each F_i^c is open by Corollary 13.1.17, so F^c is open by (iii). Hence F is closed.

Structural Commentary: Arbitrary unions are easy for open sets; finite intersections require a minimum radius.
Key Insight: The finite minimum is the reason infinite intersections of open sets need not be open. □

Definition 13.1.19. *Closure*
If $E \subseteq X$ and E^{acc} denotes the set of all limit points of E in X, then the *closure* of E is $\overline{E} := E \cup E^{\text{acc}}$.

Theorem 13.1.20. *Closure Is the Smallest Closed Superset*
If X is a metric space and $E \subseteq X$, then:

(i) $\overline{E}$ is closed;
(ii) $\overline{E} = E$ if and only if E is closed;
(iii) $\overline{E} \subseteq F$ for every closed set $F \subseteq X$ such that $E \subseteq F$.

Thus $\overline{E}$ is the smallest closed subset of X that contains E.

Proof.
(i) Let $p \notin \overline{E}$. Then $p \notin E$ and p is not a limit point of E, so $B_r(p) \cap E = \emptyset$ for some $r > 0$. If $q \in B_r(p)$, then $q \notin E$ and $B_{r-d(p,q)}(q) \subseteq B_r(p)$, so $q \notin E^{\text{acc}}$.
Hence $B_r(p) \subseteq (\overline{E})^c$, and $\overline{E}$ is closed by Corollary 13.1.17.
(ii) If $\overline{E} = E$, then E is closed by (i). Conversely, if E is closed, then $E^{\text{acc}} \subseteq E$, so $\overline{E} = E \cup E^{\text{acc}} = E$.
(iii) If F is closed and $E \subseteq F$, then $E^{\text{acc}} \subseteq F^{\text{acc}} \subseteq F$. Thus $\overline{E} = E \cup E^{\text{acc}} \subseteq F$.

Structural Commentary: Closure adds exactly the limit points forced by E.
Key Insight: Closed supersets must contain all limit points, so they must contain $\overline{E}$. □

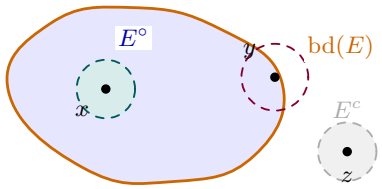


Figure 113: Interior, exterior, and boundary points are distinguished by how small balls meet E and E^c.

Definition 13.1.21. *Interior and Boundary Notation*
Let E be a subset of a metric space X. The *interior* of E is $E^\circ := \{p \in E : p \text{ is an interior point of } E\}$.
The *exterior* of E is the interior of E^c: equivalently, the points $p \in X$ for which some ball $B_r(p)$ misses E.
The *boundary* of E is $\mathrm{bd}(E) := \overline{E} \setminus E^\circ$.
Equivalently, boundary points are exactly the points of $\overline{E}$ that are not interior to E.

Definition 13.1.22. *Adherent Points*
A point $p \in X$ is *adherent* to E if every ball about p meets E. Equivalently, $p \in \overline{E}$.



Figure 114: Adherent endpoints need not lie in E; isolated points are adherent but not limit points.

Definition 13.1.23. *Bounded and Perfect Metric Sets*
Let E be a subset of a metric space X.

(i) E is *bounded* if there exist $M \in \mathbb{R}$ and $q \in X$ such that $d(p, q) < M$ for all $p \in E$.
(ii) E is *perfect* if E is closed and every point of E is a limit point of E.

Remark 13.1.24. *Boundedness and Ambient Space*
Equivalently, E is bounded iff $E \subseteq B_M(q)$ for some ball in X. Boundedness, compactness, and completeness are intrinsic to the restricted metric; openness and closedness depend on the ambient space.

Perspective 13.1.25. *Metric Sequences as Topological Tests*
Cardinality used sequences as enumerations in Definition 12.3.7. In metric spaces, sequences become convergence probes: they test local topology, compactness, and whether Cauchy approximations have internal limits.

Definition 13.1.26. *Convergent Sequence in a Metric Space*
Let (p_n) be a sequence in a metric space X.
Then (p_n) *converges* to $p \in X$ if for every $\varepsilon > 0$ there is an integer N such that $n \geq N \implies d(p_n, p) < \varepsilon$.
Notation is $p_n \to p$ or $\lim_{n\to\infty} p_n = p$. If no such point $p \in X$ exists, the sequence *diverges*.

Definition 13.1.27. *Range and Boundedness of a Sequence*
The *range* of (p_n) is the set $\{p_n : n \in \mathbb{N}\}$.
A sequence in a metric space is *bounded* if its range is bounded in the sense of Definition 13.1.23.

Definition 13.1.28. *Eventually and Frequently in a Set*
Let (p_n) be a sequence in X, and let $E \subseteq X$.

(i) (p_n) is *eventually in* E if $\exists N \in \mathbb{N}$ such that $n \geq N \implies p_n \in E$.
(ii) (p_n) is *frequently in* E if $\forall N \in \mathbb{N}\ \exists n \geq N$ such that $p_n \in E$.

Eventually means tail containment; frequently means arbitrarily late visits.

Proposition 13.1.29. *Limit Points Are Subsequence Limits*
For a sequence (p_n) in a metric space X and $p \in X$, the following are equivalent:

(i) (p_n) is frequently in every neighborhood of p.
(ii) p is a subsequential limit of (p_n).

Proof. If $p_{n_k} \to p$, then every neighborhood of p contains all sufficiently late p_{n_k}, and those indices can be chosen beyond any prescribed N.
Conversely, suppose (p_n) is frequently in every neighborhood of p. Starting with $n_0 = 0$, choose recursively $n_k > n_{k-1}$ with $d(p_{n_k}, p) < 1/k$. Then $p_{n_k} \to p$.
Structural Commentary: Frequent visits are exactly enough to extract a convergent subsequence.
Key Insight: Limits require all sufficiently late terms; limit points require arbitrarily late witnesses. □

Theorem 13.1.30. *Basic Properties of Metric Convergence*
Let (p_n) be a sequence in a metric space X.

(i) $p_n \to p$ if and only if every neighborhood of p contains p_n for all but finitely many n; equivalently, (p_n) is eventually in every neighborhood of p.
(ii) If $p_n \to p$ and $p_n \to p'$, then $p = p'$.
(iii) If $E \subseteq X$ and p is a limit point of E, then there is a sequence (p_n) in E such that $p_n \to p$.

Proof.

(i) This is the ε-N definition, since every neighborhood of p contains a ball about p.
(ii) Given $\varepsilon > 0$, choose N, N' so that $d(p_n, p) < \varepsilon/2$ for $n \geq N$ and $d(p_n, p') < \varepsilon/2$ for $n \geq N'$. For $n \geq \max\{N, N'\}$, the triangle inequality gives $d(p, p') < \varepsilon$. Hence $d(p, p') = 0$.
(iii) For each n, choose $p_n \in E$ with $p_n \neq p$ and $d(p_n, p) < 1/n$, using Definition 13.1.10. Then $p_n \to p$.

Structural Commentary: Eventual containment gives convergence; triangle inequality gives uniqueness.
Key Insight: A limit point can always be approached by a sequence of points from the set. □

Proposition 13.1.31. *Discrete-Metric Convergence*
Let X carry the discrete metric $d(x, y) = 0$ if $x = y$ and $d(x, y) = 1$ if $x \neq y$.
For $(p_n) \subset X$ and $p \in X$, $p_n \to p$ if and only if $\exists N \in \mathbb{N}$ such that $n \geq N \Longrightarrow p_n = p$.

Proof.

($\Rightarrow$) Suppose $p_n \to p$. For $\varepsilon = 1/2$, choose $N \in \mathbb{N}$ such that $n \geq N \Longrightarrow d(p_n, p) < 1/2$.
Since d takes only the values $0, 1$, $d(p_n, p) = 0$, hence $p_n = p$ for all $n \geq N$.
($\Leftarrow$) Suppose $\exists N \in \mathbb{N}$ s.t. $n \geq N \Longrightarrow p_n = p$. Then $\forall \varepsilon > 0$, $n \geq N$ gives $d(p_n, p) = 0 < \varepsilon$, so $p_n \to p$.

Structural Commentary: The discrete metric has no room for distinct points to approach one another.
Key Insight: In the discrete metric, $d(x, y) < 1 \Longleftrightarrow x = y$. □

Proposition 13.1.32. *Sequential Closure Criterion*
Let $E \subseteq X$, where X is a metric space, and let $p \in X$. Then $p \in \overline{E} \Longleftrightarrow \exists (p_n)_{n \geq 1}$ with $p_n \in E$ $\forall n$, $\quad p_n \to p$.

Proof.

($\Rightarrow$) If $p \in \overline{E}$, choose $p_n \in E \cap B_{1/n}(p)$ for $n \geq 1$ by Principle 12.3.19. Then $d(p_n, p) < 1/n$, so $p_n \to p$.
($\Leftarrow$) If $p_n \in E$ and $p_n \to p$, then every ball $B_r(p)$ contains some p_n.
Thus p is adherent to E by Definition 13.1.22, so $p \in \overline{E}$.

Structural Commentary: Closure can be read either by neighborhoods or by sequences.
Key Insight: Adherence supplies one witness at every scale. □

Proposition 13.1.33. *Density and Closure*
Let X be a metric space and let $E \subseteq X$. Then E is dense in X if and only if $\overline{E} = X$.

Proof.

($\Rightarrow$) Suppose E is dense in X. Let $p \in X$. By Definition 13.1.10, p is either a point of E or a limit point of E. Hence $p \in E \cup E^{\text{acc}} = \overline{E}$ by Definition 13.1.19. Thus $X \subseteq \overline{E}$. Since $\overline{E} \subseteq X$, $\overline{E} = X$.
($\Leftarrow$) Suppose $\overline{E} = X$. Let $p \in X$. Then $p \in \overline{E} = E \cup E^{\text{acc}}$, so p is either a point of E or a limit point of E. Therefore E is dense in X by Definition 13.1.10.

Structural Commentary: Density says every point is already in E or is forced by nearby points of E.
Key Insight: The closure packages exactly those two possibilities. □

Proposition 13.1.34. *Real-Line Density Interval Test*
Let $E \subseteq \mathbb{R}$. Then E is dense in $\mathbb{R}$ if and only if every nonempty open interval (a, b) satisfies $(a, b) \cap E \neq \emptyset$.

Proof.

($\Rightarrow$) Suppose E is dense in $\mathbb{R}$, and let (a, b) be a nonempty open interval. Choose $p \in (a, b)$ and put $r := \frac{1}{2}\min\{p - a, b - p\} > 0$. If $p \in E$, then $(a, b) \cap E \neq \emptyset$. If $p \notin E$, then p is a limit point of E by Definition 13.1.10, so $B_r(p)$ contains a point $q \in E$ with $q \neq p$. Since $B_r(p) \subseteq (a, b)$, again $(a, b) \cap E \neq \emptyset$.

($\Leftarrow$) Suppose every nonempty open interval meets E. Let $p \in \mathbb{R}$. If $p \in E$, there is nothing to prove. If $p \notin E$, then for every $\varepsilon > 0$, the interval $(p-\varepsilon, p+\varepsilon)$ contains a point $q \in E$. Since $p \notin E$, $q \neq p$, so every ball $B_\varepsilon(p)$ contains a point of E distinct from p. Thus p is a limit point of E. Hence every $p \in \mathbb{R}$ is either in E or is a limit point of E, and E is dense in $\mathbb{R}$.

Structural Commentary: On $\mathbb{R}$, balls and open intervals test the same local condition.
Key Insight: Density means no interval can avoid the set. □

Theorem 13.1.35. *Suprema Belong to Closures*
Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded above and let $y := \sup E$. Then $y \in \overline{E}$; if E is closed, then $y \in E$.

Proof. If $y \in E$, then $y \in \overline{E}$. If $y \notin E$, then for every $h > 0$, $y - h$ is not an upper bound of E, so choose $x_h \in E$ with $y - h < x_h \leq y$. Then $x_h \neq y$ and $|x_h - y| < h$, hence $B_h(y) \cap (E \setminus \{y\}) \neq \emptyset$. Thus $y \in E^{\text{acc}} \subseteq \overline{E}$. If E is closed, Theorem 13.1.20(ii) gives $\overline{E} = E$.
Structural Commentary: LUB gives points of E arbitrarily close below $\sup E$.
Key Insight: A supremum is either attained or approached. □

Definition 13.1.36. *Relative Openness*
Suppose $E \subseteq Y \subseteq X$, where X is a metric space. The set E is *open relative to Y* if for each $p \in E$ there is an $r > 0$ such that $q \in E$ whenever $q \in Y$ and $d(p,q) < r$.

Theorem 13.1.37. *Relative Opens Are Intersections*
Suppose $Y \subseteq X$. A subset E of Y is open relative to Y if and only if $E = Y \cap G$ for some open subset G of X.

Proof.
($\Rightarrow$) For each $p \in E$, choose $r_p > 0$ with $B_{r_p}(p) \cap Y \subseteq E$, and set $G := \bigcup_{p \in E} B_{r_p}(p)$. Then G is open in X, and $E \subseteq G \cap Y$. If $q \in G \cap Y$, then $q \in B_{r_p}(p) \cap Y$ for some $p \in E$, so $q \in E$. Thus $E = G \cap Y$.
($\Leftarrow$) If $E = G \cap Y$ with G open in X, then for each $p \in E$ choose $r > 0$ with $B_r(p) \subseteq G$. Hence $B_r(p) \cap Y \subseteq G \cap Y = E$, so E is open relative to Y.

Structural Commentary: Relative openness is ordinary openness viewed through the ambient subset Y.
Key Insight: Intersect open sets in X with Y to get exactly the open sets of Y. □

Example 13.1.38. *Relative Topology Depends on the Ambient Space*
Let $X = \{(x,0) : x \in \mathbb{R}\} \subseteq \mathbb{R}^2$ with the metric inherited from $\mathbb{R}^2$. The set $E = \{(x,0) : -1 < x < 1\}$ is not open as a subset of $\mathbb{R}^2$, but it is open relative to X, since $E = X \cap \big((-1,1) \times (-1,1)\big)$.
Likewise, $[0,1)$ is not closed in $\mathbb{R}$, but it is closed relative to $(-1,1)$ because $[0,1) = (-1,1) \cap [0,\infty)$.

Method 13.1.39. *Metric Proof Patterns*
(i) To prove G is open, fix $x \in G$ and find $r > 0$ with $B_r(x) \subseteq G$.
(ii) To prove F is closed, show $F^{\text{acc}} \subseteq F$, or equivalently show F^c is open.
(iii) To prove $C = \overline{E}$, show $E \subseteq C$, C is closed, and $C \subseteq F$ for every closed $F \supseteq E$.

13.2 Compactness and Completeness

Compactness extracts finite or convergent structure from infinite behavior; completeness asks whether Cauchy approximations converge internally.

Definition 13.2.1. *Open Covers and Compact Sets*
Let $E \subseteq X$, where X is a metric space.
An *open cover* of E is a collection $\{G_\alpha\}_{\alpha \in A}$ of open subsets of X such that $E \subseteq \bigcup_{\alpha \in A} G_\alpha$. A set $K \subseteq X$ is *compact* if every open cover of K has a finite subcover: $K \subseteq G_{\alpha_1} \cup \cdots \cup G_{\alpha_n}$ for some finite subcollection.



Figure 115: Compactness extracts a finite subcover.

Definition 13.2.2. *Sequential Compactness*
A metric space is *sequentially compact* if every sequence has a convergent subsequence.

Theorem 13.2.3. *Compactness Produces Convergent Subsequences*

(i) If (p_n) is a sequence in a compact metric space X, then $p_{n_i} \to p$ for some subsequence and some $p \in X$.
(ii) If $(\mathbf{x}_n)$ is bounded in $\mathbb{R}^k$, then it has a convergent subsequence.

Proof.

(i) Let $E = \{p_n : n \in \mathbb{N}\}$. If E is finite, choose a value p attained infinitely often; the corresponding constant subsequence converges. If E is infinite, Theorem 13.2.28 gives a limit point $p \in X$ of E. By Theorem 13.1.13, choose $n_i > n_{i-1}$ with $d(p_{n_i}, p) < 1/i$. Then $p_{n_i} \to p$.
(ii) The range of $(\mathbf{x}_n)$ has compact closure by Theorem 13.2.33; apply (i) in that closure.

Structural Commentary: Compactness turns arbitrary infinite behavior into an accumulating subsequence.
Key Insight: In $\mathbb{R}^k$, boundedness is enough because bounded sets sit inside compact closed cells. □

Theorem 13.2.4. *Subsequential Limits Form a Closed Set*
For a sequence (p_n) in a metric space X, the set of all subsequential limits is closed.

Proof. Let E^* be the set of subsequential limits and let q be a limit point of E^*. Set $n_0 = 0$.
For each i, choose $x_i \in E^*$ with $d(x_i, q) < 2^{-i}$. Since x_i is a subsequential limit, choose $n_i > n_{i-1}$ such that $d(p_{n_i}, x_i) < 2^{-i}$. Then $d(p_{n_i}, q) < 2^{1-i}$, so $p_{n_i} \to q$, and $q \in E^*$.
Structural Commentary: A limit of subsequential limits is reached by diagonalizing through those subsequences.
Key Insight: Choose later and closer terms at the same time. □

Definition 13.2.5. *Cauchy Sequence*
A sequence (p_n) in a metric space X is *Cauchy* if $\forall \varepsilon > 0$ $\exists$ integer N s.t. $n \geq N,\ m \geq N \implies d(p_n, p_m) < \varepsilon$.

Definition 13.2.6. *Diameter*
Let E be a nonempty subset of a metric space X. The *diameter* of E is $\operatorname{diam} E := \sup\{d(p, q) : p, q \in E\}$, with the supremum interpreted in the extended real sense when the set of distances is unbounded.

Proposition 13.2.7. *Cauchy Sequences and Tail Diameters*
For (p_n) in a metric space X, set $E_N := \{p_j : j \geq N\}$. Then (p_n) is Cauchy if and only if $\operatorname{diam} E_N \to 0$.

Proof.

($\Rightarrow$) Given $\varepsilon > 0$, choose N_0 so that $m, n \geq N_0$ implies $d(p_m, p_n) < \varepsilon/2$.
If $N \geq N_0$, then every pair in E_N has distance $< \varepsilon/2$, so $\operatorname{diam} E_N < \varepsilon$.
($\Leftarrow$) Given $\varepsilon > 0$, choose N_0 with $\operatorname{diam} E_{N_0} < \varepsilon$.
If $m, n \geq N_0$, then $p_m, p_n \in E_{N_0}$, hence $d(p_m, p_n) \leq \operatorname{diam} E_{N_0} < \varepsilon$.

Structural Commentary: Cauchy behavior is shrinking diameter of sequence tails.
Key Insight: The limit is not named; only mutual closeness is measured. □

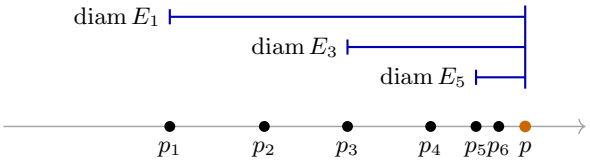


Figure 116: Tail diameters $\operatorname{diam} E_N$ shrink toward 0.

Theorem 13.2.8. *Nested Compact Sets with Shrinking Diameter*
Let X be a metric space.

(i) If $\overline{E}$ is the closure of E in a metric space, then $\operatorname{diam} \overline{E} = \operatorname{diam} E$.
(ii) If $K_1 \supseteq K_2 \supseteq K_3 \supseteq \cdots$ is a sequence of nonempty compact sets in X and $\operatorname{diam} K_n \to 0$, *then $\bigcap_{n=1}^{\infty} K_n$ consists of exactly one point.*

Proof.

(i) Since $E \subseteq \overline{E}$, $\operatorname{diam} E \leq \operatorname{diam} \overline{E}$. For $p, q \in \overline{E}$ and $\varepsilon > 0$, choose $p', q' \in E$ with $d(p, p'), d(q, q') < \varepsilon$.
Then $d(p, q) \leq \operatorname{diam} E + 2\varepsilon$, so $\operatorname{diam} \overline{E} \leq \operatorname{diam} E$.
(ii) Put $I := \bigcap_{n \geq 1} K_n$. Corollary 13.2.27 gives $I \neq \varnothing$.
If $x, y \in I$ with $x \neq y$, then $\operatorname{diam} K_n \geq d(x, y) > 0$ for all n, contradicting $\operatorname{diam} K_n \to 0$.

Structural Commentary: Closure preserves diam; $K_n \downarrow$ with $\operatorname{diam} K_n \to 0$ forces a singleton.
Key Insight: Compactness gives existence; shrinking diameter gives uniqueness. □

Theorem 13.2.9. *Cauchy Criterion and Completeness Results*

(i) In any metric space, every convergent sequence is Cauchy.
(ii) If X is a compact metric space, then every Cauchy sequence in X converges to a point of X.
(iii) In $\mathbb{R}^k$, every Cauchy sequence converges.

Proof.

(i) If $p_n \to p$, choose N with $d(p_n, p) < \varepsilon/2$ for $n \geq N$. For $m, n \geq N$, $d(p_n, p_m) < \varepsilon$.
(ii) Let $E_N := \{p_j : j \geq N\}$ and $F_N := \overline{E_N}$. Then $\operatorname{diam} F_N \to 0$, the F_N are compact and nested, and Theorem 13.2.8 gives $\bigcap_N F_N = \{p\}$. If $\operatorname{diam} F_N < \varepsilon$, then $p, p_n \in F_N$ for $n \geq N$, so $d(p_n, p) < \varepsilon$.
(iii) For Cauchy $(\mathbf{x}_n) \subseteq \mathbb{R}^k$, choose N with $\operatorname{diam}\{\mathbf{x}_j : j \geq N\} < 1$. Its range is bounded, hence has compact closure by Theorem 13.2.33; apply (ii).

Structural Commentary: Cauchy controls $\operatorname{diam} E_N$; compactness turns $\bigcap_N F_N$ into the limit.
Key Insight: In $\mathbb{R}^k$, the Cauchy criterion is equivalent to convergence. □

Lemma 13.2.10. *A Cauchy Sequence with a Convergent Subsequence Converges*
Let (p_n) be a Cauchy sequence in a metric space X. If some subsequence (p_{n_j}) converges to $p \in X$, then $p_n \to p$.

Proof. Given $\varepsilon > 0$, choose N_0 so that $m, n \geq N_0$ implies $d(p_m, p_n) < \varepsilon/2$. Choose j with $n_j \geq N_0$ and $d(p_{n_j}, p) < \varepsilon/2$. Then for every $n \geq N_0$, $d(p_n, p) \leq d(p_n, p_{n_j}) + d(p_{n_j}, p) < \varepsilon$.
Structural Commentary: The Cauchy tail follows any convergent subsequence to the same limit.
Key Insight: Mutual closeness plus one convergent thread forces the whole tail to converge. □

Definition 13.2.11. *Complete Metric Space*
A metric space X is *complete* if every Cauchy sequence in X converges to a point of X.
Compact metric spaces and $\mathbb{R}^k$ are complete (by Theorem 13.2.9).

Proposition 13.2.12. *Completeness and Closed Subspaces*
Let X be a complete metric space and let $E \subseteq X$ carry the inherited metric.
Then E is complete if and only if E is closed in X.

Proof.

($\Rightarrow$) Suppose E is complete. If $p \in \overline{E}$, choose $p_n \in E$ with $d(p_n, p) < 1/n$.
Then $p_n \to p$ in X, so (p_n) is Cauchy in E and hence $p_n \to q$ for some $q \in E$. Uniqueness of limits in X gives $p = q \in E$; thus $\overline{E} \subseteq E$, so E is closed.
($\Leftarrow$) Suppose E is closed. If $(p_n) \subseteq E$ is Cauchy, completeness of X gives $p_n \to p \in X$.
Since E is closed, $p \in E$, so E is complete.

Structural Commentary: Completeness gives the limit in X; closedness keeps it in E.
Key Insight: In a complete ambient space, E is complete exactly when $\overline{E} = E$. □

Definition 13.2.13. *Contraction*
Let X be a metric space with metric d.
A map $\varphi : X \to X$ is a *contraction* if $\exists$ a constant $0 \leq c < 1$ such that $d(\varphi(x), \varphi(y)) \leq c\, d(x, y)$ for $x, y \in X$.
Structural Commentary: Bound $c < 1$ excludes nonexpansive maps and is the hypothesis in Theorem 13.2.14.

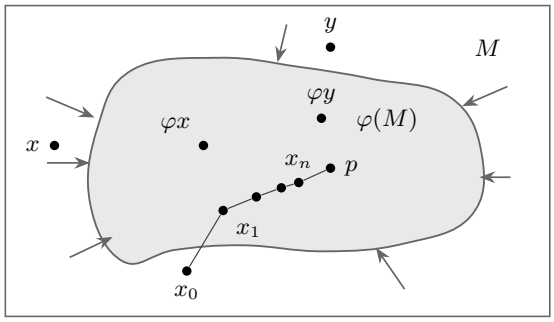


Figure 117: A contraction pulls iterates toward the unique fixed point.

Theorem 13.2.14. *Contraction Principle*
If X is a complete metric space and $\varphi : X \to X$ is a contraction, then $\exists$ a unique $x \in X$ such that $\varphi(x) = x$.

Proof. Choose $x_0 \in X$ and define $x_{n+1} = \varphi(x_n)$.
If $c < 1$ is a contraction constant, induction gives $d(x_{n+1}, x_n) \leq c^n d(x_1, x_0)$.
For $n < m$, the triangle inequality and the finite geometric sum give $d(x_n, x_m) \leq \sum_{j=n}^{m-1} d(x_j, x_{j+1}) \leq \frac{c^n}{1-c} d(x_1, x_0)$.
Thus (x_n) is Cauchy, so completeness gives $x_n \to x \in X$.
Since a contraction is continuous, $\varphi(x) = \lim \varphi(x_n) = \lim x_{n+1} = x$.
If $\varphi(x) = x$ and $\varphi(y) = y$, then $d(x, y) \leq c\, d(x, y)$, hence $d(x, y) = 0$ and $x = y$.
Structural Commentary: Iteration turns the contraction estimate into a geometric Cauchy estimate.
Key Insight: Completeness supplies the limiting point; the strict constant $c < 1$ forces uniqueness. □

Perspective 13.2.15. *Fixed Points as Existence Proofs*
A contraction proof solves $x = \varphi(x)$ by iteration in a complete metric space. This is the template behind the inverse function theorem, Theorem 15.3.4.

Example 13.2.16. *A Noncomplete Metric Space*
The rational numbers with $d(x, y) = |x - y|$ are not complete.
Let (r_n) be the decimal truncation sequence $1.4,\ 1.41,\ 1.414,\ 1.4142, \ldots$, for $\sqrt{2}$.
This is a concrete instance of the truncation squeeze in Proposition 12.2.28. If $k, n \geq N$, then $|r_k - r_n| \leq 10^{-N}$, so (r_n) is Cauchy in $\mathbb{Q}$. As a sequence in $\mathbb{R}$, it converges to $\sqrt{2}$.
If it converged to some $r \in \mathbb{Q}$ in the inherited metric, uniqueness of limits in $\mathbb{R}$ would force $r = \sqrt{2}$, contradicting Proposition 12.2.2. Thus $\mathbb{Q}$ is not complete in the inherited metric. *Structural Commentary*: The decimal approximants are Cauchy before their missing limit is available inside $\mathbb{Q}$.
Key Insight: Completeness asks whether every internally Cauchy approximation has an internal endpoint.

Perspective 13.2.17. *Completion by Cauchy Sequences*
Complete by identifying Cauchy sequences with $d(p_n, q_n) \to 0$ and setting $D([(p_n)], [(q_n)]) = \lim_{n\to\infty} d(p_n, q_n)$.
Constant sequences embed the original space; triangle inequalities and diagonals give a complete metric.

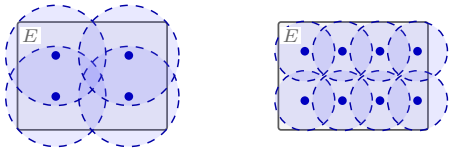


Figure 118: Finite ε- and δ-nets cover E, with $0 < \delta < \varepsilon$.

Definition 13.2.18. *Total Boundedness*
Let (X, d) be a metric space. It is *totally bounded* if, $\forall \varepsilon > 0$, $\exists$ a finite set $F \subseteq X$ such that $X \subseteq \bigcup_{x \in F} B_\varepsilon(x)$.
F is a finite ε-net. $E \subseteq X$ totally bounded if same condition holds with finite $F \subseteq E$ and X.

Theorem 13.2.19. *Total Boundedness Implies Boundedness*
Let X be a metric space. If X is totally bounded, then X is bounded.

Proof. If $X = \emptyset$, this is trivial. Otherwise cover X by finitely many 1-balls $B_1(x_i)$ and fix x_1.
Then $d(x, x_1) < 1 + \max_i d(x_i, x_1)$ for every $x \in X$.
Structural Commentary: One finite cover at scale 1 gives one global bound.
Key Insight: Total boundedness is boundedness with finite control at every scale. □

Theorem 13.2.20. *Compactness as Complete Finite Resolution*
Let X be a metric space. Then, X is compact $\iff$ X is complete and totally bounded.

Proof.

($\Rightarrow$) Let (p_k) be Cauchy in X. By Theorem 13.2.9, compactness forces (p_k) to converge, so X is complete.
Given $\varepsilon > 0$, cover $\{B_\varepsilon(x) : x \in X\}$ has finite subcover $B_\varepsilon(y_1), \ldots, B_\varepsilon(y_m)$; hence X is totally bounded.

($\Leftarrow$) Let $(x_n) \subseteq X$. Choose nested subsequences $(x_k^{(j)})_{k \geq 1}$ and points $a_j \in X$ with $x_k^{(j)} \in B_{1/j}(a_j)$ $\forall k$.
Diagonal $y_j := x_j^{(j)}$ satisfies $y_k, y_\ell \in B_{1/j}(a_j)$ for $k, \ell \geq j$, so it is Cauchy and converges by completeness; hence X is sequentially compact. Now let $\{U_\alpha\}$ cover X. If no $\delta > 0$ has every $B_\delta(x)$ contained in some U_α, choose x_n with $B_{1/n}(x_n) \nsubseteq U_\alpha$ $\forall \alpha$. Passing to $x_{n_k} \to x \in U_{\alpha_0}$, and openness gives $B_{1/n_k}(x_{n_k}) \subseteq U_{\alpha_0}$ for large k, a contradiction. Thus such δ exists.
Choose a finite $\delta/2$-net $\{z_i\}_{i=1}^m$ and α_i with $B_\delta(z_i) \subseteq U_{\alpha_i}$; then $\{U_{\alpha_i}\}_{i=1}^m$ covers X.

Structural Commentary: Finite nets plus completeness force compactness.
Key Insight: Compact metric spaces are exactly complete spaces visible at every finite scale. □

Example 13.2.21. *An Open Cover Without a Finite Subcover*
For $x \in (0,1)$, set $O_x := (x/2, 1)$. These sets cover $(0,1)$, since $y \in O_x$ whenever $0 < x < y < 1$.
No finite subcollection covers $(0,1)$: if $x' = \min\{x_1, \ldots, x_n\}$, then $(0, x'/2]$ is missed by $O_{x_1} \cup \cdots \cup O_{x_n}$.

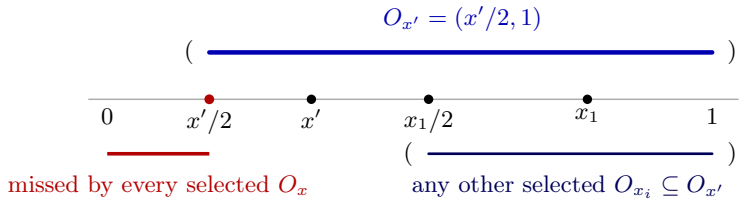


Figure 119: For finite choice, $x' = \min_i x_i$ makes $O_{x'}$ largest selected interval, but $(0, x'/2]$ remains uncovered.

Theorem 13.2.22. *Compactness Is Independent of the Ambient Space*
Suppose $K \subseteq Y \subseteq X$, where X is a metric space.
Then K is compact relative to X if and only if K is compact relative to Y.

Proof.
($\Rightarrow$) Let $\{V_\alpha\}_{\alpha \in A}$ be a relative-open cover of K in Y.
By Theorem 13.1.37, write $V_\alpha = Y \cap G_\alpha$ with G_α open in X. Since $\{G_\alpha\}_{\alpha \in A}$ covers K, compactness in X gives a finite $F \subseteq A$ such that $K \subseteq \bigcup_{\alpha \in F} G_\alpha$. As $K \subseteq Y$, $K \subseteq \bigcup_{\alpha \in F} V_\alpha$.
($\Leftarrow$) Let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of K in X. Then $\{Y \cap G_\alpha\}_{\alpha \in A}$ is a relative-open cover of K in Y.
Compactness in Y gives a finite $F \subseteq A$ such that $K \subseteq \bigcup_{\alpha \in F}(Y \cap G_\alpha) \subseteq \bigcup_{\alpha \in F} G_\alpha$.
Structural Commentary: Relative covers are ambient covers intersected with Y.
Key Insight: Openness depends on the ambient space, but compactness does not. □

Theorem 13.2.23. *Compact Sets Are Closed*
Every compact subset of a metric space is closed.

Proof. Let K be compact in X. To prove K^c is open.
Fix $p \in K^c$. For each $q \in K$, put $V_q := B_{\frac{1}{2}d(p,q)}(p)$, $W_q := B_{\frac{1}{2}d(p,q)}(q)$.
The sets W_q form an open cover of K, so compactness gives $q_1, \ldots, q_n \in K$ with $K \subseteq W_{q_1} \cup \cdots \cup W_{q_n}$.
Let $V := V_{q_1} \cap \cdots \cap V_{q_n}$, which is open by Theorem 13.1.18.
No point of V lies in any W_{q_i}, by the triangle inequality. Hence $V \cap K = \emptyset$, so p is an interior point of K^c.
Structural Commentary: Compactness makes the separation finite.
Key Insight: A finite subcover lets local point-separations become one open neighborhood of the complement. □

Theorem 13.2.24. *Closed Subsets of Compact Sets Are Compact*
Suppose $F \subseteq K \subseteq X$, where F is closed in X and K is compact. Then F is compact.

Proof. Let $\{V_\alpha\}_{\alpha \in A}$ cover F. Since F^c is open, $\{V_\alpha\}_{\alpha \in A} \cup \{F^c\}$ covers K.
Thus $K \subseteq F^c \cup \bigcup_{\alpha \in F_0} V_\alpha$ for some finite $F_0 \subseteq A$, so $F \subseteq \bigcup_{\alpha \in F_0} V_\alpha$.
Structural Commentary: The open complement of a closed set covers exactly the part of K not under discussion.
Key Insight: Cover K, then discard the complement piece. □

Corollary 13.2.25. *Closed Intersections with Compact Sets*
If F is closed in X and K is compact in X, then $F \cap K$ is compact.

Proof. By Theorem 13.2.23, K is closed in X. Since F is closed, Theorem 13.1.18 gives that $F \cap K$ is closed in X. Also $F \cap K \subseteq K$, so Theorem 13.2.24 implies $F \cap K$ is compact.
Structural Commentary: The intersection is closed inside a compact set.
Key Insight: Closed subsets of compact sets remain compact. □

Theorem 13.2.26. *Finite Intersection Property for Compact Sets*
Let $\{K_\alpha\}_{\alpha \in A}$ be compact subsets of a metric space X, with $A \neq \emptyset$.
If $\bigcap_{\alpha \in F} K_\alpha \neq \emptyset$ for every finite nonempty $F \subseteq A$, then $\bigcap_{\alpha \in A} K_\alpha \neq \emptyset$.

Proof. Choose $\alpha_0 \in A$ and suppose $\bigcap_{\alpha\in A} K_\alpha = \emptyset$. Then $K_{\alpha_0} \subseteq \bigcup_{\alpha\in A} K_\alpha^c$. Since each K_α^c is open by Theorem 13.2.23, compactness of K_{α_0} gives a finite $F \subseteq A$ with $K_{\alpha_0} \subseteq \bigcup_{\alpha\in F} K_\alpha^c$. Hence $\bigcap_{\alpha\in F\cup\{\alpha_0\}} K_\alpha = \emptyset$, a contradiction.

Structural Commentary: Complements turn failure into a finite cover.
Key Insight: For compact sets, failure of intersection is already witnessed finitely. □

Corollary 13.2.27. *Nested Nonempty Compact Sets Intersect*
If $K_1 \supseteq K_2 \supseteq K_3 \supseteq \cdots$ is a sequence of nonempty compact sets, then $\bigcap_{n=1}^{\infty} K_n \neq \emptyset$.

Proof. Let $\emptyset \neq F \subseteq \mathbb{N}$ be finite and put $m := \max F$. Since the sequence is nested, $\bigcap_{n\in F} K_n = K_m \neq \emptyset$.

Thus $\{K_n\}_{n\in\mathbb{N}}$ has the finite intersection property, so Theorem 13.2.26 gives $\bigcap_{n=1}^{\infty} K_n \neq \emptyset$.

Structural Commentary: Nestedness reduces finite intersections to one term.
Key Insight: A nested compact sequence has the finite intersection property. □

Theorem 13.2.28. *Infinite Subsets of Compact Sets Have Limit Points*
Let $K \subseteq X$ be compact in a metric space X. If $E \subseteq K$ is infinite, then $E^{\text{acc}} \cap K \neq \emptyset$.

Proof. Suppose $E^{\text{acc}} \cap K = \emptyset$. For each $q \in K$, choose $r_q > 0$ such that $B_{r_q}(q) \cap (E \setminus \{q\}) = \emptyset$, so $\#(B_{r_q}(q) \cap E) \leq 1$.
Since $\{B_{r_q}(q) : q \in K\}$ covers K, compactness gives $q_1, \ldots, q_n \in K$ with $K \subseteq \bigcup_{i=1}^{n} B_{r_{q_i}}(q_i)$.

As $E \subseteq K$, $E \subseteq \bigcup_{i=1}^{n} B_{r_{q_i}}(q_i)$, finite union of sets meeting E in at most one point. Thus E is finite, contradiction.

Structural Commentary: Compactness makes the local "at most one point" obstruction finite.
Key Insight: Infinite subsets of compact sets must accumulate somewhere inside the compact set. □

Theorem 13.2.29. *Nested Closed Intervals Intersect*
If $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$ is a sequence of closed intervals in $\mathbb{R}$, then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

Proof. Let $I_n = [a_n, b_n]$. The set $\{a_n : n \in \mathbb{N}\}$ is nonempty and bounded above by each b_m.
Let $x = \sup\{a_n : n \in \mathbb{N}\}$. Then $a_m \leq x \leq b_m$ for every m, so $x \in I_m$ for all m.
Structural Commentary: Completeness traps a common point.
Key Insight: The least upper bound of the left endpoints lies in every interval. □

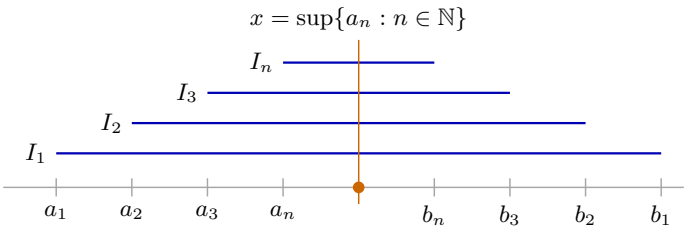


Figure 120: For nested closed intervals $[a_n, b_n]$, the supremum of the left endpoints lies between every a_n and b_n.

Theorem 13.2.30. *Nested k-Cells Intersect*
If $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$ is a sequence of k-cells in $\mathbb{R}^k$, then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

Proof. Let $I_n = \{\mathbf{x} \in \mathbb{R}^k : a_{n,j} \leq x_j \leq b_{n,j},\ 1 \leq j \leq k\}$. For each coordinate j, the intervals $[a_{n,j}, b_{n,j}]$ are nested, so Theorem 13.2.29 gives a point x_j^* lying in all of them. Then $\mathbf{x}^* = (x_1^*, \ldots, x_k^*)$ lies in every I_n.
Structural Commentary: A nested cell problem reduces coordinatewise to nested intervals.
Key Insight: Choose one persistent coordinate at a time. □

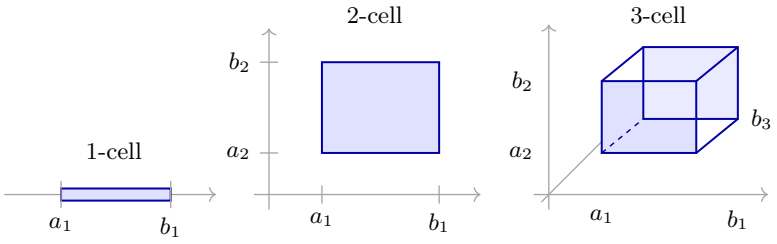


Figure 121: A 1-cell is a closed interval, a 2-cell is a coordinate rectangle, and a 3-cell is a coordinate box.

Theorem 13.2.31. *k-Cells Are Compact*
Every k-cell in $\mathbb{R}^k$ is compact.

Proof. Let $I = \{\mathbf{x} : a_j \le x_j \le b_j,\ 1 \le j \le k\}$ and set $\delta := \left(\sum_{j=1}^{k} (b_j - a_j)^2\right)^{1/2}$. Suppose an open cover $\{G_\alpha\}$ of I has no finite subcover. Bisect each coordinate interval. Among the resulting 2^k subcells, at least one cannot be finitely covered by the G_α; call it I_1. Continue inductively, obtaining nested cells $I \supseteq I_1 \supseteq I_2 \supseteq \cdots$ such that no I_n is finitely covered and every two points of I_n are within $2^{-n}\delta$ of each other.
By Theorem 13.2.30, choose $\mathbf{x}^* \in \bigcap_n I_n$. Since $\{G_\alpha\}$ covers I, some G_α contains $\mathbf{x}^*$. Because G_α is open, $B_r(\mathbf{x}^*) \subseteq G_\alpha$ for some $r > 0$. By the Archimedean property in Theorem 12.2.24, choose n with $n > \delta/r$; since $2^n \ge n$, we have $2^{-n}\delta < r$. Then $I_n \subseteq G_\alpha$, contradicting the construction of I_n.
Structural Commentary: Repeated bisection traps a point in cells whose diameters shrink to zero.
Key Insight: An open set around the trapped point eventually contains the whole small cell. □

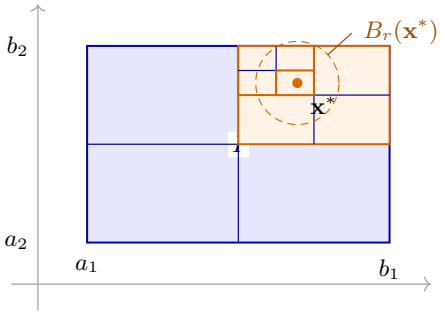


Figure 122: Repeated bisection traps a point.

Lemma 13.2.32. *Bounded Euclidean Sets Fit in k-Cells*
Every bounded subset $E \subseteq \mathbb{R}^k$ is contained in some k-cell.

Proof. If $E = \emptyset$, any k-cell contains E. Suppose $E \neq \emptyset$. By Definition 13.1.23, choose $\mathbf{q} = (q_1, \ldots, q_k) \in \mathbb{R}^k$ and $M > 0$ such that $\|\mathbf{p} - \mathbf{q}\|_2 < M$ for all $\mathbf{p} \in E$. Let $I := \{\mathbf{x} = (x_1, \ldots, x_k) : q_j - M \le x_j \le q_j + M,\ 1 \le j \le k\}$.
For $\mathbf{p} = (p_1, \ldots, p_k) \in E$, $|p_j - q_j| \le \|\mathbf{p} - \mathbf{q}\|_2 < M$ for each j, so $\mathbf{p} \in I$.
Thus $E \subseteq I$, and I is a k-cell by Definition 13.1.5.
Structural Commentary: A Euclidean ball centered at $\mathbf{q}$ sits inside the coordinate box with side half-length M.
Key Insight: Boundedness supplies one radius; coordinates turn it into interval bounds. □

Theorem 13.2.33. *Heine–Borel in $\mathbb{R}^k$*
For $E \subseteq \mathbb{R}^k$, any one of the following three properties implies the other two:

(i) E is closed and bounded;
(ii) E is compact;
(iii) every infinite subset of E has a limit point in E.

Proof.

(*i*) $\Rightarrow$ (*ii*): Since E is bounded, Lemma 13.2.32 gives $E \subseteq I$ for some k-cell I. By Theorem 13.2.31, I is compact; since E is closed, Theorem 13.2.24 gives that E is compact.
(*ii*) $\Rightarrow$ (*iii*): This is Theorem 13.2.28.

(*iii*) $\Rightarrow$ (*i*): If E is unbounded, choose $\mathbf{x}_n \in E$ with $\|\mathbf{x}_n\|_2 > n$ and set $S := \{\mathbf{x}_n : n \in \mathbb{N}\}$. If $\mathbf{x}_n \in B_R(\mathbf{y})$, then $n < \|\mathbf{y}\|_2 + R$; hence S has no limit point, contradicting (iii). Thus E is bounded. If E is not closed, choose $\mathbf{x}_0 \in E^{\text{acc}} \setminus E$ and $\mathbf{x}_n \in E$ with $0 < \|\mathbf{x}_n - \mathbf{x}_0\|_2 < 1/n$. Then $S := \{\mathbf{x}_n : n \in \mathbb{N}\}$ is infinite and $S^{\text{acc}} \subseteq \{\mathbf{x}_0\} \subseteq E^c$, contradicting (iii). Thus E is closed.

Structural Commentary: In $\mathbb{R}^k$, the three tests coincide.
Key Insight: Heine–Borel is the bridge between open-cover compactness and Euclidean bounded geometry. □

Corollary 13.2.34. *Heine–Borel on the Real Line*
For $E \subseteq \mathbb{R}$, the following are equivalent:

(i) E is closed and bounded;
(ii) E is compact;
(iii) every sequence in E has a subsequence converging to a point of E.

Proof.

(*i*) $\Rightarrow$ (*ii*) Since $E \subseteq \mathbb{R} = \mathbb{R}^1$ is closed and bounded, Theorem 13.2.33 with $k = 1$ implies E is compact.
(*ii*) $\Rightarrow$ (*iii*) Compactness and Theorem 13.2.20 give convergent subsequence; compactness keeps limit in E.
(*iii*) $\Rightarrow$ (*i*) Let $F \subseteq E$ be infinite. Choose distinct $x_n \in F$ using Principle 12.3.19. By (iii), some $x_{n_j} \to x \in E$. Since the x_n are distinct, x is a limit point of F in the sense of Definition 13.1.10. Hence every infinite subset of E has a limit point in E, so Theorem 13.2.33 gives that E is closed and bounded.

Structural Commentary: Real-line compactness equals closed-bounded sequential compactness.
Key Insight: Closed bounded sets keep sequences from escaping. □

Theorem 13.2.35. *Weierstrass Limit-Point Theorem*
Every bounded infinite subset of $\mathbb{R}^k$ has a limit point in $\mathbb{R}^k$.

Proof. By Lemma 13.2.32, a bounded set E lies in some k-cell I. By Theorem 13.2.31, I is compact. Since E is an infinite subset of I, Theorem 13.2.28 gives a limit point of E in $I \subseteq \mathbb{R}^k$.
Structural Commentary: Boundedness places the set inside a compact cell.
Key Insight: Compact cells force bounded infinite sets to accumulate. □

Theorem 13.2.36. *Perfect Sets Are Uncountable*
Every nonempty perfect subset of $\mathbb{R}^k$ is uncountable.

Proof. Assume $P = \{\mathbf{x}_1, \mathbf{x}_2, \ldots\}$. Choose a neighborhood V_1 with $V_1 \cap P \neq \emptyset$.
Inductively, since $P \cap V_n$ contains infinitely many points by Theorem 13.1.13, choose $\mathbf{y}_n \in P \cap V_n$ with $\mathbf{y}_n \neq \mathbf{x}_n$ and then a neighborhood V_{n+1} of $\mathbf{y}_n$ such that $\overline{V}_{n+1} \subseteq V_n$ and $\mathbf{x}_n \notin \overline{V}_{n+1}$.
Put $K_n := \overline{V}_n \cap P$. Then K_n is nonempty, compact, and $K_{n+1} \subseteq K_n$. By Corollary 13.2.27, choose $\mathbf{z} \in \bigcap_{n=1}^{\infty} K_n$.
For each n, $\mathbf{z} \in K_{n+1}$ but $\mathbf{x}_n \notin K_{n+1}$, so $\mathbf{z} \neq \mathbf{x}_n$ for all n, contradicting the enumeration.
Structural Commentary: Nested compact sets dodge the list.
Key Insight: Perfectness gives new points; compactness keeps one. □

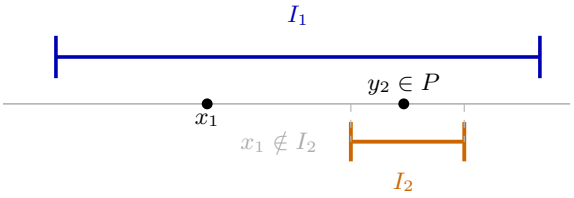


Figure 123: Choose $y_2 \in P \cap I_1$ with $y_2 \neq x_1$, then take a smaller interval $I_2 \subseteq I_1$ around y_2 that avoids x_1.

Corollary 13.2.37. *Intervals and the Real Line Are Uncountable*
Every interval $[a, b] \subseteq \mathbb{R}$ with $a < b$ is uncountable. In particular, $\mathbb{R}$ is uncountable.

Proof. The interval $I := [a, b]$ is a nonempty 1-cell, hence compact by Theorem 13.2.31 and closed by Theorem 13.2.23. For every $x \in I$ and $\varepsilon > 0$, $I \cap \big(B_\varepsilon(x) \setminus \{x\}\big) \neq \emptyset$, so I is perfect. Theorem 13.2.36 gives that I is uncountable. If $\mathbb{R}$ were countable, then $I \subseteq \mathbb{R}$ would be countable, a contradiction.
Structural Commentary: Closed intervals are perfect compact sets.
Key Insight: One closed interval already forces $\mathbb{R}$ to be uncountable. □

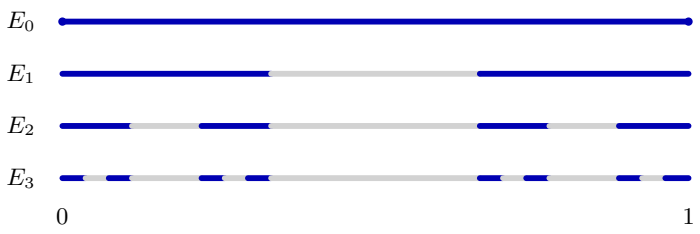


Figure 124: The first stages E_0, E_1, E_2, E_3 in the Cantor-set construction.

Example 13.2.38. *The Cantor Set*
Let $E_0 = [0,1]$. Remove the open middle third $(1/3, 2/3)$ and call the remaining set E_1. Continue by removing the open middle third from each interval component, so

$$E_1 = [0,1/3] \cup [2/3,1],$$
$$E_2 = [0,1/9] \cup [2/9,3/9] \cup [6/9,7/9] \cup [8/9,1].$$

Then E_n is a union of 2^n compact intervals of length 3^{-n} and $E_1 \supseteq E_2 \supseteq E_3 \supseteq \cdots$.
The *Cantor set* is $P := \bigcap_{n=1}^{\infty} E_n$. By Corollary 13.2.27, $P \neq \emptyset$. Since each E_n is closed and $P \subseteq E_1$, P is a closed subset of the compact set E_1, hence compact.
If $J \subseteq [0,1]$ is nonempty open interval, for some $m \in \mathbb{N}$ and $k \geq 0$, $R_{m,k} := \left(\frac{3k+1}{3^m}, \frac{3k+2}{3^m}\right) \subseteq J$ and $R_{m,k} \cap P = \emptyset$. Thus P contains no nontrivial interval. Finally, let $x \in P$ and let U be an interval neighborhood of x. If $I_n(x)$ is the component of E_n containing x, choose n with $I_n(x) \subseteq U$. An endpoint $y \neq x$ of $I_n(x)$ lies in P, so $U \cap (P \setminus \{x\}) \neq \emptyset$. Hence P is perfect.
Structural Commentary: The Cantor set is built by nested compact deletion, not by listing points.
Key Insight: It is compact and perfect, but contains no interval.

Definition 13.2.39. *F_σ, G_δ, and Nowhere-Dense Sets*
For subsets of a metric space X:

(i) A is an *F_σ set* if $A = \bigcup_{n=1}^{\infty} F_n$ for closed sets F_n.
(ii) B is a *G_δ set* if $B = \bigcap_{n=1}^{\infty} G_n$ for open sets G_n.
(iii) E is *nowhere dense* if $(\overline{E})^\circ = \emptyset$; equivalently, no nonempty open set is contained in $\overline{E}$.

For $X = \mathbb{R}$, this says exactly that $\overline{E}$ contains no nonempty open interval.

Definition 13.2.40. *First Category*
A set $E \subseteq X$ is *of first category*, *meager*, or *thin*, if covered by countably many nowhere-dense sets: $E \subseteq \bigcup_{n=1}^{\infty} E_n$.
The Baire Category Theorem states that a nonempty complete metric space is not first category in itself.

Definition 13.2.41. *Residual, Thick, and Generic Sets*
A set $G \subseteq X$ is *residual*, or *thick*, if it contains a countable intersection $\bigcap_{n=1}^{\infty} O_n$ of dense open subsets of X.
A property is *generic* in X if the set of points satisfying it is residual.
By Theorem 13.2.43, residual subsets of a complete metric space are dense.
In this sense, meager sets are topologically small and residual sets are topologically large.

Theorem 13.2.42. *Dense Open Intersections in $\mathbb{R}$*
If $G_1, G_2, G_3, \ldots$ are dense open subsets of $\mathbb{R}$, then $\bigcap_{n=1}^{\infty} G_n \neq \emptyset$.

Proof. We construct nested closed intervals. Since G_1 is dense and open, $G_1 \cap (0,1)$ contains a nonempty open interval; choose a closed interval $I_1 \subseteq G_1 \cap (0,1)$ with nonempty interior.
Given $I_n = [a_n, b_n]$, density of G_{n+1} gives a point of G_{n+1} in (a_n, b_n). Since G_{n+1} is open, choose a closed interval $I_{n+1} \subseteq G_{n+1} \cap (a_n, b_n)$ with nonempty interior and length $< 1/(n+1)$.
Then $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$, so Theorem 13.2.29 gives $x \in \bigcap_n I_n$. Since $I_n \subseteq G_n$ for every n, this x lies in $\bigcap_n G_n$.
Structural Commentary: Dense openness builds nested intervals; completeness keeps a limit point.
Key Insight: A countable dense-open intersection cannot be destroyed one interval at a time. □

Theorem 13.2.43. *Dense Open Intersections in Complete Metric Spaces*
Let (X,d) be complete, let $O_1, O_2, \ldots$ be dense open subsets of X, and let $U \subseteq X$ be nonempty and open. Then $U \cap \bigcap_{n=1}^{\infty} O_n \neq \emptyset$. Consequently, $\bigcap_{n=1}^{\infty} O_n$ is dense in X.

Proof. Choose $x_1 \in U \cap O_1$ and $0 < r_1 < 1$ with $C_1 := \overline{B}_{r_1}(x_1) \subseteq U \cap O_1$.
Inductively, if $C_n = \overline{B}_{r_n}(x_n) \subseteq U \cap O_1 \cap \cdots \cap O_n$ and $0 < r_n < 2^{-n}$, density and openness of O_{n+1} give $x_{n+1} \in B_{r_n}(x_n) \cap O_{n+1}$ and $0 < r_{n+1} < 2^{-(n+1)}$ such that $C_{n+1} := \overline{B}_{r_{n+1}}(x_{n+1}) \subseteq B_{r_n}(x_n) \cap O_{n+1}$.
Then $C_{n+1} \subseteq C_n$. If $m > n$, then $x_m \in B_{r_n}(x_n)$, so $d(x_m, x_n) < 2^{-n}$; hence (x_n) is Cauchy. Let $x_n \to x$.
Since each C_n is closed and contains all sufficiently late x_m, we have $x \in C_n$ for every n. Thus $x \in U \cap \bigcap_n O_n$.
Since every nonempty open U meets $\bigcap_n O_n$, the intersection is dense.
Structural Commentary: The interval proof becomes a nested closed-ball proof.
Key Insight: Dense open sets choose the next ball; completeness supplies the limiting point. □

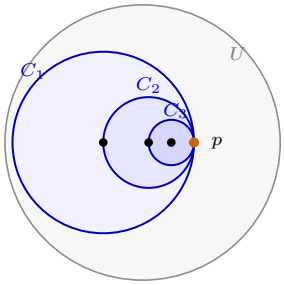


Figure 125: Nested closed balls shrink to p.

Theorem 13.2.44. *Baire Category Theorem for Complete Metric Spaces*
A nonempty complete metric space is not the union of a countable collection of nowhere-dense sets.

Proof. Suppose $X = \bigcup_n E_n$ with each E_n nowhere dense, and set $O_n := (\overline{E_n})^c$.
Then each O_n is dense open: it is open, and every nonempty open set meets it because no such set lies in $\overline{E_n}$.
By Theorem 13.2.43, choose $x \in \bigcap_n O_n$. Then $x \notin E_n$ for all n, contradicting $X = \bigcup_n E_n$.
Structural Commentary: Countably many nowhere-dense sets have dense open complements.
Key Insight: Completeness forbids countable assembly from thin pieces. □

Perspective 13.2.45. *Category in Function Spaces*
Baire category is a theorem about complete metric spaces, not just $\mathbb{R}$.
It will be reused for complete function spaces under the supremum metric.

Theorem 13.2.46. *Baire's Theorem for $\mathbb{R}$*
The real line $\mathbb{R}$ cannot be written as a countable union of nowhere-dense sets.

Proof. Assume $\mathbb{R} = \bigcup_n E_n$ with each E_n nowhere dense, and set $U_n := (\overline{E_n})^c$.
Each U_n is dense open: it is open, and every nonempty interval meets it because no such interval lies in $\overline{E_n}$.
By Theorem 13.2.42, choose $x \in \bigcap_n U_n$. Then $x \notin E_n$ for all n, contradicting $\mathbb{R} = \bigcup_n E_n$.
Structural Commentary: Nowhere-dense sets have dense open complements.
Key Insight: Countably many thin sets cannot exhaust the real line. □

Corollary 13.2.47. *Rationals and Irrationals by Category*
The set $\mathbb{Q}$ is F_σ but not G_δ. The set $\mathbb{R} \setminus \mathbb{Q}$ is G_δ but not F_σ.

Proof. Write $\mathbb{Q} = \{q_1, q_2, \ldots\}$. Since singletons are closed, $\mathbb{Q} = \bigcup_n \{q_n\}$ is F_σ, and $\mathbb{R} \setminus \mathbb{Q} = \bigcap_n (\{q_n\})^c$ is G_δ.
If $\mathbb{Q} = \bigcap_n G_n$ with each G_n open, then each G_n is dense because it contains the dense set $\mathbb{Q}$. Set $H_n := G_n \setminus \{q_n\}$.
Each H_n is dense open, so Theorem 13.2.42 gives a point of $\bigcap_n H_n$. But $\bigcap_n H_n = \emptyset$, since $\bigcap_n G_n = \mathbb{Q}$ and q_n is removed at stage n. Hence $\mathbb{Q}$ is not G_δ. Therefore $\mathbb{R} \setminus \mathbb{Q}$ is not F_σ, or else its complement $\mathbb{Q}$ would be G_δ.
Structural Commentary: Category separates the familiar countable dense set from its dense complement.
Key Insight: Dense is not the same as large in the Baire-category sense. □

Method 13.2.48. *Baire-Category Proof Patterns*

(i) *Dense-open intersection*: build nested closed intervals $I_n \subseteq G_n$ and apply Theorem 13.2.29.
(ii) *Complete metric version*: build nested closed balls $C_{n+1} \subseteq C_n$ with $r_n \to 0$ and use completeness.
(iii) *Nowhere dense*: replace E by the dense open set $(\overline{E})^c$.
(iv) *First category*: show the exceptional set is contained in $\bigcup_n E_n$ with each E_n nowhere dense.
(v) *Residual/generic*: show the desired property contains $\bigcap_n O_n$ with each O_n dense open.
(vi) *F_σ/G_δ complements*: pass between countable unions and intersections by De Morgan's laws.

13.3 Connectedness and Continuity

Connectedness prevents a space from splitting into separated pieces; continuity transports topological structure through maps. This subsection also records IVT/EVT, uniform continuity, and real-line refinements.

Definition 13.3.1. *Separated and Connected Sets*
Let A and B be subsets of a metric space X. They are *separated* if $A \cap \overline{B} = \emptyset$ and $\overline{A} \cap B = \emptyset$.
A set $E \subseteq X$ is *connected* if it is not a union of two nonempty separated sets.

Remark 13.3.2. *Disjoint Need Not Mean Separated*
Separated sets are disjoint, but disjoint sets need not be separated. For example, $[0,1]$ and $(1,2)$ are disjoint but not separated because 1 is a limit point of $(1,2)$. By contrast, $(0,1)$ and $(1,2)$ are separated.

Remark 13.3.3. *Connectedness and Clopen Sets*
A subset is *clopen* if it is both open and closed; $\sqcup$ denotes disjoint union. A metric space is disconnected iff it has a nonempty proper clopen subset: U gives $X = U \sqcup U^c$, and every separation has clopen pieces.
By Theorem 13.3.19, continuous maps pull clopen separations back to clopen separations.

Remark 13.3.4. *Path-Connectedness Uses Continuity*
The supplied Tao II Ch.2 exercises introduce path-connectedness immediately after connectedness. Here the definition and main theorem are placed later as Definition 13.3.41 and Theorem 13.3.42, after continuity has been developed.

Theorem 13.3.5. *Connected Subsets of the Real Line*
A set $E \subseteq \mathbb{R}$ is connected if and only if $x, y \in E,\ x < y \implies [x,y] \subseteq E$.

Proof.
($\Rightarrow$) If the interval property fails, choose $x, y \in E$ and $x < z < y$ with $z \notin E$. Then $A := E \cap (-\infty, z)$ and $B := E \cap (z, \infty)$ are nonempty separated sets with $E = A \cup B$, contradicting connectedness.
($\Leftarrow$) Suppose $E = A \cup B$ with A, B nonempty separated. Choose $x \in A$, $y \in B$, $x < y$, and set $S := A \cap [x,y]$, $z := \sup S$. By Theorem 13.1.35, $z \in \overline{A}$; hence $z \notin B$ and $z < y$. If $z \notin A$, then $z \in [x,y] \setminus E$.
If $z \in A$, then $z \notin \overline{B}$, so choose $z_1 \in (z,y)$ with $z_1 \notin B$; by maximality of z, also $z_1 \notin A$.
Thus $z_1 \in [x,y] \setminus E$, contradicting $[x,y] \subseteq E$.

Structural Commentary: On the real line, connectedness is exactly having no gaps between any two of its points.
Key Insight: A separation creates a missing point by taking a supremum at the boundary of one side. □

Method 13.3.6. *Compactness and Connectedness Proof Patterns*
(i) *Open-cover compactness*: for every cover $\{G_\alpha\}_{\alpha \in A}$, find finite $F \subseteq A$ with $K \subseteq \bigcup_{\alpha \in F} G_\alpha$.
(ii) *Euclidean closed-bounded compactness*: if $E \subseteq \mathbb{R}^k$ is closed and bounded, choose a compact k-cell $I \supseteq E$ and apply Theorem 13.2.24.
(iii) *Connectedness on $\mathbb{R}$*: replace connectedness of $E \subseteq \mathbb{R}$ by $x, y \in E, x < y \implies [x,y] \subseteq E$.

Definition 13.3.7. *Limit of a Function Between Metric Spaces*
Let X, Y be metric spaces, $E \subseteq X$, $f : E \to Y$, and let p be a limit point of E. For $q \in Y$, $\lim_{x \to p} f(x) = q$ means that, for every $\varepsilon > 0$, there exists $\delta > 0$ such that $x \in E$, $0 < d_X(x,p) < \delta \implies d_Y(f(x), q) < \varepsilon$.
The point p need not belong to E, and $f(p)$ is irrelevant when it does.
Key Insight: Domain E controls approach: $J \setminus \{a\}$ gives two-sided limit, (a,b) or (c,a) gives one-sided limits.

Remark 13.3.8. *Adherent-Point Limit Convention*
The adherent-point convention takes $p \in \overline{E}$ and allows $x = p$ in the δ-test; Definition 13.3.7 instead assumes p is a limit point and requires $0 < d_X(x,p)$. If $p \in E$, the unpunctured convention also tests $f(p)$, forcing $q = f(p)$; in the punctured convention this is exactly continuity at p by Theorem 13.3.15.
Key Insight: Punctured limits study nearby values; continuity adds compatibility with the value at the point.

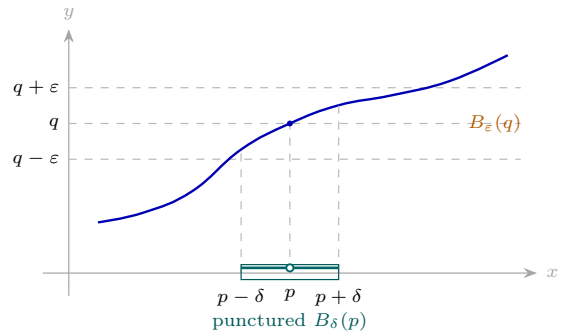


Figure 126: The epsilon–delta geometry for $\lim_{x\to p} f(x) = q$.

Theorem 13.3.9. *Sequential Characterization of Function Limits*
Let X, Y be metric spaces, $E \subseteq X$, $f : E \to Y$, and let p be a limit point of E. For $q \in Y$, $\lim_{x\to p} f(x) = q$ if and only if $f(p_n) \to q$ for every sequence $(p_n) \subseteq E \setminus \{p\}$ with $p_n \to p$.

Proof.
($\Rightarrow$) If $p_n \to p$ with $p_n \in E \setminus \{p\}$, then eventually $0 < d_X(p_n, p) < \delta$; hence $d_Y(f(p_n), q) < \varepsilon$, so $f(p_n) \to q$.
($\Leftarrow$) If the limit fails, then for some $\varepsilon > 0$ and each n there is $x_n \in E$ with $0 < d_X(x_n, p) < 1/n$ and $d_Y(f(x_n), q) \geq \varepsilon$. Then $x_n \to p$, $x_n \neq p$, but $f(x_n) \not\to q$.
Structural Commentary: Function limits can be tested by all sequences approaching the input point.
Key Insight: The negation of an epsilon–delta limit manufactures a bad sequence. □

Principle 13.3.10. *Challenge–Response Order*
Limits and continuity are controlled by quantifier order. In Definition 13.3.7, the ε-challenge at the range point must be answered by a δ-restriction near the domain point. Uniform continuity, Definition 13.3.35, strengthens the same pattern by requiring one δ to work for all relevant domain points at once.

Corollary 13.3.11. *Uniqueness of Function Limits*
If a function has a limit at p, then that limit is unique.

Proof. Let q and r be limits of f at p. Choose $p_n \in E \setminus \{p\}$ with $d_X(p_n, p) < 1/n$. Then $p_n \to p$, so Theorem 13.3.9 gives $f(p_n) \to q$ and $f(p_n) \to r$. By Theorem 13.1.30, $q = r$.
Structural Commentary: Limit uniqueness is inherited from metric sequence uniqueness.
Key Insight: One approaching sequence cannot have two different limits. □

Theorem 13.3.12. *Algebra of Function Limits*
Let X be a metric space, $E \subseteq X$, p be limit point of E.
For $f, g : E \to \mathbb{C}$, assume $\lim_{x\to p} f(x) = A$ and $\lim_{x\to p} g(x) = B$. Then:
(i) $\lim_{x\to p}(f+g)(x) = A + B$;
(ii) $\lim_{x\to p}(fg)(x) = AB$;
(iii) if $B \neq 0$ and f/g is defined near p, then $\lim_{x\to p}(f/g)(x) = A/B$.

For $f, g : E \to \mathbb{R}^k$ with limits a, b, the corresponding limits are $f + g \to a + b$, $\lambda f \to \lambda a$, and $f \cdot g \to a \cdot b$.

Proof.
(i) For any $p_n \in E \setminus \{p\}$ with $p_n \to p$, Theorem 13.3.9 gives $f(p_n) \to A$ and $g(p_n) \to B$. Hence $(f+g)(p_n) \to A + B$ by Theorem 12.4.3, so Theorem 13.3.9 gives (i).
(ii) The same sequence gives $(fg)(p_n) \to AB$ by Theorem 12.4.3; hence (ii).
(iii) If $B \neq 0$ and f/g is defined near p, then every such sequence is eventually in the domain of f/g, and $(f/g)(p_n) \to A/B$ by Theorem 12.4.3; hence (iii).
The vector laws follow by the same sequential characterization and Theorem 12.4.4.
Structural Commentary: Limit algebra for functions is pulled back to sequence algebra.
Key Insight: Sequential characterization lets previous sequence theorems do the work. □

Proposition 13.3.13. *Function Limits Are Local*
Let X, Y be metric spaces, let $E \subseteq X$, let $f : E \to Y$, let p be a limit point of E, and let $q \in Y$.
For every $r > 0$, $\lim_{x\to p;\, x\in E} f(x) = q \iff \lim_{x\to p;\, x\in E\cap B_r(p)} f(x) = q$.

Proof.

($\Rightarrow$) Since $E \cap B_r(p) \subseteq E$, any δ witnessing the limit on E also witnesses the limit on $E \cap B_r(p)$.
($\Leftarrow$) Assume the restricted limit. Given $\varepsilon > 0$, choose $\delta_0 > 0$ such that $x \in E \cap B_r(p)$ and $0 < d_X(x,p) < \delta_0$ imply $d_Y(f(x), q) < \varepsilon$. Let $\delta := \min\{\delta_0, r\}$.
If $x \in E$ and $0 < d_X(x,p) < \delta$, then $x \in E \cap B_r(p)$, so $d_Y(f(x), q) < \varepsilon$.

Structural Commentary: Values away from p cannot affect the limit at p.
Key Insight: Shrink δ until every tested point lies in the chosen local neighborhood. □

Definition 13.3.14. *Continuity at a Point and on a Set*
For metric spaces X, Y, $E \subseteq X$, $p \in E$, and $f : E \to Y$, the function f is *continuous at p* if $\forall \varepsilon > 0\ \exists \delta > 0$, $x \in E$, $d_X(x,p) < \delta \implies d_Y(f(x), f(p)) < \varepsilon$.
It is *continuous on E* if continuous at every point of E; isolated points are automatic.

Theorem 13.3.15. *Continuity as Equality with the Function Limit*
Let X, Y be metric spaces, $E \subseteq X$, $p \in E$, and $f : E \to Y$.
If p is a limit point of E, then f is continuous at p if and only if $\lim_{x \to p} f(x) = f(p)$.

Proof.

($\Rightarrow$) If $0 < d_X(x,p) < \delta$, continuity at p gives $d_Y(f(x), f(p)) < \varepsilon$, hence $\lim_{x \to p} f(x) = f(p)$.
($\Leftarrow$) If $\lim_{x \to p} f(x) = f(p)$, the same δ proves continuity for $x \neq p$; for $x = p$, $d_Y(f(p), f(p)) = 0$.

Structural Commentary: Continuity is the compatibility of the value at p with nearby values.
Key Insight: Limits ignore the point; continuity ties the point back in. □

Remark 13.3.16. *The Sequence-First View*
For functions on subsets of $\mathbb{R}$, continuity at limit points is equivalently sequential: $x_n \to p$ in the domain implies $f(x_n) \to f(p)$; isolated points are automatic by Theorems 13.3.9 and 13.3.15.

Perspective 13.3.17. *Three Views of Continuity*
For maps between metric spaces, continuity has three equivalent views: ε–δ control, preservation of sequence limits, and open inverse images; see Definition 13.3.14 and Theorems 13.3.9, 13.3.15, and 13.3.19.
Key Insight: Continuity moves local nearness forward, but tests open-set structure by pulling sets backward.

Theorem 13.3.18. *Composition of Continuous Functions*
Let $E \subseteq X$ and let $f : E \to Y$, $g : f(E) \to Z$ be maps between metric spaces.
If f is continuous at $p \in E$ and g is continuous at $f(p)$, then $g \circ f$ is continuous at p.

Proof. Given $\varepsilon > 0$, choose $\eta > 0$ for g at $f(p)$, then choose $\delta > 0$ for f at p with tolerance η.
If $x \in E$ and $d_X(x,p) < \delta$, then $d_Y(f(x), f(p)) < \eta$, hence $d_Z(g(f(x)), g(f(p))) < \varepsilon$.
Structural Commentary: The output tolerance for g becomes the input tolerance for f.
Key Insight: Continuity composes by passing epsilon through an intermediate eta. □

Theorem 13.3.19. *Continuity via Open Preimages*
For metric spaces X, Y, map $f : X \to Y$ is continuous if and only if $f^{-1}(V)$ is open in $X \forall$ open $V \subseteq Y$.

Proof.

($\Rightarrow$) If $p \in f^{-1}(V)$ and V is open, choose $\varepsilon > 0$ with $B_\varepsilon(f(p)) \subseteq V$.
Continuity gives $\delta > 0$ with $B_\delta(p) \subseteq f^{-1}(V)$, so $f^{-1}(V)$ is open.
($\Leftarrow$) For $p \in X$ and $\varepsilon > 0$, the ball $V = B_\varepsilon(f(p))$ is open by Theorem 13.1.12.
Since $p \in f^{-1}(V)$ and $f^{-1}(V)$ is open, some $B_\delta(p) \subseteq f^{-1}(V)$; this is continuity at p.

Structural Commentary: Continuity is exactly preservation of openness under inverse image.
Key Insight: Images are not Boolean-exact, but preimages are; continuity belongs to preimages. □

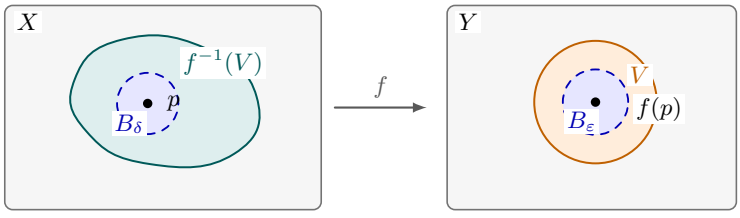


Figure 127: Continuity pulls open sets back to open sets.

Corollary 13.3.20. *Continuity via Closed Preimages*
For metric spaces X, Y, map $f : X \to Y$ is continuous if and only if $f^{-1}(C)$ is closed in $X \forall$ closed $C \subseteq Y$.

Proof.
($\Rightarrow$) If C is closed, then $Y \setminus C$ is open. By Theorem 13.3.19, $f^{-1}(Y \setminus C)$ is open; by Proposition 12.1.16, this is $X \setminus f^{-1}(C)$, so $f^{-1}(C)$ is closed.
($\Leftarrow$) If V is open, then $Y \setminus V$ is closed. Hence $f^{-1}(Y \setminus V)$ is closed, so $f^{-1}(V) = X \setminus f^{-1}(Y \setminus V)$ is open. Theorem 13.3.19 gives continuity.

Structural Commentary: Closed-set continuity is the complement form of open-set continuity.
Key Insight: $f^{-1}(Y \setminus C) = X \setminus f^{-1}(C)$ transfers open and closed formulations. □

Remark 13.3.21. *Preimages, Not Forward Images*
Theorem 13.3.19 is about inverse images. A continuous map need not send open sets to open sets or closed sets to closed sets; compactness is one hypothesis that recovers closedness of images through Theorem 13.3.31.

Definition 13.3.22. *Homeomorphism*
A *homeomorphism* from X to Y is a bijection $f : X \to Y$ such that both f and f^{-1} are continuous.
If such a map exists, X and Y are *homeomorphic*, written $X \simeq Y$.

Proposition 13.3.23. *Homeomorphisms Preserve Topology*
If $f : X \to Y$ is a homeomorphism, then $U \mapsto f(U)$ bijects open subsets of X with open subsets of Y.

Proof. If $U \subseteq X$ is open, then $f(U) = (f^{-1})^{-1}(U)$ is open in Y since f^{-1} is continuous. If $V \subseteq Y$ is open, then $f^{-1}(V)$ is open in X since f is continuous. These two assignments are inverse because f is bijective.
Structural Commentary: A homeomorphism preserves open sets in both directions.
Key Insight: Topological sameness is continuity plus continuous reversibility. □

Example 13.3.24. *Continuous Bijection Need Not Be a Homeomorphism*
Let $f : [0, 2\pi) \to S^1$, and $f(t) = (\cos t, \sin t)$. This continuous bijection is not a homeomorphism: points of S^1 approaching $(1, 0)$ from below have preimages tending to 2π, not 0.

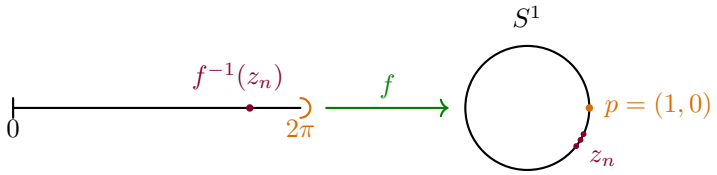


Figure 128: The parametrization $[0, 2\pi) \to S^1$ wraps continuously and bijectively, but the inverse jumps at $(1, 0)$.

Remark 13.3.25. *Continuity Only Uses the Domain Metric*
For $f : E \to Y$ with $E \subseteq X$, continuity on E depends only on the restricted metric on E, not on $X \setminus E$.

Definition 13.3.26. *Product Map*
If $f : X \to Y$ and $g : X \to Z$ are maps between metric spaces, their *product map*, or direct sum, is $f \oplus g : X \to Y \times Z$, $(f \oplus g)(x) := (f(x), g(x))$, where $Y \times Z$ carries any of product metrics in Definition 13.1.8.

Proposition 13.3.27. *Continuity of Product Maps*
The product map $f \oplus g : X \to Y \times Z$ is continuous at x_0 iff f and g are continuous at x_0; hence it is continuous on X iff both components are.

Proof. By Proposition 13.1.9, it suffices to use d_{sum}.
($\Rightarrow$) Given $\varepsilon > 0$, choose $\delta > 0$ from continuity of $f \oplus g$ at x_0.
If $d_X(x, x_0) < \delta$, then $d_Y(f(x), f(x_0)), d_Z(g(x), g(x_0)) \leq d_{\text{sum}}((f \oplus g)(x), (f \oplus g)(x_0)) < \varepsilon$.
($\Leftarrow$) Given $\varepsilon > 0$, choose δ_f, δ_g for f, g at tolerance $\varepsilon/2$ and set $\delta = \min\{\delta_f, \delta_g\}$.
Then $d_X(x, x_0) < \delta$ gives $d_{\text{sum}}((f \oplus g)(x), (f \oplus g)(x_0)) = d_Y(f(x), f(x_0)) + d_Z(g(x), g(x_0)) < \varepsilon$.

Structural Commentary: Product continuity is exactly componentwise continuity.
Key Insight: A product metric packages two output estimates into one. □

Theorem 13.3.28. *Continuous Algebra and Components*
Let X be a metric space.
(i) If $f, g : X \to \mathbb{C}$ are continuous, so are $f + g$, fg, and f/g where $g \neq 0$.

(ii) For $f = (f_1, \ldots, f_k) : X \to \mathbb{R}^k$, f is continuous if and only if each f_j is continuous.
(iii) If $f, g : X \to \mathbb{R}^k$ are continuous, so are $f + g$ and $f \cdot g$.

Proof.

(i) At isolated points continuity is automatic; at limit points, use Theorems 13.3.12 and 13.3.15.
(ii) By Theorem 12.4.4, convergence in $\mathbb{R}^k$ is exactly coordinatewise convergence; apply this to $f(x_n) \to f(x)$.
(iii) Apply (ii) to the coordinate functions of $f + g$ and $f \cdot g$, using the scalar algebra from (i).

Structural Commentary: Continuity is stable under the algebraic operations already stable for limits.
Key Insight: Componentwise control is equivalent to Euclidean control in finite dimensions. □

Example 13.3.29. *Coordinate, Polynomial, Rational, and Norm Maps*
Coordinate functions on $\mathbb{R}^k$ are continuous; hence polynomials are continuous, and rational functions are continuous where defined. Also $\mathbf{x} \mapsto \|\mathbf{x}\|_2$ is continuous since $\big|\|\mathbf{x}\|_2 - \|\mathbf{y}\|_2\big| \le \|\mathbf{x} - \mathbf{y}\|_2$.
Thus $p \mapsto \|f(p)\|_2$ is continuous whenever $f : X \to \mathbb{R}^k$ is, by Theorem 13.3.18.

Definition 13.3.30. *Bounded Mapping*
A mapping $f : E \to \mathbb{R}^k$ is *bounded* if there exists $M \in \mathbb{R}$ such that $\|f(x)\|_2 \le M$ for every $x \in E$.

Theorem 13.3.31. *Continuous Images of Compact Sets Are Compact*
If $f : X \to Y$ is continuous and X is compact, then $f(X)$ is compact.

Proof. Let $\{V_\alpha\}$ cover $f(X)$. Then $\{f^{-1}(V_\alpha)\}$ is an open cover of X by Theorem 13.3.19.
Choose $\alpha_1, \ldots, \alpha_n$ with $X \subseteq \bigcup_{i=1}^n f^{-1}(V_{\alpha_i})$; then $f(X) \subseteq \bigcup_{i=1}^n V_{\alpha_i}$.
Structural Commentary: Pull back, take a finite subcover, push forward.
Key Insight: Compactness is preserved by continuous images because inverse images preserve openness. □

Corollary 13.3.32. *Continuous Images into $\mathbb{R}^k$ Are Closed and Bounded*
If $f : X \to \mathbb{R}^k$ is continuous and X is compact, then $f(X)$ is closed and bounded. In particular, f is bounded.

Proof. By Theorem 13.3.31, $f(X)$ is compact.
By Theorem 13.2.33, compact subsets of $\mathbb{R}^k$ are closed and bounded; hence f is bounded.
Structural Commentary: Compactness in the domain becomes closed-bounded geometry in Euclidean range.
Key Insight: Continuous functions on compact spaces cannot escape to infinity. □

Theorem 13.3.33. *Extreme Value Theorem*
Let $f : X \to \mathbb{R}$ be continuous, let X be compact.
If $M := \sup_{p \in X} f(p)$, $m := \inf_{p \in X} f(p)$, then $\exists\ p, q \in X$ such that $f(p) = M$ and $f(q) = m$.

Proof. By Corollary 13.3.32, $f(X)$ is closed and bounded in $\mathbb{R}$. Thus $\sup f(X)$ and $\inf f(X)$ exist and belong to $f(X)$ by Theorems 13.1.35 and 12.2.8. Hence $M = f(p)$ and $m = f(q)$ for some $p, q \in X$.
Structural Commentary: Compactness turns suprema and infima into attained values.
Key Insight: Closed bounded images contain their endpoints. □

Theorem 13.3.34. *Compact Bijective Maps Have Continuous Inverses*
If $f : X \to Y$ is a continuous bijection from compact X to a metric space Y, then $f^{-1} : Y \to X$ is continuous.

Proof. It suffices, by Theorem 13.3.19, to show that $f(V)$ is open in Y for every open $V \subseteq X$. If V is open, then V^c is compact by Theorem 13.2.24, so $f(V^c)$ is compact, hence closed by Theorems 13.3.31 and 13.2.23. Since f is bijective, $f(V) = Y \setminus f(V^c)$ is open.
Structural Commentary: Compactness turns closed complements into closed images.
Key Insight: A continuous bijection from compact domain has no room for inverse discontinuity. □

Definition 13.3.35. *Uniform Continuity*
A mapping $f : X \to Y$ between metric spaces is *uniformly continuous* on X if $\forall\ \varepsilon > 0\ \exists\ \delta > 0$ such that $d_X(p, q) < \delta \implies d_Y(f(p), f(q)) < \varepsilon\ \forall p, q \in X$. The same δ must work for every point of X.
Structural Commentary: Uniform continuity is setwise: for each ε, one δ must work for all pairs in the set.

Proposition 13.3.36. *Sequential Test for Uniform Continuity*
Let X, Y be metric spaces and let $f : X \to Y$. The following are equivalent:
(i) f is uniformly continuous on X;
(ii) whenever (p_n) and (q_n) are sequences in X with $d_X(p_n, q_n) \to 0$, one has $d_Y(f(p_n), f(q_n)) \to 0$.

Proof.

(*i*) $\Rightarrow$ (*ii*) Given $\varepsilon > 0$, choose $\delta > 0$ such that $d_X(x, y) < \delta$ implies $d_Y(f(x), f(y)) < \varepsilon$ for all $x, y \in X$. Since $d_X(p_n, q_n) \to 0$, eventually $d_X(p_n, q_n) < \delta$, hence eventually $d_Y(f(p_n), f(q_n)) < \varepsilon$.

(*ii*) $\Rightarrow$ (*i*) Prove the contrapositive. If f is not uniformly continuous, then $\exists \varepsilon_0 > 0$ such that $\forall n \geq 1$, $\exists p_n, q_n \in X$ with $d_X(p_n, q_n) < 1/n$ and $d_Y(f(p_n), f(q_n)) \geq \varepsilon_0$. Thus $d_X(p_n, q_n) \to 0$ but $d_Y(f(p_n), f(q_n)) \not\to 0$, contradicting (ii).

Structural Commentary: Uniform continuity preserves vanishing pairwise distances.
Key Insight: Failure of one global δ manufactures two close input sequences with separated outputs. □

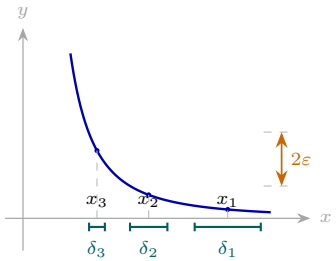


Figure 129: For $f(x) = 1/x^2$ near 0, a fixed vertical tolerance forces smaller pointwise horizontal tolerances.

Theorem 13.3.37. *Continuity on Compact Sets Is Uniform*
If $f : X \to Y$ is continuous and X is compact, then f is uniformly continuous.

Proof. Let $\varepsilon > 0$. For each $a \in X$, choose $r_a > 0$ such that $d_X(x, a) < r_a$ implies $d_Y(f(x), f(a)) < \varepsilon/2$. Compactness gives a finite cover $X \subseteq \bigcup_{i=1}^{n} B_{r_{a_i}/2}(a_i)$. Set $\delta := \frac{1}{2} \min_i r_{a_i} > 0$. If $d_X(p, q) < \delta$ and $p \in B_{r_{a_i}/2}(a_i)$, then $d_X(q, a_i) < r_{a_i}$ also; hence $f(p)$ and $f(q)$ are each within $\varepsilon/2$ of $f(a_i)$. Thus $d_Y(f(p), f(q)) < \varepsilon$.
Structural Commentary: Compactness converts pointwise radii into finitely many radii with positive minimum.
Key Insight: Uniform continuity is continuity with the point-dependence removed by a finite subcover. □

Theorem 13.3.38. *Compactness Hypotheses Are Essential*
Let E be a noncompact subset of $\mathbb{R}$. Then:

(i) there exists a continuous function on E which is not bounded;
(ii) there exists a continuous bounded function on E which has no maximum;
(iii) if E is bounded, then there exists a continuous function on E which is not uniformly continuous.

Proof.

(i) If E is unbounded, $f(x) = x$ works. If E is bounded, Theorem 13.2.33 gives $x_0 \in \overline{E} \setminus E$, and $f(x) = 1/(x - x_0)$ is continuous on E and unbounded.

(ii) If E is unbounded, $h(x) = x^2/(1 + x^2)$ has $\sup_E h = 1$ but $h < 1$. If E is bounded, with x_0 as above, $g(x) = 1/(1 + (x - x_0)^2)$ is bounded by 1 and has supremum 1 but no maximum.

(iii) If E is bounded, let $f(x) = 1/(x - x_0)$. Given $\delta > 0$, let $x \in E$ with $|x - x_0| < \delta/2$, then $y \in E$ with $|y - x_0| < \min\{\delta/2, 1/(|f(x)| + 1)\}$. Then $|x - y| < \delta$, $|f(x) - f(y)| > 1$, so f is not uniformly continuous.

Structural Commentary: Compactness is essential in the preceding theorems.
Key Insight: Noncompact sets either escape to infinity or miss an accumulation point. □

Theorem 13.3.39. *Continuous Images of Connected Sets Are Connected*
If $f : X \to Y$ is continuous and $E \subseteq X$ is connected, then $f(E)$ is connected.

Proof. If $f(E) = A \cup B$ with A, B nonempty separated, set $G := E \cap f^{-1}(A)$ and $H := E \cap f^{-1}(B)$. Then $E = G \cup H$ and G, H are nonempty. Since $G \subseteq f^{-1}(\overline{A})$ and $f^{-1}(\overline{A})$ is closed, $\overline{G} \subseteq f^{-1}(\overline{A})$; hence $\overline{G} \cap H = \emptyset$. Similarly $G \cap \overline{H} = \emptyset$, so G, H separate E, a contradiction.
Structural Commentary: A separation of the image pulls back to a separation of the domain.
Key Insight: Continuous maps cannot create a gap in the image of a connected set. □

Theorem 13.3.40. *Intermediate Value Theorem*
If $f : [a, b] \to \mathbb{R}$ is continuous and c lies strictly between $f(a)$ and $f(b)$, then $f(x) = c$ for some $x \in (a, b)$.

Proof. The interval $[a, b]$ is connected by Theorem 13.3.5, so $f([a, b])$ is connected by Theorem 13.3.39. Connected subsets of $\mathbb{R}$ contain every value between any two of their points; hence $c = f(x)$ for some $x \in [a, b]$. Since $c \neq f(a), f(b)$, $x \in (a, b)$.
Structural Commentary: The intermediate value theorem is connectedness of intervals transported by continuity.
Key Insight: Continuous real functions on intervals cannot jump over a value. □

Definition 13.3.41. *Path-Connected Set*
A set E in a metric space is *path-connected* if for every $p, q \in E$ there is a continuous map $\gamma : [a, b] \to E$ with $\gamma(a) = p$ and $\gamma(b) = q$. Such a map γ is a *path* from p to q.

Theorem 13.3.42. *Path-Connected Sets Are Connected*
Every path-connected set is connected.

Proof. Suppose $E = A \cup B$ with A, B nonempty separated.
Choose $p \in A$, $q \in B$, and a path $\gamma : [a, b] \to E$ from p to q. Since $[a, b]$ is connected, so is $\gamma([a, b])$.
But $\gamma([a, b])$ meets both A and B, which separate it, a contradiction.
Structural Commentary: A path transports connectedness from an interval to the set.
Key Insight: To connect two points by a path, the path image cannot be split into two separated pieces. □

Corollary 13.3.43. *Convex Sets Are Path-Connected*
Every convex subset of $\mathbb{R}^k$ is path-connected.

Proof. For $\mathbf{x}, \mathbf{y} \in E$, set $\gamma(t) = (1-t)\mathbf{x}+t\mathbf{y}$ on $[0, 1]$. Convexity gives $\gamma([0, 1]) \subseteq E$, with $\gamma(0) = \mathbf{x}$ and $\gamma(1) = \mathbf{y}$. Since the coordinates of γ are affine in t, γ is continuous. Hence E is path-connected.
Structural Commentary: Convexity supplies the whole straight path between any two points.
Key Insight: In Euclidean space, line segments are continuous paths. □

Definition 13.3.44. *One-Sided Limits and Types of Discontinuity*
Let $X \subseteq \mathbb{R}$, $f : X \to \mathbb{R}$, and $x_0 \in \mathbb{R}$.

(i) If x_0 is a limit point of $X \cap (x_0, \infty)$, then $f(x_0+) = q$ means $\lim_{x \to x_0;\, x \in X \cap (x_0, \infty)} f(x) = q$.

(ii) If x_0 is a limit point of $X \cap (-\infty, x_0)$, then $f(x_0-) = q$ means $\lim_{x \to x_0;\, x \in X \cap (-\infty, x_0)} f(x) = q$.

(iii) When X approaches x_0 from both sides, the two-sided punctured limit through X exists if and only if the two one-sided limits exist and are equal.

For functions on an interval, a discontinuity with both one-sided limits is of the *first kind*, or *simple*; otherwise it is of the *second kind*. Simple discontinuities are jumps, $f(x_0+) \neq f(x_0-)$, or removable value mismatches.

Example 13.3.45. *Basic Discontinuity Types*
Dirichlet's function $\mathbf{1}_{\mathbb{Q}}$ has a discontinuity of the second kind at every point.
The function $f(x) = x$ on $\mathbb{Q}$ and $f(x) = 0$ off $\mathbb{Q}$ is continuous only at 0; elsewhere its discontinuities are of the second kind. A jump in a piecewise linear function is a simple discontinuity.
Use Corollary 12.2.26: every interval contains rational and irrational points.

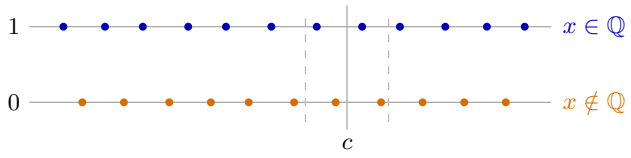


Figure 130: For $g = \mathbf{1}_{\mathbb{Q}}$, interval around c has rational inputs value 1 and irrational inputs value 0.

Definition 13.3.46. *Monotone Functions*
Let f be real-valued on (a, b). The function is *monotonically increasing* if $a < x < y < b$ implies $f(x) \leq f(y)$. Reversing the inequality gives *monotonically decreasing*. A *monotone* function is one of these two types.

Theorem 13.3.47. *One-Sided Limits of Monotone Functions*
If f is increasing on (a, b), then for every $x \in (a, b)$, $\sup_{a<t<x} f(t) = f(x-) \leq f(x) \leq f(x+) = \inf_{x<t<b} f(t)$.
Moreover, $a < x < y < b$ implies $f(x+) \leq f(y-)$. For decreasing f, reverse the inequalities.

Proof. Let $A := \sup_{a<t<x} f(t)$.
For every $\varepsilon > 0$, choose $s < x$ with $A - \varepsilon < f(s) \le A$; then $s < t < x$ implies $A - \varepsilon < f(t) \le A$, so $f(x-) = A$.
Let $B := \inf_{x<t<b} f(t)$. For every $\varepsilon > 0$, choose $s > x$ with $B \le f(s) < B + \varepsilon$; then $x < t < s$ implies $B \le f(t) < B + \varepsilon$, so $f(x+) = B$. If $x < y$, then $\inf_{x<t<y} f(t) \le \sup_{x<t<y} f(t)$ gives $f(x+) \le f(y-)$.
For decreasing f, apply the increasing case to $-f$ and multiply the resulting inequalities by -1.
Structural Commentary: Monotonicity converts one-sided limits into suprema and infima.
Key Insight: A monotone function has jumps, but not oscillatory one-sided failure. □

Corollary 13.3.48. *Monotone Functions Have No Second-Kind Discontinuities*
A monotone real function on an interval has no discontinuities of the second kind.

Proof. At every interior point, Theorem 13.3.47 gives both one-sided limits.
Hence any discontinuity is of the first kind by Definition 13.3.44.
Structural Commentary: Monotonicity rules out oscillatory discontinuities.
Key Insight: All monotone discontinuities are jumps or removable value mismatches. □

Theorem 13.3.49. *Monotone Discontinuities Are Countable*
If f is monotone on (a, b), then the set of points of (a, b) at which f is discontinuous is at most countable.

Proof. Assume f is increasing, and let E be its discontinuity set.
For each $x \in E$, Theorem 13.3.47 gives $f(x-) < f(x+)$; choose $r(x) \in \mathbb{Q}$ with $f(x-) < r(x) < f(x+)$.
If $x_1 < x_2$, then $f(x_1+) \le f(x_2-)$, so $r(x_1) \ne r(x_2)$. Thus E injects into $\mathbb{Q}$, hence is countable by Corollary 12.3.15. If f is decreasing, apply the increasing case to $-f$, which has the same discontinuity set.
Structural Commentary: Each jump contains a rational marker, and monotonicity keeps the markers distinct.
Key Insight: Countability enters by injecting discontinuities into $\mathbb{Q}$. □

Remark 13.3.50. *Monotone Discontinuities Need Not Be Isolated*
For countable $E = \{x_n : n \in \mathbb{N}\} \subset (a, b)$ and positive summable (c_n), $f(x) = \sum_{x_n<x} c_n$ is increasing, has jumps of size c_n at x_n, and is continuous off E. Thus monotone discontinuities may be dense.

Definition 13.3.51. *Extended Limits and Limits at Infinity*
Neighborhoods of $+\infty$ are intervals $(c, +\infty)$; neighborhoods of $-\infty$ are $(-\infty, c)$.
For $f : E \to \mathbb{R}$ and extended real A, x, $f(t) \to A$ as $t \to x$ means: every neighborhood U of A contains $f(t)$ for all punctured $t \in E$ sufficiently near x, i.e. $\exists V \ni x$ such that $t \in V \cap E,\ t \ne x \implies f(t) \in U$.
For real A, x, this agrees with Definition 13.3.7.
Structural Commentary: Extended limits use same neighborhood test, $\pm\infty$ allowed as input or output limits.

Theorem 13.3.52. *Algebra of Extended Limits*
Let $f, g : E \to \mathbb{R}$ and $f(t) \to A$, $g(t) \to B$ as $t \to x$, with A, B, x extended real. Then, whenever RHS is defined:

(i) if $f(t) \to A'$ as $t \to x$, then $A' = A$;
(ii) $(f + g)(t) \to A + B$;
(iii) $(fg)(t) \to AB$;
(iv) $(f/g)(t) \to A/B$;

The forms $\infty - \infty$, $0 \cdot \infty$, ∞/∞, and $A/0$ are undefined.

Proof.

(i) If $A' \ne A$, choose disjoint neighborhoods of A and A'.
Definition 13.3.51 would eventually place $f(t)$ in both, a contradiction.
(ii) For any neighborhood W of $A+B$, definedness gives neighborhoods U of A and V of B such that $u+v \in W$ for $u \in U$, $v \in V$. Apply the two limits in a common punctured neighborhood of x.
(iii) The same neighborhood argument applies to multiplication, excluding $0 \cdot \infty$.
(iv) Apply it to division; since $B \ne 0$, choose the neighborhood of B away from 0.

Structural Commentary: Extended limit algebra is ordinary limit algebra in neighborhood form.
Key Insight: The undefined forms are exactly where neighborhood control breaks. □

Method 13.3.53. *Continuity Proof Patterns*

(i) *ε–δ at p*: prove $\forall \varepsilon > 0\, \exists \delta > 0$ such that $x \in E$ and $d_X(x, p) < \delta$ imply $d_Y(f(x), f(p)) < \varepsilon$.
(ii) *Limit at p*: prove same estimate with $0 < d_X(x, p) < \delta$, target q; equivalently test all $x_n \to p$, $x_n \ne p$.
(iii) *Topological continuity*: prove $V \subseteq Y$ open $\Rightarrow f^{-1}(V)$ open, or $C \subseteq Y$ closed $\Rightarrow f^{-1}(C)$ closed.
(iv) *Compactness*: from open cover $\{V_\alpha\}$ of $f(X)$, pulled-back cover $\{f^{-1}(V_\alpha)\}$ of X extract finite subcover.
(v) *Uniform continuity*: choose local radii r_p, cover compact X by $B_{r_p/2}(p)$, and set $\delta = \frac{1}{2} \min_i r_{p_i}$.

Key Insight: Continuity is local control; compactness turns local controls into finite control.

13.4 Selected General Topology

General topology abstracts the open-set behavior of metric spaces. This selective pass records spaces, bases, products, quotients, and countability axioms for later use.

Definition 13.4.1. *Topological Space*
A *topology* on a set X is a collection $\mathcal{T} \subseteq 2^X$ such that

(i) $\emptyset, X \in \mathcal{T}$;
(ii) arbitrary unions of members of $\mathcal{T}$ belong to $\mathcal{T}$;
(iii) finite intersections of members of $\mathcal{T}$ belong to $\mathcal{T}$.

The pair $(X, \mathcal{T})$ is a *topological space*, and the members of $\mathcal{T}$ are the *open sets*. Closed sets are complements of open sets.

Proposition 13.4.2. *Metric Open Sets Form a Topology*
If (X, d) is a metric space, then the open sets from Definition 13.1.10 form a topology on X.

Proof. The empty set and X are open. Arbitrary unions and finite intersections are open by Theorem 13.1.18.
Structural Commentary: Metric balls generate an open-set system with the exact axioms of a topology.
Key Insight: General topology keeps the open-set calculus and forgets the distance function. □

Definition 13.4.3. *Basis*
Let X be a set. A collection $\mathcal{B} \subseteq 2^X$ is a *basis* for a topology if:

(i) for every $x \in X$, some $B \in \mathcal{B}$ contains x;
(ii) if $x \in B_1 \cap B_2$ with $B_1, B_2 \in \mathcal{B}$, then some $B_3 \in \mathcal{B}$ satisfies $x \in B_3 \subseteq B_1 \cap B_2$.

The topology generated by $\mathcal{B}$ consists of all unions of basis elements. A *subbasis* is a collection whose finite intersections form a basis.

Example 13.4.4. *Metric Balls as a Basis*
In a metric space, the collection of all open balls is a basis for the metric topology. On $\mathbb{R}^k$, balls with rational centers and rational radii form a countable basis.

Definition 13.4.5. *Product Topology*
For topological spaces X and Y, the *product topology* on $X \times Y$ is generated by basis sets $U \times V$, where $U \subseteq X$ and $V \subseteq Y$ are open. For a family $\{X_i\}_{i \in I}$, the product topology on $\prod_i X_i$ is generated by finite-coordinate restrictions: sets of the form $\prod_i U_i$ where each U_i is open and $U_i = X_i$ for all but finitely many i.

Proposition 13.4.6. *Product Topology and Coordinate Maps*
A map $f : Z \to X \times Y$ is continuous for the product topology if and only if both coordinate maps $\pi_X \circ f$ and $\pi_Y \circ f$ are continuous.

Proof. If f is continuous, then coordinate maps are continuous because inverse images of open sets are inverse images of product-open cylinders. Conversely, if both coordinates are continuous, then the inverse image of each basis set $U \times V$ is $(\pi_X \circ f)^{-1}(U) \cap (\pi_Y \circ f)^{-1}(V)$, which is open.
Structural Commentary: Product topology is the weakest topology making the projections continuous.
Key Insight: Continuity into a product is checked coordinatewise. □

Definition 13.4.7. *Quotient Topology*
Let $q : X \to Y$ be a surjective map from a topological space X to a set Y. The *quotient topology* on Y declares $V \subseteq Y$ open exactly when $q^{-1}(V)$ is open in X. Equivalently, it is the strongest topology on Y making q continuous.

Proposition 13.4.8. *Quotient Universal Property*
Let $q : X \to Y$ carry the quotient topology, and let $g : Y \to Z$ be a map into a topological space Z. Then g is continuous if and only if $g \circ q : X \to Z$ is continuous.

Proof. If g is continuous, then $g \circ q$ is continuous by composition. Conversely, if $g \circ q$ is continuous and $W \subseteq Z$ is open, then $q^{-1}(g^{-1}(W)) = (g \circ q)^{-1}(W)$ is open in X. By the quotient topology, $g^{-1}(W)$ is open in Y.
Structural Commentary: Quotient topology is defined so continuity out of the quotient can be tested before identifying points.
Key Insight: Pull back through the quotient map and use the defining open-set test. □

Definition 13.4.9. *Countability Axioms*
A topological space X is *first countable* if every point has a countable neighborhood basis. It is *second countable* if its topology has a countable basis. It is *separable* if it has a countable dense subset.

Proposition 13.4.10. *Second Countable Implies Separable*
If X is second countable, then X is separable.

Proof. Let $\mathcal{B} = \{B_n : n \in \mathbb{N}\}$ be a countable basis. For each nonempty B_n, choose $x_n \in B_n$.
The chosen set is countable. If U is a nonempty open set, some basis element B_n satisfies $x \in B_n \subseteq U$, so the chosen set meets U. Hence it is dense.
Structural Commentary: A countable basis yields a countable dense set.
Key Insight: Density is tested by nonempty open sets. □

14 Convergence and Function Spaces

Convergence and Function Spaces studies series, uniform limits, compactness of functions, power-series approximation, and Fourier orthogonality.

14.1 Infinite Series and Summability

Series are sequences of partial sums. Comparison, condensation, root and ratio tests, rearrangements, products, and double sums all refine that viewpoint.

Definition 14.1.1. *Series and Partial Sums*
Let (a_n) be a complex sequence, and for $p \le q$ put $\sum_{n=p}^{q} a_n := a_p + a_{p+1} + \cdots + a_q$.

The index n in the displayed finite sum is a bound variable, in the sense of Principle 12.1.2; it may be renamed when its scope is unchanged.

The *series* $\sum_{n=1}^{\infty} a_n$ (or $a_1 + a_2 + \cdots$) is the partial-sum sequence $s_n := \sum_{k=1}^{n} a_k$. If $s_n \to s$, then $\sum_{n=1}^{\infty} a_n = s$ and series *converges*; otherwise it *diverges*. The same convention applies to series starting at $n = 0$.

Motivation 14.1.2. *Partial Sums Remember Order*
An infinite sum is a limit of partial sums. For alternating harmonic series, odd and even partial sums approach the same value from opposite sides; see Corollary 14.1.24 and Theorem 14.1.34.

Perspective 14.1.3. *Formal Algebra Needs Convergence*
Finite sums are algebraic; infinite sums are partial-sum limits. Algebraic manipulations are justified only by convergence control, as in Theorem 14.1.4, Corollary 14.1.5, and Theorems 14.1.31, 14.1.34.
Key Insight: The symbol $\sum_{n=1}^{\infty}$ is a limit claim before it is an algebraic object.

Theorem 14.1.4. *Cauchy Criterion for Series*
The series $\sum a_n$ converges if and only if $\forall \varepsilon > 0$ $\exists$ an integer N s.t. $m \ge n \ge N$ implies $\left|\sum_{k=n}^{m} a_k\right| \le \varepsilon$.

Proof.
($\Rightarrow$) If $s_n := \sum_{k=1}^{n} a_k$ converges, then (s_n) is Cauchy. Thus, for $m \ge n$ large enough, $\left|\sum_{k=n}^{m} a_k\right| = |s_m - s_{n-1}| \le \varepsilon$.

($\Leftarrow$) The tail condition gives, for $m > n$ large enough, $|s_m - s_n| = \left|\sum_{k=n+1}^{m} a_k\right| \le \varepsilon$.

Hence (s_n) is Cauchy and converges by Theorem 13.2.9.

Structural Commentary: A series converges exactly when all sufficiently late finite tails are small.
Key Insight: The Cauchy test measures tails, not the eventual value of the sum. □

Corollary 14.1.5. *Terms of a Convergent Series Tend to Zero*
If $\sum a_n$ converges, then $a_n \to 0$.

Proof. Let $\varepsilon > 0$. By Theorem 14.1.4, there is N such that $m \ge n \ge N$ implies $|\sum_{k=n}^{m} a_k| \le \varepsilon$.
Taking $m = n$ gives $|a_n| \le \varepsilon$ for all $n \ge N$, hence $a_n \to 0$.
Structural Commentary: A single term is the shortest possible tail.
Key Insight: The condition $a_n \to 0$ is necessary, but not sufficient for convergence. □

Theorem 14.1.6. *Nonnegative Series and Bounded Partial Sums*
A series of nonnegative real terms converges if and only if its partial sums form a bounded sequence.

Proof.
($\Rightarrow$) Every convergent sequence is bounded, so the partial sums are bounded.

($\Leftarrow$) Since $a_n \geq 0$, $s_n = \sum_{k=1}^{n} a_k$ increases; if bounded, it converges by Theorem 12.4.12.

Structural Commentary: Nonnegative series replace cancellation by monotonicity.
Key Insight: For nonnegative terms, convergence is exactly bounded accumulation. □

Theorem 14.1.7. *Comparison Test*

(i) If $|a_n| \leq c_n$ for $n \geq N_0$, and if $\sum c_n$ converges, then $\sum a_n$ converges.
(ii) If $a_n \geq d_n \geq 0$ for $n \geq N_0$, and if $\sum d_n$ diverges, then $\sum a_n$ diverges.

Proof.

(i) Given $\varepsilon > 0$, choose $N \geq N_0$ so that $m \geq n \geq N$ implies $\sum_{k=n}^{m} c_k \leq \varepsilon$.
Then $\left|\sum_{k=n}^{m} a_k\right| \leq \sum_{k=n}^{m} |a_k| \leq \sum_{k=n}^{m} c_k \leq \varepsilon$, so Theorem 14.1.4 applies.

(ii) If $\sum a_n$ converged, then (i) applied to $d_n \leq a_n$ would force $\sum d_n$ to converge, a contradiction.

Structural Commentary: Absolute tail domination transfers Cauchy convergence.
Key Insight: Comparison is a tail test; finitely many early terms do not matter. □

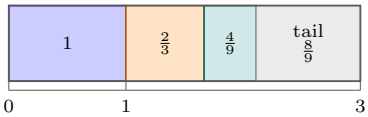


Figure 131: At $x = 2/3$, the strip widths sum to $1 + \frac{2}{3} + \frac{4}{9} + \cdots = 3$.

Theorem 14.1.8. *Geometric Series*
If $0 \leq x < 1$, then $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$. If $x \geq 1$, the series diverges.

Proof. For $x \neq 1$, $\sum_{k=0}^{n} x^k = \frac{1-x^{n+1}}{1-x}$. Let $n \to \infty$ and use Theorem 12.4.25.
If $x = 1$, the partial sums are $n + 1$, so they diverge.
Structural Commentary: The finite geometric identity converts convergence into the limit of x^{n+1}.
Key Insight: The geometric series is the model comparison series for ratio-like decay. □

Definition 14.1.9. *Real Decimals as Series*
For digits $b_N, \ldots, b_0, a_1, a_2, \ldots \in \{0, \ldots, 9\}$, with $b_N \neq 0$ unless the integer part is 0,
$b_N \cdots b_0.a_1a_2\cdots := \sum_{i=0}^{N} b_i 10^i + \sum_{j=1}^{\infty} a_j 10^{-j}$. The fractional series converges absolutely since $0 \leq a_j 10^{-j} \leq 9 \cdot 10^{-j}$ and $\sum 9 \cdot 10^{-j}$ is geometric. A minus sign denotes the negative of this value.

Proposition 14.1.10. *Terminal Nines Are Nonunique*
The same real number can have two decimal representations: $1.000\ldots = 0.999\ldots$.

Proof. By Definition 14.1.9 and Theorem 14.1.8, $1.000\ldots = 1$; $0.999\ldots = \sum_{j=1}^{\infty} 9 \cdot 10^{-j} = 9 \cdot \frac{10^{-1}}{1-10^{-1}} = 1$.
Structural Commentary: The two digit strings have the same partial-sum limit.
Key Insight: Uniqueness requires excluding terminal 9 tails. □

Perspective 14.1.11. *Decimal Expansions Are Series*
Proposition 12.2.28 gives decimal truncations that converge to a real number, while Definition 14.1.9 interprets the infinite digit string as a series. Proposition 14.1.10 explains the standard terminal-9 ambiguity.

Theorem 14.1.12. *Cauchy Condensation Test*
Suppose $a_1 \geq a_2 \geq a_3 \geq \cdots \geq 0$. Then $\sum_{n=1}^{\infty} a_n$ converges if and only if $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

Proof. Let $s_n := a_1 + \cdots + a_n$ and $t_k := a_1 + 2a_2 + \cdots + 2^k a_{2^k}$.
If $n < 2^k$, then grouping by dyadic blocks gives $s_n \leq a_1 + (a_2 + a_3) + \cdots + (a_{2^k} + \cdots + a_{2^{k+1}-1}) \leq t_k$.
If $n > 2^k$, grouping from the other side gives $2s_n \geq t_k$. Hence the nonnegative partial sums (s_n) and (t_k) are either both bounded or both unbounded; apply Theorem 14.1.6.
Structural Commentary: Monotonicity lets dyadic blocks be compared to their endpoint terms.
Key Insight: A sparse dyadic subsequence detects the whole monotone nonnegative series. □

Corollary 14.1.13. *p-Series Test*
The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ *converges if* $p > 1$ *and diverges if* $p \le 1$.

Proof. If $p \le 0$, the terms do not tend to 0, so Corollary 14.1.5 gives divergence.
If $p > 0$, Theorem 14.1.12 reduces the question to $\sum_{k=0}^{\infty} 2^k \frac{1}{(2^k)^p} = \sum_{k=0}^{\infty} 2^{(1-p)k}$.
This geometric series converges exactly when $2^{1-p} < 1$, i.e. $p > 1$, by Theorem 14.1.8.
Structural Commentary: Condensation turns powers of n into powers of 2.
Key Insight: The boundary is $p = 1$ because the condensed ratio is 2^{1-p}. □

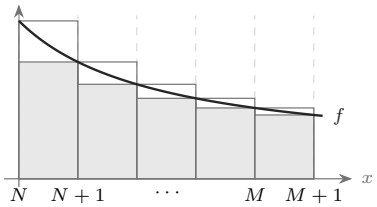


Figure 132: Left and right endpoint estimates for decreasing $f \ge 0$.

Theorem 14.1.14. *Integral Test*
Let $N \in \mathbb{N}$ *and let* $f : [N, \infty) \to [0, \infty)$ *be decreasing and Riemann integrable on compact subintervals.*
Then $\sum_{k=N}^{\infty} f(k)$ *converges iff* $\int_N^{\infty} f(x)\,dx$ *converges.*
For $M \ge N$, $\int_N^{M+1} f(x)\,dx \le \sum_{k=N}^{M} f(k) \le f(N) + \int_N^{M} f(x)\,dx$.

Proof. For a monotone function, compact-subinterval Riemann integrability follows from Theorem 15.1.45 with $\alpha(x) = x$; it is kept in the statement to make the improper-integral dependency explicit.

(⇒) For $x \in [k, k+1]$, monotonicity gives $f(x) \le f(k)$. Hence $\int_N^{M+1} f(x)\,dx \le \sum_{k=N}^{M} f(k)$. If the series converges, these truncations are bounded, so $\int_N^{\infty} f$ converges by Theorem 15.1.61.

(⇐) For $k \ge N+1$ and $x \in [k-1, k]$, monotonicity gives $f(k) \le f(x)$. Hence $\sum_{k=N}^{M} f(k) \le f(N) + \int_N^{M} f(x)\,dx$. If $\int_N^{\infty} f$ converges, the partial sums are bounded, so the series converges by Theorem 14.1.6.

Structural Commentary: A monotone sequence can be compared to neighboring unit-width areas.
Key Insight: The integral test is comparison of two infinite accumulations with the same tail size. □

Definition 14.1.15. *The Number e*
Define $e := \sum_{n=0}^{\infty} \frac{1}{n!}$, where $0! := 1$; convergence follows by comparison with the geometric series.

Theorem 14.1.16. *A Limit Formula for e*
The limit formula is $\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e$.

Proof. Let $s_m := \sum_{k=0}^{m} \frac{1}{k!}$ and $t_n := (1 + \frac{1}{n})^n$. The binomial theorem gives $t_n \le s_n$, hence $\limsup t_n \le e$.
For fixed $m \le n$, first m binomial terms give $t_n \ge 1 + 1 + \frac{1}{2!}\left(1 - \frac{1}{n}\right) + \cdots + \frac{1}{m!}\left(1 - \frac{1}{n}\right)\cdots\left(1 - \frac{m-1}{n}\right)$.
Letting $n \to \infty$ with m fixed gives $\liminf t_n \ge s_m$, and then $m \to \infty$ gives $e \le \liminf t_n$.
Structural Commentary: Bound above by the factorial series; bound below by fixed binomial terms.
Key Insight: Limsup and liminf squeeze the binomial limit to the series definition. □

Proposition 14.1.17. *Remainder Bound for the Series of e*
If $n \ge 1$ *and* $s_n = \sum_{k=0}^{n} 1/k!$, *then* $0 < e - s_n < \frac{1}{n!\,n}$.

Proof. Tail is $e - s_n = \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \cdots < \frac{1}{(n+1)!}\left(1 + \frac{1}{n+1} + \frac{1}{(n+1)^2} + \cdots\right) = \frac{1}{n!\,n}$
Structural Commentary: The factorial tail is bounded by a geometric tail.
Key Insight: Rapid convergence of $\sum \frac{1}{n!}$ gives arithmetic control of e. □

Theorem 14.1.18. *e Is Irrational*
The number e is irrational.

Proof. Suppose $e = p/q$ with p, q positive integers. By Proposition 14.1.17, $0 < q!(e - s_q) < \frac{1}{q}$. But $q!e$ and $q!s_q$ are integers, so $q!(e - s_q)$ is an integer strictly between 0 and 1, impossible.
Structural Commentary: The sharp tail bound turns rationality into an impossible integer.
Key Insight: A positive integer cannot be smaller than 1. □

Theorem 14.1.19. *Root Test*
For a series $\sum a_n$, put $\alpha := \limsup_{n\to\infty} \sqrt[n]{|a_n|}$. Then:

(i) if $\alpha < 1$, then $\sum a_n$ converges absolutely;
(ii) if $\alpha > 1$, then $\sum a_n$ diverges;
(iii) if $\alpha = 1$, the test gives no information.

Proof.

(i) Choose $\alpha < \beta < 1$. Eventually $|a_n| < \beta^n$, so $\sum |a_n|$ converges by comparison with $\sum \beta^n$.
(ii) Since $\alpha > 1$, $|a_n| > 1$ for infinitely many n, so $a_n \not\to 0$; use Corollary 14.1.5.
(iii) $\sum 1/n$ and $\sum 1/n^2$ both have $\alpha = 1$, but behave differently by Corollary 14.1.13.

Structural Commentary: The root test compares term size directly to geometric decay.
Key Insight: Values below 1 force absolute convergence; values above 1 prevent terms from tending to 0. □

Theorem 14.1.20. *Ratio Test*
Assume $a_n \neq 0$ for all sufficiently large n.

(i) If $\limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$, then $\sum a_n$ converges absolutely.
(ii) If $\liminf_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| > 1$, then $\sum a_n$ diverges.
(iii) In the remaining cases, the test gives no information.

Proof.

(i) Choose $\beta < 1$ with $|a_{n+1}/a_n| < \beta$ eventually.
Then $|a_n| = O(\beta^n)$, so $\sum |a_n|$ converges by comparison with a geometric series.
(ii) Choose $\beta > 1$ with $|a_{n+1}/a_n| > \beta$ eventually.
Then the tail $|a_n|$ is eventually increasing and bounded away from 0, so $a_n \not\to 0$.
(iii) The series $\sum 1/n$ and $\sum 1/n^2$ both have ratio limit 1, but behave differently by Corollary 14.1.13.

Structural Commentary: Ratios turn successive terms into a geometric majorant.
Key Insight: Ratios below 1 force geometric decay; ratios above 1 prevent terms from tending to zero. □

Theorem 14.1.21. *Root and Ratio Test Comparison*
For any sequence (c_n) of positive numbers, $\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \le \liminf_{n\to\infty} \sqrt[n]{c_n} \le \limsup_{n\to\infty} \sqrt[n]{c_n} \le \limsup_{n\to\infty} \frac{c_{n+1}}{c_n}$.

Proof. Let $\alpha = \limsup c_{n+1}/c_n$. If $\alpha < +\infty$ and $\beta > \alpha$, then eventually $c_{n+1} \le \beta c_n$, hence $c_n \le c_N \beta^{n-N}$ for all large n. Taking nth roots gives $\limsup \sqrt[n]{c_n} \le \beta$; let $\beta \downarrow \alpha$. If $\alpha = +\infty$, the rightmost inequality is automatic. Let $\ell = \liminf c_{n+1}/c_n$. If $\ell = 0$, the leftmost inequality is automatic. If $\gamma < \ell$ with $\gamma > 0$, then eventually $c_{n+1} \ge \gamma c_n$, hence $c_n \ge c_N \gamma^{n-N}$ for all large n. Taking nth roots gives $\liminf \sqrt[n]{c_n} \ge \gamma$; let $\gamma \uparrow \ell$.
The middle inequality is the general fact that $\liminf a_n \le \limsup a_n$.
Structural Commentary: A tail ratio bound multiplies into a root bound.
Key Insight: Whenever the ratio test proves convergence, the root test does too. □

Theorem 14.1.22. *Summation by Parts*
Given two sequences (a_n) and (b_n), put $A_n := \sum_{k=0}^{n} a_k$ and $A_{-1} := 0$.
If $0 \le p \le q$, then $\sum_{n=p}^{q} a_n b_n = \sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p$.

Proof. Since $a_n = A_n - A_{n-1}$, $\sum_{n=p}^{q} a_n b_n = \sum_{n=p}^{q} A_n b_n - \sum_{n=p}^{q} A_{n-1} b_n = \sum_{n=p}^{q} A_n b_n - \sum_{n=p-1}^{q-1} A_n b_{n+1}$.
Structural Commentary: This is the discrete analogue of integration by parts.
Key Insight: Move the difference from a_n onto b_n by rewriting $a_n = A_n - A_{n-1}$. □

Theorem 14.1.23. *Dirichlet Test*
If the partial sums A_n of $\sum a_n$ are bounded, $b_n \downarrow 0$, and $b_n \ge 0$, then $\sum a_n b_n$ converges.

Proof. Choose M with $|A_n| \le M$. If $M = 0$, the result is immediate. Otherwise choose N with $b_N \le \varepsilon/(2M)$. For $N \le p \le q$, $\left|\sum_{n=p}^{q} a_n b_n\right| \le M\left(\sum_{n=p}^{q-1}(b_n - b_{n+1}) + b_q + b_p\right) = 2Mb_p \le 2Mb_N \le \varepsilon$. Theorem 14.1.4 applies.
Structural Commentary: Bounded oscillation in A_n is damped by a monotone scalar tending to 0.
Key Insight: Summation by parts converts cancellation into a total variation estimate for b_n. □

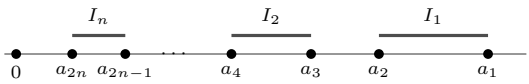


Figure 133: The intervals $I_j = [a_{2j}, a_{2j-1}]$ are disjoint inside $(0, a_1)$.

Corollary 14.1.24. *Alternating Series Test*
Suppose $|c_1| \ge |c_2| \ge |c_3| \ge \cdots$, the signs alternate, and $c_n \to 0$. Then $\sum c_n$ converges.

Proof. After shifting indices to start at 0 if desired, apply Theorem 14.1.23 with $a_n = (-1)^{n+1}$ and $b_n = |c_n|$.
Structural Commentary: Alternation gives bounded sums; decreasing magnitudes damp them.
Key Insight: Alternating convergence is Dirichlet convergence in its simplest form. □

Perspective 14.1.25. *Alternating Partial Sums as Nested Control*
The interval picture explains the boundedness in Corollary 14.1.24: the even partial sums collect disjoint lengths inside a fixed interval, while the odd partial sums differ from the even ones by the next term, which tends to 0.

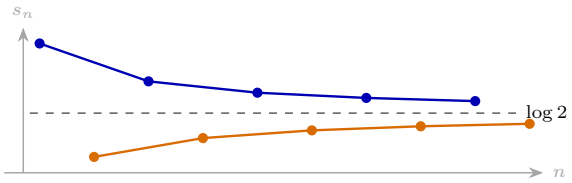


Figure 134: Odd partial sums (blue) decrease and even partial sums (orange) increase toward $\log 2$.

Definition 14.1.26. *Absolute and Nonabsolute Convergence*
The series $\sum a_n$ *converges absolutely* if $\sum |a_n|$ converges.
If $\sum a_n$ converges but $\sum |a_n|$ diverges, then $\sum a_n$ converges *nonabsolutely*.

Theorem 14.1.27. *Absolute Convergence Implies Convergence*
If $\sum a_n$ converges absolutely, then $\sum a_n$ converges.

Proof. For $m \ge n$, $\left|\sum_{k=n}^{m} a_k\right| \le \sum_{k=n}^{m} |a_k|$. Apply Theorem 14.1.4 to $\sum |a_n|$.
Structural Commentary: Absolute convergence removes cancellation from the convergence question.
Key Insight: Small absolute tails force small signed or complex tails. □

Theorem 14.1.28. *Addition and Scalar Multiplication of Series*
If $\sum a_n = A$ and $\sum b_n = B$, then $\sum(a_n + b_n) = A + B$ and $\sum ca_n = cA$ for every fixed scalar c.

Proof. Apply Theorem 12.4.3 to the corresponding partial sums.
Structural Commentary: Linearity of convergent series is just limit algebra for partial sums.
Key Insight: Add series by adding their partial-sum sequences. □

Definition 14.1.29. *Cauchy Product*
Given $\sum a_n$, $\sum b_n$, define $c_n := \sum_{k=0}^{n} a_k b_{n-k}$, $(n = 0, 1, 2, \ldots)$. Series $\sum c_n$ is *Cauchy product* of given series.

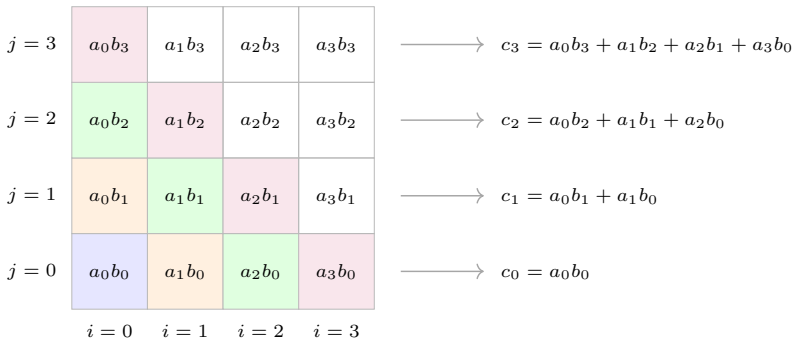


Figure 135: Diagonal sums of (a_ib_j) form the Cauchy product; finite rectangular sums are order-free.

Perspective 14.1.30. *Diagonal Ordering in a Product*
c_n is the diagonal sum of the products a_ib_j with $i + j = n$. Absolute convergence, as in Theorems 14.1.31 and 14.1.34, keeps the value independent of the accumulation order.

Theorem 14.1.31. *Mertens' Theorem for Cauchy Products*
Suppose $\sum a_n$ *converges absolutely,* $\sum a_n = A$, $\sum b_n = B$, *and* $c_n = \sum_{k=0}^{n} a_k b_{n-k}$. *Then* $\sum_{n=0}^{\infty} c_n = AB$.

Proof. Let A_n, B_n, C_n be the partial sums and set $\beta_n := B_n - B$.
Then $C_n = A_nB + \gamma_n$, and $\gamma_n := a_0\beta_n + a_1\beta_{n-1} + \cdots + a_n\beta_0$. Since $A_nB \to AB$, it suffices to show $\gamma_n \to 0$.
Put $\alpha := \sum |a_n|$. If $|\beta_j| \le \varepsilon$ for $j \ge N$, then $|\gamma_n| \le |\beta_0 a_n + \cdots + \beta_N a_{n-N}| + \varepsilon\alpha$.
Let $n \to \infty$ with N fixed. The first term tends to 0, so $\limsup |\gamma_n| \le \varepsilon\alpha$; let $\varepsilon \downarrow 0$.
Structural Commentary: Absolute convergence of one factor controls the convolution error.
Key Insight: One absolute factor makes the Cauchy product converge to AB. □

Definition 14.1.32. *Rearrangement of a Series*
If (k_n) lists each positive integer exactly once and $a'_n := a_{k_n}$, then $\sum a'_n$ is a *rearrangement* of $\sum a_n$.

Theorem 14.1.33. *Riemann Rearrangement Theorem*
If real $\sum a_n$ *converges nonabsolutely and* $-\infty \le \alpha \le \beta \le \infty$,
then some rearrangement has partial sums s'_n *with* $\liminf_{n\to\infty} s'_n = \alpha$ *and* $\limsup_{n\to\infty} s'_n = \beta$.

Proof. Write $a_n = p_n - q_n$, where $p_n = \max(a_n, 0)$ and $q_n = \max(-a_n, 0)$. Since $\sum a_n$ converges but not absolutely, $\sum p_n = \sum q_n = +\infty$, and $p_n, q_n \to 0$. List the positive nonzero p_n as $P_1, P_2, \ldots$ and the positive nonzero q_n as $Q_1, Q_2, \ldots$, preserving their original orders.
Choose finite numbers $\ell_N < u_N$ with $\ell_N \to \alpha$ and $u_N \to \beta$ in the extended sense. Starting from partial sum 0, add unused P terms until the sum is $> u_N$, then add unused terms $-Q$ until the sum is $< \ell_N$; repeat for $N = 1, 2, \ldots$. Divergence of both positive series makes each stopping time finite, and the alternating stages use every nonzero positive and negative term eventually. Insert zero terms anywhere.
After the Nth upward crossing, the partial sum lies between u_N and $u_N + P_j$ for the next positive term used; after the downward crossing, it lies between $\ell_N - Q_j$ and ℓ_N for the next negative term used. Since $P_j, Q_j \to 0$, these overshoots vanish. Hence subsequences of partial sums approach the upper targets and lower targets, while all later partial sums stay within vanishing error of $[\ell_N, u_N]$. Therefore $\liminf s'_n = \alpha$ and $\limsup s'_n = \beta$.
Structural Commentary: Conditional convergence leaves enough mass to steer partial sums.
Key Insight: Vanishing terms make the steering increasingly precise. □

Theorem 14.1.34. *Absolutely Convergent Series Are Rearrangement-Invariant*
Every rearrangement of an absolutely convergent complex series converges to the same sum.

Proof. Let s_n, s'_n be the original and rearranged partial sums.
Given $\varepsilon > 0$, choose N so that $\sum_{i=N}^{m} |a_i| \le \varepsilon$ for $m \ge N$, then choose p so $a_1, \ldots, a_N$ occur among $a'_1, \ldots, a'_p$.
For $n > p$, $s_n - s'_n$ uses only tail terms, hence $|s_n - s'_n| \le \varepsilon$. Thus s'_n has the same limit as s_n.
Structural Commentary: Absolute tails control any finite reordering of the remaining terms.
Key Insight: Absolute convergence is exactly the protection needed against rearrangement. □

Definition 14.1.35. *Absolute Summation over Countable Sets*
Let X be countable, $f : X \to \mathbb{C}$, and $g : \mathbb{N} \to X$ be a bijection, $g_n := g(n)$.
If $\sum_{n=1}^{\infty} |f(g_n)| < \infty$, then $\sum_{x\in X} f(x)$ is *absolutely convergent*, and $\sum_{x\in X} f(x) := \sum_{n=1}^{\infty} f(g_n)$.

Theorem 14.1.36. *Countable Sums Are Order-Free Under Absolute Convergence*
Let X, Y be countable, $f : X \to \mathbb{C}$ be absolutely summable, and $h : Y \to X$ be a bijection. Then $f \circ h$ is absolutely summable and $\sum_{y\in Y} f(h(y)) = \sum_{x\in X} f(x)$.
In particular, a countable absolutely convergent sum is independent of the enumeration.

Proof. Let $g : \mathbb{N} \to X$ and $k : \mathbb{N} \to Y$ be enumerations, and set $\sigma := g^{-1} \circ h \circ k : \mathbb{N} \to \mathbb{N}$.
Then σ is a bijection and $h(k_n) = g_{\sigma(n)}$. Hence $\sum_{n=1}^{\infty} |f(h(k_n))| = \sum_{n=1}^{\infty} \left|f(g_{\sigma(n)})\right| < \infty$.
Thus $\sum_{y\in Y} f(h(y)) = \sum_{n=1}^{\infty} f(g_{\sigma(n)}) = \sum_{n=1}^{\infty} f(g_n) = \sum_{x\in X} f(x)$, where the middle equality is Theorem 14.1.34.
Taking $Y = \mathbb{N}$ and h to be any enumeration of X gives enumeration independence.
Structural Commentary: Absolute convergence turns an enumeration into harmless bookkeeping.
Key Insight: Once absolute tails are controlled, the order of a countable listing no longer changes the sum. □

Theorem 14.1.37. *Fubini for Absolutely Summable Arrays*
Let $a_{mn} \in \mathbb{C}$ and suppose $L := \sum_{(m,n)\in\mathbb{N}\times\mathbb{N}} |a_{mn}| < \infty$. Then each row and column is absolutely summable.
With $r_m := \sum_{n=1}^{\infty} a_{mn}$ and $c_n := \sum_{m=1}^{\infty} a_{mn}$, the row-total and column-total series are absolutely convergent, where $\sum_{m=1}^{\infty} |r_m| < \infty$; $\sum_{n=1}^{\infty} |c_n| < \infty$, and $\sum_{m=1}^{\infty} r_m = \sum_{(m,n)\in\mathbb{N}\times\mathbb{N}} a_{mn} = \sum_{n=1}^{\infty} c_n$.

Proof. For finite $F \subseteq \mathbb{N} \times \mathbb{N}$, $\sum_{(m,n)\in F} |a_{mn}| \le L$. Hence each row and column has bounded absolute partial sums, so rows and columns converge absolutely by Theorem 14.1.6.
Moreover, for $M \ge 1$, $\sum_{m=1}^{M} |r_m| \le \sum_{m=1}^{M} \sum_{n=1}^{\infty} |a_{mn}| = \lim_{N\to\infty} \sum_{m=1}^{M} \sum_{n=1}^{N} |a_{mn}| \le L$, so $\sum_m |r_m| < \infty$.
Similarly $\sum_n |c_n| < \infty$. Let $\varepsilon > 0$. Choose finite $F \subseteq \mathbb{N}\times\mathbb{N}$ with $\sum_{(m,n)\in(\mathbb{N}\times\mathbb{N})\setminus F} |a_{mn}| < \varepsilon$, and choose a rectangle $R = \{1, \dots, M\} \times \{1, \dots, N\} \supseteq F$. The full sum, row-iterated sum, and column-iterated sum each differ from $\sum_{(m,n)\in R} a_{mn}$ by at most $\sum_{(m,n)\in(\mathbb{N}\times\mathbb{N})\setminus R} |a_{mn}| \le \varepsilon$. Let $\varepsilon \downarrow 0$.
Structural Commentary: Finite rectangles plus absolute tails justify Fubini.
Key Insight: Double sums can be rearranged only after the entire array is absolutely summable. □

Method 14.1.38. *Series Proof Patterns*

(i) *Partial sums*: translate every assertion about $\sum a_n$ into an assertion about $s_n = \sum_{k=1}^{n} a_k$.
(ii) *Cauchy tails*: prove convergence by estimating $|\sum_{k=n}^{m} a_k|$ for $m \ge n$.
(iii) *Nonnegative terms*: replace convergence by boundedness of partial sums.
(iv) *Comparison*: dominate only the tail; finitely many early terms are irrelevant.
(v) *Absolute convergence*: use $\sum |a_n|$ when cancellation should not matter.
(vi) *Countable index sets*: choose a bijection $g : \mathbb{N} \to X$; use Theorem 14.1.36 so $\sum_{x\in X} a_x$ is independent of g.
(vii) *Double sums*: prove identities on $R_{M,N} := \{1, \dots, M\} \times \{1, \dots, N\}$; pass to ∞ arrays (Theorem 14.1.37).
(viii) *Conditional convergence*: order matters; use summation by parts or rearrangements.

Key Insight: Series arguments are sequence arguments with special notation for tails.

14.2 Uniform Convergence

Uniform convergence asks when limits of functions preserve structure. The key distinction is whether N may depend on x, which controls continuity, integration, differentiation, and the supremum metric.

Perspective 14.2.1. *Function Limits Before Function-Sequence Limits*
$\lim_{t\to x} f(t)$ denotes the metric-space function limit of Definition 13.3.7.
Uniform convergence is the hypothesis that lets the limits in n and t commute.

Definition 14.2.2. *Pointwise Limits of Functions*
Let $f_n : E \to (Y, d_Y)$.

(i) $f_n \to f$ *pointwise* on E if $d_Y(f_n(x), f(x)) \to 0$ for every $x \in E$.
(ii) If Y is scalar, $\sum f_n \to f$ *pointwise* on E if $s_N(x) := \sum_{n=1}^{N} f_n(x) \to f(x)$ for every $x \in E$.

No metric on E is required; topology matters only for preserving continuity, limits, or integrals.

Example 14.2.3. *Pointwise Limits Can Lose Structure*

(i) $s_{m,n} := m/(m+n)$ gives $\lim_m \lim_n s_{m,n} = 1$ but $\lim_m \lim_n s_{m,n} = 0$.
(ii) For $f_n(x) = x^2/(1+x^2)^n$, $\sum_{n=0}^{\infty} f_n$ equals 0 at 0 and $1+x^2$ otherwise, by Theorem 14.1.8.
(iii) $f_n(x) = \sin(nx)/\sqrt{n}$ satisfies $f_n \to 0$ pointwise but $f_n'(0) = \sqrt{n} \to +\infty$.
(iv) $f_n(x) = n^2 x(1-x^2)^n$ on $[0,1]$ satisfies $f_n \to 0$ pointwise, while $\int_0^1 f_n\,dx = n^2/(2n+2) \to +\infty$.

Key Insight: Pointwise control can miss global spikes and oscillations.

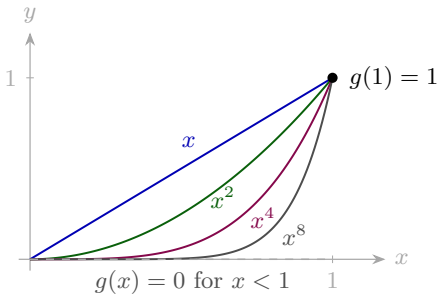


Figure 136: $x^n \to 0$ pointwise on $[0,1)$, but the endpoint prevents uniform convergence on $[0,1]$.

Example 14.2.4. *x^n Converges Pointwise But Not Uniformly*

On $[0,1]$, let $g_n(x) = x^n$. Then $g_n \to g$ pointwise, where $g(x) = \begin{cases} 0, & 0 \le x < 1, \\ 1, & x = 1. \end{cases}$.

The convergence is not uniform: for each n, some $x < 1$ has $|x^n - g(x)| = x^n > 1/2$.

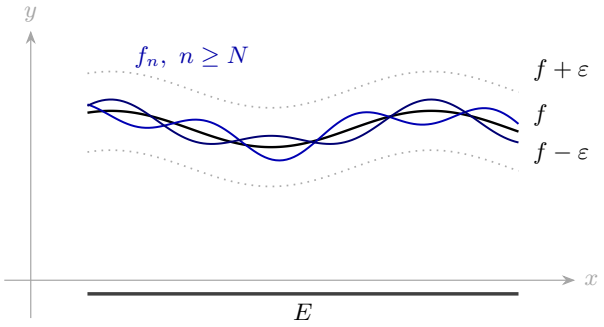


Figure 137: One index N puts all later graphs inside the same ε-strip around f on E.

Definition 14.2.5. *Uniform Convergence*

For $f_n, f : E \to (Y, d_Y)$, $f_n \rightrightarrows f$ on E if $\forall \varepsilon > 0\ \exists\ N \in \mathbb{N}$ such that $n \ge N,\ x \in E \implies d_Y(f_n(x), f(x)) \le \varepsilon$.
In the scalar-valued case, take $d_Y(u,v) = |u - v|$; $\sum f_n$ converges uniformly on E if its partial sums do.

Perspective 14.2.6. *One Tail, All Points*

Pointwise convergence lets N depend on x; uniform convergence uses one N for all x.
Equivalently, late graphs lie in one ε-strip around f.

Theorem 14.2.7. *Cauchy Criterion for Uniform Convergence*

For complex-valued $f_n : E \to \mathbb{C}$, f_n converges uniformly on E if and only if

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N} : \quad m, n \ge N,\ x \in E \implies |f_n(x) - f_m(x)| \le \varepsilon$$

Proof.

($\Rightarrow$) If $f_n \rightrightarrows f$, choose N so that $|f_k(x) - f(x)| \le \varepsilon/2$ for $k \ge N$ and $x \in E$. Then $|f_n(x) - f_m(x)| \le \varepsilon$.
($\Leftarrow$) For each x, $(f_n(x))$ is Cauchy in $\mathbb{C}$, so define $f(x) := \lim f_n(x)$.
Fix $n \ge N$ and let $m \to \infty$ in the criterion; then $|f_n(x) - f(x)| \le \varepsilon$ for all $x \in E$, hence $f_n \rightrightarrows f$.

Structural Commentary: The same N gives Cauchy control simultaneously for every $x \in E$.
Key Insight: Uniform Cauchy tails are exactly uniform convergence. □

Theorem 14.2.8. *Supremum Test for Uniform Convergence*

Suppose $f_n(x) \to f(x)\ \forall x \in E$, and $M_n := \sup_{x \in E} |f_n(x) - f(x)|$. Then $f_n \rightrightarrows f$ on E if and only if $M_n \to 0$.

Proof.
($\Rightarrow$) If $f_n \rightrightarrows f$, then $\forall \varepsilon > 0\ \exists\ N$ such that $n \geq N$ implies $|f_n(x) - f(x)| \leq \varepsilon$ for all $x \in E$. Hence $M_n \leq \varepsilon$.
($\Leftarrow$) If $M_n \to 0$, then $\forall \varepsilon > 0\ \exists\ N$ such that $n \geq N$ implies $M_n \leq \varepsilon$. Hence $|f_n(x) - f(x)| \leq \varepsilon$ for all $x \in E$.
Structural Commentary: Uniform convergence is convergence in the supremum error.
Key Insight: The pointwise errors must vanish as one global quantity. □

Theorem 14.2.9. *Weierstrass M-Test*
Let f_n be functions on E and suppose $|f_n(x)| \leq M_n$ for every $x \in E$.
If $\sum M_n$ converges, then $\sum f_n$ converges uniformly on E.

Proof. For $m \geq n$ and $x \in E$, $\left|\sum_{k=n}^{m} f_k(x)\right| \leq \sum_{k=n}^{m} M_k$.
Since $\sum M_k$ converges, Theorem 14.1.4 makes the right-hand side $< \varepsilon$ for all large m, n.
Thus the partial sums of $\sum f_k$ are uniformly Cauchy, so Theorem 14.2.7 applies.
Structural Commentary: Absolute numerical tails dominate functional tails.
Key Insight: A summable uniform majorant gives uniform convergence of a function series. □

Theorem 14.2.10. *Uniform Convergence Interchanges Limits*
Let E lie in a metric space, let x be a limit point of E, and suppose $f_n \rightrightarrows f$ on E.
If $A_n := \lim_{t \to x} f_n(t)$ exists for every n, then (A_n) converges and $\lim_{t \to x} f(t) = \lim_{n \to \infty} A_n$.

Proof. By Theorem 14.2.7, $\forall \varepsilon > 0\ \exists\ N$ such that $m, n \geq N$ and $t \in E$ imply $|f_n(t) - f_m(t)| \leq \varepsilon$.
Letting $t \to x$ gives $|A_n - A_m| \leq \varepsilon$, hence $A_n \to A$ for some $A \in \mathbb{C}$.
For any n, $|f(t) - A| \leq |f(t) - f_n(t)| + |f_n(t) - A_n| + |A_n - A|$. Given $\varepsilon > 0$, choose n so $|f(t) - f_n(t)| < \varepsilon/3$ for all $t \in E$ and $|A_n - A| < \varepsilon/3$; then take t sufficiently close to x so $|f_n(t) - A_n| < \varepsilon/3$. Thus $\lim_{t \to x} f(t) = A$.
Structural Commentary: Uniform convergence controls the outer two terms in the three-term estimate.
Key Insight: Uniformity is precisely what allows the $n \to \infty$ limit to move past the $t \to x$ limit. □

Theorem 14.2.11. *Uniform Limits Preserve Continuity*
If (f_n) is a sequence of continuous functions on E and $f_n \rightrightarrows f$ on E, then f is continuous on E.

Proof. Let $x \in E$. If x is isolated, f is continuous at x.
If x is a limit point, then $\lim_{t \to x} f_n(t) = f_n(x)$ for all n, so Theorem 14.2.10 gives $\lim_{t \to x} f(t) = \lim_{n \to \infty} f_n(x) = f(x)$.
Structural Commentary: Uniform convergence passes continuity to the limit.
Key Insight: Uniform convergence protects continuity from the failure modes in Example 14.2.3. □

Theorem 14.2.12. *Dini's Monotone Uniform Convergence Theorem*
Let K be compact and let $f_n, f : K \to \mathbb{R}$ be continuous. If $f_n(x) \downarrow f(x)$ for every $x \in K$, then $f_n \rightrightarrows f$ on K.

Proof. Set $g_n := f_n - f$. Then g_n is continuous, $g_n \downarrow 0$, and $g_n \geq 0$. Given $\varepsilon > 0$, let $K_n := \{x \in K : g_n(x) \geq \varepsilon\}$.
Then (K_n) is nested compact and $\bigcap_n K_n = \emptyset$, so Corollary 13.2.27 gives $K_N = \emptyset$ for some N.
Thus $0 \leq f_n(x) - f(x) < \varepsilon$ for all $x \in K$ and $n \geq N$.
Structural Commentary: Monotone convergence turns the bad sets into a nested compact sequence.
Key Insight: Compactness turns pointwise eventual escape from every bad set into a uniform escape. □

Definition 14.2.13. *The Supremum Metric on Bounded Continuous Functions*
For a metric space X, let $C_b(X)$ be the bounded complex-valued continuous functions and define $\|f\|_\infty := \sup_{x \in X} |f(x)|$; $d(f, g) := \|f - g\|_\infty$.
Then d is a metric on $C_b(X)$, induced by the norm of Definition 2.1.13; d-convergence is uniform convergence by Theorem 14.2.8. If X is compact, then $C(X) = C_b(X)$ by Corollary 13.3.32.

Perspective 14.2.14. *A Metric Space Whose Points Are Functions*
Uniform convergence is sequence convergence in the metric space $C_b(X)$.

Perspective 14.2.15. *Four Limit Notions*
The four limit notions are sequence limits, function limits, pointwise function limits, and uniform function limits. By Theorem 14.2.8, the last is ordinary sequence convergence in the supremum metric.

Theorem 14.2.16. *Uniform Convergence and Riemann–Stieltjes Integration*
Let α be increasing on $[a, b]$. If $f_n \in \mathcal{R}(\alpha)$ on $[a, b]$ and $f_n \rightrightarrows f$, then $f \in \mathcal{R}(\alpha)$ and $\int_a^b f\, d\alpha = \lim_{n \to \infty} \int_a^b f_n\, d\alpha$.

Proof. Put $\varepsilon_n := \sup_{a\le x\le b} |f_n(x) - f(x)|$. Then $\varepsilon_n \to 0$ by Theorem 14.2.8, and $f_n - \varepsilon_n \le f \le f_n + \varepsilon_n$. Upper and lower integrals of f are therefore squeezed between corresponding integrals of $f_n \pm \varepsilon_n$, and gap is at most $2\varepsilon_n[\alpha(b) - \alpha(a)]$. Thus $f \in \mathcal{R}(\alpha)$. Same estimate gives $\left|\int_a^b f\,d\alpha - \int_a^b f_n\,d\alpha\right| \le \varepsilon_n[\alpha(b) - \alpha(a)] \to 0$.
Structural Commentary: Uniform error becomes an integral error multiplied by the total α-variation.
Key Insight: Uniform convergence is strong enough to pass limits through the Riemann–Stieltjes integral. □

Corollary 14.2.17. *Termwise Integration of Uniformly Convergent Series*
If $f_n \in \mathcal{R}(\alpha)$ on $[a,b]$ and $\sum f_n$ converges uniformly to f, then $\int_a^b f\,d\alpha = \sum_{n=1}^{\infty} \int_a^b f_n\,d\alpha$.

Proof. Let $s_N := \sum_{n=1}^{N} f_n$. By linearity, $s_N \in \mathcal{R}(\alpha)$ and $\int_a^b s_N\,d\alpha = \sum_{n=1}^{N} \int_a^b f_n\,d\alpha$.
Since $s_N \rightrightarrows f$, Theorem 14.2.16 gives $\int_a^b f\,d\alpha = \lim_{N\to\infty} \int_a^b s_N\,d\alpha = \sum_{n=1}^{\infty} \int_a^b f_n\,d\alpha$.
Structural Commentary: Reduce the series to its uniformly convergent partial sums.
Key Insight: Uniform convergence lets integration be performed term by term. □

Theorem 14.2.18. *Uniform Convergence and Differentiation*
Let f_n be differentiable on $[a,b]$, suppose $f_n(x_0)$ converges for some $x_0 \in [a,b]$, and suppose $f_n' \rightrightarrows g$ on $[a,b]$. Then $f_n \rightrightarrows f$ for some f, and $f' = g$ on $[a,b]$ (one-sided at endpoints).

Proof. For m, n, MVT 15.1.13 gives $|f_n(x) - f_m(x)| \le |f_n(x_0) - f_m(x_0)| + (b-a) \sup_{t\in[a,b]} |f_n'(t) - f_m'(t)|$.
Thus (f_n) is uniformly Cauchy, so $f_n \rightrightarrows f$. Fix x and set $\phi_n(t) := \frac{f_n(t)-f_n(x)}{t-x}$, $\phi(t) := \frac{f(t)-f(x)}{t-x}$.
For $t \ne x$, the same estimate gives $|\phi_n(t) - \phi_m(t)| \le \|f_n' - f_m'\|_\infty$, hence $\phi_n \rightrightarrows \phi$.
By Theorem 14.2.10, $f'(x) = \lim_{t\to x} \phi(t) = \lim_{n\to\infty} f_n'(x) = g(x)$.
Structural Commentary: Derivative control anchors functions, then quotients.
Key Insight: Uniform derivative convergence plus one anchor passes derivatives to the limit. □

Method 14.2.19. *Uniform Convergence Proof Patterns*
(i) *Pointwise*: fix x, then choose $N = N(x,\varepsilon)$.
(ii) *Uniform*: choose $N = N(\varepsilon)$; equivalently $\sup_E |f_n - f| \to 0$.
(iii) *Uniform Cauchy*: prove $\sup_E |f_n - f_m| \to 0$ as $m, n \to \infty$.
(iv) *Preservation*: use a three-term estimate and choose the uniform index first.
(v) *Series of functions*: apply Weierstrass M-test when numerical absolute tails dominate all functional tails.
(vi) *Termwise operations*: require $f_n \to f$ locally uniformly, i.e. $\sup_{x\in K} |f_n - f| \to 0$ for each compact K, before $\lim_n \int_K f_n = \int_K f$; for derivatives, require $f_n' \to g$ locally uniformly and get $(\lim_n f_n)' = g$.
Key Insight: Uniformity means N is chosen before x.

14.3 Function-Space Compactness

Function-space compactness treats functions as points. Supremum metrics preview Banach spaces, Baire category describes generic behavior, and equicontinuity leads to Arzela–Ascoli.

Theorem 14.3.1. *Completeness of the Supremum Metric*
The metric space $C_b(X)$ with metric $d(f,g) = \|f - g\|_\infty$ is complete.

Proof. Let (f_n) be Cauchy in $C_b(X)$. Then Theorem 14.2.7 gives a function f with $f_n \rightrightarrows f$. By Theorem 14.2.11, f is continuous. For some n, $\|f - f_n\|_\infty < 1$, hence f_n is bounded, thus $f \in C_b(X)$ and $\|f_n - f\|_\infty \to 0$.
Structural Commentary: Uniform Cauchy convergence builds the limit inside the same function space.
Key Insight: Completeness of $\mathbb{C}$ plus uniform convergence makes $C_b(X)$ complete. □

Definition 14.3.2. *Bounded Target Metric for All Functions*
For metric spaces (X, d_X) and (Y, d_Y), let $\mathcal{F} = Y^X$ and set $\rho(y,z) := \frac{d_Y(y,z)}{1+d_Y(y,z)}$ and $d_\rho(f,g) := \sup_{x\in X} \rho(f(x), g(x))$.
Then $\rho < 1$, so d_ρ is finite on all of $\mathcal{F}$; d_ρ-convergence is uniform convergence measured by d_Y.

Theorem 14.3.3. *Completeness and Closed Subspaces for d_ρ*
If the target metric space (Y, d_Y) is complete, then:
(i) $(\mathcal{F}, d_\rho)$ is complete;
(ii) $\mathcal{F}_b := \{f \in \mathcal{F} : f(X)$ is d_Y-bounded$\}$ is closed in $\mathcal{F}$;
(iii) $C(X,Y)$ is closed in $\mathcal{F}$.

Proof.

(i) Let (f_n) be d_ρ-Cauchy. For each $x \in X$, $(f_n(x))$ is Cauchy in Y; define $f(x) := \lim_n f_n(x)$.
Given $\varepsilon > 0$, choose N with $d_\rho(f_n, f_m) < \varepsilon$ for $m, n \geq N$. Letting $m \to \infty$ gives $d_\rho(f_n, f) \leq \varepsilon$ for $n \geq N$.

(ii) If $f_n \in \mathcal{F}_b$ and $d_\rho(f_n, f) \to 0$, choose N with $d_\rho(f_N, f) < 1/2$.
Then $d_Y(f_N(x), f(x)) < 1$ for all x. Since $f_N(X)$ is bounded, so is $f(X)$.

(iii) If $f_n \in C(X, Y)$ and $d_\rho(f_n, f) \to 0$, then $f_n \to f$ uniformly in d_Y.
Given $x_0 \in X$ and $\varepsilon > 0$, choose N with $d_Y(f_N(x), f(x)) < \varepsilon/3$ for all x, then use continuity of f_N at x_0 and the triangle inequality. Thus f is continuous.

Structural Commentary: Compress the target metric; then close the useful subspaces.
Key Insight: Completeness and uniform control make boundedness and continuity closed. □

Theorem 14.3.4. *A Continuous Nowhere Differentiable Function Exists*
There exists a real continuous function on $\mathbb{R}$ which is nowhere differentiable.

Proof. Let φ be the 2-periodic extension of $|x|$ on $[-1, 1]$. Then $0 \leq \varphi \leq 1$ and $|\varphi(s) - \varphi(t)| \leq |s - t|$.
Define $f(x) := \sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n \varphi(4^n x)$, converges uniformly (Theorem 14.2.9); hence f is continuous (Theorem 14.2.11).
Fix x and $m \in \mathbb{N}$. Choose $\delta_m = \pm\frac{1}{2}4^{-m}$ so that no integer lies between $4^m x$ and $4^m(x + \delta_m)$.
Set $\gamma_{n,m} := \frac{\varphi(4^n(x+\delta_m)) - \varphi(4^n x)}{\delta_m}$. If $n > m$, then $4^n \delta_m$ is an even integer, so $\gamma_{n,m} = 0$.
If $0 \leq n \leq m$, then $|\gamma_{n,m}| \leq 4^n$, and the choice of δ_m gives $|\gamma_{m,m}| = 4^m$.
Hence $\left|\frac{f(x+\delta_m) - f(x)}{\delta_m}\right| = \left|\sum_{n=0}^{m} \left(\frac{3}{4}\right)^n \gamma_{n,m}\right| \geq 3^m - \sum_{n=0}^{m-1} 3^n = \frac{1}{2}(3^m + 1)$
Since $\delta_m \to 0$, the difference quotients are unbounded near x; since x was arbitrary, f is nowhere differentiable.
Structural Commentary: Uniform sums give continuity; scaled cusps force unbounded quotients.
Key Insight: Uniform convergence preserves continuity, not differentiability. □

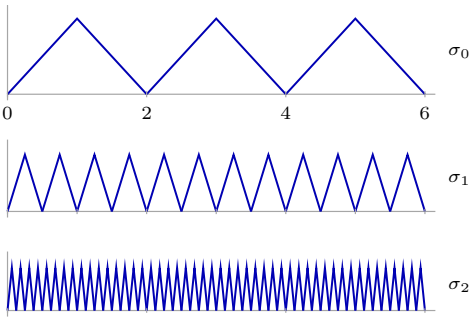


Figure 138: The first compressed sawtooth terms $\sigma_k(x) = (3/4)^k \varphi(4^k x)$: heights shrink, slopes grow.

Perspective 14.3.5. *Explicit Versus Generic Pathology*
The sawtooth series proves existence; Baire category shows nowhere differentiability is residual in $C([a, b], \mathbb{R})$.

Theorem 14.3.6. *Generic Nowhere Differentiability on Compact Intervals*
Let $a < b$, and let $C = C([a, b], \mathbb{R})$ with the supremum metric. There exists a residual set $\mathcal{G} \subseteq C$ such that:

(i) no $f \in \mathcal{G}$ has an available one-sided derivative at any point of $[a, b]$;
(ii) no $f \in \mathcal{G}$ is monotone on any nondegenerate subinterval of $[a, b]$;
(iii) the set $\{f \in C : f$ is differentiable at some point of $[a, b]\}$ is first category.

Proof.

(i) Write $\Delta_h f(x) := f(x + h) - f(x)$, and choose N_0 with $1/N_0 < b - a$. For $n \geq N_0$, define

$$R_n := \left\{ f \in C : \forall x \in [a, b - 1/n] \ \exists h > 0 \text{ with } \left|\frac{\Delta_h f(x)}{h}\right| > n \right\},$$

$$L_n := \left\{ f \in C : \forall x \in [a + 1/n, b] \ \exists h < 0 \text{ with } \left|\frac{\Delta_h f(x)}{h}\right| > n \right\},$$

$$G_n := \{f \in C : f \text{ is nonmonotone on every subinterval of length } 1/n\}.$$

Each R_n, L_n, G_n is open and dense in C (Weierstrass Approximation Theorem 14.4.15), plus small high-slope sawtooth perturbation; openness follows from compactness and uniform perturbation estimates.

(ii) Set $\mathcal{G} := \bigcap_{n\geq N_0}(R_n \cap L_n \cap G_n)$. By Definition 13.2.41, $\mathcal{G}$ is residual. If $f \in \mathcal{G}$, then the R_n and L_n conditions give one-sided difference quotients of arbitrarily large absolute value; since f is bounded, the corresponding increments tend to 0. Thus no available one-sided derivative exists.

(iii) Let $f \in \mathcal{G}$. Every nondegenerate subinterval contains a subinterval of length $1/n$ for all sufficiently large n, so the G_n conditions make f nonmonotone on every nondegenerate subinterval. If f is differentiable somewhere, then $f \notin \mathcal{G}$; hence the differentiable-somewhere set lies in $C \setminus \mathcal{G}$, which is first category.

Structural Commentary: Baire category turns dense sawtooth perturbability into a generic regularity failure.
Key Insight: The explicit construction shows existence; the category proof shows typicality. □

Definition 14.3.7. *Boundedness and Equicontinuity of Function Families*
Let $\mathcal{F}$ be a family of functions from E into a metric space (Y, d_Y); in scalar cases, take $d_Y(u,v) = |u-v|$.

(i) (f_n) is *pointwise bounded* on E if $(f_n(x))$ is bounded in Y for each $x \in E$.

(ii) A scalar sequence (f_n) is *uniformly bounded* on E if $\exists M$ such that $|f_n(x)| < M$ for all $x \in E, n \in \mathbb{N}$.

(iii) If E lies in a metric space (X, d_X), then $\mathcal{F}$ is *equicontinuous* on E if $\forall \varepsilon > 0\ \exists \delta > 0$, such that $d_X(x,y) < \delta,\ x,y \in E,\ f \in \mathcal{F} \implies d_Y(f(x), f(y)) < \varepsilon$.

(iv) The family is *pointwise equicontinuous* if for every $p \in E$ and $\varepsilon > 0$ there is $\delta = \delta(p,\varepsilon) > 0$ such that $d_X(p,y) < \delta,\ y \in E,\ f \in \mathcal{F}$ implies $d_Y(f(p), f(y)) < \varepsilon$.

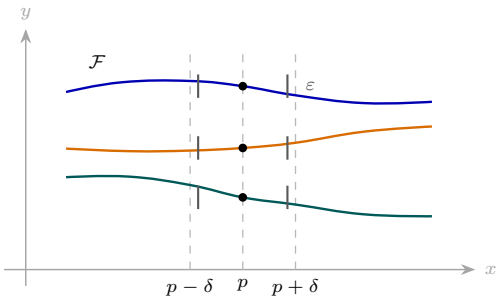


Figure 139: One input scale δ controls every curve in $\mathcal{F}$ within tolerance ε.

Theorem 14.3.8. *Pointwise Equicontinuity Becomes Uniform on Compact Sets*
Every pointwise equicontinuous family $\mathcal{F} \subset C(K, \mathbb{C})$ on compact K is equicontinuous.

Proof. Otherwise, choose $\varepsilon > 0$, $f_n \in \mathcal{F}$, and $x_n, t_n \in K$ with $d(x_n, t_n) < 1/n$ but $|f_n(x_n) - f_n(t_n)| \geq \varepsilon$.
By Theorem 13.2.3, pass to $x_n \to x_0$; then $t_n \to x_0$ too.
Pointwise equicontinuity at x_0 gives $|f_n(x_n) - f_n(t_n)| < \varepsilon$ for large n, a contradiction.
Structural Commentary: Compactness turns moving counterexamples into one base point.
Key Insight: The local $\delta(x,\varepsilon)$ becomes one uniform δ only after compactness prevents escape. □

Example 14.3.9. *Uniform Boundedness Alone Is Not Enough*
On $[0,1]$, let $f_n(x) := \frac{x^2}{x^2+(1-nx)^2}$. Then $|f_n| \leq 1$ and $f_n \to 0$ pointwise, but $f_n(1/n) = 1$; hence no uniformly convergent subsequence. The missing hypothesis is equicontinuity.

Theorem 14.3.10. *Diagonal Selection on a Countable Set*
Every pointwise bounded sequence (f_n) on countable E has a subsequence converging pointwise on E.

Proof. List $E = \{x_1, x_2, \ldots\}$. Inductively choose nested subsequences S_i so that $S_i(x_i)$ converges.
The diagonal subsequence is eventually contained in each S_i, hence converges at every x_i.
Structural Commentary: The diagonal keeps the ith convergence property after the ith step.
Key Insight: Countability permits diagonal convergence. □

Theorem 14.3.11. *Uniform Convergence Implies Equicontinuity on Compact Sets*
If K is compact, $f_n \in C_b(K)$, and $f_n \rightrightarrows f$ on K, then $\{f_n : n \in \mathbb{N}\}$ is equicontinuous.

Proof. Choose N so $\|f_n - f_N\|_\infty < \varepsilon$ for $n > N$.
By Theorem 13.3.37, choose one $\delta > 0$ giving ε-control for $f_1, \ldots, f_N$.
If $n > N$, $d(x,y) < \delta$, then $|f_n(x) - f_n(y)| \leq |f_n(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f_n(y)| < 3\varepsilon$.
Together with the finite initial family, $\{f_n\}$ is equicontinuous.
Structural Commentary: A uniformly convergent tail stays close to one uniformly continuous function.
Key Insight: Compactness handles finitely many functions; uniform convergence handles the tail. □

Theorem 14.3.12. *Arzela–Ascoli Selection Theorem*
Let K be compact. If $(f_n) \subset C_b(K)$ is pointwise bounded and equicontinuous on K, then:

(i) (f_n) is uniformly bounded on K;
(ii) (f_n) contains a uniformly convergent subsequence.

Proof.

(i) Choose $\delta > 0$ from equicontinuity with tolerance 1. Compactness gives $K \subseteq \bigcup_{j=1}^{r} B_\delta(p_j)$.
Pointwise boundedness gives M_j such that $|f_n(p_j)| \le M_j \forall n$.
If $x \in B_\delta(p_j)$, then $|f_n(x)| \le |f_n(x) - f_n(p_j)| + |f_n(p_j)| < 1 + M_j$, seq uniformly bounded by $1 + \max_j M_j$.

(ii) For each m, compactness gives a finite $1/m$-net in K; the union of these finite nets is a countable dense set $E \subset K$. By Theorem 14.3.10, some subsequence (g_i) converges pointwise on E.
Given $\varepsilon > 0$, choose $\delta > 0$ from equicontinuity with tolerance $\varepsilon/3$, and points $x_1, \dots, x_r \in E$ with $K \subseteq \bigcup_s B_\delta(x_s)$. As $(g_i(x_s))$ converges for each of finitely many x_s, $\exists N$ s.t. $i, j \ge N$ implies $|g_i(x_s) - g_j(x_s)| < \varepsilon/3 \ \forall s$. If $d(x, x_s) < \delta$, then $|g_i(x) - g_j(x)| \le |g_i(x) - g_i(x_s)| + |g_i(x_s) - g_j(x_s)| + |g_j(x_s) - g_j(x)| < \varepsilon$.
Thus (g_i) is uniformly Cauchy and converges uniformly by Theorem 14.2.7.

Structural Commentary: Dense convergence plus equicontinuity becomes uniform convergence.
Key Insight: Equicontinuity is exactly the compactness-compatible hypothesis missing from Example 14.3.9. □

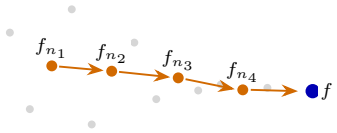


Figure 140: Arzela–Ascoli selects a uniformly convergent subsequence.

Definition 14.3.13. *σ-Compact Metric Space*
A metric space X is *σ-compact* if $X = \bigcup_{j=1}^{\infty} K_j$ for compact subsets $K_j \subseteq X$.

Theorem 14.3.14. *Arzela–Ascoli on σ-Compact Spaces*
Let X be σ-compact, Y a metric space, and (f_n) pointwise equicontinuous maps $X \to Y$. If each orbit $\{f_n(x) : n \in \mathbb{N}\}$ lies in a compact subset of Y, then some subsequence converges uniformly on compact subsets of X.

Proof. Choose an increasing compact exhaustion $X = \bigcup_j K_j$.
By Theorem 14.3.8, each restriction $f_n|_{K_j}$ is uniformly equicontinuous. Applying Theorem 14.3.12 on $K_1, K_2, \dots$ and diagonalizing gives a subsequence (g_m) converging uniformly on every K_j, hence pointwise on X.
If $K \subseteq X$ is compact, then $(g_m|_K)$ is uniformly equicontinuous, and a finite-net argument upgrades pointwise convergence on K to uniform convergence on K.
Structural Commentary: Exhaust X by compact pieces and diagonalize the compact theorem.
Key Insight: Local uniform convergence replaces global uniform convergence. □

Corollary 14.3.15. *Real-Line Local Uniform Subsequence*
If (f_n) is pointwise equicontinuous on $\mathbb{R}$ and bounded at one point, then it has a subsequence converging uniformly on compact subsets of $\mathbb{R}$.

Proof. Let $(f_n(x_0))$ be bounded. On any compact interval containing x_0, Theorem 14.3.8 gives uniform equicontinuity; finitely many δ-steps from x_0 to x bound each orbit $(f_n(x))$. Hence Theorem 14.3.14 applies on $\mathbb{R}$.
Structural Commentary: One bounded anchor propagates across compact intervals.
Key Insight: Pointwise equicontinuity turns local boundedness into compactwise compactness. □

Method 14.3.16. *Function-Space Compactness Proof Patterns*

(i) *Completeness*: prove a function sequence is uniformly Cauchy, build the pointwise limit, and use uniform convergence to keep continuity or boundedness.
(ii) *Baire category*: show a desired property contains a countable intersection of dense open subsets of a complete function space.
(iii) *Compactness/equicontinuity*: pass from finite nets to all of K.
(iv) *Diagonal selection*: get pointwise convergence on a countable dense set, then upgrade using equicontinuity.
(v) *Locally uniform*: exhaust a σ-compact domain by compact sets and diagonalize.

Key Insight: Compactness in function spaces is finite-net control plus equicontinuity.

14.4 Power Series and Approximation

Power series are controlled infinite polynomials, and approximation is uniform closure. This subsection covers radii, Taylor structure, Abel behavior, Weierstrass, Stone–Weierstrass, and elementary analytic functions.

Motivation 14.4.1. *Power Series as Controlled Infinite Polynomials*
Power-series manipulations rest on uniform convergence: Definition 14.1.1, Theorems 14.4.4, 14.2.9, and 14.2.18.

Definition 14.4.2. *Core Objects: Power Series, Approximation, and Gamma*

(i) A real *power series about a* is $\sum_{n=0}^{\infty} c_n(x-a)^n$; a local representation of f by one makes f *analytic* at a.
(ii) A *polynomial approximant* to f on a compact set K is a polynomial P chosen so that $\|f-P\|_{\infty,K}$ is small.
(iii) A function algebra is a collection closed under addition, multiplication, and scalar multiplication; Stone–Weierstrass asks when such algebras are uniformly dense in $C(K)$.
(iv) For $x>0$, the *gamma function* is $\Gamma(x) := \int_0^\infty t^{x-1}e^{-t}\,dt$ when the improper integral converges.

Key Insight: Power series are local analytic objects; approximation theorems turn algebraic families into dense function families; Γ extends factorials by integration.

Definition 14.4.3. *Power Series and Radius of Convergence*
Given a complex sequence (c_n), the series $\sum_{n=0}^{\infty} c_n z^n$ is a *power series*. If $\alpha := \limsup_{n\to\infty} \sqrt[n]{|c_n|}$, define $R := 1/\alpha$, with $R=+\infty$ when $\alpha=0$ and $R=0$ when $\alpha=+\infty$. The number R is the *radius of convergence*.

Theorem 14.4.4. *Convergence Inside the Circle*
For $\sum c_n z^n$, radius R: converge if $|z|<R$, diverge if $|z|>R$; boundary $|z|=R$ is undecided.

Proof. For $a_n = c_n z^n$, $\limsup_{n\to\infty} \sqrt[n]{|a_n|} = |z| \limsup_{n\to\infty} \sqrt[n]{|c_n|} = \frac{|z|}{R}$, with extended meaning if $R=0$ or $R=+\infty$. If $|z|<R$, this value is <1; if $|z|>R$, it is >1. Apply Theorem 14.1.19.
Structural Commentary: Radius controls the interior and exterior, not the boundary.
Key Insight: The root test is the radius-of-convergence theorem in disguise. □

Theorem 14.4.5. *A Boundary Test for Power Series*
If $\sum c_n z^n$ has radius 1, $c_n \downarrow 0$, and $c_n \geq 0$, then it converges for $|z|=1$, except possibly at $z=1$.

Proof. For $|z|=1$ and $z\neq 1$, put $a_n = z^n$ and $b_n = c_n$.
The partial sums of $\sum z^n$ are bounded because $\left|\sum_{m=0}^{n} z^m\right| = \left|\frac{1-z^{n+1}}{1-z}\right| \leq \frac{2}{|1-z|}$. Theorem 14.1.23 applies.
Structural Commentary: Boundary convergence away from $z=1$ comes from bounded geometric partial sums.
Key Insight: Monotone coefficients can make the unit-circle boundary converge without absolute convergence. □

Perspective 14.4.6. *Centered Series Are Translated Series*
$\sum c_n(x-a)^n$ is $\sum c_n y^n$ with $y = x-a$. Its radius is R from Definition 14.4.3: absolute convergence for $|x-a|<R$, divergence for $|x-a|>R$, and separate endpoint tests for $|x-a|=R$.

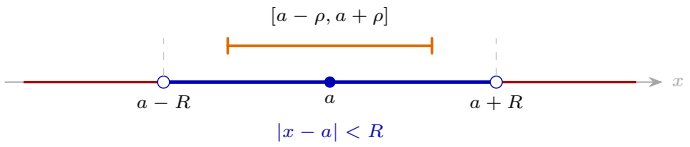


Figure 141: Inside $|x-a|<R$: absolute and locally uniform; outside divergence; endpoints undecided.

Definition 14.4.7. *Real Analytic at a Point*
Let $E\subseteq\mathbb{R}$, let $f: E\to\mathbb{R}$, and let a be an interior point of E. The function f is *real analytic at a* if there are $r>0$ and coefficients (c_n) such that $(a-r, a+r)\subseteq E$, the power series $\sum c_n(x-a)^n$ has radius of convergence at least r, and $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$ for all $x\in(a-r,a+r)$.
If E is open and f is real analytic at every point of E, then f is *real analytic on E*.

Theorem 14.4.8. *Power Series May Be Differentiated Term by Term*
Let $f(x) = \sum_{n\geq 0} c_n x^n$ have radius $R>0$, and $I_R = (-R,R)$. For $0<\rho<R$, let $K_\rho = [-\rho,\rho]$, a closed interval contained in I_R. Then $\sum c_n x^n$ converges uniformly on each K_ρ, and $f\in C^\infty(I_R)$.
For $k\geq 1$, $f^{(k)}(x) = \sum_{n\geq k} \frac{n!}{(n-k)!} c_n x^{n-k}$, $(x\in I_R)$, and $f^{(k)}(0) = k!c_k$.

Proof. For $0 < \rho < R$, Theorem 14.4.4 and the root-test argument give $\sum |c_n|\,\rho^n < \infty$; hence the Weierstrass M-test, Theorem 14.2.9, gives uniform convergence on K_ρ.
Since $\sqrt[n]{n} \to 1$, polynomial factors in n do not change the radius. Thus each differentiated series converges uniformly on K_ρ, and Theorem 14.2.18, iterated, gives the formula and $f^{(k)}(0) = k!c_k$.
Structural Commentary: On each K_ρ, termwise operations are justified.
Key Insight: The derivative coefficients grow only by polynomial factors, which do not change the radius. □

Corollary 14.4.9. *Termwise Calculus for Centered Power Series*
Let $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$ *have radius* $R > 0$. *On* $(a-R, a+R)$ *the series converges locally uniformly and defines a continuous function. For* $|x-a| < R$, $f'(x) = \sum_{n=1}^{\infty} nc_n(x-a)^{n-1}$ *with local uniform convergence of the derivative series. If* $[y,z] \subset (a-R, a+R)$, *then* $\int_y^z f(x)\,dx = \sum_{n=0}^{\infty} c_n \frac{(z-a)^{n+1}-(y-a)^{n+1}}{n+1}$.

Proof. Set $g(u) = \sum c_n u^n$ and $u = x - a$.
Theorem 14.4.8 applied to g on $(-R, R)$ gives convergence, continuity, derivative formula after translating.
For $[y,z] \subset (a-R, a+R)$, choose $\rho < R$ with $[y,z] \subset [a-\rho, a+\rho]$; then Corollary 14.2.17 lets the integrals of partial sums pass to the uniform limit.
Structural Commentary: All local termwise operations happen on closed intervals strictly inside the radius.
Key Insight: Translation reduces centered power series to the radius-R theorem about $\sum c_n u^n$. □

Corollary 14.4.10. *Analytic Functions Are Their Taylor Series*
If $f(x) = \sum c_n(x-a)^n$ *near* a, *then* f *is* C^∞, *expansion* f *is unique, and* $f(x) = f(a) + \sum_{n=1}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$.

Proof. Corollary 14.4.9, applied repeatedly, gives $c_0 = f(a)$ and $c_k = f^{(k)}(a)/k!$.
Hence two expansions about a have the same coefficients.
Structural Commentary: Derivatives determine the local series.
Key Insight: Once a power-series representation exists, the coefficients cannot vary. □

Perspective 14.4.11. *Taylor Coefficients Are Only the First Half*
Power-series coefficients are forced by derivatives (Corollary 14.4.10); for a function not already known to be analytic, proving the Taylor series represents the function needs a remainder estimate (Theorem 15.1.26).

Theorem 14.4.12. *Abel's Endpoint Theorem*
Let $s_n := \sum_{k=0}^{n} c_k$ *with* $s_n \to s$, *and let* $f(x) = \sum_{n\geq 0} c_n x^n$ *on* $(-1,1)$. *Then* $f(x) \to s$, *for* $x \to 1^-$.

Proof. With $s_{-1} = 0$, summation by parts, Theorem 14.1.22, gives for $|x| < 1$, $f(x) = (1-x)\sum_{n=0}^{\infty} s_n x^n$.
Since $(1-x)\sum_{n\geq 0} x^n = 1$, then $f(x) - s = (1-x)\sum_{n\geq 0}(s_n - s)x^n$. Choose N so that $|s_n - s| < \varepsilon/2$ for $n > N$.
The tail is at most $\varepsilon/2$, and the finite part tends to 0 after multiplication by $1-x$.
Structural Commentary: Endpoint convergence becomes one-sided continuity.
Key Insight: The factor $(1-x)$ averages the partial sums s_n with geometric weights. □

Theorem 14.4.13. *Abel Product Theorem for Cauchy Products*
Suppose $\sum a_n = A$, $\sum b_n = B$, $\sum c_n = C$, *and* $c_n = \sum_{k=0}^{n} a_k b_{n-k}$. *Then* $C = AB$.

Proof. For $0 \leq x < 1$, set $f(x) := \sum_{n\geq 0} a_n x^n$; $g(x) := \sum_{n\geq 0} b_n x^n$; and $h(x) := \sum_{n\geq 0} c_n x^n$.
Since convergent series have bounded terms, f, g, h converge absolutely for $|x| < 1$. Thus the Cauchy product gives $fg = h$ on $[0,1)$. By Theorem 14.4.12, $f \to A$, $g \to B$, and $h \to C$ as $x \to 1^-$. Hence $AB = C$.
Structural Commentary: Prove $fg = h$ inside, then pass to $x = 1$.
Key Insight: Inside $|x| < 1$, the Cauchy product is safe; Abel then moves the identity to $x = 1$. □

Theorem 14.4.14. *Re-Expansion and Identity for Power Series*
Let $f(x) = \sum_{n\geq 0} c_n x^n$ *have radius* $R > 0$, *and set* $S = (-R, R)$.

(i) If $a \in S$, *then* $f(x) = \sum_{m\geq 0} \frac{f^{(m)}(a)}{m!}(x-a)^m$, $(|x-a| < R - |a|)$

(ii) Let $F = \sum_{n\geq 0} a_n x^n$ *and* $G = \sum_{n\geq 0} b_n x^n$ *on* S.
If $\mathcal{Z}_{F,G} := \{x \in S : F(x) = G(x)\}$ *has a limit point in* S, *then* $a_n = b_n$ *for all* $n \geq 0$, *hence* $F = G$ *on* S.

Proof.

(i) Write $x^n = ((x-a)+a)^n$. If $|x-a|+|a| < R$, then $\sum_{n\geq 0} |c_n| \left(|x-a|+|a|\right)^n < \infty$, so the resulting double sum may be rearranged. Theorem 14.4.8 gives the coefficients $f^{(m)}(a)/m!$.

(ii) Subtract the series and call the difference f. Let $Z = \{x \in S : f(x) = 0\}$, and let A be the set of limit points of Z in S. Then $A \neq \emptyset$. For $x_0 \in A$, part (i) gives $f(x) = \sum d_n (x-x_0)^n$.
If d_k were the first nonzero coefficient, then $f(x) = (x-x_0)^k g(x)$ with $g(x_0) \neq 0$, contradicting $x_0 \in A$. Hence $f = 0$ near x_0, so A is open. Also $S \setminus A$ is open; by connectedness, Theorem 13.3.5, $A = S$. Since Z is closed, $f = 0$ on S, and evaluating derivatives at 0 gives equality of coefficients.

Structural Commentary: Power-series equality is local-to-global on an interval.
Key Insight: A limit point of zeros forces all Taylor coefficients to vanish. □

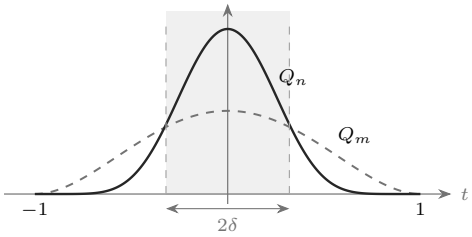


Figure 142: The normalized kernels $Q_n(t) = c_n(1-t^2)^n$ concentrate near 0 while retaining integral 1.

Theorem 14.4.15. *Weierstrass Polynomial Approximation*
If $f : [a,b] \to \mathbb{C}$ is continuous, then polynomials P_n satisfy $P_n \rightrightarrows f$ on $[a,b]$. If f is real-valued, take P_n real.

Proof. By an affine change of variables and subtraction of the endpoint interpolant, assume $[a,b] = [0,1]$ and $f(0) = f(1) = 0$; extend f by 0 outside $[0,1]$. For $n \geq 1$, let $Q_n(x) := c_n(1-x^2)^n$ and $\int_{-1}^{1} Q_n(x)\,dx = 1$.
Source estimates give $c_n < \sqrt{n}$ and $Q_n(t) \leq \sqrt{n}(1-\delta^2)^n$ for $\delta \leq |t| \leq 1$.
Define $P_n(x) := \int_{-1}^{1} f(x+t)Q_n(t)\,dt$, $(0 \leq x \leq 1)$. By change of variables, $P_n(x) = \int_0^1 f(u)Q_n(u-x)\,du$, so P_n is a polynomial in x (real if f is real). Given $\varepsilon > 0$, choose $\delta > 0$ so $|u-v| < \delta$ implies $|f(u)-f(v)| < \varepsilon/2$, and set $M = \sup|f|$. Since $Q_n \geq 0$ and $\int Q_n = 1$, $|P_n(x) - f(x)| \leq \int_{-1}^{1} |f(x+t)-f(x)|\,Q_n(t)\,dt \leq \frac{\varepsilon}{2} + 4M\sqrt{n}(1-\delta^2)^n$.
The last term tends to 0, so $P_n \rightrightarrows f$ on $[0,1]$. Undoing the reductions proves the theorem.
Structural Commentary: The kernels Q_n form polynomial approximate identities.
Key Insight: Uniform continuity controls the near part, and concentration of Q_n kills the far part. □

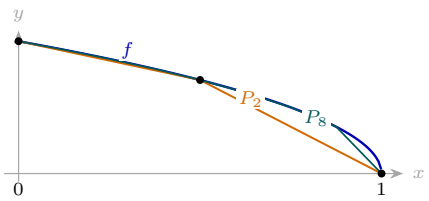


Figure 143: Finer piecewise-linear interpolants approach $f(x) = \sqrt{1-x}$ uniformly on $[0,1]$.

Corollary 14.4.16. *Polynomial Approximation of $|x|$*
For every interval $[-a,a]$, there are real polynomials P_n such that $P_n(0) = 0$ and $P_n \rightrightarrows |x|$ on $[-a,a]$.

Proof. By Theorem 14.4.15, choose real polynomials P_n^* converging uniformly to $|x|$ on $[-a,a]$.
Since $P_n^*(0) \to 0$, the polynomials $P_n(x) := P_n^*(x) - P_n^*(0)$ have $P_n(0) = 0$ and still converge uniformly to $|x|$.
Structural Commentary: Normalize the approximants at the base point.
Key Insight: Subtracting the value at 0 fixes the normalization without changing the uniform limit. □

Definition 14.4.17. *Algebras and Uniform Closure*
Let $\mathcal{A}$ be a family of complex-valued functions on a set E.

(i) $\mathcal{A}$ is an *algebra* if it is closed under $+$, products, and scalar multiplication (real scalars in the real case).
(ii) $\mathcal{A}$ is *uniformly closed* if $f_n \in \mathcal{A}$ and $f_n \rightrightarrows f$ on E imply $f \in \mathcal{A}$.
(iii) The *uniform closure* of $\mathcal{A}$ is $\{f : f_n \in \mathcal{A},\ f_n \rightrightarrows f \text{ on } E\}$.

Theorem 14.4.18. *Uniform Closures Are Closed Algebras*
The uniform closure of an algebra of bounded functions is a uniformly closed algebra.

Proof. If $f, g \in \mathcal{B}$, choose $f_n, g_n \in \mathcal{A}$ with $f_n \rightrightarrows f$ and $g_n \rightrightarrows g$.
Boundedness and uniform convergence give $f_n + g_n \rightrightarrows f + g$; $f_n g_n \rightrightarrows fg$; and $cf_n \rightrightarrows cf$. Thus $\mathcal{B}$ is an algebra.
If $h_n \in \mathcal{B}$ and $h_n \rightrightarrows h$, approximate each h_n uniformly by some $a_n \in \mathcal{A}$ within $1/n$; then $a_n \rightrightarrows h$, so $h \in \mathcal{B}$.
Structural Commentary: Uniform limits preserve algebraic operations for bounded functions.
Key Insight: The uniform closure of an algebra has already absorbed all further uniform limits. □

Definition 14.4.19. *Separation, Nonvanishing, and Self-Adjointness*
Let $\mathcal{A}$ be a family of functions on E.

(i) $\mathcal{A}$ *separates points* on E if for every distinct $x_1, x_2 \in E$ there is $f \in \mathcal{A}$ with $f(x_1) \neq f(x_2)$.
(ii) $\mathcal{A}$ *vanishes at no point* of E if for every $x \in E$ there is $g \in \mathcal{A}$ with $g(x) \neq 0$.
(iii) A complex algebra $\mathcal{A}$ is *self-adjoint* if $f \in \mathcal{A}$ implies $\overline{f} \in \mathcal{A}$, where $\overline{f}(x) := \overline{f(x)}$.

Theorem 14.4.20. *Two-Point Interpolation in a Separating Algebra*
If an algebra $\mathcal{A}$ on E separates points and vanishes nowhere, then for distinct $x_1, x_2 \in E$ and scalars c_1, c_2, there is $f \in \mathcal{A}$ with $f(x_i) = c_i$, $(i = 1, 2)$.

Proof. Choose $g, h, k \in \mathcal{A}$ with $g(x_1) \neq g(x_2)$, $h(x_1) \neq 0$, and $k(x_2) \neq 0$.
Let $u := gk - g(x_1)k$ and $v := gh - g(x_2)h$. Then $u(x_1) = v(x_2) = 0$ and $u(x_2)v(x_1) \neq 0$.
Hence $f := \frac{c_1 v}{v(x_1)} + \frac{c_2 u}{u(x_2)}$ lies in $\mathcal{A}$ and satisfies $f(x_i) = c_i$.
Structural Commentary: Separation and nonvanishing build two functions that isolate the two points.
Key Insight: Once two point-isolating functions are in the algebra, interpolation is linear algebra. □

Perspective 14.4.21. *Interpolation Is Linear Algebra*
The two-point interpolation step can also be written as a 2×2 linear system. Its invertibility is exactly the finite-dimensional criterion from Theorem 1.3.27, with the determinant test supplied by Theorem 4.1.5. The proof above chooses explicit point-isolating functions instead of reopening the calculation.

Theorem 14.4.22. *Stone–Weierstrass Theorem, Real Form*
$\mathcal{A} \subset C(K, \mathbb{R})$ is algebra on compact K that separates points, vanishes nowhere, then uniform closure is $C(K, \mathbb{R})$.

Proof. Let $\mathcal{B}$ be the uniform closure of $\mathcal{A}$; by Theorem 14.4.18, $\mathcal{B}$ is a uniformly closed algebra.
If $f \in \mathcal{B}$, then $|f| \in \mathcal{B}$: for $a = \|f\|_\infty$, approximate $|y|$ on $[-a, a]$ by polynomials p_n with $p_n(0) = 0$ using Corollary 14.4.16; then $p_n(f) \in \mathcal{B}$ and $p_n(f) \rightrightarrows |f|$.
Hence $\max(f, g), \min(f, g) \in \mathcal{B}$ whenever $f, g \in \mathcal{B}$, as $\max(f, g) = \frac{f+g}{2} + \frac{|f-g|}{2}$; $\min(f, g) = \frac{f+g}{2} - \frac{|f-g|}{2}$.
Fix $f \in C(K, \mathbb{R})$, $x \in K$, $\varepsilon > 0$. $\forall y \in K$, Theorem 14.4.20 gives $h_y \in \mathcal{B}$, $h_y(x) = f(x)$, $h_y(y) = f(y)$.
Compactness gives $y_1, \ldots, y_r$ s.t. $g_x := \max(h_{y_1}, \ldots, h_{y_r})$ satisfies $g_x(x) = f(x)$ and $g_x > f - \varepsilon$ on K.
Compactness gives $x_1, \ldots, x_m$ such that $h := \min(g_{x_1}, \ldots, g_{x_m})$ satisfies $|h - f| < \varepsilon$ on K.
Thus every $f \in C(K, \mathbb{R})$ is a uniform limit of members of $\mathcal{B}$, so $f \in \mathcal{B}$.
Structural Commentary: Local supports plus finite max/min give global approximation.
Key Insight: Separation plus compactness gives uniform approximation. □

Theorem 14.4.23. *Stone–Weierstrass Theorem, Complex Form*
If $\mathcal{A} \subset C_b(K)$ is self-adjoint, separates points, and vanishes nowhere on compact K, then $\overline{\mathcal{A}} = C_b(K)$.

Proof. Let $\mathcal{A}_\mathbb{R} := \{f \in \mathcal{A} : f \text{ real-valued}\}$.
If $f = u + iv \in \mathcal{A}$, self-adjointness gives $u = \frac{f+\overline{f}}{2} \in \mathcal{A}_\mathbb{R}$ and $v = \frac{f-\overline{f}}{2i} \in \mathcal{A}_\mathbb{R}$.
Thus $\mathcal{A}_\mathbb{R}$ separates points; rescaling a nonzero value and taking real parts shows it vanishes nowhere.
By Theorem 14.4.22, $\overline{\mathcal{A}_\mathbb{R}} = C(K, \mathbb{R})$. Approximate the real and imaginary parts separately, then combine.
Structural Commentary: Self-adjointness lets the complex algebra expose enough real functions.
Key Insight: The complex theorem is the real theorem applied to real and imaginary parts. □

Theorem 14.4.24. *Exponential and Logarithm from the Power Series*
Define $\mathsf{E}(z) := \sum_{n=0}^{\infty} \frac{z^n}{n!}$ for $z \in \mathbb{C}$.
Then, for $z, w \in \mathbb{C}$, $x \in \mathbb{R}$, $\mathsf{E}(z+w) = \mathsf{E}(z)\mathsf{E}(w)$, $\mathsf{E}(x) > 0$, $\mathsf{E}'(x) = \mathsf{E}(x)$, $\mathsf{E}(x) = e^x$, and E is strictly increasing on $\mathbb{R}$. Its inverse $\mathsf{L} : (0, \infty) \to \mathbb{R}$ satisfies $\mathsf{L}'(y) = \frac{1}{y}$, $\mathsf{L}(uv) = \mathsf{L}(u) + \mathsf{L}(v)$ for $u, v > 0$.
For $x > 0$ and $\alpha \in \mathbb{R}$, set $x^\alpha := \mathsf{E}(\alpha \mathsf{L}(x))$; then $(x^\alpha)' = \alpha x^{\alpha-1}$.

Proof. The ratio test (Theorem 14.1.20), gives convergence for all $z \in \mathbb{C}$.
Absolute convergence and Cauchy product give $\mathsf{E}(z)\mathsf{E}(w) = \sum_{n\geq 0} \frac{(z+w)^n}{n!} = \mathsf{E}(z+w)$, hence $\mathsf{E}(z)\mathsf{E}(-z) = 1$.
Theorem 14.4.8 gives $\mathsf{E}' = \mathsf{E}$ on $\mathbb{R}$. Thus $\mathsf{E}(x) > 0$ $\forall$ real x and $\mathsf{E}' > 0$, so E is strictly increasing.
Since $\mathsf{E}(1) = e$, the addition formula gives $\mathsf{E}(p) = e^p$ for $p \in \mathbb{Q}$; continuity extends this to $x \in \mathbb{R}$.
Let $\mathsf{L} = \mathsf{E}^{-1}$. Differentiating $\mathsf{L}(\mathsf{E}(x)) = x$ gives $\mathsf{L}'(y) = 1/y$.
If $u = \mathsf{E}(x)$ and $v = \mathsf{E}(y)$, then $\mathsf{L}(uv) = \mathsf{L}(\mathsf{E}(x+y)) = \mathsf{L}(u) + \mathsf{L}(v)$.
Finally, the chain rule, Theorem 15.1.6, gives $(x^\alpha)' = \alpha x^{\alpha-1}$.
Structural Commentary: Power series define E; inversion gives L and real powers.
Key Insight: The addition formula is the bridge from coefficients to the logarithm law. □

Perspective 14.4.25. *Real Powers by Rational Approximation*
One route defines x^α for $x > 0$ by choosing $q_n \in \mathbb{Q}$ with $q_n \to \alpha$ and setting $x^\alpha = \lim_{n\to\infty} x^{q_n}$ after proving existence and independence of (q_n). The power-series construction sets $x^\alpha = \mathsf{E}(\alpha\mathsf{L}(x))$; both routes must agree with rational powers and be choice-independent.

Perspective 14.4.26. *Logarithmic Integral Route*
With the Riemann integral, set $L(y) := \int_1^y \frac{1}{t}\,dt$ for $y > 0$. Theorem 15.1.57 gives $L'(y) = 1/y$; endpoint limits make L invertible, and Theorem 15.1.6 then recovers $E' = E$ and the product laws.

Perspective 14.4.27. *From Exponentials to Factorials*
Theorems 14.4.24, 15.1.59 supply real-power laws and integration-by-parts pattern used for Γ in Theorem 14.4.30.

Theorem 14.4.28. *Trigonometric Functions from the Exponential*
For real x, define $\mathsf{C}(x) := \frac{\mathsf{E}(ix)+\mathsf{E}(-ix)}{2}$ and $\mathsf{S}(x) := \frac{\mathsf{E}(ix)-\mathsf{E}(-ix)}{2i}$.
Then $\mathsf{E}(ix) = \mathsf{C}(x) + i\mathsf{S}(x)$, $\mathsf{C}^2 + \mathsf{S}^2 = 1$; $\mathsf{C}' = -\mathsf{S}$, and $\mathsf{S}' = \mathsf{C}$.
If $x_0 = \min\{x > 0 : \mathsf{C}(x) = 0\}$ and $\pi := 2x_0$, then $\mathsf{E}(\pi i) = -1$, $\mathsf{E}(2\pi i) = 1$, $\mathsf{E}(z + 2\pi i) = \mathsf{E}(z)$ for $z \in \mathbb{C}$.
Thus C, S are 2π-periodic, and $\gamma(t) := \mathsf{E}(it)$ traces the unit circle once on $[0, 2\pi]$.

Proof. The definitions give $\mathsf{E}(ix) = \mathsf{C} + i\mathsf{S}$. Since $\mathsf{E}(ix)\mathsf{E}(-ix) = 1$, $|\mathsf{E}(ix)| = 1$, hence $\mathsf{C}^2 + \mathsf{S}^2 = 1$.
Differentiating the definitions and using Theorem 14.4.8 gives $\mathsf{C}' = -\mathsf{S}$ and $\mathsf{S}' = \mathsf{C}$.
If C had no positive zero, then $\mathsf{C} > 0$ and $\mathsf{S}' > 0$ on $(0, \infty)$.
Thus, for $0 < x < y$, $\mathsf{S}(x)(y - x) < \int_x^y \mathsf{S}(t)\,dt = \mathsf{C}(x) - \mathsf{C}(y) \leq 2$, impossible for large y. Hence x_0 exists.
On $(0, \pi/2)$, $\mathsf{C} > 0$, so S increases from 0 to $\mathsf{S}(\pi/2) = 1$; thus $\mathsf{E}(\pi i/2) = i$, and the addition formula gives $\mathsf{E}(\pi i) = -1$, $\mathsf{E}(2\pi i) = 1$, and the period.
Finally, $|\mathsf{E}(it)| = 1$ and the source quadrant argument give the one-time traversal of the unit circle.
Structural Commentary: Trigonometry comes from the path $\mathsf{E}(it)$.
Key Insight: The identity $\mathsf{E}(ix)\mathsf{E}(-ix) = 1$ puts the exponential path on the unit circle. □

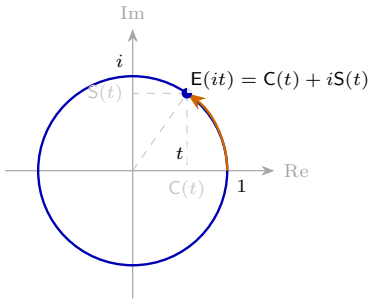


Figure 144: $\mathsf{E}(it) = \mathsf{C}(t) + i\mathsf{S}(t)$ traces the unit circle, defining sine and cosine analytically.

Theorem 14.4.29. *Algebraic Completeness of $\mathbb{C}$*
Every nonconstant complex polynomial has a complex root.

Proof. Let $P(z) = \sum_{j=0}^n a_j z^j$, $n \geq 1$, and normalize $a_n = 1$.
Since $|P(z)| \to \infty$, Theorem 13.3.33 gives z_0 with $\mu := |P(z_0)| = \min_{\mathbb{C}} |P|$. If $\mu = 0$, done.
Assume $\mu > 0$ and set $Q(w) := P(z_0 + w)/P(z_0)$. Then $Q(0) = 1$ and $|Q(w)| \geq 1$.
Write $Q(w) = 1 + \sum_{j=k}^n b_j w^j$, where $b_k \neq 0$ and k is minimal. By Theorem 14.4.28, choose θ with $b_k e^{ik\theta} = -|b_k|$.
Then $Q(re^{i\theta}) = 1 - |b_k|\, r^k + r^{k+1}H(r)$, with H bounded near 0, so $\left|Q(re^{i\theta})\right| < 1$ for small $r > 0$, contradiction.

Structural Commentary: Minimize, translate, then perturb.
Key Insight: Choose the phase that lowers the first nonzero term. □

Theorem 14.4.30. *Gamma Function, Beta Function, and Stirling Skeleton*
For $x > 0$, $\Gamma(x+1) = x\Gamma(x)$; $\Gamma(n+1) = n!$ for $n \in \mathbb{N}$; $\log\Gamma$ is convex on $(0,\infty)$.
Moreover, Γ is the unique $f : (0,\infty) \to (0,\infty)$ with $f(1) = 1$, $f(x+1) = xf(x)$, and $\log f$ convex.
For $x, y > 0$, $\mathrm{B}(x,y) := \int_0^1 t^{x-1}(1-t)^{y-1}\,dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$.
Consequently, $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$; $\int_{-\infty}^{\infty} e^{-s^2}\,ds = \sqrt{\pi}$; $\Gamma(x) = \frac{2^{x-1}}{\sqrt{\pi}}\Gamma\left(\frac{x}{2}\right)\Gamma\left(\frac{x+1}{2}\right)$; and $\lim_{x\to\infty} \frac{\Gamma(x+1)}{(x/e)^x\sqrt{2\pi x}} = 1$.

Proof. Integration by parts gives $\Gamma(x+1) = x\Gamma(x)$; $\Gamma(1) = 1$ gives $\Gamma(n+1) = n!$.
Holder's inequality gives log-convexity. For uniqueness, let $\varphi = \log f$.
If $0 < x < 1$, convexity and $\varphi(x+1) = \varphi(x) + \log x$ give $0 \le \varphi(x) - \log\frac{n!\,n^x}{x(x+1)\cdots(x+n)} \le x\log\left(1+\frac{1}{n}\right) \to 0$.
Thus f is uniquely determined on $(0,1)$, hence on $(0,\infty)$.
For fixed $y > 0$, $\mathrm{B}(1,y) = 1/y$, $\mathrm{B}(x+1,y) = \frac{x}{x+y}\mathrm{B}(x,y)$, and Holder gives log-convexity in x.
Hence $F_y(x) := \frac{\Gamma(x+y)}{\Gamma(y)}\mathrm{B}(x,y)$ satisfies the three uniqueness conditions, so $F_y = \Gamma$.
The consequences follow from $t = \sin^2\theta$, $t = s^2$, the duplication identity, and the substitution $t = x(1+u)$ in the asymptotic proof of Stirling's formula.
Structural Commentary: Functional equation plus convexity gives uniqueness.
Key Insight: Bohr–Mollerup uniqueness turns beta-function identities into gamma identities. □

Method 14.4.31. *Power-Series and Approximation Proof Patterns*

(i) *Power series*: work on $K_\rho \Subset I_R$; use uniform convergence before termwise limits or derivatives.
(ii) *Endpoint theorems*: prove identities inside the radius and pass to endpoints with Abel-type control.
(iii) *Approximate identities*: use kernels $Q_n \ge 0$, $\int Q_n = 1$, with tails $\to 0$ off 0.
(iv) *Uniform closure*: show $a_n \in \mathcal{A}$ and $a_n \rightrightarrows f$.
(v) *Lattice closure*: derive $|f|$, then $\max, \min$ from algebraic formulas.
(vi) *Stone–Weierstrass*: interpolate locally; use compactness and finite max/min to globalize.
(vii) *Gamma function*: use $\Gamma(x+1) = x\Gamma(x)$, $\Gamma(1) = 1$, and log-convexity for uniqueness.

Key Insight: Uniform convergence and compactness are what make infinite algebra behave like finite algebra.

14.5 Fourier Structure

Fourier structure studies trigonometric series through orthogonality, projection, kernels, and convergence, with inner-product geometry reappearing throughout.

Definition 14.5.1. *Core Fourier Objects*

(i) A *trigonometric polynomial* is a finite sum $a_0 + \sum_{n=1}^{N}(a_n\cos nx + b_n\sin nx)$, equivalently $\sum_{n=-N}^{N}\alpha_n e^{inx}$.
(ii) If f is Riemann integrable on $[-\pi,\pi]$, its *Fourier coefficients* are $\widehat{f}(m) := \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-imx}\,dx$, $m \in \mathbb{Z}$, and $f(x) \sim_{\text{coef}} \sum_{m=-\infty}^{\infty}\widehat{f}(m)e^{imx}$ records coefficients, not convergence.
(iii) A sequence (ϕ_n) on $[a,b]$ is *orthonormal* if $\int_a^b \phi_n(x)\overline{\phi_m(x)}\,dx = 0$ $(n \ne m)$ and $\int_a^b |\phi_n(x)|^2\,dx = 1$. Its Fourier coefficients are $c_n := \int_a^b f(t)\overline{\phi_n(t)}\,dt$.

Key Insight: Fourier coefficients are orthonormal coordinates; convergence is a separate analytic question.

Definition 14.5.2. *Period-One Fourier Normalization*
Let $C(\mathbb{R}/\mathbb{Z};\mathbb{C})$ denote the continuous functions $f : \mathbb{R} \to \mathbb{C}$ satisfying $f(x+1) = f(x)$ for all x.
For $f, g \in C(\mathbb{R}/\mathbb{Z};\mathbb{C})$, set $\langle f,g\rangle_{\mathbb{T}} := \int_0^1 f(x)\overline{g(x)}\,dx$; $\|f\|_{L^2(\mathbb{T})} := \sqrt{\langle f,f\rangle_{\mathbb{T}}}$; and $e_n(x) := e^{2\pi inx}$ for $n \in \mathbb{Z}$.
The *period-one Fourier coefficient* is $\widehat{f}_{\mathbb{T}}(n) := \langle f, e_n\rangle_{\mathbb{T}} = \int_0^1 f(x)e^{-2\pi inx}\,dx$.
A period-one trigonometric polynomial is a finite sum $\sum_{n=-N}^{N} c_n e_n$.

Perspective 14.5.3. *Two Equivalent Fourier Clocks*
The period-one and 2π-periodic conventions differ by $t = 2\pi x$; only normalizing constants change.
Key Insight: Changing Fourier clocks changes constants, not the projection geometry.

Proposition 14.5.4. *Characters Are an Orthonormal System*
For $m, n \in \mathbb{Z}$, $\langle e_n, e_m\rangle_{\mathbb{T}} = 1$ if $n = m$ and 0 otherwise.
Hence, if $P = \sum_{n=-N}^{N} c_n e_n$, then $c_n = \langle P, e_n\rangle_{\mathbb{T}}$ and $\|P\|^2_{L^2(\mathbb{T})} = \sum_{n=-N}^{N}|c_n|^2$.

Proof. If $n = m$, the integrand is 1. If $n \neq m$, then $\int_0^1 e^{2\pi i(n-m)x}\,dx = 0$.
The coefficient and norm identities are the finite orthonormal-coordinate identities from Theorems 2.1.33 and 2.1.38, applied inside the finite-dimensional span of $e_{-N}, \dots, e_N$.
Structural Commentary: Fourier coefficients are orthonormal coordinates in a finite trigonometric subspace.
Key Insight: The exponential characters make cross terms vanish. □

Definition 14.5.5. *Periodic Convolution and Fejer Kernels*
For $f, g \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$, define their *periodic convolution* as $(f * g)(x) := \int_0^1 f(y)g(x-y)\,dy$.
For $N \geq 1$, the *Fejer kernel* is $F_N(x) := \sum_{n=-N}^{N} \left(1 - \frac{|n|}{N}\right) e_n(x) = \frac{1}{N}\left|\sum_{n=0}^{N-1} e_n(x)\right|^2$.

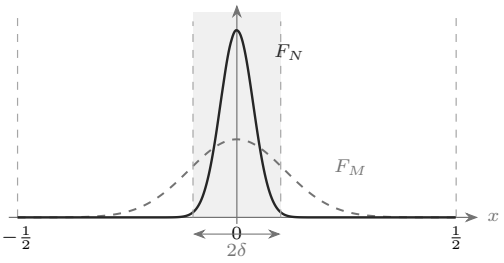


Figure 145: The Fejer kernels are nonnegative trigonometric polynomial approximate identities on $\mathbb{R}/\mathbb{Z}$.

Theorem 14.5.6. *Fejer Approximation on $\mathbb{R}/\mathbb{Z}$*
For $N \geq 1$, F_N is a nonnegative trigonometric polynomial with $\int_0^1 F_N = 1$. For each $0 < \delta < 1/2$, $F_N \to 0$ uniformly on $[\delta, 1-\delta]$. If $f \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$, then $f * F_N$ is a trigonometric polynomial and $f * F_N \rightrightarrows f$.

Proof. From Definition 14.5.5, F_N is a finite linear combination of e_n and $F_N = \frac{1}{N}\left|\sum_{n=0}^{N-1} e_n\right|^2 \geq 0$.
Since $\int_0^1 e_n = 0$ for $n \neq 0$, the e_0 coefficient gives $\int_0^1 F_N = 1$.
For $x \notin \mathbb{Z}$, $F_N(x) = \frac{\sin^2(\pi N x)}{N \sin^2(\pi x)}$. If $m_\delta := \min_{\delta \leq x \leq 1-\delta} \sin^2(\pi x) > 0$, then $0 \leq F_N(x) \leq (N m_\delta)^{-1}$ on $[\delta, 1-\delta]$.
Also $f * e_n = \widehat{f_{\mathbb{T}}}(n) e_n$, so $f * F_N$ is a trigonometric polynomial. For uniform convergence, write $(f * F_N)(x) = \int_0^1 f(x-t)F_N(t)\,dt$. Let $M = \|f\|_\infty$ and choose δ so $|f(x-t) - f(x)| < \varepsilon$ whenever $t \in [0, \delta] \cup [1-\delta, 1]$.
By $F_N \geq 0$ and $\int_0^1 F_N = 1$, uniformly in x, $|(f * F_N)(x) - f(x)| \leq \varepsilon + 2M \sup_{\delta \leq t \leq 1-\delta} F_N(t) \to \varepsilon$.
Structural Commentary: Fejer kernels concentrate mass near the identity element of $\mathbb{R}/\mathbb{Z}$.
Key Insight: Positivity plus concentration turns convolution into uniform approximation. □

Perspective 14.5.7. *Fourier and Plancherel Roadmap*
Path: orthonormal characters, Fejer density, best L^2 projection, then Parseval/Plancherel.
Key Insight: Fourier theory is finite-dimensional orthogonal projection plus an approximation theorem.

Perspective 14.5.8. *Fourier Coefficients as Projections*
Fourier coefficients are projections onto orthonormal directions, modeled finitely by Theorems 2.2.3, 2.1.33, and 2.1.36; infinite directions make convergence separate.

Theorem 14.5.9. *Fourier Coefficients Give Best Mean-Square Approximation*
Let $\langle u, v\rangle := \int_a^b u\overline{v}$ and $\|u\|_2^2 := \langle u, u\rangle$.
If (ϕ_n) is orthonormal, $c_m = \langle f, \phi_m\rangle$, $s_n = \sum_{m=1}^{n} c_m\phi_m$, and $t_n = \sum_{m=1}^{n} \gamma_m\phi_m$, then $\|f - s_n\|_2 \leq \|f - t_n\|_2$, with equality iff $\gamma_m = c_m$ for $m = 1, \dots, n$. In particular, $\sum_{m=1}^{\infty} |c_m|^2 \leq \|f\|_2^2$ for $c_m \to 0$.

Proof. For sums over $m = 1, \dots, n$, orthonormality gives
$\|f - t_n\|_2^2 = \|f\|_2^2 - 2\,\mathrm{Re}\sum c_m\overline{\gamma_m} + \sum |\gamma_m|^2 = \|f\|_2^2 - \sum |c_m|^2 + \sum |\gamma_m - c_m|^2$.
Thus the minimum occurs exactly when $\gamma_m = c_m$.
Taking $t_n = s_n$ gives $\sum_{m=1}^{n} |c_m|^2 = \|s_n\|_2^2 \leq \|f\|_2^2$; let $n \to \infty$ to get Bessel and $c_m \to 0$.
Structural Commentary: Orthogonality makes the error split into independent coefficient errors.
Key Insight: Fourier coefficients are the coordinates of the orthogonal projection. □

Theorem 14.5.10. *Dirichlet Kernel and Local Fourier Convergence*
Let f be Riemann integrable and 2π-periodic, with $\widehat{f}(m) := \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-imt}\,dt$.

Let $s_N(f;x) := \sum_{m=-N}^{N} \widehat{f}(m)e^{imx}$ *and* $D_N(t) := \sum_{m=-N}^{N} e^{imt} = \frac{\sin((N+\frac{1}{2})t)}{\sin(t/2)}$.
Then $s_N(f;x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)D_N(t)\,dt$.
If, for some x, $|f(x+t)-f(x)| \le M|t|$ *whenever* $|t|<\delta$, *then* $s_N(f;x)\to f(x)$.
Hence $f=g$ *near* x *implies* $s_N(f;x)-s_N(g;x)\to 0$.

Proof. The displayed form of D_N is the finite geometric sum; substituting $\widehat{f}(m)$ gives the convolution formula.
Also $\frac{1}{2\pi}\int_{-\pi}^{\pi} D_N = 1$. Set $q_x(t) := \frac{f(x-t)-f(x)}{\sin(t/2)}$, for $0<|t|\le\pi$,$q_x(0) := 0$.
Then $s_N(f;x)-f(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} q_x(t)\sin\left(N+\frac{1}{2}\right)t\,dt$.
The hypothesis makes q_x bounded near 0; away from 0 it is Riemann integrable.
Since $\sin((N+\frac{1}{2})t) = \sin Nt\cos(t/2)+\cos Nt\sin(t/2)$, Theorem 14.5.9 sends the two Fourier coefficients to 0.
Apply the result to $f-g$ for localization.
Structural Commentary: The Dirichlet kernel rewrites Fourier partial sums as a local oscillatory integral.
Key Insight: Pointwise convergence under a local Lipschitz condition depends only on nearby values of f. □

Corollary 14.5.11. *Uniform Approximation by Trigonometric Polynomials*
If $f\in C(\mathbb{R})$ *is* 2π*-periodic, then for every* $\varepsilon>0$ *there is a trigonometric polynomial* P *such that* $\|f-P\|_\infty<\varepsilon$.

Proof. Let $\mathbb{T} := \{z\in\mathbb{C} : |z|=1\}$ and set $F(e^{ix}) := f(x)$.
The algebra $\mathcal{T} := \operatorname{span}\{z^n : n\in\mathbb{Z}\}\subset C(\mathbb{T})$ is self-adjoint, separates points, and vanishes nowhere.
By Theorem 14.4.23, $\overline{\mathcal{T}} = C(\mathbb{T})$. Pulling approximants back by $z=e^{ix}$ gives the desired P.
Structural Commentary: Periodic approximation is Stone–Weierstrass on the unit circle.
Key Insight: The functions e^{inx} are the algebraic coordinates on the circle. □

Theorem 14.5.12. *Parseval's Theorem for Trigonometric Fourier Series*
Let f,g *be Riemann integrable and* 2π*-periodic. Let* $\langle u,v\rangle_2 := \frac{1}{2\pi}\int_{-\pi}^{\pi} u(x)\overline{v(x)}\,dx$ *and* $\|u\|_2^2 := \langle u,u\rangle_2$.
If $f\sim_{\text{coef}}\sum_{n\in\mathbb{Z}}\widehat{f}(n)e^{inx}$, *and* $g\sim_{\text{coef}}\sum_{n\in\mathbb{Z}}\widehat{g}(n)e^{inx}$,
then $\|f-s_N(f)\|_2\to 0$; $\langle f,g\rangle_2 = \sum_{n\in\mathbb{Z}}\widehat{f}(n)\overline{\widehat{g}(n)}$; *and* $\|f\|_2^2 = \sum_{n\in\mathbb{Z}}\left|\widehat{f}(n)\right|^2$.

Proof. If f is continuous, Corollary 14.5.11 gives trigonometric polynomials P with $\|f-P\|_2$ arbitrarily small.
If P has degree N_0, then Theorem 14.5.9 gives $\|f-s_N(f)\|_2\le\|f-P\|_2$ for $N\ge N_0$.
For Riemann integrable f, approximate f in $\|\cdot\|_2$ by a continuous 2π-periodic h.
Then $\|f-s_N(f)\|_2\le\|f-h\|_2+\|h-s_N(h)\|_2+\|s_N(h-f)\|_2$ and $\|s_N(h-f)\|_2\le\|h-f\|_2$ (Theorem 14.5.9).
Hence $s_N(f)\to f$ in $\|\cdot\|_2$. Finally, $\langle s_N(f),g\rangle_2 = \sum_{|n|\le N}\widehat{f}(n)\overline{\widehat{g}(n)}$.
Let $N\to\infty$ and use Schwarz' inequality; the last identity is $g=f$.
Structural Commentary: Density upgrades Bessel to Parseval.
Key Insight: Bessel gives one inequality; density of trigonometric polynomials gives equality. □

Method 14.5.13. *Fourier Proof Patterns*

(i) *Characters*: use orthogonality of e_n to compute coefficients and finite-dimensional projections.
(ii) *Kernels*: rewrite partial sums or averages as convolutions with Dirichlet or Fejer kernels.
(iii) *Approximation*: use Fejer positivity or Stone–Weierstrass to approximate continuous periodic functions by trigonometric polynomials.
(iv) *L^2 control*: project onto finite orthonormal spans, then pass to limits by density.
(v) *Localization*: express the error as an oscillatory integral and use local regularity near the point.

Key Insight: Fourier theory is finite-dimensional orthogonal projection plus an approximation theorem.

15 Differential and Geometric Analysis

15.1 One-Variable Differentiation and Integration

One-variable calculus studies functions on intervals through derivatives, mean-value theorems, Riemann integration, and the FTC, then points toward vector-valued estimates and linear approximation.

Definition 15.1.1. *Derivative of a Real Function*
Let f be real-valued on $[a,b]$ and let $x\in[a,b]$. For $a<t<b$ and $t\ne x$, set $\phi_x(t) := \frac{f(t)-f(x)}{t-x}$.
If $\lim_{t\to x}\phi_x(t)$ exists, denote *derivative* by $f'(x)$; then f is *differentiable at* x.
If this holds for every $x\in E\subseteq[a,b]$, then f is *differentiable on* E.



Figure 146: Domain matters: isolated points give no punctured approach; $[x_0, \infty)$ gives a right-hand one.

Perspective 15.1.2. *Derivative Depends on the Domain*
For $f : X \to \mathbb{R}$, $f'(x_0)$ is the limit of $\frac{f(x)-f(x_0)}{x-x_0}$ as $x \to x_0$ in $X \setminus \{x_0\}$; see Definition 13.3.7. Hence x_0 must be a limit point of X: isolated points give no derivative, while half-interval domains give one-sided derivatives.

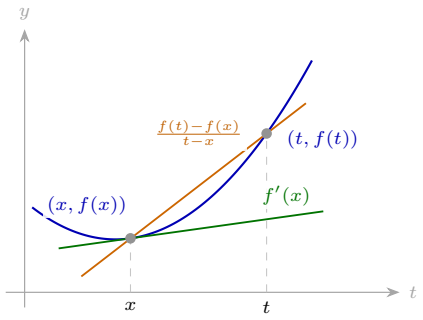


Figure 147: The derivative is the limiting slope of secant lines through $(x, f(x))$.

Proposition 15.1.3. *First-Order Approximation Form*
Let $X \subseteq \mathbb{R}$, let $x_0 \in X$ be a limit point, let $f : X \to \mathbb{R}$, and let $L \in \mathbb{R}$. Then $f'(x_0) = L$ relative to X iff $\forall \varepsilon > 0$ $\exists \delta > 0$ such that $|f(x) - f(x_0) - L(x - x_0)| \leq \varepsilon |x - x_0|$ whenever $x \in X$ and $|x - x_0| < \delta$.

Proof. For $x \neq x_0$, $\frac{|f(x)-f(x_0)-L(x-x_0)|}{|x-x_0|} = \left|\frac{f(x)-f(x_0)}{x-x_0} - L\right|$, while $x = x_0$ makes approx. estimate automatic.

($\Rightarrow$) If $f'(x_0) = L$, then the displayed right-hand side is at most ε for all $x \in X \setminus \{x_0\}$ sufficiently near x_0. Multiplying by $|x - x_0|$ gives the approximation estimate.

($\Leftarrow$) If the approximation estimate holds, then division by $|x - x_0|$ gives $\left|\frac{f(x)-f(x_0)}{x-x_0} - L\right| \leq \varepsilon$ for all $x \in X \setminus \{x_0\}$ sufficiently near x_0. Hence $f'(x_0) = L$.

Structural Commentary: Differentiability is first-order linear approximation.
Key Insight: Multiply or divide by $|x - x_0|$. □

Theorem 15.1.4. *Differentiability Implies Continuity*
If f is differentiable at x, then f is continuous at x.

Proof. For $t \neq x$, $f(t) - f(x) = \frac{f(t)-f(x)}{t-x}(t - x)$. As $t \to x$, the difference quotient tends to $f'(x)$ and $t - x \to 0$. By Theorem 13.3.12, $f(t) - f(x) \to 0$, so $f(t) \to f(x)$. Theorem 13.3.15 gives continuity at x.
Structural Commentary: A differentiable function has first-order control near the point.
Key Insight: The quotient stays near a finite number while $t - x$ shrinks to zero. □

Theorem 15.1.5. *Algebra Rules for Derivatives*
If f and g are differentiable at x, then so are $f+g$ and fg; if $g(x) \neq 0$, then f/g is differentiable at x. Moreover,

$$(f + g)'(x) = f'(x) + g'(x),$$
$$(fg)'(x) = f'(x)g(x) + f(x)g'(x),$$
$$\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - g'(x)f(x)}{g(x)^2}.$$

Proof. The sum rule is limit algebra. For $h = fg$, $h(t) - h(x) = f(t)\big(g(t) - g(x)\big) + g(x)\big(f(t) - f(x)\big)$. Dividing by $t - x$ and letting $t \to x$ gives the product rule, using Theorem 15.1.4 for $f(t) \to f(x)$.
For $h = f/g$, continuity of g makes $g(t) \neq 0$ near x, and $\frac{h(t)-h(x)}{t-x} = \frac{1}{g(t)g(x)}\left[g(x)\frac{f(t)-f(x)}{t-x} - f(x)\frac{g(t)-g(x)}{t-x}\right]$. Letting $t \to x$ gives the quotient rule.
Structural Commentary: Product and quotient rules are difference-quotient identities plus continuity.
Key Insight: Difference-quotient identities plus continuity give the algebra rules. □

Theorem 15.1.6. *Chain Rule*
Let f be continuous on $[a,b]$ with $f'(x)$ existing at $x \in [a,b]$, and let g be defined on an interval containing $f([a,b])$ and differentiable at $f(x)$. If $h(t) = g(f(t))$, then $h'(x) = g'(f(x))f'(x)$.

Proof. Let $y = f(x)$. For some $u(t) \to 0$ and $v(s) \to 0$,

$$f(t) - f(x) = (t-x)\big(f'(x) + u(t)\big),$$
$$g(s) - g(y) = (s-y)\big(g'(y) + v(s)\big).$$

With $s = f(t)$, continuity gives $s \to y$, and for $t \neq x$, $\frac{h(t)-h(x)}{t-x} = \big(g'(y) + v(f(t))\big)\big(f'(x) + u(t)\big) \to g'(y)f'(x)$.
Structural Commentary: Each differentiability statement is a linear approximation plus a vanishing error.
Key Insight: Composition multiplies the two first-order parts; the error terms disappear. □

Example 15.1.7. *Oscillation at a Point*
Assuming the usual derivative of $\sin x$, define

$$f_1(x) = \begin{cases} x\sin(1/x), & x \neq 0, \\ 0, & x = 0, \end{cases} \qquad f_2(x) = \begin{cases} x^2\sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

For f_1, the quotient at 0 is $\sin(1/t)$, so $f_1'(0)$ does not exist. For f_2, $|(f_2(t) - f_2(0))/t| \leq |t|$, so $f_2'(0) = 0$. However, for $x \neq 0$, $f_2'(x) = 2x\sin(1/x) - \cos(1/x)$, so f_2' is not continuous at 0.

Definition 15.1.8. *Local Extrema*
Let f be a real-valued function on a metric space X. The function has a *local maximum* at $p \in X$ if there is $\delta > 0$ such that $f(q) \leq f(p)$ whenever $d(p,q) < \delta$. Local minima are defined by reversing the inequality.

Remark 15.1.9. *Local and Global Extrema*
Global extrema are local, but local extrema depend on domain: isolated points are automatic extrema. Derivative tests therefore need an interior interval point; compactness supplies global extrema (Theorem 13.3.33).

Theorem 15.1.10. *Interior Extremum Has Zero Derivative*
If f has a local extremum at $x \in (a,b)$ and $f'(x)$ exists, then $f'(x) = 0$.

Proof. For a local maximum, $f(t) \leq f(x)$ near x.
Thus the quotient $(f(t) - f(x))/(t-x)$ is nonnegative for $t < x$ and nonpositive for $t > x$. Since the two-sided limit exists, $f'(x) \geq 0$ and $f'(x) \leq 0$, so $f'(x) = 0$. For a local minimum, apply the local-maximum case to $-f$. Since $(-f)'(x) = -f'(x)$, we get $-f'(x) = 0$, hence $f'(x) = 0$.
Structural Commentary: Interior extrema force opposite one-sided quotient signs.
Key Insight: A two-sided derivative compatible with both signs must be zero. □

Theorem 15.1.11. *Rolle's Theorem*
If $f : [a,b] \to \mathbb{R}$ continuous on $[a,b]$, differentiable on (a,b), and $f(a) = f(b)$, then $\exists x \in (a,b)$ s.t. $f'(x) = 0$.

Proof. If f is constant, any $x \in (a,b)$ works. Otherwise, by Theorem 13.3.33, f attains a nonendpoint extremum at some $x \in (a,b)$. Theorem 15.1.10 gives $f'(x) = 0$.
Structural Commentary: Equal endpoints force any nonconstant height change to turn around inside.
Key Insight: Compactness gives the extremum; differentiability makes the interior extremum stationary. □

Theorem 15.1.12. *Cauchy Mean Value Theorem*
Let f, g be continuous on $[a,b]$, differentiable on (a,b). $\exists\, x \in (a,b)$, $\big(f(b) - f(a)\big)g'(x) = \big(g(b) - g(a)\big)f'(x)$.

Proof. Set $h(t) := \big(f(b) - f(a)\big)g(t) - \big(g(b) - g(a)\big)f(t)$.
Then $h(a) = h(b)$, h is continuous on $[a,b]$, and h is differentiable on (a,b). By Theorem 15.1.11, $h'(x) = 0$ for some $x \in (a,b)$. Since $h'(x) = \big(f(b) - f(a)\big)g'(x) - \big(g(b) - g(a)\big)f'(x)$, the identity follows.
Structural Commentary: Build one auxiliary function whose endpoint values agree.
Key Insight: Equal endpoints force an interior horizontal tangent. □

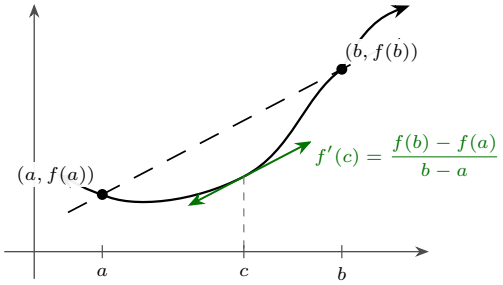


Figure 148: The Mean Value Theorem finds an interior tangent slope equal to the endpoint secant slope.

Corollary 15.1.13. *Mean Value Theorem*
If f is continuous on $[a,b]$, differentiable in (a,b), then $\exists\ x \in (a,b)$ such that $f(b)-f(a) = (b-a)f'(x)$.

Proof. Let $g(t) = t$. Then $g'(x) = 1$, $g(b) - g(a) = b - a$.
Hence Theorem 15.1.12 gives $f(b) - f(a) = (b-a)f'(x)$ for some $x \in (a,b)$.
Structural Commentary: This is Cauchy's theorem with $g(t) = t$.
Key Insight: A secant slope is achieved as a tangent slope somewhere inside. □

Corollary 15.1.14. *Bounded Derivative Gives Lipschitz Control*
If $f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$, differentiable on (a,b), and $|f'| \le M$ on (a,b), then f is M-Lipschitz on $[a,b]$, hence uniformly continuous.

Proof. For $x < y$, Corollary 15.1.13 gives $f(y) - f(x) = (y-x)f'(c)$ for some $c \in (x,y)$,
so $|f(y) - f(x)| \le M(y-x)$. Symmetry handles $x > y$, and $x = y$ is trivial.
Structural Commentary: A derivative bound controls every secant slope.
Key Insight: Local slope bounds become global Lipschitz bounds through the Mean Value Theorem. □

Theorem 15.1.15. *Derivative Sign and Monotonicity*
Suppose f is differentiable in (a,b).

(i) If $f'(x) \ge 0$ for all $x \in (a,b)$, then f is monotonically increasing.
(ii) If $f'(x) = 0$ for all $x \in (a,b)$, then f is constant.
(iii) If $f'(x) \le 0$ for all $x \in (a,b)$, then f is monotonically decreasing.

Proof. For $x_1 < x_2$ in (a,b), Corollary 15.1.13 on $[x_1,x_2]$ gives $f(x_2) - f(x_1) = (x_2 - x_1)f'(c)$ for some $c \in (x_1,x_2)$.

(i) If $f' \ge 0$, then $f(x_2) - f(x_1) \ge 0$.
(ii) If $f' = 0$, then $f(x_2) - f(x_1) = 0$.
(iii) If $f' \le 0$, then $f(x_2) - f(x_1) \le 0$.

Structural Commentary: Monotonicity is a global reading of the mean-value identity.
Key Insight: The sign of every tangent slope controls every secant slope. □

Theorem 15.1.16. *Derivatives Have the Intermediate Value Property*
If f is differentiable on $[a,b]$ in the one-sided endpoint sense and $f'(a) < \lambda < f'(b)$,
Then $f'(x) = \lambda$ for some $x \in (a,b)$. The same conclusion holds when $f'(b) < \lambda < f'(a)$.

Proof. Set $g(t) = f(t) - \lambda t$. Since $g'(a) < 0 < g'(b)$, there are $t_1, t_2 \in (a,b)$ with $g(t_1) < g(a)$ and $g(t_2) < g(b)$.
Hence the minimum of g on $[a,b]$ occurs at some $x \in (a,b)$. By Theorem 15.1.10, $0 = g'(x) = f'(x) - \lambda$.
Apply the same argument to $-f$ for the reversed case.
Structural Commentary: A derivative need not be continuous, but it cannot jump over values.
Key Insight: Tilt the graph by subtracting λt and force an interior extremum. □

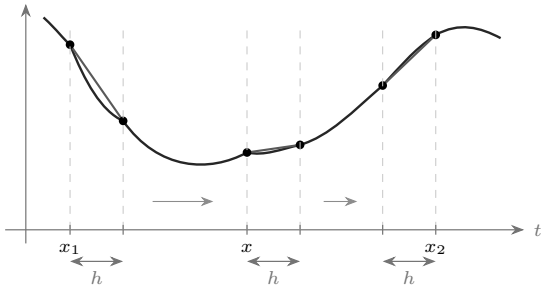


Figure 149: Sliding fixed-width secants in Darboux's theorem.

Perspective 15.1.17. *Sliding Secants*
For fixed small $h > 0$, the secant-slope function $\sigma_h(x) := \frac{f(x+h)-f(x)}{h}$ is continuous. Thus Theorem 13.3.40 finds intermediate secant slopes, and Corollary 15.1.13 converts them to derivative values.
Key Insight: Slide secants, then apply the Mean Value Theorem.

Corollary 15.1.18. *Derivatives Have No Simple Discontinuities*
If f is differentiable on $[a,b]$, then f' cannot have a simple discontinuity on $[a,b]$.

Proof. Suppose a simple discontinuity occurs at an interior point c; endpoint cases are one-sided.
If $f'(c-) \neq f'(c+)$, choose λ strictly between these limits, $\lambda \neq f'(c)$. For $x < c < y$ close to c, Theorem 15.1.16 gives $z \in (x,y)$ with $f'(z) = \lambda$, contradicting the one-sided limits. If $f'(c-) = f'(c+) = L \neq f'(c)$, choose λ between L and $f'(c)$ and apply the same theorem on a sufficiently small one-sided interval with endpoint c.
Structural Commentary: Darboux behavior rules out jump discontinuities for derivatives.
Key Insight: Derivatives may oscillate, but they do not jump. □

Theorem 15.1.19. *Derivative of an Inverse Function*
Let $f : I \to \mathbb{R}$ be one-to-one on an open interval I, let $x_0 \in I$, and put $y_0 = f(x_0)$. Suppose f is differentiable at x_0, $f'(x_0) \neq 0$, and $g = f^{-1} : f(I) \to I$ is continuous at y_0. Then g is differentiable at y_0 and $g'(y_0) = \frac{1}{f'(x_0)}$.

Proof. For $y \in f(I)$ with $y \neq y_0$, write $x = g(y)$. Then $x \neq x_0$ and $\frac{g(y)-g(y_0)}{y-y_0} = \frac{x-x_0}{f(x)-f(x_0)} = \left(\frac{f(x)-f(x_0)}{x-x_0}\right)^{-1}$.
As $y \to y_0$, the assumed continuity of g gives $x = g(y) \to x_0$.
The difference quotient for f tends to $f'(x_0) \neq 0$, so the reciprocal tends to $1/f'(x_0)$ by Theorem 13.3.12.
Structural Commentary: Matched inverse quotients are reciprocal.
Key Insight: Continuity of the inverse converts the limit $y \to y_0$ back into $x \to x_0$. □

Corollary 15.1.20. *One-Dimensional Inverse Function Theorem*
If $f : (a,b) \to (c,d)$ is a differentiable bijection and $f' \neq 0$,
then f is a homeomorphism and f^{-1} is differentiable, with $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$.

Proof. By Theorem 15.1.16, f' has constant sign. Corollary 15.1.13 makes f strictly monotone, hence its continuous interval inverse is continuous. Apply Theorem 15.1.19 pointwise.
Structural Commentary: Darboux gives constant sign; the inverse derivative formula is local.
Key Insight: Nonvanishing derivative is the one-dimensional obstruction to differentiating the inverse. □

Theorem 15.1.21. *L'Hospital's Rule*
Let f, g be differentiable on (a,b), $g' \neq 0$, and $-\infty \leq a < b \leq +\infty$.
If $\frac{f'}{g'} \to A \in \overline{\mathbb{R}}$ as $x \to a$, and either $f, g \to 0$ or $g \to +\infty$, then $\frac{f}{g} \to A$ as $x \to a$.
The same holds as $x \to b$, and for $g \to -\infty$.

Proof. For $a < x < y < b$, Theorem 15.1.12 gives $t \in (x,y)$ with $\frac{f(x)-f(y)}{g(x)-g(y)} = \frac{f'(t)}{g'(t)}$.
If $f, g \to 0$, let $x \to a$ with y fixed. If $g \to +\infty$, use $\frac{f(x)}{g(x)} = \frac{f(x)-f(y)}{g(x)-g(y)}\frac{g(x)-g(y)}{g(x)} + \frac{f(y)}{g(x)}$ and let $x \to a$.
In both cases, bounds on f'/g' near a transfer to f/g, forcing the limit.
Structural Commentary: CMVT transfers derivative bounds to quotient bounds.
Key Insight: Derivative quotients govern the original indeterminate quotient. □

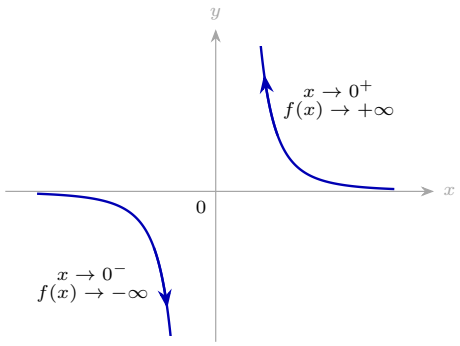


Figure 150: For $f(x) = x^{-3}$, the one-sided limits at 0 are opposite infinities.

Definition 15.1.22. *Higher Derivatives*
If f has derivative f' on an interval and f' is differentiable, then $(f')'$ is denoted f'' and called the *second derivative*. Continuing recursively gives $f^{(n)}$, the derivative of order n.

Definition 15.1.23. *Smoothness Classes*
On an interval, C^0 means continuous, C^r means $f^{(r)}$ exists and continuous, C^∞ means C^r for every finite r. Analytic functions (Definition 14.4.7) form the class C^ω.

Perspective 15.1.24. *Regularity Hierarchy*
The inclusions are $C^\omega \subseteq C^\infty \subseteq \cdots \subseteq C^2 \subseteq C^1 \subseteq C^0$.
The inclusions follow from Theorem 14.4.8 and Definition 15.1.23; they are generally proper.
Key Insight: Higher regularity is extra structure, not just a different notation for continuity.

Theorem 15.1.25. *Higher-Order Leibniz Product Rule*
If $f, g : (a,b) \to \mathbb{R}$ are r times differentiable, $r \geq 1$, then $(fg)^{(r)}(x) = \sum_{k=0}^{r} \binom{r}{k} f^{(k)}(x) g^{(r-k)}(x)$.

Proof. Induct on r. The case $r = 1$ is Theorem 15.1.5.
Differentiating the order-r formula gives $(fg)^{(r+1)} = \sum_{k=0}^{r} \binom{r}{k} f^{(k+1)} g^{(r-k)} + \sum_{k=0}^{r} \binom{r}{k} f^{(k)} g^{(r-k+1)}$.
Reindex and use Pascal's identity.
Structural Commentary: Each derivative chooses whether it lands on f or on g.
Key Insight: The binomial coefficients count those choices. □

Theorem 15.1.26. *Taylor's Theorem with Lagrange Remainder*
Let $f : [a,b] \to \mathbb{R}$, let $n \geq 1$, suppose $f^{(n-1)}$ is continuous on $[a,b]$, and suppose $f^{(n)}$ exists on (a,b).
For distinct $\alpha, \beta \in [a,b]$, set $P(t) := \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!}(t-\alpha)^k$.
Then for some x between α and β, $f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!}(\beta-\alpha)^n$.

Proof. Choose M so that $f(\beta) = P(\beta) + M(\beta-\alpha)^n$, and set $G(t) := f(t) - P(t) - M(t-\alpha)^n$.
Then $G(\alpha) = G'(\alpha) = \cdots = G^{(n-1)}(\alpha) = G(\beta) = 0$. Repeated applications of Corollary 15.1.13 give $G^{(n)}(x) = 0$ for some x between α and β. Since $G^{(n)}(x) = f^{(n)}(x) - n!M$, the formula follows.
Structural Commentary: The auxiliary error vanishes at β and to order $n-1$ at α.
Key Insight: Repeated Rolle-style arguments turn endpoint cancellation into a precise remainder. □

Example 15.1.27. *Smooth but Not Analytic*
Define $e(x) = \begin{cases} e^{-1/x}, & x > 0, \\ 0, & x \leq 0. \end{cases}$. By repeated L'Hospital, Theorem 15.1.21, $e \in C^\infty$ and $e^{(r)}(0) = 0$ for all r.
Its Taylor series at 0 is zero, but $e(x) > 0$ for $x > 0$, so e is not analytic at 0 (Definition 14.4.7).
Key Insight: A function can be infinitely tangent to the x-axis without being locally equal to its Taylor series.

Method 15.1.28. *Newton and Secant Iterations*
To solve $f(x) = 0$:

(i) Newton uses tangent-line zero: $x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$, $n \geq 1$ when $f'(x_{n-1}) \neq 0$.

(ii) Secant uses two-point slope: $x_n = x_{n-1} - \frac{f(x_{n-1})(x_{n-2}-x_{n-1})}{f(x_{n-2})-f(x_{n-1})}$, $n \geq 2$ when the denominator is nonzero.

These define iterates only; local convergence is proved with Taylor's theorem (Theorem 15.1.26).



Figure 151: Newton: tangent-line zero. Secant: two-point line zero.

Definition 15.1.29. *Vector-Valued Derivative*
For $\mathbf{f} : [a, b] \to \mathbb{R}^k$, set $\mathbf{f}'(x) = \mathbf{L}$ when $\lim_{t \to x} \left\| \frac{\mathbf{f}(t) - \mathbf{f}(x)}{t - x} - \mathbf{L} \right\|_2 = 0$.
For $\mathbf{f} = (f_1, \ldots, f_k)$, this is equivalent to componentwise differentiability, with $\mathbf{f}'(x) = (f_1'(x), \ldots, f_k'(x))$.

Theorem 15.1.30. *Vector-Valued Mean Value Inequality*
If $\mathbf{f} : [a, b] \to \mathbb{R}^k$ is continuous on $[a, b]$ and differentiable on (a, b).
Then for some $x \in (a, b)$, $\|\mathbf{f}(b) - \mathbf{f}(a)\|_2 \leq (b - a) \|\mathbf{f}'(x)\|_2$.

Proof. Set $\mathbf{z} := \mathbf{f}(b) - \mathbf{f}(a)$ and $\varphi(t) := \mathbf{z} \cdot \mathbf{f}(t)$.
Corollary 15.1.13 gives $\varphi(b) - \varphi(a) = (b-a)\varphi'(x) = (b-a)\mathbf{z} \cdot \mathbf{f}'(x)$ for some $x \in (a, b)$. Since $\varphi(b) - \varphi(a) = \|\mathbf{z}\|_2^2$, Theorem 12.2.39 gives $\|\mathbf{z}\|_2^2 \leq (b - a) \|\mathbf{z}\|_2 \|\mathbf{f}'(x)\|_2$, and the result follows by cancellation, with $\mathbf{z} = 0$ trivial.
Structural Commentary: A scalar projection yields the vector norm bound.
Key Insight: Project onto the displacement vector, then use Schwarz to return to norms. □

Method 15.1.31. *Differentiation Proof Patterns*

(i) *Domain*: require x_0 to be a limit point of X; isolated points have no derivative by Perspective 15.1.2.
(ii) *Definition*: factor $f(t) - f(x) = (t - x)Q_x(t)$ and take $Q_x(t) \to f'(x)$.
(iii) *Approximation*: use Proposition 15.1.3 to replace quotient limits by first-order error bounds.
(iv) *Algebra*: prove quotient identities first; use continuity to pass to limits.
(v) *Mean values*: choose h with equal endpoint values and apply Theorem 15.1.10.
(vi) *Taylor*: set $G = f - P - M(t - \alpha)^n$ and iterate the mean value theorem.

Key Insight: Difference quotients and auxiliary functions drive one-variable differentiation.

The integration side of one-variable calculus is organized by Riemann and Riemann–Stieltjes sums; the fundamental theorem connects the two halves.

This subsection develops Riemann and Riemann–Stieltjes integration by partitions, then links integrals to antiderivatives through the Fundamental Theorem of Calculus. Point partitions and interval-envelope partitions are equivalent via common refinement.

Definition 15.1.32. *Partitions and Riemann Upper/Lower Sums*
A *partition* of $[a, b]$ is $\Pi = \{x_0, \ldots, x_n\}$ with $a = x_0 \leq x_1 \leq \cdots \leq x_n = b$, $\Delta x_i := x_i - x_{i-1}$.
For bounded real f on $[a, b]$, set $M_i := \sup_{x_{i-1} \leq x \leq x_i} f(x)$, $m_i := \inf_{x_{i-1} \leq x \leq x_i} f(x)$,
and $U(\Pi, f) := \sum_{i=1}^{n} M_i \Delta x_i$, $L(\Pi, f) := \sum_{i=1}^{n} m_i \Delta x_i$.
Define $\overline{\int_a^b} f(x)\,dx := \inf_\Pi U(\Pi, f)$, $\underline{\int_a^b} f(x)\,dx := \sup_\Pi L(\Pi, f)$, with Π ranging over all partitions.
If these agree, f is *Riemann integrable*, written $f \in \mathcal{R}$, and the common value is $\int_a^b f(x)\,dx$ or $\int_a^b f$.
For $m \leq f \leq M$, $m(b - a) \leq L(\Pi, f) \leq U(\Pi, f) \leq M(b - a)$, upper and lower integrals exist for bounded f.

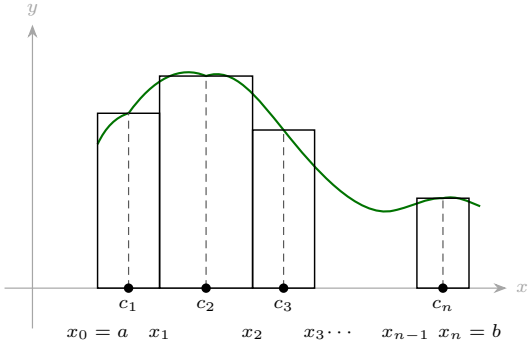


Figure 152: A tagged Riemann sum uses one sample value $f(c_i)$ on each subinterval $[x_{i-1}, x_i]$.

Definition 15.1.33. *Tagged Partitions and Riemann Sums*
A *tagged partition* of $[a,b]$ is a partition $\Pi = \{x_0, \dots, x_n\}$ together with tags $c_i \in [x_{i-1}, x_i]$.
For the ordinary Riemann integral, the corresponding *Riemann sum* is $R(f, \Pi, \{c_i\}) := \sum_{i=1}^{n} f(c_i)\Delta x_i$.
For every tagged partition, $L(\Pi, f) \le R(f, \Pi, \{c_i\}) \le U(\Pi, f)$.

Perspective 15.1.34. *Tagged Sums and Gauges*
Gauge integration keeps tagged sums but replaces one mesh bound δ with a local bound $\delta(c_i)$.
Thus fineness may vary by tag, giving a stronger framework for FTC-type results.

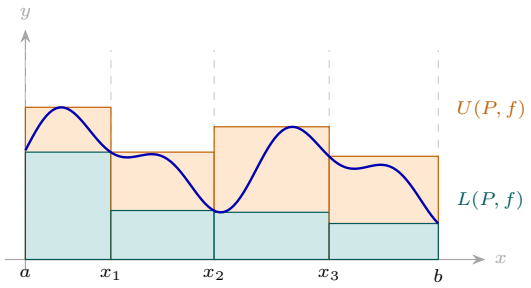


Figure 153: Upper and lower sums record the oscillation of f on each partition interval.

Definition 15.1.35. *Riemann–Stieltjes Integral*
Let α be increasing on $[a,b]$. For $\Pi = \{x_0, \dots, x_n\}$, put $\Delta\alpha_i := \alpha(x_i) - \alpha(x_{i-1}) \ge 0$.
For bounded real f, with M_i, m_i as in Definition 15.1.32, set $U(\Pi, f, \alpha) := \sum_{i=1}^{n} M_i \Delta\alpha_i$; $L(\Pi, f, \alpha) := \sum_{i=1}^{n} m_i \Delta\alpha_i$.
Define $\overline{\int_a^b f\, d\alpha} := \inf_{\Pi} U(\Pi, f, \alpha)$; $\underline{\int_a^b f\, d\alpha} := \sup_{\Pi} L(\Pi, f, \alpha)$. If value match, f is *Riemann integrable with respect to* α, written $f \in \mathcal{R}(\alpha)$, and the common value is $\int_a^b f\, d\alpha$, also written $\int_a^b f(x)\, d\alpha(x)$.
Taking $\alpha(x) = x$ gives the ordinary Riemann integral; the variable of integration is dummy.

Remark 15.1.36. *Stieltjes Increments as α-Lengths*
Stieltjes sums replace ordinary length Δx_i by $\Delta\alpha_i$, the length measured by α.
Thus $\alpha(x) = x$ gives Riemann sums, while jumps of α place point mass as in Theorem 15.1.53.

Definition 15.1.37. *Refinements*
A partition Π^* is a *refinement* of Π if $\Pi^* \supseteq \Pi$. Given two partitions Π_1, Π_2, their *common refinement* is $\Pi_1 \cup \Pi_2$.

Theorem 15.1.38. *Refinement Monotonicity*
If Π^* is a refinement of Π, then $L(\Pi, f, \alpha) \le L(\Pi^*, f, \alpha)$ and $U(\Pi^*, f, \alpha) \le U(\Pi, f, \alpha)$.

Proof. It suffices to add one point $x^* \in (x_{i-1}, x_i)$. If w_1, w_2 are infima on $[x_{i-1}, x^*]$ and $[x^*, x_i]$, then $w_1, w_2 \ge m_i$ and $L(\Pi^*, f, \alpha) - L(\Pi, f, \alpha) = (w_1 - m_i)\big[\alpha(x^*) - \alpha(x_{i-1})\big] + (w_2 - m_i)\big[\alpha(x_i) - \alpha(x^*)\big] \ge 0$.
Iterate for additional points. The upper-sum inequality is the same argument with suprema.
Structural Commentary: Refinement raises lower sums and lowers upper sums.
Key Insight: Splitting an interval tightens the local oscillation bounds. □

Perspective 15.1.39. *Piecewise-Constant Envelopes*
A partition Π gives lower and upper step envelopes with values m_i and M_i; their integrals are exactly $L(\Pi, f)$ and $U(\Pi, f)$. Common refinements make these envelope sums compatible across partitions.

Theorem 15.1.40. *Lower Integral Below Upper Integral*
For every bounded real f and increasing α on $[a,b]$, $\underline{\int_a^b f\, d\alpha} \le \overline{\int_a^b f\, d\alpha}$.

Proof. Given partitions Π_1, Π_2, let $\Pi^* = \Pi_1 \cup \Pi_2$.
By Theorem 15.1.38, $L(\Pi_1, f, \alpha) \le L(\Pi^*, f, \alpha) \le U(\Pi^*, f, \alpha) \le U(\Pi_2, f, \alpha)$.
Take the supremum over Π_1 and then the infimum over Π_2.
Structural Commentary: Common refinements compare sums from different partitions.
Key Insight: The lower envelope never rises above the upper envelope. □

Theorem 15.1.41. *Upper–Lower Criterion*
A bounded real f is in $\mathcal{R}(\alpha)$ on $[a,b]$ if and only if $\forall \varepsilon > 0\ \exists$ a partition Π with $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$.

Proof.

($\Rightarrow$) If $f \in \mathcal{R}(\alpha)$, choose Π_1, Π_2 whose lower and upper sums are within $\varepsilon/2$ of $\int_a^b f\,d\alpha$.
Their common refinement Π satisfies the displayed inequality by Theorem 15.1.38.

($\Leftarrow$) If such Π exist, then $L(\Pi, f, \alpha) \le \underline{\int_a^b f\,d\alpha} \le \overline{\int_a^b f\,d\alpha} \le U(\Pi, f, \alpha)$ forces upper and lower integrals to agree.

Structural Commentary: Integrability is exactly the ability to make one partition squeeze the oscillation.
Key Insight: The gap $U - L$ is the quantitative defect of integrability. □

Perspective 15.1.42. *Darboux Gaps and Riemann Sums*
For $\alpha(x) = x$, a small Darboux gap $U(\Pi, f) - L(\Pi, f)$ traps every tagged sum for Π near the integral. This is the local reason the Darboux definition above and the tagged Riemann-sum definition give the same number. The same estimate works with Stieltjes weights $\Delta\alpha_i$ whenever the upper–lower gap is small.

Theorem 15.1.43. *Tagged Sums*
Let $\Pi = \{x_0, \dots, x_n\}$ satisfy $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$, and choose tags $t_i \in [x_{i-1}, x_i]$.

(i) If $s_i, t_i \in [x_{i-1}, x_i]$, then $\sum_{i=1}^n |f(s_i) - f(t_i)|\,\Delta\alpha_i < \varepsilon$.

(ii) If $f \in \mathcal{R}(\alpha)$, then $\left|\sum_{i=1}^n f(t_i)\Delta\alpha_i - \int_a^b f\,d\alpha\right| < \varepsilon$.

Proof.

(i) Since $|f(s_i) - f(t_i)| \le M_i - m_i$, summing gives $\sum_{i=1}^n |f(s_i) - f(t_i)|\,\Delta\alpha_i \le U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$.

(ii) The tagged sum and $\int_a^b f\,d\alpha$ both lie between $L(\Pi, f, \alpha)$ and $U(\Pi, f, \alpha)$, so their distance is $< \varepsilon$.

Structural Commentary: A small Darboux gap makes tags and the integral close.
Key Insight: Tags are harmless after oscillation has been controlled. □

Theorem 15.1.44. *Continuous Functions Are Riemann–Stieltjes Integrable*
If f is continuous on $[a, b]$ and α is increasing on $[a, b]$, then $f \in \mathcal{R}(\alpha)$.

Proof. If $\alpha(b) = \alpha(a)$, the Darboux gap is zero.
Otherwise choose $\eta > 0$ with $\eta[\alpha(b) - \alpha(a)] < \varepsilon$. Uniform continuity gives partition Π with $M_i - m_i \le \eta\ \forall i$, so $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) = \sum_{i=1}^n (M_i - m_i)\Delta\alpha_i \le \eta[\alpha(b) - \alpha(a)] < \varepsilon$. Apply Theorem 15.1.41.
Structural Commentary: Uniform continuity controls oscillation uniformly across the partition.
Key Insight: Continuity turns local small intervals into a small total Darboux gap. □

Theorem 15.1.45. *Monotone Functions and Continuous Integrators*
If f is monotone on $[a, b]$ and α is continuous and increasing on $[a, b]$, then $f \in \mathcal{R}(\alpha)$.

Proof. Assume first that f is increasing. If $\alpha(b) = \alpha(a)$, then every $\Delta\alpha_i = 0$, so $U(\Pi, f, \alpha) = L(\Pi, f, \alpha) = 0$ for every partition Π. Choose n and a partition Π such that $\Delta\alpha_i = [\alpha(b) - \alpha(a)]/n$ for all i.
Existence follows from continuity of α and Theorem 13.3.40.
Then $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) = \frac{\alpha(b)-\alpha(a)}{n} \sum_{i=1}^n \big(f(x_i) - f(x_{i-1})\big) = \frac{\alpha(b)-\alpha(a)}{n}\big(f(b) - f(a)\big) < \varepsilon$ for n large.
If f is decreasing, use the same partition. Then the suprema and infima on $[x_{i-1}, x_i]$ are $f(x_{i-1})$ and $f(x_i)$, so the same computation gives $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) = \frac{\alpha(b)-\alpha(a)}{n}\big(f(a) - f(b)\big)$, which is $< \varepsilon$ for large n.
Structural Commentary: Equal α-increments convert the monotone oscillations into a telescoping sum.
Key Insight: A monotone function's total oscillation is finite and can be spread thinly. □

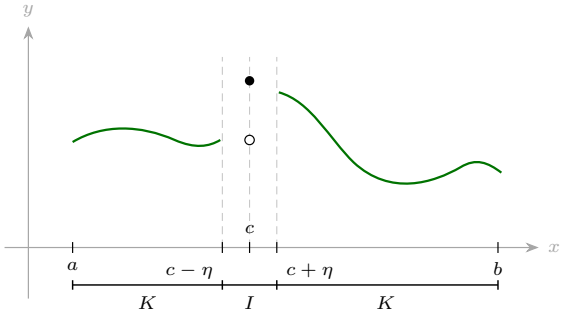


Figure 154: Isolate the discontinuity in a short interval; control the compact pieces outside it.

Theorem 15.1.46. *Finitely Many Discontinuities*
Suppose f is bounded on $[a,b]$, the discontinuity set $E := \{x \in [a,b] : f \text{ is discontinuous at } x\}$ is finite, and α is continuous at each point of E. Then $f \in \mathcal{R}(\alpha)$.

Proof. Let $M = \sup_{[a,b]} |f|$ and let $\eta > 0$. By continuity of α on the finite set E, choose disjoint intervals $I_j = [u_j, v_j]$ covering E, with each point of $E \cap (a,b)$ in some (u_j, v_j), such that $\sum_j \big(\alpha(v_j) - \alpha(u_j)\big) < \eta$.
Set $K = [a,b] \setminus \bigcup_j (u_j, v_j)$. Then $f|_K$ is uniformly continuous by Theorem 13.3.37.
Choose a partition Π containing all u_j, v_j, with no points inside the intervals (u_j, v_j), and fine enough on K that $M_i - m_i \le \eta$ whenever $[x_{i-1}, x_i] \subset K$.
Hence $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) \le \eta\big[\alpha(b) - \alpha(a)\big] + 2M \sum_j \big(\alpha(v_j) - \alpha(u_j)\big) < \eta\big[\alpha(b) - \alpha(a) + 2M\big]$.
Since η is arbitrary, Theorem 15.1.41 applies.
Structural Commentary: Isolate bad points in small α-intervals; use uniform continuity elsewhere.
Key Insight: Finitely many discontinuities do not matter unless α also jumps there. □

Example 15.1.47. *Dirichlet's Function Is Not Riemann Integrable*
Let $g = \mathbf{1}_{\mathbb{Q}}$ on $[0,1]$. Every nondegenerate subinterval contains rationals and irrationals by Corollary 12.2.26, so for every partition Π, $M_i = 1$ and $m_i = 0$ on each nonzero-width subinterval.
Thus $U(\Pi, g) = 1$ and $L(\Pi, g) = 0$, hence $g \notin \mathcal{R}$.

Remark 15.1.48. *Small Discontinuity Sets*
The same idea underlies Theorem 16.3.7: small discontinuity sets, later called measure-zero sets in Definition 16.1.18, do not obstruct integrability. Dirichlet's function fails because every point is discontinuous.

Theorem 15.1.49. *Continuous Composition Preserves Integrability*
Let α be increasing on $[a,b]$. Suppose $f \in \mathcal{R}(\alpha)$, $m \le f(x) \le M$ on $[a,b]$, and $\phi : [m,M] \to \mathbb{R}$ is continuous. Define $h : [a,b] \to \mathbb{R}$ by $h(x) := \phi(f(x))$. Then $h \in \mathcal{R}(\alpha)$.

Proof. Let $C := \sup_{y \in [m,M]} |\phi(y)|$ and let $\varepsilon > 0$. Choose $\eta > 0$ such that $\eta[\alpha(b) - \alpha(a) + 2C] < \varepsilon$. Since ϕ is uniformly continuous on $[m,M]$, choose $\gamma > 0$ such that $|s - t| < \gamma \implies |\phi(s) - \phi(t)| < \eta$ $(s, t \in [m,M])$.
Choose a partition $\Pi = \{x_0, \dots, x_n\}$ with $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \gamma\eta$.
Let M_i, m_i denote upper and lower values for f on $[x_{i-1}, x_i]$, M_i^h, m_i^h denote corresponding values for h.
Split $A := \{i : M_i - m_i < \gamma\}$, $B := \{i : M_i - m_i \ge \gamma\}$. For $i \in A$, $M_i^h - m_i^h \le \eta$; for $i \in B$, $M_i^h - m_i^h \le 2C$.
Also $\gamma \sum_{i \in B} \Delta\alpha_i \le \sum_{i \in B} (M_i - m_i)\Delta\alpha_i \le U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \gamma\eta$, so $\sum_{i \in B} \Delta\alpha_i < \eta$.
$U(\Pi, h, \alpha) - L(\Pi, h, \alpha) \le \eta \sum_{i \in A} \Delta\alpha_i + 2C \sum_{i \in B} \Delta\alpha_i < \eta[\alpha(b) - \alpha(a) + 2C] < \varepsilon$. Theorem 15.1.41 gives $h \in \mathcal{R}(\alpha)$.
Structural Commentary: Integrability survives continuous remeasurement of the values of f.
Key Insight: Large oscillation intervals have small total α-weight. □

Theorem 15.1.50. *Basic Properties of the Integral*
Let all functions be bounded and real on $[a,b]$, and let all integrators be increasing.

(i) If $f_1, f_2 \in \mathcal{R}(\alpha)$, $c \in \mathbb{R}$, then $f_1 + f_2, cf_1 \in \mathcal{R}(\alpha)$, and $\int_a^b (f_1 + f_2)\,d\alpha = \int_a^b f_1\,d\alpha + \int_a^b f_2\,d\alpha$; $\int_a^b cf_1\,d\alpha = c\int_a^b f_1\,d\alpha$.

(ii) If $f_1 \le f_2$ and both functions are in $\mathcal{R}(\alpha)$, then $\int_a^b f_1\,d\alpha \le \int_a^b f_2\,d\alpha$.

(iii) If $f \in \mathcal{R}(\alpha)$ and $a < c < b$, then f is integrable on $[a,c]$ and $[c,b]$, and $\int_a^c f\,d\alpha + \int_c^b f\,d\alpha = \int_a^b f\,d\alpha$.

(iv) If $f \in \mathcal{R}(\alpha)$ and $|f| \le M$, then $\left|\int_a^b f\,d\alpha\right| \le M\big[\alpha(b) - \alpha(a)\big]$

(v) If $f \in \mathcal{R}(\alpha_1)$ and $f \in \mathcal{R}(\alpha_2)$, then $f \in \mathcal{R}(\alpha_1 + \alpha_2)$ and $\int_a^b f\,d(\alpha_1 + \alpha_2) = \int_a^b f\,d\alpha_1 + \int_a^b f\,d\alpha_2$. If $c > 0$ and $f \in \mathcal{R}(\alpha)$, then $f \in \mathcal{R}(c\alpha)$ and $\int_a^b f\,d(c\alpha) = c\int_a^b f\,d\alpha$.

Proof.

(i) $L(\Pi, f_1, \alpha) + L(\Pi, f_2, \alpha) \le L(\Pi, f_1 + f_2, \alpha) \le U(\Pi, f_1 + f_2, \alpha) \le U(\Pi, f_1, \alpha) + U(\Pi, f_2, \alpha)$, where Π is common refinement, and Theorem 15.1.41; scalar multiplication is the same estimate with signs handled by swapping upper and lower sums when $c < 0$.
(ii) If $f_1 \le f_2$, then $L(\Pi, f_2 - f_1, \alpha) \ge 0$ for every Π; apply (i) to $f_2 - f_1$.
(iii) Refine so that $c \in \Pi$. Then the lower and upper sums split into the sums over $[a,c]$ and $[c,b]$.
(iv) Apply (ii) to $-M \le f \le M$ and use the integrals of the constant functions $\pm M$.
(v) Since $\Delta(\alpha_1 + \alpha_2)_i = \Delta\alpha_{1,i} + \Delta\alpha_{2,i}$ and $\Delta(c\alpha)_i = c\,\Delta\alpha_i$, corresponding lower and upper sums split exactly.

Structural Commentary: The algebra of integrals is inherited from the algebra of upper and lower sums.
Key Insight: Common refinements let several integrable functions be approximated by the same partition. □

Theorem 15.1.51. *Products and Absolute Values*
If $f, g \in \mathcal{R}(\alpha)$ on $[a,b]$, then $fg \in \mathcal{R}(\alpha)$, $|f| \in \mathcal{R}(\alpha)$, and $\left|\int_a^b f\,d\alpha\right| \le \int_a^b |f|\,d\alpha$.

Proof. By Theorem 15.1.50, $f \pm g \in \mathcal{R}(\alpha)$. Applying Theorem 15.1.49 to $\phi(t) = t^2$ gives $(f \pm g)^2 \in \mathcal{R}(\alpha)$, hence $fg \in \mathcal{R}(\alpha)$ from $4fg = (f+g)^2 - (f-g)^2$.
Applying the same theorem to $\phi(t) = |t|$ gives $|f| \in \mathcal{R}(\alpha)$, and $-|f| \le f \le |f|$ gives the inequality.
Structural Commentary: Products reduce to squares; absolute values to composition.
Key Insight: Once continuous composition is available, the usual algebra closes. □

Definition 15.1.52. *Unit Step Function*
The *unit step function* I is $I(x) = \begin{cases} 0, & x \le 0, \\ 1, & x > 0. \end{cases}$.

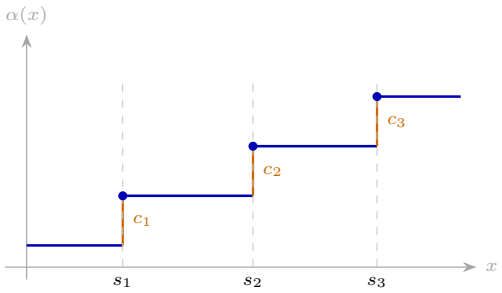


Figure 155: A step integrator concentrates mass at jumps: $\int f\,d\alpha = \sum_i c_i f(s_i)$.

Theorem 15.1.53. *Single Jump Integrator*
If $a < s < b$, f is bounded on $[a,b]$ and continuous at s, and $\alpha(x) = I(x-s)$, then $\int_a^b f\,d\alpha = f(s)$.

Proof. Use partitions $\Pi = \{a, s, x_2, b\}$ with $s < x_2 < b$. The only nonzero $\Delta\alpha_i$ occurs on $[s, x_2]$, so $L(\Pi, f, \alpha)$ and $U(\Pi, f, \alpha)$ are the infimum and supremum of f on $[s, x_2]$. As $x_2 \downarrow s$, continuity at s squeezes both to $f(s)$.
Structural Commentary: A jump of α makes the integral sample f at the jump.
Key Insight: Stieltjes integration includes point masses without changing the Darboux framework. □

Theorem 15.1.54. *Pure Step Integrators Give Series*
Suppose $c_n \ge 0$, $\sum c_n$ converges, and (s_n) is a sequence of distinct points in (a,b).
Define $\alpha(x) := \sum_{n=1}^{\infty} c_n I(x - s_n)$. If f is continuous on $[a,b]$, then $\int_a^b f\,d\alpha = \sum_{n=1}^{\infty} c_n f(s_n)$.

Proof. By comparison, α is increasing. For $N \in \mathbb{N}$, put $\alpha_N(x) := \sum_{n=1}^{N} c_n I(x - s_n)$ and $\beta_N := \alpha - \alpha_N$..

Theorems 15.1.53 and 15.1.50 give $\int_a^b f\,d\alpha_N = \sum_{n=1}^{N} c_n f(s_n)$. Also $f \in \mathcal{R}(\beta_N)$ by Theorem 15.1.44.

If $|f| \le M$, then Theorem 15.1.50 gives $\left|\int_a^b f\,d\beta_N\right| \le M[\beta_N(b) - \beta_N(a)] = M \sum_{n>N} c_n$.

Letting $N \to \infty$ and using additivity in the integrator gives the formula.
Structural Commentary: Finite step integrators sum point samples; summable tails vanish by the integral bound.
Key Insight: A Stieltjes integral can be an ordinary series in disguise. □

Theorem 15.1.55. *Differentiable Integrators Reduce to Riemann Integrals*
Assume α increases monotonically, $\alpha' \in \mathcal{R}$ on $[a,b]$, and f is bounded and real on $[a,b]$.
Then $f \in \mathcal{R}(\alpha) \iff f\alpha' \in \mathcal{R}$, and in that case $\int_a^b f\,d\alpha = \int_a^b f(x)\alpha'(x)\,dx$.

Proof. Let $M = \sup_{[a,b]} |f|$. Choose Π with $U(\Pi, \alpha') - L(\Pi, \alpha') < \varepsilon$.
For any refinement and any tags $s_i \in [x_{i-1}, x_i]$, MVT gives points $t_i \in [x_{i-1}, x_i]$ with $\Delta\alpha_i = \alpha'(t_i)\Delta x_i$.
Hence $|\sum_i f(s_i)\Delta\alpha_i - \sum_i f(s_i)\alpha'(s_i)\Delta x_i| \le M \sum_i |\alpha'(t_i) - \alpha'(s_i)|\,\Delta x_i \le M\varepsilon$. Thus every tagged Stieltjes sum for $f\,d\alpha$ is within $M\varepsilon$ of the corresponding tagged Riemann sum for $f\alpha'\,dx$, and conversely. By the tagged-sum criterion, one family has a limit exactly when the other does, and the two limits are equal.
Structural Commentary: The increment $d\alpha$ is replaced by $\alpha'(x)\,dx$ when α' is Riemann integrable.
Key Insight: The mean value theorem turns Stieltjes increments into ordinary Riemann weights. □

Theorem 15.1.56. *Change of Variable*
Let $\varphi : [A, B] \to [a, b]$ be strictly increasing, continuous, and onto.
Let α be increasing on $[a, b]$, $f \in \mathcal{R}(\alpha)$, and set $\beta(y) := \alpha(\varphi(y))$ and $g(y) := f(\varphi(y))$.
Then $g \in \mathcal{R}(\beta)$ and $\int_A^B g\, d\beta = \int_a^b f\, d\alpha$. If also $\alpha(x) = x$ and $\varphi' \in \mathcal{R}$, then $\int_a^b f(x)\, dx = \int_A^B f(\varphi(y))\varphi'(y)\, dy$.

Proof. Partitions $Q = \{y_i\}$ of $[A, B]$ correspond to partitions $\Pi = \{\varphi(y_i)\}$ of $[a, b]$. On corresponding intervals, g has same range as f and $\Delta\beta_i = \Delta\alpha_i$, so $U(Q, g, \beta) = U(\Pi, f, \alpha)$ and $L(Q, g, \beta) = L(\Pi, f, \alpha)$. The upper–lower criterion gives the Stieltjes formula. If $\alpha(x) = x$, then $\beta = \varphi$; apply Theorem 15.1.55 to the left side.
Structural Commentary: A monotone change of variable relabels the same partition intervals.
Key Insight: Stieltjes change of variable is equality of Darboux sums under reparametrization. □

Theorem 15.1.57. *Integral Functions*
Let $f \in \mathcal{R}$ on $[a, b]$ and define $F(x) := \int_a^x f(t)\, dt$. Then F is continuous.
If f is continuous at x_0, then $F'(x_0) = f(x_0)$, with one-sided derivatives at endpoints.

Proof. Let $|f| \le M$. For $x < y$, Theorem 15.1.50 gives $|F(y) - F(x)| = \left|\int_x^y f(t)\, dt\right| \le M(y - x)$ so F continuous.
If f continuous at x_0, for $x \ne x_0$ near x_0, $\left|\frac{F(x)-F(x_0)}{x-x_0} - f(x_0)\right| \le \sup_{u\in[x,x_0]} |f(u) - f(x_0)| \to 0$. $F'(x_0) = f(x_0)$.
Structural Commentary: Boundedness controls increments; local continuity gives the derivative.
Key Insight: Averaging a continuous function over shrinking intervals returns its value. □

Theorem 15.1.58. *Fundamental Theorem of Calculus*
If $f \in \mathcal{R}$ on $[a, b]$ and $\exists$ a differentiable function F on $[a, b]$ such that $F' = f$, then $\int_a^b f(x)\, dx = F(b) - F(a)$.

Proof. Choose $\Pi = \{x_0, \ldots, x_n\}$ with $U(\Pi, f) - L(\Pi, f) < \varepsilon$.
By MVT, choose $t_i \in [x_{i-1}, x_i]$ so that $F(b) - F(a) = \sum_i \big(F(x_i) - F(x_{i-1})\big) = \sum_i f(t_i)\Delta x_i$.

Theorem 15.1.43 puts this tagged sum within ε of $\int_a^b f$. Let $\varepsilon \to 0$.
Structural Commentary: Mean-value increments convert antiderivatives into tagged Riemann sums.
Key Insight: The integral is the common limit of all sufficiently fine tagged sums. □

Theorem 15.1.59. *Integration by Parts*
Suppose F, G are differentiable on $[a, b]$, $F' = f \in \mathcal{R}$, and $G' = g \in \mathcal{R}$.
Then $\int_a^b F(x)g(x)\, dx = F(b)G(b) - F(a)G(a) - \int_a^b f(x)G(x)\, dx$.

Proof. Set $H := FG$. Since F, G are continuous, $F, G \in \mathcal{R}$; hence $Fg, fG \in \mathcal{R}$ by Theorem 15.1.51.
Thus $H' = Fg + fG \in \mathcal{R}$, and Theorem 15.1.58 gives $F(b)G(b) - F(a)G(a) = H(b) - H(a) = \int_a^b Fg + \int_a^b fG$.
Structural Commentary: Integration by parts is the product rule integrated over the interval.
Key Insight: Move one derivative from G onto F by applying the fundamental theorem to FG. □

Definition 15.1.60. *Improper Riemann Integrals*
Let $f \in \mathcal{R}$ on every compact subinterval of its domain.
Set $\int_a^b f(x)\, dx := \lim_{d\to b^-} \int_a^d f(x)\, dx$ at a right endpoint, and $\int_a^b f(x)\, dx := \lim_{c\to a^+} \int_c^b f(x)\, dx$ at a left endpoint.
If both endpoints are improper, choose $\xi \in (a, b)$ and set $\int_a^b f(x)\, dx := \int_a^\xi f(x)\, dx + \int_\xi^b f(x)\, dx$, provided both one-sided integrals exist and the sum is not indeterminate.
A finite value means the improper integral *converges*; otherwise it diverges to $+\infty$ or $-\infty$.

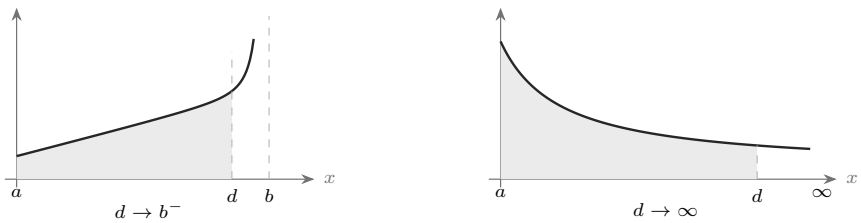


Figure 156: Improper integrals are limits of ordinary integrals over compact truncations.

Theorem 15.1.61. *Nonnegative Improper Integrals*
Let $f \ge 0$ and $f \in \mathcal{R}$ on every $[a, d]$, $a < d < b$. Then $\int_a^b f(x)\, dx = \sup\left\{\int_a^d f(x)\, dx : a < d < b\right\}$.
Thus $\int_a^b f$ exists in $[0, +\infty]$ and converges exactly when this supremum is finite.

Proof. For $a < d_1 < d_2 < b$, $\int_a^{d_2} f - \int_a^{d_1} f = \int_{d_1}^{d_2} f \geq 0$.
Hence $d \mapsto \int_a^d f$ is increasing, so its extended limit exists and equals its supremum.
Structural Commentary: Nonnegative area can only accumulate.
Key Insight: Improper convergence for $f \geq 0$ is boundedness of all finite truncations. □

Perspective 15.1.62. *Principal Value Is Different*
Improper integrals over $(-\infty, \infty)$ require separate one-sided limits.
A symmetric limit such as $\lim_{A\to\infty} \int_{-A}^{A} \sin x\, dx$ may still exist (Cauchy principal value, not Definition 15.1.60).

Definition 15.1.63. *Improper Riemann–Stieltjes Integrals on $\mathbb{R}$*
Let $\alpha : \mathbb{R} \to \mathbb{R}$ be increasing and $f \in \mathcal{R}(\alpha)$ on every finite interval.
Define, when the limits exist, $\int_0^\infty f\, d\alpha := \lim_{b\to\infty} \int_0^b f\, d\alpha$; and $\int_{-\infty}^0 f\, d\alpha := \lim_{a\to-\infty} \int_a^0 f\, d\alpha$.
If their sum is not indeterminate, set $\int_{-\infty}^\infty f\, d\alpha := \int_{-\infty}^0 f\, d\alpha + \int_0^\infty f\, d\alpha$.
Finite value means f is *α-integrable on $\mathbb{R}$*. Write $\alpha(\pm\infty) := \lim_{t\to\pm\infty} \alpha(t)$ when these extended limits are used.

Theorem 15.1.64. *Nonnegative and Bounded-Mass Stieltjes Integrals*
Let $\alpha : \mathbb{R} \to \mathbb{R}$ be increasing, and suppose $f \in \mathcal{R}(\alpha)$ on every finite interval.

(i) If $f \geq 0$, then either f is α-integrable on $\mathbb{R}$ or $\int_{-\infty}^\infty f\, d\alpha = +\infty$.
(ii) If $-\infty < \alpha(-\infty) < \alpha(\infty) < +\infty$ and f is bounded, then f is α-integrable on $\mathbb{R}$.

Proof.

(i) By additivity and order preservation, $\int_0^b f\, d\alpha$ and $\int_a^0 f\, d\alpha$ are monotone under outward truncation. Hence both extended one-sided limits exist, so their sum is finite or $+\infty$.

(ii) Let $|f| \leq M$. For finite $a < b$, Theorem 15.1.50 gives $\left|\int_a^b f\, d\alpha\right| \leq M\big[\alpha(b) - \alpha(a)\big]$. Since $\alpha(\infty)$ and $\alpha(-\infty)$ are finite, the right and left tails have vanishing α-increment. Thus the one-sided truncations are Cauchy and have finite limits.

Structural Commentary: Positivity gives monotone limits; finite mass bounds tails.
Key Insight: A bounded increasing integrator behaves like finite total mass. □

Definition 15.1.65. *Distribution Functions and Densities*
A *distribution function* is an increasing right-continuous $F : \mathbb{R} \to \mathbb{R}$ with $F(-\infty) = 0$ and $F(\infty) = 1$.
A *density* for F is an integrable $g \geq 0$ with $\int_{-\infty}^\infty g = 1$ and $F(t) = \int_{-\infty}^t g(x)\, dx$, in which case continuous g gives $F'(t) = g(t)$ by Theorem 15.1.57.

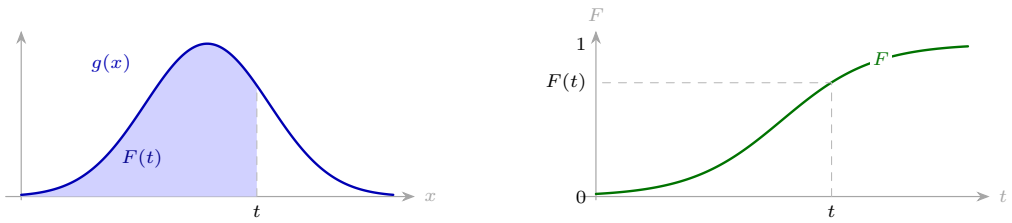


Figure 157: A normal density accumulates area into the distribution function $\Phi(t) = \int_{-\infty}^t g(x)\, dx$.

Definition 15.1.66. *Vector-Valued Riemann–Stieltjes Integral*
For $\mathbf{f} = (f_1, \ldots, f_k) : [a, b] \to \mathbb{R}^k$, define $\mathbf{f} \in \mathcal{R}(\alpha)$ iff $f_j \in \mathcal{R}(\alpha)$ $\forall j$, and set $\int_a^b \mathbf{f}\, d\alpha := \left(\int_a^b f_1\, d\alpha, \ldots, \int_a^b f_k\, d\alpha\right)$.

Theorem 15.1.67. *Vector-Valued Integral Results*

(i) If $\mathbf{f}, \mathbf{F} : [a, b] \to \mathbb{R}^k$, $\mathbf{f} \in \mathcal{R}$, and $\mathbf{F}' = \mathbf{f}$, then $\int_a^b \mathbf{f}(t)\, dt = \mathbf{F}(b) - \mathbf{F}(a)$.
(ii) If $\mathbf{f} : [a, b] \to \mathbb{R}^k$ and $\mathbf{f} \in \mathcal{R}(\alpha)$, then $\|\mathbf{f}\| \in \mathcal{R}(\alpha)$ and $\left\|\int_a^b \mathbf{f}\, d\alpha\right\| \leq \int_a^b \|\mathbf{f}\|\, d\alpha$

Proof.

(i) Write $\mathbf{f} = (f_1, \ldots, f_k)$ and $\mathbf{F} = (F_1, \ldots, F_k)$. For each j, Theorem 15.1.58 gives $\int_a^b f_j(t)\, dt = F_j(b) - F_j(a)$. The vector identity follows componentwise.

(ii) Write $\mathbf{f} = (f_1, \dots, f_k)$. Since $\|\mathbf{f}\| = (f_1^2 + \cdots + f_k^2)^{1/2}$, Theorems 15.1.51 and 15.1.49 give $\|\mathbf{f}\| \in \mathcal{R}(\alpha)$. Let $\mathbf{y} := \int_a^b \mathbf{f}\, d\alpha$. If $\mathbf{y} = 0$, the inequality is immediate.

Otherwise, $\|\mathbf{y}\|^2 = \sum_{j=1}^{k} y_j \int_a^b f_j\, d\alpha = \int_a^b \sum_{j=1}^{k} y_j f_j\, d\alpha \le \int_a^b \|\mathbf{y}\|\,\|\mathbf{f}\|\, d\alpha = \|\mathbf{y}\| \int_a^b \|\mathbf{f}\|\, d\alpha$, by Theorem 12.2.39 and order preservation. Divide by $\|\mathbf{y}\|$.

Structural Commentary: Coordinatewise integration; Schwarz controls norms.
Key Insight: Integrate the projection of $\mathbf{f}$ onto the resulting vector. □

Definition 15.1.68. *Curves and Rectifiability*
A *curve* in $\mathbb{R}^k$ is a continuous map $\gamma : [a, b] \to \mathbb{R}^k$; it is an *arc* if one-to-one and a *closed curve* if $\gamma(a) = \gamma(b)$. Curves are maps, not just ranges. For a partition $\Pi = \{x_0, \dots, x_n\}$, set $\ell(\Pi, \gamma) := \sum_{i=1}^{n} \|\gamma(x_i) - \gamma(x_{i-1})\|$. The *length* is $\ell(\gamma) := \sup_\Pi \ell(\Pi, \gamma)$; if $\ell(\gamma) < \infty$, γ is *rectifiable*.

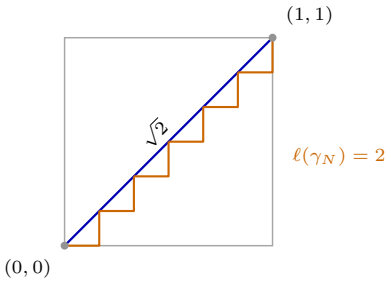


Figure 158: A staircase path can approach the diagonal in position while its polygonal length stays 2, not $\sqrt{2}$.

Perspective 15.1.69. *Length Is Not Just Shape*
Length is a supremum over partitions of one curve. Passing lengths through curve limits needs more than positional convergence; Theorem 15.1.70 uses C^1 speed control.

Theorem 15.1.70. *Length of a Continuously Differentiable Curve*
If $\gamma' : [a, b] \to \mathbb{R}^k$ is continuous, then γ is rectifiable and $\ell(\gamma) = \int_a^b \|\gamma'(t)\|\, dt$.

Proof. For $\Pi = \{x_0, \dots, x_n\}$, Theorem 15.1.67(i) gives $\|\gamma(x_i) - \gamma(x_{i-1})\| = \left\| \int_{x_{i-1}}^{x_i} \gamma'(t)\, dt \right\| \le \int_{x_{i-1}}^{x_i} \|\gamma'(t)\|\, dt$.
Hence $\ell(\Pi, \gamma) \le \int_a^b \|\gamma'\|$. Conversely, choose Π so fine that $\|\gamma'(s) - \gamma'(t)\| < \varepsilon$ on each subinterval.
Choose $t_i \in [x_{i-1}, x_i]$. Then $\int_{x_{i-1}}^{x_i} \|\gamma'(t)\|\, dt \le \|\gamma'(t_i)\|\, \Delta x_i + \varepsilon \Delta x_i$.
Also $\gamma(x_i) - \gamma(x_{i-1}) = \int_{x_{i-1}}^{x_i} \gamma'(t)\, dt$, so $\|\gamma'(t_i)\|\, \Delta x_i \le \|\gamma(x_i) - \gamma(x_{i-1})\| + \varepsilon \Delta x_i$.
Thus $\int_{x_{i-1}}^{x_i} \|\gamma'(t)\|\, dt \le \|\gamma(x_i) - \gamma(x_{i-1})\| + 2\varepsilon \Delta x_i$.
Summing gives $\int_a^b \|\gamma'\| \le \ell(\gamma) + 2\varepsilon(b - a)$; let $\varepsilon \downarrow 0$.
Structural Commentary: Upper bounds are immediate; uniform continuity gives matching lower bounds.
Key Insight: For C^1 curves, length is accumulated speed. □

Method 15.1.71. *Integration Proof Patterns*
(i) *Definitions*: fix $\Pi = \{x_i\}_{i=0}^n$ before defining $M_i, m_i, \Delta x_i, \Delta \alpha_i$.
(ii) *Refinement*: prove the one-point insertion case; iterate.
(iii) *Integrability*: find Π with $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$.
(iv) *Bad sets*: cover E by $I_j = [u_j, v_j]$ with $\sum_j [\alpha(v_j) - \alpha(u_j)] < \varepsilon$; control $K = [a, b] \setminus \bigcup_j (u_j, v_j)$ uniformly.
(v) *Properties*: pass to a common refinement before comparing sums.
(vi) *Calculus*: use the mean value theorem to form tagged sums, then apply Theorem 15.1.43.
(vii) *Improper integrals*: truncate endpoints; use monotonicity or tail bounds.
Key Insight: Riemann–Stieltjes integration is Darboux approximation plus a choice of interval weights $\Delta \alpha_i$.

15.2 Several-Variable Differentiation

Several-variable differentiation studies best linear approximations and local geometry, using linear maps, operator norms, determinants, invertibility, and metric estimates.

Definition 15.2.1. *Total Derivative*
Let $E \subseteq \mathbb{R}^n$ be open, $\mathbf{f} : E \to \mathbb{R}^m$, $\mathbf{x} \in E$, and $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ denote space of linear maps (Definition 1.3.1).
If $\exists A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ s.t. $\lim_{\mathbf{h}\to\mathbf{0}} \frac{\|\mathbf{f}(\mathbf{x}+\mathbf{h})-\mathbf{f}(\mathbf{x})-A\mathbf{h}\|_2}{\|\mathbf{h}\|_2} = 0$, then $\mathbf{f}$ is *differentiable at* $\mathbf{x}$, and A denoted $D\mathbf{f}(\mathbf{x})$ or $\mathbf{f}'(\mathbf{x})$.
Equivalently, $\mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) = D\mathbf{f}(\mathbf{x})\mathbf{h} + \mathbf{r}(\mathbf{h})$, and $\frac{\|\mathbf{r}(\mathbf{h})\|_2}{\|\mathbf{h}\|_2} \to 0$.
If this holds for every $\mathbf{x} \in E$, then $\mathbf{f}$ is differentiable on E.

Perspective 15.2.2. *Derivative First, Matrix Second*
The total derivative is a linear map first; the Jacobian is its matrix in chosen bases. This keeps coordinate formulas subordinate to the intrinsic estimate $\|A\mathbf{h}\|_2 \le \|A\|_{\text{op}} \|\mathbf{h}\|_2$ from Definition 4.3.4.

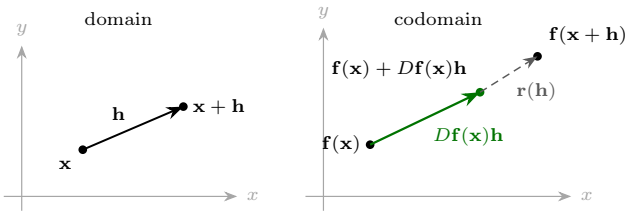


Figure 159: $D\mathbf{f}(\mathbf{x})\mathbf{h}$ is the linear part; $\mathbf{r}(\mathbf{h}) = o(\|\mathbf{h}\|_2)$ is the error.

Theorem 15.2.3. *Basic Consequences of Total Differentiability*
If $\mathbf{f} : E \to \mathbb{R}^m$ is differentiable at $\mathbf{x} \in E$, then $D\mathbf{f}(\mathbf{x})$ is unique and $\mathbf{f}$ is continuous at $\mathbf{x}$.

Proof. If A_1, A_2 both satisfy Definition 15.2.1, set $B = A_1 - A_2$. Then $\|B\mathbf{h}\|_2 / \|\mathbf{h}\|_2 \to 0$ as $\mathbf{h} \to \mathbf{0}$.
For fixed $\mathbf{h} \neq \mathbf{0}$, applying this to $t\mathbf{h}$ and using linearity makes the quotient independent of t, so $B\mathbf{h} = \mathbf{0}$.
Hence $B = 0$. For continuity, write $\mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) = D\mathbf{f}(\mathbf{x})\mathbf{h} + \mathbf{r}(\mathbf{h})$.
The operator-norm estimate gives $\|D\mathbf{f}(\mathbf{x})\mathbf{h}\|_2 \le \|D\mathbf{f}(\mathbf{x})\|_{\text{op}} \|\mathbf{h}\|_2$, and remainder is $o(\|\mathbf{h}\|_2)$, so $\mathbf{f}(\mathbf{x}+\mathbf{h}) \to \mathbf{f}(\mathbf{x})$.
Structural Commentary: The derivative is a rigid first-order linear part.
Key Insight: A remainder smaller than $\|\mathbf{h}\|$ cannot hide a second linear map. □

Example 15.2.4. *Derivative of a Linear Map*
If $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ and $\mathbf{f}(\mathbf{x}) = A\mathbf{x}$, then $D\mathbf{f}(\mathbf{x}) = A$ for every $\mathbf{x} \in \mathbb{R}^n$.
The remainder in Definition 15.2.1 is identically zero.

Theorem 15.2.5. *Chain Rule*
If $\mathbf{f} : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at $\mathbf{x}_0$, $\mathbf{g} : G \subseteq \mathbb{R}^m \to \mathbb{R}^k$ is differentiable at $\mathbf{f}(\mathbf{x}_0)$, and $\mathbf{f}(E) \subseteq G$, then $\mathbf{g} \circ \mathbf{f}$ is differentiable at $\mathbf{x}_0$ and $D(\mathbf{g} \circ \mathbf{f})(\mathbf{x}_0) = D\mathbf{g}(\mathbf{f}(\mathbf{x}_0))\, D\mathbf{f}(\mathbf{x}_0)$.

Proof. Let $A = D\mathbf{f}(\mathbf{x}_0)$, $B = D\mathbf{g}(\mathbf{f}(\mathbf{x}_0))$, $\mathbf{u}(\mathbf{h}) = \mathbf{f}(\mathbf{x}_0+\mathbf{h}) - \mathbf{f}(\mathbf{x}_0) - A\mathbf{h}$, and $\mathbf{v}(\mathbf{k}) = \mathbf{g}(\mathbf{f}(\mathbf{x}_0)+\mathbf{k}) - \mathbf{g}(\mathbf{f}(\mathbf{x}_0)) - B\mathbf{k}$.
For $\mathbf{k} = \mathbf{f}(\mathbf{x}_0+\mathbf{h}) - \mathbf{f}(\mathbf{x}_0)$, differentiability of $\mathbf{f}$ gives $\mathbf{k} \to \mathbf{0}$ and $(\mathbf{g}\circ\mathbf{f})(\mathbf{x}_0+\mathbf{h}) - (\mathbf{g}\circ\mathbf{f})(\mathbf{x}_0) - BA\mathbf{h} = B\mathbf{u}(\mathbf{h}) + \mathbf{v}(\mathbf{k})$, where $\|\mathbf{u}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$ and $\|\mathbf{v}(\mathbf{k})\|_2 = o(\|\mathbf{k}\|_2) = o(\|\mathbf{h}\|_2)$. Divide by $\|\mathbf{h}\|_2$.
Structural Commentary: The first-order parts compose as linear maps.
Key Insight: The output increment of $\mathbf{f}$ is small enough to feed into the remainder estimate for $\mathbf{g}$. □

Theorem 15.2.6. *Componentwise Differentiability*
Let $\mathbf{f} = (f_1, \ldots, f_m) : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$ and let $\pi_i : \mathbb{R}^m \to \mathbb{R}$ be the ith coordinate projection. Then $\mathbf{f}$ is differentiable at $\mathbf{x} \in E$ iff each f_i is differentiable at $\mathbf{x}$, and $Df_i(\mathbf{x}) = \pi_i \circ D\mathbf{f}(\mathbf{x})$ when these derivatives exist.

Proof.
($\Rightarrow$) If $\mathbf{f}$ is differentiable, write $\mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) = D\mathbf{f}(\mathbf{x})\mathbf{h} + \mathbf{r}(\mathbf{h})$ with $\|\mathbf{r}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$.
Applying π_i gives $f_i(\mathbf{x}+\mathbf{h}) - f_i(\mathbf{x}) = \pi_i(D\mathbf{f}(\mathbf{x})\mathbf{h}) + \pi_i(\mathbf{r}(\mathbf{h}))$ and $|\pi_i(\mathbf{r}(\mathbf{h}))| \le \|\mathbf{r}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$.
Thus $Df_i(\mathbf{x}) = \pi_i \circ D\mathbf{f}(\mathbf{x})$.
($\Leftarrow$) Suppose $Df_i(\mathbf{x}) = A_i \in \mathcal{L}(\mathbb{R}^n, \mathbb{R})$ for each i. Define $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ by $A\mathbf{h} = (A_1\mathbf{h}, \ldots, A_m\mathbf{h})$.
With $r_i(\mathbf{h}) = f_i(\mathbf{x}+\mathbf{h}) - f_i(\mathbf{x}) - A_i\mathbf{h}$, each $r_i(\mathbf{h}) = o(\|\mathbf{h}\|_2)$.
So $\mathbf{r}(\mathbf{h}) = (r_1(\mathbf{h}), \ldots, r_m(\mathbf{h}))$ satisfies $\|\mathbf{r}(\mathbf{h})\|_2 \le \sum_i |r_i(\mathbf{h})| = o(\|\mathbf{h}\|_2)$. Definition 15.2.1 gives $D\mathbf{f}(\mathbf{x}) = A$.
Structural Commentary: Component derivatives assemble into one linear map.
Key Insight: Components test the derivative; the total derivative packages the tests together. □

Definition 15.2.7. *Partial, Gradient, and Directional Derivatives*
Let $E \subseteq \mathbb{R}^n$ be open, $\mathbf{f} = (f_1, \ldots, f_m) : E \to \mathbb{R}^m$, and let $\mathbf{e}_1, \ldots, \mathbf{e}_n$ be the standard basis of $\mathbb{R}^n$.
If indicated limits exist, *partial derivative* is $(D_j f_i)(\mathbf{x}) := \lim_{t\to 0} \frac{f_i(\mathbf{x}+t\mathbf{e}_j) - f_i(\mathbf{x})}{t}$ for $1 \le i \le m$, $1 \le j \le n$.

Directional derivative is $(D_{\mathbf{u}}f)(\mathbf{x}) := \lim_{t\to 0} \frac{f(\mathbf{x}+t\mathbf{u})-f(\mathbf{x})}{t}$ for $\mathbf{u} \in \mathbb{R}^n$; when only the direction is meant, one often assumes $\|\mathbf{u}\|_2 = 1$. If $m = 1$ and all partials exist, the *gradient* is $(\nabla f)(\mathbf{x}) := \sum_{j=1}^{n} (D_j f)(\mathbf{x})\mathbf{e}_j$.

Perspective 15.2.8. *Partials Are Coordinate Tests*
Partials and directional derivatives test selected lines; the total derivative controls all small increments by one linear map. Theorem 15.2.9 extracts partials, and Theorem 15.2.14 gives a continuity-based converse.

Theorem 15.2.9. *Matrix of the Total Derivative*
If $\mathbf{f} : E \to \mathbb{R}^m$ is differentiable at $\mathbf{x}$, then every $(D_j f_i)(\mathbf{x})$ exists and, in standard bases,

$$[D\mathbf{f}(\mathbf{x})] = \begin{bmatrix} (D_1 f_1)(\mathbf{x}) & \cdots & (D_n f_1)(\mathbf{x}) \\ \vdots & & \vdots \\ (D_1 f_m)(\mathbf{x}) & \cdots & (D_n f_m)(\mathbf{x}) \end{bmatrix}$$

If $m = 1$ and f is differentiable at $\mathbf{x}$, then $(D_{\mathbf{u}}f)(\mathbf{x}) = (\nabla f)(\mathbf{x}) \cdot \mathbf{u}$ for every $\mathbf{u} \in \mathbb{R}^n$.

Proof. Set $\mathbf{h} = t\mathbf{e}_j$ in Definition 15.2.1; limit is $D\mathbf{f}(\mathbf{x})\mathbf{e}_j$, so components form column j (Proposition 1.3.53). For $m = 1$, set $\mathbf{h} = t\mathbf{u}$, identify linear functional with dot product against $\nabla f(\mathbf{x})$ (Theorem 2.1.41).
Structural Commentary: Partials are coordinate columns of $D\mathbf{f}(\mathbf{x})$.
Key Insight: Differentiability controls all directions at once; partial derivatives record its coordinate columns. □

Remark 15.2.10. *Row and Column Conventions*
With the coordinate-column convention from Proposition 1.3.53, the Jacobian matrix above is the standard-basis matrix of the linear map $D\mathbf{f}(\mathbf{x})$, and its jth column is $D\mathbf{f}(\mathbf{x})\mathbf{e}_j$.

Theorem 15.2.11. *Mean-Value Inequality on Convex Sets*
Let $E \subseteq \mathbb{R}^n$ be open and convex.
If $\mathbf{f} : E \to \mathbb{R}^m$ is differentiable and $\|D\mathbf{f}\|_{op} \le M$ on E, then for all $\mathbf{a}, \mathbf{b} \in E$, $\|\mathbf{f}(\mathbf{b}) - \mathbf{f}(\mathbf{a})\|_2 \le M \|\mathbf{b} - \mathbf{a}\|_2$.
If $D\mathbf{f} = 0$ on E, then $\mathbf{f}$ is constant.

Proof. Fix $\mathbf{a}, \mathbf{b} \in E$, set $\gamma(t) = (1-t)\mathbf{a} + t\mathbf{b}$, and let $\mathbf{g} = \mathbf{f} \circ \gamma$. Then $\gamma([0,1]) \subseteq E$ and $\|\mathbf{g}'(t)\|_2 = \|D\mathbf{f}(\gamma(t))(\mathbf{b} - \mathbf{a})\|_2 \le M \|\mathbf{b} - \mathbf{a}\|_2$. Apply Theorem 15.1.30; the constant case is $M = 0$.
Structural Commentary: Restricting to a line segment reduces the estimate to one variable.
Key Insight: A uniform bound on the derivative bounds every secant displacement. □

Definition 15.2.12. *C^1 Mapping*
Let $E \subseteq \mathbb{R}^n$ be open. A differentiable map $\mathbf{f} : E \to \mathbb{R}^m$ is *continuously differentiable*, or C^1, if $\mathbf{x} \mapsto D\mathbf{f}(\mathbf{x})$ is continuous from E into $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ in operator norm.

Perspective 15.2.13. *C^1 Uses the Operator-Norm Metric*
C^1 continuity measures $\mathbf{x} \mapsto D\mathbf{f}(\mathbf{x})$ in operator norm. In coordinates this is spectral-norm continuity, equivalent here to continuity of the derivative matrix entries.

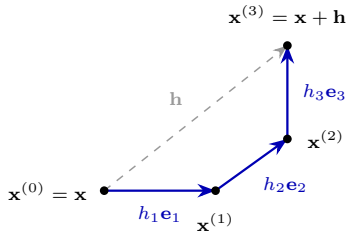


Figure 160: Coordinate increments telescope $\mathbf{x}$ to $\mathbf{x} + \mathbf{h}$.

Theorem 15.2.14. *C^1 Means Continuous Partials*
A map $\mathbf{f} : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$ is C^1 iff all $D_j f_i$ exist and are continuous on E.

Proof. If $\mathbf{f}$ is C^1, Theorem 15.2.9 makes each $D_j f_i$ a continuous matrix entry of $D\mathbf{f}$.
Conversely reduce componentwise to $m = 1$. Fix $\mathbf{x} \in E$ and take $\mathbf{h}$ small enough that the coordinate polygon from $\mathbf{x}$ to $\mathbf{x} + \mathbf{h}$ lies in E. Put $\mathbf{x}^{(0)} = \mathbf{x}$ and $\mathbf{x}^{(j)} = \mathbf{x} + h_1\mathbf{e}_1 + \cdots + h_j\mathbf{e}_j$. By one-variable MVT on each coordinate segment, $f(\mathbf{x}+\mathbf{h}) - f(\mathbf{x}) = \sum_{j=1}^{n} h_j (D_j f)(\xi_j)$ for points ξ_j on corresponding segments. Continuity of

partials at $\mathbf{x}$ gives $\left|f(\mathbf{x}+\mathbf{h}) - f(\mathbf{x}) - \sum_{j=1}^{n} h_j (D_j f)(\mathbf{x})\right| \le \sum_{j=1}^{n} |h_j|\,|(D_j f)(\xi_j) - (D_j f)(\mathbf{x})| = o(\|\mathbf{h}\|_2)$.
Hence $Df(\mathbf{x})$ has row matrix $[(D_1 f)(\mathbf{x}) \ \cdots \ (D_n f)(\mathbf{x})]$. Since operator norm is controlled by the Euclidean norm of the matrix entries, continuity of the partials gives continuity of $\mathbf{x} \mapsto Df(\mathbf{x})$.
Structural Commentary: Continuous partial derivatives assemble into a continuous total derivative.
Key Insight: Coordinate-by-coordinate increments work only because the partials vary continuously. □

Higher derivatives repeat the same idea: once $D\mathbf{f}$ is itself a map into the finite-dimensional vector space $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$, we differentiate it again. The linear-algebra viewpoint matters here because the second derivative is naturally bilinear, not merely an array of second partial derivatives.

Definition 15.2.15. *Higher Derivatives and C^r*
Let $E \subseteq \mathbb{R}^n$ be open, $\mathbf{f} : E \to \mathbb{R}^m$, $\mathcal{L}^r(\mathbb{R}^n, \mathbb{R}^m)$ be space of r-linear maps $(\mathbb{R}^n)^r \to \mathbb{R}^m$. Set $D^1\mathbf{f} := D\mathbf{f}$. Recursively, if $D^{r-1}\mathbf{f} : E \to \mathcal{L}^{r-1}(\mathbb{R}^n, \mathbb{R}^m)$ differentiable at $\mathbf{x}$, let $D^r\mathbf{f}(\mathbf{x})(\mathbf{v}_1, \ldots, \mathbf{v}_r) := \big(D(D^{r-1}\mathbf{f})(\mathbf{x})\mathbf{v}_1\big)(\mathbf{v}_2, \ldots, \mathbf{v}_r)$. For real-valued f, write $D_{ij} f := D_i(D_j f)$ when defined. The map $\mathbf{f}$ is C^r if derivatives through order r exist and $D^r\mathbf{f} : E \to \mathcal{L}^r(\mathbb{R}^n, \mathbb{R}^m)$ is continuous; it is C^∞ if it is C^r for all r.

Theorem 15.2.16. *Second Derivatives and Second Partials*
Let $\mathbf{f} = (f_1, \ldots, f_m) : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$ and $p \in E$.
(⇒) *If $D^2\mathbf{f}(p)$ exists, then each $D_{ij} f_\ell(p)$ exists and $D^2 f_\ell(p)(\mathbf{e}_i, \mathbf{e}_j) = D_{ij} f_\ell(p)$.*
(⇐) *If all $D_{ij} f_\ell$ exist near p and are continuous at p, then $D^2\mathbf{f}(p)$ exists.*

Proof.
(⇒) Apply Theorem 15.2.6 to $D\mathbf{f} : E \to \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$. The entries of $D\mathbf{f}$ are $D_j f_\ell$ by Theorem 15.2.9; differentiating the (ℓ, j) entry in the $\mathbf{e}_i$ direction gives $D_{ij} f_\ell(p)$ and the displayed formula.
(⇐) Matrix entries of $D\mathbf{f}$ are functions $D_j f_\ell$. Since first partials $D_i(D_j f_\ell) = D_{ij} f_\ell$ exist near p and are continuous at p, coordinate criterion at p gives differentiability of $D\mathbf{f}$ there. Hence $D^2\mathbf{f}(p)$ exists.

Structural Commentary: Second partials are matrix entries of the derivative of the first-derivative matrix.
Key Insight: Continuous second partials make the first-derivative matrix differentiable. □

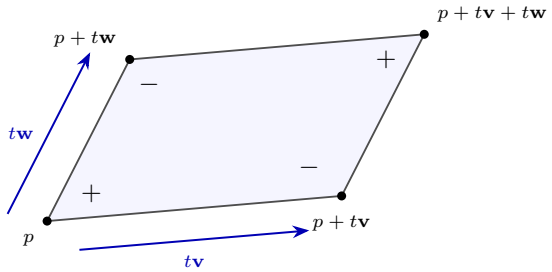


Figure 161: The signed vertex sum on a small parallelogram is symmetric in $\mathbf{v}$ and $\mathbf{w}$.

Theorem 15.2.17. *Symmetry of the Second Derivative*
If $D^2\mathbf{f}(p)$ exists, then $D^2\mathbf{f}(p)(\mathbf{v}, \mathbf{w}) = D^2\mathbf{f}(p)(\mathbf{w}, \mathbf{v})$ for all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$.

Proof. Reduce to scalar f by Theorem 15.2.6.
For small t, set $\Delta(t, \mathbf{v}, \mathbf{w}) := f(p + t\mathbf{v} + t\mathbf{w}) - f(p + t\mathbf{v}) - f(p + t\mathbf{w}) + f(p)$.
Then $\Delta(t, \mathbf{v}, \mathbf{w}) = \Delta(t, \mathbf{w}, \mathbf{v})$; see Figure 161. The one-variable mean value theorem and differentiability of Df at p give $D^2 f(p)(\mathbf{v}, \mathbf{w}) = \lim_{t \to 0} \frac{\Delta(t, \mathbf{v}, \mathbf{w})}{t^2}$. The same formula with $\mathbf{v}, \mathbf{w}$ reversed proves symmetry.
Structural Commentary: A pointwise second derivative is already forced to be a symmetric bilinear form.
Key Insight: The parallelogram difference is symmetric in the two directions. □

Corollary 15.2.18. *Mixed Partials from D^2*
If $D^2\mathbf{f}(p)$ exists, then $D_{ij} f_\ell(p) = D_{ji} f_\ell(p)$ for all components and all i, j.

Proof. By Theorem 15.2.16, $D_{ij} f_\ell(p) = D^2 f_\ell(p)(\mathbf{e}_i, \mathbf{e}_j)$; and $D_{ji} f_\ell(p) = D^2 f_\ell(p)(\mathbf{e}_j, \mathbf{e}_i)$.
Theorem 15.2.17 makes the two right-hand sides equal.
Structural Commentary: Mixed partials are coordinate values of one bilinear map.
Key Insight: Symmetry of $D^2 f_\ell(p)$ gives equality of mixed entries. □

Corollary 15.2.19. *Symmetry of Higher Derivatives*
If $D^r\mathbf{f}(p)$ exists, then the r-linear map $D^r\mathbf{f}(p)$ is unchanged by permuting its arguments.

Proof. It suffices to prove invariance under adjacent transpositions. Fix all arguments except positions $k, k+1$. Resulting two-variable second derivative is $(\mathbf{u},\mathbf{v}) \longmapsto D^r\mathbf{f}(p)(\mathbf{v}_1,\dots,\mathbf{v}_{k-1},\mathbf{u},\mathbf{v},\mathbf{v}_{k+2},\dots,\mathbf{v}_r)$, so Theorem 15.2.17 allows $\mathbf{u}$ and $\mathbf{v}$ to be swapped. Since adjacent transpositions generate all permutations, $D^r\mathbf{f}(p)$ is symmetric.
Structural Commentary: Higher derivatives are iterated second-derivative symmetries.
Key Insight: Adjacent swaps generate every permutation. □

The next rectangle argument is the classical coordinate route to equality of mixed partials. It is weaker in hypothesis than global C^2, but it proves the same compatibility of coordinate second derivatives.

Theorem 15.2.20. *Rectangle Mean Value Formula*
Let $E \subseteq \mathbb{R}^2$ be open, and suppose $D_1 f$ and $D_{21} f$ exist on E. For a closed coordinate rectangle $Q \subseteq E$ with opposite vertices (a,b) and $(a+h,b+k)$, $h,k \neq 0$, set $\Delta(f,Q) := f(a+h,b+k) - f(a+h,b) - f(a,b+k) + f(a,b)$. Then some $(x,y) \in \operatorname{int} Q$ satisfies $\Delta(f,Q) = hk(D_{21}f)(x,y)$.

Proof. Set $u(t) = f(t,b+k) - f(t,b)$.
By Corollary 15.1.13, for some $x \in (a,a+h)$, $\Delta(f,Q) = u(a+h) - u(a) = h\big[(D_1 f)(x,b+k) - (D_1 f)(x,b)\big]$.
Apply same corollary to $s \mapsto (D_1 f)(x,s)$; for some $y \in (b,b+k)$, the bracket equals $k(D_{21}f)(x,y)$.
Hence $(x,y) \in \operatorname{int} Q$ and the formula follows.
Structural Commentary: The rectangle difference is a two-step one-variable mean value calculation.
Key Insight: Mixed second derivatives measure the first-order change of a first partial across the other coordinate. □

Theorem 15.2.21. *Equality of Mixed Partials*
Let (a,b) lie in an open set $E \subseteq \mathbb{R}^2$. If $D_1 f$, $D_2 f$, and $D_{21} f$ exist on E, and $D_{21} f$ is continuous at (a,b), then $D_{12} f(a,b)$ exists and $(D_{12}f)(a,b) = (D_{21}f)(a,b)$. Hence $D_{12} f = D_{21} f$ when $f \in C^2(E)$.

Proof. Let $A = (D_{21}f)(a,b)$.
Given $\varepsilon > 0$, continuity of $D_{21} f$ and Theorem 15.2.20 give, for all small $h,k \neq 0$, $\left|\frac{\Delta(f,Q)}{hk} - A\right| < \varepsilon$.
Fix such an h and let $k \to 0$. Since $D_2 f$ exists at $(a+h,b)$ and (a,b), $\frac{\Delta(f,Q)}{hk} \to \frac{(D_2 f)(a+h,b) - (D_2 f)(a,b)}{h}$.
Thus this last quotient differs from A by at most ε for all sufficiently small $h \neq 0$.
Since ε was arbitrary, the quotient tends to A as $h \to 0$, so $D_{12} f(a,b)$ exists and equals A.
Structural Commentary: Continuity of $D_{21} f$ controls local rectangle quotients.
Key Insight: The missing mixed partial is recovered by taking the two one-variable limits in the reverse order. □

Perspective 15.2.22. *Mixed Partials Need Hypotheses*
$D_{12} f$ and $D_{21} f$ are different iterated limits. Theorems 15.2.17 and 15.2.21 give hypotheses for equality.

Theorem 15.2.23. *Differentiation Under a Riemann–Stieltjes Integral*
Let α be increasing on $[a,b]$, let $\varphi : [a,b] \times [c,d] \to \mathbb{R}$, and put $\varphi^t(x) := \varphi(x,t)$.
Assume $\varphi^t \in \mathcal{R}(\alpha)$ for all t and $D_2\varphi(\cdot,t) \rightrightarrows D_2\varphi(\cdot,s)$ on $[a,b]$ as $t \to s \in (c,d)$.
If $F(t) := \int_a^b \varphi(x,t)\,d\alpha(x)$, then $(D_2\varphi)^s \in \mathcal{R}(\alpha)$ and $F'(s) = \int_a^b (D_2\varphi)(x,s)\,d\alpha(x)$.

Proof. For $t \neq s$, set $\psi_t = (\varphi^t - \varphi^s)/(t-s)$. Corollary 15.1.13 and the uniform hypothesis give $\psi_t \rightrightarrows (D_2\varphi)^s$.
Since $\psi_t \in \mathcal{R}(\alpha)$, Theorem 14.2.16 gives $\frac{F(t)-F(s)}{t-s} = \int_a^b \psi_t\,d\alpha \longrightarrow \int_a^b (D_2\varphi)(x,s)\,d\alpha(x)$.
Structural Commentary: Uniform parameter control lets limits pass under the integral.
Key Insight: The theorem is exactly the commutation of a derivative limit with a Riemann–Stieltjes integral. □

Example 15.2.24. *A Gaussian Fourier Integral*
Define $f(t) = \int_{-\infty}^{\infty} e^{-x^2}\cos(xt)\,dx$ and $g(t) = -\int_{-\infty}^{\infty} xe^{-x^2}\sin(xt)\,dx$.
Then $f'(t) = g(t)$, and integration by parts gives $2f'(t) + tf(t) = 0$. Since $f(0) = \sqrt{\pi}$, $f(t) = \sqrt{\pi}\exp\left(-\frac{t^2}{4}\right)$.

Definition 15.2.25. *Jacobian*
If $\mathbf{f} : E \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is differentiable at $\mathbf{x}$, define $J_{\mathbf{f}}(\mathbf{x}) := \det D\mathbf{f}(\mathbf{x}) = \frac{\partial(y_1,\dots,y_n)}{\partial(x_1,\dots,x_n)}$ when $f = (y_1,\dots,y_n)$.
By Theorem 4.1.5, $D\mathbf{f}(\mathbf{x})$ is invertible if and only if $J_{\mathbf{f}}(\mathbf{x}) \neq 0$. The absolute value $|J_{\mathbf{f}}(\mathbf{x})|$ is the local version of determinant volume scaling from Theorem 15.4.1, used again in the change-of-variables theorem.

15.3 Inverse, Implicit, and Rank Theorems

Theorem 15.3.1. *Finite-Dimensional Linear Maps Are Lipschitz*
Let $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$. Then A is $\|A\|_{op}$-Lipschitz: $\|A\mathbf{x}\|_2 \le \|A\|_{op} \|\mathbf{x}\|_2$ for $\mathbf{x} \in \mathbb{R}^n$.
If $A \in \mathcal{L}(\mathbb{R}^n)$ is invertible, then $\frac{1}{\|A^{-1}\|_{op}} \|\mathbf{x}\|_2 \le \|A\mathbf{x}\|_2 \le \|A\|_{op} \|\mathbf{x}\|_2$, so A is a homeomorphism of $\mathbb{R}^n$.

Proof. The upper estimate follows from Definition 4.3.4: apply it to $\mathbf{x}/\|\mathbf{x}\|_2$ when $\mathbf{x} \neq \mathbf{0}$, while $\mathbf{x} = \mathbf{0}$ is trivial.
If A is invertible, apply the same estimate to A^{-1} with $\mathbf{y} = A\mathbf{x}$: $\|\mathbf{x}\|_2 = \|A^{-1}\mathbf{y}\|_2 \le \|A^{-1}\|_{\text{op}} \|A\mathbf{x}\|_2$.
Thus A and A^{-1} are continuous.
Structural Commentary: Linear maps have finite stretch; invertible ones have positive lower stretch.
Key Insight: The operator norm turns linear algebra into uniform metric estimates. □

Theorem 15.3.2. *Invertible Operators Form an Open Set*
Let $\Omega \subseteq \mathcal{L}(\mathbb{R}^n)$ be the set of invertible linear operators. If $A \in \Omega$, $B \in \mathcal{L}(\mathbb{R}^n)$, $\|B - A\|_{op} \|A^{-1}\|_{op} < 1$, then $B \in \Omega$. Consequently, Ω is open in $\mathcal{L}(\mathbb{R}^n)$ and the map $A \mapsto A^{-1}$ is continuous on Ω.

Proof. Put $\alpha = 1/\|A^{-1}\|_{\text{op}}$ and $\beta = \|B - A\|_{\text{op}}$.
For every $\mathbf{x} \in \mathbb{R}^n$, $\alpha \|\mathbf{x}\|_2 \le \|A\mathbf{x}\|_2 \le \|(A-B)\mathbf{x}\|_2 + \|B\mathbf{x}\|_2 \le \beta \|\mathbf{x}\|_2 + \|B\mathbf{x}\|_2$.
Since $\beta < \alpha$, this shows $B\mathbf{x} \neq 0$ for $\mathbf{x} \neq 0$; hence B is injective and therefore invertible by Theorem 1.3.27.
Replacing $\mathbf{x}$ by $B^{-1}\mathbf{y}$ gives $\|B^{-1}\|_{\text{op}} \le (\alpha - \beta)^{-1}$.
Finally, $B^{-1} - A^{-1} = B^{-1}(A - B)A^{-1}$, Proposition 4.3.3 gives the continuity of inversion.
Structural Commentary: Small operator-norm perturbations of an invertible map remain invertible.
Key Insight: The inverse changes continuously because the difference of inverses factors through $A - B$. □

Perspective 15.3.3. *Linear Inverse Estimate*
For invertible $A \in \mathcal{L}(\mathbb{R}^n)$, Theorem 15.3.1 gives $c\|\mathbf{h}\|_2 \le \|A\mathbf{h}\|_2 \le \|A\|_{\text{op}} \|\mathbf{h}\|_2$; and $c := \frac{1}{\|A^{-1}\|_{\text{op}}}$.
The inverse function theorem preserves this lower stretch under small derivative changes.

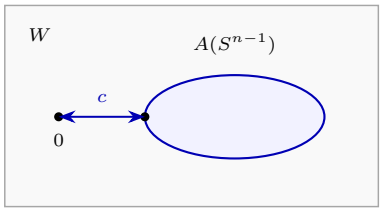


Figure 162: Invertibility keeps $A(S^{n-1})$ a positive distance from 0.

Theorem 15.3.4. *Inverse Function Theorem*
Let $\mathbf{f} : E \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be C^1, let $\mathbf{a} \in E$, and suppose $D\mathbf{f}(\mathbf{a})$ is invertible. With $\mathbf{b} = \mathbf{f}(\mathbf{a})$, $\mathbf{f}$ restricts to a bijection $U \to V$ between open neighborhoods of $\mathbf{a}$ and $\mathbf{b}$; inverse $\mathbf{g} : V \to U$ is C^1, and $D\mathbf{g}(\mathbf{y}) = \big(D\mathbf{f}(\mathbf{g}(\mathbf{y}))\big)^{-1}$, $\mathbf{y} \in V$.

Proof. Put $A = D\mathbf{f}(\mathbf{a})$ and choose $0 < \theta < 1$.
By continuity of $D\mathbf{f}$, choose $r > 0$ with $\overline{B}_r(\mathbf{a}) \subset E$ and $\|I - A^{-1}D\mathbf{f}(\mathbf{x})\|_{\text{op}} \le \theta$ on this closed ball.
For fixed $\mathbf{y}$, solutions of $\mathbf{f}(\mathbf{x}) = \mathbf{y}$ are fixed points of $\varphi_{\mathbf{y}}(\mathbf{x}) = \mathbf{x} + A^{-1}(\mathbf{y} - \mathbf{f}(\mathbf{x}))$.
Theorem 15.2.11 gives $\|\varphi_{\mathbf{y}}(\mathbf{x}) - \varphi_{\mathbf{y}}(\mathbf{z})\|_2 \le \theta \|\mathbf{x} - \mathbf{z}\|_2$. If $\|A^{-1}(\mathbf{y} - \mathbf{b})\|_2 \le (1-\theta)r$, then $\varphi_{\mathbf{y}}$ maps $\overline{B}_r(\mathbf{a})$ into itself. Since $\overline{B}_r(\mathbf{a})$ is a closed subset of complete $\mathbb{R}^n$, Proposition 13.2.12 makes it complete.
The contraction principle, Theorem 13.2.14, gives a unique $\mathbf{x} \in \overline{B}_r(\mathbf{a})$ with $\mathbf{f}(\mathbf{x}) = \mathbf{y}$. Thus, after shrinking to open neighborhoods U of $\mathbf{a}$ and V of $\mathbf{b}$, $\mathbf{f} : U \to V$ is bijective with inverse $\mathbf{g}$.
For $\mathbf{y} = \mathbf{f}(\mathbf{x})$ and $\mathbf{y} + \mathbf{k} = \mathbf{f}(\mathbf{x} + \mathbf{h})$ in V, the same derivative bound gives $\|\mathbf{h}\|_2 \le C\|\mathbf{k}\|_2$ for a local constant C. Moreover $\mathbf{g}(\mathbf{y} + \mathbf{k}) - \mathbf{g}(\mathbf{y}) - \big(D\mathbf{f}(\mathbf{x})\big)^{-1}\mathbf{k} = -\big(D\mathbf{f}(\mathbf{x})\big)^{-1}\big[\mathbf{f}(\mathbf{x} + \mathbf{h}) - \mathbf{f}(\mathbf{x}) - D\mathbf{f}(\mathbf{x})\mathbf{h}\big]$.
Since $\mathbf{f}$ is differentiable at $\mathbf{x}$ and $\|\mathbf{h}\|_2 = O(\|\mathbf{k}\|_2)$, division by $\|\mathbf{k}\|_2$ gives $D\mathbf{g}(\mathbf{y}) = (D\mathbf{f}(\mathbf{x}))^{-1}$.
Continuity of $D\mathbf{g}$ follows from continuity of $D\mathbf{f}$, continuity of $\mathbf{g}$, and Theorem 15.3.2.
Structural Commentary: Local inversion is a contraction built from A^{-1}.
Key Insight: Small derivative variation preserves local invertibility. □

Corollary 15.3.5. *Local Diffeomorphisms Are Open*
If $\mathbf{f} : E \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is C^1 and $D\mathbf{f}$ is everywhere invertible, then $\mathbf{f}$ is an open map (locally one-to-one).

Proof. Let $W \subseteq E$ be open and let $\mathbf{y} \in \mathbf{f}(W)$. Choose $\mathbf{x} \in W$ with $\mathbf{f}(\mathbf{x}) = \mathbf{y}$. Applying Theorem 15.3.4 to $\mathbf{f}|_W$ at $\mathbf{x}$ gives open neighborhoods $U \subseteq W$ of $\mathbf{x}$ and V of $\mathbf{y}$ with $\mathbf{f}(U) = V$. Hence $V \subseteq \mathbf{f}(W)$, so $\mathbf{y}$ is interior to $\mathbf{f}(W)$. Since $\mathbf{y}$ was arbitrary, $\mathbf{f}(W)$ is open. Local one-to-one-ness is the same theorem applied pointwise.
Structural Commentary: Openness is checked one image point at a time.
Key Insight: A local inverse turns nearby domain points into a full image neighborhood. □

Definition 15.3.6. *Block Splitting of a Linear Map*
For $A \in \mathcal{L}(\mathbb{R}^{n+m}, \mathbb{R}^n)$ and $(\mathbf{h}, \mathbf{k}) \in \mathbb{R}^n \times \mathbb{R}^m$, define $A_x\mathbf{h} := A(\mathbf{h}, \mathbf{0})$ and $A_y\mathbf{k} := A(\mathbf{0}, \mathbf{k})$.
Then $A_x \in \mathcal{L}(\mathbb{R}^n)$, $A_y \in \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$, and $A(\mathbf{h}, \mathbf{k}) = A_x\mathbf{h} + A_y\mathbf{k}$.

Theorem 15.3.7. *Linear Implicit Function Theorem*
If $A \in \mathcal{L}(\mathbb{R}^{n+m}, \mathbb{R}^n)$, A_x invertible, then $\forall \mathbf{k} \in \mathbb{R}^m$ $\exists$ unique $\mathbf{h} \in \mathbb{R}^n$ s.t. $A(\mathbf{h}, \mathbf{k}) = 0$, that $\mathbf{h} = -(A_x)^{-1}A_y\mathbf{k}$.

Proof. Fix $\mathbf{k} \in \mathbb{R}^m$ and set $\mathbf{h}_0 := -(A_x)^{-1}A_y\mathbf{k}$.
By Definition 15.3.6, $A(\mathbf{h}_0, \mathbf{k}) = A_x\mathbf{h}_0 + A_y\mathbf{k} = -A_y\mathbf{k} + A_y\mathbf{k} = 0$, so a solution exists.
If $\mathbf{h} \in \mathbb{R}^n$ also satisfies $A(\mathbf{h}, \mathbf{k}) = 0$, then $A_x\mathbf{h} + A_y\mathbf{k} = 0$, hence $A_x\mathbf{h} = -A_y\mathbf{k}$.
Applying $(A_x)^{-1}$ gives $\mathbf{h} = -(A_x)^{-1}A_y\mathbf{k} = \mathbf{h}_0$. Thus the solution exists and is unique.
Structural Commentary: Splitting A separates the unknown variable from the parameter.
Key Insight: Invertibility of A_x turns constraint $A(\mathbf{h}, \mathbf{k}) = 0$ into a uniquely solvable linear equation for $\mathbf{h}$. □

The implicit theorem below is the nonlinear counterpart of the block equation theorem. The invertible block is the set of variables one can solve for, just as in Theorem 15.3.7.

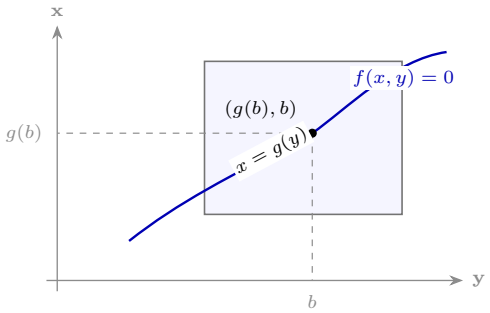


Figure 163: Near a regular point, the zero locus is the graph of an implicit function.

Theorem 15.3.8. *Implicit Function Theorem*
Let $\mathbf{f} : E \subseteq \mathbb{R}^{n+m} \to \mathbb{R}^n$ be C^1, $\mathbf{f}(\mathbf{a}, \mathbf{b}) = 0$, $A = D\mathbf{f}(\mathbf{a}, \mathbf{b})$, and let A_x, A_y be the block maps of Definition 15.3.6. If A_x is invertible, then near $\mathbf{b}$, the equation $\mathbf{f}(\mathbf{x}, \mathbf{y}) = 0$ has unique local solution $\mathbf{x} = \mathbf{g}(\mathbf{y})$, where $\mathbf{g}$ is C^1, $\mathbf{g}(\mathbf{b}) = \mathbf{a}$, $D\mathbf{g}(\mathbf{b}) = -(A_x)^{-1}A_y$.

Proof. Set $\Phi(\mathbf{x}, \mathbf{y}) = (\mathbf{f}(\mathbf{x}, \mathbf{y}), \mathbf{y})$. Then $D\Phi(\mathbf{a}, \mathbf{b})(\mathbf{h}, \mathbf{k}) = (A_x\mathbf{h} + A_y\mathbf{k}, \mathbf{k})$, whose kernel is zero since A_x is invertible; hence $D\Phi(\mathbf{a}, \mathbf{b})$ is invertible by Theorem 1.3.27. Theorem 15.3.4 makes Φ locally invertible.
Points with $\mathbf{f}(\mathbf{x}, \mathbf{y}) = 0$ are exactly those sent to $(0, \mathbf{y})$, so $\Phi^{-1}(0, \mathbf{y}) = (\mathbf{g}(\mathbf{y}), \mathbf{y})$ gives the unique local solution.
Differentiating $\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y}) = 0$ at $\mathbf{b}$ gives $A_x D\mathbf{g}(\mathbf{b}) + A_y = 0$.
Structural Commentary: Add $\mathbf{y}$ as a coordinate, then invert.
Key Insight: Invertibility of the x-block is precisely the linear ability to solve for x. □

Perspective 15.3.9. *Rank Theorem Linear Model*
Theorem 1.2.22 reduces a linear map to its kernel–range split.
The constant-rank theorem is the local C^1 analogue after coordinate changes.

Theorem 15.3.10. *Rank Theorem*
Let $\mathbf{F} : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be C^1 with constant rank r. At $\mathbf{a} \in E$, put $A = D\mathbf{F}(\mathbf{a})$, $Y_1 = \operatorname{range}(A)$, and let P be the orthogonal projection onto Y_1 (Definition 2.2.8). Locally, after a C^1 coordinate change H with C^1 inverse on a small coordinate neighborhood V_0, there is a C^1 map $\varphi : A(V_0) \subseteq Y_1 \to \ker(P)$ such that $\mathbf{F}(H(\mathbf{x})) = A\mathbf{x} + \varphi(A\mathbf{x})$. In coordinates adapted to $\ker A$ and $\operatorname{range}(A)$, this is the same local geometry as a projection onto r coordinates.

Proof. If $r = 0$, Theorem 15.2.11 makes $\mathbf{F}$ locally constant.
Assume $r > 0$. Choose $S : Y_1 \to \mathbb{R}^n$ with $AS = I_{Y_1}$, possible by choosing preimages of a basis of Y_1, and set $G(\mathbf{x}) = \mathbf{x} + SP(\mathbf{F}(\mathbf{x}) - A\mathbf{x})$. Since $DG(\mathbf{a}) = I$, Theorem 15.3.4 gives a local inverse $H = G^{-1}$.
Let $\Psi = \mathbf{F} \circ H$. From $AG = P\mathbf{F}$ we get $P\Psi(\mathbf{x}) = A\mathbf{x}$. Hence $PD\Psi(\mathbf{x}) = A$. The rank of $D\Psi(\mathbf{x})$ is r because $DH(\mathbf{x})$ is invertible and $\mathbf{F}$ has constant rank r; the rank of $PD\Psi(\mathbf{x})$ is also r.

Therefore P is injective on range $D\Psi(\mathbf{x})$, and $D\Psi(\mathbf{x})\mathbf{v} = 0$ whenever $A\mathbf{v} = 0$.
On a small convex coordinate neighborhood, any two points with the same $A\mathbf{x}$ differ by a vector in $\ker(A)$ and can be joined inside that affine fiber. The preceding derivative vanishing and Theorem 15.2.11 show that Ψ is constant on such fibers. Thus $(I - P)\Psi(\mathbf{x})$ depends only on $A\mathbf{x}$; call it $\varphi(A\mathbf{x})$. Then $\mathbf{F}(H(\mathbf{x})) = \Psi(\mathbf{x}) = A\mathbf{x} + \varphi(A\mathbf{x})$, and φ is C^1 because Ψ is C^1 and the dependence is through the coordinate $A\mathbf{x}$.
Structural Commentary: Constant rank gives a local graph normal form.
Key Insight: Same rank, same local level-set geometry. □

Rank-nullity (Theorem 1.3.19) says that the domain directions split into rank directions and fiber directions. The rank theorem says a constant-rank map has, locally, the same split after C^1 changes of coordinates.

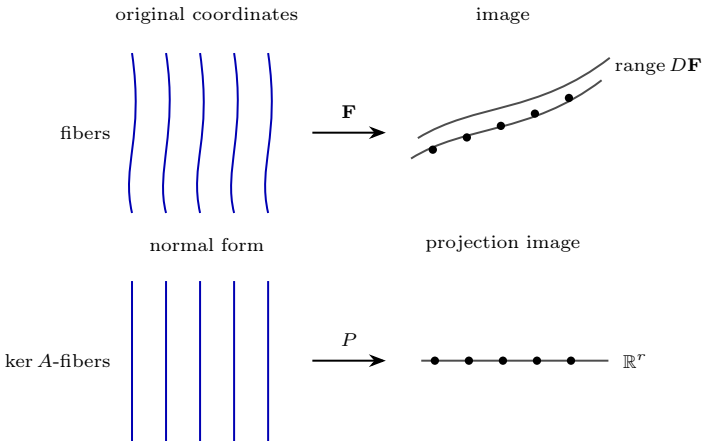


Figure 164: After local coordinate changes, a constant-rank map looks like projection onto the rank directions.

Theorem 15.3.11. *One-Constraint Lagrange Multiplier*
Let $E \subseteq \mathbb{R}^n$ be open, $f, g : E \to \mathbb{R}$ be C^1, and $S = \{\mathbf{x} : g(\mathbf{x}) = c\}$.
If $p \in S$, $\nabla g(p) \neq \mathbf{0}$, and $f|_S$ has a local extremum at p, then $\exists \lambda \in \mathbb{R}$ with $\nabla f(p) = \lambda \nabla g(p)$.

Proof. If $Df(p) = 0$, take $\lambda = 0$. Otherwise suppose $Df(p)$ is not a multiple of $Dg(p)$.
Complete $Dg(p), Df(p)$ to a dual basis $Dg(p), \quad Df(p), \quad \ell_3, \ldots, \ell_n$ by Theorem 1.4.3,
and set $\Phi(\mathbf{x}) = \big(g(\mathbf{x}) - c,\ f(\mathbf{x}) - f(p),\ \ell_3(\mathbf{x} - p), \ldots, \ell_n(\mathbf{x} - p)\big)$.
Then $D\Phi(p)$ is invertible, so Theorem 15.3.4 makes Φ locally invertible. In these coordinates, S is locally the slice with first coordinate 0, while the second coordinate is $f - f(p)$. Local invertibility gives nearby points of S with this second coordinate of both signs, contradicting extremality.
Hence $Df(p) = \lambda Dg(p)$; Riesz representation, Theorem 2.1.41, gives the gradient form.
Structural Commentary: The constraint kills first-order variation along S.
Key Insight: If Df were independent of Dg, f would be a free constraint coordinate. □

Method 15.3.12. *Several-Variable Differentiation Proof Patterns*

(i) *Linear approximation*: write $\mathbf{f}(\mathbf{x} + \mathbf{h}) - \mathbf{f}(\mathbf{x}) = D\mathbf{f}(\mathbf{x})\mathbf{h} + \mathbf{r}(\mathbf{h})$ with $\|\mathbf{r}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$.
(ii) *Norm estimates*: use $\|A\mathbf{h}\|_2 \leq \|A\|_{\text{op}} \|\mathbf{h}\|_2$ and the product estimate from Proposition 4.3.3.
(iii) *Line-segment reduction*: on convex sets, restrict to $\gamma(t) = (1 - t)\mathbf{a} + t\mathbf{b}$ and apply one-variable results.
(iv) *Local inversion*: if $A = D\mathbf{f}(\mathbf{a})$ is invertible, write $D\mathbf{f}(\mathbf{x}) = A + E_{\mathbf{x}}$ with $\|E_{\mathbf{x}}\|_{\text{op}}$ small and solve by contraction (Theorem 13.2.14); linear block equations use Theorem 15.3.7.
(v) *Coordinates*: read partial derivatives as the entries of the matrix of $D\mathbf{f}(\mathbf{x})$, using Theorem 15.2.9.
(vi) *Higher derivatives*: regard $D^r\mathbf{f}(p)$ as an r-linear map; recover coordinate partials via Theorem 15.2.16.
(vii) *Rank normal forms*: for $A = D\mathbf{F}(\mathbf{a})$, split domain by $\ker A$ and image by range A; use the quotient-by-kernel picture from Theorem 1.2.22 and project with $P_{\text{range } A}$.

Key Insight: Several-variable calculus is one-variable limiting plus finite-dimensional linear algebra.

15.4 Differential Forms and Manifolds

This subsection moves from Euclidean-region integration and change of variables to differential forms, pullbacks, and oriented parametrized surfaces.

Theorem 15.4.1. *Linear Maps Scale Volume by $|\det T|$*
Let $T \in \mathcal{L}(\mathbb{R}^n)$ and let $\Omega \subseteq \mathbb{R}^n$ be a sufficiently regular measurable set. Then $\text{vol}(T(\Omega)) = |\det T| \,\text{vol}(\Omega)$.

Proof. It is enough to prove the formula first for invertible T. Let $T = SP$ by polar decomposition, where S is an isometry and $P = \sqrt{T^*T}$ is positive definite. Isometries preserve Euclidean volume, so $\operatorname{vol}(T(\Omega)) = \operatorname{vol}(P(\Omega))$. By the real spectral theorem, choose orthonormal coordinates with $P = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ and $\lambda_j > 0$. Rectangles are sent to rectangles with side lengths multiplied by λ_j, hence their volumes are multiplied by $\prod_j \lambda_j = \det P$. Finite unions of disjoint rectangles have the same scaling by additivity, and the usual inner-outer rectangle approximation of a sufficiently regular measurable set gives $\operatorname{vol}(P(\Omega)) = (\det P)\operatorname{vol}(\Omega)$.
If T is singular, then $T(\Omega)$ lies in a proper subspace after an orthogonal change of coordinates, so its n-dimensional volume is 0; also $|\det T| = 0$. Finally, Proposition 4.2.29 gives $\det P = |\det T|$ in the invertible case.
Structural Commentary: Polar decomposition reduces volume scaling to diagonal stretching.
Key Insight: The determinant is the product of independent stretch factors. □

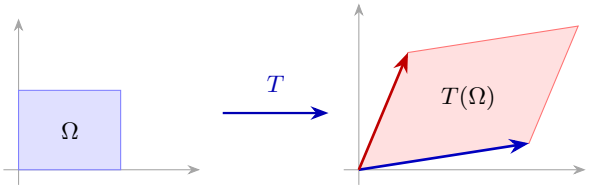


Figure 165: A linear map scales volume by $|\det T|$.

Method 15.4.2. *Change-of-Variables Jacobian Method*
For a differentiable map $\sigma : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$:

(i) compute the derivative matrix $D\sigma(\mathbf{x})$;
(ii) compute the local volume factor $|\det D\sigma(\mathbf{x})|$;
(iii) replace $d\mathbf{y}$ by $|\det D\sigma(\mathbf{x})|\, d\mathbf{x}$ when $\mathbf{y} = \sigma(\mathbf{x})$;
(iv) transform the domain bounds in the new coordinates.

Example 15.4.3. *Polar and Spherical Jacobian Factors*
For polar coordinates $\sigma(r, \theta) = (r\cos\theta, r\sin\theta)$, $D\sigma(r,\theta) = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}$, and $|\det D\sigma(r,\theta)| = r$.
Thus area integrals acquire the factor $r\, dr\, d\theta$.
For spherical coordinates $\sigma(\rho, \varphi, \theta) = (\rho\sin\varphi\cos\theta, \rho\sin\varphi\sin\theta, \rho\cos\varphi)$, and $|\det D\sigma(\rho,\varphi,\theta)| = \rho^2\sin\varphi$.
Thus volume integrals acquire the factor $\rho^2\sin\varphi\, d\rho\, d\varphi\, d\theta$.

Definition 15.4.4. *k-Cells and Multiple Integrals*
A *k-cell* is $I^k := \prod_{i=1}^{k} [a_i, b_i] \subseteq \mathbb{R}^k$ or $I^j := \prod_{i=1}^{j} [a_i, b_i]$.
For $f \in C(I^k)$, set $f_k = f$ and, for $j = k, \dots, 1$, define $f_{j-1}(x_1, \dots, x_{j-1}) := \int_{a_j}^{b_j} f_j(x_1, \dots, x_j)\, dx_j$.
The *integral of f over I^k* is $\int_{I^k} f(\mathbf{x})\, d\mathbf{x} := f_0$, also written $\int_{I^k} f$.

For bounded functions on rectangles, the Darboux picture from Definition 15.1.32 generalizes by replacing intervals with grid rectangles.

Definition 15.4.5. *Rectangular Grids and Riemann Sums*
For $R = [a,b] \times [c,d] \subseteq \mathbb{R}^2$ and partitions $P : a = x_0 < \cdots < x_m = b$; and $Q : c = y_0 < \cdots < y_n = d$, put $G = P \times Q$, where $R_{ij} := [x_{i-1}, x_i] \times [y_{j-1}, y_j]$ and $|R_{ij}| := (x_i - x_{i-1})(y_j - y_{j-1})$.
For sample points $(s_{ij}, t_{ij}) \in R_{ij}$, $R(f, G, S) := \sum_{i=1}^{m} \sum_{j=1}^{n} f(s_{ij}, t_{ij})\, |R_{ij}|$. For bounded real f, with $m_{ij} := \inf_{R_{ij}} f$ and $M_{ij} := \sup_{R_{ij}} f$, set $L(f, G) := \sum_{i,j} m_{ij}\, |R_{ij}|$ and $U(f, G) := \sum_{i,j} M_{ij}\, |R_{ij}|$.
The lower and upper integrals over R are $\sup_G L(f, G)$ and $\inf_G U(f, G)$. If equal, their common value is $\int_R f$.

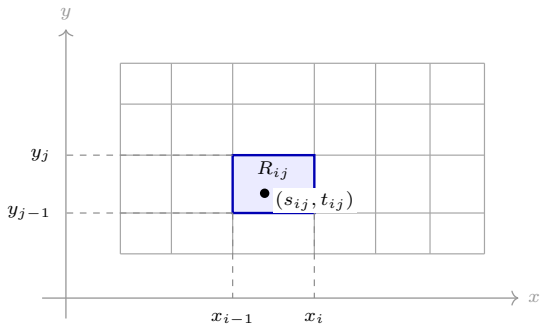


Figure 166: A grid rectangle R_{ij} and sample point (s_{ij}, t_{ij}).

Theorem 15.4.6. *Fubini on a Rectangle, Darboux Form*
Let $R = [a,b] \times [c,d]$ and let $f : R \to \mathbb{R}$ be Riemann integrable.
For $y \in [c,d]$, set $f_y(x) := f(x,y)$ and $\underline{F}(y) := \sup_P L(P, f_y)$; $\overline{F}(y) := \inf_P U(P, f_y)$, where P ranges over partitions of $[a,b]$. Then $\underline{F}, \overline{F}$ are Riemann integrable on $[c,d]$ and $\int_R f = \int_c^d \underline{F}(y)\,dy = \int_c^d \overline{F}(y)\,dy$.
If $\int_a^b f(x,y)\,dx$ exists for every y, this is the usual iterated integral formula.

Proof. For $G = P \times Q$, horizontal slice sums give $L(f,G) \le L(Q,\underline{F}) \le U(Q,\overline{F}) \le U(f,G)$.
As f is integrable on R, the outer terms can be made arbitrarily close. The upper–lower criterion, Theorem 15.1.41, squeezes $\underline{F}$ and $\overline{F}$ to the common value $\int_R f$.
Structural Commentary: Product grids turn area sums into slice sums.
Key Insight: Fubini is a squeeze between lower and upper sums before it is an equality of iterated integrals. □

Theorem 15.4.7. *Order Independence on Cells*
For $f \in C(I^k)$, all iterated orders defining $\int_{I^k} f$ give the same value.

Proof. Let L, L' be two iterated orders. For $h(\mathbf{x}) = h_1(x_1)\cdots h_k(x_k)$, $L(h) = L'(h) = \prod_{i=1}^{k} \int_{a_i}^{b_i} h_i(x_i)\,dx_i$, so $L(g) = L'(g)$ for every finite sum g of such products. These sums are dense in $C(I^k)$ by Theorem 14.4.22.
Choose g with $\|f - g\|_\infty < \varepsilon/V$, where $V = \prod_{i=1}^{k} (b_i - a_i)$.
Since $|L(r)|, |L'(r)| \le V \|r\|_\infty$, then $|L(f) - L'(f)| \le |L(f-g)| + |L'(f-g)| < 2\varepsilon$.
Structural Commentary: Product functions plus uniform approximation prove order independence.
Key Insight: Fubini-type order independence is first an algebraic fact, then a uniform approximation fact. □

Perspective 15.4.8. *Order Interchange Requires Hypotheses*
Finite sums are order-free; limiting processes are not.
Theorems 15.4.6 and 15.4.7 give hypotheses for interchanging integral order.

Definition 15.4.9. *Support and Simplex Integrals*
For $f : \mathbb{R}^k \to \mathbb{R}$ or $\mathbb{C}$, define $\operatorname{supp}(f) := \overline{\{\mathbf{x} : f(\mathbf{x}) \neq 0\}}$.
If f is continuous with compact support and $\operatorname{supp}(f) \subseteq I^k$, set $\int_{\mathbb{R}^k} f := \int_{I^k} f$. This is independent of the containing cell I^k. The *standard k-simplex* is $Q^k := \{\mathbf{x} \in \mathbb{R}^k : x_i \ge 0,\ x_1 + \cdots + x_k \le 1\}$.
For $f \in C(Q^k)$, extend f by 0 to unit cube; resulting order-independent iterated integral is denoted $\int_{Q^k} f$.

Definition 15.4.10. *Primitive Mappings and Flips*
Let $E \subseteq \mathbb{R}^n$ be open.
A map $G : E \to \mathbb{R}^n$ is *primitive* if, for some coordinate m and real function g, $G(\mathbf{x}) = \mathbf{x} + \big(g(\mathbf{x}) - x_m\big)\mathbf{e}_m$.
If g is differentiable at $\mathbf{a}$, then $DG(\mathbf{a})$ is the identity matrix with row m replaced by $\big((D_1 g)(\mathbf{a}), \ldots, (D_n g)(\mathbf{a})\big)$, hence $J_G(\mathbf{a}) = \det DG(\mathbf{a}) = (D_m g)(\mathbf{a})$ by Theorem 4.1.5.
A *flip* is the linear map that swaps two standard-basis vectors and fixes the rest.

Theorem 15.4.11. *Local Decomposition into Primitive Mappings*
Let $\mathbf{F} : E \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be C^1, let $\mathbf{0} \in E$, and suppose $\mathbf{F}(\mathbf{0}) = \mathbf{0}$ and $D\mathbf{F}(\mathbf{0})$ is invertible.
Then, near $\mathbf{0}$, $\mathbf{F}(\mathbf{x}) = B_1 \cdots B_{n-1} G_n \circ \cdots \circ G_1(\mathbf{x})$, where each G_i is a primitive C^1 map with $G_i(\mathbf{0}) = \mathbf{0}$ and $DG_i(\mathbf{0})$ invertible, and each B_i is a flip or the identity.

Proof. Let $P_m \mathbf{x} := x_1\mathbf{e}_1 + \cdots + x_m\mathbf{e}_m$. Put $\mathbf{F}_1 = \mathbf{F}$ and inductively assume that $\mathbf{F}_m$ is defined near $\mathbf{0}$, $\mathbf{F}_m(\mathbf{0}) = \mathbf{0}$, $D\mathbf{F}_m(\mathbf{0})$ is invertible, and $P_{m-1}\mathbf{F}_m = P_{m-1}$. Then $\mathbf{F}_m(\mathbf{x}) = P_{m-1}\mathbf{x} + \sum_{i=m}^{n} \alpha_i(\mathbf{x})\mathbf{e}_i$.

Since $D\mathbf{F}_m(\mathbf{0})\mathbf{e}_m \neq \mathbf{0}$, some $k \in \{m, \dots, n\}$ has $(D_m\alpha_k)(\mathbf{0}) \neq 0$. Let B_m swap coordinates m and k (or be the identity if $k = m$), and set $G_m(\mathbf{x}) := \mathbf{x} + \big(\alpha_k(\mathbf{x}) - x_m\big)\mathbf{e}_m$.
By Definition 15.4.10, G_m is primitive and $DG_m(\mathbf{0})$ is invertible. The inverse function theorem, Theorem 15.3.4, gives a local inverse for G_m, so define $\mathbf{F}_{m+1} := B_m\mathbf{F}_m \circ G_m^{-1}$. Then $P_m\mathbf{F}_{m+1} = P_m$, and the induction continues. After $m = n - 1$, $\mathbf{F}_n$ is primitive with invertible derivative.
Taking $G_n = \mathbf{F}_n$ and using $B_m^2 = I$ gives $\mathbf{F} = \mathbf{F}_1 = B_1 \cdots B_{n-1}G_n \circ G_{n-1} \circ \cdots \circ G_1$ near $\mathbf{0}$.
Structural Commentary: Invertible C^1 maps factor locally into coordinate changes and flips.
Key Insight: Gaussian elimination on $D\mathbf{F}$ becomes nonlinear via the inverse function theorem. □

Theorem 15.4.12. *Partition of Unity on a Compact Set*
Let $K \subseteq \mathbb{R}^n$ be compact and let $\{V_\alpha\}$ cover K by open sets. Then there are $\psi_1, \dots, \psi_s \in C^1(\mathbb{R}^n)$ such that

(i) $0 \leq \psi_i \leq 1$ *for all i;*
(ii) $\operatorname{supp}(\psi_i) \subseteq V_\alpha$ *for some α;*
(iii) $\sum_{i=1}^s \psi_i = 1$ *on K.*

If $f \in C(\mathbb{R}^n)$ and $\operatorname{supp}(f) \subseteq K$, then $f = \sum_{i=1}^s \psi_i f$, with each summand supported in one member of the cover.

Proof.

(i) For each $\mathbf{x} \in K$, choose balls with $\overline{B(\mathbf{x})} \subseteq W(\mathbf{x}) \subseteq \overline{W(\mathbf{x})} \subseteq V_{\alpha(\mathbf{x})}$.
By compactness, Definition 13.2.1, finitely many $B(\mathbf{x}_i)$ cover K.
(ii) Choose $\varphi_i \in C^1(\mathbb{R}^n)$ with $\varphi_i = 1$ on $B(\mathbf{x}_i)$ and $\operatorname{supp}(\varphi_i) \subseteq W(\mathbf{x}_i)$. Set $\psi_i = (1 - \varphi_1) \cdots (1 - \varphi_{i-1})\varphi_i$.
(iii) Then $0 \leq \psi_i \leq 1$, each ψ_i is supported in some V_α, and $\displaystyle\sum_{i=1}^{s} \psi_i = 1 - \prod_{i=1}^{s}(1 - \varphi_i)$, which equals 1 on K.
Multiplying by f gives the final assertion.

Structural Commentary: Compactness finite; product cutoffs sum to one.
Key Insight: A partition of unity turns local change-of-variable information into a global formula. □

The nonlinear change-of-variables theorem is the local volume-scaling formula from Theorem 15.4.1 applied cell by cell after a fine grid has been warped by a C^1 diffeomorphism.

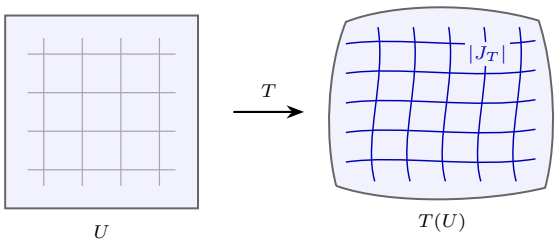


Figure 167: A C^1 change of variables turns a fine grid into curvilinear cells; $|J_T|$ records local volume scaling.

Theorem 15.4.13. *Change of Variables in $\mathbb{R}^k$*
Let $T : E \to \mathbb{R}^k$ be one-to-one and C^1 on open $E \subseteq \mathbb{R}^k$, with $J_T \neq 0$ on E.
If $f \in C(\mathbb{R}^k)$ and $\operatorname{supp}(f)$ is compact in $T(E)$, then $\int_{\mathbb{R}^k} f(\mathbf{y})\, d\mathbf{y} = \int_{\mathbb{R}^k} f(T(\mathbf{x}))\, |J_T(\mathbf{x})|\, d\mathbf{x}$.

Proof. Let $\mathcal{C}$ be the class of maps for which formula holds for all continuous functions with compact support in the image. Primitive maps lie in $\mathcal{C}$ by one-variable substitution, and flips lie in $\mathcal{C}$ by Theorem 15.4.7.
If $P, Q \in \mathcal{C}$, then $P \circ Q \in \mathcal{C}$, since Theorems 15.2.5 and 4.1.5 give $J_{P\circ Q}(\mathbf{x}) = J_P(Q(\mathbf{x}))J_Q(\mathbf{x})$.
After translating coordinates, Theorem 15.4.11 therefore gives the formula on a neighborhood of each point of E. By Theorem 15.3.4, these neighborhoods have open images covering $\operatorname{supp}(f)$.
Choose a partition of unity from Theorem 15.4.12, write $f = \sum_i \psi_i f$ with each $\psi_i f$ supported in one such image, apply the local formula to each summand, and sum.
Structural Commentary: Model cases plus partitions of unity prove the theorem.
Key Insight: The absolute Jacobian is local volume scaling, globalized by compact support. □

Method 15.4.14. *Multiple-Integration Proof Patterns*
For a multiple-integration identity:

(i) *Products*: prove product cases, then pass by uniform approximation;
(ii) *Grids*: compare product-grid upper and lower sums with one-variable slice sums;
(iii) *Support*: enclose compact support in a cell and check cell-independence;
(iv) *Local form*: reduce C^1 coordinate changes to primitive maps and flips;
(v) *Determinants*: use multiplicativity and Theorem 4.1.5;

(vi) *Globalize*: apply the partition of unity, Theorem 15.4.12.

Key Insight: Change of variables is local determinant geometry plus compactness.

Differential forms package oriented integration on parametrized cells. They use determinant sign and laws (Definition 4.1.1, Theorems 4.1.5, 4.1.12) and change of variables (Theorem 15.4.13); conceptually, forms test oriented cells.

Perspective 15.4.15. *Forms as Functionals*
The notation $\int f\,dx$ treats dx as directed coordinate variation. Differential forms make this precise: a 1-form such as $f\,dx + g\,dy$ assigns a number to an oriented path; dx records net x-variation, and $f\,dx$ weights it by f.

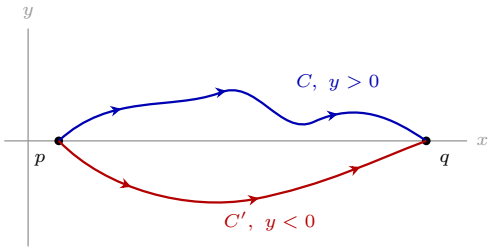


Figure 168: $y\,dx$ records signed, y-weighted x-variation.

Definition 15.4.16. *k-Surfaces and Differential Forms*
Let $E \subseteq \mathbb{R}^n$ be open. A *k-surface* in E is a C^1 map $\Phi : D \to E$ from a compact *parameter domain* $D \subseteq \mathbb{R}^k$, here a k-cell or the simplex Q^k of Definition 15.4.9.
For compact D, C^r means extension to an open neighborhood of D.
A *k-form* in E, $k \geq 1$, is $\omega = \sum a_{i_1\cdots i_k}(\mathbf{x})\,dx_{i_1} \wedge \cdots \wedge dx_{i_k}$, with continuous real coefficients on E, and assigns to each k-surface Φ the number $\int_D \sum a_{i_1\cdots i_k}(\Phi(\mathbf{u}))\frac{\partial(x_{i_1},\ldots,x_{i_k})}{\partial(u_1,\ldots,u_k)}\,d\mathbf{u}$.
A 0-form is a continuous function; a form is C^r if all coefficients are C^r.

Remark 15.4.17. *Parametrized Cells, Not Just Images*
Parametrizations may fail to be injective or regular. Forms therefore act on parametrized cells, not just images; this matters for orientation, multiplicity, and cancellation.

Proposition 15.4.18. *Basic Forms and Standard Presentation*
Swapping neighboring factors in $dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ changes sign, and repeated indices give 0. For an increasing k-index $I = (i_1, \ldots, i_k)$, write $dx_I := dx_{i_1} \wedge \cdots \wedge dx_{i_k}$.
Every k-form has a unique standard presentation $\omega = \sum_I b_I(\mathbf{x})\,dx_I$, where I ranges over increasing k-indices.

Proof. The sign rules follow from the determinant row-swap and repeated-row rules in Theorem 4.1.5, applied to the Jacobian in Definition 15.4.16. The sign needed to sort an index tuple is exactly the permutation sign of Definition 4.1.1; hence every term either vanishes or reduces to an increasing index, giving existence.
For uniqueness, suppose $\omega = \sum_I b_I(\mathbf{x})\,dx_I$ is a zero k-form in standard presentation. If $b_J(\mathbf{v}) \neq 0$ for some increasing $J = (j_1, \ldots, j_k)$, continuity gives a small cell $D = [-h, h]^k$ such that, for $\Phi(\mathbf{u}) = \mathbf{v} + \sum_{r=1}^{k} u_r \mathbf{e}_{j_r}$, the coefficient $b_J(\Phi(\mathbf{u}))$ has constant nonzero sign on D. In the defining integral, the J-Jacobian is 1 and every other standard Jacobian is 0, so $0 = \int_\Phi \omega = \int_D b_J(\Phi(\mathbf{u}))\,d\mathbf{u}$, a contradiction. Thus every b_I is zero.
Structural Commentary: Wedge signs are determinant signs.
Key Insight: Standard presentation removes ordered-index redundancy. □

Theorem 15.4.19. *Naturality Under Reparametrization*
Let $\Phi : D \to E \subseteq \mathbb{R}^n$ be a C^1 k-surface and let $R : D' \to D$ be a C^1 diffeomorphism.
For every k-form ω, $\int_{\Phi\circ R} \omega = \operatorname{sgn}(\det DR) \int_\Phi \omega$, with constant sign on D'.

Proof. It suffices to take $\omega = f\,dx_I$. By Theorem 4.1.12, the relevant minor of $D(\Phi \circ R)$ contributes the corresponding minor of $D\Phi$ times $\det DR$. Change of variables (Theorem 15.4.13) cancels $|\det DR|$ and leaves $\operatorname{sgn}(\det DR)$; linearity handles general forms.
Structural Commentary: Reparametrization changes coordinates, not the oriented cell.
Key Insight: The same determinant sign controls orientation and the sign of the form integral. □

Definition 15.4.20. *Product and Exterior Derivative*
For basic forms, define $dx_I \wedge dx_J$ by concatenating indices and reducing to standard presentation by Proposition 15.4.18; extend bilinearly. For a C^1 0-form f, set $df := \sum_{i=1}^{n} (D_i f)(\mathbf{x})\, dx_i$.
For a C^1 k-form $\omega = \sum_I b_I(\mathbf{x})\, dx_I$, set $d\omega := \sum_I (db_I) \wedge dx_I$.

Theorem 15.4.21. *Exterior Derivative Laws*
If ω, λ are C^1 forms in E and ω is a k-form, then $d(\omega \wedge \lambda) = (d\omega) \wedge \lambda + (-1)^k \omega \wedge d\lambda$. If ω is C^2, then $d^2\omega = 0$.

Proof. By bilinearity, it suffices to take $\omega = f\, dx_I$ and $\lambda = g\, dx_J$.
The ordinary product rule gives $d(fg) = f\, dg + g\, df$; moving dg past k factors in dx_I gives the displayed sign.
For a C^2 0-form f, $d^2 f = \sum_{i,j} (D_{ij} f)(\mathbf{x})\, dx_i \wedge dx_j = 0$, by Theorem 15.2.21 and antisymmetry.
The general case follows from $\omega = \sum_I b_I\, dx_I$ and the product rule just proved.
Structural Commentary: The exterior derivative is built to make mixed partial cancellation visible.
Key Insight: Symmetric second derivatives meet antisymmetric wedge products, so $d^2 = 0$. □

Definition 15.4.22. *Change of Variables for Forms*
Let $T : E \subseteq \mathbb{R}^n \to V \subseteq \mathbb{R}^m$ be C^1, with $T = (t_1, \ldots, t_m)$. For a k-form $\omega = \sum_I b_I(\mathbf{y})\, dy_I$ in V, define its *pullback* by $\omega_T := T^*\omega := \sum_I b_I(T(\mathbf{x}))\, dt_{i_1} \wedge \cdots \wedge dt_{i_k}$, and $dt_i := \sum_{j=1}^{n} (D_j t_i)(\mathbf{x})\, dx_j$.

Pullback is the differential-form analogue of the dual map in Definition 1.4.4: cells move forward by composition with T, while forms move backward so they can be evaluated before or after that forward motion.

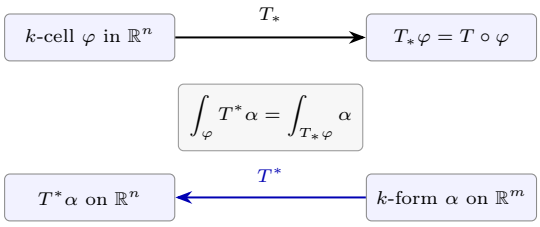


Figure 169: Cells push forward with T; forms pull back with T^*, so $T^*\alpha(\varphi) = \alpha(T_*\varphi)$.

Theorem 15.4.23. *Pullback Commutes with the Basic Operations*
With $T : E \to V$ as in Definition 15.4.22,

(i) $(\omega + \lambda)_T = \omega_T + \lambda_T$ *for forms of the same order;*
(ii) $(\omega \wedge \lambda)_T = \omega_T \wedge \lambda_T$*;*
(iii) $d(\omega_T) = (d\omega)_T$ *when ω is C^1 and T is C^2;*
(iv) if $S : V \to W$ is C^1, then $(\omega_S)_T = \omega_{S \circ T}$.

Moreover, if Φ is a k-surface in E and ω is a k-form in V, then $\int_{T \circ \Phi} \omega = \int_\Phi \omega_T$.

Proof.
(i) Addition is the linearity of Definition 15.4.22.
(ii) Pullback replaces each dy_i by dt_i; bilinearity gives $(\omega \wedge \lambda)_T = \omega_T \wedge \lambda_T$.
(iii) Let $\omega = \sum_I b_I\, dy_I$. Since $d(b_I \circ T) = (db_I)_T$ (Theorem 15.2.5) and $d(dt_i) = d^2 t_i = 0$, $d(\omega_T) = (d\omega)_T$.
(iv) If $S = (s_1, \ldots, s_p)$, both $(\omega_S)_T$ and $\omega_{S \circ T}$ replace dz_q by $d(s_q \circ T)$, by the chain rule.
For the integral identity, expand both sides over D; the Jacobian minors agree by Theorem 4.1.12, since $D(T \circ \Phi) = DT(\Phi) D\Phi$.
Structural Commentary: Pullback translates forms before integration.
Key Insight: Forms are defined so that composition of parameterizations becomes an algebraic substitution. □

15.5 Stokes Theory

Stokes theory relates exterior differentiation to oriented boundary, with Green's theorem, classical Stokes, and divergence as special cases.

Definition 15.5.1. *Chains and Boundaries*
For $\mathbf{p}_0, \ldots, \mathbf{p}_k \in \mathbb{R}^n$, oriented affine k-simplex $\sigma = [\mathbf{p}_0, \ldots, \mathbf{p}_k]$ is k-surface on Q^k: $\sigma(\mathbf{u}) = \mathbf{p}_0 + \sum_{i=1}^{k} u_i(\mathbf{p}_i - \mathbf{p}_0)$.
Permuting vertices multiplies σ by the permutation sign. An *affine k-chain* is a finite formal sum of oriented

affine k-simplices; integration is extended linearly.
For $k \geq 1$, define $\partial_{\mathrm{ch}}[\mathbf{p}_0, \ldots, \mathbf{p}_k] := \sum_{j=0}^{k} (-1)^j [\mathbf{p}_0, \ldots, \mathbf{p}_{j-1}, \mathbf{p}_{j+1}, \ldots, \mathbf{p}_k]$, and extend ∂_{ch} linearly. If $\Phi = T \circ \sigma$ is a C^2 oriented simplex, set $\partial_{\mathrm{ch}}\Phi := T(\partial_{\mathrm{ch}}\sigma)$, again extended linearly to chains.

Theorem 15.5.2. *Stokes' Theorem*
If Ψ is a C^2 k-chain in open $V \subseteq \mathbb{R}^m$ and ω is a C^1 $(k-1)$-form in V, then $\int_\Psi d\omega = \int_{\partial_{\mathrm{ch}}\Psi} \omega$.

Proof. By linearity, take $\Psi = \Phi = T \circ \sigma$, where σ is the standard oriented k-simplex.
By Theorem 15.4.23, $\int_\Phi d\omega = \int_\sigma (d\omega)_T = \int_\sigma d(\omega_T)$; and $\int_{\partial_{\mathrm{ch}}\Phi} \omega = \int_{\partial_{\mathrm{ch}}\sigma} \omega_T$, remains to prove formula on σ.
By linearity, reduce to $\lambda = f\, dx_1 \wedge \cdots \wedge \widehat{dx_r} \wedge \cdots \wedge dx_k$.
The boundary formula in Definition 15.5.1 leaves only the two faces in the x_r direction, and the resulting identity is the one-variable fundamental theorem of calculus inside the iterated integral.
Structural Commentary: Pullback reduces Stokes' theorem to a computation on the standard simplex.
Key Insight: The boundary signs are exactly the signs produced by the missing wedge factor. □

Definition 15.5.3. *Closed and Exact Forms*
A k-form ω in E is *exact* if $\omega = d\lambda$ for some $(k-1)$-form λ in E, and *closed* if ω is C^1 with $d\omega = 0$.
By Theorem 15.4.21, exact C^1 forms are closed.

Perspective 15.5.4. *Closed Is Local, Exact Is Global*
Exact $\Rightarrow$ closed is just $d^2 = 0$; the converse depends on the domain.
On $\mathbb{R}^2 \setminus \{\mathbf{0}\}$ the standard warning example is $\omega = \frac{-y}{x^2+y^2}\, dx + \frac{x}{x^2+y^2}\, dy$, which is closed but not exact.
Holes can obstruct global antiderivatives even when local derivative tests vanish.

Theorem 15.5.5. *Poincare Lemma on Convex Sets*
If $E \subseteq \mathbb{R}^n$ is convex and open, then every closed C^1 k-form in E, $k \geq 1$, is exact.
The same holds on every open set C^2-equivalent to a convex open set.

Proof. Choose $\mathbf{a} \in E$ and let $H_t(\mathbf{x}) = \mathbf{a} + t(\mathbf{x} - \mathbf{a})$, $0 \leq t \leq 1$. Convexity gives $H_t(E) \subseteq E$. For a k-form ω, define the $(k-1)$-form as $(K\omega)_{\mathbf{x}}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}) := \int_0^1 t^{k-1} \omega_{H_t(\mathbf{x})}(\mathbf{x} - \mathbf{a}, \mathbf{v}_1, \ldots, \mathbf{v}_{k-1})\, dt$.
A coordinate differentiation under the integral gives the homotopy identity $\omega - H_0^*\omega = dK\omega + Kd\omega$. Since $k \geq 1$, the constant map H_0 pulls every k-form back to 0. If $d\omega = 0$, then $\omega = d(K\omega)$, so ω is exact.
If E is C^2-equivalent to a convex open set C by $\Phi : C \to E$, then $\Phi^*\omega$ is closed by Theorem 15.4.23. The convex case gives $\Phi^*\omega = d\eta$ on C. Applying $(\Phi^{-1})^*$ and using the same pullback law gives $\omega = d((\Phi^{-1})^*\eta)$ on E.
Structural Commentary: Convexity lets coefficient-level antiderivatives be chosen consistently.
Key Insight: Closed is local symmetry; convexity supplies the global primitive. □

Theorem 15.5.6. *Vector Analysis Dictionary in $\mathbb{R}^3$*
For $\mathbf{F} = F_1\mathbf{e}_1 + F_2\mathbf{e}_2 + F_3\mathbf{e}_3$ on open $E \subseteq \mathbb{R}^3$,
set $\lambda_{\mathbf{F}} := F_1\, dx + F_2\, dy + F_3\, dz$ and $\omega_{\mathbf{F}} := F_1\, dy \wedge dz + F_2\, dz \wedge dx + F_3\, dx \wedge dy$. Then
$\mathbf{F} = \nabla u \iff \lambda_{\mathbf{F}} = du$; and $\nabla \times \mathbf{F} = 0 \iff d\lambda_{\mathbf{F}} = 0$; and
$\mathbf{F} = \nabla \times \mathbf{G} \iff \omega_{\mathbf{F}} = d\lambda_{\mathbf{G}}$; and $\nabla \cdot \mathbf{F} = 0 \iff d\omega_{\mathbf{F}} = 0$.
Hence $\nabla \times (\nabla u) = 0$ and $\nabla \cdot (\nabla \times \mathbf{G}) = 0$; on sets C^2-equivalent to convex open sets, the converses hold.

Proof. Expand $d\lambda_{\mathbf{F}}$ and $d\omega_{\mathbf{F}}$ by Definition 15.4.20; their coefficients are exactly $\nabla \times \mathbf{F}$ and $\nabla \cdot \mathbf{F}$. Potentials and vector potentials are exactness of $\lambda_{\mathbf{F}}$ and $\omega_{\mathbf{F}}$; the converses follow from Theorem 15.5.5.
Structural Commentary: Gradient, curl, and divergence are exterior derivative in degrees $0, 1, 2$.
Key Insight: The familiar vector identities are $d^2 = 0$ written in $\mathbb{R}^3$ notation. □

Theorem 15.5.7. *Green, Classical Stokes, and Divergence*
The following are special cases of Theorem 15.5.2.

(i) If $\Omega \subseteq \mathbb{R}^2$ has positively oriented boundary, $\alpha, \beta \in C^1$ near Ω, then $\int_{\mathrm{bd}\,\Omega} (\alpha\, dx + \beta\, dy) = \int_\Omega \left(\frac{\partial \beta}{\partial x} - \frac{\partial \alpha}{\partial y} \right) dA$
(ii) If Φ is a C^2 2-surface in $\mathbb{R}^3$ and $\mathbf{F} \in C^1$ near Φ, then $\int_\Phi (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dA = \int_{\partial_{\mathrm{ch}}\Phi} \mathbf{F} \cdot \mathbf{t}\, ds$.
(iii) If $\Omega \subseteq \mathbb{R}^3$ has positively oriented boundary and $\mathbf{F} \in C^1$ near Ω, then $\int_\Omega (\nabla \cdot \mathbf{F})\, dV = \int_{\mathrm{bd}\,\Omega} \mathbf{F} \cdot \mathbf{n}\, dA$

Proof.
(i) Put $\lambda = \alpha\, dx + \beta\, dy$. Then $d\lambda = (\partial\beta/\partial x - \partial\alpha/\partial y)\, dA$.
(ii) Put $\lambda = \lambda_{\mathbf{F}}$. By Theorem 15.5.6, $d\lambda = \omega_{\nabla \times \mathbf{F}}$.
(iii) Put $\omega = \omega_{\mathbf{F}}$. Then $d\omega = (\nabla \cdot \mathbf{F})\, dx \wedge dy \wedge dz$.

Theorem 15.5.2, together with the line, area, and volume element translations, gives the three displayed formulas.
Structural Commentary: Forms translate Stokes into the classical vector formulas.
Key Insight: Boundary plus d is the content; vector notation is translation. □

Method 15.5.8. *Differential-Forms Proof Patterns*

(i) *Signs*: compute wedge signs by determinant swaps and permutation parity.
(ii) *Standard presentation*: remove repeated indices and sort the rest.
(iii) *Pullback*: substitute dt_i for dy_i and apply the chain rule.
(iv) *Stokes*: pull back to the standard simplex, prove the coordinate case, extend linearly.
(v) *Closed versus exact*: use $d^2 = 0$ for necessity; use convexity or C^2-equivalence for primitives.
(vi) *Vector analysis*: translate fields to forms, apply Stokes, translate back.

Key Insight: Forms package signs, orientation, and parameter changes.

16 Measure and Integration Structure

16.1 Sigma-Algebras and Measures

Measure theory first builds measurable sets, then measurable functions and integrals. Outer measure covers from outside; measurability splits test sets without losing measure.

Definition 16.1.1. *Rings, σ-Rings, and Set Functions*
A *ring* of sets is closed under finite unions and relative complements.
A *σ-ring* is also closed under countable unions.
A *set function* on $\mathcal{A}$ is an extended-real map on $\mathcal{A}$ whose range does not contain both $+\infty$ and $-\infty$.
It is *additive* if $A \cap B = \emptyset$ implies $\phi(A \cup B) = \phi(A) + \phi(B)$.
It is *countably additive* if the same rule holds for pairwise disjoint countable unions: $\phi\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \phi(A_n)$.
Structural Commentary: Rings support finite additivity; σ-rings support countable limits.

Proposition 16.1.2. *Basic Laws for Additive Set Functions*
Let ϕ be additive on a ring $\mathcal{A}$. Then:

(i) $\phi(\emptyset) = 0$;
(ii) $\phi(A_1 \cup \cdots \cup A_n) = \phi(A_1) + \cdots + \phi(A_n)$ *for pairwise disjoint* $A_i \in \mathcal{A}$;
(iii) $\phi(A \cup B) + \phi(A \cap B) = \phi(A) + \phi(B)$;
(iv) if ϕ is nonnegative and $A \subseteq B$, then $\phi(A) \le \phi(B)$;
(v) if $B \subseteq A$ and $|\phi(B)| < +\infty$, then $\phi(A \setminus B) = \phi(A) - \phi(B)$.

Proof.

(i) Apply additivity to the empty disjoint union.
(ii) Induct on n, applying additivity to $(A_1 \cup \cdots \cup A_{n-1}) \cup A_n$.
(iii) By disjoint decompositions, $A \cup B = (A \setminus B) \cup (B \setminus A) \cup (A \cap B)$; $A = (A \setminus B) \cup (A \cap B)$; $B = (B \setminus A) \cup (A \cap B)$
(iv) If $A \subseteq B$, then $B = A \cup (B \setminus A)$ disjointly, so nonnegativity gives $\phi(A) \le \phi(B)$.
(v) If $B \subseteq A$, then $A = B \cup (A \setminus B)$ disjointly; since $|\phi(B)| < +\infty$, subtract $\phi(B)$.

Structural Commentary: Finite additivity plus disjoint decompositions controls all elementary identities.
Key Insight: Rewrite overlaps as disjoint pieces before applying ϕ. □

Theorem 16.1.3. *Continuity from Below*
Suppose ϕ is countably additive on a ring $\mathcal{A}$.
If $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$, and $A = \bigcup_{n=1}^{\infty} A_n$ with all $A_n, A \in \mathcal{A}$, then $\phi(A_n) \to \phi(A)$.

Proof. Put $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \ge 2$. Then B_n are pairwise disjoint, $A_n = B_1 \cup \cdots \cup B_n$, and $A = \bigcup_n B_n$. Countable additivity turns $\phi(A_n)$ and $\phi(A)$ into partial sums and full sum of same series.
Structural Commentary: Increasing unions are disjoint tail decompositions in disguise.
Key Insight: Countable additivity converts set convergence into series convergence. □

Definition 16.1.4. *Elementary Sets and Regularity in $\mathbb{R}^p$*
An *interval* in $\mathbb{R}^p$ is a coordinate box $a_i \le x_i \le b_i$, with endpoints optionally omitted; degenerate and empty cases are allowed. Define $m(I) := \prod_{i=1}^{p} (b_i - a_i)$ independent of endpoint inclusion.
A finite union of intervals is an *elementary set*; write $\mathcal{E}$ for all such sets.
If $A = I_1 \cup \cdots \cup I_n$ is disjoint, set $m(A) := m(I_1) + \cdots + m(I_n)$.

A nonnegative additive set function μ on $\mathcal{E}$ is *regular* if for every $A \in \mathcal{E}$ and $\varepsilon > 0$ there are elementary sets $F \subseteq A \subseteq G$, with F closed and G open, such that $\mu(G) - \varepsilon \le \mu(A) \le \mu(F) + \varepsilon$. The source verifies that $\mathcal{E}$ is a ring, every elementary set has a disjoint interval decomposition, and m is well defined, additive, and regular.

Perspective 16.1.5. *Box Volume and Linear Algebra*
For a box $B = \prod_i (a_i, b_i)$ or $\prod_i [a_i, b_i]$, $m(B) = \prod_i (b_i - a_i)$, the diagonal case of determinant volume scaling. Measure extends this product rule beyond boxes.
Key Insight: Lebesgue measure starts from determinant-style coordinate volume.

Definition 16.1.6. *Outer Measure and μ-Measurable Sets*
Let μ be additive, regular, nonnegative, and finite on $\mathcal{E}$.
For $E \subseteq \mathbb{R}^p$, define $\mu^*(E) := \inf \sum_{n=1}^{\infty} \mu(A_n)$, where infimum is over open elementary covers $E \subseteq \bigcup_{n=1}^{\infty} A_n$.
For $A, B \subseteq \mathbb{R}^p$, set $A \triangle B := (A \setminus B) \cup (B \setminus A)$ and $d_\mu(A, B) := \mu^*(A \triangle B)$.
A set is *finitely μ-measurable* if it is a d_μ-limit of elementary sets.
It is *μ-measurable* if it is a countable union of such sets; denote the classes by $\mathcal{M}_F(\mu)$ and $\mathcal{M}(\mu)$.

Definition 16.1.7. *Box-Cover Outer Measure Model*
An *open box* in $\mathbb{R}^n$ is $B = \prod_{i=1}^{n} (a_i, b_i) = \{(x_1, \dots, x_n) \in \mathbb{R}^n : x_i \in (a_i, b_i)\ \forall i\}$, $b_i \ge a_i$, $\mathrm{vol}_n(B) := \prod_{i=1}^{n} (b_i - a_i)$.
For $\Omega \subseteq \mathbb{R}^n$, the *box-cover outer measure* is $m_n^*(\Omega) := \inf \left\{ \sum_{j \in J} \mathrm{vol}_n(B_j) : (B_j)_{j \in J} \text{ covers } \Omega,\ J \text{ at most countable} \right\}$.
This is Definition 16.1.6 for Lebesgue box volume.

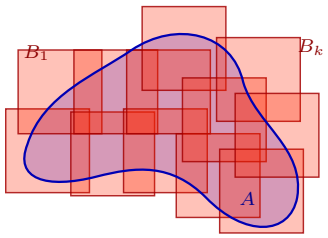


Figure 170: Cover $A \subseteq \bigcup_k B_k$ by squares; minimize $\sum_k |B_k|$.

Perspective 16.1.8. *Outer Measure as Outside Approximation*
Outer measure covers A from outside; $d_\mu(A, B) = \mu^*(A \triangle B)$ measures closeness.
Key Insight: Countable subadditivity collects near-optimal covers.

Method 16.1.9. *Lebesgue Measure Construction*

(i) *Cover from outside*
For $A \subseteq \mathbb{R}^n$, define $m_n^*(A) := \inf \left\{ \sum_{j \in J} \mathrm{vol}_n(B_j) : (B_j)_{j \in J} \text{ covers } A,\ J \text{ at most countable} \right\}$.
This is defined for every set, but it is only countably subadditive in general.

(ii) *Keep the clean splitters*
Call A measurable when every test set $X \subseteq \mathbb{R}^n$ splits additively: $m_n^*(X) = m_n^*(X \cap A) + m_n^*(X \cap A^c)$.
These measurable sets form a σ-algebra.

(iii) *Restrict the outer measure*
Define Lebesgue measure on measurable sets by $m(A) := m_n^*(A)$.
On this restricted class, m is countably additive.

Key Insight: Outer measure sizes all sets; clean splitting restores additivity.

Theorem 16.1.10. *Extension to a Measure*
Let μ be additive, regular, nonnegative, and finite on $\mathcal{E}$. Form μ^* from open elementary covers, set $d_\mu(A, B) = \mu^*(A \triangle B)$, and let $\mathcal{M}(\mu)$ be the countable unions of d_μ-limits of elementary sets. Then:

(i) $\mu^*(A) = \mu(A)$ for $A \in \mathcal{E}$;
(ii) μ^* is countably subadditive;
(iii) $\mathcal{M}(\mu)$ is a σ-ring;
(iv) μ^* is countably additive on $\mathcal{M}(\mu)$.

Proof.

(i) Regularity gives $\mu^*(A) \le \mu(A)$; compact closed approximants inside A give the reverse inequality.
(ii) Cover each E_n within $2^{-n}\varepsilon$ of $\mu^*(E_n)$; the combined cover proves subadditivity.

(iii) If $A_k \to A$, $B_k \to B$ in d_μ, $A_k, B_k \in \mathcal{E}$, then $d_\mu(A_k \diamond B_k, A \diamond B) \le d_\mu(A_k, A) + d_\mu(B_k, B) \to 0$, $\diamond \in \{\cup, \cap, \setminus\}$. Thus $\mathcal{M}_F(\mu)$ is a ring. If $A = \bigcup_k A_k$ with $A_k \in \mathcal{M}_F(\mu)$, then $A \setminus B = \bigcup_k (A_k \setminus B)$, so $\mathcal{M}(\mu)$ is a σ-ring.

(iv) For $A = \bigsqcup_n A_n$, $A_n \in \mathcal{M}_F(\mu)$, subadditivity gives $\le$,

$$\text{while finite additivity gives } \mu^*(A) \ge \mu^*\left(\bigsqcup_{n=1}^{N} A_n\right) = \sum_{n=1}^{N} \mu^*(A_n).$$

Let $N \to \infty$, then decompose general disjoint families in $\mathcal{M}(\mu)$ into finite-measurable pieces.

Structural Commentary: Elementary sets are completed under the distance measured by symmetric difference.
Key Insight: Approximate by elementary sets, prove identities there, then pass through $d_\mu(A, B)$. □

Definition 16.1.11. *Extended and Lebesgue Measure*
The *extended measure* is $\mu^*|_{\mathcal{M}(\mu)}$, denoted again by μ.
For the elementary volume m, it is *Lebesgue measure* on $\mathbb{R}^p$.

Definition 16.1.12. *Abstract Outer Measure and Clean Division*
An *abstract outer measure* on M is a monotone, countably subadditive function $\omega : 2^M \to [0, +\infty]$ with $\omega(\emptyset) = 0$.
A set $E \subseteq M$ is *ω-measurable* if it cleanly splits every $X \subseteq M$: $\omega(X) = \omega(X \cap E) + \omega(X \cap E^c)$.
For Lebesgue outer measure, this is Caratheodory measurability.

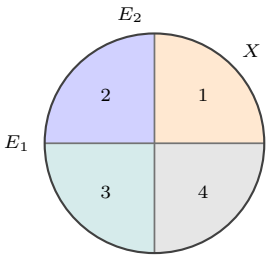


Figure 171: Two cuts split X into four regions; $E_1 \setminus E_2$ is region 2.

Theorem 16.1.13. *Caratheodory Measurable Sets*
For any abstract outer measure ω on M, the ω-measurable sets form a σ-algebra, and ω is countably additive there. Zero sets are measurable and may be added to or removed from measurable sets without changing measure.

Proof. If $\omega(Z) = 0$, then for every $X \subseteq M$, $\omega(X) \le \omega(X \cap Z) + \omega(X \cap Z^c) \le 0 + \omega(X)$, so Z is measurable; the same estimate handles changes by zero sets.
Complements are immediate, clean splitting is stable under differences, hence finite unions. If $E = \bigsqcup_i E_i$ with each E_i measurable, finite additivity on partial unions and subadditivity give $\omega(X) = \sum_{i=1}^{\infty} \omega(X \cap E_i) + \omega(X \cap E^c)$.
With $X = E$ this is countable additivity; for arbitrary X it proves E is measurable.
Disjointizing reduces general countable unions to this case.
Structural Commentary: Measurable sets preserve countable additivity.
Key Insight: Measurable cuts turn subadditivity into equality. □

Lemma 16.1.14. *Open Sets Are Countable Unions of Rational Boxes*
Every open set $G \subseteq \mathbb{R}^n$ can be written as a finite or countable union of open boxes with rational endpoints.

Proof. The rational boxes are countable: each is determined by $2n$ rationals, and Corollary 12.3.15 with Theorem 12.3.13 makes finite rational tuples countable. Let Σ be the rational boxes contained in G. Then $\bigcup_{B \in \Sigma} B \subseteq G$.
If $\mathbf{x} \in G$, choose $r > 0$ with $B_r(\mathbf{x}) \subseteq G$. By Corollary 12.2.26, pick rationals a_i, b_i satisfying $x_i - r/n < a_i < x_i < b_i < x_i + r/n$. Then $\mathbf{x} \in \prod_i (a_i, b_i) \subseteq B_r(\mathbf{x}) \subseteq G$, so $\mathbf{x} \in \bigcup_{B \in \Sigma} B$.
Structural Commentary: Rational boxes give a countable Euclidean base.
Key Insight: Openness supplies a ball; density of $\mathbb{Q}$ supplies rational box endpoints inside it. □

Theorem 16.1.15. *Lebesgue Measure Basic Properties*
Let $\mathcal{L}^n$ be the Lebesgue measurable subsets of $\mathbb{R}^n$, let $\mathcal{B}(\mathbb{R}^n)$ be the Borel σ-algebra, and set $m = m_n^*|_{\mathcal{L}^n}$.
Then $\mathcal{B}(\mathbb{R}^n) \subseteq \mathcal{L}^n$, and for $E, E_k \in \mathcal{L}^n$, E^c, $\bigcup_{k=1}^{\infty} E_k$, $\bigcap_{k=1}^{\infty} E_k \in \mathcal{L}^n$.

For $E, F, E_k \in \mathcal{L}^n$ and $t \in \mathbb{R}^n$, $0 \le m(E)$, $E \subseteq F \implies m(E) \le m(F)$, and $m(\bigcup_{k=1}^{\infty} E_k) \le \sum_{k=1}^{\infty} m(E_k)$.

Hence $E_i \cap E_j = \emptyset$ $(i \ne j) \implies m\left(\bigsqcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} m(E_k)$; $m([0,1]^n) = 1$; $E + t \in \mathcal{L}^n$; and $m(E + t) = m(E)$.

Proof. Theorems 16.1.10 and 16.1.13 make $\mathcal{L}^n$ a σ-algebra and make m nonnegative and countably additive on it; monotonicity and subadditivity follow from the outer measure m_n^*. Open boxes are elementary, hence measurable, so Lemma 16.1.14 puts every open set in $\mathcal{L}^n$; therefore $\mathcal{B}(\mathbb{R}^n) = \sigma(\text{open sets}) \subseteq \mathcal{L}^n$, and closed sets follow by complements. Definition 16.1.7 gives $m([0,1]^n) = 1$. Since box volume is translation invariant, translating covers gives $m_n^*(A+t) = m_n^*(A)$ for every $A \subseteq \mathbb{R}^n$; the Caratheodory equality is translation invariant, hence $E + t \in \mathcal{L}^n$ and $m(E+t) = m(E)$.
Structural Commentary: The wish list is recovered only after restricting outer measure to measurable sets.
Key Insight: The box-cover construction gives size; the measurable-set condition restores additivity. □

Perspective 16.1.16. *Why Not Measure Every Subset?*
Vitali-type constructions show that m_1^* cannot be additive on all subsets of $\mathbb{R}$: disjoint sets can violate countable additivity, and refinements can violate finite additivity.
Measurability keeps only the clean splitters, where additivity survives.
Key Insight: Outer measure exists for all sets, but measure is reserved for sets that split every test set cleanly.

Corollary 16.1.17. *Measure Continuity*
Let μ be a measure. If $E_k \uparrow E$, then $\mu(E_k) \uparrow \mu(E)$. If $F_k \downarrow F$ and $\mu(F_1) < +\infty$, then $\mu(F_k) \downarrow \mu(F)$.

Proof. Disjointize $E_k \uparrow E$ by $E_1' = E_1$ and $E_k' = E_k \setminus E_{k-1}$; countable additivity makes $\mu(E_k)$ the kth partial sum for $\mu(E)$. For $F_k \downarrow F$, apply this to $F_1 \setminus F_k \uparrow F_1 \setminus F$ and subtract from finite $\mu(F_1)$.
Structural Commentary: Monotone set limits are disjoint series or complements of such series.
Key Insight: Downward continuity needs one finite anchor so subtraction is legal. □

Definition 16.1.18. *Null Sets and Real-Line Measure Zero*
In a measure space, N is *null* if it is measurable and $\mu(N) = 0$.
A set $A \subseteq \mathbb{R}$ has *measure zero* if, for every $\varepsilon > 0$, open intervals O_n cover A and satisfy $\sum_{n=1}^{\infty} |O_n| \le \varepsilon$.
For Lebesgue measure, null sets are also *zero sets*.
A subset of a compact interval has *content zero* if the same condition holds with finitely many intervals.

Proposition 16.1.19. *Basic Measure-Zero Examples on $\mathbb{R}$*
Every finite $A \subseteq \mathbb{R}$ has content 0; every countable $A \subseteq \mathbb{R}$ has measure 0; Cantor set P has content 0.

Proof. Let $\varepsilon > 0$. If $A = \{a_1, \dots, a_N\}$ with $N \ge 1$, choose open intervals $I_j \ni a_j$ with $|I_j| < \varepsilon/N$; then $A \subseteq \bigcup_{j=1}^N I_j$ and $\sum_{j=1}^N |I_j| < \varepsilon$. The case $A = \emptyset$ is trivial.
If $A = \{a_j : j \in \mathbb{N}\}$, choose open intervals $I_j \ni a_j$ with $|I_j| < \varepsilon 2^{-j}$; then $A \subseteq \bigcup_j I_j$ and $\sum_j |I_j| < \varepsilon$.
For P, choose N with $(2/3)^N < \varepsilon/2$. Write $E_N = \bigcup_{k=1}^{2^N} J_k$, where each J_k is closed and $|J_k| = 3^{-N}$. Choose open $I_k \supseteq J_k$ with $|I_k| < 3^{-N} + \varepsilon/2^{N+1}$. Since $P \subseteq E_N$, the I_k cover P, and $\sum_{k=1}^{2^N} |I_k| < \left(\frac{2}{3}\right)^N + \frac{\varepsilon}{2} < \varepsilon$.
Structural Commentary: Zero size means arbitrarily small interval covers.
Key Insight: Lists split ε by terms; Cantor stages have length $(2/3)^N \to 0$. □

Remark 16.1.20. *Borel Sets, Null Sets, and Regular Approximation*
Open, closed, and Borel subsets of $\mathbb{R}^p$ are measurable. Each $A \in \mathcal{M}(\mu)$ admits closed–open approximations $F \subseteq A \subseteq G$ with $\mu(G \setminus A)$ and $\mu(A \setminus F)$ arbitrarily small. In Lebesgue measure, one may take $F_\sigma \subseteq A \subseteq G_\delta$ with $\mu(G_\delta \setminus F_\sigma) = 0$. Null sets form a σ-ring, and countable subsets of $\mathbb{R}^p$ are null.
Structural Commentary: Measurable sets are Borel sets modified by null sets.
Key Insight: Regularity explains why open and closed approximations are enough for integration.

16.2 Measurable Functions

Definition 16.2.1. *Borel versus Lebesgue Measurability*
The *Borel σ-algebra* on $\mathbb{R}^p$ is generated by open sets; its elements are *Borel sets*. A function is *Borel measurable* if inverse images of Borel sets are Borel. Lebesgue measurability contains all Borel sets and all subsets of Lebesgue null sets; this completion goes beyond the Borel σ-algebra.

Perspective 16.2.2. *Topology versus Completion*
Borel measurability tracks topology; Lebesgue measurability completes measure by adding subsets of null sets.
Key Insight: Borel sets are topological; Lebesgue sets are complete.

Definition 16.2.3. *Measure Spaces and Measurable Functions*
A *measure space* is a set X together with a σ-ring $\mathcal{M}$ of subsets of X and a nonnegative countably additive set function μ on $\mathcal{M}$. If $X \in \mathcal{M}$, then X is a *measurable space*.
For $\bowtie \in \{>, \ge, <, \le\}$, write $\{f \bowtie a\} := \{x \in X : f(x) \bowtie a\}$.
An extended-real-valued f on X is *measurable* if $\{f > a\} \in \mathcal{M}$ for every $a \in \mathbb{R}$.

Proposition 16.2.4. *Open-Set Test for Euclidean Targets*
Let $\Omega \subseteq \mathbb{R}^n$ be Lebesgue measurable and let $f : \Omega \to \mathbb{R}^m$. The following tests are equivalent:

(i) $f^{-1}(V)$ *is measurable for every open* $V \subseteq \mathbb{R}^m$*;*
(ii) $f^{-1}(B)$ *is measurable for every open box* $B \subseteq \mathbb{R}^m$*;*
(iii) every coordinate function $f_j : \Omega \to \mathbb{R}$ *is measurable in the threshold-set sense of Definition 16.2.3.*

In particular, every continuous $f : \Omega \to \mathbb{R}^m$ *is measurable.*

Proof.

$(i) \Rightarrow (ii)$. Open boxes are open.
$(ii) \Rightarrow (i)$. By Lemma 16.1.14, write $V = \bigcup_k B_k$ with rational boxes B_k.
Then $f^{-1}(V) = \bigcup_k f^{-1}(B_k)$ is measurable.
$(ii) \Rightarrow (iii)$. By the previous implication, $f^{-1}(U)$ is measurable for every open $U \subseteq \mathbb{R}^m$.
Hence $\{f_j > a\} = f^{-1}(\{y : y_j > a\})$ is measurable.
$(iii) \Rightarrow (ii)$. If $B = \prod_{j=1}^m (a_j, b_j)$, then $f^{-1}(B) = \bigcap_{j=1}^m \big(\{f_j > a_j\} \cap \{f_j < b_j\}\big)$,
where $\{f_j < b_j\} = \Omega \setminus \bigcap_{\ell=1}^{\infty} \{f_j > b_j - 1/\ell\}$. Thus each factor is measurable, hence so is $f^{-1}(B)$.

If f is continuous, Theorem 13.3.19 makes $f^{-1}(V)$ open relative to Ω.
By Theorem 13.1.37, it is $W \cap \Omega$ for some open $W \subseteq \mathbb{R}^n$, hence measurable by Theorem 16.1.15.
Structural Commentary: Euclidean-valued measurability is just countable box algebra applied to inverse images.
Key Insight: Open sets reduce to rational boxes; vector functions reduce to coordinate threshold tests. □

Theorem 16.2.5. *Closure Properties of Measurable Functions*
Let $(X, \mathcal{M})$ be a measurable space.

(i) The four tests $\{f > a\}, \{f \geq a\}, \{f < a\}, \{f \leq a\} \in \mathcal{M}$ *for all* $a \in \mathbb{R}$ *are equivalent.*
(ii) f *measurable implies* $|f|$ *measurable.*
(iii) If f_n *are measurable, then* $\sup_n f_n, \inf_n f_n, \limsup_n f_n, \liminf_n f_n$ *are measurable.*
(iv) If f, g *real-valued measurable,* $F : \mathbb{R}^2 \to \mathbb{R}$ *continuous, then* $F(f, g)$ *measurable; so are* $f + g$ *and* fg*.*

Proof.

(i) Use $\{f \geq a\} = \bigcap_{m \geq 1} \{f > a - 1/m\}$; $\{f < a\} = X \setminus \{f \geq a\}$; $\{f \leq a\} = \bigcap_{m \geq 1} \{f < a + 1/m\}$.
(ii) For $a > 0$, $\{|f| < a\} = \{f < a\} \cap \{f > -a\}$; for $a \leq 0$ the set is empty.
(iii) $\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\}$. Also $\limsup_n f_n = \inf_m \sup_{n \geq m} f_n$; $\liminf_n f_n = \sup_m \inf_{n \geq m} f_n$; inf follow from (i).
(iv) For $G_a = \{(u, v) : F(u, v) > a\}$, write $G_a = \bigcup_k (\alpha_k, \beta_k) \times (\gamma_k, \delta_k)$.
Then $\{F(f, g) > a\} = \bigcup_k \big(\{\alpha_k < f < \beta_k\} \cap \{\gamma_k < g < \delta_k\}\big) \in \mathcal{M}$.

Structural Commentary: Measurability is countable set algebra applied to level sets.
Key Insight: Once threshold sets are measurable, ordinary limiting and algebraic operations stay measurable. □

Definition 16.2.6. *Simple Functions*
Let $(X, \mathcal{M})$ be a measurable space. For $E \subseteq X$, set $\mathbf{1}_E(x) := \begin{cases} 1, & x \in E, \\ 0, & x \notin E. \end{cases}$ Real-valued s is *simple* if its range is finite. With distinct values c_i and sets $E_i = \{s = c_i\}$, $s = \sum_{i=1}^n c_i \mathbf{1}_{E_i}$ and s is measurable iff $E_i \in \mathcal{M}$ for all i.

Theorem 16.2.7. *Approximation by Simple Functions*
Every real f on X is the pointwise limit of simple functions s_n.
If f is measurable, choose s_n measurable; if $f \geq 0$, choose $0 \leq s_n \uparrow f$.

Proof. For $f \geq 0$, set $E_{ni} = \big\{x : \frac{i-1}{2^n} \leq f(x) < \frac{i}{2^n}\big\}$; $F_n = \{f \geq n\}$; and $s_n = \sum_{i=1}^{n2^n} \frac{i-1}{2^n} \mathbf{1}_{E_{ni}} + n\mathbf{1}_{F_n}$.
Then s_n is simple, measurable when f is, and $0 \leq s_n \uparrow f$. For general f, approximate f^+ and f^- and subtract.
Structural Commentary: Simple functions quantize the range of f.
Key Insight: Dyadic range partitions give pointwise approximation while preserving measurability. □

Definition 16.2.8. *Almost Everywhere*
For measurable E, write $f \sim_{\text{a.e.}} g$ on E if $\mu(\{x \in E : f(x) \neq g(x)\}) = 0$. A property holds *almost everywhere* (a.e.) on E if its failure set in E has measure zero. Then $\sim_{\text{a.e.}}$ is an equivalence relation; if $f \sim_{\text{a.e.}} g$ on E, integrals over measurable $A \subseteq E$ agree whenever they exist. If $f \in \mathscr{L}^1(\mu)$ on E, then f is finite a.e. on E.

Definition 16.2.9. *Convergence Almost Everywhere and in Measure*
A sequence f_n *converges almost everywhere* to f on E if $f_n(x) \to f(x)$ for all $x \in E \setminus N$ for some null set N.
On a measure space, f_n *converges in measure* to f on E if for every $\varepsilon > 0$, $\mu\{x \in E : |f_n(x) - f(x)| > \varepsilon\} \to 0$.

Method 16.2.10. *Measure Proof Patterns*

(i) *Disjointize.* If $A_1 \subseteq A_2 \subseteq \cdots$, set $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$.
Then $A_n = \bigcup_{k=1}^{n} B_k$ and $\bigcup_n A_n = \bigcup_n B_n$ are disjoint decompositions.

(ii) *Approximate.* Prove identities on $\mathcal{E}$; if $A_n \to A$ and $B_n \to B$ in $d_\mu(A,B) = \mu^*(A \triangle B)$, pass through the symmetric-difference estimates for $\cup, \cap, \setminus$.

(iii) *Threshold.* To prove real-valued f measurable, verify $\{f > a\} \in \mathcal{M}$ for all $a \in \mathbb{R}$; for Euclidean-valued maps, use Proposition 16.2.4.

(iv) *Topological bridge.* Use Theorem 13.3.19: open inverse images reduce continuous constructions to countable unions of interval or rectangle preimages.

Key Insight: Measure theory alternates between countable set algebra and approximation by elementary sets.

16.3 Lebesgue Integration

Lebesgue integration starts with simple functions, then extends by monotone approximation and positive–negative decomposition.

Definition 16.3.1. *Lebesgue Integral*
Let $(X, \mathcal{M}, \mu)$ be a measure space. Simple measurable functions are as in Definition 16.2.6. For $s = \sum_{i=1}^{n} c_i \mathbf{1}_{E_i} \geq 0$ simple measurable and $E \in \mathcal{M}$, define $\mathcal{I}_E(s) := \sum_{i=1}^{n} c_i \mu(E \cap E_i)$.
For $f \geq 0$ measurable, let $\mathcal{S}^+(f) := \{s : 0 \leq s \leq f,\ s \text{ simple measurable}\}$ and define $\int_E f\, d\mu := \sup_{s \in \mathcal{S}^+(f)} \mathcal{I}_E(s)$.
For measurable real f, put $f^+ = \max(f, 0)$ and $f^- = -\min(f, 0)$.
If at least one of $\int_E f^+\, d\mu, \int_E f^-\, d\mu$ is finite, set $\int_E f\, d\mu := \int_E f^+\, d\mu - \int_E f^-\, d\mu$.
If both are finite, f is *Lebesgue integrable* on E, written $f \in \mathscr{L}^1(\mu)$ on E.

Simple functions are finite linear combinations of indicators of measurable sets: coefficients carry size, while the level sets carry measurability.

Perspective 16.3.2. *Undergraph Picture for Nonnegative Functions*
For $f : \mathbb{R} \to [0, +\infty)$, set $\mathcal{U}f = \{(x, y) \in \mathbb{R} \times [0, +\infty) : 0 \leq y < f(x)\}$.
Geometrically, $\int f = m(\mathcal{U}f)$; formally, Definition 16.3.1 defines the integral.

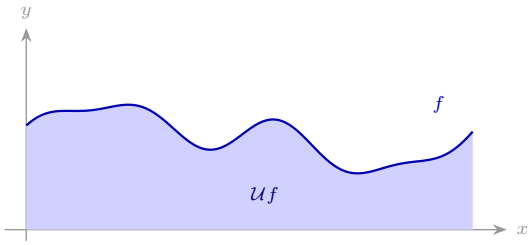


Figure 172: The integral of $f \geq 0$ is modeled by the measure of its undergraph.

Proposition 16.3.3. *Basic Integral Properties*
Let $E \in \mathcal{M}$.

(i) If f is measurable and bounded on E, and $\mu(E) < +\infty$, then $f \in \mathscr{L}^1(\mu)$ on E.
(ii) If $a \leq f(x) \leq b$ on E and $\mu(E) < +\infty$, then $a\mu(E) \leq \int_E f\, d\mu \leq b\mu(E)$.
(iii) If $f, g \in \mathscr{L}^1(\mu)$ on E and $f \leq g$ on E, then $\int_E f\, d\mu \leq \int_E g\, d\mu$.
(iv) If $f \in \mathscr{L}^1(\mu)$ on E, then $cf \in \mathscr{L}^1(\mu)$ on E for finite c, and $\int_E cf\, d\mu = c \int_E f\, d\mu$.
(v) If $\mu(E) = 0$ and f is measurable, then $\int_E f\, d\mu = 0$.
(vi) If $f \in \mathscr{L}^1(\mu)$ on E, $A \in \mathcal{M}$, and $A \subseteq E$, then $f \in \mathscr{L}^1(\mu)$ on A.

Proof.

(i) If $|f| \leq M$ on E, then $0 \leq f^{\pm} \leq M\mathbf{1}_E$ on E, so $\int_E f^{\pm}\, d\mu \leq M\mu(E) < +\infty$.
(ii) Since $a\mathbf{1}_E \leq f \leq b\mathbf{1}_E$ on E, order for the positive integral gives $a\mu(E) \leq \int_E f\, d\mu \leq b\mu(E)$.

(iii) Since $g - f \geq 0$ on E, $\int_E (g-f)\,d\mu \geq 0$, hence $\int_E f\,d\mu \leq \int_E g\,d\mu$.
(iv) For $c \geq 0$, scale the simple approximants. For $c < 0$, use $(cf)^+ = (-c)f^-$ and $(cf)^- = (-c)f^+$.
(v) If $\mu(E) = 0$, then $\mathcal{I}_E(s) = 0$ for every $s \geq 0$ simple, so $\int_E f^\pm\,d\mu = 0$.
(vi) Since $A \subseteq E$, $\mathcal{I}_A(s) \leq \mathcal{I}_E(s)$ for $s \geq 0$ simple; hence $\int_A f^\pm\,d\mu \leq \int_E f^\pm\,d\mu < +\infty$.

Structural Commentary: Estimate $f^\pm$ by simpler functions.
Key Insight: Monotonicity makes bounds, null sets, and restrictions immediate. □

Theorem 16.3.4. *The Integral as a Countably Additive Set Function*
For $f \geq 0$ measurable, or $f \in \mathscr{L}^1(\mu)$, set function $\phi_f(A) := \int_A f\,d\mu$ for $A \in \mathcal{M}$ is countably additive on $\mathcal{M}$.

Proof. For $\mathbf{1}_B$, $\phi_{\mathbf{1}_B}(A) = \mu(A \cap B)$; linearity gives simple f.
If $f \geq 0$ and $A = \bigcup_n A_n$, then for $s \in \mathcal{S}^+(f)$, $\mathcal{I}_A(s) = \sum_n \mathcal{I}_{A_n}(s) \leq \sum_n \phi_f(A_n)$, so $\phi_f(A) \leq \sum_n \phi_f(A_n)$.
Finite additivity gives $\sum_{n=1}^{N} \phi_f(A_n) = \phi_f(\bigcup_{n=1}^{N} A_n) \leq \phi_f(A)$; let $N \to \infty$. For signed f, use $\phi_f = \phi_{f^+} - \phi_{f^-}$.
Structural Commentary: Integrating over the set variable produces a new measure.
Key Insight: Countable additivity is first true for indicators, then simple functions, then suprema. □

Theorem 16.3.5. *Absolute Integrability and Domination*
If $f \in \mathscr{L}^1(\mu)$ on E, then $|f| \in \mathscr{L}^1(\mu)$ and $\left|\int_E f\,d\mu\right| \leq \int_E |f|\,d\mu$.
If f is measurable and $|f| \leq g \in \mathscr{L}^1(\mu)$ on E, then $f \in \mathscr{L}^1(\mu)$ on E.

Proof. Use $|f| = f^+ + f^-$, $f \leq |f|$, and $-f \leq |f|$. If $|f| \leq g$, then $0 \leq f^\pm \leq g$, so $\int_E f^\pm\,d\mu < +\infty$.
Structural Commentary: Lebesgue integrability is absolute integrability.
Key Insight: A single integrable majorant controls both signs. □

Proposition 16.3.6. *Linearity of the Lebesgue Integral*
If $f, g \in \mathscr{L}^1(\mu)$ on E and $a, b \in \mathbb{R}$, then $af + bg \in \mathscr{L}^1(\mu)$ on E and $\int_E (af + bg)\,d\mu = a\int_E f\,d\mu + b\int_E g\,d\mu$.

Proof. For $f, g \geq 0$, choose simple $s_n \uparrow f$ and $t_n \uparrow g$. Simple-function additivity gives the formula for $s_n + t_n$, and monotone convergence passes to $f + g$. The signed case follows by positive–negative parts and Proposition 16.3.3; scalar multiplication is included there.
Structural Commentary: Simple additivity plus monotone convergence gives linearity.
Key Insight: Reduce additivity to simple functions before splitting signs. □

Theorem 16.3.7. *Comparison with the Riemann Integral*
On $[a, b]$ with Lebesgue measure μ, $f \in \mathcal{R}$ implies $f \in \mathscr{L}^1(\mu)$ and the integrals agree. If f is bounded with discontinuity set D, then $f \in \mathcal{R} \iff \mu(D) = 0$.

Proof. For refining partitions, let $L_n \uparrow L$ and $U_n \downarrow U$ be lower and upper step envelopes.
Monotone convergence identifies $\int L$ and $\int U$ with the lower and upper Riemann integrals; if $f \in \mathcal{R}$, then $L = U$ a.e., so the common Lebesgue integral equals the Riemann integral.
For bounded f, put $\omega_f(x) := \lim_{r \downarrow 0} \operatorname{diam} f\big([a, b] \cap (x - r, x + r)\big)$, where $D_k := \{x : \omega_f(x) \geq 1/k\}$.
Then f is continuous at x iff $\omega_f(x) = 0$, and its discontinuity set is $D = \bigcup_{k \geq 1} D_k$.

($\Rightarrow$) Fix k and $\varepsilon > 0$. Choose P with $U(P, f) - L(P, f) < \varepsilon/k$.
Intervals of P whose interiors meet D_k have total length $< \varepsilon$; adding the finitely many partition points, null by Proposition 16.1.19, covers D_k by arbitrarily short intervals.
Thus $\mu(D_k) = 0$ by Definition 16.1.18, and $\mu(D) = 0$.

($\Leftarrow$) Suppose $\mu(D) = 0$ and $|f| \leq B$. If $B = 0$, there is nothing to prove. Given $\varepsilon > 0$, choose k with $(b - a)/k < \varepsilon/2$ and cover D_k by intervals of total length $< \varepsilon/(4B)$.
On the compact complement, $\omega_f < 1/k$, so a finite refinement makes the good intervals contribute $< \varepsilon/2$ to $U - L$; the covered intervals contribute $< 2B \cdot \varepsilon/(4B) = \varepsilon/2$.
Hence $U(P, f) - L(P, f) < \varepsilon$, and Theorem 15.1.41 with $\alpha(x) = x$ gives $f \in \mathcal{R}$.

Structural Commentary: Darboux gap measures oscillation sets.
Key Insight: Lebesgue measure turns "bad discontinuities" into a null-set condition. □

Perspective 16.3.8. *Lebesgue's Criterion as a Roadmap*
Theorem 16.3.7 gives the threshold: a bounded function is Riemann integrable exactly when its discontinuities are null. Finite, countable, and Cantor-type exceptions are allowed; Dirichlet's function fails.

16.4 Convergence Theorems

This subsection collects the main limit-interchange tools: monotone convergence, Fatou, and dominated convergence.

Theorem 16.4.1. *Lebesgue Convergence Theorems*
Let $E \in \mathcal{M}$.

(i) Monotone convergence: If $0 \le f_n \uparrow f$ on E, then $\int_E f_n \, d\mu \to \int_E f \, d\mu$
(ii) Series form: If $f_n \ge 0$ and $f = \sum_{n=1}^{\infty} f_n$ on E, then $\int_E f \, d\mu = \sum_{n=1}^{\infty} \int_E f_n \, d\mu$
(iii) Fatou: If $f_n \ge 0$, then $\int_E \liminf_{n\to\infty} f_n \, d\mu \le \liminf_{n\to\infty} \int_E f_n \, d\mu$
(iv) Dominated convergence: If $f_n \to f$ on E and $|f_n| \le g \in \mathscr{L}^1(\mu)$ on E, then $\lim_{n\to\infty} \int_E f_n \, d\mu = \int_E f \, d\mu$

Proof.

(i) For $s \in \mathcal{S}^+(f)$ and $0 < c < 1$, set $E_n = \{f_n \ge cs\}$.
Then $E_n \uparrow E$, so Theorem 16.1.3 applied to $\phi_s(A) = \int_A s \, d\mu$ gives $\liminf_n \int_E f_n \, d\mu \ge c \int_E s \, d\mu$.
Let $c \uparrow 1$ and take $\sup_s$; the reverse inequality is monotonicity.

(ii) Apply (i) to $S_N = \sum_{n=1}^{N} f_n \uparrow f$.

(iii) Let $g_n = \inf_{i \ge n} f_i$. Then $0 \le g_n \uparrow \liminf_i f_i$ and $\int_E g_n \, d\mu \le \inf_{i \ge n} \int_E f_i \, d\mu$; use (i).

(iv) Fatou applied to $f_n + g$ gives $\int_E f \, d\mu \le \liminf_n \int_E f_n \, d\mu$.
Fatou applied to $g - f_n$ gives $\limsup_n \int_E f_n \, d\mu \le \int_E f \, d\mu$.

Structural Commentary: The major convergence theorems all reduce limit interchange to monotone convergence.
Key Insight: Monotone convergence is the engine; Fatou and domination are its comparison forms. □

Perspective 16.4.2. *Domination Is the Safeguard*
Pointwise convergence lose mass; Theorem 16.4.1 prevents by keeping every f_n under one integrable envelope.

16.5 Product Measures and Fubini Theory

Product measure builds measurable rectangles and reads higher-dimensional integrals through slices.

Definition 16.5.1. *Products and Slices*
For $A \subseteq \mathbb{R}^n$, $B \subseteq \mathbb{R}^k$, set $A \times B \subseteq \mathbb{R}^{n+k}$.
For $E \subseteq \mathbb{R}^n \times \mathbb{R}^k$ define $E_x := \{y \in \mathbb{R}^k : (x, y) \in E\}$ as *x-slice*, and $E^y := \{x \in \mathbb{R}^n : (x, y) \in E\}$ as *y-slice*.
For $f : E \to \mathbb{R}$, write $f_x(y) = f^y(x) := f(x, y)$ where defined.

Theorem 16.5.2. *Measurable Products*
If $A \subseteq \mathbb{R}^n$, $B \subseteq \mathbb{R}^k$ Lebesgue measurable, then $A \times B$ measurable, $m(A \times B) = m(A)m(B)$; $0 \cdot \infty = \infty \cdot 0 = 0$.

Proof. For boxes I, J, $m(I \times J) = m(I)m(J)$; disjoint box decompositions extend this to open sets.
First assume A, B are bounded with finite measure.
By regularity choose closed $F_A \subseteq A$, $F_B \subseteq B$ and open $G_A \supseteq A$, $G_B \supseteq B$ such that $m(G_A \backslash F_A)$, $m(G_B \backslash F_B) < \varepsilon$.
Then $F_A \times F_B \subseteq A \times B \subseteq G_A \times G_B$, and the open/closed product formula squeezes $m(A \times B)$ to $m(A)m(B)$.
For general A, B, set $A_N = A \cap [-N, N]^n$ and $B_N = B \cap [-N, N]^k$. Since $A_N \times B_N \uparrow A \times B$, continuity from below gives $m(A \times B) = \lim_N m(A_N \times B_N) = \lim_N m(A_N)m(B_N) = m(A)m(B)$ with $0 \cdot \infty = \infty \cdot 0 = 0$.
Structural Commentary: Product measure comes from boxes and regularity.
Key Insight: Regularity turns rough measurable products into box products plus zero-set errors. □

Theorem 16.5.3. *Zero Slices*
If $E \subseteq \mathbb{R}^n \times \mathbb{R}^k$ is Lebesgue measurable, then E is null iff E_x is null for a.e. x.

Proof.

(⇒) If $m(E) = 0$, cover E by an open W with small measure.
For $X_\alpha(W) = \{x : m(W_x) > \alpha\}$, $m(W) \ge \alpha m(X_\alpha(W))$; letting $m(W) \to 0$ gives $m(E_x) = 0$ for a.e. x.

(⇐) Remove the exceptional null base so every slice is null. For compact $K \subseteq E$, cover each K_x by a small-measure slice neighborhood; compactness gives finitely many product neighborhoods covering K, whose total measure is arbitrarily small by Theorem 16.5.2. Thus $m(K) = 0$, hence $m(E) = 0$ by regularity.

Structural Commentary: Nullity can be checked fiberwise after removing an exceptional base set.
Key Insight: Product measure turns small slice covers into small total measure. □

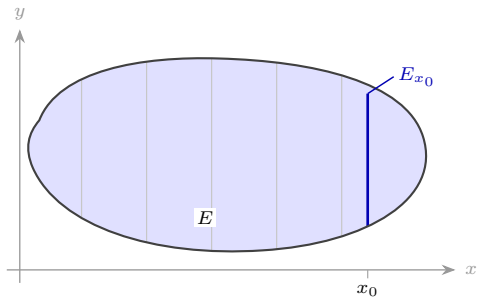


Figure 173: A planar set can be read by vertical slices; Fubini integrates the slice data over the base variable.

Theorem 16.5.4. *Cavalieri's Principle for Lebesgue Measure*
If $E \in \mathcal{L}^{n+k}$, then $E_x \in \mathcal{L}^k$ for a.e. x, and the a.e.-defined function $f_E(x) := m_k(E_x)$ has a measurable representative with $m_{n+k}(E) = \int_{\mathbb{R}^n} f_E \, dm_n$. The analogous statement holds for E^y.

Proof. Let $P(E)$ denote the displayed assertion. For $E = A \times B$, $f_E = m_k(B)\mathbf{1}_A$, so $P(E)$ is Theorem 16.5.2. Fix a bounded box $Q = I \times J$. The class $\{E \subseteq Q : P(E)\}$ contains finite unions of rectangles and is monotone: if $E_j \uparrow E$, then $f_{E_j} \uparrow f_E$ and monotone convergence applies; if $E_j \downarrow E$, then $f_{E_j} \downarrow f_E \le m_k(J)\mathbf{1}_I$ and dominated convergence applies. Hence the monotone class theorem gives $P(E)$ for every Borel $E \subseteq Q$. Exhaustion by $Q_N = [-N, N]^{n+k}$ and monotone convergence give all Borel E.
For Lebesgue E, choose Borel B with $m_{n+k}(E \triangle B) = 0$ (Remark 16.1.20). Then $m_k((E \triangle B)_x) = 0$ for a.e. x (Theorem 16.5.3), so $f_E = f_B$ a.e. and $m_{n+k}(E) = m_{n+k}(B)$. The y-slice statement is identical.
Structural Commentary: Slicing turns a measure in $\mathbb{R}^{n+k}$ into a measurable family of measures in $\mathbb{R}^k$.
Key Insight: Prove boxes, close under monotone limits, then ignore null symmetric differences. □

Perspective 16.5.5. *Fubini Proof Ladder*
Prove Fubini first for indicators, extend to simple functions, pass to nonnegative functions by monotone convergence, then split $f = f^+ - f^-$ under absolute integrability.
Key Insight: Fubini is a slice theorem, proof runs through indicators, simple functions, monotone convergence.

Proposition 16.5.6. *Undergraphs and Level-Set Measurability*
Let $f : \mathbb{R}^n \to [0, \infty]$, set $F_a := \{x : f(x) > a\}$, and write $\mathcal{U}f = \{(x, t) \in \mathbb{R}^n \times [0, \infty) : 0 < t < f(x)\}$. Then f is Lebesgue measurable iff $F_q \in \mathcal{L}^n$ $\forall q \in \mathbb{Q}_{>0}$. In that case, $\mathcal{U}f \in \mathcal{L}^{n+1}$ and $(\mathcal{U}f)^t = F_t$, $t > 0$.

Proof. For $a \ge 0$, $F_a = \bigcup_{\substack{q \in \mathbb{Q} \\ q > a}} F_q$; for $a < 0$, $F_a = \mathbb{R}^n$.
Hence rational positive thresholds are equivalent to measurability.
If they are measurable, then $\mathcal{U}f = \bigcup_{\substack{q,r \in \mathbb{Q} \\ 0<q<r}} F_r \times (q, r) \in \mathcal{L}^{n+1}$. Finally, $(\mathcal{U}f)^t = \{x : 0 < t < f(x)\} = F_t$ for $t > 0$.
Structural Commentary: A nonnegative function may be studied as the region lying beneath its graph.
Key Insight: Horizontal slices of an undergraph are exactly strict superlevel sets. □

Theorem 16.5.7. *Fubini–Tonelli for Nonnegative Functions*
Let $f : \mathbb{R}^n \times \mathbb{R}^k \to [0, \infty]$ be Lebesgue measurable. Then f_x is measurable for a.e. x; with $F(x) := \int_{\mathbb{R}^k} f_x \, dm_k$ there and arbitrary values elsewhere, F is measurable and $\int_{\mathbb{R}^{n+k}} f \, dm_{n+k} = \int_{\mathbb{R}^n} F \, dm_n$.
The same holds for $G(y) := \int_{\mathbb{R}^n} f^y \, dm_n$, with $\int_{\mathbb{R}^{n+k}} f \, dm_{n+k} = \int_{\mathbb{R}^k} G \, dm_k$.

Proof. Put $U := \mathcal{U}f \subseteq \mathbb{R}^{n+k} \times [0, \infty)$. Proposition 16.5.6 gives $U \in \mathcal{L}^{n+k+1}$. By Cavalieri, Theorem 16.5.4, U_x is measurable for a.e. x, $H(x) := m_{k+1}(U_x)$ is measurable, and $m_{n+k+1}(U) = \int_{\mathbb{R}^n} H \, dm_n$.
For those x, $U_x = \mathcal{U}(f_x)$, so f_x is measurable and $H(x) = \int_{\mathbb{R}^k} f_x \, dm_k$.
Also $m_{n+k+1}(U) = \int_{\mathbb{R}^{n+k}} f \, dm_{n+k}$ by the undergraph formula. The y-formula is identical.
Structural Commentary: Tonelli's theorem is Cavalieri's principle applied one dimension higher.
Key Insight: $(\mathcal{U}f)_x = \mathcal{U}(f_x)$. □

Theorem 16.5.8. *Fubini for L^1 Functions*
If $f \in L^1(\mathbb{R}^2)$, then $f(x, \cdot) \in L^1(\mathbb{R})$ for a.e. x and $f(\cdot, y) \in L^1(\mathbb{R})$ for a.e. y. Defining $F(x) := \int_{\mathbb{R}} f(x, y) \, dy$; $G(y) := \int_{\mathbb{R}} f(x, y) \, dx$ on those sets and as 0 elsewhere gives $F, G \in L^1(\mathbb{R})$ and $\int_{\mathbb{R}} F(x) \, dx = \int_{\mathbb{R}^2} f = \int_{\mathbb{R}} G(y) \, dy$

Proof. Let $E_x = \{y : (x, y) \in E\}$ and $E^y = \{x : (x, y) \in E\}$. For measurable $E \subseteq \mathbb{R}^2$, the box/regularity argument and Theorems 16.5.2–16.5.3 give $\int_{\mathbb{R}} m(E_x) \, dx = m(E) = \int_{\mathbb{R}} m(E^y) \, dy$.
Hence Fubini holds for $\mathbf{1}_E$, and by linearity for every nonnegative simple function.
Let $0 \le h$ be measurable and choose $0 \le s_n \uparrow h$ by Theorem 16.2.7.

Monotone convergence gives $\int_{\mathbb{R}}\big(\int_{\mathbb{R}} h(x,y)\,dy\big)\,dx = \lim_n \int_{\mathbb{R}}\big(\int_{\mathbb{R}} s_n(x,y)\,dy\big)\,dx = \lim_n \int_{\mathbb{R}^2} s_n = \int_{\mathbb{R}^2} h$ and the same argument with x, y interchanged. Now let $f \in L^1(\mathbb{R}^2)$.
Applying the nonnegative case to $|f|$ yields $\int_{\mathbb{R}}\big(\int_{\mathbb{R}} |f(x,y)|\,dy\big)\,dx = \int_{\mathbb{R}^2} |f| < \infty$, so $f(x,\cdot) \in L^1(\mathbb{R})$ for a.e. x; similarly $f(\cdot,y) \in L^1(\mathbb{R})$ for a.e. y. With $f = f^+ - f^-$, define, up to null-set choices, $F = F^+ - F^-$, $F^{\pm}(x) = \int_{\mathbb{R}} f^{\pm}(x,y)\,dy$, and analogously $G = G^+ - G^-$. Then $\int_{\mathbb{R}} |F| \le \int_{\mathbb{R}}(F^+ + F^-) = \int_{\mathbb{R}^2} |f| < \infty$, and similarly $G \in L^1(\mathbb{R})$. Subtracting the nonnegative identities for f^+ and f^- gives $\int_{\mathbb{R}} F = \int_{\mathbb{R}^2} f = \int_{\mathbb{R}} G$.
Structural Commentary: Prove identity for $\mathbf{1}_E$, extend by linearity and monotone convergence, apply to $f^{\pm}$.
Key Insight: $\int |f| < \infty$ makes the positive and negative slice integrals finite a.e., so subtraction is legitimate. □

16.6 Radon-Nikodym and Differentiation

This subsection collects differentiation-of-measure material; signed-measure and Radon–Nikodym results can expand here later.

Definition 16.6.1. *Averages and Absolute Continuity*
For $f \in \mathscr{L}^1(E)$, measurable $A \subseteq E$ with $m(A) > 0$, measurable B, and cube or ball Q with $m(Q) > 0$, set $[f : A] := \frac{1}{m(A)} \int_A f$; and $[B : Q] := \frac{m(B \cap Q)}{m(Q)}$.
For $f \in \mathscr{L}^1([a,b])$, an indefinite integral has form $F(x) = F(a) + \int_a^x f(t)\,dt$.
A function $F : [a,b] \to \mathbb{R}$ is *absolutely continuous* if $\forall \varepsilon > 0\ \exists \delta > 0\ \forall \{(a_j, b_j)\}_{j=1}^N$ finite disjoint in $[a,b]$, $\sum_j (b_j - a_j) < \delta \Longrightarrow \sum_j |F(b_j) - F(a_j)| < \varepsilon$.

Definition 16.6.2. *Bounded Variation*
For $F : [a,b] \to \mathbb{R}$, define total variation $V_a^b(F) := \sup \left\{ \sum_{j=1}^N |F(x_j) - F(x_{j-1})| : a = x_0 < x_1 < \cdots < x_N = b \right\}$.
The function F has *bounded variation* on $[a,b]$ if $V_a^b(F) < \infty$.

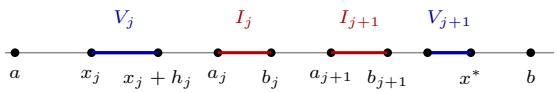


Figure 174: Interval bookkeeping for absolute continuity.

Perspective 16.6.3. *Absolute Continuity as Interval Bookkeeping*
If $F(x) = \int_a^x f(t)\,dt$, by absolute continuity, disjoint intervals with small total length force small F total change.

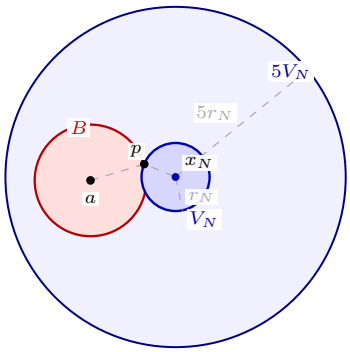


Figure 175: A comparable missed ball lies inside the fivefold dilation of the selected ball.

Perspective 16.6.4. *Vitali Covering Control*
In a Vitali selection, choose a pairwise disjoint subfamily by decreasing scale. If a candidate ball is not selected, it meets an earlier selected ball of comparable radius, so the fivefold dilation of that selected ball contains it.
Key Insight: Disjoint selected balls make measures add, while fixed dilations recover the missed part with bounded overlap cost.

Definition 16.6.5. *Vitali Coverings and Density Points*
For A in a metric space M, $\mathcal{V}$ is a *Vitali covering* of A if $\forall p \in A\ \forall r > 0\ \exists V \in \mathcal{V}$ such that $p \in V$, $\operatorname{diam}(V) < r$.
For measurable $E \subseteq \mathbb{R}^n$, define its *density-point set* $\operatorname{dp}(E) := \{p \in \mathbb{R}^n : \lim_{Q \downarrow p} [E : Q] = 1\}$, where Q ranges over cubes, equivalently balls, containing p and shrinking to p.

Theorem 16.6.6. *Vitali Covering Lemma for Balls and Cubes*
Let $m^*(E) < \infty$, and let $\mathcal{V}$ be a Vitali covering of $E \subseteq \mathbb{R}^n$ by closed balls or axis-parallel cubes.
Then there exists a countable pairwise disjoint $(V_j) \subseteq \mathcal{V}$ such that $m^*\left(E \setminus \bigcup_{j=1}^{\infty} V_j\right) = 0$.

Proof. Fix open $O \supseteq E$ with $m(O) < m^*(E) + \varepsilon$, and replace $\mathcal{V}$ by $\{V \in \mathcal{V} : V \subseteq O\}$, still a Vitali cover.
Choose greedily by dyadic diameter scales a maximal disjoint family $(V_j) \subseteq O$. Let V^* be a fixed dilation such that $W \cap V \neq \varnothing, \quad \operatorname{diam} W \leq 2 \operatorname{diam} V \Longrightarrow W \subseteq V^*$, $m(V^*) \leq C_n m(V)$.
For $R = E \setminus \bigcup_j V_j$ and each N, closedness gives, for every $p \in R$, a small $W \in \mathcal{V}$ with $p \in W$ and $W \cap \bigcup_{j \leq N} V_j = \varnothing$. Maximality yields $W \cap V_j \neq \varnothing$ for some $j > N$ with comparable diameter, hence $R \subseteq \bigcup_{j>N} V_j^*$.
Thus $m^*(R) \leq C_n \sum_{j>N} m(V_j) \to 0$, since $\sum_j m(V_j) = m(\bigsqcup_j V_j) \leq m(O) < \infty$. Hence $m^*(R) = 0$.
Structural Commentary: Vitali selection gives a disjoint cover modulo null sets.
Key Insight: $R \subseteq \bigcup_{j>N} V_j^*$ and disjoint tails vanish. □

Theorem 16.6.7. *Lebesgue Density Theorem*
For Lebesgue measurable $E \subseteq \mathbb{R}^n$, $m(E \triangle \operatorname{dp}(E)) = 0$; and $\lim_{Q \downarrow p}[E : Q] = \mathbf{1}_E(p)$ for a.e. p, where Q ranges over cubes, equivalently balls, shrinking to p.

Proof. Assume first $m(E) < \infty$. For $0 < a < 1$, set $E_a = \left\{p \in E : \liminf_{Q \downarrow p}[E : Q] < a\right\}$.
The cubes $Q \ni p$ with $[E : Q] < a$ form a Vitali cover of E_a. Given $\varepsilon > 0$, restrict to cubes in an open $O \supseteq E_a$ with $m(O) < m(E_a) + \varepsilon$, and choose disjoint (Q_j) covering E_a modulo null sets by Theorem 16.6.6.
Then $m(E_a) \leq \sum_j m(E_a \cap Q_j) \leq \sum_j m(E \cap Q_j) < a \sum_j m(Q_j) \leq a\, m(O) < a(m(E_a) + \varepsilon)$.
Letting $\varepsilon \downarrow 0$ gives $m(E_a) \leq a\, m(E_a)$, so $m(E_a) = 0$. Hence, outside $\bigcup_{a \in \mathbb{Q} \cap (0,1)} E_a$, points of E have density 1.
Apply to bounded pieces of E^c to get density 0 a.e. off E; apply same bounded exhaustion to remove $m(E) < \infty$.
Structural Commentary: Measurable sets are locally 0 or 1 a.e.
Key Insight: Vitali turns bad density into $m(E_a) \leq a\, m(E_a)$. □

Corollary 16.6.8. *Density Points Commute with Slicing Almost Everywhere*
Let $E \subseteq \mathbb{R}^n \times \mathbb{R}^k$ be measurable. For a.e. $x \in \mathbb{R}^n$, $\operatorname{dp}(E_x) = \{y : (x, y) \in \operatorname{dp}(E)\}$ up to a null subset of $\mathbb{R}^k$.

Proof. Set $D = \operatorname{dp}(E)$ and $N = E \triangle D$.
By Theorem 16.6.7, $m_{n+k}(N) = 0$; hence Theorem 16.5.3 gives $m_k(N_x) = 0$ for a.e. x.
Intersecting with the Fubini good set, E_x, D_x are measurable and $m_k(E_x \triangle D_x) = m_k((E \triangle D)_x) = 0$.
Applying Theorem 16.6.7 in $\mathbb{R}^k$ gives $m_k(E_x \triangle \operatorname{dp}(E_x)) = 0$.
Therefore $m_k(\operatorname{dp}(E_x) \triangle D_x) = 0$; and $m_k(\operatorname{dp}(E_x) \triangle D_x) = 0$. This is the claimed equality modulo null sets.
Structural Commentary: Almost-everywhere local fullness is compatible with almost-everywhere slicing.
Key Insight: Null ambient errors have null slices for almost every base point. □

Lemma 16.6.9. *Large Averages Have Small Outer Measure*
For a cube $Q_0 \subseteq \mathbb{R}^n$, $g \geq 0$ in $\mathscr{L}^1(Q_0)$, and $\lambda > 0$,
set $A_\lambda := \{p \in Q_0 : \forall r > 0\ \exists$ cube $Q \subseteq Q_0,\ p \in Q,\ \operatorname{diam} Q < r,\ [g : Q] > \lambda\}$ for $m^*(A_\lambda) \leq \frac{1}{\lambda} \int_{Q_0} g$.

Proof. Let $\mathcal{V}_\lambda = \{Q \subseteq Q_0 : Q$ is a cube and $[g : Q] > \lambda\}$. By definition, $\mathcal{V}_\lambda$ is a Vitali cover of A_λ; choose disjoint $(Q_j) \subseteq \mathcal{V}_\lambda$ with $m^*(A_\lambda \setminus \bigcup_j Q_j) = 0$. Then $\lambda m^*(A_\lambda) \leq \lambda \sum_j m(Q_j) < \sum_j \int_{Q_j} g \leq \int_{Q_0} g$.
Structural Commentary: Weak type for local averages.
Key Insight: Disjoint Vitali cubes turn pointwise large averages into one integral bound. □

Theorem 16.6.10. *Lebesgue Average Value Theorem*
For $f \in \mathscr{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, $\lim_{Q \downarrow p}[f : Q] = f(p)$ for a.e. $p \in \mathbb{R}^n$ with Q cubes, equivalently balls, shrinking to p.

Proof. Fix a cube Q_0; it suffices to prove the claim for a.e. $p \in Q_0^\circ$.
For $\alpha > 0$, let $B_\alpha = \left\{p \in Q_0^\circ : \limsup_{Q \downarrow p} |[f : Q] - f(p)| > \alpha\right\}$.
Choose $h \in C(Q_0)$ with $\|f - h\|_{L^1(Q_0)} < \eta$, and set $g = |f - h|$. Since $[h : Q] \to h(p)$, $B_\alpha \subseteq \{g > \alpha/3\} \cup A_{\alpha/3}(g)$.
By Markov and Lemma 16.6.9, $m^*(B_\alpha) \leq \frac{3}{\alpha} \int_{Q_0} g + \frac{3}{\alpha} \int_{Q_0} g < \frac{6\eta}{\alpha}$.
Let $\eta \downarrow 0$; then $m(B_\alpha) = 0$. Count over $\alpha \in \mathbb{Q}_{>0}$ and exhaust $\mathbb{R}^n$ by cube interiors.
Structural Commentary: Local averages recover the original integrable function almost everywhere.
Key Insight: Continuous approximation handles the good part; Vitali's estimate makes the bad part small. □

Theorem 16.6.11. *Lebesgue's Antiderivative Theorem*
For $f \in \mathscr{L}^1([a,b])$, define $F(x) = c + \int_a^x f(t)\,dt$. Then $F \in AC([a,b])$ and $F'(x) = f(x)$ for a.e. $x \in [a,b]$.

Proof. Given $\varepsilon > 0$, choose $\delta > 0$ so that $m(A) < \delta$ implies $\int_A |f| < \varepsilon$.
If (α_j, β_j) are disjoint and $\sum_j (\beta_j - \alpha_j) < \delta$, then $\sum_j |F(\beta_j) - F(\alpha_j)| \le \sum_j \int_{\alpha_j}^{\beta_j} |f| = \int_{\bigcup_j (\alpha_j,\beta_j)} |f| < \varepsilon$, so $F \in AC([a,b])$. For $x, x+h \in [a,b]$, let $I_h(x) = (\min\{x, x+h\}, \max\{x, x+h\})$.
Then $\frac{F(x+h)-F(x)}{h} = [f : I_h(x)]$. Extending f by 0 outside $[a,b]$, Theorem 16.6.10 in one dimension gives $[f : I_h(x)] \to f(x)$ for a.e. x. Hence $F' = f$ a.e.
Structural Commentary: The indefinite Lebesgue integral is the correct antiderivative for integrable functions.
Key Insight: Differentiating the integral is the same as shrinking the averaging interval. □

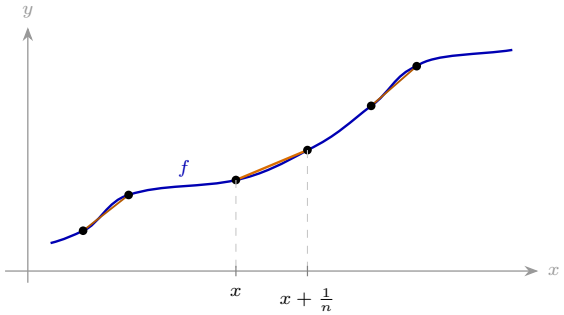


Figure 176: $g_n(x)$ is the slope of the short right secant from x to $x + \frac{1}{n}$.

Perspective 16.6.12. *Right Secants and Differentiation*
Right quotients $g_n(x) = n\big(f(x + 1/n) - f(x)\big)$ encode local slope data.
Key Insight: Secants turn differentiation into slope estimates.

Theorem 16.6.13. *Lebesgue's Last Theorem and Consequences*
On $[a,b]$:

(i) F monotone $\Rightarrow F'$ exists a.e.;
(ii) F Lipschitz $\Rightarrow F'$ exists a.e.;
(iii) $F \in BV([a,b]) \Rightarrow F'$ exists a.e.;
(iv) $F \in AC([a,b]) \Rightarrow F \in BV([a,b])$, hence F' exists a.e.

Proof.

(i) Assume F nondecreasing. Let $D^+F(x) = \limsup_{h \downarrow 0} \frac{F(x+h)-F(x)}{h}$ and $D_-F(x) = \liminf_{h \downarrow 0} \frac{F(x+h)-F(x)}{h}$.
For $\alpha < \beta$ in $\mathbb{Q}$, set $A_{\alpha,\beta} = \{D_-F < \alpha < \beta < D^+F\}$. The right-secant Vitali selection of Figure 176 gives $m(A_{\alpha,\beta}) = 0$: otherwise selected intervals with slopes $< \alpha$ and $> \beta$ contradict monotonicity of total rise. Also $m^*(\{D^+F > M\}) \le \frac{F(b)-F(a)}{M} \to 0$, so $D^+F = D_-F < \infty$ a.e.; hence F' exists a.e.

(ii) If F is L-Lipschitz, set $G_+(x) = F(x) + Lx$ and $G_-(x) = Lx - F(x)$. Then $G_\pm$ are nondecreasing, by (i) $\exists$ null sets $N_\pm$ s.t. $G'_\pm$ exist off $N_\pm$. For $x \notin N_+ \cup N_-$, $F'(x) = \frac{G'_+(x) - G'_-(x)}{2}$, so F' exists a.e.

(iii) If $F \in BV([a,b])$, write its Jordan decomposition $F = F(a) + V^+ - V^-$ with $V^\pm$ nondecreasing. By (i), $\exists$ null sets $N_\pm$ s.t. $(V^\pm)'$ exist off $N_\pm$. For $x \notin N_+ \cup N_-$, $F'(x) = (V^+)'(x) - (V^-)'(x)$, so F' exists a.e.

(iv) If $F \in AC([a,b])$, choose δ for $\varepsilon = 1$, and partition $a = x_0 < \cdots < x_N = b$ with $x_i - x_{i-1} < \delta$. Grouping any refinement by these subintervals gives $V_a^b(F) \le N$, so $F \in BV([a,b])$. By (iii), $\exists Z \subset [a,b]$ with $m(Z) = 0$ s.t. $F'(x)$ exists for all $x \in [a,b] \setminus Z$.

Structural Commentary: Monotonicity drives all cases.
Key Insight: Vitali controls secants; decompose to monotone. □

16.7 Lp Spaces

This block currently uses L^2 as the representative L^p space, with orthonormal expansions and Hilbert structure continued in Functional and Operator Analysis.

Definition 16.7.1. *Complex Integrals and L^2*
For $f = u + iv$, measurability means measurability of u and v.
On $E \in \mathcal{M}$, $f \in \mathscr{L}^1(\mu)$ on $E \iff f$ is measurable and $\int_E |f|\,d\mu < +\infty$.

On X, $f \in L^2(\mu) \iff f$ is complex measurable and $\int_X |f|^2 \, d\mu < +\infty$; $\|f\|_2 := \left(\int_X |f|^2 \, d\mu\right)^{1/2}$.
Modulo equality a.e., define $\langle f, g\rangle_{L^2} := \int_X f\overline{g} \, d\mu$. This is the infinite-dimensional analogue of the inner-product structure from Definition 2.1.3; the norm it induces is $\|f\|_2$.

Theorem 16.7.2. *L^2 Geometry*
For $f, g \in L^2(\mu)$, $\int_X |fg| \, d\mu \le \|f\|_2 \|g\|_2$; $f + g \in L^2(\mu)$; and $\|f + g\|_2 \le \|f\|_2 + \|g\|_2$.
Modulo equality a.e., $L^2(\mu)$ is complete for $\|\cdot\|_2$.

Proof. For real λ, $0 \le Q(\lambda) := \int_X (|f| + \lambda |g|)^2 \, d\mu = \|f\|_2^2 + 2\lambda \int_X |fg| \, d\mu + \lambda^2 \|g\|_2^2$, so the discriminant gives Schwarz; Minkowski follows from $\|f + g\|_2^2 \le \|f\|_2^2 + 2\|f\|_2 \|g\|_2 + \|g\|_2^2$.
For completeness, let (f_n) be Cauchy and choose n_k with $\left\|f_{n_{k+1}} - f_{n_k}\right\|_2 \le 2^{-k}$. Put $h_N = \sum_{k<N} \left|f_{n_{k+1}} - f_{n_k}\right|$.
Then $\|h_N\|_2 \le \sum_{k<N} 2^{-k}$, so Fatou gives $h = \lim h_N \in L^2$; hence $f_{n_k} \to f$ a.e. for some f.
Again by Fatou, $\|f - f_n\|_2 \le \liminf_k \|f_{n_k} - f_n\|_2$, so $f_n \to f$ in L^2.
Structural Commentary: L^2 is Hilbert-space geometry in measure form.
Key Insight: The same Schwarz mechanism survives because the integral is positive and additive. □

Method 16.7.3. *Lebesgue Proof Patterns*

(i) *Approximate*: for $f \ge 0$, use $s \in \mathcal{S}^+(f)$ and $\int_E f \, d\mu = \sup_{s \in \mathcal{S}^+(f)} \mathcal{I}_E(s)$.
(ii) *Promote*: prove for $\mathbf{1}_A$, extend to $s = \sum_i c_i \mathbf{1}_{A_i}$, then pass through $s_n \uparrow f$.
(iii) *Ignore null sets*: if $f \sim_{\text{a.e.}} g$ on E, replace f by g inside integrals over measurable $A \subseteq E$.
(iv) *Split signs*: use $f = f^+ - f^-$ and $h = u + iv$.
(v) *Exchange limits*: reduce Fatou and dominated convergence to $0 \le f_n \uparrow f$.
(vi) *Slice*: prove boxes first; use regularity and null slices to pass to products and Fubini.

Key Insight: Reduce to simple functions, monotone convergence, null sets, and product boxes.

17 Functional and Operator Analysis

Functional and Operator Analysis treats infinite-dimensional linear algebra through Banach spaces, Hilbert spaces, duality, weak topologies, compact operators, spectral theory, and distributions.

17.1 Normed and Banach Spaces

Normed spaces make vector spaces metric; Banach spaces add completeness. This is the setting for infinite series, fixed points, bounded linear maps, and operator norms.

Definition 17.1.1. *Normed and Banach Spaces*
Let $\mathcal{X}$ be a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. A *norm* is a function $\|\cdot\| : \mathcal{X} \to [0, \infty)$ such that, for all $x, y \in \mathcal{X}$ and $\lambda \in \mathbb{F}$, $\|x\| = 0 \iff x = 0$; $\|\lambda x\| = |\lambda| \|x\|$; and $\|x + y\| \le \|x\| + \|y\|$.
The metric $d(x, y) := \|x - y\|$ makes $\mathcal{X}$ a *normed space*. If this metric is complete, $\mathcal{X}$ is a *Banach space*.

Definition 17.1.2. *Bounded Linear Maps and Operator Norm*
For normed spaces $\mathcal{X}, \mathcal{Y}$, a linear map $T : \mathcal{X} \to \mathcal{Y}$ is *bounded* if there exists $C \ge 0$ such that $\|Tx\|_{\mathcal{Y}} \le C \|x\|_{\mathcal{X}}$ for all $x \in \mathcal{X}$. The *operator norm* is $\|T\|_{\text{op}} := \sup_{\|x\|_{\mathcal{X}} \le 1} \|Tx\|_{\mathcal{Y}} = \sup_{x \ne 0} \frac{\|Tx\|_{\mathcal{Y}}}{\|x\|_{\mathcal{X}}}$.

Theorem 17.1.3. *Bounded Equals Continuous for Linear Maps*
For a linear map $T : \mathcal{X} \to \mathcal{Y}$ between normed spaces, the following are equivalent:

(i) T is bounded.
(ii) T is continuous at 0.
(iii) T is continuous everywhere.

Proof.

(i) $\Rightarrow$ (iii). If T is bounded, then $\|T(x - y)\| \le \|T\|_{\text{op}} \|x - y\|$, so T is continuous everywhere.
(iii) $\Rightarrow$ (ii). Since $0 \in \mathcal{X}$, continuity everywhere gives continuity at 0.
(ii) $\Rightarrow$ (i). Choose $\delta > 0$ such that $\|x\| < \delta$ implies $\|Tx\| < 1$.
For $x \ne 0$, apply this to $\delta x / (2\|x\|)$ to get $\|Tx\| \le 2\delta^{-1} \|x\|$; the case $x = 0$ is trivial.

Structural Commentary: Linearity turns local continuity at one point into uniform norm control.
Key Insight: In normed linear spaces, continuity and boundedness are the same operator property. □

Example 17.1.4. *Standard Banach Space Examples*
The complete supremum spaces from Theorem 14.3.1 are Banach spaces under $\|\cdot\|_\infty$. The L^2 space in Theorem 16.7.2 is a Banach space under $\|\cdot\|_2$ and, more specifically, a Hilbert space.

17.2 Hilbert Spaces

Hilbert spaces are complete inner-product spaces. They extend orthogonal geometry to function and sequence spaces, supporting projection, Fourier expansion, and Riesz representation.

Definition 17.2.1. *Hilbert Space*
An inner-product space $\mathcal{H}$ is a *Hilbert space* if the norm $\|x\| := \sqrt{\langle x, x\rangle}$ makes $\mathcal{H}$ complete.
A subset $M \subseteq \mathcal{H}$ is *orthonormal* if $\langle e_i, e_j\rangle = \delta_{ij}$ for all indexed vectors in M.

Theorem 17.2.2. *Hilbert Projection Theorem*
Let $C \subseteq \mathcal{H}$ be nonempty, closed, and convex. For every $x \in \mathcal{H}$ there is a unique $p \in C$ minimizing $\|x - p\|$. If $M \subseteq \mathcal{H}$ is a closed subspace, then $x = p + (x - p)$ with $p \in M$ and $x - p \in M^{\perp}$.

Proof. Let $d = \inf_{q\in C} \|x - q\|$ and choose $q_n \in C$ with $\|x - q_n\| \to d$.
Since C is convex, $(q_n + q_m)/2 \in C$. The parallelogram identity gives
$\|q_n - q_m\|^2 = 2\|x - q_n\|^2 + 2\|x - q_m\|^2 - 4\left\|x - \frac{q_n+q_m}{2}\right\|^2 \le 2\|x - q_n\|^2 + 2\|x - q_m\|^2 - 4d^2$, so (q_n) is Cauchy.
Completeness and closedness give $q_n \to p \in C$, and continuity of the norm gives $\|x - p\| = d$. If p_1, p_2 both minimize, the same identity with $q_n = p_1$, $q_m = p_2$ gives $\|p_1 - p_2\| = 0$. Now let $C = M$ and $r = x - p$.
For $y \in M$ and $t \in \mathbb{R}$, minimality of $p + ty \in M$ gives $0 \le \|r - ty\|^2 - \|r\|^2 = -2t \operatorname{Re}\langle r, y\rangle + t^2\|y\|^2$, hence $\operatorname{Re}\langle r, y\rangle = 0$. Replacing y by iy in the complex case gives $\langle r, y\rangle = 0$. Thus $r \in M^{\perp}$ and $x = p + r$.
Structural Commentary: Convexity makes minimizing sequences Cauchy; closedness keeps the limit.
Key Insight: The nearest point is characterized by an orthogonal error. □

Theorem 17.2.3. *Riesz Representation for Hilbert Spaces*
$\forall$ bounded linear functional $\varphi : \mathcal{H} \to \mathbb{F}$ $\exists$ unique $y \in \mathcal{H}$ s.t. $\varphi(x) = \langle x, y\rangle$, $x \in \mathcal{H}$. Moreover, $\|\varphi\|_{op} = \|y\|$.

Proof. If $\varphi = 0$, take $y = 0$. Otherwise let $M = \ker\varphi$, a closed subspace, and choose $a \notin M$. By Theorem 17.2.2, $a = p + u$ with $p \in M$ and $0 \ne u \in M^{\perp}$, so $\varphi(u) = \varphi(a) \ne 0$.
For $x \in \mathcal{H}$, $x - \frac{\varphi(x)}{\varphi(u)}u \in M$, hence $\langle x, u\rangle = \frac{\varphi(x)}{\varphi(u)}\|u\|^2$. Set $y = \frac{\overline{\varphi(u)}}{\|u\|^2}u$. Since the second slot is conjugate linear, $\langle x, y\rangle = \varphi(x)$ for all x.
If z also represents φ, then $\langle x, y - z\rangle = 0$ for all x; taking $x = y - z$ gives $y = z$. Finally, Cauchy-Schwarz gives $\|\varphi\|_{\text{op}} \le \|y\|$, while $y \ne 0$ gives $\|\varphi\|_{\text{op}} \ge |\varphi(y)| / \|y\| = \|y\|$; the case $y = 0$ is trivial.
Structural Commentary: The kernel is a hyperplane; its orthogonal direction represents the functional.
Key Insight: Bounded functionals are exactly inner products with fixed vectors. □

Theorem 17.2.4. *Orthonormal Systems in L^2*
Let (ϕ_n) be orthonormal in $L^2(\mu)$: $\int_X \phi_n\overline{\phi_m}\,d\mu = \delta_{nm}$.

(i) For $f \in L^2(\mu)$, $c_n = \int_X f\overline{\phi_n}\,d\mu$ defines $f \sim_{\text{coef}} \sum c_n\phi_n$.

(ii) If $\sum |c_n|^2 < +\infty$, then $s_n = \sum_{k=1}^{n} c_k\phi_k$ converges in L^2 to some f.

(iii) If (ϕ_n) is complete and $f \sim_{\text{coef}} \sum c_n\phi_n$, then $\int_X |f|^2\,d\mu = \sum_{n=1}^{\infty} |c_n|^2$.

For the trigonometric system on $[-\pi, \pi]$, this is L^2 Parseval and a.e. uniqueness.

Proof.

(i) For $c_n = \int_X f\overline{\phi_n}\,d\mu$, finite orthonormality gives $\left\|f - \sum_{k=1}^{N} c_k\phi_k\right\|_2^2 = \|f\|_2^2 - \sum_{k=1}^{N} |c_k|^2$.
Thus $\sum_{n=1}^{\infty} |c_n|^2 \le \|f\|_2^2$, and these coefficients are denoted $f \sim_{\text{coef}} \sum c_n\phi_n$.

(ii) If $\sum |c_n|^2 < +\infty$ and $m < n$, then $\|s_n - s_m\|_2^2 = \sum_{k=m+1}^{n} |c_k|^2$.
Thus (s_n) is Cauchy, so Theorem 16.7.2 gives $s_n \to g$ in L^2.

(iii) If (ϕ_n) is complete, Bessel gives $\sum |c_n|^2 < +\infty$, so (ii) gives $s_n \to g$.
$\forall j$, $\langle f - g, \phi_j\rangle = c_j - \lim_n \langle s_n, \phi_j\rangle = 0$; completeness gives $f = g$. Hence $\|f\|_2^2 = \lim_n \|s_n\|_2^2 = \sum_{n=1}^{\infty} |c_n|^2$.

Structural Commentary: Complete orthonormal systems coordinatize L^2 by square-summable sequences.
Key Insight: Parseval is Pythagoras after passing from finite coordinates to an L^2 limit. □

17.3 Core Functional Analytic Theorems

The core theorems explain why complete normed spaces behave rigidly under linear maps. Baire category turns pointwise control, surjectivity, and closed graphs into operator statements.

Theorem 17.3.1. *Hahn–Banach Extension Theorem*
Let $\mathcal{X}$ be a real normed space, let $M \subseteq \mathcal{X}$ be a linear subspace, and let $p : \mathcal{X} \to \mathbb{R}$ be sublinear. If $\varphi : M \to \mathbb{R}$ is linear and $\varphi(x) \le p(x)$ for all $x \in M$, then φ extends to a linear functional $\Phi : \mathcal{X} \to \mathbb{R}$ with $\Phi(x) \le p(x)$ for all $x \in \mathcal{X}$. In the normed complex form, every bounded functional on a subspace extends with the same norm.

Theorem 17.3.2. *Uniform Boundedness Principle*
Let $\mathcal{X}$ be Banach and let $\mathcal{Y}$ be normed. If $\mathcal{F} \subseteq \mathcal{L}(\mathcal{X}, \mathcal{Y})$ is pointwise bounded, meaning $\sup_{T \in \mathcal{F}} \|Tx\| < \infty$ for every $x \in \mathcal{X}$, then $\sup_{T \in \mathcal{F}} \|T\|_{op} < \infty$.

Theorem 17.3.3. *Open Mapping Theorem*
A surjective bounded linear map $T : \mathcal{X} \to \mathcal{Y}$ between Banach spaces is open. Equivalently, T maps neighborhoods of 0 onto neighborhoods of 0.

Theorem 17.3.4. *Closed Graph Theorem*
Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces. A linear map $T : \mathcal{X} \to \mathcal{Y}$ is bounded if and only if its graph

$$\Gamma(T) := \{(x, Tx) : x \in \mathcal{X}\}$$

is closed in $\mathcal{X} \times \mathcal{Y}$.

17.4 Weak Topologies and Duality

Weak topologies replace norm control with scalar tests by functionals, exposing compactness and duality that strong topology can hide.

Definition 17.4.1. *Dual Space and Weak Convergence*
For a normed space $\mathcal{X}$, its continuous dual is $\mathcal{X}^* := \mathcal{L}(\mathcal{X}, \mathbb{F})$. A sequence (x_n) *converges weakly* to x, written $x_n \rightharpoonup x$, if

$$\varphi(x_n) \to \varphi(x) \qquad \text{for every } \varphi \in \mathcal{X}^*.$$

Definition 17.4.2. *Weak-* Convergence*
If $(\varphi_n) \subseteq \mathcal{X}^*$ and $\varphi \in \mathcal{X}^*$, then φ_n *converges weak-** to φ if

$$\varphi_n(x) \to \varphi(x) \qquad \text{for every } x \in \mathcal{X}.$$

This is weak convergence in the dual tested only against evaluation at points of the original space.

Definition 17.4.3. *Reflexivity*
The canonical map $J : \mathcal{X} \to \mathcal{X}^{**}$ is defined by $Jx(\varphi) = \varphi(x)$. A Banach space is *reflexive* if J is onto.

Theorem 17.4.4. *Banach–Alaoglu Theorem*
The closed unit ball of $\mathcal{X}^*$ is compact in the weak- topology. Consequently, bounded nets of functionals have weak-* cluster points.*

17.5 Compact Operators

Compact operators carry bounded sets to relatively compact sets. Their spectral and solvability theory therefore resembles matrix theory inside infinite-dimensional spaces.

Definition 17.5.1. *Compact and Finite-Rank Operators*
Let $\mathcal{X}, \mathcal{Y}$ be normed spaces. A linear map $T : \mathcal{X} \to \mathcal{Y}$ is *compact* if $T(B)$ has compact closure in $\mathcal{Y}$ whenever $B \subseteq \mathcal{X}$ is bounded.
The map has *finite rank* if $\dim \operatorname{range}(T) < \infty$.

Proposition 17.5.2. *Finite Rank Operators Are Compact*
If $\mathcal{X}$ is normed, $\mathcal{Y}$ is Banach, and $T : \mathcal{X} \to \mathcal{Y}$ is bounded with finite-dimensional range, then T is compact.

Theorem 17.5.3. *Norm Limits of Compact Operators*
If $T_n : \mathcal{X} \to \mathcal{Y}$ are compact operators and $T_n \to T$ in operator norm, with $\mathcal{Y}$ Banach, then T is compact.

Theorem 17.5.4. *Fredholm Alternative*
Let $K : \mathcal{X} \to \mathcal{X}$ be compact on a Banach space. For $\lambda \neq 0$, the equation

$$(\lambda I - K)x = y$$

has a solution for every y exactly when the homogeneous equation $(\lambda I - K)x = 0$ has only the zero solution. In this case $(\lambda I - K)^{-1}$ is bounded.

17.6 Spectral Theory of Operators

For bounded operators, spectrum replaces the finite-dimensional eigenvalue list. The resolvent tracks invertibility, while the spectrum records every obstruction to it.

Definition 17.6.1. *Spectrum and Resolvent*
Let $T \in \mathcal{L}(\mathcal{X})$ be a bounded operator on a Banach space. The *resolvent set* is

$$\rho(T) := \{\lambda \in \mathbb{C} : \lambda I - T \text{ is bijective and } (\lambda I - T)^{-1} \in \mathcal{L}(\mathcal{X})\}.$$

The *spectrum* is $\sigma(T) := \mathbb{C} \setminus \rho(T)$.

Definition 17.6.2. *Spectral Radius*
The *spectral radius* of $T \in \mathcal{L}(\mathcal{X})$ is

$$r(T) := \sup\{|\lambda| : \lambda \in \sigma(T)\}.$$

Theorem 17.6.3. *Spectrum of a Complex Bounded Operator*
If $\mathcal{X}$ is a nonzero complex Banach space and $T \in \mathcal{L}(\mathcal{X})$, then $\sigma(T)$ is a nonempty compact subset of $\mathbb{C}$ and $\sigma(T) \subseteq \{\lambda : |\lambda| \le \|T\|_{op}\}$.

Theorem 17.6.4. *Spectral Radius Formula*
For $T \in \mathcal{L}(\mathcal{X})$ on a complex Banach space,

$$r(T) = \lim_{n\to\infty} \|T^n\|_{op}^{1/n}.$$

Perspective 17.6.5. *Hilbert Operator Spectral Theory*
For bounded self-adjoint and normal operators on Hilbert spaces, spectral theory recovers a functional-calculus form of diagonalization. Compact self-adjoint operators have an orthonormal eigenbasis away from the kernel, giving the closest infinite-dimensional analogue of the finite spectral theorem.

17.7 Distributions and Sobolev Preview

Distributions test against smooth compactly supported functions; Sobolev spaces package weak derivatives in normed or Hilbert spaces as a bridge to PDE.

Definition 17.7.1. *Test Functions and Distributions*
Let $\Omega \subseteq \mathbb{R}^n$ be open. A *test function* is a smooth function $\varphi \in C_c^\infty(\Omega)$. A *distribution* on Ω is a continuous linear functional on the test-function space, written $u \in \mathcal{D}'(\Omega)$ and paired as $\langle u, \varphi\rangle$.

Definition 17.7.2. *Weak Derivative*
Let $u \in L^1_{\mathrm{loc}}(\Omega)$ and let α be a multi-index. A function $v \in L^1_{\mathrm{loc}}(\Omega)$ is the *weak derivative* $D^\alpha u$ if

$$\int_\Omega u\, D^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_\Omega v\, \varphi \, dx \qquad \text{for every } \varphi \in C_c^\infty(\Omega).$$

Definition 17.7.3. *Sobolev Spaces*
For $1 \le p \le \infty$ and an integer $k \ge 0$, the Sobolev space $W^{k,p}(\Omega)$ consists of functions $u \in L^p(\Omega)$ whose weak derivatives $D^\alpha u$ lie in $L^p(\Omega)$ for every $|\alpha| \le k$. The norm is

$$\|u\|_{W^{k,p}} := \sum_{|\alpha|\le k} \|D^\alpha u\|_{L^p}$$

with the usual supremum modification when $p = \infty$.

Part V
Probability and Stochastic Structure

18 Probability Spaces and Convergence

18.1 Probability Spaces and Random Variables

18.2 Expectation and Integration in Probability

18.3 Conditional Probability and Conditional Expectation

18.4 Modes of Probabilistic Convergence

18.5 Laws of Large Numbers

18.6 Central Limit Theory

18.7 Probability on Metric Spaces

19 Stochastic Processes

19.1 Process-Level Probability

19.2 Filtrations and Adapted Processes

19.3 Martingales in Discrete and Continuous Time

19.4 Markov Chains and Markov Processes

19.5 Stationary Processes and Ergodicity

19.6 Gaussian Processes

19.7 Point Processes and Renewal Structure

20 Stochastic Calculus and Dynamics

20.1 Brownian Motion

20.2 Stochastic Integration

20.3 Itô Calculus

20.4 Stochastic Differential Equations

20.5 Change of Measure and Girsanov Theory

20.6 Diffusions and PDE Connections

20.7 Stochastic Flows and Advanced Dynamics

Part VI
Dynamical and PDE Systems

21 Dynamical Systems and ODEs

21.1 ODE Foundations and Flows
21.2 Linear Systems and Stability
21.3 Nonlinear Stability Theory
21.4 Bifurcation Theory
21.5 Dynamical Systems Structure
21.6 Semigroups and Evolution Viewpoint

22 PDE and Evolution Equations

22.1 First-Order PDE and Characteristics
22.2 Elliptic Equations
22.3 Parabolic Equations
22.4 Hyperbolic Equations
22.5 Sobolev Spaces for PDE
22.6 Weak Solutions and Variational Methods
22.7 Evolution Equations

Part VII
Numerical and Computational Mathematics

23 Numerical Analysis

23.1 Floating Point and Error Analysis
23.2 Nonlinear Equations and Optimization Basics
23.3 Interpolation and Approximation
23.4 Numerical Differentiation and Quadrature
23.5 Numerical Stability and Convergence

24 Numerical Differential Systems

24.1 Numerical ODEs
24.2 Geometric Numerical Integration
24.3 Finite Difference PDE Methods
24.4 Finite Element Methods
24.5 Numerical SDEs
24.6 Computational Stability Across ODE/PDE/SDE

25 Monte Carlo and Stochastic Computation

25.1 Monte Carlo Foundations
25.2 Variance Reduction
25.3 Markov Chain Monte Carlo
25.4 Sequential Monte Carlo
25.5 Rare-Event Simulation
25.6 Stochastic Optimization

Part VIII
Statistical and Learning Systems

26 Statistical Learning

26.1 Statistical Inference Foundations

26.2 Risk Minimization

26.3 Linear and Generalized Linear Models

26.4 Regularization and Model Selection

26.5 Kernels and Nonlinear Methods

26.6 Trees, Ensembles, and Boosting

26.7 Generalization Theory

27 Deep Learning and Representation Systems

27.1 Neural Network Foundations

27.1.1 Deep Feedforward Networks

Remark 27.1.1. *Deep Feedforward Networks (DFNs)*
Defines a mapping $\boldsymbol{y} = f(\boldsymbol{x}; \boldsymbol{\theta})$ and learns value of parameters $\boldsymbol{\theta}$ that results in best function approximation.

Remark 27.1.2. *Linear Models*
Linear models may fit data efficiently and reliably, either in closed form or with convex optimisation.
Model capacity limited to linear functions, does not model interaction between any two input variables

Remark 27.1.3. *Nonlinear Models*
To represent nonlinear functions of $\boldsymbol{x}$, apply linear model to transformed input $\phi(\boldsymbol{x})$ where ϕ is a nonlinear transformation. Kernel trick may be applied to obtain nonlinear learning algorithm. To choose mapping ϕ:

i. Choose generic ϕ, such as that used by kernel machines based on RBF kernel.
If $\phi(\boldsymbol{x})$ is of high enough dimension, can find enough capacity to fit training set, but generalisations to test set remains poor. Mappings are based on principle of local smoothness and do not encode enough prior information to solve advanced problems.
ii. Manually engineer ϕ. Requires decades of human effort for each task, and practitioners specialising in different domains (i.e, speech recognition, computer vision) with little transfer between domains.
iii. Learn ϕ through deep learning. Model is $y = f(\boldsymbol{x}; \boldsymbol{\theta}, \boldsymbol{w}) = \phi(\boldsymbol{x}; \boldsymbol{\theta})^T \boldsymbol{w}$, where parameters $\boldsymbol{\theta}$ can be used to learn ϕ from broad class of functions, and parameters $\boldsymbol{w}$ that map from $\phi(\boldsymbol{x})$ to desired output.
Do not require training problem to be convex, and only require finding the right general function.

Definition 27.1.4. *Cost Functions*

i. Learning Conditional Distributions with Maximum Likelihood: cost function is negative log-likelihood, which is the cross-entropy between training data and model distribution:
$$J(\boldsymbol{\theta}) = -\mathbb{E}_{\boldsymbol{x},\boldsymbol{y}\sim\hat{p}_{\text{data}}} \log p_{\text{model}}(\boldsymbol{y} \mid \boldsymbol{x})$$
Specific form of cost function changes from model to model, depending on form of $\log p_{\text{model}}$.
Method removes the burden of designing cost functions for each model, as specifying a model $p(\boldsymbol{y} \mid \boldsymbol{x})$ automatically determines a cost function $\log p(\boldsymbol{y} \mid \boldsymbol{x})$.
Gradient of cost function for neural networks must be large and predictable enough to serve as good guide for the learning algorithm. The negative log-likelihood helps to avoid saturation of the function.
ii. Learning Conditional Statistics: to learn just one conditional statistic of $\boldsymbol{y}$ given $\boldsymbol{x}$.
With sufficiently powerful neural network, this can represent any function f from a wide class of function, limited by features of continuity and boundedness. Hence the cost function is a functional. Learning is choosing a functional rather than a set of parameters.
By calculus of variation , solving the optimisation problem yields the below function,
$$f^* = \arg\min_f \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}\sim p_{\text{data}}} \|\boldsymbol{y} - f(\boldsymbol{x})\|^2$$
$$f^* = \mathbb{E}_{\boldsymbol{y}\sim p_{\text{data}}(\boldsymbol{y} \mid \boldsymbol{x})}[\boldsymbol{y}]$$

If infinitely many samples from the true data-generating distribution could be trained, then minimising the mean squared error cost function gives a function that predicts mean of $\boldsymbol{y}$ for each value of $\boldsymbol{x}$.
A second result from calculus of variations is:

$$f^* = \arg\min_f \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}\sim p_{\text{data}}} \|\boldsymbol{y} - f(\boldsymbol{x})\|_1$$

which is a function that predicts the median value of $\boldsymbol{y}$ for each $\boldsymbol{x}$. This is the mean absolute error.
Note that mean squared error and mean absolute error often lead to poor results when used with gradient-based optimisation. Output units that saturate may produce very small gradients when combined with these cost functions. Hence the reason that cross-entropy cost function is more popular.

Definition 27.1.5. *Output Units*

i. Linear Units: base on affine transformation with no nonlinearity.
Given features $\boldsymbol{h}$, a layer of output units produces vector $\hat{\boldsymbol{y}} = \boldsymbol{W}^T\boldsymbol{h} + \boldsymbol{b}$. Linear output layers are used to produce mean of conditional Gaussian distribution:

$$p(\boldsymbol{y} \mid \boldsymbol{x}) = \mathcal{N}(\boldsymbol{y}; \hat{\boldsymbol{y}}, \boldsymbol{I})$$

Maximising log-likelihood is equivalent to minimising mean squared error.
Linear units do not saturate, hence may be used for wide variety of optimisation algorithms.

ii. Sigmoid Units: define Bernoulli distribution y conditioned on $\boldsymbol{x}$. Neural net to predict $P(y = 1 \mid \boldsymbol{x})$, which lies in interval $[0, 1]$. The sigmoid unit is defined by

$$\hat{y} = \sigma(\boldsymbol{w}^T\boldsymbol{h} + b)$$

where σ is the logistic sigmoid function.
Note that the cost function used with maximum likelihood is $-\log P(y \mid \boldsymbol{x})$, preventing saturation.
The loss function for MLE of Bernoulli parametrised by sigmoid is

$$J(\boldsymbol{\theta}) = -\log P(y \mid \boldsymbol{x}) = -\log \sigma((2y-1)z) = \zeta((1-2y)z)$$

Function saturates only when $(1-2y)z$ is very negative, i.e., when model has the right answer.

iii. Softmax Units: used to represent probability distribution over n different classes.
A linear layer predicts unnormalised log probabilities:

$$\boldsymbol{z} = \boldsymbol{W}^T\boldsymbol{h} + \boldsymbol{b}$$

where $z_i = \log \tilde{P}(y = i \mid \boldsymbol{x})$. Softmax function is then

$$\text{softmax}(\boldsymbol{z})_i = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$$

The function is to maximise $\log P(y = i; \boldsymbol{z}) = \log \text{softmax}(\boldsymbol{z})_i = z_i - \log \sum_j \exp(z_j)$.
Note that the input z_i always has direct contribution to cost function, and the term cannot saturate.
Un-regularised maximum likelihood will drive the model to learn parameters that drive the softmax to predict fraction of counts for each outcome observed in the training set:

$$\text{softmax}(z(\boldsymbol{x};\boldsymbol{\theta}))_i \approx \frac{\sum_{j=1}^{m} \mathbf{1}_{y^{(j)}=i, \boldsymbol{x}^{(j)}=x}}{\sum_{j=1}^{m} \mathbf{1}_{\boldsymbol{x}^{(j)}=x}}$$

Note that objective functions other than log-likelihood does not work well with softmax function. Squared error is poor loss function for softmax units, and can fail to train the model to change its output.
Softmax function can saturate, and many functions based on softmax also saturate, unless they are able to invert the saturating activating function.

iv. Other Output Types: generally, given a conditional distribution $p(\boldsymbol{y} \mid \boldsymbol{x};\ \boldsymbol{\theta})$, principle of maximum likelihood suggests using $-\log p(\boldsymbol{y} \mid \boldsymbol{x};\ \boldsymbol{\theta})$ as the cost function.
Neural networks represent a function $f(\boldsymbol{x};\boldsymbol{\theta}) = \boldsymbol{\omega}$. The outputs of the function are not direct predictions of value $\boldsymbol{y}$, but the parameters for distribution over $\boldsymbol{y}$. The loss function is then $-\log p(\boldsymbol{y};\boldsymbol{\omega}(\boldsymbol{x}))$.

Remark 27.1.6. *Learning Distribution Parameters*

i. Heteroscedastic Model: to predict different variance in $\boldsymbol{y}$ for different values of $\boldsymbol{x}$, formulate the Gaussian distribution using precision rather than variance. In multi-variate case, the diagonal precision matrix is used, $\text{diag}(\boldsymbol{\beta})$. The log-likelihood of Gaussian distribution parametrised by $\boldsymbol{\beta}$ involves only multiplication by β_i and addition of $\log \boldsymbol{\beta}_i$. The gradient of these operations are well-behaved.
Let $\boldsymbol{\alpha}$ be raw activation of the model used to determine diagonal precision. The softplus function may be used to obtain a positive precision vector $\boldsymbol{\beta} = \zeta(\boldsymbol{\alpha})$. Same strategy applies equally if using variance variance or standard deviation rather than precision.

If covariance is full and conditional, then parametrisation must be chosen that guarantees positive-definiteness of predicted covariance matrix.

$$\boldsymbol{\Sigma}(\boldsymbol{x}) = \boldsymbol{B}(\boldsymbol{x})\boldsymbol{B}^T(\boldsymbol{x})$$

where $\boldsymbol{B}$ is unconstrained square matrix. Note that if matrix is full rank, then computing likelihood requires $O(d^3)$ a $d \times d$ matrix for the determinant and inverse of $\boldsymbol{\Sigma}(\boldsymbol{x})$.

ii. Mixture Density Networks: to perform multimodal regression (predict real values that come from conditional distribution $p(\boldsymbol{y} \mid \boldsymbol{x})$) that can have several different peaks in $\boldsymbol{y}$ for the same $\boldsymbol{x}$.

$$p(\boldsymbol{y} \mid \boldsymbol{x}) = \sum_{i=1}^{n} p(c = i \mid \boldsymbol{x})\mathcal{N}(\boldsymbol{y}; \boldsymbol{\mu}^{(i)}(\boldsymbol{x}), \boldsymbol{\Sigma}^{(i)}(\boldsymbol{x}))$$

The neural network will have three outputs:

1. Mixture components $p(c = i \mid \boldsymbol{x})$, forming a multinoulli distribution over n different components with latent variable c, obtained by softmax over n-dimensional vector.
2. Means $\boldsymbol{\mu}^{(i)}(\boldsymbol{x})$
3. Covariances $\boldsymbol{\Sigma}^{(i)}(\boldsymbol{x})$

Gradient-based optimisation of conditional Gaussian mixtures can be unreliable as the divisions can be numerically unstable. May be solved by clipping gradients, or scaling gradients heuristically.

Definition 27.1.7. *Hidden Units*

Even if hidden units are not differentiable at all input points, gradient descent still performs well enough as the training algorithms do not usually arrive at local minimum of cost function, but reduce its value significantly. Most hidden units has input vector $\boldsymbol{x}$, computes an affine transformation $\boldsymbol{z} = \boldsymbol{W}^T\boldsymbol{x} + \boldsymbol{b}$, then applying element-wise non-linear function $g(\boldsymbol{z})$.

i. Rectified Linear Units (ReLUs): uses activation function $g(z) = \max\{0, z\}$.
Typically used on top of an affine transformation $\boldsymbol{h} = g(\boldsymbol{W}^T\boldsymbol{x} + \boldsymbol{b})$. In initialisation, set all elements of $\boldsymbol{b}$ to small positive values, so that ReLUs will be initially active for most inputs in training set.
Generalisations of ReLUs have non-zero slope α_i for $z_i < 0$: $h_i = \max(0, z_i) + \alpha_i \min(0, z_i)$.

1. Absolute Value Rectification: sets $\alpha_i = -1$ to obtain $g(z) = |z|$. Used for object recognition from images, to seek features that are invariant under polarity reversal of input illumination.
2. Leaky ReLU: fixes α_i to small positive value
3. Parametric ReU: treats α_i as a learnable parameter

ii. Maxout Units: divide $\boldsymbol{z}$ into groups of k values. Each maxout unit then outputs maximum element of one of these groups: $g(\boldsymbol{z})_i = \max_{j \in \mathbb{G}^{(i)}} z_j$, where $\mathbb{G}^{(i)}$ is indices of inputs for group i, which is $\{(i-1)k+1, \ldots, ik\}$. This allows learning of piecewise linear function that responds to multiple directions in input $\boldsymbol{x}$ space .
Maxout units learn the activation function itself. With large k, maxout unit can learn to approximate any convex function. Each maxout unit is parametrised by k weight vectors, hence need more regularisation than ReLUs. Benefits include:

1. Can work well without regularisation if training set is large and number of pieces per unit is low.
2. Can gain statistical and computation advantages by requiring fewer parameters.
3. Have redundancy that resists catastrophic forgetting, where neural networks forgot how to perform tasks that were trained on in the past.

iii. Logistic Sigmoid and Hyperbolic Tangent: the logistic sigmoid activation function is $g(z) = \sigma(z)$, and the hyperbolic tangent activation function is $g(z) = \tanh(z)$. Note that $\tanh(z) = 2\sigma(2z) - 1$.
Sigmoidal units saturate across most of the domain, which makes gradient-based learning very difficult. Hence the use in hidden units in feedforward networks is now discouraged.
If sigmoidal activation function must be used, hyperbolic tangent activation function performs better, as it resembles identity function more closely. Training $\hat{y} = \boldsymbol{w}^T \tanh(\boldsymbol{U}^T \tanh(\boldsymbol{V}^T\boldsymbol{x}))$ resembles training a linear model $\hat{y} = \boldsymbol{w}^T\boldsymbol{U}^T\boldsymbol{V}^T\boldsymbol{x}$ as long as the activations of the network can be kept small.
Sigmoidal functions are more common in recurrent networks, probabilities models, auto-encoders

iv. Linear Unit: consider neural network layer with n inputs, p outputs, $\boldsymbol{h} = g(\boldsymbol{W}^T\boldsymbol{x} + \boldsymbol{b})$. Replace with two layers, one using weight matrix $\boldsymbol{U}$ and the other using weight matrix $\boldsymbol{V}$. If first layer has no activation function, then the factored approach is to compute $\boldsymbol{h} = g(\boldsymbol{V}^T\boldsymbol{U}^T\boldsymbol{x} + \boldsymbol{b})$. If $\boldsymbol{U}$ produces q outputs, then $\boldsymbol{U}$ and $\boldsymbol{V}$ together only contains $(n + p)q$ parameters, while $\boldsymbol{W}$ contains np parameters. For small q, this is considerable saving in parameters, while the cost is constraining linear transformation to be low rank. This is an efficient way of reducing number of parameters in the model.

v. Softmax: naturally represent probability distribution over discrete variable with k possible values. Used only in more advanced architectures that explicitly learn to manipulate memory.

vi. Radial Basis Function (RBF): function becomes more active as $\boldsymbol{x}$ approaches a template $\boldsymbol{W}_{:,i}$. As it

saturates to 0 for most $\boldsymbol{x}$, it can be difficult to optimise.

$$h_t = \exp\left(-\frac{1}{\sigma_i^2}\|\boldsymbol{W}_{:,i} - \boldsymbol{x}\|^2\right)$$

vii. Softplus: smooth version of rectifier for functional approximation and for conditional distributions of undirected probabilistic models. Usage is generally discouraged.

$$g(a) = \zeta(a) = \log(1 + e^a)$$

viii. Hard tanh: shaped similarly to tanh but bounded.

$$g(a) = \max(-1, \min(1, a))$$

Theorem 27.1.8. *Universal Approximation Theorem*
A feedforward network with linear output layer and at least one hidden layer with any 'squashing' activation function can approximate any Borel measurable function from one finite-dimensional space to another with any desired non-zero amount of error, provided the network is given enough hidden units.

Method 27.1.9. *Architecture Design*
Neural network layers are arranged in a chain structure, with each layer being a function of preceding layer.

$$\boldsymbol{h}^{(1)} = g^{(1)}(\boldsymbol{W}^{(1)T}\boldsymbol{x} + \boldsymbol{b}^{(1)})$$
$$\boldsymbol{h}^{(k)} = g^{(k)}(\boldsymbol{W}^{(k)T}\boldsymbol{h}^{(k-1)} + \boldsymbol{b}^{(k)}), \quad k \geq 2$$

The main considerations are the depth of network and width of each layer. Deeper networks use far fewer units per layers and far fewer parameters, and often generalise to the test set, but are harder to optimise.

Remark 27.1.10. *Depth of Network and Universal Approximation Theorem*
A feedforward network with single layer is sufficient to represent any function, but the layer may be infeasibly large and fail to learn and generalise correctly. Using deeper models can reduce number of units required to represent the desired function and can reduce generalisation error.
Shallow networks with broad family of non-polynomial activation functions have universal approximation properties. Piecewise linear networks can represent functions with number of regions that is exponential to depth. The number of linear regions carved out by deep rectifier network with d inputs, depth l, and n units per hidden layer is $O(\binom{n}{d}^{d(d-1)} n^d)$. For maxout networks with k filters per unit, this is $O(k^{(k-1)+d})$.

Remark 27.1.11. *Backpropagation*
To calculate the gradient of loss function with respect to each of individual parameters of the neural network. Model training begins with random initialisation of weights and biases.

i. Forward pass: input is sampled from training data. Nodes receive input vector and passes their value (multiplied by random initial weight) to nodes of first hidden layer. The hidden units take weighted sum of these output values as an input to the activation function, whose output is used for next hidden layer.
ii. Error computation: the final output of the network is compared to the ground truth, difference is calculated for the error value.
iii. Backwards pass: the error value computed earlier is used to compute the gradient of loss function. The gradient is then propagated back through the network, and the weights are updated according to their contribution to the error. The learning rate determines the size of weight updates.
iv. Weights update: the weights are updated in opposite direction of the gradient

Algorithm 1: Backpropagation Learning Algorithm

Require:
A set of training examples $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$
A multilayer network with L layers, weights w_{ij}^l, and activation function f
Loss function $J(y, o)$
Learning rate $0 < \alpha < 1$
Number of epochs *epochs*.

Initialize all weights:
for w_{ij}^l in the network **do**
  $w_{ij}^l \leftarrow$ small random number
end for

for $i = 1$ to *epochs* **do**
  for all training examples $(x, y) \in \mathcal{D}$ **do**

Propagate the inputs forward to compute the outputs
for all neurons i in the input layer **do**
$a_i^0 \leftarrow x_i$
end for

for $l = 2$ to L **do**
for all neurons i in layer l **do**
$z_i^l \leftarrow \sum_j w_{ij}^l \, a_j^{l-1}$
$a_i^l \leftarrow f(z_i^l)$
end for
end for

Propagate deltas backward from output layer to input layer
for all neurone i in the output layer **do**
Let $o_i \equiv a_i^L$ *be output of neurone* i
$\delta_i^L \leftarrow \frac{\partial J(y_i, o_i)}{\partial o_i} \, f'(z_i^L)$
end for

for $l = L - 1$ to 1 **do**
for all neurons i in layer l **do**
$\delta_i^l \leftarrow f'(z_i^l) \sum_j (w_{ji}^{l+1} \, \delta_j^{l+1})$
end for
end for

Update weights using the deltas
for all w_{ij}^l in the network **do**
$w_{ij}^l \leftarrow w_{ij}^l - \alpha \, \delta_i^l \, a_j^{l-1}$
end for
end for
end for

Example 27.1.12. *Backpropagation Computation Example (Matt Mazur (2015))*

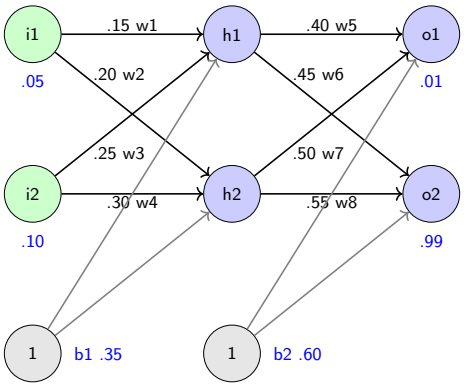


Figure 177: Backpropagation Example

For the forward pass, compute the total net input to each hidden layer neurone, squash total net input with activation function (logistic function), then repeat process with output layer neurones.
The total net input for h_1 is:

$$\begin{aligned} \text{net}_{h1} &= w_1 \cdot i_1 + w_2 \cdot i_2 + b_1 \cdot 1 \\ &= 0.15 \cdot 0.05 + 0.2 \cdot 0.1 + 0.35 \cdot 1 = 0.3775 \end{aligned}$$

Squash with logistic function to get output of h_2:

$$\text{out}_{h1} = 1/(1 + e^{-\text{net}_{h1}}) = 1/(1 + e^{-0.3775}) = 0.593269992$$

For h_2, repeating the same process, the output is then $\text{out}_{h2} = 0.596884378$.

Repeating the process for outer layer neurones,

$$\begin{aligned}\text{net}_{o1} &= w_5 \cdot \text{out}_{h1} + w_6 \cdot \text{out}_{h2} + b_2 \cdot 1 \\ &= 0.4 \cdot 0.593269992 + 0.45 \cdot 0.596884378 + 0.6 \cdot 1 = 1.105905967 \\ \text{out}_{o1} &= 1/(1 + e^{-\text{net}_{o1}}) = 1/(1 + e^{-1.105905967}) = 0.75136507\end{aligned}$$

For o_2, repeating the same process, the output is then $\text{out}_{o2} = 0.772928465$.
The total error for each output neurone may be computed using squared error function $E_{\text{total}} = \sum \frac{1}{2}(Y - \hat{Y})^2$.
For o_1, the target output is 0.01, neurone output is 0.75136507, hence error is

$$E_{o1} = \frac{1}{2}(0.01 - 0.75136507)^2 = 0.274811083$$

Repeating the process for o_2, the total error for neural network is then

$$E_{\text{total}} = E_{o1} + E_{o2} = 0.274811083 + 0.023560026 = 0.298371109$$

For the backpropagation step, update each of the weights in network to reduce the error for each output neurone.

$$\begin{aligned}E_{\text{total}} &= \frac{1}{2}(Y_{o1} - \hat{Y}_{o1})^2 + \frac{1}{2}(Y_{o2} - \hat{Y}_{o2})^2 \\ \frac{\partial E_{\text{total}}}{\partial \text{out}_{o1}} &= 2 \cdot \frac{1}{2}(Y_{o1} - \hat{Y}_{o1}) \cdot -1 + 0 = -(0.01 - 0.75136507) = 0.74136507\end{aligned}$$

Also, note that

$$\begin{aligned}\text{out}_{o1} &= 1/(1 + e^{-\text{net}_{o1}}) \\ \frac{\partial \text{out}_{o1}}{\partial \text{net}_{o1}} &= \text{out}_{o1}(1 - \text{out}_{o1}) = 0.75136507(1 - 0.75136507) = 0.186815602 \\ \text{net}_{o1} &= w_5 \cdot \text{out}_{h1} + w_6 \cdot \text{out}_{h2} + b_2 \cdot 1 \\ \frac{\text{net}_{o1}}{w_5} &= 1 \cdot \text{out}_{h1} \cdot w_5^0 + 0 + 0 = \text{out}_{h1} = 0.593269992\end{aligned}$$

Putting it all together,

$$\begin{aligned}\frac{\partial E_{\text{total}}}{\partial w_5} &= \frac{\partial E_{\text{total}}}{\partial \text{out}_{o1}} \cdot \frac{\partial \text{out}_{o1}}{\partial \text{net}_{o1}} \cdot \frac{\partial \text{out}_{o1}}{\partial w_5} \\ &= -(Y_{o1} - \hat{Y}_{o1}) \cdot \text{out}_{o1}(1 - \text{out}_{o1}) \cdot textout_{o1} \\ &= 0.74136507 \cdot 0.186815602 \cdot 0.593269992 = 0.0821267041\end{aligned}$$

To decrease the error, subtract the value from current weight multiplied by learning rate (i.e., $\alpha = 0.5$).

$$w_5^+ = w_5 - \alpha \frac{\partial E_{\text{total}}}{\partial w_5} = 0.4 - 0.5 \cdot 0.0821267041$$

Repeating for the rest of the weights, we get

$$w_6^+ = 0.408666186, \quad w_7^+ = 0.511301270, \quad w_8^+ = 0.561370121$$

For the hidden layers, note that

$$\begin{aligned}\frac{\partial E_{\text{total}}}{\partial w_1} &= \frac{\partial E_{\text{total}}}{\partial \text{out}_{h1}} \cdot \frac{\partial \text{out}_{h1}}{\partial \text{net}_{h1}} \cdot \frac{\partial \text{net}_{h1}}{\partial w_1} \\ &= \left(\frac{\partial E_{o1}}{\partial \text{out}_{h1}} + \frac{\partial E_{o2}}{\partial \text{out}_{h1}}\right) \cdot \frac{\partial \text{out}_{h1}}{\partial \text{net}_{h1}} \cdot \frac{\partial \text{net}_{h1}}{\partial w_1}\end{aligned}$$

Note that out_{h1} affects both out_{o1} and out_{o2}.

$$\begin{aligned}\frac{\partial E_{o1}}{\partial \text{out}_{h1}} &= \frac{\partial E_{o1}}{\partial \text{net}_{h1}} \cdot \frac{\partial \text{net}_{h1}}{\partial \text{out}_{h1}} \\ \frac{\partial E_{o1}}{\partial \text{net}_{h1}} &= \frac{\partial E_{o1}}{\partial \text{out}_{h1}} \cdot \frac{\partial \text{out}_{o1}}{\partial \text{net}_{h1}} = 0.74136507 \cdot 0.186815602 = 0.138498562 \\ \frac{\partial \text{net}_{h1}}{\partial \text{out}_{h1}} &= w_5 = 0.4 \\ \frac{\partial E_{o1}}{\partial \text{out}_{h1}} &= 0.138498562 \cdot 0.4 = 0.055399425\end{aligned}$$

Following same process, $\frac{\partial E_{o2}}{\partial \text{out}_{h1}} = -0.019049119$. Hence

$$\frac{\partial E_{\text{total}}}{\partial \text{out}_{h1}} = \frac{\partial E_{o1}}{\partial \text{out}_{h1}} + \frac{\partial E_{o2}}{\partial \text{out}_{h1}} = 0.05539425 - 0.019049119 = 0.036350306$$

Next, figure out $\frac{\partial \text{out}_{h1}}{\partial \text{net}_{h1}}$ and $\frac{\partial \text{net}_{h1}}{\partial w_1}$.

$$\frac{\partial \text{out}_{h1}}{\partial \text{net}_{h1}} = \text{out}_{h1} \cdot (1 - \text{out}_{h1}) = 0.59326999(1 - 0.59326999) = 0.2410.241300709$$
$$\frac{\partial \text{net}_{h1}}{\partial w_1} = i_1 = 0.05$$

Putting it together,

$$\frac{\partial E_{\text{total}}}{\partial w_1} = \frac{\partial E_{\text{total}}}{\partial \text{out}_{h1}} \cdot \frac{\partial \text{out}_{h1}}{\partial \text{net}_{h1}} \cdot \frac{\partial \text{net}_{h1}}{\partial w_1} = 0.036350306 \cdot 0.241300709 \cdot 0.05 = 0.000438658$$

This is also written as

$$\frac{\partial E_{\text{total}}}{\partial w_1} = \left(\sum_o \frac{\partial E_{\text{total}}}{\partial \text{out}_o} \cdot \frac{\partial \text{out}_o}{\partial \text{net}_o} \cdot \frac{\partial \text{net}_o}{\partial \text{out}_{h_1}} \right) \cdot \frac{\partial \text{out}_{h1}}{\partial \text{net}_{h1}} \cdot \frac{\partial \text{net}_{h1}}{\partial w_1}$$

Next, update w_1 and rest of the weights

$$w_1^+ = w_1 - \alpha \frac{\partial E_{\text{total}}}{\partial w_1} = 0.15 - 0.5 \cdot 0.000438658 = 0.149780716$$
$$w_2^+ = 0.19956143, \quad w_3^+ = 0.24975114, \quad w_4^+ = 0.29950229$$

Hence all the weights has been updated. After the first round of backpropagation, total error has decreased to 0.291027924. Repeating the process $10,000$ times, the error reduces to 0.0000351085.

27.2 Optimization for Deep Learning

Remark 27.2.1. *Optimisation*
A cost function $J(\boldsymbol{\theta})$ is reduced to improve some performance measure P.
Optimisation algorithms include some specialisation on specific structure of machine learning objective functions.
Cost function is average over training set,

$$J(\boldsymbol{\theta}) = \mathbb{E}_{(\boldsymbol{x},y)\sim\hat{p}_{\text{data}}} L(f(\boldsymbol{x};\boldsymbol{\theta}), y)$$

where L is per-example loss function, $f(\boldsymbol{x};\boldsymbol{\theta})$ is predicted output given input $\boldsymbol{x}$, $\hat{p}_{\text{data}}$ is empirical distribution. For supervised learning, y is target output.
The goal is to usually to minimise objective function where expectation is taken on data generating distribution p_{data} rather than over the finite training set:

$$J^*(\boldsymbol{\theta}) = \mathbb{E}_{(\boldsymbol{x},y)\sim p_{\text{data}}} L(f(\boldsymbol{x};\boldsymbol{\theta}), y)$$

Note that machine learning algorithm usually minimises a surrogate loss function, and halts when convergence criterion based on early stopping is satisfied.

Definition 27.2.2. *Empirical Risk Minimisation*
To minimise expected loss on training set, where $\hat{p}(\boldsymbol{x}, y)$ is the empirical distribution. The empirical risk is

$$\mathbb{E}_{\boldsymbol{x},y\sim\hat{p}_{\text{data}}(\boldsymbol{x},y)}[L(f(\boldsymbol{x},\boldsymbol{\theta}), y)] = \frac{1}{m}\sum_{i=1}^{m} L(f(\boldsymbol{x}^{(i)};\boldsymbol{\theta}), y^{(i)})$$

where m is number of training samples. Note that empirical risk minimisation is prone to overfitting, as models with high capacity may memorise the training set.

Remark 27.2.3. *Batch and Minibatch Algorithms*
Objective algorithms for machine learning computes each update to the parameters based on expected value of cost function using only a subset of terms of full cost function.
For maximum likelihood estimation problems, note that

$$\boldsymbol{\theta}_{\text{ML}} = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{m} \log p_{\text{model}}(\boldsymbol{x}^{(i)}, y^{(i)}; \boldsymbol{\theta})$$
$$J(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{x},y\sim\hat{p}_{\text{data}}} \log p_{\text{model}}(\boldsymbol{x}, y; \boldsymbol{\theta})$$

The gradient is then

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{x},y\sim\hat{p}_{\text{data}}} \nabla_{\boldsymbol{\theta}} \log\log p_{\text{model}}(\boldsymbol{x}, y; \boldsymbol{\theta})$$

If the optimisation algorithm uses the entire training set, then it is a batch gradient method.
If a single sample is used at a time, then this is a stochastic/online gradient method.
For algorithms that uses more than one but less than all samples, then this is a minibatch gradient method.

Remark 27.2.4. *Challenges: Ill-Conditioning of Hessian Matrix H*
May cause stochastic gradient descent to be stuck where very small step increases the cost function.

Note that a gradient descent step of $-\epsilon \boldsymbol{g}$ will add

$$\frac{1}{2}\epsilon^2 \boldsymbol{g}^T \boldsymbol{H}\boldsymbol{g} - \epsilon \boldsymbol{g}^T \boldsymbol{g}$$

to the cost. This will be a problem when $\frac{1}{2}\epsilon^2 \boldsymbol{g}^T \boldsymbol{H}\boldsymbol{g} > \epsilon \boldsymbol{g}^T \boldsymbol{g}$.
Monitor squared gradient norm $\boldsymbol{g}^T \boldsymbol{g}$ and $\boldsymbol{g}^T \boldsymbol{H}\boldsymbol{g}$ term. When ill-conditioning happens, gradient norm does not shrink significantly but $\boldsymbol{g}^T \boldsymbol{H}\boldsymbol{g}$ term grows more than order of magnitude, and learning rate shrinks.

Remark 27.2.5. *Challenges: Local Minimums*
Neural networks and models with multiple parametrised latent variables have multiple local minima due to *model identifiability* problem. Model is identifiable if a sufficiently large training set can rule out all but one setting of model parameters. Models with latent variables are not identifiable as equivalent models are obtained by exchanging latent variables with each other (*weight space symmetry*). If there are m layers with n units each, then there are $n!^m$ ways to arrange the hidden units, making the model unidentifiable.
There may be an extremely large or uncountably infinite amount of local minima in a cost function. If the local minima have a high cost compared to global minimum, then this is problematic.

Remark 27.2.6. *Challenges: Plateaus, Saddle Points, Flat Regions*
For a function $f : \mathbb{R}^n \to \mathbb{R}$, expected ratio of saddle points to local minima grows exponentially with n. Hessian matrix at a saddle point has a mixture of positive and negative eigenvalues.
Degenerate regions with flat gradient may correspond to high value of objective function.

Remark 27.2.7. *Challenges: Cliffs and Exploding Gradients*
On extremely steep cliff structure, gradient update step may move parameters extremely far, losing most of the optimisation work done previously. The consequences may be mitigated using gradient clipping heuristics, which reduces the step size to be small enough.

Remark 27.2.8. *Challenges: Long-Term Dependencies*
Occurs when computational graph becomes very deep. Let a computational graph contain a path that repeatedly multiply a matrix $\boldsymbol{W}$. After t steps, this is $\boldsymbol{W}^t$. Let the eigen-decomposition of $\boldsymbol{W}$ be $\boldsymbol{W} = \boldsymbol{V}\text{diag}(\boldsymbol{\lambda})\boldsymbol{V}^{-1}$. Then we can see that $\boldsymbol{W}^t = (\boldsymbol{V}\text{diag}(\boldsymbol{\lambda})\boldsymbol{V}^{-1})^t = \boldsymbol{V}\text{diag}(\lambda)^t\boldsymbol{V}^{-1}$.
There will be a vanishing and exploding gradient problem if gradients on the graph are scaled according to $\text{diag}(\boldsymbol{\lambda})^t$. Note that recurrent networks uses same $\boldsymbol{W}$ at each time step, hence may face the problem.

Remark 27.2.9. *Challenges: Inexact Gradients*
Objective function for minimisation is intractable due to noisy/biased estimate of these quantities; in this case its gradient is intractable as well. Choose surrogate loss function that is easier to approximate than true loss.

Remark 27.2.10. *Challenges: Poor Correspondence between Local and Global Structure*
As gradient descent algorithms make small local moves, optimisation on local downhill moves may fail if local surface does not point toward the global solution. In higher dimensional space, learning algorithms may incur high costs or long training time to circumnavigate the manifold. Hence to find good initial points for problems.

Remark 27.2.11. *Stochastic Gradient Descent (SGD)*
Crucial parameter is the learning rate, which may be changed over time; at iteration k this is ϵ_k.
The SGD gradient estimation introduces a source of noise that does not vanish even at minimum. A sufficient condition to guarantee converge of SGD is

$$\sum_{k=1}^{\infty} \epsilon_k = \infty, \quad \sum_{k=1}^{\infty} \epsilon_k^2 < \infty$$

In practice, it is common to decay learning rate linearly until iteration τ:

$$\epsilon_k = (1-\alpha)\epsilon_0 + \alpha\epsilon_\tau, \quad \alpha = \frac{k}{\tau}$$

After iteration τ, leave ϵ constant.
Compute time per update does not growth with number of training examples.

Algorithm 2: Stochastic Gradient Descent (SGD) Update at Training Iteration k

Require:
Learning rate ϵ_k, Initial parameter $\boldsymbol{\theta}$

while stopping criterion not met **do**
Sample a minibatch of m samples from the training set $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ with targets $\boldsymbol{y}^{(i)}$
Compute gradient estimate $\hat{\boldsymbol{g}} \leftarrow +\frac{1}{m}\nabla_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \epsilon\hat{\boldsymbol{g}}$

end while

Remark 27.2.12. *Momentum Algorithm*
Algorithm introduces variable $\boldsymbol{v}$ that is set to an exponentially decaying average of negative gradient.
Hyperparameter $\alpha \in [0, 1)$ determines how quickly the contributions of previous gradients exponentially decay.
The update rule is given by

$$\boldsymbol{v} \leftarrow \alpha \boldsymbol{v} - \epsilon \nabla_{\boldsymbol{\theta}} \left(\frac{1}{m} \sum_{i=1}^{m} L(\boldsymbol{f}(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)}) \right)$$
$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \boldsymbol{v}$$

The larger α is relative to ϵ, the more previous gradients affect current direction.
If gradient is $\boldsymbol{g}$, it will accelerate in direction of $-\boldsymbol{g}$ until reaching terminal velocity where size of each step is

$$\frac{\epsilon \|\boldsymbol{g}\|}{1 - \alpha}$$

Algorithm 3: Stochastic Gradient Descent (SGD) with Momentum

Require:
Learning rate ϵ_k, Momentum parameter α

while stopping criterion not met **do**
Sample a minibatch of m samples from the training set $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ with targets $\boldsymbol{y}^{(i)}$
Compute gradient estimate $\boldsymbol{g} \leftarrow +\frac{1}{m} \nabla_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Compute velocity update: $\boldsymbol{v} \leftarrow \alpha \boldsymbol{v} - \epsilon \boldsymbol{g}$
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \boldsymbol{v}$
end while

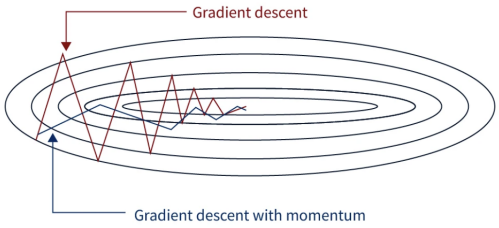


Figure 178: Stochastic Gradient Descent (SGD) Algorithm with and without Momentum

Remark 27.2.13. *Nesterov Momentum*
Inspired by Nesterov's accelerated gradient method, the update rules are:

$$\boldsymbol{v} \leftarrow \alpha \boldsymbol{v} - \epsilon \nabla_{\boldsymbol{\theta}} \left(\frac{1}{m} \sum_{i=1}^{m} L(\boldsymbol{f}(\boldsymbol{x}^{(i)}; \boldsymbol{\theta} + \alpha \boldsymbol{v}), \boldsymbol{y}^{(i)}) \right)$$
$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \boldsymbol{v}$$

Note, gradient is evaluated after current velocity is applied.
Hence, Nesterov momentum is adding a correction factor to the standard method of momentum.

Algorithm 4: Stochastic Gradient Descent (SGD) with Nesterov Momentum

Require:
Learning rate ϵ_k, Momentum parameter α
Initial parameter $\boldsymbol{\theta}$, Initial velocity $\boldsymbol{v}$

while stopping criterion not met **do**
Sample a minibatch of m samples from the training set $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ with targets $\boldsymbol{y}^{(i)}$
Apply interim update: $\tilde{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta} + \alpha \boldsymbol{v}$
Compute gradient (at interim point): $\boldsymbol{g} \leftarrow \frac{1}{m} \nabla_{\tilde{\boldsymbol{\theta}}} \sum_i L(f(\boldsymbol{x}^{(i)}; \tilde{\boldsymbol{\theta}}), \boldsymbol{y}^{(i)})$
Compute velocity update: $\boldsymbol{v} \leftarrow \alpha \boldsymbol{v} - \epsilon \boldsymbol{g}$
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \boldsymbol{v}$
end while

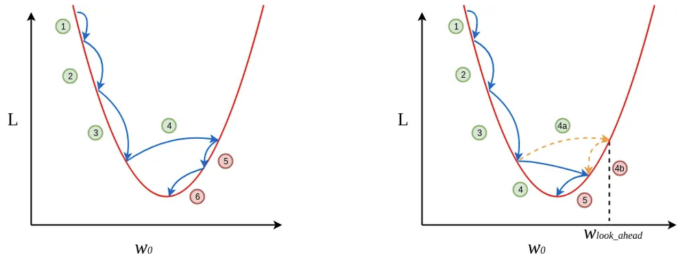


Figure 179: Momentum vs Nesterov Gradient Descent

Remark 27.2.14. *Parameter Initialisation Challenges*
Training deep neural networks is difficult as most algorithms are strongly affected by choice of initialisation.
Designing initialisation strategies is difficult as neural network optimisation is not yet well understood, as the nice properties from initialisation may not be preserved as training proceeds.
Some initial points may be beneficial for optimisation, but detrimental for generalisation.

Remark 27.2.15. *Parameter Initialisation Strategies*
Initial parameters need to 'break symmetry' between different units. May explicitly search for large set of basis functions mutually different from each other, but this may incur noticeable computational cost.
Random initialisation from high-entropy distribution over high-dimensional space is computationally cheaper.
Larger initial weights will yield stronger symmetry breaking effect and avoid redundant units. However, if weights are set too large, may result in chaos (extreme sensitivity to small perturbations of input). Exploding gradient may be mitigated by gradient clipping. The bias (i.e., parameters encoding conditional variance of a prediction) for each unit is set to heuristically chosen constraints.

Remark 27.2.16. *Weight Initialisation Heuristics*
Initialising parameters of neural network $\boldsymbol{\theta}$ to $\boldsymbol{\theta}_0$ is similar to imposing a Gaussian prior $p(\boldsymbol{\theta})$ with mean $\boldsymbol{\theta}_0$, hence it make sense to choose $\boldsymbol{\theta}_0$ near 0. Initialisation of $\boldsymbol{\theta}_0$ to large values means prior specifies which units should interact with each other, and how they should interact.
For a fully connected neural network with m inputs and n outputs, weights may be initialised by:

i. Sampling from $U\left(-\frac{1}{\sqrt{m}}, \frac{1}{\sqrt{m}}\right)$
ii. Sampling from normalised initialisation $W_{i,j} \sim U\left(-\sqrt{\frac{6}{m+n}}, \sqrt{\frac{6}{m+n}}\right)$ (Glorot and Bengio). Compromise between goal of initialising all layers to have same activation variance and that of same gradient variance. Derived from assumption that network is chain of matrix multiplications with no nonlinearities.
iii. Sampling from random orthogonal matrices, with scaling or gain factor g that accounts for nonlinearity at each layer, with specific values for different types of nonlinear activation functions. Assumes neural network is sequence of matrix multiplications without nonlinearities. Guarantees the total number of training iterations required to reach convergence is independent of depth.
Increasing scaling factor g pushes network toward regime where activations increase in norm in forward propagation, and gradients increase in norm in backward propagation.
For scaling rules that set all of initial weights to have same standard deviation, individual weight becomes extremely small when layers become large.
iv. Sparse initialisation (Martens), where each unit is initialised to have exactly k non-zero weight. Keep total number of input to unit independent from number of inputs m without making magnitude of individual weight elements shrink with m.
Imposes very strong prior on weights that are chosen to have large Gaussian values.

If computational resources allow, treat initial scale of weights for each layer as a hyper-parameter, and use hyper-parameter search algorithm. Choice of dense or sparse initialisation may also be a hyper-parameter.

Remark 27.2.17. *Bias Initialisation Heuristics*
Setting bias to zero is compatible with most weight initialisation schemes. The few other situations are:

i. If bias is for output unit, initialise bias to obtain right marginal statistics of the output. Assume initial weights are small enough that output of unit is determined only by bias. The bias will be inverse of activation function applied to marginal statistics of the output in training set.
If output is a highly skewed distribution over classes with marginal probability of class i given by element c_i of some vector $\boldsymbol{c}$, then bias vector $\boldsymbol{b}$ is such that $\text{softmax}(\boldsymbol{b}) = \boldsymbol{c}$. The models have output layers that should resemble input data $\boldsymbol{x}$, hence bias should match marginal distribution over $\boldsymbol{x}$.
ii. If bias is to be set to avoid causing too much saturation at initialisation (i.e., with ReLU). Not compatible with weight initialisation schemes that do not expect strong input form bias (i.e., with random walk init).

iii. If a unit controls whether other units are able to participate in a function. There is unit with output u and another unit $h \in [0, 1]$, multiplied together to produce output uh. Then h is a gate that determines if $uh \approx u$ or $uh \approx 0$. Hence, to set bias for h such that $h \approx 1$ at initialisation, else u does not learn.

Remark 27.2.18. *Variance or Precision Parameter Initialisation Heuristics*
For linear regression with conditional variance estimate using model

$$p(y \mid \boldsymbol{x}) = \mathcal{N}(y \mid \boldsymbol{w}^T \boldsymbol{x} + b, 1/\beta)$$

where β is a precision parameter. The variance of precision parameters may be initialised to 1.
Another approach is to assume initial weights are close enough to zero, and bias may be set while ignoring effect of the weights, then set bias to produce the correct marginal mean of the output, and set variance parameters to the marginal variance of the output in the training set.

Remark 27.2.19. *Model Parameter Initialisation with Machine Learning*
Initialise supervised model with parameters learned by unsupervised model trained on same inputs.
Alternatively, perform supervised training on a related task, which may yield initialisation that offers faster convergence than random initialisation. Information on distribution is encoded into initial parameters of model.

Remark 27.2.20. *Adaptive Gradient (AdaGrad) Algorithm*
Algorithm individually adapts learning rates of all model parameters by scaling them inversely proportional to square root of sum of all of their historical squared values. Has greater progress in more gently sloped directions of parameter space. On deep neural networks, accumulation of squared gradients from beginning of training can result in premature and excessive decrease in effective learning rate.

Algorithm 5: Adaptive Gradient (AdaGrad) Algorithm

Require:
Global Learning rate ϵ
Initial parameter $\boldsymbol{\theta}$
Small constant δ, perhaps 10^{-7} for numerical stability

Initialise gradient accumulation variable $\boldsymbol{r} = \boldsymbol{0}$
while stopping criterion not met **do**
Sample a minibatch of m samples from the training set $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ with targets $\boldsymbol{y}^{(i)}$
Compute gradient: $\boldsymbol{g} \leftarrow \frac{1}{m} \nabla_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Accumulate squared gradient: $\boldsymbol{r} \leftarrow \boldsymbol{r} + \boldsymbol{g} \odot \boldsymbol{g}$
Compute update: $\Delta\boldsymbol{\theta} \leftarrow -\frac{\epsilon}{\delta + \sqrt{\boldsymbol{r}}} \odot \boldsymbol{g}$ (Division and square root applied element-wise)
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \Delta\boldsymbol{\theta}$
end while

Remark 27.2.21. *Root Mean Squared Propagation (RMSProp) Algorithm*
Modifies AdaGrad to perform better in non-convex setting by changing gradient accumulation into exponentially weighted moving average, which discards history from extreme past so it can converge rapidly after finding a convex bowl. Use of moving average introduces a new hyper-parameter ρ that controls length scale of the moving average. RMSProp is effective and practical for deep neural networks.

Algorithm 6: Root Mean Squared Propagation (RMSProp) Algorithm

Require:
Global Learning rate ϵ
Decay rate ρ
Initial parameter $\boldsymbol{\theta}$
Small constant δ, perhaps 10^{-6} to stabilise division by small numbers

Initialise accumulation variable $\boldsymbol{r} = \boldsymbol{0}$
while stopping criterion not met **do**
Sample a minibatch of m samples from the training set $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ with targets $\boldsymbol{y}^{(i)}$
Compute gradient: $\boldsymbol{g} \leftarrow \frac{1}{m} \nabla_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Accumulate squared gradient: $\boldsymbol{r} \leftarrow \rho\boldsymbol{r} + (1 - \rho)\boldsymbol{g} \odot \boldsymbol{g}$
Compute update: $\Delta\boldsymbol{\theta} \leftarrow -\frac{\epsilon}{\sqrt{\delta + \boldsymbol{r}}} \odot \boldsymbol{g}$ ($\frac{\epsilon}{\sqrt{\delta + \boldsymbol{r}}}$ applied element-wise)
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \Delta\boldsymbol{\theta}$
end while

Algorithm 7: Root Mean Squared Propagation (RMSProp) Algorithm

Require:
Global Learning rate ϵ
Decay rate ρ
Momentum coefficient α
Initial parameter $\boldsymbol{\theta}$
Initial velocity $\boldsymbol{v}$

Initialise accumulation variable $\boldsymbol{r} = \boldsymbol{0}$
while stopping criterion not met **do**
Sample a minibatch of m samples from the training set $\{\boldsymbol{x}^{(1)}, \dots, \boldsymbol{x}^{(m)}\}$ with targets $\boldsymbol{y}^{(i)}$
Compute interim update $\tilde{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta} + \alpha \boldsymbol{v}$
Compute gradient: $\boldsymbol{g} \leftarrow \frac{1}{m} \nabla_{\tilde{\boldsymbol{\theta}}} \sum_i L(f(\boldsymbol{x}^{(i)}; \tilde{\boldsymbol{\theta}}), \boldsymbol{y}^{(i)})$
Accumulate gradient: $\boldsymbol{r} \leftarrow \rho \boldsymbol{r} + (1-\rho) \boldsymbol{g} \odot \boldsymbol{g}$
Compute velocity update: $\boldsymbol{v} \leftarrow \alpha \boldsymbol{v} - \frac{\epsilon}{\sqrt{\boldsymbol{r}}} \odot \boldsymbol{g}$ ($\frac{1}{\sqrt{\boldsymbol{r}}}$ applied element-wise)
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \boldsymbol{v}$
end while

Remark 27.2.22. *Adaptive Moment Estimation (Adam) Algorithm*
Variant on combination of RMSProp and momentum. Note, momentum is incorporated directly as an estimate of first order moment of gradient; bias corrections are made to estimates of both first-order moments and second-order moments to account for initialisation at origin.
Algorithm is fairly robust to choice of hyper-parameters.

Algorithm 8: Adaptive Moment Estimation (Adam) Algorithm

Require:
Step size ϵ (Suggested default: 0.001)
Exponential decay rates for moment estimates ρ_1, ρ_2 in $[0, 1)$ (Suggested defaults: 0.9 and 0.999)
Small constant δ used for numerical stabilisation (Suggested default: 10^{-8})
Initial parameters $\boldsymbol{\theta}$

Initialise 1st and 2nd momentum variables $\boldsymbol{s} = \boldsymbol{0}$, $\boldsymbol{r} = \boldsymbol{0}$
Initialise time step $t = 0$
while stopping criterion not met **do**
Sample a minibatch of m samples from the training set $\{\boldsymbol{x}^{(1)}, \dots, \boldsymbol{x}^{(m)}\}$ with targets $\boldsymbol{y}^{(i)}$
Compute gradient: $\boldsymbol{g} \leftarrow \frac{1}{m} \nabla_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Update time step $t \leftarrow t + 1$
Update biased first and second moment estimate: $\boldsymbol{s} \leftarrow \rho_1 \boldsymbol{s} + (1-\rho_1)\boldsymbol{g}$ and $\boldsymbol{r} \leftarrow \rho_2 \boldsymbol{r} + (1-\rho_2)\boldsymbol{g} \odot \boldsymbol{g}$
Correct bias in first and second moment: $\hat{\boldsymbol{s}} \leftarrow \frac{\boldsymbol{s}}{1-\rho_1^t}$ and $\hat{\boldsymbol{r}} \leftarrow \frac{\boldsymbol{r}}{1-\rho_2^t}$
Compute update: $\Delta\boldsymbol{\theta} = -\epsilon \frac{\hat{\boldsymbol{s}}}{\sqrt{\hat{\boldsymbol{r}}} + \delta}$ (Operations applied element-wise)
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \Delta\boldsymbol{\theta}$
end while

Remark 27.2.23. *Newton's Method*
Based on second-order Taylor's series approximating $J(\boldsymbol{\theta})$ near the point $\boldsymbol{\theta}_0$, ignoring higher derivatives

$$J(\boldsymbol{\theta}) \approx J(\boldsymbol{\theta}_0) + (\boldsymbol{\theta} - \boldsymbol{\theta}_0)^T \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_0) + \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)^T \boldsymbol{H} (\boldsymbol{\theta} - \boldsymbol{\theta}_0)$$

where $\boldsymbol{H}$ is the Hessian of J with respect to $\boldsymbol{\theta}$ evaluated at $\boldsymbol{\theta}_0$.
Solving for critical point will get the Newton parameter update rule

$$\boldsymbol{\theta}^* = \boldsymbol{\theta}_0 - \boldsymbol{H}^{-1} \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_0)$$

Note that for deep learning, the surface of objective function is non-convex. Regularising the Hessian will solve many of the issues such as saddle points etc.

$$\boldsymbol{\theta}^* = \boldsymbol{\theta}_0 - [H(f(\boldsymbol{\theta}_0)) + \alpha \boldsymbol{I}]^{-1} \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_0)$$

Due to requirement for inversion of the $k \times k$ matrix with time complexity $O(k^3)$ at every training iteration, only networks with very small number of parameters may be trained with this method.

Algorithm 9: Newton's Method with Objective $J(\boldsymbol{\theta}) = \frac{1}{m} \sum_{i=1}^{m} L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), y^{(i)})$

Require:
Initial parameter $\boldsymbol{\theta}_0$
Training set of m examples

while stopping criterion not met **do**
Compute gradient $\boldsymbol{g} \leftarrow \frac{1}{m}\nabla_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Compute Hessian: $\boldsymbol{H} \leftarrow \frac{1}{m}\nabla^2_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Compute Hessian Inverse: $\boldsymbol{H}^{-1}$
Compute update: $\Delta\boldsymbol{\theta} = -\boldsymbol{H}^{-1}\boldsymbol{g}$
Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \Delta\boldsymbol{\theta}$
end while

Remark 27.2.24. *Conjugate Gradient Method*
Method effectively avoids computation of inversion Hessian by descending conjugate directions, which is a search direction conjugate to the previous line search direction (will not undo progress made in that direction).
Let current and previous search directions be $\boldsymbol{d}_t, \boldsymbol{d}_{t-1}$. At training iteration t, the next search direction $\boldsymbol{d}_t$ is

$$\boldsymbol{d}_t = \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) + \beta_t \boldsymbol{d}_{t-1}$$

note that β_t may be computed via Fletcher Reeves:

$$\beta_t = \frac{\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_t)^T \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_t)}{\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_{t-1})^T \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_{t-1})}$$

or may be computed with Polak-Ribiere:

$$\beta_t = \frac{(\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_t) - \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_{t-1}))^T \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_t)}{\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_{t-1})^T \nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_{t-1})}$$

The non-linear conjugate gradient method is a variation which uses occasional resets where the method is restarted with line search along with the unaltered gradient.

Algorithm 10: Conjugate Gradient Method

Require:
Initial parameter $\boldsymbol{\theta}_0$
Training set of m examples
Initialise $\boldsymbol{\rho}_0 = 0$
Initialise $g_0 = 0$
Initialise $t = 1$

while stopping criterion not met **do**
Initialise the gradient $\boldsymbol{g}_t = \boldsymbol{0}$
Compute gradient $\boldsymbol{g} \leftarrow \frac{1}{m}\nabla_{\boldsymbol{\theta}} \sum_i L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}), \boldsymbol{y}^{(i)})$
Compute Polak-Ribiere metric: $\beta_t = \frac{(\boldsymbol{g}_t - \boldsymbol{g}_{t-1})^T \boldsymbol{g}_t}{\boldsymbol{g}_{t-1}^T \boldsymbol{g}_{t-1}}$
(Nonlinear conjugate gradient: optionally reset $\beta_t = 0$, i.e., if t is a multiple of some constant k)
Compute search direction: $\boldsymbol{\rho}_t = -\boldsymbol{g}_t + \beta_t \boldsymbol{\rho}_{t-1}$
Perform line search to find: $\epsilon^* = \arg\min_\epsilon \frac{1}{m} \sum_{i=1}^m L(f(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}_t + \epsilon\boldsymbol{\rho}_t), \boldsymbol{y}^{(i)})$
(On truly quadratic cost function, analytically solve for ϵ^*)
Apply update: $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \epsilon^{\theta}\boldsymbol{\rho}_t$
Apply time step $t \leftarrow t + 1$
end while

Remark 27.2.25. *Nonlinear Conjugate Gradients*
Method of conjugate gradients is still applicable for non-linear objective function.
Conjugate directions are no longer assumed to remain at minimum of the objective for previous directions. Has occasional resets where method of conjugate gradients is restarted with line search along unaltered gradient.

Remark 27.2.26. *Broyden-Fletcher-Goldfarb-Shanno (BGFS) Algorithm*
Similar to conjugate gradient, but uses more direct approach to approximation of newton's update.
Method approximate the inverse of hessian $\boldsymbol{H}$ with matrix $\boldsymbol{M}_t$, iteratively refined by low rank updates to better approximate $\boldsymbol{H}^{-1}$. Direction of gradient descent is then determined by $\boldsymbol{\rho}_t = \boldsymbol{M}_t \boldsymbol{g}_t$. Line search is performed in this direction to determine step size ϵ^*. Final update of parameter is then $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \epsilon^* \boldsymbol{\rho}_t$.
Note, BGFS store matrix $\boldsymbol{M}$ which requires $O(n^2)$ memory, making it impractical for most modern models.

Remark 27.2.27. *Limited Memory BFGS (L-BFGS) Algorithm*
Memory costs of BFGS may be decreased by not storing complete inverse Hessian approximation $\boldsymbol{M}$. L-BFGS computes approximation $\boldsymbol{M}$ with same method as BFGS, but assumes at beginning that $\boldsymbol{M}^{(t-1)}$ is the identity matrix, rather than storing the approximation between steps.

If used with exact line searches, directions defined by L-BFGS are mutually conjugate, and this procedure remains well-behaved when minimum of line search is reached only approximately.
L-BFGS may be generalised to store some vectors used to update $\boldsymbol{M}$ at each time step, costing $O(n)$ per step.

Remark 27.2.28. *Meta-Algorithm: Batch Normalisation*
An update from optimisation algorithm may result in multiple-order effects, and in very deep networks, even higher-order interactions can be significant.
Batch normalisation provides an elegant way of re-parametrising any deep network, which reduces the problem of coordinating updates across multiple layers. Batch normalisation may be applied to any input/hidden layer.
Let $\boldsymbol{H}$ be minibatch of activations of layer to normalise, with activations for each example appearing in a row.
To normalise $\boldsymbol{H}$, replace it with $\boldsymbol{H}' = \frac{\boldsymbol{H}-\boldsymbol{\mu}}{\boldsymbol{\sigma}}$ where $\boldsymbol{\mu}$ is a vector of mean of each unit and σ is a vector of standard deviation of each unit. Then H_{ij} is normalised by subtracting μ_j and dividing σ_j. Rest of network then operates on $\boldsymbol{H}'$ just as in original network on $\boldsymbol{H}$.
At training, the mean and standard deviations are computed as

$$\boldsymbol{\mu} = \frac{1}{m}\sum_i \boldsymbol{H}_{i,;}, \quad \boldsymbol{\sigma} = \sqrt{\delta + \frac{1}{m}\sum_i (\boldsymbol{H}-\boldsymbol{\mu})_i^2}$$

where δ is a small positive value (i.e., 10^{-8}) to avoid undefined gradient of $\sqrt{z}$ at $z = 0$.
These are then back-propagated to normalise $\boldsymbol{H}$, hence gradient will never increase the standard deviation or mean of h_i. Batch normalisation re-parametrises the model to make some units always standardised.
At test time, μ and σ may be replaced by running averages from training time. This allows model to be evaluated on a single example, without needing to use definitions of μ and $\boldsymbol{\sigma}$ that depends on entire minibatch.

Remark 27.2.29. *Batch Normalisation but Maintain Network Express Power*
Replace batch of hidden activations $\boldsymbol{H}$ with $\boldsymbol{\gamma}\boldsymbol{H}' + \boldsymbol{\beta}$ rather than $\boldsymbol{H}'$, where $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ are learned parameters to allow new variable to have any mean and standard deviation. With original $\boldsymbol{H}'$, mean of $\boldsymbol{H}$ was determined by complicated interaction between parameters in layers below $\boldsymbol{H}$. With $\boldsymbol{\gamma}\boldsymbol{H}' + \boldsymbol{\beta}$, the mean is determined solely by $\boldsymbol{\beta}$, which is much easier to learn with gradient descent.
Most neural network layers are of form $\phi(\boldsymbol{X}\boldsymbol{W} + \boldsymbol{b})$, where ϕ is some fixed nonlinear activation function. Then $\boldsymbol{X}\boldsymbol{W} + \boldsymbol{b})$ should be replaced by normalised version of $\boldsymbol{X}\boldsymbol{W}$, and bias term omitted as it becomes redundant with β parameter applied from batch normalisation re-parametrisation. Statistics of input are non-Gaussian and less amenable to standardisation by linear operations.

Remark 27.2.30. *Meta-Algorithm: Coordinate Descent*
Solving optimisation problems by minimising the objective function $f(\boldsymbol{x})$ with respect to a single variable x_i, and rapidly cycling through all variables. *Block coordinate descent* refers to minimising with respect to a subset of variables simultaneously.
Coordinate descent works when different variables can be clearly separated into groups that play relatively isolated roles, or when optimisation with respect to one group of variables is significantly more efficient than optimisation with respect to all of the variables.

Example 27.2.31. *Sparse Encoding with Coordinate Descent*
Consider the cost function

$$J(\boldsymbol{H}, \boldsymbol{W}) = \sum_{i,j} |H_{ij}| + \sum_{i,j} (\boldsymbol{X} - \boldsymbol{W}^T\boldsymbol{H})_{i,j}^2$$

To find weight matrix $\boldsymbol{W}$ that can linearly decode a matrix of activation values $\boldsymbol{H}$ to reconstruct training set $\boldsymbol{X}$. Most applications also include weight decay or constraint on norms of columns of $\boldsymbol{W}$ to prevent the pathological solution with extremely small $\boldsymbol{H}$ and large $\boldsymbol{W}$.
Note J is not convex, but inputs may be divided into the dictionary parameters $\boldsymbol{W}$ and the code representations $\boldsymbol{H}$. Minimising object function with respect to either set is a convex problem.

Remark 27.2.32. *Polyak Averaging*
Average several points in trajectory through parameter space visited by optimisation algorithm.
If t iterations of gradient descent visit points $\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(t)}$, then the output is $\hat{\boldsymbol{\theta}}^{(t)} = \frac{1}{t}\sum_i \boldsymbol{\theta}^{(i)}$.
The optimisation algorithm may leap back and forth across a valley several times without visiting the minimum point, and the average of all points on either side of the valley should be close to minimum value.
This is developed solely for stochastic gradient descent (SGD) to stabilise the estimates (Granziol et al. (2020)).
For non-convex problems, an exponentially decaying running average is used:

$$\hat{\boldsymbol{\theta}}^{(t)} = \alpha\hat{\boldsymbol{\theta}}^{(t-1)} + (1-\alpha)\boldsymbol{\theta}^{(t)}$$

Remark 27.2.33. *Supervised Pre-training*
Pre-training refers to training simple models on simple tasks before training the desired model on desired task.

Greedy algorithms break problem into many components, then solve for optimal version of each component in isolation, which is must cheaper than algorithms that solve for best joint solution. Fine-tuning on joint optimisation algorithm searches for optimal solution to the full problem.
Greedy supervised training breaks supervised learning problems into other simpler supervised learning problems.
Transfer learning refers to pre-training a deep network on a set of tasks, then initialise a same-size network with first k layers of first net. The layers of second networks (upper layers initialised randomly) are then jointly trained to perform a different set of tasks with fewer training examples than for first set of tasks.
FitNets approach begins by training a network with low depth and high width, which becomes a teacher for a second network (the student), which is much deeper and thinner. Student network is trained to not only predict the output on original task, but also on value of middle layer of teacher network.

Remark 27.2.34. *Neural Network Design Choice: Linear Transformation*
Model units have generally moved towards using more linear functions, which allow model to have nice properties that make optimisation easier. The model's local gradient information corresponds reasonably well to moving towards a distant solution.

Remark 27.2.35. *Neural Network Design Choice: Linear Paths and Skip Connections*
Linear paths and skip connections between layers reduce length of shortest path from lower layer's parameters to the output, mitigating vanishing gradient problem.
Adding extra copies of output that are attached to intermediate hidden layers is an alternative to pre-training, where these 'auxiliary heads' are trained to perform the same task as primary output at the top of the network to ensure that the lower layers receive a large gradient. After training, these may be discarded.

Remark 27.2.36. *Continuation Methods*
Family of strategies that can make optimisation easier by choosing initial points to ensure that local optimisation spends most of its time in well-behaved regions. Method constructs series of objective functions $\{J^{(0)}, \dots, J^{(n)}\}$ which are increasingly difficult to minimise (well-behaved over less regions of $\boldsymbol{\theta}$ space) for the true cost function $J(\boldsymbol{\theta})$. The cost functions are designed such that a solution to one is a good initial point for the next.
Continuation methods aim to reach a global minimum despite many local minima. Easier cost functions are constructed through the blurring operation, which approximates

$$J^{(i)}(\boldsymbol{\theta}) = \mathbb{E}_{\theta' \sim \mathcal{N}(\boldsymbol{\theta}'; \boldsymbol{\theta}, \sigma^{(i)2})} J(\boldsymbol{\theta})'$$

via sampling. Some non-convex functions become approximately convex when blurred, but preserved enough information about local global minimum. However, approach might not work when

i. so many incremental cost functions are required that cost of entire procedure remains high;
ii. NP-hard optimisation problems remain NP-hard, even when continuation methods are applicable;
iii. function might not become convex, no matter how much it is blurred;
iv. function may become convex due to blurring, but minimum of this blurred function may track a local rather than global minimum of original cost function.

Remark 27.2.37. *Curriculum Learning*
Planning a learning process to begin by learning simple concepts, and progress to learning more complex topics that depend on these simpler concepts. Earlier objective functions $J^{(i)}$ are made easier by increasing the influence of simpler examples.
Stochastic curriculum is a variation of the approach where a random mix of easy and difficult examples is always presented to the learner, but average proportion of more difficult examples is gradually increased.

27.3 Regularization and Generalization

Definition 27.3.1. *Regularisation* refers to adding a parameter norm penalty $\Omega(\boldsymbol{\theta})$ to the objective function J. The regularised objective function is then

$$\tilde{J}(\boldsymbol{\theta}; \boldsymbol{X}, \boldsymbol{y}) = J(\boldsymbol{\theta}; \boldsymbol{X}, \boldsymbol{y}) + \alpha\Omega(\boldsymbol{\theta})$$

where $\alpha in [0, \infty)$ is a hyper-parameter that weights the contribution of norm penalty term.
For neural networks, the parameter norm penalty Ω is chosen such that it penalises only the weights of the affine transformation at each layer and leaves the biases un-regularised.

Definition 27.3.2. *L^2 (Ridge) Regularisation*
The regularisation term added to objective function is

$$\Omega(\boldsymbol{\theta}) = \frac{1}{2}\|\boldsymbol{w}\|_2^2$$

Remark 27.3.3. *Behaviour of Weight Decay (L^2) Regularisation*
Assuming no bias parameter, a model have the following objective function and parameter gradient:

$$\tilde{J}(\boldsymbol{w};\boldsymbol{X},\boldsymbol{y}) = \frac{\alpha}{2}\boldsymbol{w}^T\boldsymbol{w} + J(\boldsymbol{w};\boldsymbol{X},\boldsymbol{y})$$
$$\nabla_{\boldsymbol{w}}\tilde{J}(\boldsymbol{w};\boldsymbol{X},\boldsymbol{y}) = \alpha\boldsymbol{w} + \nabla_{\boldsymbol{w}}J(\boldsymbol{w};\boldsymbol{X},\boldsymbol{y})$$

On a single gradient step, the update is as follows:

$$\boldsymbol{w} \leftarrow (1-\epsilon\alpha)\boldsymbol{w} - \epsilon\nabla_{\boldsymbol{w}}J(\boldsymbol{w};\boldsymbol{X},\boldsymbol{y})$$

The addition of weight decay has modified learning rule to multiplicatively shrink weight vector by constant factor on each step before gradient update.
Using quadratic approximation to objective function at minimal unregularised training cost, approximation is

$$\hat{J}(\bar{\theta}) = J(\boldsymbol{w}^*) + \frac{1}{2}(\boldsymbol{w}-\boldsymbol{w}^*)\boldsymbol{H}(\boldsymbol{w}-\boldsymbol{w}^*), \qquad \boldsymbol{w}^* = \arg\min_{\boldsymbol{w}} J(\boldsymbol{w})$$

where $\boldsymbol{H}$ is Hessian matrix of J with respect to $\boldsymbol{w}$ evaluated at $\boldsymbol{w}^*$. Minimum of $\hat{J}$ occurs where gradient is

$$\nabla_{\boldsymbol{w}}\hat{J}(\boldsymbol{w}) = \boldsymbol{H}(\boldsymbol{w}-\boldsymbol{w}^*) = \boldsymbol{0}$$

Adding weight decay gradient, solve for minimum of regularised version of $\hat{J}$. Let $\tilde{\boldsymbol{w}}$ be minimum, then

$$\alpha\tilde{\boldsymbol{w}} + \boldsymbol{H}(\tilde{\boldsymbol{w}}-\boldsymbol{w}^*) = 0$$
$$(\boldsymbol{H}+\alpha\boldsymbol{I})\tilde{\boldsymbol{w}} = \boldsymbol{H}\boldsymbol{w}^*$$
$$\tilde{\boldsymbol{w}} = (\boldsymbol{H}+\alpha\boldsymbol{I})^{-1}\boldsymbol{H}\boldsymbol{w}^*$$

As $\alpha \to 0$, regularised solution $\tilde{\boldsymbol{w}} \to \boldsymbol{w}^*$. Note $\boldsymbol{H}$ is real and symmetric, hence decompose into diagonal matrix $\boldsymbol{A}$ and orthonormal basis of eigenvectors $\boldsymbol{Q}$ such that $\boldsymbol{H} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^T$, to get

$$\tilde{\boldsymbol{w}} = (\boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^T + \alpha\boldsymbol{I})^{-1}\boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^T\boldsymbol{w}^* = \boldsymbol{Q}(\boldsymbol{\Lambda}+\alpha\boldsymbol{I})^{-1}\boldsymbol{\Lambda}\boldsymbol{Q}^T\boldsymbol{w}^*$$

The effect of weight decay rescale $\boldsymbol{w}^*$ along axes defined by eigenvectors of $\boldsymbol{H}$. The component of $\boldsymbol{w}^*$ aligned with i-th eigenvector of $\boldsymbol{H}$ is rescaled by factor $\frac{\lambda_i}{\lambda_i+\alpha}$. For components where $\lambda_i \gg \alpha$, the effects of regularisation is relatively small. For components where $\lambda_i \ll \alpha$, components will be shrunk to nearly zero magnitude.

Definition 27.3.4. *L^1 Regularisation*
The L^1 regularisation is the sum of absolute values of individual parameters.

$$\Omega(\boldsymbol{\theta}) = \|\boldsymbol{w}\|_1 = \sum_i |w_i|$$

Definition 27.3.5. *Behaviour of L^1 Regularisation*
Assuming no bias parameter, a model has the following objective function and parameter gradient:

$$\tilde{J}(\boldsymbol{w};\boldsymbol{X},\boldsymbol{y}) = \alpha\|\boldsymbol{w}\|_1 + J(\boldsymbol{w};\boldsymbol{X},\boldsymbol{y})$$
$$\nabla_{\boldsymbol{w}}\tilde{J}(\boldsymbol{w};\boldsymbol{X},\boldsymbol{y}) = \alpha\ \mathrm{sign}(\boldsymbol{w}) + \nabla_{\boldsymbol{w}}J(\boldsymbol{w};\boldsymbol{X},\boldsymbol{y})$$

Note that the regularisation contribution to gradient is a constant factor with sign equal to $\mathrm{sign}(w_i)$. Minimum of $\tilde{J}$ occurs at where

$$\nabla_{\boldsymbol{w}}\tilde{J}(\boldsymbol{w}) = \boldsymbol{H}(\boldsymbol{w}-\boldsymbol{w}^*)$$

Assuming the Hessian is diagonal, $\boldsymbol{H} = \mathrm{diag}([H_{1,1},\dots,H_{n,n}])$, where each $H_{i,i} > 0$. The quadratic approximation of regularised objective function is then

$$\tilde{J}(\boldsymbol{w};\boldsymbol{X},\boldsymbol{y}) = J(\boldsymbol{w}^*;\boldsymbol{X},\boldsymbol{y}) + \sum_i \left[\frac{1}{2}H_{i,i}(\boldsymbol{w}_i-\boldsymbol{w}_i^*)^2 + \alpha\,|w_i|\right]$$
$$w_i = \mathrm{sign}(w_i^*)\max\left\{|w_i^*| - \frac{\alpha}{H_{i,i}}, 0\right\}$$

For the case where $w_i^* > 0$ for all i, then

i. if $w_i^* \le \frac{\alpha}{H_{i,i}}$, the optimal value is $w_i = 0$ as contribution of $J(\boldsymbol{w};\boldsymbol{X},\boldsymbol{y})$ to regularised objective $\tilde{J}(\boldsymbol{w};\boldsymbol{X},\boldsymbol{y})$ is overwhelmed in direction i by L^1 regularisation which pushes w_i to zero.
ii. if $w_i^* > \frac{\alpha}{H_{i,i}}$, regularisation shifts optimal value of w_i in the direction by $\frac{\alpha}{H_{i,i}}$.

For the case where $w_i^* < 0$, this happens similarly, but with L^1 penalty decreasing w_i by $\frac{\alpha}{H_{i,i}}$, with min value 0. Note that L^1 produces a more sparse solution, which is a feature selection mechanism.

Remark 27.3.6. *Norm Penalties as Constrained Optimisation*
Let cost function regularised by parameter norm penalty be

$$\tilde{J}(\boldsymbol{\theta};\boldsymbol{X},\boldsymbol{y}) = J(\boldsymbol{\theta};\boldsymbol{X},\boldsymbol{y}) + \alpha\Omega(\boldsymbol{\theta})$$

Construct a generalised generalised Lagrange function,

$$\mathcal{L}(\boldsymbol{\theta}; \boldsymbol{X}, \boldsymbol{y}) = J(\boldsymbol{\theta}; \boldsymbol{X}, \boldsymbol{y}) + \alpha(\Omega(\boldsymbol{\theta}) - k)$$

The solution to the constrained problem is then

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \max_{\alpha, \alpha \geq 0} \mathcal{L}(\boldsymbol{\theta}, \alpha)$$

Note that optimal value α^* will shrink $\Omega(\boldsymbol{\theta})$, but not such that it is less than k.
Fixing α^*, the problem is then a function of $\boldsymbol{\theta}$,

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}, \alpha^*) = \arg\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta}; \boldsymbol{X}, \boldsymbol{y}) + \alpha^* \Omega(\boldsymbol{\theta})$$

This is the regularised training problem of minimising $\tilde{J}$. The parameter norm penalty is imposing a constraint on the weights. If explicit constraints are to be used rather than penalties, use stochastic gradient descent on $J(\boldsymbol{\theta})$, then take the projected $\boldsymbol{\theta}$ to the nearest point that satisfies $\Omega(\boldsymbol{\theta}) < k$. This method is used if k is predefined by user, and time is not to be spent on searching for α value that corresponds to this k.
Explicit constraints and re-projection work better in circumstances where non-convex optimisation is involved, and this may avoid getting stuck in local minima. Explicit constraints also impose stability, and together with high learning rate allows rapid exploration of parameter space.

Remark 27.3.7. *Dataset Augmentation*
Effective technique for object classification. Operations such as translation of images by few pixels in each direction can greatly improved generalisation, even if the model has been designed to be partially translation invariant using convolution and pooling techniques. Rotation and scaling of images are also quite effective.
For speech recognition tasks, noice injection will improve performance. To improve robustness, train models with random noise applied to inputs. Noise injection also works when applied to the hidden units (data augmentation at multiple levels of abstraction). Enabling dropout constructs new inputs by multiplying by noise.

Remark 27.3.8. *Noise Robustness*
For some models, addition of noise with infinitesimal variance at input of model is equivalent to imposing penalty on norm of the weights. Generally, noise injection can be much more powerful than simply shrinking the parameters, especially when noise is added to hidden units.
Noise may also be added to weights, which is primarily used in recurrent neural networks. This is a stochastic implementation of Bayesian inference over the weights (model weights are uncertain and representable via a probability distribution that reflects this uncertainty).
Adding noise to weights may also result in more stability in function to be learned. Let the cost function be

$$J = \mathbb{E}_{p(x,y)}[(\hat{y}(\boldsymbol{x}) - y)^2]$$

where $\hat{y}(\boldsymbol{x})$ is function to be trained that maps set of features $\boldsymbol{x}$ to a scalar using least-squares cost function J. Assume each input presentation include a random perturbation $\epsilon_{\boldsymbol{W}} \sim \mathcal{N}(\boldsymbol{\epsilon}; \boldsymbol{0}, \eta \boldsymbol{I})$ in the network weight of a standard l-layer MLP. Let the perturbed model be $\hat{y}_{\boldsymbol{\epsilon_W}}(\boldsymbol{x})$. With injection o noise, objective function becomes

$$\tilde{J}_{\boldsymbol{W}} = \mathbb{E}_{p(\boldsymbol{x}, y, \boldsymbol{\epsilon_W})}\left[(\hat{y}_{\boldsymbol{\epsilon_W}}(\boldsymbol{x}) - y)^2\right] = \mathbb{E}_{p(\boldsymbol{x}, y, \boldsymbol{\epsilon_W})}\left[\hat{y}^2_{\boldsymbol{\epsilon_W}}(\boldsymbol{x}) - 2y\hat{y}_{\boldsymbol{\epsilon_W}} + y^2\right]$$

For small η, the minimisation of J with added noise (with covariance $\eta \boldsymbol{I}$) is equivalent to minimisation of J with additional regularisation term $\eta \mathbb{E}_{p(\boldsymbol{x}), y}[\|\nabla_{\boldsymbol{W}} \hat{y}(\boldsymbol{x})\|^2]$. This encourages parameters to go to regions of parameter space where small perturbations of weights have relatively small influence on output.

Remark 27.3.9. *Label Smoothing*
Datasets may have some mistake in y labels. Regularise a model based on softmax with k output values by replacing hard 0 and 1 classification targets with targets of $\frac{\epsilon}{k-1}$ and $1-\epsilon$ respectively, assuming that the training set label y is correct with probability $1 - \epsilon$. Cross-entropy can then be used on these soft targets.

Remark 27.3.10. *Semi-Supervised Learning*
Unlabelled examples $P(\boldsymbol{x})$ and $P(\boldsymbol{x}, \boldsymbol{y})$ are used to estimate $P(\boldsymbol{y} \mid \boldsymbol{x})$ or predict $\boldsymbol{y}$ from $\boldsymbol{x}$.
Construct a generative model of either $P(\boldsymbol{x})$ or $P(\boldsymbol{x}, \boldsymbol{y})$ which shares parameters with a discriminative model of $P(\boldsymbol{y} \mid \boldsymbol{x})$. The supervised criterion $\log P(\boldsymbol{y} \mid \boldsymbol{x})$ may then be traded with the unsupervised or generative one ($\log P(\boldsymbol{x})$ or $\log P(\boldsymbol{x}, \boldsymbol{y})$). The generative criterion then expresses a particular form of prior belief about the solution to the supervised learning problem (structure of $P(\boldsymbol{x})$ is connected to structure of $P(\boldsymbol{y} \mid \boldsymbol{x})$ such that is is captured by the shared parametrisation).

Remark 27.3.11. *Multi-Task Learning*
Improve generalisations by pooling examples (soft constraints imposed on parameters) from several tasks.
Different supervised tasks (predicting $\boldsymbol{y}^{(i)}$ given $\boldsymbol{x}$) share the same input $\boldsymbol{x}$, as well as some intermediate-level representation $\boldsymbol{h}^{(\text{shared})}$ capturing a common pool of factors. Model can be divided in two parts:

i. Task-specific parameters

ii. Generic parameters, shared across all the tasks

Improved generalisation and associated error bounds can be achieved due to shared parameters, but this only happens if some assumptions about statistical relationship between different tasks are valid.

Remark 27.3.12. *Early Stopping*
To obtain a model with better validation set error by returning to the parameter setting at the point in time with lowest validation set error. Model validation is done in parallel to model training.
Early stopping has the effect of restricting the optimisation procedure to a relatively small volume of parameter space in neighbourhood of initial parameter value $\boldsymbol{\theta}_o$. Assume τ optimisation steps are taken with learning rate ϵ. The product $\epsilon\tau$ is a measure of effective capacity. Assuming gradient is bounded, restricting $\epsilon\tau$ limits the volume of parameter space reachable from $\boldsymbol{\theta}_o$, hence $\epsilon\tau$ behaves as the reciprocal of weight decay coefficients.

Algorithm 11: Early Stopping Algorithm

Require:
Number of steps n between evaluations
Patience p, the number of times to observe worsening validation error before stopping
Initial parameters $\boldsymbol{\theta}_o$

$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}_o, \quad i \leftarrow 0, \quad j \leftarrow 0, \quad v \leftarrow \infty, \quad \boldsymbol{\theta}^* \leftarrow \boldsymbol{\theta}, \quad i^* \leftarrow i$
while $j < p$ **do**
  Update $\boldsymbol{\theta}$ by running training algorithm for n steps
  $i \leftarrow i + n, \quad v' \leftarrow \text{ValidationSetError}(\boldsymbol{\theta})$
  if v' < v **then**
    $j \leftarrow 0, \quad \boldsymbol{\theta}^* \leftarrow \boldsymbol{\theta}, \quad i^* \leftarrow i, \quad v \leftarrow v'$
  end if
  $j \leftarrow j + 1$
end while
Best parameters are $\boldsymbol{\theta}^*$, best number of training steps is i^*

Remark 27.3.13. *Parameter Tying and Parameter Sharing*
Used to express prior knowledge on suitable values of model parameters (precise values are not known, but there should be some dependencies between parameters). To regularise parameters to be close, may use:

i. Parameter norm penalty: assume model A with parameters $\boldsymbol{w}^{(A)}$ and model B with parameters $\boldsymbol{w}^{(B)}$. Two models map input to two different but related outputs $\hat{y}^{(A)} = f(\boldsymbol{w}^{(A)}, \boldsymbol{x})$ and $\hat{y}^{(B)} = g(\boldsymbol{w}^{(B)}, \boldsymbol{x})$. The model parameters $w_i^{(A)}$ should be close to $w_i^{(B)}$ $\forall i$. Using regularisation, a parameter norm penalty may be imposed, such as using L^2 penalty to get $\Omega(\boldsymbol{w}^{(A)}, \boldsymbol{w}^{(B)}) = \left\|\boldsymbol{w}^{(A)} - \boldsymbol{w}^{(B)}\right\|_2^2$.
ii. Parameter sharing: using constraints. Only a subset of the parameters (input set) need to be stored in memory. May lead to significant reduction in memory footprint of the model (i.e., in CCNs)

Remark 27.3.14. *Sparse Representations*
Representation sparsity is where elements of the representation are zero, accomplished by same sort of mechanisms used in parameter regularisation.
Given a linear regression $\boldsymbol{y} = \boldsymbol{B}\boldsymbol{h}$, where $\boldsymbol{y} \in \mathbb{R}^m, \boldsymbol{B} \in \mathbb{R}^{m \times n}, \boldsymbol{h} \in \mathbb{R}^n$, and $\boldsymbol{h}$ is the representation of data $\boldsymbol{x}$, add to loss function J a norm penalty on the representation ($\Omega(\boldsymbol{h})$)):

$$\tilde{J}(\boldsymbol{\theta}; \boldsymbol{X}, \boldsymbol{y}) = J(\boldsymbol{\theta}; \boldsymbol{X}, \boldsymbol{y}) + \alpha\Omega(\boldsymbol{h})$$

where $\alpha \in [0, \infty)$ weights relative contribution of norm penalty, with larger values of α for more regularisation. L^1 penalty on elements of representation induces sparsity, $\Omega(\boldsymbol{h}) = \|\boldsymbol{h}\|_1 = \sum_i |h_i|$. For representations with elements constrained to lie on unit interval, may use Student-t prior or KL divergence penalties.
Other approaches such as hard constraint on activation values may also work. *Orthogonal matching pursuit* encodes an input $\boldsymbol{x}$ with representation $\boldsymbol{h}$ that solves the constrained optimisation problem

$$\arg \min_{\boldsymbol{h}, \|\boldsymbol{h}\|_0 < k} \|\boldsymbol{x} - \boldsymbol{W}\boldsymbol{h}\|^2$$

where $\|\boldsymbol{h}\|_0$ is the number of non-zero entries of $\boldsymbol{h}$. If $\boldsymbol{W}$ is constrained to be orthogonal, this can be solved efficiently with OMP-k method, where k is number of non-zero features allowed.

Remark 27.3.15. *Bootstrap Aggregating (Bagging)*
Construct k different datasets, each with same number of examples as original dataset, but constructed by sampling with replacement from original dataset. Each dataset may miss some examples but also contain duplicate examples. Each model i is then trained on the dataset i. Differences in random initialisation, random selection of mini-batches, differences in hyperparameters, or different outcomes of non-deterministic implementation of neural networks are enough to allow each model to make partially independent errors.

Remark 27.3.16. *Boosting*
Construct k different datasets from original dataset. Using ith dataset, train ith weak learner. Test the ith weak learner with validation dataset, and each datapoint with wrong prediction is sent to $(i+1)$th dataset. The $(i+1)$th dataset is then used to train the $(i+1)$th weak learner. Continue until reaching the total number of subsets for the total prediction.

Remark 27.3.17. *Dropout*
Dropout trains the ensemble consisting of all sub-networks that can be formed by removing non-output units from the base network by multiplying its output value by zero. To train a neural network with dropout, use minibatch-based learning algorithms (i.e., Stochastic Gradient Descent). Each time an example is loaded into minibatch, randomly (and independently) sample a different binary mask $\boldsymbol{\mu}$ to apply to all of input and hidden units in the network, where the probability of sampling a mask value of one is the dropout rate.
Let $J(\boldsymbol{\theta}, \boldsymbol{\mu})$ be cost of model defined by parameters $\boldsymbol{\theta}$ and mask $\boldsymbol{\mu}$. Dropout training is minimising $\mathbb{E}_{\boldsymbol{\mu}} J(\boldsymbol{\theta}, \boldsymbol{\mu})$. Each model inherits different subset of parameters from parent neural network, and a tiny fraction of possible sub-networks are trained for a single step. Parameter sharing allows sub-networks to arrive at good parameter settings. Each sub-network given by $\boldsymbol{\mu}$ defines a probability distribution $p(y \mid \boldsymbol{x}, \boldsymbol{\mu})$. The arithmetic mean over all masks is then given by

$$\sum_{\boldsymbol{\mu}} p(\boldsymbol{\mu}) p(y \mid \boldsymbol{x}, \boldsymbol{\mu})$$

where $p(\boldsymbol{\mu})$ is the distribution used to sample $\boldsymbol{\mu}$ during training.
Using dropout as a regularisation technique reduces effective capacity of a model. To offset this, increase the size of the model. Typically optimal validation error is much lower when using dropout, but this comes at cost of larger model and more iterations of training.
For very large datasets, costs of using dropout and larger models may outweigh benefits of regularisation. For sparse datasets (extremely few labelled training examples available), Bayesian neural networks outperform dropout. When additional unlabelled data is available, unsupervised learning may gain advantage over dropout.

Remark 27.3.18. *Adversarial Training*
Adversarial example may be constructed by optimisation to search for an input $\boldsymbol{x}'$ near $\boldsymbol{x}$ such that the model output is very different at $\boldsymbol{x}'$.
The primary cause is excessive linearity in some neural networks. Changing input by ϵ causes linear function with weights $\boldsymbol{w}$ to change by as much as $\epsilon \|\boldsymbol{w}\|_1$, which can be very large if $\boldsymbol{w}$ is high-dimensional.
Adversarial examples assists in semi-supervised learning. At point $\boldsymbol{x}$ that is not associated with a label, model assigns label $\hat{y}$ (may not be true label). Seek adversarial example $\boldsymbol{x}'$ that causes classifier to output label $y' \neq \hat{y}$. Classifier may then be trained to assign same label to $\boldsymbol{x}$ and $\boldsymbol{x}'$, learning function robust to small changes.

Remark 27.3.19. *Tangent Distance Algorithm*
Non-parametric nearest-neighbour algorithm, where examples on the same manifold share the same category. The nearest-neighbour distance between points $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ is the distance between manifolds M_1 and M_2. Computationally cheap alternative is to approximate M_i by its tangent plane at $\boldsymbol{x}_i$ and measure the distance between two tangents, or between a tangent plane and a point.

Remark 27.3.20. *Tangent Propagation Algorithm*
Trains neural net classifier with extra penalty to make each output $f(\boldsymbol{x})$ of neural net locally invariant to factors of variation, which correspond to movement along manifold which examples from same class concentrate. Local invariance is achieved by requiring $\nabla_{\mathbf{x}} f(\boldsymbol{x})$ to be orthogonal to known manifold tangent vectors $\boldsymbol{v}^{(i)}$ at $\boldsymbol{x}$, or equivalently the directional derivative of f at $\boldsymbol{x}$ in directions $\boldsymbol{v}^{(i)}$ be small by adding regularisation penalty:

$$\Omega(f) = \sum_i \left(\left(\nabla_{\boldsymbol{x}} f(\boldsymbol{x}) \right)^T \boldsymbol{v}^{(i)} \right)^2$$

Tangent vectors are derived a priori from effect of transformations (i.e., translation, rotation, scaling).
Note this regularises model to resist infinitesimal perturbation, and the infinitesimal approach poses difficulties for models based on rectified linear units.
Data augmentation is the non-infinitesimal version of Tangent Propagation Algorithm, where network is explicitly trained to correctly classify distinct inputs created from more than infinitesimal transformation.

Remark 27.3.21. *Double Back-propagation Algorithm*
Algorithm regularises Jacobian to be small. Require model to be invariant to all directions of change in input so long as the change is small. Adversarial training is the non-infinitesimal version of Double Back-propagation Algorithm, where inputs were found near original inputs and model is trained to produce same input.

Remark 27.3.22. *Manifold Tangent Classifier*
Eliminates need to know tangent vectors a priori. Auto-encoders may estimate manifold tangent vectors, and

the classifier makes use of this technique to avoid requiring user-specified tangent vectors. The steps are:

i. Use auto-encoder to learn manifold structure by unsupervised learning
ii. Use these tangents to regularise a neural net classifier as in Tangent Propagation Algorithm.

27.4 Representation Learning

27.5 Sequence Models

27.6 Generative Models

27.7 Modern Architecture and Systems View

27.7.1 Convolutional Neural Networks

Definition 27.7.1. *Convolutional Neural Networks (CNNs)*
Neural network for processing data with a known, grid-like topology. Uses convolution instead of general matrix multiplication in at least one of their layers.

Definition 27.7.2. *Convolution*
The convolution of two functions f and g, denoted by $*$, is the integral of product of two functions after one is reflected about the y-axis and shifted.

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau$$

The first argument $f(\tau)$ is the input, and second argument $g(t-\tau)$ is the kernel. The output is feature map.

Definition 27.7.3. *Discrete Convolution*
For complex-valued functions f and g defined on set of $\mathbb{Z}$ integers, the discrete convolution of f and g is

$$(f * g)[n] = \sum_{m=-\infty}^{\infty} f[m]g[n-m]$$

If function g_N is periodic with period N, then the convolution is also periodic and identical to

$$(f * g_N)[n] = \sum_{m=0}^{N-1} \left(\sum_{k=-\infty}^{\infty} f[m+kN] \right) g_N[n-m]$$

Note that convolution is commutative, as the kernel is flipped relative to the input (as m increases, the index into input increases, but index into kernel decreases), but this is not an important property of neural network implementation. Neural network libraries implement cross-correlation instead.

Definition 27.7.4. *Cross-Correlation*
A measure of similarity of two series as a function of the displacement of one relative to the other. Also known as *sliding inner or dot product*. This is defined as

$$(f \star g)(\tau) = \int_{-\infty}^{\infty} \overline{f(\tau)}g(t+\tau)dt$$

This is related to convolution by

$$[f(t) \star gt](t) = [\overline{f(-t)} * g(t)](t)$$

Remark 27.7.5. *Motivation: Sparse Interactions*
The kernel is smaller than input, but is able to detect meaningful features. Hence fewer parameters are to be stored, reducing memory requirements of the model. Units in deeper layers may indirectly interact with a larger portion of the input, hence efficiently describe complicated interactions between many variables.

Remark 27.7.6. *Motivation: Parameter Sharing*
Each member of kernel is used at every position of the input except at boundary, which depends on design decisions regarding the boundary. Only one set of parameter is required to be learned. The runtime of forward propagation is at $O(k \times n)$ for $k \lll m$ connections for each input, given m input and n outputs.

Remark 27.7.7. *Motivation: Equivariant Representations*
An *equivariant* function is such that if the input changes, the output changes in the same way.
Applying a transformation, then convolution, is equivalent to applying convolution, then the transformation.

Remark 27.7.8. *Typical CNN Layers*
The layers of a typical CNN are:

i. Convolution: layer performs several convolutions in parallel to produce set of linear activations

ii. Detector: each linear activation is run through a non-linear activation function
iii. Pooling: modify the output of the layer further with summary statistics of nearby outputs

Example 27.7.9. *Example Types of Pooling*

i. Max Pooling: reports maximum output for each region of the feature map
ii. Average Pooling: reports average for each region of the feature map
iii. L^2 Pooling: calculates Euclidean norm (L^2 norm) for each region of the feature map

Remark 27.7.10. *Pooling on Invariance*
Pooling makes the representation approximately invariant to small translations of input.
Use of pooling is analogous to adding an infinitely strong prior that the function the layer learns must be invariant to small translations. As pooling summarises responses over whole neighbourhood, fewer pooling units than detector units may be used. To handle inputs of varying size, the size of offset between pooling regions may be changed so that the classification layer always receives the same number of summary statistics.

Remark 27.7.11. *Multi-Channel Operations*
Not commutative unless each operation has the same number of output and input channels.
Assume there exists 4-D kernel tensor $\mathcal{K}$ with element $K_{i,j,k,l}$ giving connection between strength between a unit in channel i of the output and a unit in channel j of the input, with offset of k rows and l columns between the output unit and input unit. Assume input consists of observed data $\mathcal{V}$ with element $V_{i,j,k}$ giving the value of input unit within channel i at row j and column k. Assume output consists of $\mathcal{Z}$ in same format as $\mathcal{V}$. If $\mathbb{Z}$ is produced by convolving $\mathcal{K}$ across $\mathcal{V}$ without flipping $\mathcal{K}$, then

$$Z_{i,j,k} = \sum_{l,m,n} V_{l,j+m-1,k+n-1} K_{i,l,m,n}$$

where summation over l, m, n is over all values for which tensor indexing operations inside summation is valid.

Remark 27.7.12. *Downsampled Convolution*
To reduce computational cost by sampling only every s pixels (*stride*) in each direction of output, then

$$Z_{i,j,k} = c(\mathcal{K}, \mathcal{V}, s)_{i,j,k} = \sum_{l,m,n} [\mathcal{V}_{l,(j-1)\times s+m,(k-1)\times s+n} \mathcal{K}_{i,l,m,n}]$$

Remark 27.7.13. *Zero Padding*
Zero-padding is adding zeroes to end of input sequence $\mathcal{V}$ so that the total number of samples is equal to the next higher multiple of two.

i. Valid Convolution: no zero-padding used. If input image has width m and kernel has width k, the output will be width $m - k + 1$. Rate of shrinkage can be dramatic if kernel used are large.
ii. Same Convolution: just enough zero-padding is added to keep size of output equal to size of input. Border pixels may be underrepresented in the model.
iii. Full Convolution: zeroes are added for every pixel to be visited k times in each direction, resulting in an input image of width $m + k - 1$.

Remark 27.7.14. *Locally Connected Layers (Unshared Convolution)*
Similar operation to discrete convolution with small kernel, but without parameter sharing. Adjacency matrix of MLP is the same, but every connection has its own weight as specified by 6-D tensor $\mathcal{W}$, where the indices of $\mathcal{W}$ are the output channel i, the output row j, the output column k, the input channel l, the row offset within the the input m, and the column offset within the input n. The liner part of the layer is then

$$Z_{i,j,k} = \sum_{l,m,n} [\mathcal{V}_{l,j+m-1,k+n-1} w_{i,j,k,l,m,n}]$$

Locally connected layers are useful when it is known that each feature should be function of a small part of space but would not occur across all of space.

Remark 27.7.15. *Tiled Convolution*
Compromise between convolutional layer and locally connected layer. A set of kernels will be rotated through space. Immediate neighbouring locations will have different filters, but memory requirements for storing the parameters will increase only by a factor of the size of this set of kernels.
Let k be a 6-D tensor, where two of the dimensions correspond to different locations in output matrix. Output locations cycle through a set of t different choices of kernel stack in each direction.

$$Z_{i,j,k} = \sum_{l,m,n} \mathcal{V}_{l,j+m-1,k+n-1} \mathcal{K}_{i,l,m,n,j \bmod t+1,k \bmod t+1}$$

28 Time Series and Sequential Systems

These notes follow Hamilton's time-series framework Hamilton (1994).

28.1 Time Series Foundations

28.1.1 Stationary Time Series

Definition 28.1.1. A *linear first-order difference equation* is defined as

$$y_t = \phi y_{t-1} + w_t$$

where y_t is the target variable (with y_{t-1} the lag 1 variable), w_t is an input variable at time t

The difference equation may be solved by recursive substitution to arrive at

$$y_t = \phi^{t+1} y_{-1} + \phi^t w_0 + \phi^{t-1} w_1 + \phi^{t-2} w_2 + \cdots + \phi w_{t-1} + w_t$$

The *dynamic multiplier* calculates effect of w_t on y_{t+j}, and is given by

$$\frac{\partial y_{t+j}}{\partial w_t} = \phi^j$$

If $|\phi| < 1$, the system is stable. If $|\phi| > 1$, then the system is explosive.

We may generalise the process to *p-th order difference equation*, i.e.,

$$y_t = \phi_t y_{t-1} + \phi_2 t_{t-2} + \cdots + \phi_p y_{t-p} + w_t$$

We may rewrite this as a first-order difference equation in a $(p \times 1)$ vector $\boldsymbol{\xi}_t$:

$$\boldsymbol{\xi}_t = \begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix}$$

Define the $(p \times p)$ matrix $\boldsymbol{F}$ by

$$\boldsymbol{F} = \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 & \cdots & \phi_{p-1} & \phi_p \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix}$$

Finally, define the $(p \times 1)$ vector $\boldsymbol{v}_t$ by

$$\boldsymbol{v}_t = \begin{bmatrix} w_t \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Then the system of p equations $\boldsymbol{\xi}_t = \boldsymbol{F}\boldsymbol{\xi}_{t-1} + \boldsymbol{v}_t$ is identical to the first order difference equation, and by recursive substitution, we have the following for the case of p-th order difference equation:

$$\boldsymbol{\xi}_{t+j} = \boldsymbol{F}^{j+1}\boldsymbol{\xi}_{t-1} + \boldsymbol{F}^j \boldsymbol{v}_t + \boldsymbol{F}^{j-1}\boldsymbol{v}_{t+1} + \cdots + \boldsymbol{F}\boldsymbol{v}_{t+j-1} + \boldsymbol{v}_{t+j}$$

Hence the dynamic multiplier is then $\frac{\partial y_{t+j}}{\partial w_t} f_{11}^{(j)}$, where $f_{11}^{(j)}$ is the $(1,1)$ element of $\boldsymbol{F}^j$.

28.1.2 Univariate Time Series Models

Trading activities through an exchange can be described by a sequence of time stamps ('ticks') $t_0 < t_1 < \cdots < t_n$, and 'marks' y_i at time t_i, where t_i denote market open, t_n denote market close. The marks y_i is a characteristic of the order book at time of ith activity. Events with marks associated with the ticks can be described mathematically as a marked point process.

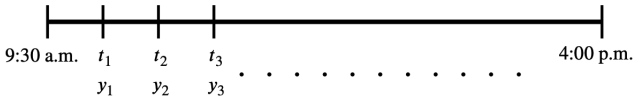


Figure 180: Ticks and Marks

A data aggregation method is where aggregation is conducted when there is a change in marker. An alternative aggregation method would be to divide the time span T for exchange hours into K intervals, so regularly spaced intervals are of size $\Delta t = T/K$. The time series method to be discussed will apply to all aggregated data.

Let $p_{it} = \ln P_{it}$ denote price of ith asset at time t; $p_t = (p_{1t}, p_{2t}, \dots, p_{nt})$ denote price vector for n assets; y_{it} denote vector of characteristics of ith asset at time t. These quantities are aggregated from high frequency data. Consider r factors $f_t = (f_{1t}, f_{2t}, \dots, f_{rt})$ that may include market and industry factors, and asset characteristics. Trading rules can be broadly grouped as follows:

i. *Statistical Arbitrage*: $E(p_{i,t+1} \mid p_{i,t}, p_{i,t-1}, \dots, y_{i,t}, y_{i,t-1}, \dots)$, that predicts the price of ith stock at $t+1$ based on past trading information (also known as time series momentum)
ii. *Momentum*: $E(p_{t+1} \mid p_t, p_{t-1}, \dots, y_t, y_{t-1}, \dots)$, that predicts the cross-sectional momentum of a subset of stocks based on past trading characteristics. For portfolio formation and rebalancing, pairs trading
iii. *Fair Value*: $E(p_{t+1} \mid p_t, p_{t-1}, \dots, y_t, y_{t-1}, \dots, f_t, f_{t-1}, \dots)$, predicts price using all relevant quantities. Factors normally include market, Fama-French; at a more macro level than timescale considered for price prediction, but may still be useful.

Hence price and volatility prediction can be formulated as a time series prediction problem. Autocorrelations and partial autocorrelations can be used to build autoregressive and ARCH models with some predictive power.

Let $Y_1, Y_2, \dots, Y_T$ be a sequence of random variables with a joint probability distribution. A sequence of observations of stochastic process $\{Y_t, t = 1, \dots, T\}$ is a realisation of the process. A time series $\{Y_t\}$ is *stationary* if for every integer m, the set of variables $Y_{t_1}, Y_{t_2}, \dots, Y_{t_m}$ depends only on the distance between times $t_1, t_2, \dots, t_m$. Thus $E(Y_t) = \mu$, $Var(Y_t) = \sigma^2$ are constant for all t. The *auto-covariance* function is defined as

$$\gamma(s) = Cov(Y_t, Y_{t-s}) = E[(Y_t - \mu)(Y_{t-s} - \mu)] \quad \forall s = 0, \pm 1, \dots$$

The *auto-correlation* of process at lag s is defined as

$$\rho(s) = Corr(Y_t, Y_{t-s}) = \frac{Cov(Y_t, Y_{t-s})}{(Var(Y_t)\ Var(Y_{t-s}))^{1/2}} = \frac{\gamma(s)}{\gamma(0)}, \quad \forall s = 0, \pm 1, \dots$$

Some examples of stationary stochastic processes are as follows:

Example 28.1.2. *(White Noise)* Sequence of discrete independent random variables (r.v.) $\{\epsilon_t\}$ with $E[\epsilon_t] = 0$ and $E[\epsilon_t^2] = \sigma^2$. Set $Y_t = \mu + \epsilon_t$, then $E[Y_t] = \mu$, $Cov(Y_t, Y_{t-s}) = Var(Y_t) = \sigma^2$ if $s = 0$, and $Cov(Y_t, Y_{t-s}) = 0$ if $s \neq 0$. Hence the process is stationary.

Example 28.1.3. *(Moving Average)* Let $\{\epsilon_t\}$ be independent r.v., with process $\{Y_t\}$ where $Y_t = \mu + \epsilon_t + \epsilon_{t_1}$ for $t = 0, 1, 2, \dots$, and μ is constant. Then $E[Y_t] = \mu\ \forall t$, and

$$Cov(Y_t, Y_{t-s}) = \gamma(s) = \begin{cases} 2\sigma^2 & \text{if } s = 0 \\ \sigma^2 & \text{if } s = 1 \\ 0 & \text{if } s > 1 \end{cases}$$

Hence $\{Y_t\}$ is stationary, with $\rho(s) = \gamma(s)/\gamma(0)$ such that $\rho(0) = 1$, $\rho(1) = 1/2$, $\rho(s) = 0$ for $|s| > 1$.

Some examples of non-stationary stochastic processes are as follows:

Example 28.1.4. *(Random Walk with Drift)* Let $\{\epsilon_t\}$ be sequence of independent r.v. with $E[\epsilon_t] = 0$, $E[\epsilon_t^2] = \sigma^2$, and define process $\{Y_t\}$ by $Y_t = Y_{t-1} + \delta + \epsilon_t$ with $Y_0 = 0$. Then the process can be summarised as

$$Y_t = \delta t + \sum_{j=1}^{t} \epsilon_j$$

Note that $E[Y_t] = \delta t$ and $Var[Y_t] = t\sigma^2$, hence process $\{Y_t\}$ is not stationary.

Given T observations from stationary process $\{Y_t\}$, the *sample mean* is $\overline{Y} = \frac{1}{T}\sum_{t=1}^{T} Y_t$. The *sample auto-covariance* function is defined by $\hat{\gamma}(s) = \frac{1}{T}\sum_{t=1}^{T-s}(Y_t - \overline{Y})(Y_{T-s} - \overline{Y})$ for $s = 0, 1, \dots$. The *sample auto-correlation* function (ACF) is defined as $\hat{\rho}(s) = \frac{\hat{\gamma}(s)}{\hat{\gamma}(0)} = r(s)$. The *sample variance* is $Var(\overline{Y}) = \frac{\gamma(0)}{T}\left[1 + 2\sum_{u=1}^{Y-1}\left(\frac{T-u}{T}\right)\rho(u)\right]$; note that it has to account for auto-correlations.

Definition 28.1.5. A stochastic process $\{Y_t\}$ is a *linear process* if it can be represented as

$$Y_t = \mu + \sum_{j=0}^{\infty} \Psi_j \epsilon_{t-j}$$

where ϵ_t are independent with mean 0, variance σ_ϵ^2, and $\sum_{j=0}^{\infty} < \infty$.

28.2 ARMA and ARIMA Structure

28.3 Spectral Methods

28.4 State-Space Models

28.5 Nonlinear and Volatility Models

28.6 Long Memory and Multiscale Structure

28.7 Sequential Inference and Filtering

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